

TREATISE ON THE CIRCLE

Volume I – Foundations

Euclidean Geometry, Analytic Geometry, Trigonometry and Robotic Applications



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Complete document on the circle

Preface

This is the first volume of the *Treatise on the Circle*, and it constitutes its foundations: everything the later volumes will use without proving again is to be found here. The guiding thread is that the circle is not one topic among many, but an object that reappears, always recognisable, at every level of mathematics and of engineering — from the taut rope of Neolithic builders to the path interpolated by the numerical controller of an industrial robot. The volume climbs that scale from the bottom: from the history of π to the primitive definitions, from the theorems on chords and angles to circle inversion, from the measurement of length and area to analytic geometry and trigonometry, to Gauss's circle problem in number theory, up to the representation of the circle inside a computer, the circular motion of a machine tool, and the sizing of a circular settler in a water treatment plant.

Target audience. Three different readers, and the structure of the volume takes all three into account. The secondary-school student, who in Chapters 2–4, 6 and 9 finds the entire school syllabus on the circle with every formula proved and not merely stated. The undergraduate, who in Chapters 5, 7, 8, 11 and 12 finds circle inversion, analytic geometry, non-circular trigonometry, numerical methods and Gauss's circle problem treated with the rigour that will be demanded of him. The practising engineer — mechanical designer, CNC programmer, robotics integrator, water treatment plant engineer — who in Chapters 13 and 14 (plus Chapter 11) finds the mathematics behind the G02/G03, MOVc, MoveC and CIRC instructions he uses every day, the reasons why a programmed arc comes out of the machine slightly smaller than the one drawn, and the circular sizing of a settler. A reader may skip the chapters that do not concern him without losing the thread: the logical dependencies are declared at the beginning of each chapter.

How it is organised. Each chapter follows the same pattern: an opening box declaring purpose and prerequisites; the theory, in which every definition is followed by its consequences and every statement by its complete proof (and, where the literature offers more than one independent argument, by one or more alternative proofs, because the same result may hide more than one idea); observations connecting the result to the rest of the treatise or to an industrial application; a set of exercises divided into numerical ones and symbolic-and-proof ones, each with a fully worked solution; a multiple-choice self-assessment quiz with a commented key; a chapter Formula sheet; and finally the verified bibliographic references. The volume closes with a final quiz, a glossary, a one-page cheat sheet and the index.

Everything proved, or declared imported. This is the rule that governs the whole treatise, and this volume applies it without exceptions. Every theorem, lemma, proposition and corollary is followed by a proof written out in full; when a result is taken from outside — because proving it would require tools not yet available at that point — the fact is stated explicitly in a box, together with what is missing in order to conclude and with the source. A proof that claims to be complete and then silently assumes a step is not a proof. For the same reason every numerical value appearing in the text has been computed twice, by independent routes, and the check is shown to the reader.

Prerequisites. None, for Chapters 1–4: elementary plane geometry and secondary-school algebra suffice, and the axioms actually used are listed in Chapter 2. Chapter 6 requires the least-upper-bound property of $\mathbb{R}$; Chapter 7 requires Cartesian coordinates; Chapter 11 some notion of floating-point arithmetic; Chapter 12 the notion of area from Chapter 6 and elementary arithmetic (divisors, residues modulo 4), no prior number theory; Chapter 13 no previous

knowledge of robotics, which is built from scratch out of geometry; Chapter 14 no previous knowledge of plant engineering, for the same reason. Differential and integral calculus is *not* a prerequisite of this volume: where it is needed, it is recalled as a statement and deferred to Volume II.

Difficulty and reading paths. The exercises of each chapter are divided into two groups: the *numerical* ones, which ask that a formula or a construction be applied to concrete data and serve to fix the method; and the *symbolic and demonstrative* ones, which ask that a statement be proved or a general formula derived, and are appreciably more demanding. All of them have the complete solution immediately after the text, designed for self-checking. The more advanced sections — the Mohr–Mascheroni theorem and the Poncelet–Steiner theorem in Chapter 5, the isoperimetric inequality and the proof of Archimedes’ postulate in Chapter 6, Feuerbach’s theorem in Chapter 10 — may be accepted as stated on a first reading without compromising the understanding of the rest: they are flagged by a box declaring their prerequisites.

Where the discussion continues. Volume II takes up the circle again with the tools of calculus: parametrisation and curvature, integrals, the complex exponential, Fourier series, applications to electrical engineering and to statistics. Volume III continues with differential and global geometry, the sphere, the topology of S^1 , transforms and signals. Every time a topic of this first volume finds its natural sequel there, the forward reference is given in the text.

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CHAPTER 1

History of the Circle and the Constant π

1.1. How this chapter is organised

The circle is the oldest and most universal figure in mathematics. Its presence in human material culture goes back at least to the Neolithic: the wheel, thrown pottery, ceremonial enclosures, solar calendars. Almost every documented civilisation has had, sooner or later, to answer the same practical question — *how long is the rim, how much is inside* — and the answers it gave say a great deal about how it thought.

This chapter follows those answers culture by culture. For each, where the sources allow, it gives:

- (a) the **primary source** (tablet, papyrus, treatise, artefact);
- (b) the **period method**, reconstructed in the terms in which it was conceived rather than hastily restated in modern notation;
- (c) a **modern note** quantifying how close that result came to the true value.

Definition 1.1: Relative error of an approximation of π . Given an approximation p of π , the **relative error** is

$$\varepsilon(p) := \frac{|p - \pi|}{\pi},$$

which we shall express as a percentage or in scientific notation. The number of **correct decimal digits** is the number of decimals of p that agree with those of $\pi = 3.14159\,26535\,89793 \dots$ before the first discrepancy.

Warning. Three historiographical cautions, to be kept in mind throughout.

Using circles is not computing π . Many cultures built enclosures, dwellings, mandalas and circular calendars of great precision without ever isolating the ratio C/d as an object of study. These are two different achievements, and it is wrong to credit the first with the merit of the second.

The “implicit π ” is our reconstruction. When a source gives a rule for the area (say, “the field is the square on eight ninths of the diameter”), we can invert it and recover the value of π it presupposes. That number is ours, not theirs: no Egyptian scribe was thinking of a constant.

Absence of evidence is not evidence of absence. For entire regions of the world the written record has been lost or never existed. Where we have no numerical values we shall say so explicitly, rather than filling the gap with conjecture.

1.2. Before writing: circles without numbers

1.2.1 Neolithic circular enclosures

The oldest deliberately built circles that survive are ceremonial enclosures and solar observatories, all of them predating writing.

- **Nabta Playa** (Nubian desert, southern Egypt, ca. 4800 BCE): a small circle of standing stones with pairs of slabs defining a sight line towards sunrise at the summer solstice.
- **Goseck circle** (Saxony-Anhalt, ca. 4900 BCE): three concentric rings of ditch and palisade, outer diameter about 75 m, with three oriented gateways; it is the oldest known solar observatory in Europe.
- **Newgrange** (Ireland, ca. 3200 BCE): a circular mound about 85 m in diameter with a passage aligned to sunrise at the winter solstice; the kerbstones carry incised triple spirals.
- **Stonehenge** (Wiltshire, phases from 3000 to 1600 BCE): the Aubrey circle has 56 pits on a circumference some 87 m across; the sarsen circle has 30 uprights on about 33 m.

***Remark 1.1: What we may and may not infer.** Laying out a 30 m circle on the ground with a rope and a peg requires no knowledge of π whatsoever: it requires a rope. Dividing its rim into 30 or 56 equal parts, on the other hand, requires a technique of repeated bisection, and that is genuine practical geometry. In the 1960s Alexander Thom argued that many British rings are not circles but “flattened circles” built on Pythagorean triples with a common unit (the megalithic yard, 0.829 m); the thesis is today largely discounted by archaeologists, who attribute the irregularities to terrain and construction technique. We report it because it is a good example of how an archaeological datum can be over-interpreted.*

1.3. Mesopotamia and Elam

1.3.1 The standard Babylonian rule

Primary sources: Old Babylonian mathematical tablets (ca. 1900–1600 BCE), among them BM 85194 and the metrological texts from Nippur.

Babylonian scribes did not start from the diameter but from the **circumference**, the quantity one measures directly with a rope round a ditch or a silo. Their rule for the area is

$$A = \frac{C^2}{12}.$$

Step-by-step proof. What value of π does it presuppose? Since $C = 2\pi r$ and $A = \pi r^2$, we have identically

$$\frac{C^2}{4\pi} = \frac{4\pi^2 r^2}{4\pi} = \pi r^2 = A.$$

Comparing with the Babylonian rule $A = C^2/12$ gives $4\pi = 12$, that is

$$\pi_{\text{Babylon}} = 3.$$

□

1.3.2 The Susa tablet

Primary source: the Susa tablet (Elam, south-western Iran, ca. 1900–1600 BCE), recovered in the French excavations and published in 1961.

The tablet does not give π : it gives the ratio of the *perimeter of the inscribed regular hexagon* to the circumference containing it, in sexagesimal notation, as

$$0;57,36 = \frac{57}{60} + \frac{36}{60^2} = 0.96.$$

Step-by-step proof. Step 1. The side of the regular hexagon inscribed in a circle of radius r is exactly r (the six central triangles are equilateral). The hexagon's perimeter is therefore $6r$.

Step 2. The circumference is $2\pi r$. The ratio stated on the tablet is thus

$$\frac{6r}{2\pi r} = \frac{3}{\pi} = 0.96.$$

Step 3. Solving,

$$\pi_{\text{Susa}} = \frac{3}{0.96} = \frac{25}{8} = 3.125.$$

□

***Remark 1.2: Modern note — Mesopotamia.** The everyday value $\pi = 3$ has a relative error of 4.51%: on a silo 10 m across it misses the area by some 3.5 m². The Susa value $25/8 = 3.125$ brings this down to 0.53%, an improvement of nearly tenfold. What is most remarkable, though, is the method: starting from the inscribed hexagon is precisely the first step of the procedure Archimedes would carry through fifteen centuries later. The Babylonians stopped at the first term of the sequence; Archimedes doubled it four times.*

1.4. Pharaonic Egypt

1.4.1 The Rhind Papyrus

Primary source: the Rhind Papyrus (British Museum EA 10057–10058), copied by the scribe Ahmose about 1650 BCE from an original some two centuries older.

Problem 50 states the rule: *for a circular field of diameter 9 khet, take away a ninth of the diameter and build the square on the remainder*. Problem 48 accompanies the rule with a figure in which an octagon is inscribed in a square, perhaps the visual justification of the procedure.

Step-by-step proof. Step 1. The rule says $A = (d - \frac{d}{9})^2 = (\frac{8}{9}d)^2$.

Step 2. The true area is $A = \pi r^2 = \pi d^2/4$. Equating,

$$\frac{\pi d^2}{4} = \frac{64}{81}d^2 \implies \pi_{\text{Rhind}} = \frac{256}{81} = 3.16049\dots$$

Step 3 (a possible origin). Inscribe in a square of side 9 the octagon obtained by cutting the four corners with right isosceles triangles of leg 3. Its area is $81 - 4 \cdot \frac{9}{2} = 63$, very close to $64 = 8^2$. Replacing 63 by the perfect square 64 makes the rule immediately computable with Egyptian unit fractions. □

1.4.2 The Moscow Papyrus

Primary source: the Moscow Mathematical Papyrus (E 4676), ca. 1850 BCE.

Problem 10 computes the surface of a basket or hemisphere of diameter $4\frac{1}{2}$, applying the same “take away a ninth” twice. Whether the object is a hemisphere or a half-cylinder is still debated; on either reading the implied value of π remains $256/81$.

Remark 1.3: Modern note — Egypt. $256/81 = 3.16049$ has a relative error of 0.602%, in excess. This is a surprising precision for 1650 BCE, and better than the standard Babylonian value by almost a factor of eight. It must be noted, however, that it is a practical rule, not an estimate with an error bound: the papyrus does not say, and could not say, how far from the truth it is. The difference between a good recipe and a certified estimate is the conceptual leap that only Greece would make.

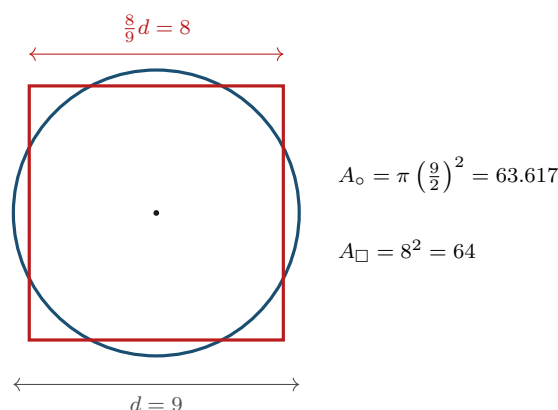


Figure 1.1: The Egyptian rule of the Rhind papyrus (problem 50): the area of the circle is taken to equal that of the square whose side is $\frac{8}{9}$ of the diameter. For $d = 9$ the square has side 8 and area 64, against the true area $\pi(9/2)^2 = 63.617$: the relative error is 0.60%. The rule amounts to taking $\pi \simeq (16/9)^2 = 256/81 = 3.1605$, and is the best documented approximation before the Greeks. Note that the scribe gives no justification whatsoever: it is an operational recipe, not a theorem.

1.5. The Levant and the Jewish tradition

Primary sources: *1 Kings* 7:23 and *2 Chronicles* 4:2 (redacted between the eighth and sixth centuries BCE).

The text describes the “molten sea” of Solomon’s Temple: ten cubits from brim to brim, and a line of thirty cubits to compass it round about. The implicit ratio is $30/10 = 3$.

Remark 1.4: Modern note — the molten sea. The value 3 has a relative error of 4.51%. In this particular case, though, the literal reading is probably unfair: the text also gives the thickness of the rim (a handbreadth) and the flared, “lily-blossom” profile; measuring the circumference inside and the diameter outside yields a ratio appreciably smaller than π without any error of measurement at all. It is a good exercise in critical reading: before charging a source with a mathematical mistake, it is worth asking what exactly was measured.

In the later rabbinic tradition the value 3 was retained as a *legal norm* for computing ritual measures, in full awareness that it was an approximation. The treatise *Mishnat ha-Middot* (probably second century CE, the oldest known Hebrew geometrical text) uses 3 in practical calculation. In the twelfth century Moses Maimonides, in his commentary on the Mishnah (*Eruvin* 1:5), makes a striking observation: the ratio of circumference to diameter *cannot be*

known exactly, not through any defect of instruments but by the nature of the number itself, and it is for this reason that the sages adopted the approximate value 3. Six centuries before Lambert, this is the irrationality of π stated as a principle.

1.6. The Aegean and Anatolia: Minoans, Mycenaeans, Hittites

For these civilisations we possess no mathematical texts: Linear A remains undeciphered and Linear B records warehouse accounts, not geometry. What we have is constructional evidence.

- **Minoans** (Crete, 1900–1450 BCE): the *kouloures*, circular pits at the palace of Knossos, and seals with radial decoration; circular tholos tombs.
- **Mycenaeans** (1600–1100 BCE): the tholos tombs, of which the “Treasury of Atreus” at Mycenae (ca. 1250 BCE) is the greatest, with a corbelled dome on a circular plan 14.5 m across; Grave Circle A, a circular enclosure some 28 m in diameter.
- **Hittites** (Anatolia, 1650–1180 BCE): the fortifications of Hattusa, with towers and arched postern gates.

Remark 1.5: Modern note — the Aegean. The corbelled dome of the Treasury of Atreus requires, at every course of blocks, that a circle of predetermined radius be laid out and divided into a whole number of ashlar: it is a problem of dividing the circumference, solved thirty-three times over with tolerances of a few centimetres. We know of no Mycenaean value of π , and it would be wrong to infer one; but we may record constructional competence of the first order.

1.7. Greece: from measurement to proof

It is in Greece that the circle ceases to be a problem of measurement and becomes a problem of proof. The transition is not immediate: it takes two centuries.

1.7.1 The first attempts (fifth century BCE)

- **Anaxagoras of Clazomenae** (ca. 430 BCE) worked on squaring the circle, according to Plutarch while imprisoned in Athens. It is the earliest attestation of the problem.
- **Antiphon** proposed inscribing a polygon and doubling its sides indefinitely until it “co-incides” with the circle. The argument is logically defective — a polygon never becomes a circle — but it contains the germ of the method of exhaustion.
- **Bryson of Heraclea** corrected it decisively: he took *inscribed and circumscribed* polygons together, observing that the area of the circle lies between the two. This is the birth of estimation by trapping.
- **Hippocrates of Chios** (ca. 440 BCE) proved that certain **lunes** (regions bounded by two circular arcs) can be squared exactly with straightedge and compass, and established that the areas of circles are to one another as the squares on their diameters.

Remark 1.6: Why Hippocrates’ lunes do not settle the problem. Hippocrates squared a curvilinear figure with straightedge and compass: a genuine and startling result, which for centuries sustained the hope of squaring the circle as well. The hope was misplaced, though not through any defect of the method: the squarable lunes are those in which the two arcs stand in an angular ratio that makes the transcendental contributions cancel, whereas for the

whole circle the contribution remains. Only Lindemann's proof (1882, Section 1.22) would make clear why.

1.7.2 Eudoxus and Euclid: the method of exhaustion

Eudoxus of Cnidus (ca. 370 BCE) supplied the rigorous foundation, later transmitted by Euclid in Book XII of the *Elements*.

Theorem 1.1: Elements XII, 2 (Euclid, ca. 300 BCE). Circles are to one another as the squares on their diameters:

$$\frac{A_1}{A_2} = \frac{d_1^2}{d_2^2}.$$

Proof. (Modern version; Euclid's own proof is by double reductio ad absurdum with the exhaustion lemma, and is reproduced in full in Chapter 6.)

By Theorem 6.5 (Chapter 6), the area of a circle of radius r is $A = \pi r^2$, with π the same constant for every circle (Theorem 6.2). Writing $d_i = 2r_i$ for the two diameters,

$$\frac{A_1}{A_2} = \frac{\pi r_1^2}{\pi r_2^2} = \frac{r_1^2}{r_2^2} = \frac{(d_1/2)^2}{(d_2/2)^2} = \frac{d_1^2}{d_2^2},$$

having cancelled π (legitimate since $\pi \neq 0$) and the factor $1/4$.

Non-circularity note. The chain of Chapter 6 (rectifiability $\rightarrow$ constancy of π by homothety $\rightarrow$ Archimedes by exhaustion) nowhere uses Proposition XII,2: the derivation above is therefore legitimate and not circular. $\square$

Remark 1.7: What Euclid proves and what he does not. Book III of the *Elements* systematises, in axiomatic form, chords, tangents, inscribed angles and the power of a point: it is the first deductive treatment of the circle in history. But Euclid never gives a numerical value for π , and not through oversight: Proposition XII,2 states exactly that the ratio A/d^2 is the same for all circles, that is, that a constant exists. What that constant may be is, for Euclid, a question of a different kind — and indeed the exact answer is not expressible in his language of ratios between constructible magnitudes.

1.7.3 Archimedes: the first estimate with a certified error

Primary source: Archimedes of Syracuse, *Measurement of a Circle* (ca. 250 BCE), transmitted through the commentary of Eutocius of Ascalon (sixth century CE).

Theorem 1.2: Measurement of a Circle, Proposition 1. The area of a circle equals that of a right triangle whose legs are the radius and the rectified circumference:

$$A = \frac{1}{2} r C.$$

Proof. The statement coincides with Theorem 6.5 (Chapter 6), where it is proved in full by Archimedes' double reductio ad absurdum, resting on the exhaustion lemma (Lemma 6.4, i.e. Euclid *Elements* X.1) and on the estimates for inscribed and circumscribed polygons (Lemma 6.5). It is recorded here in its *historical* place, to show that the link $A = \frac{1}{2} r C$ between area and length predates any notion of integral by nearly two millennia; the complete logical verification is deferred to the chapter that has the tools for it. $\square$

Theorem 1.3: Measurement of a Circle, Proposition 3.

$$3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}, \quad \text{that is} \quad 3.14084 < \pi < 3.14286.$$

Step-by-step proof. Step 1. One begins with the inscribed and circumscribed regular hexagon ($n = 6$) and doubles the number of sides four times: $6 \rightarrow 12 \rightarrow 24 \rightarrow 48 \rightarrow 96$.

Step 2. At each doubling Archimedes applies the recursive relations between the perimeters, equivalent in modern notation to $P_n = 2nr \sin(\pi/n)$ and $Q_n = 2nr \tan(\pi/n)$, with $P_n < C < Q_n$.

Step 3. Every square root arising in the computation is replaced by a fraction, always *on the correct side* so that the inequality is preserved. The crucial case is

$$\frac{265}{153} < \sqrt{3} < \frac{1351}{780},$$

bounds whose provenance Archimedes does not explain and which remain one of the small enigmas in the history of mathematics.

Step 4. With $n = 96$ one obtains the two stated bounds. □

Example 1.1: Archimedes' computation redone in full. Let us redo Archimedes' four doublings in modern notation, to see exactly where his two numbers come from. We work with the ratios *perimeter over diameter*, which both tend to π :

$$a_n = \frac{\text{circumscribed perimeter}}{d} = n \tan \frac{\pi}{n}, \quad b_n = \frac{\text{inscribed perimeter}}{d} = n \sin \frac{\pi}{n}, \quad b_n < \pi < a_n.$$

Step 1 — the two doubling relations. One has

$$a_{2n} = \frac{2a_nb_n}{a_n + b_n} \quad (\text{harmonic mean}), \quad b_{2n} = \sqrt{a_{2n}b_n} \quad (\text{geometric mean}).$$

These are exactly the ruler-and-compass constructions Archimedes performs at each bisection: halving the central angle amounts to taking first the harmonic and then the geometric mean. Their virtue is that both are computed with four operations and one square root — no trigonometric function whatever.

Step 2 — the starting point, the hexagon ($n = 6$). The side of the inscribed hexagon equals the radius, so the perimeter is $6r = 3d$:

$$b_6 = 3 \text{ exactly}, \quad a_6 = 6 \tan 30^\circ = \frac{6}{\sqrt{3}} = 2\sqrt{3} = 3.4641016.$$

(Here we recognise Liu Hui's observation reported in Chapter 10: $b_6 = 3$ explains the value $\pi \approx 3$ of the older texts.)

Step 3 — the four doublings.

n	sides	b_n (inscribed)	a_n (circumscribed)
6	6	3.0000000	3.4641016
12	12	3.1058285	3.2153903
24	24	3.1326286	3.1596599
48	48	3.1393502	3.1460862
96	96	3.1410319	3.1427146

One doubling in detail (from 6 to 12):

$$a_{12} = \frac{2 \cdot 3.4641016 \cdot 3}{3.4641016 + 3} = \frac{20.784610}{6.4641016} = 3.2153903,$$

$$b_{12} = \sqrt{3.2153903 \cdot 3} = \sqrt{9.6461709} = 3.1058285.$$

Step 4 — the exact outcome of the procedure.

$$3.1410319 < \pi < 3.1427146.$$

Step 5 — why Archimedes announces *worse* numbers. His bounds are

$$3 + \frac{10}{71} = 3.1408451, \quad 3 + \frac{1}{7} = 3.1428571,$$

that is, a bracket *wider* than the one just computed:

$$3.1408451 < \mathbf{3.1410319} < \pi < \mathbf{3.1427146} < 3.1428571. \quad \checkmark$$

This is not an error, it is **rigour**. Archimedes has no decimals: at every square root he must substitute a fraction, and he always substitutes it *on the side that preserves the inequality* — from below when bounding the inscribed figure, from above when bounding the circumscribed one. Each rounding widens the bracket slightly; after the four doublings the total widening is about 2×10^{-4} , and that is precisely what separates the two computed numbers from the two announced ones.

Step 6 — the cost of one more digit. The polygon error decreases as $1/n^2$: to gain one decimal digit (a factor of 10 on the error) the number of *sides* must be multiplied by $\sqrt{10} \approx 3.16$, i.e. about 1.66 doublings. Taking the estimate from 3 to 6 correct digits (a factor of 1000 on the error) requires $n \approx 96 \cdot \sqrt{1000} \approx 96 \cdot 31.6 \approx 3000$ sides, i.e. **about five** further doublings, not ten: it is precisely the polygon of $96 \cdot 2^5 = 3072$ sides with which, in the Chinese tradition, Liu Hui will obtain $\pi \approx 3.14159$ (later in this chapter). With $n = 10^5$ sides the relative error falls below 10^{-9} , i.e. nine to ten correct digits. Each doubling nonetheless requires a square root computed by hand on fractions with dozens of digits. This is why the polygon method remains the best available for **eighteen centuries**, and is superseded only by the infinite series of the seventeenth.

***Remark 1.8: Modern note — Archimedes.** The point is not the value but the shape of the result. Archimedes does not say “ π is about 3.14”; he says that π lies with certainty in an explicit interval of width $1/497 \approx 0.002$. It is the first estimate in history equipped with a proved margin of error, and it sets a standard mathematics has never since abandoned. The midpoint of the interval, ≈ 3.14185 , has relative error 8.2×10^{-5} ; the celebrated $22/7$, which is merely the upper bound, has error 4.0×10^{-4} , that is 0.040%. It should also be noted that the method is indefinitely refinable: nothing in principle forbids further doubling. That is exactly what the Chinese, the Persians and the Dutch would do, with increasing patience, for the next eighteen centuries.*

1.7.4 Apollonius, Hero, Ptolemy

- **Apollonius of Perga** (ca. 200 BCE) wrote, according to Eutocius, an *Okytokion* (“quick delivery”) containing a value of π better than the Archimedean one. The work is lost and the value has not come down to us.
- **Hero of Alexandria** (first century CE), in the *Metрика*, uses $22/7$ for practical computation and reports estimates attributed to Archimedes tighter than those that survive.
- **Claudius Ptolemy** (ca. 150 CE), in the *Almagest*, adopts the sexagesimal value

$$\pi \approx 3; 8, 30 = 3 + \frac{8}{60} + \frac{30}{60^2} = \frac{377}{120} = 3.141\bar{6}.$$

Remark 1.9: Modern note — Ptolemy. $377/120$ has relative error 2.36×10^{-5} , that is 0.0024%: three correct decimal places. Ptolemy derived it from his table of chords, built by means of the theorem that bears his name and by repeated bisection of the angle, which makes it an estimate trigonometric rather than polygonal in nature. It would remain the best value in use around the Mediterranean for over a millennium.

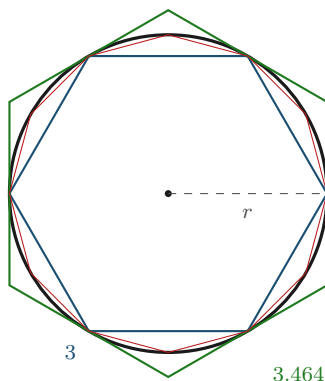


Figure 1.2: Archimedes' method of exhaustion. The perimeter of every inscribed polygon is smaller than the circumference, that of every circumscribed polygon larger: this yields a pair of estimates squeezing π from both sides. With hexagons (blue and green) one starts from $3 < \pi < 2\sqrt{3} = 3.464$; doubling the sides gives the dodecagons (the inscribed one in red), which yield $3.106 < \pi < 3.215$. Archimedes doubled four times, reaching 96-gons and the celebrated estimate $3\frac{10}{71} < \pi < 3\frac{1}{7}$, i.e. $3.1408 < \pi < 3.1429$. The decisive novelty is not the numerical value but the method: for the first time π is enclosed between two proved bounds.

1.8. Rome

Rome produced no theoretical mathematics of the circle, but made systematic engineering use of it, and the Latin technical sources preserve working values.

- **Vitruvius**, *De Architectura* X,9, describing the odometer, assumes that a wheel 4 feet in diameter covers $12\frac{1}{2}$ feet per revolution: implicitly $\pi = 12.5/4 = 3.125$.
- **Columella**, *De Re Rustica* V,2, gives the area of the circle as $\frac{11}{14}d^2$, that is $\pi = \frac{44}{14} = \frac{22}{7}$.
- The corpus of the *agrimensores* (land surveyors) applies $22/7$ to centuriation and to the measurement of fields.

Remark 1.10: Modern note — Rome. *Vitruvius'* 3.125 (error 0.53%) coincides with the *Susa* value of two thousand years earlier; *Columella's* $22/7$ (error 0.040%) is Archimedes' upper bound adopted as a working value. It is an instructive case: Roman culture did not improve the estimate, but knew how to choose, among the values available, the one with the best practical return — $22/7$ can be computed in the head and errs by less than half a millimetre per metre.

1.9. India

The Indian tradition spans nearly three millennia and culminates, in fourteenth-century Kerala, in a result Europe would reach only two hundred and fifty years later.

1.9.1 The Śulbasūtra (ca. 800–500 BCE)

Primary sources: the *Baudhāyana*, *Āpastamba* and *Kātyāyana Śulbasūtra*, manuals for the construction of Vedic altars.

The problem is not to measure the circle but to *transform* it: to build a circular altar equal in area to a given square altar, and conversely. The two constructions, being approximate, presuppose different values of π .

Step-by-step proof. From square to circle (Baudhāyana 2.9): carry half the diagonal from the centre towards the side and add a third of the part that projects beyond. This gives

$$r = s \frac{2 + \sqrt{2}}{6} \approx 0.569 s,$$

whence, imposing $\pi r^2 = s^2$, one obtains $\pi \approx 3.088$.

From circle to square (Baudhāyana 2.10): take as side $s = \frac{13}{15}d$, whence $\pi = 4\left(\frac{13}{15}\right)^2 = \frac{676}{225} \approx 3.00444$. $\square$

1.9.2 The Jaina tradition and the early astronomers

The *Sūryaprajñapti* and other Jaina texts (ca. fourth century BCE) adopt

$$\pi = \sqrt{10} = 3.16227 \dots$$

a value destined for great fortune, reappearing in China with Zhang Heng and in the Islamic world as “the geometers’ π ”.

1.9.3 Āryabhaṭa (499 CE)

Primary source: the *Āryabhaṭīya*, *Gaṇitapāda* 10.

The verse reads, in translation: *add four to one hundred, multiply by eight and add sixty-two thousand; this is approximately the circumference of a circle whose diameter is twenty thousand.*

Step-by-step proof.

$$(100 + 4) \times 8 + 62000 = 62832, \quad \pi \approx \frac{62832}{20000} = 3.1416.$$

$\square$

Remark 1.11: Modern note — Āryabhaṭa and the word āsanna. 3.1416 has relative error 2.34×10^{-6} : three correct decimal places (the fourth, 6 against the 5 of $\pi = 3.14159 \dots$, disagrees). The most significant detail, however, is not numerical: Āryabhaṭa qualifies the result with the adjective āsanna, “approached, approximate”. A number of scholars read in this an awareness that the exact value is not expressible as a ratio of integers — which, if accepted, would make it the oldest known formulation of the suspicion of irrationality, twelve centuries before Lambert and, in spirit, contemporary with Maimonides’ remark.

1.9.4 Brahmagupta and Bhāskara II

- **Brahmagupta** (628), *Brahmasphuṭasiddhanta*: explicitly distinguishes the *practical* value $\pi = 3$ from the “exact” (that is, for him, better) value $\pi = \sqrt{10}$.

- **Bhāskara II** (1150), *Līlāvati*: gives $\frac{3927}{1250} = 3.1416$ as the accurate value and $\frac{22}{7}$ as the rough one, and develops tables of the sine (*jyā*) for circular arcs.

1.9.5 The Kerala school: infinite series, two and a half centuries early

Primary sources: the work of **Mādhava of Saṅgamagrāma** (ca. 1340–1425) has not come down to us directly, but is extensively cited by Nīlakaṇṭha Somayāji (*Tantrasaṅgraha*, 1501) and in the *Yuktibhāṣā* of Jyeṣṭhadeva (ca. 1530), written in Malayalam.

Mādhava obtained the power series expansion of the arctangent and, as a special case, the series that in Europe would be attributed to Gregory and Leibniz (1671–1674):

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Remark 1.12: Mādhava’s correction term. *Mādhava knew perfectly well that this series is useless for computation, and did something far more sophisticated: he constructed correction terms to be added to the truncation, with successive orders of correction of increasing accuracy. He further obtained series that converge rapidly in themselves, among them the one reported by Nīlakaṇṭha,*

$$\pi = 3 + \frac{4}{2 \cdot 3 \cdot 4} - \frac{4}{4 \cdot 5 \cdot 6} + \frac{4}{6 \cdot 7 \cdot 8} - \cdots,$$

whose terms decrease like $1/n^3$ instead of like $1/n$.

Remark 1.13: Modern note — Kerala. *The value of π transmitted as Mādhava’s is*

$$\pi \approx 3.14159265359,$$

with ten correct decimal places by strict truncation (the eleventh, 9 against the 8 of $\pi = 3.14159265358979\dots$, disagrees: the value is obtained by rounding; relative error $\sim 10^{-12}$). The historical significance, however, goes beyond the number. Between 1350 and 1500 the Kerala school possessed: the series expansions of arctan, sin and cos; the notion of truncation with an estimate of the remainder; and convergence acceleration. These are the ingredients of the infinitesimal calculus, two and a half centuries before Newton and Leibniz, developed in a cultural setting that did not connect them to the problems of the tangent and of motion — and it is probably this missing connection, rather than any difference of mathematical ability, that explains why the calculus was born in Europe and not in India.

1.10. China

1.10.1 From the value 3 to Liu Hui

Primary sources: the *Zhoubi Suanjing* (ca. first century BCE), the *Jiuzhang Suanshu* (*Nine Chapters on the Mathematical Art*, ca. 100 CE) and Liu Hui’s commentary (263 CE).

The *Nine Chapters* use the traditional value $\pi = 3$ (the *gūlū*). The first corrections date from the Han period:

- **Liu Xin** (ca. 5 CE): $\pi \approx 3.1547$, derived from the calibration of an imperial measuring vessel.
- **Zhang Heng** (ca. 130 CE): $\pi = \sqrt{10} \approx 3.1623$, and elsewhere $\frac{92}{29} \approx 3.1724$.

- **Wang Fan** (ca. 250 CE): $\frac{142}{45} \approx 3.1556$.

1.10.2 Liu Hui: the bisection algorithm and its acceleration

In his commentary on the *Nine Chapters* Liu Hui constructs the first rigorous Chinese algorithm, independent of Archimedes and in some respects superior to it.

Step-by-step proof. Step 1 (recursion on the side). Let ℓ_n be the side of the regular n -gon inscribed in the unit circle. On doubling the number of sides,

$$\ell_{2n} = \sqrt{2 - \sqrt{4 - \ell_n^2}},$$

a formula Liu Hui obtains by applying the *gougu* theorem (the Chinese Pythagorean theorem) twice to the triangle formed by half the side, the apothem and the radius.

Step 2 (from the areas). Writing S_n for the area of the inscribed n -gon, we have $S_{2n} = \frac{n}{2} \ell_n$.

Step 3 (the acceleration — the genuinely original contribution). Liu Hui proves the double inequality

$$S_{2n} < \pi < S_{2n} + (S_{2n} - S_n),$$

that is, he estimates the missing area by the increment just obtained. This allows him to bracket π *without ever constructing the circumscribed polygon*, halving the work relative to Archimedes.

Step 4. With the 192-gon he obtains $3.141024 < \pi < 3.142708$; pushing on to 3072 sides he reaches $\pi \approx 3.14159$. $\square$

1.10.3 Zu Chongzhi: seven decimal places

Primary source: the *Zhui Shu* (*Method of Interpolation*), lost; the result is preserved in the *Suishu*, the official history of the Sui dynasty.

Zu Chongzhi (429–500) and his son Zu Gengzhi established

$$3.1415926 < \pi < 3.1415927,$$

and proposed two fractions: the “approximate” value $yuelü = \frac{22}{7}$ and the “precise” value $milü = \frac{355}{113}$.

Remark 1.14: Modern note — why $355/113$ is special. $355/113 = 3.14159292\dots$ has relative error 8.5×10^{-8} : six correct decimal places with a three-digit denominator. This is no lucky accident. The continued fraction expansion of π is

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, \dots],$$

and its successive convergents are

$$3, \quad \frac{22}{7}, \quad \frac{333}{106}, \quad \frac{355}{113}, \quad \frac{103993}{33102}, \quad \dots$$

The theory of continued fractions guarantees that every convergent is the best rational approximation to π among all those with denominator no larger than its own. The unusually large partial quotient that follows, 292, makes the jump from $355/113$ to the next convergent enormous: $355/113$ is therefore extraordinarily good for its size, and no fraction with denominator below 16,604 beats it. Zu Chongzhi had no theory of continued fractions; he found the right convergent all the same. His result would stand unbeaten worldwide for some nine hundred years.

1.11. The Islamic world

The Islamic world inherits Archimedes, the Indian *siddhāntas* and the Persian tradition simultaneously, and fuses them.

- **al-Khwārizmī** (ca. 825), *Kitāb al-jabr*: lists three values according to use — $\frac{22}{7}$ for practical men, $\sqrt{10}$ for geometers, and 3.1416 (of Indian provenance) for astronomers.
- **al-Bīrūnī** (ca. 1030) discusses critically the inherited values and the methods for obtaining them.
- **Umar Khayyām** (ca. 1100) works on conics and on the parallel postulate, opening the road that would lead to the non-Euclidean geometries.

1.11.1 al-Kāshī (1424): the culmination of the polygonal method

Primary source: Ghiyāth al-Dīn Jamshīd al-Kāshī, *al-Risāla al-muḥīṭiyya* (*Treatise on the Circumference*), Samarkand, 1424.

Step-by-step proof. The stated aim. To compute the circumference of the universe with an error smaller than the thickness of a horse’s hair. That is, al-Kāshī fixes a *tolerance* at the outset and then determines how many sides are needed to meet it.

Method. Repeated bisection starting from the hexagon, through 28 doublings, up to the polygon of

$$3 \cdot 2^{28} = 805,306,368 \text{ sides,}$$

with all computation carried out in sexagesimal arithmetic.

Result. 2π to nine exact sexagesimal places, equivalent to

$$2\pi = 6.28318\,53071\,79586\,5,$$

that is, **sixteen correct decimal places** of π . □

Remark 1.15: Modern note — al-Kāshī. The relative error is of order 10^{-17} . Two aspects deserve attention. The first is the engineering stance: al-Kāshī does not ask “how far can I get” but “how much precision do I need, and what does it cost” — the reasoning of modern numerical analysis. The second is that this record would stand for some 180 years, until Ludolph van Ceulen, and that with it the polygonal method reaches its practical limit: each further decimal place requires roughly tripling the number of sides. The breakthrough would come not from greater patience but from a different idea — infinite series.

1.12. East Asia: the *wasan* tradition

Closed to foreign trade during the Edo period, Japan developed a high-level indigenous mathematics, the *wasan*.

- **Seki Takakazu** (ca. 1642–1708) applied to the computation of π a convergence-acceleration procedure (*enri*, “the principle of the circle”) essentially equivalent to Richardson extrapolation: from the perimeters of successive polygons he extracted the limit far more accurately than any single term, obtaining about ten correct decimal places.
- **Takebe Katahiro** (1664–1739) obtained forty-one correct decimal places in 1722 by expanding the function $(\arcsin x)^2$ in series: independently, a discovery in power-series analysis.

- **Matsunaga Ryōhitsu** (ca. 1739) carried the computation to fifty places.
- The votive *sangaku* tablets, hung in Shinto and Buddhist temples between the seventeenth and nineteenth centuries, collect thousands of problems in the geometry of the circle — typically configurations of mutually tangent circles inscribed in figures — offered both as devotion and as public challenge.

Remark 1.16: Modern note — Japan. Takebe’s forty-one places (1722) far exceed the European polygonal record of van Ceulen (35 places, 1615), and are obtained by a conceptually more advanced method. The Japanese case shows that convergence acceleration — the idea that from a slow sequence one may extract the limit far more quickly — was discovered independently in at least three cultures (Kerala, Japan, Europe).

1.13. Africa beyond Egypt

Outside pharaonic Egypt, ancient and medieval Africa has left us no mathematical texts on the circle. What we possess is architectural and textile evidence, which documents constructional competence but no numerical values.

- **Aksum** (Ethiopia, first to tenth centuries CE): monolithic stelae up to 33 m tall, tower architecture with circular elements; in the Ethiopian monasteries the Greek geometrical tradition was preserved through Coptic transmission.
- **Great Zimbabwe** (eleventh to fifteenth centuries): the Great Enclosure has a curvilinear perimeter wall some 250 m long, built dry-stone from granite blocks with continuous-curvature junctions.
- **Yoruba** (Nigeria) and the **Bantu** world: village layouts in concentric circles about a communal centre, and *adire* textiles with radial symmetry.
- **Berbers** (Maghreb): carpets with 6- and 8-pointed stars inscribed in circles, that is, division of the circumference into equal parts.

Warning. The **Ishango bone** (Democratic Republic of the Congo, ca. 20,000 BCE) is often cited in this context: it is a most important artefact for the history of counting, but it bears series of notches, not circular geometry. Citing it as evidence of knowledge about the circle is a widespread error best not repeated.

Remark 1.17: Modern note — sub-Saharan Africa. There exist, on present knowledge, no documented values of π for pre-colonial sub-Saharan Africa. This tells us something about the surviving sources — transmission was largely oral and the media perishable — and nothing about the mathematical capacities of the cultures involved. The ethnomathematical research of Claudia Zaslavsky (Africa Counts, 1973) and Paulus Gerdes has documented in these traditions symmetries, tessellations and systems of proportion of considerable complexity: simply not in the form of a numerical treatise.

1.14. The pre-Columbian Americas

1.14.1 The mound builders of Ohio

Primary source: the Newark complex (Ohio), Hopewell culture, ca. 100 BCE–400 CE; surveys by Squier and Davis (1848) and later.

The complex includes a **Great Circle** some 366 m in diameter and an **Octagon** joined to a second circle. The dimensions of the various elements are multiples of a common unit (about 321 m, derived from the side of the Octagon), and the alignments of the Octagon follow the eight extreme positions of moonrise and moonset in the 18.6-year cycle.

***Remark 1.18: Modern note — Hopewell.** Laying out a 366 m circle with a departure from circularity of a few metres, and joining it to a commensurable regular octagon, is a non-trivial surveying problem: it requires a baseline, its repeated bisection, and the transfer of equal lengths over broken ground. The 18.6-year lunar cycle, moreover, is not observable within a single season: identifying it implies systematic observation handed down across generations. No value of π is documented, and none should be inferred.*

1.14.2 Mesoamerica and the Andes

- **Maya:** the *Caracol* at Chichén Itzá is an observatory on a circular plan; the codices (the Dresden in particular) contain tables of the 583.92-day Venus cycle, accurate to within an hour of the modern value. The achievement is *periodicity*, not the measurement of the circle.
- **Aztecs:** the Sun Stone organises the reckoning of time in regularly subdivided concentric rings.
- **Inca:** the *ceque* system of Cusco articulates the territory into 41 radial lines issuing from the Coricancha, each with a series of shrines; it is a radial scheme integrating topography, calendar and social organisation. The *Intihuatana* at Machu Picchu serves as a gnomon.
- **Nazca** (Peru, 500 BCE–500 CE): among the geoglyphs are spirals of great regularity, laid out with cords and pegs.
- **Medicine wheels** of the Great Plains: the Bighorn wheel (Wyoming) has a circle some 25 m across with 28 spokes and peripheral cairns aligned to the summer solstice and to the rising of first-magnitude stars.

1.15. Oceania and Australia

- **Polynesian navigation:** the “star compass” divides the horizon into 32 houses (or 16, according to the school), each defined by the rising or setting point of a known star; $360^\circ/32 = 11.25^\circ$ per sector. The *etak* system keeps account of position by referring it to a reference island that “moves” against the stars: in effect, a change of reference frame.
- **Aboriginal Australians:** concentric circles in painting and rock engraving denoting waterholes, camps and centres of creation; the *songlines* as cyclic routes memorised in sung form.

***Remark 1.19: Modern note — Oceania.** Dividing the horizon into 32 equal sectors is a partition of the circumference into congruent parts, transmitted orally and checked every night against the real sky. No computation of π appears, but the conceptual apparatus — angle, sector, arc — is fully present.*

1.16. Europe beyond the Mediterranean

- **Celts** (500 BCE–500 CE): La Tène art is built on spirals, triskeles and compass-drawn circles joined tangentially; the Celtic cross inscribes an equal-armed cross in a circle,

thereby realising the division into four quadrants.

- **The Germanic and Norse world:** the Trundholm sun chariot (Denmark, ca. 1400 BCE) carries a disc decorated with concentric circles; the sun cross recurs from the Bronze Age onwards.
- **Slavs and Balts:** the *kolovrat*, a solar wheel with curved spokes, and rosette motifs obtained by direct compass construction (six arcs of radius equal to the radius of the circle).

***Remark 1.20: The six-petal rosette as an implicit theorem.** The six-petal motif, found from central Europe to Scandinavia and throughout the Mediterranean world, is obtained by setting the compass on a point of the circle at the same opening as the radius and repeating six times. That one returns exactly to the starting point is the very property used by the Susa tablet and by the first step of Archimedes: the side of the inscribed hexagon equals the radius. It is a theorem, proved in silence by every craftsman who ever drew that motif.*

1.17. Central Asia and the Tibetan plateau

- **Scythians and Saka** (700–300 BCE): goldwork of rotational composition and kurgan burial mounds on circular plans.
- **Tibet:** the *maṇḍala* as a cosmological scheme built on concentric circles and squares with fourfold symmetry; executing one requires the proportional division of circles and the construction of inscribed squares according to canons fixed in writing in the ritual manuals.
- **Mongolia:** the *ger* (yurt) is a structure on a circular plan in which the roof poles, all equal, converge on a central ring (*toono*); the whole dwelling is a division of the circumference into equal parts turned into an object of daily use.

1.18. Medieval and Renaissance Europe

With the resumption of contact with the Islamic world Europe recovers Archimedes, and takes up the counting of digits exactly where al-Kāshī had left it.

- **Leonardo of Pisa (Fibonacci)**, *Practica Geometriae* (1220): $\pi \approx \frac{864}{275} = 3.14\overline{18}$, error 0.0072%.
- **Nicholas of Cusa** (ca. 1450) attempted the quadrature repeatedly; Regiomontanus refuted his constructions, but one Cusan lemma would prove useful to Huygens two centuries later.
- **François Viète** (1593) computes nine decimal places with a polygon of $3 \cdot 2^{17} = 393,216$ sides, and publishes the first *infinite product* in history for π :

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2 + \sqrt{2}}}{2} \cdot \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2} \dots$$

- **Adriaan van Roomen** (1593): fifteen decimal places.
- **Ludolph van Ceulen** (1540–1610): twenty places in 1596, later thirty-five, with a polygon of 2^{62} sides; the result was carved on his tombstone in Leiden, and in German π is still called the *Ludolphsche Zahl*.

- **Willebrord Snel** (1621) and **Christiaan Huygens** (1654) refine the polygonal inequalities: Snel obtains from a polygon of 2^{30} sides the same precision for which van Ceulen had needed 2^{62} , and Huygens supplies the rigorous proof in *De circuli magnitudine inventa*.

Theorem 1.4: Viète's product (1593).

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \dots$$

Proof. From the sine duplication formula (Chapter 8), $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$; iterating it n times,

$$\sin x = 2^n \sin \frac{x}{2^n} \prod_{k=1}^n \cos \frac{x}{2^k}.$$

With $x = \pi/2$:

$$1 = 2^n \sin \frac{\pi}{2^{n+1}} \prod_{k=1}^n \cos \frac{\pi}{2^{k+1}}.$$

As $n \rightarrow \infty$, setting $u = \pi/2^{n+1} \rightarrow 0$, $2^n \sin \frac{\pi}{2^{n+1}} = \frac{\pi}{2} \cdot \frac{\sin u}{u} \rightarrow \frac{\pi}{2}$ (fundamental limit, Theorem 8.5), hence

$$\prod_{k \geq 1} \cos \frac{\pi}{2^{k+1}} = \frac{2}{\pi}.$$

The individual factors are computed by repeated bisection starting from $\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$, using $\cos \frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$ (first-quadrant angles, Corollary 8.3): $\cos \frac{\pi}{8} = \sqrt{\frac{1+\sqrt{2}/2}{2}} = \frac{\sqrt{2+\sqrt{2}}}{2}$, and so on, reproducing exactly the stated formula. $\square$

Numerical check. The product of the first 10 factors equals 0.636620, against $2/\pi = 0.636620$ (checked, independent script).

Remark 1.21: Modern note — the end of an era. Van Ceulen devoted much of his adult life to thirty-five decimal places. Fifty years later, with the arctangent series, the same result would become an afternoon's work; today it is a matter of microseconds. The comparison is instructive but must be read correctly: van Ceulen was not less able; he used the best tool he had. The decisive advance was not in computing power but in representation — moving from the polygon, which approximates the figure, to the series, which approximates the number.

1.19. A comparative view: how close they came

The tables below collect the documented values examined in this chapter, in chronological order, with the relative error computed against the true value. Cultures for which no numerical value is documented do not appear: their absence from the table is not a judgement but a fact about the sources.

Remark 1.22: Three thresholds, not a continuous scale. Read as a whole, Table 1.1 shows not gradual progress but three qualitative leaps.

First threshold — from recipe to bounded estimate (Archimedes, 250 BCE). Before: practical rules, of which nobody knows how good they are. After: proved intervals. The value improves little; what changes is the epistemic status of the result.

Culture / author	Date	Value	Decimal	Rel. error
Babylon (rule $A = C^2/12$)	1900 BCE	3	3.0000	$4.5 \cdot 10^{-2}$
Susa (hexagon/circle)	1900 BCE	25/8	3.1250	$5.3 \cdot 10^{-3}$
Egypt (Rhind Papyrus)	1650 BCE	256/81	3.1605	$6.0 \cdot 10^{-3}$
Śulbasūtra (circle from sq.)	800 BCE	—	3.0883	$1.7 \cdot 10^{-2}$
Śulbasūtra (sq. from circle)	800 BCE	—	3.0044	$4.4 \cdot 10^{-2}$
Hebrew Bible (molten sea)	600 BCE	3	3.0000	$4.5 \cdot 10^{-2}$
Jaina texts	400 BCE	$\sqrt{10}$	3.1623	$6.6 \cdot 10^{-3}$
Archimedes (upper bound)	250 BCE	22/7	3.1429	$4.0 \cdot 10^{-4}$
Archimedes (interval midpoint)	250 BCE	—	3.1419	$8.2 \cdot 10^{-5}$
Vitruvius (odometer)	25 BCE	25/8	3.1250	$5.3 \cdot 10^{-3}$
Liu Xin	5 CE	—	3.1547	$4.2 \cdot 10^{-3}$
Columella	60 CE	22/7	3.1429	$4.0 \cdot 10^{-4}$
Zhang Heng	130 CE	$\sqrt{10}$	3.1623	$6.6 \cdot 10^{-3}$
Ptolemy (<i>Almagest</i>)	150 CE	377/120	3.1417	$2.4 \cdot 10^{-5}$
Wang Fan	250 CE	142/45	3.1556	$4.4 \cdot 10^{-3}$
Liu Hui (3072 sides)	263 CE	—	3.14159	$< 10^{-6}$
Zu Chongzhi (<i>mǐlǔ</i>)	480 CE	355/113	3.1415929	$8.5 \cdot 10^{-8}$
Āryabhaṭa	499 CE	62832/20000	3.1416	$2.3 \cdot 10^{-6}$

Table 1.1: Values of π documented in antiquity, in chronological order

Author	Date	Correct decimal places	Method
Brahmagupta	628	0 ($\sqrt{10}$)	Indian tradition
Bhāskara II	1150	3	Indian tradition
Fibonacci	1220	3	polygons
Mādhava (Kerala)	1400	10	series with correction term
al-Kāshī	1424	16	polygon of $3 \cdot 2^{28}$ sides
Viète	1593	9	polygon of $3 \cdot 2^{17}$ sides
van Roomen	1593	15	polygons
van Ceulen	1615	35	polygon of 2^{62} sides
Seki Takakazu	1680	10	<i>enri</i> acceleration
Machin	1706	100	arctangents
Takebe Katahiro	1722	41	series for $(\arcsin x)^2$
Shanks	1873	527 (of 707 computed)	arctangents

Table 1.2: From the Middle Ages to the threshold of the mechanical era: correct digits

Second threshold — from polygon to series (Kerala, 1400; Europe, 1670). The polygon approximates the figure and pays for every digit with a tripling of the labour. The series approximates the number, and every term contributes digits of its own. The polygonal method exhausts its possibilities with al-Kāshī and van Ceulen.

Third threshold — from computation to structure (Lambert 1761, Lindemann 1882). The question is no longer to compute π but to say what kind of number it is; and it is this, not one more digit in a billion, that closes the problem of squaring the circle opened by Anaxagoras twenty-three centuries earlier.

Remark 1.23: Historiographical sources. For a philologically accurate reconstruction see: Beckmann, *A History of Pi* (1971); Arndt & Haenel, *Pi Unleashed* (2001); for the Chinese tradition Martzloff, *A History of Chinese Mathematics* (1997); for the Indian one Plofker, *Mathematics in India* (2009); for the Islamic world Berggren, *Episodes in the Mathematics of Medieval Islam* (1986); for Japan Smith & Mikami, *A History of Japanese Mathematics* (1914).

1.20. The number π : definition and value

Definition 1.2: The number π . For every circle with diameter $d > 0$ and circumference (perimeter) C , the ratio C/d is constant, independent of the circle chosen. This constant is denoted π :

$$\pi := \frac{C}{d} = 3.14159\,26535\,89793\,23846\,\dots$$

$$\pi = \frac{C}{d} = \frac{C}{2r}$$

The symbol π (from the initial of the Greek word “periphery”) was introduced by William Jones in 1706 and popularised by Euler from 1737 onwards, becoming universal standard.

Theorem 1.5: Independence of the ratio C/d from the radius. For every pair of circles $C_1(O_1, r_1)$ and $C_2(O_2, r_2)$ with $r_1 \neq r_2$, the perimeter/diameter ratio is the same.

Proof. All circles are similar to each other (there exists a homothety centred at O_1 with ratio r_2/r_1 that maps C_1 to C_2). In a similarity with ratio k , every length is multiplied by k : thus $C_2 = k C_1$ and $d_2 = k d_1$, therefore $C_2/d_2 = C_1/d_1$. $\square$

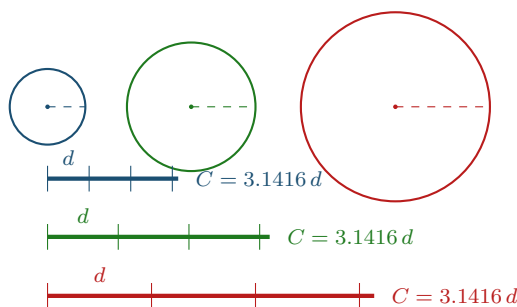


Figure 1.3: What the definition of π actually asserts. Circles of different radius have different circumferences and different diameters, but their ratio is always the same number. Below, the three circumferences have been *unrolled* into straight segments, measured in units of their own diameter: the three segments all have the same length, $3.1416\dots$ diameters. It is this constancy — not the numerical value — that is the mathematical content of the theorem, and it must be proved: it follows from the fact that any two circles are similar, with similarity ratio equal to the ratio of the radii, and that length is invariant under similarity up to the scale factor.

1.21. Empirical and Computational Demonstrations of π

Beyond rigorous demonstrations (Section 1.22), it is instructive to observe *empirical* methods that estimate π by experimental or computational means, useful also as a simulation exercise for a programmer.

1.21.1 Archimedes' Method (polygons)

Step-by-step proof. Step 1. Regular polygon inscribed in n sides: perimeter $P_n = 2nr \sin(\pi/n)$.

Step 2. Regular polygon circumscribed in n sides: perimeter $Q_n = 2nr \tan(\pi/n)$.

Step 3. For every n : $P_n < 2\pi r < Q_n$. Doubling n iteratively ($6 \rightarrow 12 \rightarrow 24 \rightarrow 48 \rightarrow 96$) the interval $[P_n/2r, Q_n/2r]$ narrows around π . With $n = 96$: $3.1410 < \pi < 3.1427$. $\square$

n	$P_n/(2r)$	$Q_n/(2r)$	interval width
6	3.0000	3.4641	0.4641
12	3.1058	3.2154	0.1096
24	3.1326	3.1597	0.0271
48	3.1393	3.1461	0.0068
96	3.1410	3.1427	0.0017

Table 1.3: Convergence of Archimedes' method of exhaustion

1.21.2 Monte Carlo Method

Definition 1.3: Monte Carlo Estimation of π . One generates N points (x_i, y_i) uniform in the square $[-1, 1]^2$ (area 4). Let N_{in} be the number of points with $x_i^2 + y_i^2 \leq 1$ (interior to the unit circle, area π). Then $\hat{\pi} = 4N_{\text{in}}/N \xrightarrow{N \rightarrow \infty} \pi$ almost surely (law of large numbers).

Proof. The probability that a uniform point in the square falls in the circle is $p = \text{Area}(\text{circle})/\text{Area}(\text{square}) = \pi/4$. The variable N_{in} is binomial $\text{Bin}(N, p)$; by the strong law of large numbers, $N_{\text{in}}/N \rightarrow p = \pi/4$ a.s., so $4N_{\text{in}}/N \rightarrow \pi$. The speed of convergence is $O(1/\sqrt{N})$ (central limit theorem): to double the correct digits one needs $100\times$ more samples — a method that is notably slow. $\square$

Warning. The Monte Carlo method is pedagogically elegant but computationally inefficient for obtaining many digits of π : the convergence $O(1/\sqrt{N})$ is enormously slower than hypergeometric series (e.g. Chudnovsky algorithm, exponential convergence) used in world records of computation.

1.21.3 Buffon's Needle

Theorem 1.6: Buffon's Needle (1733). A needle of length ℓ is thrown on a plane ruled by parallel equidistant lines with spacing $t \geq \ell$. The probability that the needle intersects a line is $p = \frac{2\ell}{\pi t}$, from which $\pi \approx \frac{2\ell N}{t N_{\text{hit}}}$ over N throws with N_{hit} successes.

Proof. Let $x \in [0, t/2]$ be the distance of the needle's centre from the nearest line (uniform) and $\theta \in [0, \pi]$ the needle's angle with the lines (uniform, independent of x). The needle intersects the line if and only if $x \leq (\ell/2) \sin \theta$. Thus

$$p = \frac{1}{(\pi)(t/2)} \int_0^\pi \int_0^{(\ell/2) \sin \theta} dx d\theta = \frac{1}{\pi t/2} \int_0^\pi \frac{\ell}{2} \sin \theta d\theta = \frac{\ell}{\pi t} [-\cos \theta]_0^\pi = \frac{2\ell}{\pi t}.$$

$\square$

Remark 1.24: Two dates, two events. Buffon posed the problem in 1733; the closed-form formula proved here was published by Buffon himself only in 1777, in his *Essai d'arithmétique morale* — the date used elsewhere in the treatise (Chapter 6) for the same theorem. The two dates are not in contradiction: they refer to two different moments of the same story.

1.21.4 Modern Algorithms: the Chudnovsky Method

The empirical methods above are pedagogically useful but computationally inadequate for modern world records in computing π . Today's tool is the family of **hypergeometric series of Ramanujan type**, of which the most used is the Chudnovsky brothers' series (1988).

Definition 1.4: Chudnovsky Series.

$$\frac{1}{\pi} = 12 \sum_{k=0}^{\infty} \frac{(-1)^k (6k)! (13591409 + 545140134k)}{(3k)! (k!)^3 640320^{3k+3/2}}.$$

Remark 1.25: Speed of convergence and the binary splitting algorithm. Each term of the series adds approximately 14.18 correct decimal digits of π (geometric convergence, unlike the $O(1/\sqrt{N})$ convergence of Monte Carlo or $O(1/n^2)$ of Archimedes' method). To be practical on billions of digits, the sum is not computed term by term but using the binary splitting algorithm, which reduces the computational cost to $O(M(n) \log n)$ with $M(n)$ the cost of multiplication of integers with n digits. With this method all major computation records have been achieved: 2.7 trillion digits (2009), 10 trillion (2011), 22.4 trillion (2016), 50 trillion (2020), up to 100 trillion digits (2022); the record at the time of writing (September 2026) is higher, see Chapter 6.

Method	Order of convergence	Correct digits
Monte Carlo	$O(1/\sqrt{N})$	fractional, very slow
Buffon's Needle	$O(1/\sqrt{N})$	comparable to Monte Carlo
Archimedes (polygons)	$O(1/n^2)$	double roughly every 3–4 doublings of n
Chudnovsky	geometric	≈ 14 per series term

Table 1.4: Qualitative comparison of the convergence speed of π computation methods seen in this chapter

1.22. Irrationality and Transcendence of π

Theorem 1.7: Irrationality of π (Niven's proof, 1947). $\pi \notin \mathbb{Q}$: there do not exist $p, q \in \mathbb{Z}$, $q \neq 0$, with $\pi = p/q$.

INFO

Prerequisites. This proof uses higher-order derivatives and integration by parts, recalled here by statement only: differential and integral calculus is developed in full in Volume II.

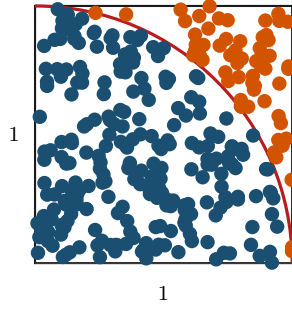


Figure 1.4: Estimating π by the Monte Carlo method. One generates $N = 300$ random points, uniform in the unit square, and counts those falling inside the quarter circle (in blue); the others are in orange. Since the quarter circle has area $\pi/4$ and the square area 1, the fraction of interior points estimates $\pi/4$. With the points in the figure one counts 238 of them, whence $\pi \simeq 4 \cdot 238/300 = 3.1733$, with an error of 1.01%. Convergence is slow: the error goes as $1/\sqrt{N}$, so gaining one decimal digit requires multiplying the points by a hundred. The method is therefore useless for computing π , but it illustrates well the link between area, probability and measure.

Proof. Suppose for contradiction that $\pi = a/b$, $a, b \in \mathbb{N}$. Define

$$f(x) = \frac{x^n(a-bx)^n}{n!}, \quad F(x) = \sum_{k=0}^n (-1)^k f^{(2k)}(x).$$

Step 1 (integrality). Write $x^n(a-bx)^n = \sum_{j=n}^{2n} c_j x^j$ with $c_j \in \mathbb{Z}$ (it is a polynomial with integer coefficients, the product of x^n with the binomial expansion of $(a-bx)^n$). Then $f^{(k)}(0) = 0$ for $k < n$ and for $k > 2n$ (the terms of degree $j \neq k$ vanish at 0 after k differentiations, and there are no terms of degree $> 2n$), while for $n \leq k \leq 2n$

$$f^{(k)}(0) = \frac{k!}{n!} c_k \in \mathbb{Z},$$

because $k \geq n$ makes $k!/n!$ a product of consecutive integers, hence an integer. Moreover $f(a/b-x) = f(x)$ (substituting, $a-b(a/b-x) = bx$, so $f(a/b-x) = \frac{(a/b-x)^n (bx)^n}{n!} = \frac{x^n (a-bx)^n}{n!} = f(x)$, using $b^n(a/b-x)^n = (a-bx)^n$), so differentiating k times and setting $x = 0$: $f^{(k)}(\pi) = (-1)^k f^{(k)}(0) \in \mathbb{Z}$. Hence $f^{(k)}(0), f^{(k)}(\pi) \in \mathbb{Z}$ for every k .

Differential identity: $\frac{d}{dx}[F'(x) \sin x - F(x) \cos x] = f(x) \sin x$, hence $\int_0^\pi f(x) \sin x \, dx = F(\pi) + F(0) \in \mathbb{Z}$.

Contradiction: for $0 < x < \pi$ we have $0 < f(x) \sin x \leq \pi^n a^n / n! \rightarrow 0$ as $n \rightarrow \infty$; choosing n sufficiently large the integral is a number in $(0, 1)$ but must be an integer: contradiction. $\square$

Remark 1.26: Transcendence of π : what we import. *Statement (Lindemann, 1882).* π is not a root of any non-zero polynomial with rational coefficients.

Imported, source F-01 (F. Lindemann, Ueber die Zahl π , *Mathematische Annalen* 20 (1882), 213–225). The outline of the deduction, for readers who already know the Lindemann–Weierstrass theorem (if $\alpha_1, \dots, \alpha_n$ are distinct algebraic numbers, $e^{\alpha_1}, \dots, e^{\alpha_n}$ are linearly independent over $\mathbb{Q}$): if π were algebraic, so would $i\pi$ be; but then $e^0 = 1$ and $e^{i\pi} = -1$ would have to be $\mathbb{Q}$ -independent, whereas $1 \cdot 1 + 1 \cdot (-1) = 0$ is a non-trivial rational relation: contradiction. The Lindemann–Weierstrass theorem itself is not proved in this treatise (it requires the complex analysis of Volume II).

Geometric consequence, also imported: *squaring the circle with straightedge and compass is impossible, because constructions with straightedge and compass produce only algebraic numbers (Wantzel 1837, source F-02), while the side of the equivalent square, $r\sqrt{\pi}$, is transcendental by the above.*

Remark 1.27: Two thousand years of attempts: from Anaxagoras to Lambert.

We report Lambert's result as **imported** (1761; for a modern account of the proof see M. Laczkovich, *On Lambert's proof of the irrationality of π* , *American Mathematical Monthly* 104(5) (1997), 439–443, source F-08): the proof is not reproduced here. This may be skipped on first reading.

The problem of squaring the circle (constructing, with straightedge and compass, a square with area equal to that of a given circle) dates back at least to Anaxagoras of Clazomenae, who, according to tradition, occupied himself with it while imprisoned in Athens around 430 B.C. For over two thousand years the problem remained open, generating a vast literature of attempts (and as many false announcements of solution). The first decisive step towards impossibility came with Johann Heinrich Lambert, who in 1761 proved the irrationality of π using a completely different method from Niven's (which is from 1947, nearly two centuries later): Lambert developed the tangent function into a **continued fraction**,

$$\tan r = \frac{r}{1 - \frac{r^2}{3 - \frac{r^2}{5 - \frac{r^2}{7 - \dots}}}}$$

and proved that if $r \neq 0$ is rational then $\tan r$ is irrational; since $\tan(45^\circ) = 1$ is rational, the angle of 45° (that is $\pi/4$ in radians) cannot be rational, so π is irrational. Lambert's proof, though elegant, did not yet exclude squaring: as noted above, irrationality alone is not enough, because some irrational numbers (for example $\sqrt{2}$) are nonetheless constructible with straightedge and compass. Only **transcendence**, proved by Lindemann one hundred twenty-one years later, finally closed the question.

1.23. Machin's Formula: Practical Computation of π before Calculators

Theorem 1.8: Machin's Formula (1706).

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}.$$

Proof. We use the addition formula for tangent, $\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$ (Chapter 8). Let $\alpha = \arctan(1/5)$. Then

$$\tan(2\alpha) = \frac{2/5}{1 - 1/25} = \frac{10}{24} = \frac{5}{12}, \quad \tan(4\alpha) = \frac{2 \cdot 5/12}{1 - 25/144} = \frac{10/12}{119/144} = \frac{120}{119}.$$

Now let $\beta = \arctan(1/239)$:

$$\tan(4\alpha - \beta) = \frac{120/119 - 1/239}{1 + (120/119)(1/239)} = \frac{120 \cdot 239 - 119}{119 \cdot 239 + 120} = \frac{28561}{28561} = 1.$$

Since $4\alpha - \beta$ has tangent 1 and is in the right interval (numerical check: $4\alpha \approx 4 \times 11.31^\circ = 45.24^\circ$, $\beta \approx 0.24^\circ$, so $4\alpha - \beta \approx 45^\circ$), we conclude $4\alpha - \beta = \pi/4$, which is the claim. $\square$

Remark 1.28: Why two “small” arctangents are convenient. Leibniz’s series $\pi/4 = \arctan 1 = 1 - 1/3 + 1/5 - \dots$ converges too slowly to be practical (roughly 10^k terms are needed for k correct digits). Taylor’s series for $\arctan x$ converges much more rapidly for small x : using $x = 1/5$ and $x = 1/239$ (both $\ll 1$), Machin’s formula was the tool with which, before electronic computers, William Shanks calculated by hand (1873) the first 707 digits of π — of which, discovered only in 1946, only the first 527 were correct: a transcription error had propagated through the rest of the calculation.

Remark 1.29: “Machin-like” formulas. There exist dozens of variants (Machin-like formulas), each based on the same idea of combining arctangents of small arguments. For example, Gauss used $\pi/4 = 12 \arctan(1/18) + 8 \arctan(1/57) - 5 \arctan(1/239)$. These formulas were the basis of nearly all world records for computing π until the advent of hypergeometric series of Ramanujan type discussed in this chapter (Chudnovsky algorithm).

1.24. Wallis’ Product: π as an Infinite Product

Long before Machin, the English mathematician John Wallis published in 1655 (in his work *Arithmetica Infinitorum*) one of the first expressions of π as a purely algebraic infinite process, obtained by a completely different route (analysis, not trigonometry of continued fractions or arctangents).

Theorem 1.9: Wallis’ Product (1655).

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots$$

Proof. Define, for $n \geq 0$, the **Wallis integrals** $I_n = \int_0^{\pi/2} \sin^n x \, dx$.

Reduction formula. Integrating by parts ($u = \sin^{n-1} x$, $dv = \sin x \, dx$):

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx = (n-1)(I_{n-2} - I_n),$$

(the boundary term vanishes at both endpoints), hence

$$I_n = \frac{n-1}{n} I_{n-2}, \quad n \geq 2.$$

Initial values: $I_0 = \pi/2$, $I_1 = 1$. Iterating the reduction formula:

$$I_{2m} = \frac{(2m-1)!!}{(2m)!!} \cdot \frac{\pi}{2}, \quad I_{2m+1} = \frac{(2m)!!}{(2m+1)!!},$$

where $n!!$ denotes the double factorial ($n!! = n(n-2)(n-4)\cdots$).

Monotonicity and limit of the ratio. On $(0, \pi/2)$ we have $0 < \sin x < 1$, so $\sin^{n+1} x \leq \sin^n x$ pointwise, and therefore $I_{n+1} \leq I_n$ for every n . In particular $I_{2m+1} \leq I_{2m} \leq I_{2m-1}$, and dividing by I_{2m+1} :

$$1 \leq \frac{I_{2m}}{I_{2m+1}} \leq \frac{I_{2m-1}}{I_{2m+1}} = \frac{2m+1}{2m} \xrightarrow{m \rightarrow \infty} 1$$

(the last equality follows directly from the reduction formula applied to $I_{2m+1} = \frac{2m}{2m+1} I_{2m-1}$). By the squeeze theorem, $I_{2m}/I_{2m+1} \rightarrow 1$.

From limit to product. Define the partial product

$$P_m := \prod_{n=1}^m \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} = \frac{(2m)!!}{(2m-1)!!} \cdot \frac{(2m)!!}{(2m+1)!!} = \frac{[(2m)!!]^2}{(2m-1)!!(2m+1)!!}.$$

Using $(2m+1)!! = (2m+1)(2m-1)!!$:

$$P_m = \frac{[(2m)!!]^2}{(2m+1)[(2m-1)!!]^2}.$$

Comparing with the formulas for I_{2m} and I_{2m+1} derived above, one verifies directly that $P_m = \frac{\pi}{2} \cdot \frac{I_{2m+1}}{I_{2m}}$. Since $I_{2m+1}/I_{2m} \rightarrow 1$ (reciprocal of the limit shown above), we conclude

$$\lim_{m \rightarrow \infty} P_m = \frac{\pi}{2} \cdot 1 = \frac{\pi}{2}.$$

□

Remark 1.30: A bridge between two centuries and two methods. Wallis' product (1655) precedes Machin's formula (1706) by over half a century, and is conceptually independent of it: it uses only elementary integral calculus (repeated integration by parts), without any reference to continued fractions or arctangents. Historically, about eighty years after Wallis, Euler deduced Wallis' product as a special case of his celebrated infinite product factorization formula for the sine function, $\sin(\pi z) = \pi z \prod_{n=1}^{\infty} (1 - z^2/n^2)$, evaluated at $z = 1/2$: an instructive example of how the same numerical result can emerge from completely disconnected proof paths — elementary integration on one hand, complex analysis on the other.

Exercise 1.1. Calculate the partial product $P_2 = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5}$ and compare it with $\pi/2 \approx 1.5708$.

Solution.

$$P_2 = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} = \frac{2 \cdot 2 \cdot 4 \cdot 4}{1 \cdot 3 \cdot 3 \cdot 5} = \frac{64}{45} \approx 1.4222,$$

already in the right order of magnitude but still far from $\pi/2 \approx 1.5708$: the convergence of Wallis' product is known to be rather slow (comparable in practice to that of Leibniz's series), which makes it today a result of historical and conceptual importance more than a practical computational tool — unlike Machin's formula (previous section) or the Chudnovsky series, both explicitly designed for speed of convergence.

1.25. The Basel problem, and how π is computed today

Remark 1.31: The Basel problem. Another identity involving π , of an altogether different nature from the arctangent series seen so far, is the sum of the reciprocals of the squares:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Posed by Pietro Mengoli in 1650 and solved by Euler in 1735 (whence the name, from the home city of Euler and the Bernoullis), this result requires differential and integral calculus (a Fourier series expansion, or the infinite product for the sine) to be proved: it is **proved in Volume II**. The same holds for the Gregory–Leibniz–Mādhava series $\pi/4 = 1 - 1/3 + 1/5 - \dots$ already encountered (its rigorous proof uses term-by-term integration of a power series).

Remark 1.32: How the digits of π are computed today. The modern records (Chapter 6) do not use Machin’s arctangent series, too slow, but two families of much faster-converging methods, presented here only as a panorama, without proof:

Gauss–Legendre algorithm (arithmetic-geometric mean). One iterates $a_{n+1} = (a_n + b_n)/2$, $b_{n+1} = \sqrt{a_n b_n}$, $t_{n+1} = t_n - 2^n(a_n - a_{n+1})^2$, starting from $a_0 = 1$, $b_0 = 1/\sqrt{2}$, $t_0 = 1/4$: then $\pi \approx (a_n + b_n)^2/(4t_n)$, with the number of correct digits **doubling** at every iteration (quadratic convergence, faster than any series). Rediscovered independently by Salamin and by Brent in 1976, although the link between π and the arithmetic-geometric mean goes back to Gauss.

BBP formula (Bailey–Borwein–Plouffe, 1997).

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right),$$

remarkable because it allows isolated hexadecimal (or binary) digits of π to be computed without computing the preceding ones, a property no earlier formula possessed.

1.26. Exercises

Exercise 1.2. The Babylonian rule for the area of the circle is $A = C^2/12$. A circular ditch has measured circumference $C = 60$ cubits. Compute the area by the Babylonian rule, then the true area, and determine the percentage error the scribe commits.

Solution. Babylonian rule: $A = 60^2/12 = 300$ square cubits.

True area: from $C = 2\pi r$ we get $r = 60/(2\pi) = 9.5493$, hence $A = \pi r^2 = C^2/(4\pi) = 3600/12.5664 = 286.479$.

Error: $(300 - 286.479)/286.479 = 4.72\%$ in excess. Note that the error on the *area* (4.72%) differs slightly from that on π (4.51%): the area depends on $1/\pi$, not on π , and the relative errors of a number and of its reciprocal agree only to first order.

Exercise 1.3. Verify “Step 3” of the reconstruction of the Rhind Papyrus: in a square of side 9, cut the four corners with right isosceles triangles of leg 3 and compute the area of the resulting octagon. By how much does it differ from 64? And what would the implied π be if, instead of rounding to 64, one used the exact value?

Solution. Area of the octagon: $81 - 4 \cdot \frac{1}{2} \cdot 3 \cdot 3 = 81 - 18 = 63$. It differs from 64 by one unit, that is by 1.6%.

With the exact value 63 we would have $\pi d^2/4 = 63$ with $d = 9$, whence $\pi = 252/81 = 3.1111$, an error of 0.97% in defect. Rounding to 64 — apparently a loss of precision — in fact *improves* the result (0.60% in excess): a fortunate case in which two errors partly cancel.

Exercise 1.4. (Source: Beckmann 1971) — Verify numerically the Chinese approximation $355/113$ against π and compute the relative error; compare it with the error of the Babylonian fraction $25/8$. By what factor is it better?

Solution. $355/113 = 3.14159292035\dots$, so $|355/113 - \pi| = 2.667 \times 10^{-7}$ and $\varepsilon = 8.49 \times 10^{-8}$. $25/8 = 3.125$, so $|25/8 - \pi| = 1.6593 \times 10^{-2}$ and $\varepsilon = 5.28 \times 10^{-3}$. The ratio of the two errors is $5.28 \times 10^{-3} / 8.49 \times 10^{-8} \approx 6.2 \times 10^4$: Zu Chongzhi's fraction is about *sixty-two thousand times* more accurate, while having a denominator only fourteen times larger.

Exercise 1.5. Compute the first four convergents of the continued fraction $\pi = [3; 7, 15, 1, 292, \dots]$ and verify that they coincide with the historical values 3 , $22/7$, $333/106$, $355/113$. Explain, by looking at the next partial quotient, why $355/113$ is so good.

Solution. With the recursion for convergents $p_k = a_k p_{k-1} + p_{k-2}$, $q_k = a_k q_{k-1} + q_{k-2}$ and $a = (3, 7, 15, 1, 292)$:

$$\frac{3}{1}, \quad \frac{22}{7}, \quad \frac{333}{106}, \quad \frac{355}{113}, \quad \frac{103993}{33102}.$$

The error of a convergent satisfies $|x - p_k/q_k| < 1/(q_k q_{k+1})$. Here $a_4 = 292$ is unusually large, so $q_4 = 33102$ is enormous compared with $q_3 = 113$: the error of $355/113$ is less than $1/(113 \cdot 33102) \approx 2.7 \times 10^{-7}$. It is precisely the large partial quotient 292 that makes $355/113$ exceptional.

Exercise 1.6. al-Kāshī used a polygon of $3 \cdot 2^{28}$ sides. Knowing that the error of the polygonal method decreases as $O(1/n^2)$, estimate how many sides would be needed to obtain 30 correct decimal places, and comment on the result.

Solution. With $n = 3 \cdot 2^{28} \approx 8.05 \times 10^8$ one obtains about 16 digits, that is an error $\approx 10^{-16}$. If the error goes as C/n^2 , passing from 10^{-16} to 10^{-30} requires reducing it by 10^{14} , that is multiplying n by 10^7 : some 8×10^{15} sides would be needed. Every doubling of the number of sides requires a square-root extraction in high-precision arithmetic: the cost grows unsustainably. This is the structural reason why the polygonal method could not progress beyond al-Kāshī, and why the move to series was not an incremental improvement but a change of paradigm.

Exercise 1.7. Using Archimedes' method, calculate the interval $[P_n/2r, Q_n/2r]$ for $n = 192$ and verify that the width has reduced by roughly a factor of 4 compared to $n = 96$.

Solution. With $x = \pi/192 = 0.01636246$: $\sin x \approx x - x^3/6 = 0.01636173$ and $\tan x \approx x + x^3/3 = 0.01636392$. Therefore

$$P_{192}/2r = 192 \sin x \approx 3.14145, \quad Q_{192}/2r = 192 \tan x \approx 3.14187,$$

interval of width ≈ 0.00042 . Relative to the width 0.0017 for $n = 96$ (Table 1.3), the ratio is $0.0017/0.00042 \approx 4.0$: doubling n the width reduces by a factor ≈ 4 , consistent with an error decaying as $O(1/n^2)$ (the error of $\sin x$ and $\tan x$ relative to x is cubic in $x \propto 1/n$, so $\tan x - \sin x$ is $O(1/n^3)$, but multiplied by n points gives $O(1/n^2)$ on the overall width).

Exercise 1.8. Write (in words, in pseudocode) a Monte Carlo algorithm to estimate π and estimate how many samples N are needed to obtain an expected error less than 10^{-3} , using the estimate $O(1/\sqrt{N})$.

Solution. Pseudocode:

Listing 1.1: Monte Carlo for pi

```
N_in = 0
repeat N times:
    x = random_uniform_number(-1, 1)
    y = random_uniform_number(-1, 1)
    if x*x + y*y <= 1:
        N_in = N_in + 1
pi_estimated = 4 * N_in / N
```

Estimate of N : the variable $4N_{\text{in}}/N$ has standard deviation $\sigma = 4\sqrt{p(1-p)/N}$ with $p = \pi/4 \approx 0.7854$, so $p(1-p) \approx 0.1685$. Imposing $\sigma < 10^{-3}$:

$$4\sqrt{\frac{0.1685}{N}} < 10^{-3} \implies N > \frac{16 \cdot 0.1685}{10^{-6}} \approx 2.7 \times 10^6.$$

Therefore roughly 2.7 million samples are needed for a typical deviation less than one thousandth: numerical confirmation of the slowness $O(1/\sqrt{N})$ discussed in the warning box above.

Exercise 1.9. In Niven's proof, explain why it is essential that $f^{(k)}(0)$ and $f^{(k)}(\pi)$ are *integers* and not merely rational.

Solution. The contradiction at the end is based on the fact that $\int_0^\pi f(x) \sin x \, dx = F(\pi) + F(0)$ is, by construction, an *integer*, but is also strictly between 0 and 1 for large n — and no integer lies in $(0, 1)$: the set $\mathbb{Z}$ is *discrete*, with points spaced by at least 1. If $F(\pi) + F(0)$ were merely a **rational** number (not necessarily an integer), there would be no contradiction: the rationals are *dense* in $\mathbb{R}$, so there exist rationals arbitrarily close to 0 or indeed interior to $(0, 1)$ without violating anything. It is precisely the discreteness of $\mathbb{Z}$ (unlike the density of $\mathbb{Q}$) that is the logical tool making the contradiction possible.

Exercise 1.10. Compute the first three terms of Nīlakaṇṭha's series $\pi = 3 + \frac{4}{2 \cdot 3 \cdot 4} - \frac{4}{4 \cdot 5 \cdot 6} + \frac{4}{6 \cdot 7 \cdot 8} - \dots$ and compare the accuracy obtained with that of the first three terms of the Leibniz series.

Solution. *Step 1 — Nīlakaṇṭha.* $3 + 4/24 - 4/120 + 4/336 = 3 + 0.16667 - 0.03333 + 0.01190 = 3.14524$; relative error (Definition 1.1) $\varepsilon = 1.2 \times 10^{-3}$.

Step 2 — Leibniz. $4(1 - 1/3 + 1/5) = 4 \times 0.86667 = 3.46667$; relative error $\varepsilon = 1.0 \times 10^{-1}$.

Step 3 — independent check with exact fractions. Reducing to the common denominator 1680 gives $4/24 = 280/1680$, $4/120 = 56/1680$ and $4/336 = 20/1680$, so Nīlakaṇṭha's partial sum is exactly

$$3 + \frac{280 - 56 + 20}{1680} = 3 + \frac{61}{420} = \frac{1321}{420} = 3.145238\dots,$$

while Leibniz's is $4(1 - \frac{1}{3} + \frac{1}{5}) = 4 \cdot \frac{13}{15} = \frac{52}{15} = 3.466667\dots$. The two exact fractions confirm the decimals of Steps 1–2. ✓

Step 4 — comparison. For the same number of terms Nilakanṭha's series is almost *ninety times* more accurate: with the exact fractions $\varepsilon_L/\varepsilon_N = 1.035 \times 10^{-1}/1.160 \times 10^{-3} = 89.2$. The reason is that its terms decrease like $1/n^3$ against $1/n$. This is why the Kerala school could really *use* its series to compute, whereas for Europe the Leibniz series would remain a theoretical result.

Exercise 1.11. Check Machin's formula $\pi/4 = 4\arctan(1/5) - \arctan(1/239)$ numerically by computing the two arctangents with the Gregory–Leibniz series $\arctan x = x - x^3/3 + x^5/5 - \dots$, and say how many terms each requires for eight correct decimal digits.

Solution. For $x = 1/5 = 0.2$:

$$0.2 - \frac{0.008}{3} + \frac{0.00032}{5} - \frac{0.0000128}{7} + \frac{0.000000512}{9} = 0.2 - 0.00266667 + 0.000064 - 0.00000183 + 0.00000006 = 0.19739556$$

The next term is $0.2^{11}/11 = 1.9 \cdot 10^{-9}$: five terms therefore suffice for eight decimals. For $x = 1/239 = 0.004184100$ the second term is $x^3/3 = 2.44 \cdot 10^{-8}$ and the third $x^5/5 = 2.6 \cdot 10^{-13}$: two terms suffice, and $\arctan(1/239) = 0.00418408$. Hence

$$4 \cdot 0.19739556 - 0.00418408 = 0.78958224 - 0.00418408 = 0.78539816,$$

while $\pi/4 = 0.78539816$: the eight digits agree. The advantage of the formula is plain: the direct series for $\arctan 1 = \pi/4$ would need 10^8 terms for the same accuracy, these two need seven in all.

Exercise 1.12. Compute the partial Wallis product $P_N = \prod_{n=1}^N \frac{4n^2}{4n^2-1}$ for $N = 10$, compare it with $\pi/2$ and check the asymptotic estimate $P_N \simeq \frac{\pi}{2}(1 - \frac{1}{4N})$.

Solution. The successive factors are $4/3, 16/15, 36/35, 64/63, 100/99, 144/143, 196/195, 256/255, 324/323, 400/399$, and the partial products are

$$1.333333; 1.422222; 1.462857; 1.486077; 1.501088; 1.511585; 1.519337; 1.525295; 1.530017; 1.533852.$$

Hence $P_{10} = 1.533852$ against $\pi/2 = 1.570796$: the relative error is 2.35%. The asymptotic estimate gives $\frac{\pi}{2}(1 - 1/40) = 1.570796 \cdot 0.975 = 1.531526$, differing from the exact value by 0.0023, that is by a term of order $1/N^2$ as expected. Convergence is therefore extremely slow: one more decimal digit requires multiplying N by ten.

1.27. Self-assessment quiz

Multiple-choice questions for a quick check before tackling the exercises. The answer key is at the end of the section.

- The value of π implicit in the Rhind papyrus (Egypt, ca. 1650 BC) is
 - 3
 - $256/81 \approx 3.1605$
 - $22/7 \approx 3.1429$

-
- D) $355/113 \approx 3.1416$
2. Archimedes estimated π by means of
- A) experimental measurements on wheels
 - B) inscribed and circumscribed regular polygons
 - C) an infinite series
 - D) integral calculus
3. The number π is
- A) rational
 - B) irrational but algebraic
 - C) transcendental
 - D) algebraic of degree 2
4. The irrationality of π was first proved by
- A) Archimedes
 - B) Lambert (1761)
 - C) Lindemann (1882)
 - D) Euler
5. The transcendence of π was proved by
- A) Lambert (1761)
 - B) Wallis (1655)
 - C) Lindemann (1882)
 - D) Machin (1706)
6. Squaring the circle with ruler and compass is impossible because
- A) π is irrational
 - B) π is transcendental
 - C) the circle is not a polygon
 - D) the construction would require infinitely many steps
7. Machin-like formulas are used to
- A) define π
 - B) prove the irrationality of π
 - C) compute π with rapidly convergent arctangent series
 - D) construct the circle with ruler and compass
8. The Wallis product expresses
- A) π as a limit of polygons
 - B) $\pi/2$ as an infinite product

C) π as a continued fraction

D) $\pi^2/6$ as a series sum

BEST PRACTICE

Answer key. 1: B — 2: B — 3: C — 4: B — 5: C — 6: B — 7: C — 8: B.

1.28. Chapter Formula sheet

The number π (Definition 1.2; Theorem 1.5)

$$\pi := \frac{C}{d} = \frac{C}{2r} = 3.14159\,26535\,89793\,23846\,\dots$$

Relative error of an approximation (Definition 1.1)

$$\varepsilon(p) = \frac{|p - \pi|}{\pi}$$

Euclid, *Elements* XII.2 (Theorem 1.1)

$$\frac{A_1}{A_2} = \frac{d_1^2}{d_2^2}$$

Archimedes, *Measurement of a Circle* (Theorems 1.2 and 1.3)

$$A = \frac{1}{2} r C, \quad 3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}$$

Inscribed and circumscribed polygons with n sides (Step 2 of Theorem 1.3)

$$P_n = 2nr \sin \frac{\pi}{n} < C < Q_n = 2nr \tan \frac{\pi}{n}$$

Monte Carlo estimate (Definition 1.3)

$$\hat{\pi} = \frac{4N_{\text{in}}}{N} \xrightarrow[N \rightarrow \infty]{} \pi, \quad \text{convergence } O(1/\sqrt{N})$$

Buffon's needle (Theorem 1.6), spacing $t \geq \ell$

$$p = \frac{2\ell}{\pi t}, \quad \pi \approx \frac{2\ell N}{t N_{\text{hit}}}$$

Chudnovsky series (Definition 1.4)

$$\frac{1}{\pi} = 12 \sum_{k=0}^{\infty} \frac{(-1)^k (6k)! (13591409 + 545140134k)}{(3k)! (k!)^3 640320^{3k+3/2}}$$

Machin's formula (Theorem 1.8)

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$$

Wallis' product (Theorem 1.9)

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{2n}{2n-1} \cdot \frac{2n}{2n+1}$$

Wallis integrals (proof of Theorem 1.9)

$$I_n = \int_0^{\pi/2} \sin^n x \, dx, \quad I_n = \frac{n-1}{n} I_{n-2} \quad (n \geq 2), \quad I_0 = \frac{\pi}{2}, \quad I_1 = 1$$

Arithmetic nature of π (Theorem 1.7 and Observation 1.26)

$$\pi \notin \mathbb{Q} \quad (\text{Lambert 1761; Niven 1947}), \quad \pi \text{ transcendental (Lindemann 1882)}$$

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CHAPTER 2

Fundamental Definitions

INFO

Aim of the chapter. To build, starting from a declared and minimal set of primitive notions, the whole vocabulary of the circle: circumference, centre, radius, chord, arc, tangent, secant, sector, circular segment, power. Every definition is formulated *without using concepts introduced later*, so that Chapter 3 can prove the theorems on chords and tangents with no circular dependency whatsoever.

How to read it. Sections 2.1–2.7 are indispensable and should be read in order. Sections 2.12 and 2.13 are further explorations: they may be postponed to a second reading without affecting Chapter 3.

2.1. Primitive objects, axioms and conventions

Any rigorous treatment must declare where it starts from. It is not possible to define *everything*: some notions must be assumed as primitive, and some properties must be assumed without proof. This section lists exactly what we take as known; everything else in the volume will be proved.

2.1.1 Primitive notions

We assume as primitive, that is, undefined:

- the **plane** $\mathcal{P}$, its **points** and its **lines** ($\mathcal{P}$ will always be used for the plane: the symbol π is reserved, throughout the treatise, for the constant $\pi = 3.14159\dots$);
- the relation of **incidence** between points and lines;
- the relation of **order** among collinear points (“ B lies between A and C ”);
- the **congruence** of segments and of angles.

We shall work in the Cartesian plane $\mathbb{R}^2$ whenever an analytic verification is needed; this is legitimate because $\mathbb{R}^2$ with the Euclidean distance is a *model* of plane Euclidean geometry, that is, it satisfies all the axioms we now list.

2.1.2 Axioms and results assumed

Remark 2.1: List of the results assumed without proof. In what follows we shall freely use the following facts, all of them proved in elementary geometry courses and here assumed:

- (A1) Through two distinct points there passes one and only one line.
- (A2) Through a point there passes one and only one line perpendicular to a given line.
- (A3) Congruence criteria for triangles: SSS, SAS, ASA; and for right triangles the leg-hypotenuse criterion.
- (A4) In an isosceles triangle the base angles are congruent, and conversely. The altitude to the base is also a median and a bisector.

- (A5) *The Pythagorean Theorem and its converse.*
 (A6) *The AA similarity criterion.*
 (A7) *The perpendicular bisector of a segment is the locus of the points equidistant from its endpoints.*
 (A8) *Triangle inequality: $|AC| \leq |AB| + |BC|$, with equality if and only if B belongs to the segment $\overline{AC}$.*
 (A9) *Sum of the interior angles: π for a triangle, 2π for a convex quadrilateral.*
 (A10) *Continuity postulate (Dedekind): if a line has a point inside and a point outside a circle, then it meets the circle; similarly for two circles.*

Warning. Axiom (A10) looks obvious but is not: Euclid uses it implicitly from Proposition I.1 onwards without ever stating it, and it is one of the classical gaps of the *Elements*, filled only by modern axiomatics (Hilbert, 1899). In Chapter 3 we shall use it *explicitly* in the proof of Theorem 3.8: without it, that proof would have a hole.

2.1.3 Notational conventions

Symbol	Meaning
$A, B, O, P, T, \dots$	points (capital Latin letters)
r, s, t, ℓ	lines (small Latin letters)
$\overline{AB}$	segment with endpoints A and B
$ AB $	length of the segment $\overline{AB}$, i.e. the distance from A to B
$\angle ABC$	angle with vertex B and sides BA, BC (unoriented)
$C(O, r)$	circle with centre O and radius r
$D(O, r), D^\circ(O, r)$	closed disc, open disc
$\widehat{AB}$	arc with endpoints A and B
$\mu(\widehat{AB})$	measure of the arc (in radians)
$d(P, \ell)$	distance from the point P to the line ℓ
$\text{pow}_C(P)$	power of the point P with respect to C
$\vec{u}, \vec{v}$	vectors (lower-case letters with an arrow)
$\vec{u} \cdot \vec{v}$	dot product of $\vec{u}$ and $\vec{v}$
$ \vec{u} $	modulus (length) of the vector $\vec{u}$

INFO

Two frameworks: synthetic and Cartesian. This chapter, like the rest of the volume, works with two parallel languages: the **synthetic** one (points, lines, angles, congruence, axioms A1–A10) and the **Cartesian** one (coordinates, vectors, equations), introduced wherever an algebraic check or a classification of lines by direction is needed. The two languages describe the same Euclidean plane, and their equivalence — every synthetic fact has a Cartesian counterpart and vice versa — is discussed in Chapter 7.

Remark 2.2: Two conventions to be fixed at once. (1) Circumference versus disc. In this treatise circumference (English *circle*) always denotes the **curve**, a one-dimensional object; disc denotes the **plane region** bounded by it. Italian uses *circonferenza* for the curve and *cerchio* for the region; the title of the work uses the latter in its common, inclusive sense, but in the mathematical body the distinction is rigorous. Beware: English *circle* translates Italian *circonferenza*, not *cerchio*.

(2) Angular measure. Formulas are written in radians; numerical examples use degrees, with explicit conversion. Angles are unoriented (hence in $[0, \pi]$) unless stated otherwise:

where oriented angles or segments are needed — only in the signed formulation of the power of a point — this will be stated expressly.

2.2. Distance and perpendicularity

The circle is a *metric* object: its definition requires the ability to measure distances. Let us therefore fix the notion of distance before anything else.

Definition 2.1: Distance between two points. The **distance** between $A = (x_A, y_A)$ and $B = (x_B, y_B)$ is

$$|AB| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}.$$

It satisfies: $|AB| \geq 0$ with equality if and only if $A = B$; $|AB| = |BA|$; the triangle inequality (A8).

Definition 2.2: Distance from a point to a line. The **distance** from a point P to a line ℓ is

$$d(P, \ell) = \min_{Q \in \ell} |PQ|,$$

that is, the smallest among all the distances from P to the points of ℓ .

The preceding definition would be useless if that minimum did not exist or could not be located. The following theorem, elementary but indispensable, guarantees both and will be used in every subsequent chapter.

Theorem 2.1: The foot of the perpendicular realises the minimum distance. Let ℓ be a line and $P \notin \ell$ a point. If H denotes the foot of the perpendicular drawn from P to ℓ , then

$$|PH| < |PQ| \quad \text{for every } Q \in \ell, Q \neq H.$$

In particular the minimum in Definition 2.2 exists, is unique, and is attained exactly at H : $d(P, \ell) = |PH|$.

Proof. The point H exists and is unique by axiom (A2). Let $Q \in \ell$ with $Q \neq H$. The triangle PHQ has a right angle at H (by construction), hence $\overline{PQ}$ is its **hypotenuse** and $\overline{PH}$ a leg. By the Pythagorean Theorem (A5),

$$|PQ|^2 = |PH|^2 + |HQ|^2 > |PH|^2,$$

since $|HQ| > 0$ because $Q \neq H$. Taking square roots (a strictly increasing function on the non-negative reals) yields $|PQ| > |PH|$.

It follows that the value $|PH|$ is smaller than every other value $|PQ|$: it is therefore the minimum, and it is attained at a single point. Finally, if $P \in \ell$ one sets $d(P, \ell) = 0$ and the statement is trivial with $H = P$. $\square$

Remark 2.3: Why this theorem matters. Whenever we say “the distance from the centre to the chord” or “the distance from the centre to the line” we shall always mean, without further warning, the length of the perpendicular segment. Theorem 2.1 is what makes this abbreviation legitimate and unambiguous.

2.3. Circle and disc

Definition 2.3: Circle and disc. Given a point $O \in \mathbb{R}^2$ (the **centre**) and a real number $r > 0$ (the **radius**):

Circle (the curve): $C(O, r) = \{P \in \mathbb{R}^2 : |OP| = r\}$

Closed disc: $D(O, r) = \{P \in \mathbb{R}^2 : |OP| \leq r\}$

Open disc: $D^\circ(O, r) = \{P \in \mathbb{R}^2 : |OP| < r\}$

$C(O, r) = \partial D(O, r)$ (the circle is the topological boundary of the disc)

Definition 2.4: Interior, exterior and boundary points. A point P is said to be **interior** to the circle $C(O, r)$ if $|OP| < r$; **exterior** if $|OP| > r$; and it is said to *lie on* the circle if $|OP| = r$. The set of interior points is the open disc $D^\circ(O, r)$; that of the exterior points is the complement of the closed disc.

Theorem 2.2: Trichotomy of the points of the plane. For every point P of the plane and every circle $C(O, r)$ exactly one of the three alternatives holds: P is interior, $P \in C(O, r)$, P is exterior. The three conditions are mutually exclusive and exhaustive.

Proof. The real number $|OP|$ is well defined and non-negative. By the trichotomy of the order on the real numbers, exactly one of $|OP| < r$, $|OP| = r$, $|OP| > r$ holds. These three conditions are, by Definition 2.4, precisely the three alternatives of the statement. $\square$

Remark 2.4: A trivial but not useless theorem. The preceding proof is two lines long, but the fact it states is what allows us, in Chapter 3, to argue by elimination: “ H is neither exterior nor on the circle, hence it is interior”. Without the exhaustiveness of the trichotomy, such an argument would not be legitimate.

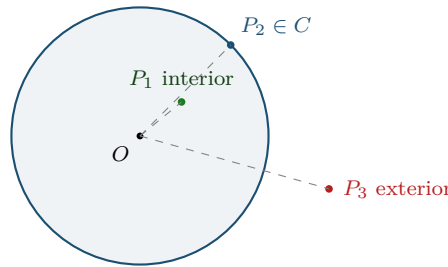


Figure 2.1: — Trichotomy: every point of the plane is interior, on the circle, or exterior, according to the comparison between $|OP|$ and r .

2.4. Centre, radius, diameter, antipodal points

Definition 2.5: Centre. The **centre** O is the point equidistant from all the points of the circle.

Definition 2.6: Radius. The **radius** is, indifferently: (a) the real number $r = |OP|$ for every $P \in C(O, r)$; or (b) the segment $\overline{OP}$ joining the centre to a point of the circle. The context always makes clear which of the two meanings is in use.

Theorem 2.3: All radii are congruent. If $P, Q \in C(O, r)$, then $\overline{OP} \cong \overline{OQ}$.

Proof. By Definition 2.3, $P \in C(O, r)$ means $|OP| = r$ and $Q \in C(O, r)$ means $|OQ| = r$. Hence $|OP| = |OQ| = r$: the two segments have the same length, that is, they are congruent. $\square$

Remark 2.5: The one fact we shall really use. Theorem 2.3 is so immediate as to look superfluous, and yet it is the engine of almost every proof in Chapter 3: whenever a triangle appears with two sides equal to the radius, that triangle is isosceles and (A4) applies. It is worth stating once and for all rather than repeating it implicitly.

Theorem 2.4: Uniqueness of centre and radius. If $C(O_1, r_1) = C(O_2, r_2)$ as sets of points, then $O_1 = O_2$ and $r_1 = r_2$. The centre and the radius of a circle are therefore uniquely determined by the circle itself.

Proof. Suppose for contradiction that $O_1 \neq O_2$ and consider the line ℓ through O_1 and O_2 (it exists and is unique by (A1)). On ℓ , let A and B be the two points at distance r_1 from O_1 , one on each side of ℓ (they exist and are distinct because $r_1 > 0$: this is the very definition of $C(O_1, r_1) \cap \ell$, with no need for the continuity postulate). By construction $A, B \in C(O_1, r_1)$; since $C(O_1, r_1) = C(O_2, r_2)$ as sets of points, also $A, B \in C(O_2, r_2)$. O_1 lies on ℓ between A and B with $|O_1A| = |O_1B| = r_1$: hence O_1 is the midpoint of $\overline{AB}$ (the unique point of line AB equidistant from the endpoints, by (A7): the axis of $\overline{AB}$, the locus of points equidistant from A, B , meets the line AB itself in exactly one point, being perpendicular to it). By the same argument applied to $C(O_2, r_2)$ (O_2 also lies on ℓ , and since $A, B \in C(O_2, r_2)$ we have $|O_2A| = |O_2B| = r_2$, so O_2 too is the point of line AB equidistant from the endpoints), O_2 is also the midpoint of $\overline{AB}$. But that midpoint is unique (A7): hence $O_1 = O_2$, against the assumption $O_1 \neq O_2$. Hence $O_1 = O_2 =: O$. Taking then any P on the circle, $r_1 = |OP| = r_2$. $\square$

Definition 2.7: Chord. A **chord** is a segment $\overline{AB}$ whose endpoints $A \neq B$ both belong to the circle.

Definition 2.8: Diameter and antipodal points. A **diameter** is a chord passing through the centre; by extension the number $d = 2r$ is also called the diameter. Two points $A, B \in C(O, r)$ are said to be **antipodal** (or diametrically opposite) if $\overline{AB}$ is a diameter, that is, if O is the midpoint of $\overline{AB}$.

Remark 2.6: Why the length of the diameter is $2r$. If $\overline{AB}$ is a diameter, O belongs to the segment $\overline{AB}$ and $|OA| = |OB| = r$; since O lies between A and B , axiom (A8) gives the equality $|AB| = |AO| + |OB| = 2r$. That O is indeed the midpoint then follows from $|AO| = |OB|$.

Definition 2.9: Open arcs determined by two points. Every point of $C(O, r)$ is written uniquely as $P(\theta) = O + r(\cos \theta, \sin \theta)$ with $\theta \in [0, 2\pi)$. Given two distinct points $A = P(\theta_A)$, $B = P(\theta_B)$ (with, say, $\theta_A < \theta_B$), we call the **open arcs** determined by A, B the two sets

$$S_1 = \{P(\theta) : \theta_A < \theta < \theta_B\}, \quad S_2 = \{P(\theta) : \theta_B < \theta < \theta_A + 2\pi\}$$

(in the second, angles are understood modulo 2π). Each is the continuous, injective image of

an open interval, hence a set of points “in one piece” in the elementary sense that suffices for this treatise; they are disjoint from each other and their union with $\{A, B\}$ is the whole circle. These two sets are precisely what, elsewhere in the chapter, is called “the two connected components of $C(O, r) \setminus \{A, B\}$ ”: the general notion of connected component (for arbitrary sets) belongs to topology and is not needed here, because the explicit parametrisation $\theta \mapsto P(\theta)$ already shows everything that is required.

Theorem 2.5: A diameter divides the circle into two congruent semicircles. Let A, B be two antipodal points of $C(O, r)$. Then the two open arcs S_1, S_2 determined by A, B (Definition 2.9) are interchanged by the reflection in the line AB . They are therefore congruent.

Proof. Place, without loss of generality, O at the origin and $A = (-r, 0)$, $B = (r, 0)$: the line AB is then the x axis, and in the parametrisation $P(\theta) = (r \cos \theta, r \sin \theta)$ we have $A = P(\pi)$, $B = P(0)$. By Definition 2.9 the two open arcs are

$$S^+ = \{P(\theta) : 0 < \theta < \pi\}, \quad S^- = \{P(\theta) : \pi < \theta < 2\pi\}.$$

The reflection σ in the x axis sends (x, y) to $(x, -y)$, hence $\sigma(P(\theta)) = P(-\theta) = P(2\pi - \theta)$. If $0 < \theta < \pi$ then $\pi < 2\pi - \theta < 2\pi$: therefore $\sigma(S^+) = S^-$ and, by the same computation, $\sigma(S^-) = S^+$. Since σ is an isometry, the two components are congruent. $\square$

Definition 2.10: Semicircle. Each of the two components of Theorem 2.5, taken together with the two antipodal points bounding it, is called a **semicircle**.

2.5. Chord, arc, measure of arcs

Defining “arc” calls for a caution that many texts neglect: we cannot say “the part of the circle between two points” without specifying which of the two parts, nor can we define it through angular measure if angular measure has not yet been introduced. We proceed therefore in two steps: first the arc as a *set of points*, then its *measure*.

Definition 2.11: Arc. Let $A, B \in C(O, r)$ be two distinct points. Each of the two open arcs S_1, S_2 determined by A, B (Definition 2.9), taken together with the points A and B , is called an **arc with endpoints A and B** and is denoted $\widehat{AB}$. When they must be distinguished, a third point is specified: $\widehat{APB}$ is the arc containing P .

Warning. The notation $\widehat{AB}$ is **ambiguous** unless one declares which of the two arcs is meant. In the text we shall always use one of these three remedies: (i) saying explicitly “minor arc” or “major arc”; (ii) naming a third point of the arc; (iii) declaring that the statement holds for both. This is the commonest source of error in the didactic literature.

Definition 2.12: Central angle and intercepted arc. A **central angle** is an angle whose vertex is the centre O and whose sides are two radii OA, OB . The **intercepted arc** of the central angle $\angle AOB$ is the arc $\widehat{AB}$ consisting of the points of the circle that belong to the angle (understood as a convex region of the plane, when $\angle AOB < \pi$).

Definition 2.13: Minor arc and major arc. Let $A, B \in C(O, r)$ be two distinct, non-antipodal points. The line AB does not pass through O (otherwise $\overline{AB}$ would be a diameter), hence it divides the plane into two open half-planes, one of which contains O . Of the two arcs $\widehat{AB}$ (Definition 2.11), the one contained in the closed half-plane that does *not* contain O is called **minor**, and the other **major**. If A and B are antipodal the two arcs are the two semicircles of Theorem 2.5: the minor/major distinction does not apply.

Definition 2.14: Measure of an arc. The **measure** of an arc $\widehat{AB}$, written $\mu(\widehat{AB})$, is defined as follows: if A, B are not antipodal and $\widehat{AB}$ is the **minor** arc (Definition 2.13), it is the measure of the convex angle $\angle AOB$; if it is the **major** arc, it is $2\pi - \angle AOB$; if A, B are antipodal, the measure of each semicircle is π .

Definition 2.15: Congruent arcs. Two arcs of the same circle, or of congruent circles, are said to be **congruent** if they have the same measure.

Warning. The hypothesis “of the same circle or of congruent circles” is not decorative. An arc of measure $\pi/2$ on a circle of radius 1 and one of measure $\pi/2$ on a circle of radius 100 have the same *angular measure* but wildly different lengths ($\pi/2$ against 50π). Confusing the angular measure of an arc with its length is a frequent error: see Section 2.9.

Definition 2.16: Inscribed angle. An **inscribed angle** is an angle whose vertex V lies on the circle and whose sides are two chords issuing from V (or a chord and a tangent). The intercepted arc is the one contained in the angle and not containing V .

Remark 2.7: What is proved here and what is not. In this chapter central and inscribed angles are only defined: their properties — in particular the central angle theorem, according to which the central angle is twice any inscribed angle standing on the same arc — are proved in Chapter 4. Chapter 3 will use them as an explicitly declared prerequisite, without proving them: this avoids any circularity, because the proof in Chapter 4 uses no result from Chapter 3.

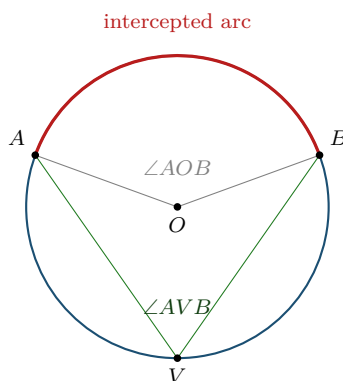


Figure 2.2: — Central angle $\angle AOB$ and inscribed angle $\angle AVB$ standing on the same arc (marked in bold). The relation between their amplitudes is proved in Chapter 4, not here.

2.6. Sector, circular segment, annulus

Definition 2.17: Circular sector. The **circular sector** determined by two radii OA , OB is the plane region bounded by the two radii and by one of the two arcs $\widehat{AB}$.

Definition 2.18: Circular segment. The **circular segment** determined by a chord $\overline{AB}$ is the plane region bounded by the chord and by one of the two arcs $\widehat{AB}$. The circular segment relative to the minor arc is called *minor*; the other, *major*.

Warning. The word “segment” has two distinct meanings: the *line segment* (the portion of a line between two points) and the *circular segment* defined here. In the text we shall always write “circular segment” in full when the latter is meant. In Italian the same ambiguity arises between *segmento* and *segmento circolare*.

Definition 2.19: Annulus. Given two concentric circles $C(O, r_1)$ and $C(O, r_2)$ with $0 < r_1 < r_2$, the **annulus** is the region $\{P : r_1 \leq |OP| \leq r_2\}$, that is, $D(O, r_2) \setminus D^\circ(O, r_1)$. Its area equals $\pi(r_2^2 - r_1^2)$.

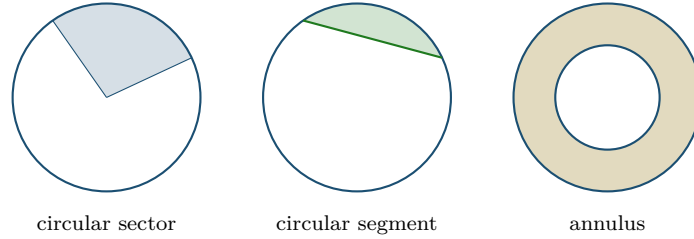


Figure 2.3: — The three plane regions associated with the circle. The sector is bounded by two radii, the circular segment by a chord, the annulus by two concentric circles.

2.7. Lines and circle: secant, tangent, external

Definition 2.20: Secant, tangent and external line. Let ℓ be a line and $C(O, r)$ a circle. The line ℓ is said to be:

- **secant** if $\ell \cap C(O, r)$ contains exactly two distinct points;
- **tangent** if $\ell \cap C(O, r)$ contains exactly one point, called the **point of tangency**;
- **external** if $\ell \cap C(O, r) = \emptyset$.

Definition 2.21: Tangent. A line t is **tangent** to the circle $C(O, r)$ if $t \cap C(O, r)$ contains exactly one point T , called the **point of tangency**.

Remark 2.8: Three candidate definitions of tangent. In the literature the tangent is defined in at least three different ways:

- (D1) a line having exactly one point in common with the circle;
- (D2) a line perpendicular to a radius at its endpoint on the circle;
- (D3) a line whose distance from the centre equals the radius.

The three conditions are equivalent, but this equivalence is a **theorem**, not a definition: it must be proved. We adopt (D1) as the official definition, following Euclid (*Elements*, Def. III.2), because it is the only one that does not presuppose knowledge of the centre.

The equivalence $(D1) \Leftrightarrow (D3)$ is proved immediately below (Theorem 2.6) by purely metric means; the equivalences with $(D2)$ are Theorems 3.8 and 3.9 of Chapter 3. **It is essential that $(D1) \Leftrightarrow (D3)$ be proved without using perpendicularity:** otherwise one would fall into the circular argument, denounced by all critical texts, of “proving” that the tangent is perpendicular to the radius after having assumed it in the classification of lines.

Theorem 2.6: Classification of lines by their distance from the centre. Let $C(O, r)$ be a circle, ℓ a line and $\delta = d(O, \ell)$. Then:

$$\delta < r \iff \ell \text{ is secant}, \quad \delta = r \iff \ell \text{ is tangent}, \quad \delta > r \iff \ell \text{ is external}.$$

Moreover, in the secant case, if M denotes the projection of O onto ℓ , the two intersection points are symmetric with respect to M and lie at distance $\sqrt{r^2 - \delta^2}$ from it.

Proof. Let M be the foot of the perpendicular from O to ℓ , so that $|OM| = \delta$ by Theorem 2.1. Let u be a unit vector in the direction of ℓ : every point of ℓ can be written uniquely as

$$P(s) = M + s u, \quad s \in \mathbb{R},$$

and $|s|$ is exactly the distance of $P(s)$ from M .

Since $\vec{OM} \perp \ell$, the vector $\vec{OM}$ is orthogonal to u , hence $\vec{OM} \cdot u = 0$. We therefore compute

$$|OP(s)|^2 = |\vec{OM} + s u|^2 = |\vec{OM}|^2 + 2s(\vec{OM} \cdot u) + s^2|u|^2 = \delta^2 + s^2.$$

(This is simply the Pythagorean Theorem (A5) applied to the right triangle $OMP(s)$, written in vector form.)

The point $P(s)$ therefore belongs to the circle if and only if

$$\delta^2 + s^2 = r^2 \iff s^2 = r^2 - \delta^2.$$

We discuss the three cases according to the sign of $r^2 - \delta^2$:

Case $\delta < r$.

Then $r^2 - \delta^2 > 0$ and the equation has exactly two distinct real solutions $s = \pm\sqrt{r^2 - \delta^2}$: the line meets the circle at two distinct points, symmetric with respect to M and at distance $\sqrt{r^2 - \delta^2}$ from it. Hence ℓ is secant.

Case $\delta = r$.

Then $r^2 - \delta^2 = 0$ and the only solution is $s = 0$, that is, the single point M : the line meets the circle at exactly one point. Hence ℓ is tangent, with point of tangency M .

Case $\delta > r$.

Then $r^2 - \delta^2 < 0$ and the equation $s^2 = r^2 - \delta^2$ has no real solutions: no point of ℓ lies on the circle. Hence ℓ is external.

The three hypotheses $\delta < r$, $\delta = r$, $\delta > r$ are exhaustive and mutually exclusive (trichotomy of the order on the reals), and so are the three conclusions (secant, tangent, external, by Definition 2.20): the implications proved are therefore automatically *equivalences*. Indeed, if for instance ℓ is tangent, it cannot be that $\delta < r$ (it would be secant) nor that $\delta > r$ (it would be external), hence $\delta = r$. $\square$

Remark 2.9: Line–circle classification. Summarising Theorem 2.6: for every line ℓ and circle $C(O, r)$, setting $\delta = d(O, \ell)$, one has $\delta > r \Rightarrow \ell \cap C = \emptyset$ (**external**); $\delta = r \Rightarrow$ **tangent**; $\delta < r \Rightarrow$ **secant**. This trichotomy is exhaustive and mutually exclusive. Note that the proof nowhere used the perpendicularity between the tangent and the radius: that result

will be deduced from here in Chapter 3, not presupposed.

Corollary 2.1: Length of the intercepted chord. If a line at distance $\delta < r$ from the centre meets $C(O, r)$ at the points A and B , then

$$|AB| = 2\sqrt{r^2 - \delta^2}.$$

Proof. By Theorem 2.6 the two points correspond to $s = \pm\sqrt{r^2 - \delta^2}$, hence their mutual distance is $|s_1 - s_2| = 2\sqrt{r^2 - \delta^2}$. $\square$

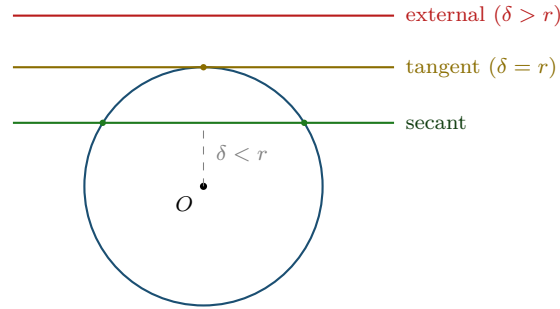


Figure 2.4: — Classification of lines according to their distance δ from the centre. The dashed segment is the perpendicular realising the distance (Theorem 2.1).

Definition 2.22: Power of a point. Given P and $C(O, r)$, the **power** of P with respect to C is the real number

$$\text{pow}_C(P) = |OP|^2 - r^2.$$

By Theorem 2.2 it is negative, zero or positive according as P is interior, on the circle, or exterior.

Remark 2.10: A definition that looks arbitrary. As things stand, Definition 2.22 is only a convenient abbreviation. Chapter 3 will show that this number has a deep geometric meaning: it is the (signed) product of the distances from P to the two points where any line through P meets the circle — a product which, surprisingly, does not depend on the chosen line.

2.8. The overall picture

Definition 2.23: Circle inscribed in and circumscribed about a polygon. A circle is **circumscribed** about a polygon if it passes through all of its vertices; it is **inscribed** if it is tangent to all of its sides.

Definition 2.24: Incentre and circumcentre of a triangle. The **circumcentre** O of a triangle is the centre of its circumscribed circle (existence and uniqueness proved in Chapter 3, Corollary 3.2, since the three vertices are not collinear). The **incentre** I is the centre of its inscribed circle (existence and uniqueness proved in Chapter 10, together with the metric properties of both centres).

2.9. Chord, arc, sector, segment: formulas

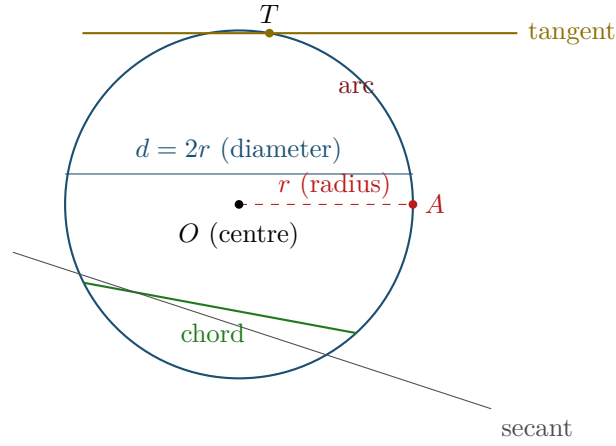


Figure 2.5: — All the fundamental elements of the circle in a single figure

For a central angle θ (in radians):

Arc length:	$l = r\theta$	$\iff \theta = l/r$
Area of the circular sector:	$A_s = \frac{1}{2}r^2\theta$	$\iff \theta = 2A_s/r^2$
Chord length:	$ AB = 2r \sin(\theta/2)$	$\iff \theta = 2 \arcsin(AB /2r)$
Area of the circular segment:	$A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta)$	

Proof. The first two formulas express the direct proportionality between the measure of the central angle, the arc length and the area of the sector: to the full angle 2π correspond the whole circumference $2\pi r$ and the whole disc πr^2 , hence

$$\frac{l}{2\pi r} = \frac{\theta}{2\pi} \implies l = r\theta, \quad \frac{A_s}{\pi r^2} = \frac{\theta}{2\pi} \implies A_s = \frac{1}{2}r^2\theta.$$

(The legitimacy of the proportionality follows from the additivity of lengths and areas with respect to the sum of adjacent central angles.)

For the chord: the triangle OAB is isosceles with $|OA| = |OB| = r$ and vertex angle θ . If M denotes the midpoint of $\overline{AB}$, by (A4) OM is an altitude and a bisector, hence the right triangle OMA has angle $\theta/2$ at O and hypotenuse r ; therefore $|MA| = r \sin(\theta/2)$ and $|AB| = 2|MA| = 2r \sin(\theta/2)$.

For the circular segment: it is the difference between the sector and the triangle OAB , whose area is $\frac{1}{2}|OA||OB|\sin \theta = \frac{1}{2}r^2 \sin \theta$. Hence $A_{\text{seg}} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta$. $\square$

Remark 2.11: Inverse formula for the segment and asymptotic behaviour. $A_{\text{seg}} = A_s - A_{\triangle}$, where $A_{\triangle} = \frac{1}{2}r^2 \sin \theta$ is the area of the isosceles triangle OAB . As $\theta \rightarrow 0$: $A_{\text{seg}} \sim r^2\theta^3/12$ (Taylor expansion of $\theta - \sin \theta$), so the segment vanishes far more rapidly than the sector when the angle is small.

Warning. Compare the two formulas $l = r\theta$ (arc length) and $\mu(\widehat{AB}) = \theta$ (arc measure): the first depends on the radius, the second does not. “Congruent arcs” means *same measure*, and on different circles this by no means implies the same length. In Chapter 3 this distinction

is crucial in every statement containing the hypothesis “on the same circle”.

Definition 2.25: Sagitta of an arc. Given a chord $\overline{AB}$ and one of the two arcs it determines, the **sagitta** h is the distance between the midpoint of the arc and the midpoint of the chord. If d is the distance from the centre to the chord and θ the central angle of the chosen arc, $h = r - d = r(1 - \cos(\theta/2))$ for the minor arc ($h = r + d$ for the major arc).

2.10. Other recurring definitions

Definition 2.26: Cyclic (inscribed) quadrilateral. A quadrilateral $ABCD$ is **inscribed** in a circle, or **cyclic**, if its four vertices all belong to the same circle. The property that characterises it — opposite angles supplementary if and only if convex and cyclic — is proved in Chapter 4 (P-12bis).

Definition 2.27: Chord of contact. From a point P outside a circle γ , draw the two tangents to γ , touching γ at points T_1, T_2 (Theorem 3.11, Chapter 3). The segment $\overline{T_1T_2}$ is called the **chord of contact** of P with respect to γ .

Definition 2.28: Relative positions of two circles. Given two distinct circles $C(O_1, r_1)$, $C(O_2, r_2)$, with $d = |O_1O_2|$, their relative position is classified by comparing d with $|r_1 - r_2|$ and $r_1 + r_2$:

external $d > r_1 + r_2$: no common points, each outside the other;

externally tangent

$d = r_1 + r_2$: one common point, the interior discs disjoint;

secant $|r_1 - r_2| < d < r_1 + r_2$: two common points;

internally tangent

$d = |r_1 - r_2|$ (with $r_1 \neq r_2$): one common point, one circle inside the other;

one inside the other

$d < |r_1 - r_2|$: no common points;

concentric $d = 0$: degenerate case of “one inside the other”.

Definition 2.29: Angle between two intersecting circles. If two circles are secant (Definition 2.28) and meet at a point P , their **angle** at P is the angle between the tangent lines to the two circles at P (equivalently, the angle between the two radii O_1P and O_2P , each tangent being perpendicular to its own radius). Two secant circles are called **orthogonal** if this angle is a right angle.

2.11. Ruler-and-compass constructions

The definitions of this chapter are not merely descriptive: every element of the circle is *constructible* with an (ungraduated) ruler and compass, following the Euclidean tradition. The three constructions below are the building blocks used implicitly throughout the rest of the treatise.

Warning. A declared anticipation. The three constructions that follow justify their last step by citing theorems (Corollary 3.1, Theorem 3.9, Corollary 3.3, Theorem 3.11) proved only in Chapter 3, which follows this one. It is the same kind of anticipation already flagged in Observation 2.7: the constructions themselves (the ruler-and-compass steps) are elementary and depend on nothing later; only the *justification of why they work* uses results the reader will meet a few pages on.

Construction 2.1: — Centre of a given circle. **Data:** a circle C whose centre is unknown.

Step 1. Choose three points A, B, P on C and draw the chords AB and BP .

Step 2. Construct the perpendicular bisector of AB , and likewise that of BP , with ruler and compass (intersection of two arcs of equal radius centred at the two endpoints of each chord).

Step 3. By Corollary 3.1 (Chapter 3), each perpendicular bisector passes through the centre: the intersection point of the two bisectors is the centre O sought.

Why the two bisectors meet. If they were parallel, they would both be perpendicular to the same direction, hence $AB \parallel BP$; but these two chords have the point B in common, so they would coincide as lines, and A, B, P would be collinear: impossible for three distinct points of a circle (a line meets it in at most two points, by Theorem 2.6). $\square$

Construction 2.2: — Tangent at a point of the circle. **Data:** circle $C(O, r)$ and point $T \in C$.

Step 1. Draw the radius OT .

Step 2. Construct the perpendicular to OT through T (with ruler and compass, marking two arcs centred at T on the extension of OT and then the perpendicular bisector of the segment so obtained).

Step 3. By Theorem 3.9 this line is tangent, and by Corollary 3.3 it is the only tangent at T . $\square$

Construction 2.3: — Tangents from an external point (Thales' circle). **Data:** circle $C(O, r)$ and external point P .

Step 1. Draw the segment OP and construct its midpoint M .

Step 2. Draw the auxiliary circle with centre M and radius $|MO| = |MP|$ ("Thales' circle" on the diameter OP).

Step 3. This auxiliary circle meets C at two points T_1, T_2 : by Thales' theorem, $\angle OT_1P = \angle OT_2P = 90^\circ$, hence $PT_1 \perp OT_1$ and $PT_2 \perp OT_2$, that is (Theorem 3.9) PT_1 and PT_2 are the two tangents sought, in agreement with Theorem 3.11. $\square$

2.12. The circle as a locus: equivalent definitions

Remark 2.12: Three ways of "seeing" a circle. Definition 2.3 (points equidistant from a centre) is only one of the possible characterisations of the circle as a locus. The following theorem shows how the same curve arises from an apparently very different condition.

Theorem 2.7: Circle of Apollonius (locus of constant ratio of distances). Given two distinct points A, B and a constant $k > 0$, $k \neq 1$, the locus of the points P such that $PA/PB = k$ is a circle (called the circle of Apollonius relative to A, B, k), whose centre lies on the line AB .

Proof. Set $A = (-a, 0)$, $B = (a, 0)$ and $P = (x, y)$. The condition $PA = k PB$ becomes, on squaring,

$$(x + a)^2 + y^2 = k^2[(x - a)^2 + y^2].$$

Expanding and collecting:

$$(1 - k^2)(x^2 + y^2) + 2a(1 + k^2)x + a^2(1 - k^2) = 0.$$

Since $k \neq 1$, one may divide by $(1 - k^2) \neq 0$, obtaining

$$x^2 + y^2 + \frac{2a(1 + k^2)}{1 - k^2}x + a^2 = 0,$$

of the form $x^2 + y^2 + Dx + F = 0$ (no isolated term in y), with $D = \frac{2a(1 + k^2)}{1 - k^2}$ and $F = a^2$. It remains to check that this is the equation of a *non-degenerate* circle, that is, $D^2/4 - F > 0$:

$$\frac{D^2}{4} - F = \frac{a^2(1 + k^2)^2}{(1 - k^2)^2} - a^2 = a^2 \frac{(1 + k^2)^2 - (1 - k^2)^2}{(1 - k^2)^2} = \frac{4a^2k^2}{(1 - k^2)^2} > 0$$

(the last equality by expanding the two squares, which differ by $4k^2$). Hence the equation is exactly that of a circle, with centre $\left(-\frac{a(1 + k^2)}{1 - k^2}, 0\right)$ and radius $\frac{2ak}{|1 - k^2|} > 0$, centred on the x axis, that is, on the line AB . $\square$

Synthetic proof (Apollonius). The internal and external bisectors of the angle $\widehat{APB}$ cut the line AB at the points M (interior to the segment AB) and N (exterior), with $MA/MB = NA/NB = k$: this is the **angle bisector theorem**, which we prove here with the law of sines. In triangles APM and BPM , which share the angle $\widehat{APM} = \widehat{BPM} = \widehat{P}/2$ at P ,

$$\frac{MA}{\sin(\widehat{P}/2)} = \frac{PA}{\sin \widehat{AMP}}, \quad \frac{MB}{\sin(\widehat{P}/2)} = \frac{PB}{\sin \widehat{BMP}},$$

and $\widehat{AMP}$, $\widehat{BMP}$ are supplementary (adjacent angles on the line AB), hence have the same sine: dividing side by side, $MA/MB = PA/PB = k$. The same argument with the external bisector (the angle between PA and the extension of PB) gives $NA/NB = k$.

The internal and external bisectors of the same angle are perpendicular, hence P sees the segment MN at a right angle: by the converse of Thales' theorem (Theorem 4.5, Chapter 4), P lies on the circle of diameter MN . The points M, N depend only on A, B, k (not on P), so **every** point P of the locus $PA/PB = k$ lies on that circle: the locus is contained in it. This synthetic argument thus shows that the locus $PA/PB = k$ is **contained** in the circle of diameter MN ; that the converse also holds (every point of that circle satisfies $PA/PB = k$), that is, that the locus coincides with the whole circle, is already established by the preceding analytic proof, which directly gives the equality of the two equations. $\square$

Remark 2.13: Why $k = 1$ is excluded. If $k = 1$, the condition $PA = PB$ becomes the locus of the points equidistant from A and B : by definition, the perpendicular bisector of AB , that is, a **line**, not a circle (the limiting case $k \rightarrow 1$ corresponds to a circle whose radius tends to infinity, “degenerating” into a line). This circle of Apollonius is the same object which, generalised to three pairs of points/lines/circles, gives its name to the problem of Apollonius treated in Chapter 5.

2.13. Pencils of coaxal circles

The circle of Apollonius of the previous section is a particular case of a wider and recurrent family: a whole *pencil* (one-parameter family) of circles generated by two given circles.

Definition 2.30: Pencil of circles. Given two distinct circles $\Gamma_1 : x^2 + y^2 + D_1x + E_1y + F_1 = 0$ and $\Gamma_2 : x^2 + y^2 + D_2x + E_2y + F_2 = 0$, the **pencil** generated by Γ_1, Γ_2 is the one-parameter family $\lambda \in \mathbb{R} \setminus \{-1\}$

$$\Gamma_\lambda : \frac{1}{1+\lambda} \left[(x^2 + y^2 + D_1x + E_1y + F_1) + \lambda (x^2 + y^2 + D_2x + E_2y + F_2) \right] = 0,$$

normalised so as always to have leading coefficient 1 (it is a genuine circle equation, not merely one up to a factor). The case $\lambda = -1$ is excluded from the family and treated separately: as $\lambda \rightarrow -1$ the expression in square brackets tends to $\Gamma_1 - \Gamma_2$, the **radical axis** of Γ_1, Γ_2 (a line, not a circle): it is the degenerate member of the pencil, common to *every* pencil (secant, tangent or non-secant), not only to those with no base points.

Theorem 2.8: Base points and radical axis of the pencil. Every circle Γ_λ of the pencil passes through all the points common to Γ_1 and Γ_2 (the **base points** of the pencil, if they exist); moreover, all the circles of the pencil have the same radical axis with respect to any pair of them, coinciding with the radical axis of Γ_1, Γ_2 .

Proof. If P satisfies both the equation of Γ_1 and that of Γ_2 (that is, $\Gamma_1(P) = \Gamma_2(P) = 0$), then substituting into Γ_λ gives $\Gamma_\lambda(P) = 0 + \lambda \cdot 0 = 0$ for every λ : P belongs to every circle of the pencil. As for the radical axis, observe that the difference $\Gamma_\lambda - \Gamma_\mu$ (for $\lambda \neq \mu$) is, after dividing by $(\lambda - \mu)$, exactly $\Gamma_1 - \Gamma_2$ up to a non-zero constant factor (a direct algebraic check by expanding the linear combinations): since by Definition 2.30 the radical axis of two circles is the locus $\Gamma_1(P) = \Gamma_2(P)$, that is, $\Gamma_1(P) - \Gamma_2(P) = 0$, and this same equation (up to rescaling) turns out to be also $\Gamma_\lambda(P) - \Gamma_\mu(P) = 0$ for every pair of parameters, the radical axis is the same for every pair of circles of the pencil. $\square$

Example 2.1: A non-secant pencil and its limit points. Consider

$$\Gamma_1 : x^2 + y^2 - 25 = 0 \quad (\text{centre } (0;0), r_1 = 5),$$

$$\Gamma_2 : (x - 10)^2 + y^2 = 9 \quad \text{i.e.} \quad x^2 + y^2 - 20x + 91 = 0 \quad (\text{centre } (10;0), r_2 = 3).$$

Step 1 — what type of pencil? The distance of the centres is $d = 10$, whereas $r_1 + r_2 = 8$ and $|r_1 - r_2| = 2$. Since $d > r_1 + r_2$, the two circles are **external** and do not meet: the pencil is **non-secant** and possesses two limit points.

Step 2 — the equation of the pencil. By Definition 2.30, the members of the pencil are

$$\frac{1}{1+\lambda} [\Gamma_1 + \lambda \Gamma_2] = 0 \iff \frac{1}{1+\lambda} [(x^2 + y^2 - 25) + \lambda(x^2 + y^2 - 20x + 91)] = 0 \iff x^2 + y^2 - \frac{20\lambda}{1+\lambda}x + \frac{91\lambda - 25}{1+\lambda} = 0$$

Each member therefore has centre $\left(\frac{10\lambda}{1+\lambda}; 0 \right)$ — all on the line of centres, as it must be — and, over the common denominator $(1+\lambda)^2$,

$$r^2(\lambda) = \left(\frac{10\lambda}{1+\lambda} \right)^2 - \frac{91\lambda - 25}{1+\lambda} = \frac{100\lambda^2 - (91\lambda - 25)(1+\lambda)}{(1+\lambda)^2} = \frac{9\lambda^2 - 66\lambda + 25}{(1+\lambda)^2}.$$

Step 3 — the limit points. They are the degenerate members, those with $r(\lambda) = 0$, that is, with vanishing numerator:

$$9\lambda^2 - 66\lambda + 25 = 0 \implies \lambda = \frac{66 \pm \sqrt{4356 - 900}}{18} = \frac{66 \pm \sqrt{3456}}{18} = \frac{66 \pm 58.7878}{18},$$

$$\lambda_1 = 6.932653, \quad \lambda_2 = 0.400680.$$

The corresponding centres — which, having zero radius, *are* the limit points — are (checked)

$$L_1 = \left(\frac{10\lambda_1}{1 + \lambda_1}; 0 \right) = (8.73939; 0), \quad L_2 = \left(\frac{10\lambda_2}{1 + \lambda_2}; 0 \right) = (2.86061; 0).$$

(These are the same two points as in the other worked example that uses an unnormalised parametrisation: only the λ value labelling them changes, owing to the different normalisation of Definition 2.30 relative to an unnormalised parametrisation.)

Step 4 — the discriminant means something. The discriminant $3456 > 0$ is what guarantees two real limit points. Were the two circles *secant*, it would come out negative and the pencil would have no limit points but two base points — consistently with Remark 2.14 below.

Step 5 — the radical axis. Subtracting the two equations: $-20x + 116 = 0$, i.e. $x = 5.8$. It is a vertical line, perpendicular to the line of centres as predicted. Note that the two limit points $L_2 = 2.86$ and $L_1 = 8.74$ lie on *opposite sides* of it, and that

$$\frac{L_1 + L_2}{2} = \frac{8.73939 + 2.86061}{2} = 5.80,$$

that is, the radical axis passes through their midpoint.

Step 6 — the link with Chapter 5. These two points are exactly the roots of equation (5.2) of Lemma 5.10: choosing L_1 or L_2 as the pole of an inversion turns Γ_1 and Γ_2 into **concentric** circles. Check with the lemma's formula, $d = 10$, $r_1 = 5$, $r_2 = 3$:

$$10p^2 + (9 - 25 - 100)p + 10 \cdot 25 = 0 \iff 5p^2 - 58p + 125 = 0,$$

$$p = \frac{58 \pm \sqrt{3364 - 2500}}{10} = \frac{58 \pm 29.3939}{10},$$

that is $p_1 = 8.73939$ and $p_2 = 2.86061$: the same two points. ✓ Two chapters, two different algebraic routes, one result.

Remark 2.14: Three types of pencil. According to the relative position of Γ_1, Γ_2 , the pencil is classified into three types: **secant pencil** (if Γ_1, Γ_2 meet at two real points, all the circles of the pencil pass through those two base points); **tangent pencil** (if Γ_1, Γ_2 are tangent at a point, all the circles of the pencil are tangent to one another at that same point); **non-secant pencil** (if Γ_1, Γ_2 do not meet, the pencil has no real base points, and instead admits two **limit points**, as in Example 2.1). In **all three** cases, as $\lambda \rightarrow -1$ the member Γ_λ of Definition 2.30 degenerates into the radical axis (a line): this is not a peculiarity of the non-secant pencil, but the degenerate member common to every pencil.

Remark 2.15: Connection with the circle of Apollonius. As $k > 0$ varies in Theorem 2.7, the circles of Apollonius relative to the same pair A, B themselves form a non-secant pencil (none of these circles, for $k \neq 1$, passes through A or B): the degenerate case $k = 1$ — the perpendicular bisector of AB , a line — is exactly the limiting radical axis of this pencil, in perfect agreement with Theorem 2.8. A second pencil, **orthogonal** to the first at

every point (each circle of one meets each circle of the other at a right angle), is given by the circles through A and B themselves (a secant pencil with base points A, B): this pair of orthogonal pencils underlies the bipolar coordinate system, useful for instance in the study of the electrostatic fields generated by two point charges, by direct mathematical analogy.

2.14. Glossary and table of notation

Centre	point equidistant from every point of the circle.
Radius	distance from the centre to the circle, or the segment itself.
Diameter	the greatest chord, passing through the centre, $d = 2r$.
Antipodal points	endpoints of one and the same diameter.
Chord	segment between two points of the circle.
Arc	one of the two open arcs determined by two of its points (Definition 2.9), taken with the two endpoints.
Minor/major arc	arc of measure less/greater than π .
Semicircle	arc determined by two antipodal points; measure π .
Measure of an arc	measure in radians of the corresponding central angle; it is <i>not</i> the arc length.
Central angle	angle with vertex at the centre and sides two radii.
Inscribed angle	angle with vertex on the circle and sides two chords.
Tangent	line with exactly one point in common with the circle.
Secant	line with two points in common with the circle.
External	line with no point in common with the circle.
Circular sector	region bounded by two radii and an arc.
Circular segment	region bounded by a chord and an arc.
Annulus	region between two concentric circles.
Power of a point	the number $ OP ^2 - r^2$.

2.14.1 Italian–English terminological table

International usage is not a word-for-word translation of the Italian one: the following table prevents the commonest errors when consulting the literature.

Italiano	English	Note
circonferenza	circle	in English <i>circle</i> is the <i>curve</i>
cerchio, disco	disc	
corda	chord	plural <i>radii</i>
raggio	radius	
diametro	diameter	
arco	arc	
arco minore/maggiore	minor/major arc	
semicirconferenza	semicircle	
secante	secant (line)	
tangente	tangent (line)	
punto di tangenza	point of tangency, of contact	
angolo al centro	central angle	
angolo inscritto	inscribed angle	
arco intercettato	intercepted arc	
segmento di tangente	tangent segment	
potenza di un punto	power of a point	
asse di un segmento	perpendicular bisector	false friend: not <i>axis</i>
asse radicale	radical axis	here <i>axis</i> is correct
settore circolare	circular sector	
segmento circolare	circular segment	false friend with <i>line segment</i>
corona circolare	annulus	

Warning. The two false friends to watch are *asse* (translated *perpendicular bisector* when it is the perpendicular bisector of a segment, and *axis* when it is the radical axis or an axis of symmetry) and *segmento* (which is a *line segment* in elementary geometry, but a *circular segment* for the region bounded by a chord and an arc). A wrong translation of these two terms makes bibliographic search impossible.

2.15. Exercises

Exercise 2.1. Given $r = 5$ cm and $\theta = 120^\circ$, compute the arc length, the area of the sector, the chord length and the area of the circular segment (convert θ into radians first).

Solution. $\theta = 120^\circ = 2\pi/3$ rad.

$$l = r\theta = 5 \cdot \frac{2\pi}{3} = \frac{10\pi}{3} \approx 10.47 \text{ cm}, \quad A_s = \frac{1}{2}r^2\theta = \frac{1}{2} \cdot 25 \cdot \frac{2\pi}{3} = \frac{25\pi}{3} \approx 26.18 \text{ cm}^2,$$

$$|AB| = 2r \sin(\theta/2) = 10 \sin 60^\circ = 5\sqrt{3} \approx 8.66 \text{ cm},$$

$$A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta) = \frac{25}{2} \left(\frac{2\pi}{3} - \sin 120^\circ \right) = 12.5 (2.0944 - 0.8660) \approx 15.35 \text{ cm}^2.$$

Exercise 2.2. Prove that, for a fixed circle, as $\theta \in (0, \pi]$ increases the chord $|AB| = 2r \sin(\theta/2)$ is a strictly increasing function, whereas for $\theta \in (\pi, 2\pi)$ it is decreasing: interpret this fact geometrically.

Solution. $\frac{d}{d\theta} [2r \sin(\theta/2)] = r \cos(\theta/2)$. For $\theta \in (0, \pi)$, $\theta/2 \in (0, \pi/2)$ and $\cos(\theta/2) > 0$: the function increases. For $\theta \in (\pi, 2\pi)$, $\theta/2 \in (\pi/2, \pi)$ and $\cos(\theta/2) < 0$: the function decreases. Geometrically, as θ grows from 0 to π the two points A, B move apart until they become

diametrically opposite (maximum chord = $2r$, the diameter); for $\theta > \pi$ the central angle “passes” the diameter and the two points begin to approach each other again from the other side, a fact also visible from the identity $\text{chord}(\theta) = \text{chord}(2\pi - \theta)$ (symmetry of the sine function about π).

Exercise 2.3. A line is 3 cm from the centre of a circle of radius 5 cm. Classify the line (external/tangent/secant) and, if secant, compute the length of the intercepted chord.

Solution. $d = 3 < r = 5$: by the trichotomy of Remark 2.9, the line is **secant**. Chord length (Corollary 2.1):

$$|AB| = 2\sqrt{r^2 - d^2} = 2\sqrt{25 - 9} = 2\sqrt{16} = 8 \text{ cm.}$$

Exercise 2.4. Verify directly, by a computation in coordinates, Theorem 2.4 in the following particular case: show that there exist no $O_2 \neq (0, 0)$ and $r_2 > 0$ such that $C(O_2, r_2)$ coincides with the circle $x^2 + y^2 = 1$.

Solution. Let $O_2 = (p, q)$ and suppose that $C(O_2, r_2)$ coincides with $x^2 + y^2 = 1$. Then every point (x, y) with $x^2 + y^2 = 1$ satisfies $(x - p)^2 + (y - q)^2 = r_2^2$, that is

$$x^2 + y^2 - 2px - 2qy + p^2 + q^2 = r_2^2 \implies 1 - 2px - 2qy + p^2 + q^2 = r_2^2.$$

This relation must hold for *every* point of the unit circle. Taking $(1, 0)$ and $(-1, 0)$ gives $-2p = 2p$, that is $p = 0$; taking $(0, 1)$ and $(0, -1)$ gives $q = 0$. Hence $O_2 = (0, 0)$ and, substituting, $r_2^2 = 1$, that is $r_2 = 1$: centre and radius coincide with the original ones, as the theorem asserts.

Exercise 2.5. (*On the risk of circularity.*) A student states: “The line ℓ is tangent because it has only one point in common; hence it is perpendicular to the radius; hence its distance from the centre is r ”. Identify the step that, as the chapter now stands, is not yet justified, and show how to reach the same conclusion without that step.

Solution. The unjustified step is the second: “has only one point in common $\implies$ is perpendicular to the radius”. This is exactly the conclusion of Theorem 3.8, proved only in Chapter 3; assuming it here would be circular, because that theorem rests precisely on the classification of lines.

The same conclusion is instead reached without perpendicularity: by Theorem 2.6, if $\delta < r$ the line would be secant (two points) and if $\delta > r$ it would be external (no point). Since by hypothesis the line has exactly one point in common, it can be neither secant nor external, hence $\delta = r$. The deduction uses only the metric comparison and the trichotomy, never perpendicularity.

Exercise 2.6. In a circle of radius $r = 8$ cm a chord of length 12 cm is drawn. Compute the distance of the chord from the centre, the central angle, the lengths of the two arcs and the areas of the two circular segments.

Solution. Distance: $d = \sqrt{8^2 - 6^2} = \sqrt{28} = 5.2915$ cm. Angle: from $\sin(\theta/2) = 6/8 = 0.75$ we get $\theta/2 = 48.590^\circ$, hence $\theta = 97.181^\circ = 1.69612$ rad. Arcs: $l_{\min} = r\theta = 13.569$ cm and

$l_{\max} = 8(2\pi - 1.69612) = 36.697$ cm, whose sum is $50.265 = 2\pi \cdot 8 \checkmark$. Minor segment:

$$A = \frac{1}{2}r^2(\theta - \sin \theta) = 32(1.69612 - 0.99216) = 22.527 \text{ cm}^2,$$

and the major segment $= \pi \cdot 64 - 22.527 = 178.535 \text{ cm}^2$.

Exercise 2.7. An annulus has radii $R = 10$ and $r = 6$. Compute its area and determine the radius ρ of the circle dividing it into two parts of equal area.

Solution. $A = \pi(100 - 36) = 64\pi = 201.062$. Imposing $\pi(\rho^2 - 36) = 32\pi$ gives $\rho^2 = 68$, that is $\rho = 8.2462$. Note that ρ is *not* the arithmetic mean $(10 + 6)/2 = 8$ but the quadratic mean $\sqrt{(100 + 36)/2} = \sqrt{68}$: area grows as the square of the radius, and bisecting the area is bisecting the squares.

Exercise 2.8. Prove that the perpendicular drawn from the centre to a chord is its perpendicular bisector, and deduce that two chords of equal length are equidistant from the centre.

Solution. Let A and B be the endpoints of the chord and H the foot of the perpendicular from O . The triangles OHA and OHB are right-angled at H , share the leg OH and have hypotenuses $OA = OB = r$; by Pythagoras $HA^2 = r^2 - OH^2 = HB^2$, hence $HA = HB$ and H is the midpoint: the perpendicular is the bisector. Conversely, if two chords have the same length $2c$, their distances from the centre both equal $\sqrt{r^2 - c^2}$, hence are equal.

Exercise 2.9. From a point P with $|OP| = 13$ the two tangents are drawn to a circle of centre O and radius 5. Compute the length of the tangent segments, the angle between the two tangents and the distance between the two contact points.

Solution. The triangle OTP is right-angled at T , hence $|PT| = \sqrt{169 - 25} = 12$. The angle $\widehat{OPT}$ satisfies $\sin \widehat{OPT} = 5/13$, that is $\widehat{OPT} = 22.6199^\circ$, and the angle between the tangents is twice as much: 45.2397° . The chord of contact is perpendicular to OP and its half-length is $r|PT|/|OP| = 5 \cdot 12/13$, hence the distance between the contact points is $120/13 = 9.2308$.

Exercise 2.10. Two circles have radii $r_1 = 7$ and $r_2 = 4$ and centres at distance d . Classify their mutual positions as d varies and compute, for $d = 8$, the length of the common chord.

Solution. The positions are: internal without contact for $d < 3$; internally tangent for $d = 3$; secant for $3 < d < 11$; externally tangent for $d = 11$; external for $d > 11$. For $d = 8$ they are secant. The distance from the first centre to the line of the common chord is

$$x = \frac{d^2 + r_1^2 - r_2^2}{2d} = \frac{64 + 49 - 16}{16} = \frac{97}{16} = 6.0625,$$

hence the half-chord is $\sqrt{49 - 36.7539} = 3.4994$ and the common chord measures 6.9988.

Exercise 2.11. Given the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 4x + 3 = 0$, determine the radical axis, say whether the pencil they generate has base points, and compute them.

Solution. The second has centre $(2, 0)$ and radius 1, because $x^2 + y^2 - 4x + 3 = (x - 2)^2 + y^2 - 1$. Subtracting the two equations gives $4x - 3 - 4 = 0$, that is $x = 7/4$: this is the radical axis, perpendicular to the line of centres. Since $|O_1 O_2| = 2$ and $|r_1 - r_2| = 1 < 2 < 3 = r_1 + r_2$, the two circles are secant and the pencil has two base points: substituting $x = 7/4$ in the first gives $y^2 = 4 - 49/16 = 15/16$, hence the base points are $(7/4, \pm\sqrt{15}/4) = (1.75, \pm 0.96825)$.

Exercise 2.12. Prove that the locus of points P with $|PA|/|PB| = k$, where $k > 0$ and $k \neq 1$, is a circle (the Apollonius circle), and determine it for $A = (0, 0)$, $B = (4, 0)$, $k = 2$.

Solution. The condition $|PA|^2 = k^2|PB|^2$ reads $(1 - k^2)(x^2 + y^2) + 2k^2 \mathbf{b} \cdot P - k^2|\mathbf{b}|^2 = 0$ with $\mathbf{b} = B - A$; since $k \neq 1$ the coefficient of $x^2 + y^2$ is non-zero and one may divide by it, obtaining an equation of the form $x^2 + y^2 + ax + by + c = 0$, that is a circle (possibly degenerate, but not here, since the locus contains points). In the case proposed:

$$x^2 + y^2 = 4[(x - 4)^2 + y^2] \iff 3x^2 + 3y^2 - 32x + 64 = 0 \iff x^2 + y^2 - \frac{32}{3}x + \frac{64}{3} = 0,$$

with centre $(16/3, 0) = (5.3333, 0)$ and radius $\sqrt{256/9 - 64/3} = \sqrt{64/9} = 8/3$. Check on the two points of the x axis: for $x = 8$ we get $8/4 = 2 \checkmark$; for $x = 8/3$ we get $(8/3)/(4/3) = 2 \checkmark$.

2.16. Self-assessment quiz

Multiple-choice questions for a quick check before tackling the exercises. The answer key is at the end of the section.

1. In this treatise the word *circle* denotes
 - A) the plane region enclosed
 - B) the curve
 - C) the upper boundary only
 - D) either one indifferently
2. A line whose distance from the centre is *greater* than the radius is
 - A) secant
 - B) tangent
 - C) external
 - D) a diameter
3. The classification of lines in Chapter 2 is proved
 - A) by assuming that the tangent is perpendicular to the radius
 - B) by purely metric means, without using perpendicularity
 - C) experimentally
 - D) only for the secant case
4. The measure of an arc
 - A) coincides with its length

-
- B) is measured in metres
C) does not depend on the radius
D) depends on the radius
5. The length of a chord subtended by a central angle θ is
A) $r\theta$
B) $2r \sin(\theta/2)$
C) $2r\theta$
D) $r \sin \theta$
6. The power of a point P with respect to $C(O, r)$ is negative if and only if
A) P is exterior
B) P is on the circle
C) P is interior
D) P coincides with the centre
7. The Italian term *asse* of a segment is rendered in English as
A) axis
B) perpendicular bisector
C) median
D) axle
8. The annulus is the region between
A) two parallel chords
B) two radii and an arc
C) two concentric circles
D) a chord and an arc
9. The locus of the points P with $PA/PB = k$, $k \neq 1$, is
A) the perpendicular bisector of AB
B) a circle (of Apollonius)
C) an ellipse
D) a line parallel to AB
10. The centre and the radius of a circle
A) may be chosen in several ways
B) are uniquely determined
C) depend on the coordinate system
D) are defined only if the radius is an integer

BEST PRACTICE

Answer key. 1: B — 2: C — 3: B — 4: C — 5: B — 6: C — 7: B — 8: C — 9: B — 10: B.

2.17. Chapter Formula sheet

Distance between two points (Definition 2.1)

$$|AB| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}, \quad d(P, \ell) = \min_{Q \in \ell} |PQ|$$

Circle, closed disc, open disc (Definition 2.3)

$$C(O, r) = \{P : |OP| = r\}, \quad D(O, r) = \{P : |OP| \leq r\}, \quad D^\circ(O, r) = \{P : |OP| < r\}$$

Position of a point (Definition 2.4; Theorem 2.2)

$$|OP| < r \iff \text{interior}, \quad |OP| = r \iff P \in C(O, r), \quad |OP| > r \iff \text{exterior}$$

Line and circle, with $\delta = d(O, \ell)$ (Theorem 2.6; Corollary 2.1)

$$\delta < r \iff \text{secant}, \quad \delta = r \iff \text{tangent}, \quad \delta > r \iff \text{external}$$

$$|AB| = 2\sqrt{r^2 - \delta^2}$$

Arc, sector, chord, circular segment (Section 2.9), with θ in radians

$$\begin{aligned} l = r\theta &\iff \theta = l/r \\ A_s = \frac{1}{2}r^2\theta &\iff \theta = 2A_s/r^2 \\ |AB| = 2r \sin(\theta/2) &\iff \theta = 2 \arcsin(|AB|/2r) \\ A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta) &\quad (A_{\text{seg}} \sim r^2\theta^3/12 \text{ as } \theta \rightarrow 0) \end{aligned}$$

Measure of an arc (Definition 2.14)

$$\mu(\widehat{AB}) = \theta \quad (\text{minor arc}), \quad \mu_{\text{major}} = 2\pi - \theta$$

Annulus (Definition 2.19)

$$\{P : r_1 \leq |OP| \leq r_2\}, \quad A = \pi(r_2^2 - r_1^2)$$

Power of a point (Definition 2.22)

$$\text{pow}_C(P) = |OP|^2 - r^2$$

Circle of Apollonius (Theorem 2.7), with $A = (-a, 0)$, $B = (a, 0)$, $k > 0$, $k \neq 1$

$$\frac{PA}{PB} = k \iff (1 - k^2)(x^2 + y^2) + 2a(1 + k^2)x + a^2(1 - k^2) = 0$$

Pencil of circles (Definition 2.30; Theorem 2.8)

$$\Gamma_\lambda : (x^2 + y^2 + D_1x + E_1y + F_1) + \lambda(x^2 + y^2 + D_2x + E_2y + F_2) = 0, \quad \lambda \neq -1$$

Sources and bibliography

- [1] Euclid, *Elements*, Book III, Definitions 1–11 (trans. T.L. Heath, *The Thirteen Books of Euclid's Elements*, vol. II, Cambridge University Press, 1908; repr. Dover, 1956). In particular Definition III.2 (tangent line) and Definition III.10 (sector).
- [2] R. Hartshorne, *Geometry: Euclid and Beyond*, Springer, UTM, 2000, Ch. 1–3. For the axiomatic framework, the role of the continuity postulate and the critical discussion of the gaps in the *Elements*.
- [3] H.S.M. Coxeter, S.L. Greitzer, *Geometry Revisited*, MAA, New Mathematical Library 19, 1967, Ch. 1 (classical constructions).
- [4] D. Pedoe, *Circles: A Mathematical View*, MAA, rev. ed. 1995, Ch. 1 (equivalent definitions of the circle, circle of Apollonius).
- [5] D. E. Joyce, *Euclid's Elements*, hypertext edition with guides, Clark University, <https://mathcs.clarku.edu/~djoyce/elements/> (accessed 2 September 2026).

Further reading (not primary sources):

- [6] Wikipedia, *Compass-and-straightedge construction*, https://en.wikipedia.org/wiki/Compass-and-straightedge_construction (accessed 2026).
- [7] Wikipedia, *Coaxal circles*, https://en.wikipedia.org/wiki/Coaxal_circles (accessed 2026).
- [8] E.W. Weisstein, *Coaxal Circles*, MathWorld, <https://mathworld.wolfram.com/CoaxalCircles.html> (accessed 2026).

Theorems on Chords and Tangents

INFO

Aim of the chapter. To prove, completely and without circular dependencies, all the elementary theorems relating a circle to its chords and its tangents, up to the unifying theorem of the *power of a point*.

Method. Every result is stated with hypothesis and conclusion kept apart, proved in full (no “it is easily seen”), accompanied by the reference to the corresponding proposition of Euclid where one exists, and checked against the limiting cases. Converses are treated as *distinct* theorems, with independent proofs: deducing a converse from the direct theorem is a logical error, not a shortcut.

Logical order. The chain of dependencies is declared in Section 3.1 and is free of cycles: every theorem uses only results that precede it.

3.1. Prerequisites: what we assume and where it comes from

For a proof to be verifiable, the reader must be able to check that no step uses, directly or indirectly, the result being proved. The following table lists everything this chapter assumes.

Code	Assumed statement	Status
(A1)–(A10)	axioms listed in Remark 2.1	assumed
T-D01	the foot of the perpendicular realises the minimum distance (Theorem 2.1)	proved, Ch. 2
T-D04	trichotomy of points (Theorem 2.2)	proved, Ch. 2
T-D05	classification of lines (Theorem 2.6)	proved, Ch. 2
P-11	the central angle is twice any inscribed angle standing on the same arc (Euclid III.20)	proved, Ch. 4
P-12	every angle inscribed in a semicircle is right — Thales’ theorem (Euclid III.31), and its converse: if a point sees a segment under a right angle, it lies on the circle of that diameter (Theorem 4.5)	proved, Ch. 4
P-12bis	in an inscribed quadrilateral the opposite angles are supplementary (Theorem 4.6)	proved, Ch. 4
P-13	isometries preserve distances, incidences and the number of intersections	assumed
P-14	criterion for the intersection of two circles: $C(O_1, r_1)$ and $C(O_2, r_2)$ meet in two points if and only if $ r_1 - r_2 < O_1O_2 < r_1 + r_2$ (theorem on relative positions)	proved, Ch. 2

Warning. The delicate point of the whole chapter. Theorem 2.6 of Chapter 2 (a line is tangent if and only if its distance from the centre equals the radius) was proved by purely metric means, *without ever using the perpendicularity between tangent and radius*. This is essential: below we shall deduce perpendicularity *from* that classification. Had the classification been justified by assuming perpendicularity — as happens in many popular texts — the whole chapter would be circular.

Likewise, P-11 and P-12 are proved in Chapter 4 with tools (central angle, isosceles triangle) that use no result of the present chapter: assuming them here is therefore legitimate.

3.2. Metric and symmetry properties of chords

3.2.1 T-C01 — The perpendicular from the centre bisects the chord

Theorem 3.1: Perpendicularity of the radius to the chord (Euclid III.3). Let $\overline{AB}$ be a chord of $C(O, r)$ and let M be the point where the perpendicular drawn from O to AB meets the chord. Then M is the midpoint of $\overline{AB}$:

$$|AM| = |MB|.$$

Moreover the line OM bisects both arcs $\widehat{AB}$.

Proof. Hypothesis. $A, B \in C(O, r)$ with $A \neq B$; $OM \perp AB$ with $M \in AB$. **Conclusion.** $|AM| = |MB|$.

Step 1 — the two triangles. Consider the triangles OMA and OMB . Both have a right angle at M , by construction of the perpendicular.

Step 2 — the sides. We have $|OA| = |OB| = r$ by Theorem 2.3 (all radii are congruent); the side OM is common to the two triangles.

Step 3 — congruence. The triangles OMA and OMB are right-angled, have congruent hypotenuses ($|OA| = |OB|$) and a common leg (OM): by the leg-hypotenuse criterion (A3) they are congruent.

Step 4 — conclusion. From $OMA \cong OMB$ it follows in particular that $|MA| = |MB|$, that is, M is the midpoint of $\overline{AB}$.

Step 5 — the arcs. From the congruence it also follows that $\angle AOM = \angle MOB$. Extending the ray OM until it meets the circle at a point N (on the side of M with respect to O), the central angles $\angle AON$ and $\angle NOB$ are congruent; by Definition 2.14 the two arcs $\widehat{AN}$ and $\widehat{NB}$ have the same measure, that is, N bisects the minor arc. The same argument with the antipodal point of N bisects the major arc. $\square$

Alternative proof — via the isosceles triangle. The triangle OAB has $|OA| = |OB| = r$, hence it is **isosceles** with base $\overline{AB}$. By (A4), in an isosceles triangle the altitude to the base is also a median: but OM is by hypothesis precisely the altitude relative to AB (since $OM \perp AB$ with $M \in AB$), hence it is also a median, that is, M is the midpoint. This version is shorter, but it hides inside axiom (A4) the work that the first proof carries out explicitly. $\square$

Remark 3.1: The case of a diametral chord. If $\overline{AB}$ is a diameter, the perpendicular from O to AB meets the chord at $M = O$ itself, and the conclusion $|AM| = |MB|$ becomes $|AO| = |OB| = r$: true by the definition of radius. The theorem therefore remains valid, though in a trivial form. It is important to state this: many texts announce the theorem for “arbitrary” chords and then prove it with a figure in which $M \neq O$, leaving the limiting case implicit.

Remark 3.2: A detail almost all sources omit. The statement says “the point where the perpendicular meets the chord”: one must know that this point exists and falls inside the segment $\overline{AB}$, not on its extension. This follows from Theorem 2.1: M is the point of AB closest to O , and since $|OA| = |OB| = r$ while $|OM| \leq r$ with equality only if M coincides with A or B . But if the foot M coincided with A , we would have $OA \perp AB$, i.e. triangle OAB would be right-angled at A : by the Pythagorean theorem (A5), $|OB|^2 =$

$|OA|^2 + |AB|^2$, i.e. $r^2 = r^2 + |AB|^2$, hence $|AB| = 0$, i.e. $A = B$ — absurd, since $\overline{AB}$ is a chord (Definition 2.7, $A \neq B$ by definition). Likewise M cannot coincide with B . Hence M lies strictly inside the segment.

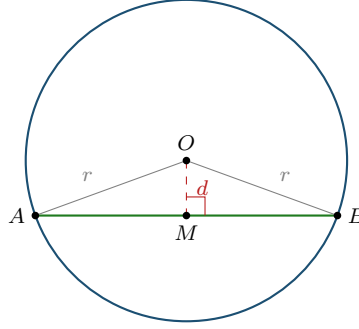


Figure 3.1: — T-C01 and T-C02. The triangles OMA and OMB have congruent hypotenuses (radii) and the leg OM in common. The dashed line marks the auxiliary construction; the right angle at M is a hypothesis in T-C01 and a conclusion in T-C02.

3.2.2 T-C02 — Converse: from the midpoint to the perpendicular

Theorem 3.2: Converse: the line from the centre to the midpoint is perpendicular. Let $\overline{AB}$ be a non-diametral chord of $C(O, r)$ and let M be its midpoint. Then $OM \perp AB$.

Proof. Hypothesis. $A, B \in C(O, r)$, $A \neq B$, $O \notin AB$; $|MA| = |MB|$ with $M \in \overline{AB}$.

Conclusion. $\angle OMA = \angle OMB = \pi/2$.

Step 1. Consider the triangles OMA and OMB . They have: $|OA| = |OB| = r$ (Theorem 2.3); $|MA| = |MB|$ by hypothesis; OM in common.

Step 2. By the SSS criterion (A3), $OMA \cong OMB$.

Step 3. From the congruence it follows that $\angle OMA = \angle OMB$.

Step 4. The two angles $\angle OMA$ and $\angle OMB$ are **adjacent**: they share the side OM and their sides MA , MB are opposite on the line AB (because M lies *between* A and B , being their midpoint). Hence their sum is a straight angle: $\angle OMA + \angle OMB = \pi$.

Step 5. From Steps 3 and 4: $2\angle OMA = \pi$, that is $\angle OMA = \pi/2$. Therefore $OM \perp AB$. $\square$

Warning. Why an independent proof is needed. Theorem 3.1 states “perpendicular $\Rightarrow$ bisects”; Theorem 3.2 states “bisects $\Rightarrow$ perpendicular”. The second implication does *not* follow from the first: from $P \Rightarrow Q$ one cannot infer $Q \Rightarrow P$. Note indeed that the two proofs use *different* congruence criteria (leg–hypotenuse in the first case, SSS in the second).

Excluded case. If $\overline{AB}$ is a diameter, its midpoint is $M = O$ and the “segment OM ” degenerates to a point: perpendicularity is undefined. This is why the hypothesis “non-diametral” appears explicitly in the statement.

3.2.3 T-C03 — The perpendicular bisector of a chord passes through the centre

Corollary 3.1: *The perpendicular bisector of a chord passes through the centre (Euclid III.1, cor.). The perpendicular bisector of every chord of a circle passes through the centre O .*

Proof. Let $\overline{AB}$ be a chord. By Theorem 2.3 we have $|OA| = |OB| = r$, hence O is equidistant from the endpoints of the chord. By axiom (A7), the perpendicular bisector of $\overline{AB}$ is *exactly* the locus of the points equidistant from A and B : therefore O lies on it. $\square$

Remark 3.3: *A stronger proof than the congruence one. One could have deduced this corollary from Theorem 3.2 ($OM \perp AB$ with M the midpoint, hence the line OM is the perpendicular bisector). The proof given here is preferable for two reasons: (i) it is shorter; (ii) it also covers the case of a diametral chord, which Theorem 3.2 must exclude. Indeed, if $\overline{AB}$ is a diameter, O is the midpoint and is trivially equidistant from the endpoints, hence it lies on the perpendicular bisector.*

Corollary 3.2: *Circle through three non-collinear points (Euclid IV.5). Through three non-collinear points A, B, C there passes one and only one circle.*

Proof. Existence. Let a be the perpendicular bisector of $\overline{AB}$ and b that of $\overline{BC}$. If a and b were parallel, they would be perpendicular to the same direction, hence AB and BC would be parallel; having the point B in common, they would coincide as lines, that is, A, B, C would be collinear, contrary to hypothesis. Hence a and b meet at a point O .

By (A7), $O \in a$ implies $|OA| = |OB|$, and $O \in b$ implies $|OB| = |OC|$; hence $|OA| = |OB| = |OC| =: r > 0$ (positive because the points are distinct). The circle $C(O, r)$ passes through the three points.

Uniqueness. If a circle $C(O', r')$ also passed through A, B, C , then O' would be equidistant from A and B , hence $O' \in a$ by (A7), and likewise $O' \in b$: therefore $O' = O$, since $a \cap b$ is a single point (two non-parallel lines meet at exactly one point). From $O' = O$ it follows that $r' = |O'A| = |OA| = r$. $\square$

Remark 3.4: *The practical use: finding the centre. Corollary 3.1 is what makes Construction 2.1 of Chapter 2 correct: given three points on a circle of unknown centre, the perpendicular bisectors of two chords meet at the centre. On the shop floor this is the standard procedure for locating the axis of a bore or a shaft from three points measured on its profile.*

3.2.4 T-C04 — Chords and central angles

Theorem 3.3: *Congruent chords and central angles (Euclid III.26–27). Let $\overline{AB}$ and $\overline{A'B'}$ be two chords of the same circle $C(O, r)$ (or of congruent circles). Then*

$$|AB| = |A'B'| \iff \angle AOB = \angle A'OB'.$$

Proof. ($\Rightarrow$) Suppose $|AB| = |A'B'|$. The triangles OAB and $OA'B'$ have $|OA| = |OA'| = |OB| = |OB'| = r$ (Theorem 2.3) and $|AB| = |A'B'|$ by hypothesis: by the SSS criterion (A3) they are congruent, hence $\angle AOB = \angle A'OB'$.

($\Leftarrow$) Suppose $\angle AOB = \angle A'OB'$. The triangles OAB and $OA'B'$ have two pairs of congruent sides ($|OA| = |OA'| = r$, $|OB| = |OB'| = r$) and the included angle congruent: by the SAS

criterion (A3) they are congruent, hence $|AB| = |A'B'|$. $\square$

Warning. The hypothesis “same circle” is not decorative. Without it the statement is *false*. Counterexample: a chord of length 1 in a circle of radius 1 subtends a central angle of $2 \arcsin(1/2) = \pi/3 \approx 60^\circ$; a chord of the same length 1 in a circle of radius 100 subtends an angle of $2 \arcsin(1/200) \approx 0.57^\circ$. Same chord, completely different central angles.

3.2.5 T-C05 — Chords and arcs

Theorem 3.4: Arcs subtended by congruent chords (Euclid III.28–29). *On the same circle (or on congruent circles), two chords are congruent if and only if the minor arcs they subtend are congruent:*

$$|AB| = |A'B'| \iff \mu(\widehat{AB}_{\min}) = \mu(\widehat{A'B'}_{\min}).$$

The same holds for the major arcs.

Proof. By Theorem 3.3 we have $|AB| = |A'B'| \iff \angle AOB = \angle A'OB'$. By Definition 2.14, the measure of the minor arc is by definition the measure of the central angle intercepting it; hence $\angle AOB = \angle A'OB' \iff \mu(\widehat{AB}_{\min}) = \mu(\widehat{A'B'}_{\min})$. Chaining the two equivalences gives the conclusion.

For the major arcs: again by Definition 2.14, $\mu(\widehat{AB}_{\max}) = 2\pi - \mu(\widehat{AB}_{\min})$. Since $x \mapsto 2\pi - x$ is injective, equality of the minor arcs is equivalent to equality of the major ones. $\square$

Remark 3.5: Why derive it rather than assume it. Many texts present T-C04 and T-C05 as two independent statements, proving the same thing twice. Having connected them explicitly through the definition of arc measure makes it plain that the mathematical content is one only: the bijective correspondence between chords and central angles. Arcs add no information, they are merely another notation for the same angles.

3.2.6 T-C06 and T-C07 — Chords and distance from the centre

Theorem 3.5: Congruent chords and distance from the centre (Euclid III.14–15). *Let $\overline{AB}$ and $\overline{CD}$ be two chords of the same circle $C(O, r)$, and let $d_1 = d(O, AB)$ and $d_2 = d(O, CD)$ be their distances from the centre. Then the following **trichotomy** holds:*

$$|AB| = |CD| \iff d_1 = d_2, \quad |AB| > |CD| \iff d_1 < d_2.$$

In particular, congruent chords are equidistant from the centre and, of two unequal chords, the longer is the one nearer the centre.

Proof. Step 1 — the fundamental formula. Let $\overline{AB}$ be a chord and M the foot of the perpendicular from O to AB , so that $|OM| = d_1$ (Theorem 2.1). By Theorem 3.1, M is the midpoint of $\overline{AB}$, hence $|MA| = |AB|/2$. The triangle OMA is right-angled at M , with hypotenuse $|OA| = r$: by the Pythagorean Theorem (A5),

$$r^2 = d_1^2 + \left(\frac{|AB|}{2}\right)^2 \iff |AB| = 2\sqrt{r^2 - d_1^2}.$$

Step 2 — strict monotonicity. Consider the function

$$f : [0, r] \rightarrow [0, 2r], \quad f(d) = 2\sqrt{r^2 - d^2}.$$

For $0 \leq d_1 < d_2 \leq r$ we have $d_1^2 < d_2^2$, hence $r^2 - d_1^2 > r^2 - d_2^2 \geq 0$; since the square root is strictly increasing on the non-negative reals, $f(d_1) > f(d_2)$. Hence f is **strictly decreasing** on $[0, r]$.

Step 3 — conclusion. A strictly decreasing function is in particular injective, and reverses inequalities. Therefore: $|AB| = |CD| \iff f(d_1) = f(d_2) \iff d_1 = d_2$; and $|AB| > |CD| \iff f(d_1) > f(d_2) \iff d_1 < d_2$. $\square$

Alternative proof — purely by congruence. The equality case (T-C06) can also be proved without Pythagoras, as Euclid does in III.14. Let M and N be the feet of the perpendiculars from O to the two chords. If $|AB| = |CD|$, then by Theorem 3.1 $|MA| = |AB|/2 = |CD|/2 = |NC|$. The triangles OMA and ONC are right-angled, have congruent hypotenuses ($|OA| = |OC| = r$) and a congruent leg ($|MA| = |NC|$): by the leg-hypotenuse criterion (A3) they are congruent, hence $|OM| = |ON|$, that is $d_1 = d_2$.

Conversely, if $d_1 = d_2$ the same triangles are congruent by the same criterion (leg $OM = ON$ and hypotenuse r), whence $|MA| = |NC|$ and therefore $|AB| = 2|MA| = 2|NC| = |CD|$.

This version has the merit of requiring no algebraic manipulation, but it does not provide the case of the inequalities, for which the metric route of Step 2 remains the most direct. $\square$

$$|AB| = 2\sqrt{r^2 - d^2} \iff d = \sqrt{r^2 - (|AB|/2)^2}$$

3.2.7 T-C08 — The diameter is the greatest chord

Theorem 3.6: The diameter is the chord of maximum length (Euclid III.15).

Every non-diametral chord of $C(O, r)$ has length strictly less than $2r$; the maximum $2r$ is attained exactly by the diametral chords.

Proof. First proof (from T-C07). By Theorem 3.5, $|AB| = f(d) = 2\sqrt{r^2 - d^2}$ with $d = d(O, AB) \in [0, r)$, and f is strictly decreasing. Its maximum on $[0, r)$ is therefore attained at $d = 0$ and equals $f(0) = 2r$. Now, $d = 0$ means $O \in AB$, that is, the chord passes through the centre: it is a diameter. Every other chord has $d > 0$ and hence $|AB| < 2r$. $\square$

Second proof — via the triangle inequality. Let $\overline{AB}$ be a non-diametral chord, that is, with $O \notin \overline{AB}$. Applying the triangle inequality (A8) to the triangle AOB :

$$|AB| \leq |AO| + |OB| = r + r = 2r,$$

and the inequality is **strict** precisely because, by (A8), equality holds if and only if O belongs to the segment $\overline{AB}$ — a case excluded here by hypothesis. If instead $O \in \overline{AB}$, the chord is a diameter and the equality $|AB| = 2r$ holds.

This second proof is remarkable for its economy of means: it uses neither the Pythagorean Theorem, nor the monotonicity of a function, nor Theorem 3.1. It uses a single axiom. It is the rare instance in which the shortest proof is also the most elementary and the most general. $\square$

3.2.8 T-C09 — Intersecting chords

Theorem 3.7: Intersecting chords (Euclid III.35). Let $\overline{AB}$ and $\overline{CD}$ be two chords of $C(O, r)$ meeting at a point P interior to the circle (so that P lies between A and B , and between C and D). Then

$$|PA| \cdot |PB| = |PC| \cdot |PD|.$$

Proof. Step 1 — auxiliary construction. Draw the segments $\overline{AC}$ and $\overline{DB}$, obtaining the triangles PAC and PDB .

Step 2 — first angle. The angles $\angle APC$ and $\angle DPB$ are vertically opposite (the lines AB and CD meet at P), hence congruent.

Step 3 — second angle. The angles $\angle PAC = \angle BAC$ and $\angle PDB = \angle CDB$ are inscribed angles standing on the *same* arc $\widehat{CB}$ (the one containing neither A nor D): by prerequisite P-11 (inscribed angles on the same arc are congruent, being each half of the same central angle), they are congruent.

Step 4 — similarity. The triangles PAC and PDB have two pairs of congruent angles: by the AA criterion (A6) they are similar, with the correspondence $P \leftrightarrow P$, $A \leftrightarrow D$, $C \leftrightarrow B$.

Step 5 — proportion and conclusion. From the similarity, corresponding sides are proportional:

$$\frac{|PA|}{|PD|} = \frac{|PC|}{|PB|} \implies |PA| \cdot |PB| = |PC| \cdot |PD|.$$

□

Direct proof (Euclid III.35, original form). Let M be the midpoint of the chord $\overline{AB}$: by Theorem 3.1, $OM \perp AB$. Then

$$|PA| \cdot |PB| = (|MA| - |MP|)(|MA| + |MP|) = |MA|^2 - |MP|^2.$$

By Pythagoras in the right triangles OMA and OMP (both with the right angle at M): $|MA|^2 = |OA|^2 - |OM|^2 = r^2 - |OM|^2$ and $|MP|^2 = |OP|^2 - |OM|^2$. Subtracting, the term $|OM|^2$ cancels:

$$|PA| \cdot |PB| = (r^2 - |OM|^2) - (|OP|^2 - |OM|^2) = r^2 - |OP|^2.$$

The same computation, with the midpoint of $\overline{CD}$, gives $|PC| \cdot |PD| = r^2 - |OP|^2$: the two products coincide, both equal to $r^2 - |OP|^2$ (which is $-\text{pow}(P)$ for P interior — a particular case of the general result we shall meet in Theorem 3.19). No similarity is used: only Pythagoras, twice. □

The same method for the exterior case (two secants). With P exterior and M the midpoint of $\overline{AB}$ (now A and B on the same side of P , with A between P and B): $|PA| \cdot |PB| = (|MP| - |MA|)(|MP| + |MA|) = |MP|^2 - |MA|^2 = (|OP|^2 - |OM|^2) - (r^2 - |OM|^2) = |OP|^2 - r^2$, the same value for every secant through P — the same fact we shall meet, by a different route, in Theorem 3.18 — again without similarity. □

Source: Heath, *The Thirteen Books of Euclid's Elements*, vol. II, III.35 and II.5.

Warning. The correspondence of vertices must be written correctly. The similarity is $PAC \sim PDB$, not $PAC \sim PBD$. Many expositions write the wrong correspondence and then obtain the right proportion “by accident”, swapping the terms. The check is

immediate: in a similarity the corresponding vertices are those with congruent angles, and $\angle PAC = \angle PDB$ pairs A with D .

Limiting cases. (i) If one of the two chords is a diameter, nothing changes in the proof. (ii) If $P = O$, both chords are diameters and the identity becomes $r \cdot r = r \cdot r$. (iii) If the two chords coincide the statement is trivially true.

3.3. Fundamental properties of tangents

3.3.1 T-T01 — The radius is perpendicular to the tangent

Theorem 3.8: Perpendicularity of the tangent (Euclid III.18). *The line tangent to a circle at a point T is perpendicular to the radius OT .*

Proof. Hypothesis. t is tangent to $C(O, r)$ at T , that is (Definition 2.21) $t \cap C(O, r) = \{T\}$.

Conclusion. $OT \perp t$.

We argue by contradiction. Suppose OT is *not* perpendicular to t .

Step 1. Let H be the foot of the perpendicular drawn from O to t (it exists and is unique by (A2)). By the absurd assumption, $H \neq T$.

Step 2. By Theorem 2.1, H is the point of t closest to O , and the inequality is strict for every other point: in particular

$$d(O, t) = |OH| < |OT| = r.$$

Step 3. By Theorem 2.6, a line whose distance from the centre is *strictly less* than the radius is secant: it meets the circle at **two** distinct points.

Step 4 — contradiction. But by hypothesis t meets the circle at a single point. Absurd.

Conclusion. The assumption was false: hence $H = T$, that is, OT is precisely the perpendicular to t , i.e. $OT \perp t$. $\square$

Warning. The step almost all sources skip is Step 3. Many texts write: “ H is inside the disc, hence the line crosses it and meets the circle at two points”, taking this as evident. But “a line having an interior point meets the circle” is exactly the continuity postulate (A10), or — as here — a consequence of Theorem 2.6. It is the most widespread gap in the didactic literature on this theorem, and it is the reason why Chapter 2 proved the classification of lines *first* and by an independent route.

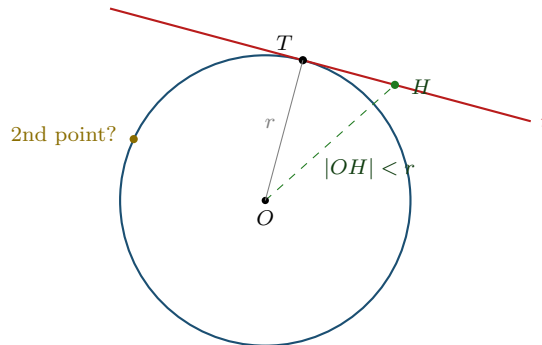


Figure 3.2: — T-T01, the configuration of the contradiction. If OT were not perpendicular to t , the foot H of the perpendicular would lie at distance $|OH| < r$ from the centre: by the classification of lines (Theorem 2.6) the line would be secant and would meet the circle at a second point, contrary to the hypothesis of tangency.

Alternative proof — analytic (double root). Place O at the origin and, without loss of generality, $T = (r, 0)$. A general line through T with direction $u = (\cos \varphi, \sin \varphi)$ is parametrised as $T + su$, $s \in \mathbb{R}$. Intersection with the circle $x^2 + y^2 = r^2$ requires

$$|T + su|^2 = r^2 \implies r^2 + 2rs \cos \varphi + s^2 = r^2 \implies s(s + 2r \cos \varphi) = 0.$$

The two solutions are $s = 0$ (the point T itself) and $s = -2r \cos \varphi$. For the line to be **tangent** (that is, to meet the circle at a single point, with multiplicity two, and not at a second distinct point), the two solutions must coincide: $-2r \cos \varphi = 0$, that is $\cos \varphi = 0$, hence $\varphi = 90^\circ$. The direction $u = (\cos 90^\circ, \sin 90^\circ) = (0, 1)$ is exactly perpendicular to the radius $OT = (r, 0)$ (direction $(1, 0)$). This proof, of a purely algebraic nature (double root of the quadratic equation, in perfect analogy with the vanishing discriminant we shall meet later, in the Alternative proof of Theorem 3.19), requires neither contradiction nor the continuity postulate, and is the natural one from the viewpoint of the analytic geometry of Chapter 7. $\square$

3.3.2 T-T02 — Converse

Theorem 3.9: Converse: the perpendicular to the radius is tangent (Euclid III.16). Let $T \in C(O, r)$ and let t be the line through T perpendicular to the radius OT . Then t is tangent to the circle at T .

Proof. Hypothesis. $T \in C(O, r)$, $t \ni T$, $t \perp OT$. **Conclusion.** $t \cap C(O, r) = \{T\}$.

Step 1. Let $P \in t$ with $P \neq T$. The triangle OTP has a right angle at T (by hypothesis $t \perp OT$), hence $\overline{OP}$ is its hypotenuse and $\overline{OT}$ a leg.

Step 2. By the Pythagorean Theorem (A5),

$$|OP|^2 = |OT|^2 + |TP|^2 = r^2 + |TP|^2 > r^2,$$

since $|TP| > 0$ because $P \neq T$. Hence $|OP| > r$.

Step 3. By Definition 2.4, P is *exterior* to the circle, hence $P \notin C(O, r)$.

Step 4 — conclusion. Every point of t other than T is exterior to the circle; and T lies on it by hypothesis. Hence $t \cap C(O, r) = \{T\}$: by Definition 2.21, t is tangent. $\square$

Remark 3.6: A simpler proof than the direct theorem. Note that this proof is direct and constructive: it does not argue by contradiction and does not use the continuity postulate. It is more elementary than Theorem 3.8, and this is why Euclid places it first (III.16 precedes III.18). It also shows, as a by-product, that $d(O, t) = |OT| = r$: the tangent realises exactly the limiting case of the classification of lines.

3.3.3 T-T03 — Uniqueness of the tangent at a point

Corollary 3.3: Existence and uniqueness of the tangent at a point (Euclid III.19). Through every point T of a circle $C(O, r)$ there passes one and only one tangent line, and this is the unique line through T perpendicular to OT .

Proof. Existence. By (A2) there exists the line t through T perpendicular to OT ; by Theorem 3.9 it is tangent.

Uniqueness. Let t' be any tangent at T . By Theorem 3.8, $t' \perp OT$. Hence t' is a line through T perpendicular to OT ; but by (A2) such a line is unique, therefore $t' = t$. $\square$

3.3.4 T-T04 — Equivalence of the three definitions of tangent

Theorem 3.10: The three characterisations of the tangent are equivalent. For a line t and a circle $C(O, r)$ the following three conditions are equivalent:

- (D1) $t \cap C(O, r)$ contains exactly one point;
- (D2) there exists $T \in t \cap C(O, r)$ with $t \perp OT$;
- (D3) $d(O, t) = r$.

Proof. (D1) $\Leftrightarrow$ (D3). This is exactly the central case of Theorem 2.6, proved in Chapter 2 by purely metric means.

(D1) $\Rightarrow$ (D2). This is Theorem 3.8: if T denotes the unique common point, then $t \perp OT$.

(D2) $\Rightarrow$ (D1). This is Theorem 3.9.

The three implications close the cycle of equivalences. □

Remark 3.7: Why this is the logical crux of the chapter. The chain just closed is exactly the point where circularity is risked. The correct order is: first (D1) $\Leftrightarrow$ (D3) without perpendicularity (Chapter 2), then (D1) $\Rightarrow$ (D2) using (D3), and finally (D2) $\Rightarrow$ (D1) independently. If instead one defined the tangent by (D2) and then “proved” (D1) $\Rightarrow$ (D2), one would obtain a tautology disguised as a theorem. From now on we may use the three characterisations interchangeably.

3.3.5 T-T05 — Tangents from an external point

Theorem 3.11: Tangents from an external point (Euclid III.17). Let P be a point exterior to $C(O, r)$. Then:

- (a) there exist tangent lines to $C(O, r)$ through P ;
- (b) there are exactly two of them, with points of tangency $T_1 \neq T_2$;
- (c) the two tangent segments are congruent: $|PT_1| = |PT_2|$, and

$$|PT_i| = \sqrt{|OP|^2 - r^2}.$$

Proof. Set $\rho = |OP| > r$ (P is exterior) and let M be the midpoint of $\overline{OP}$.

(a) *Existence.* Consider the auxiliary circle $\Sigma = C(M, \rho/2)$, with diameter $\overline{OP}$ (“Thales’ circle”). Let us check that Σ and $C(O, r)$ meet: two circles whose centres are at distance $|OM| = \rho/2$ and whose radii are r and $\rho/2$ meet at two points if and only if

$$\left| r - \frac{\rho}{2} \right| < \frac{\rho}{2} < r + \frac{\rho}{2}.$$

The right-hand inequality is obvious ($r > 0$). For the left-hand one: if $r \geq \rho/2$ it becomes $r - \rho/2 < \rho/2$, that is $r < \rho$, true by hypothesis; if $r < \rho/2$ it becomes $\rho/2 - r < \rho/2$, that is $r > 0$, true. Hence $\Sigma \cap C(O, r)$ consists of two points T_1, T_2 .

For each of them, $\angle OT_i P$ is an angle inscribed in Σ standing on the diameter $\overline{OP}$: by prerequisite P-12 (Thales’ theorem) $\angle OT_i P = \pi/2$, that is $PT_i \perp OT_i$. Since $T_i \in C(O, r)$, Theorem 3.9 guarantees that PT_i is tangent to $C(O, r)$.

(b) *Exactly two.* Let t be any tangent through P , with point of tangency T . By Theorem 3.8, $OT \perp PT$, hence $\angle OTP = \pi/2$. But then T sees the segment $\overline{OP}$ under a right angle, and by the converse of Thales’ theorem (P-12) T lies on the circle Σ with diameter $\overline{OP}$. Hence $T \in \Sigma \cap C(O, r) = \{T_1, T_2\}$: there are no other points of tangency, and since the tangent at a point is unique (Corollary 3.3), there are no other tangents.

(c) *Congruence.* The triangles $OT_1 P$ and $OT_2 P$ are right-angled at T_1 and T_2 . By the

Pythagorean Theorem (A5), applied to each,

$$|PT_i|^2 = |OP|^2 - |OT_i|^2 = \rho^2 - r^2 \quad (i = 1, 2),$$

a quantity not depending on i : hence $|PT_1| = |PT_2| = \sqrt{\rho^2 - r^2}$, well defined because $\rho > r$. $\square$

$$|PT| = \sqrt{|OP|^2 - r^2}$$

Remark 3.8: The limiting case $|OP| \rightarrow r$. As P approaches the circle, $\rho \rightarrow r$ and $|PT_i| \rightarrow 0$: the two points of tangency merge into P itself and the two tangents coalesce into the unique tangent at P (Corollary 3.3). If instead P is interior ($\rho < r$), the quantity $\rho^2 - r^2$ is negative and no tangent through P exists: consistent with the fact that every line through an interior point is secant (Theorem 2.6).

3.3.6 T-T06 — The axis of symmetry of the configuration

Theorem 3.12: The line OP is an axis of symmetry. Under the hypotheses and with the notation of Theorem 3.11:

- (a) the line OP bisects the angle $\angle T_1PT_2$ formed by the two tangents;
- (b) the line OP also bisects the angle $\angle T_1OT_2$;
- (c) the line OP is the perpendicular bisector of the segment $\overline{T_1T_2}$ (the chord of contact, Definition 2.27); in particular $OP \perp T_1T_2$.

Proof. Step 1. The triangles OT_1P and OT_2P have: $|OT_1| = |OT_2| = r$; $|PT_1| = |PT_2|$ (Theorem 3.11(c)); OP in common. By the SSS criterion (A3) they are congruent.

Step 2 — (a) and (b). From the congruence follow $\angle OPT_1 = \angle OPT_2$ and $\angle POT_1 = \angle POT_2$: the line OP bisects both angles.

Step 3 — (c). We have $|OT_1| = |OT_2| = r$, hence O is equidistant from T_1 and T_2 ; and $|PT_1| = |PT_2|$, hence so is P . By axiom (A7), both points O and P lie on the perpendicular bisector of the segment $\overline{T_1T_2}$. But through two distinct points there passes a single line (A1): hence the line OP is that perpendicular bisector, and in particular it is perpendicular to T_1T_2 at its midpoint. $\square$

Alternative proof — by axial symmetry. The line OP is an axis of symmetry of the whole figure: indeed every line through the centre O is an axis of symmetry of the circle itself (reflection in such a line maps $C(O, r)$ onto itself, since it preserves distances from O), and P lies on this axis by construction, so the reflection also fixes P . Let σ be this reflection. The tangent line PT_1 is mapped by σ to a line still passing through P (since $\sigma(P) = P$) and still tangent to the circle (by P-13 isometries preserve the property of tangency, that is, of “meeting the circle at a single point”, since they preserve distances and hence the number of intersections). But from P there exists, by Theorem 3.11(b), a unique other tangent besides PT_1 , namely PT_2 : hence $\sigma(PT_1) = PT_2$, and in particular $\sigma(T_1) = T_2$ (the unique point of tangency of each line is mapped to the unique point of tangency of the other). Since σ is an isometry, $|PT_1| = |P\sigma(T_1)| = |PT_2|$ immediately. Moreover, since σ interchanges the two rays PT_1 and PT_2 while fixing P , by definition OP is precisely the bisector of the angle $\angle T_1PT_2$. This proof, based solely on the axial symmetry of the circle about each of its diameters, avoids the Pythagorean Theorem used in the preceding proof. $\square$

3.4. Angles associated with tangents and chords

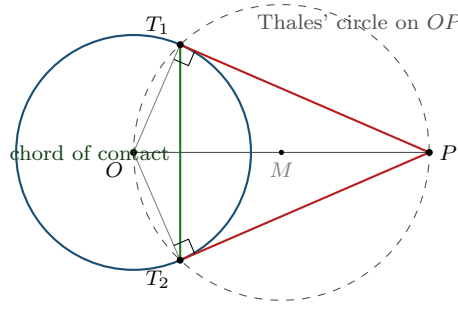


Figure 3.3: — T-T05 and T-T06. The dashed auxiliary circle, with diameter $\overline{OP}$, locates the two points of tangency: by Thales' theorem the angles at T_1 and T_2 are right. The line OP is an axis of symmetry of the figure and the perpendicular bisector of the chord of contact $\overline{T_1T_2}$.

3.4.1 T-T07 — Angle between tangent and chord

Theorem 3.13: Tangent–chord angle, or alternate segment theorem (Euclid III.32). Let t be the tangent to $C(O, r)$ at T and let $\overline{TA}$ be a chord issuing from T . Then each of the two angles that t makes with TA is congruent to the inscribed angle standing on the arc $\widehat{TA}$ contained in the alternate segment, that is, in the half-plane opposite to that angle.

Proof. Denote by α one of the two angles formed by t and TA , and let D be a point of the arc $\widehat{TA}$ lying in the half-plane opposite to α with respect to the line TA . We must show that $\angle TDA = \alpha$.

We treat separately the three possible cases.

Case 1: α acute. Draw the diameter $\overline{TD'}$ issuing from T , where D' is the antipodal point of T .

1. By Theorem 3.8, $OT \perp t$, hence the angle between t and the diameter TD' is right: $\angle(t, TD') = \pi/2$.
2. Since the chord TA lies, with respect to t , on the same side as the diameter, we have $\angle D'TA = \pi/2 - \alpha$ (the angle α and the angle $\angle D'TA$ add up to the right angle).
3. The triangle TAD' is inscribed in the circle with one side equal to the diameter $\overline{TD'}$: by prerequisite P-12 (Thales' theorem), $\angle TAD' = \pi/2$.
4. In the triangle TAD' the sum of the interior angles is π (A9), hence

$$\angle AD'T = \pi - \angle TAD' - \angle D'TA = \pi - \frac{\pi}{2} - \left(\frac{\pi}{2} - \alpha\right) = \alpha.$$

5. Finally $\angle AD'T$ and $\angle ADT$ are inscribed angles standing on the same arc $\widehat{TA}$ (the one containing both D and D'): by P-11 they are congruent. Hence $\angle TDA = \alpha$.

Case 2: α right. Then $TA \perp t$; but also $OT \perp t$, hence by (A2) the chord TA lies on the line OT , that is, $\overline{TA}$ is a diameter. The angle inscribed in a semicircle is right by P-12, hence again $\angle TDA = \pi/2 = \alpha$.

Case 3: α obtuse. Let α' be the other angle formed by t and TA ; it is acute and supplementary to α ($\alpha + \alpha' = \pi$). By Case 1, α' is congruent to the inscribed angle $\angle TD_1A$ in its own alternate segment. On the other hand, taking D and D_1 on the two opposite arcs, the quadrilateral TD_1AD is inscribed in the circle, hence its opposite angles are supplementary (Euclid III.22, an immediate consequence of P-11): $\angle TDA + \angle TD_1A = \pi$. Therefore

$$\angle TDA = \pi - \angle TD_1A = \pi - \alpha' = \alpha,$$

as required. □

Warning. Both angles must be dealt with. The tangent and the chord form *two* supplementary angles, and each corresponds to its own alternate segment. Expositions that draw a single figure with the acute angle leave half the theorem to the reader. In the proof above, Case 3 covers the obtuse angle explicitly, and Case 2 the limiting case of the diameter, in which the two angles coincide.

Theorem 3.14: Converse of the tangent–chord theorem (Euclid III.32, conv.). Let $T \in C(O, r)$, let $\overline{TA}$ be a chord and let s be a line through T such that the angle between s and TA is congruent to the inscribed angle in the alternate segment relative to $\widehat{TA}$. Then s is tangent at T .

Proof. Let t be the tangent at T , which exists and is unique by Corollary 3.3. By Theorem 3.13, t makes with TA , on the side considered, an angle congruent to the inscribed angle in the alternate segment. By hypothesis s too makes with TA , on the same side, an angle congruent to the same inscribed angle. Hence s and t make congruent angles with the ray TA , on the same side and with the same vertex T : by the uniqueness of the ray making a given angle with a given ray in a given half-plane, s and t coincide. Therefore s is tangent. $\square$

3.4.2 T-T08 — Angle formed by two tangents

Theorem 3.15: Angle between two tangents from an external point. Let P be exterior to $C(O, r)$ and let PT_1, PT_2 be the two tangents. Then:

- (a) $\angle T_1PT_2 + \angle T_1OT_2 = \pi$ (the two angles are supplementary);
- (b) equivalently, if $\mu_{\min}$ and $\mu_{\max}$ denote the measures of the two arcs determined by T_1 and T_2 ,

$$\angle T_1PT_2 = \frac{1}{2}(\mu_{\max} - \mu_{\min}).$$

Proof. (a) Consider the quadrilateral OT_1PT_2 . Let us check that it is convex by showing that its two diagonals, OP and T_1T_2 , separate each other. On one hand, T_1 and T_2 lie on opposite sides of the line OP : the reflection σ about OP swaps T_1 and T_2 (Theorem 3.12), and $T_1 \neq T_2$ do not lie on OP (if they did, $\sigma(T_1) = T_1$ would give $T_1 = T_2$, absurd); an axial reflection swaps two distinct points, neither on the axis, if and only if they lie on opposite sides of the axis (an elementary property of reflections: the axis is the locus of fixed points, and the reflection reverses orientation transverse to the axis), hence T_1, T_2 lie on opposite sides of OP . On the other hand, O and P lie on opposite sides of the line T_1T_2 (the *chord of contact*): by Theorem 3.12(c), OP is the axis of T_1T_2 and meets it perpendicularly at its midpoint H . The triangle OT_1P is right-angled at T_1 (Theorem 3.8), and H , the foot of the perpendicular from T_1 to the hypotenuse OP (since $T_1H \subset T_1T_2 \perp OP$), lies strictly inside segment OP : in a right triangle the angles at O and P are both acute (they sum to $\pi/2$), and the foot of the altitude from the right-angle vertex falls inside the opposite side exactly when the two angles adjacent to that side are both acute. Hence the two diagonals OP and T_1T_2 meet at their respective interior points, which is the convexity condition for the quadrilateral OT_1PT_2 . Its angles at T_1 and T_2 are right, by Theorem 3.8. By axiom (A9) the sum of the interior angles of a convex quadrilateral is 2π , hence

$$\angle T_1OT_2 + \frac{\pi}{2} + \angle T_1PT_2 + \frac{\pi}{2} = 2\pi \implies \angle T_1OT_2 + \angle T_1PT_2 = \pi.$$

(b) Set $\theta = \angle T_1OT_2$. By Definition 2.14, the minor arc between T_1 and T_2 has measure $\mu_{\min} = \theta$ and the major one $\mu_{\max} = 2\pi - \theta$. Then

$$\frac{\mu_{\max} - \mu_{\min}}{2} = \frac{(2\pi - \theta) - \theta}{2} = \pi - \theta = \angle T_1PT_2,$$

where the last equality is point (a). The two formulations are therefore the same statement written in two ways. $\square$

Remark 3.9: A check on a known case. If $|OP| = 2r$, the right triangle OT_1P has hypotenuse $2r$ and leg $|OT_1| = r$, hence $\sin(\angle OPT_1) = r/2r = 1/2$ and $\angle OPT_1 = 30^\circ$. The angle between the tangents is then $\angle T_1PT_2 = 60^\circ$, and by point (a) the central angle equals $180^\circ - 60^\circ = 120^\circ$: consistent, because in the triangle OT_1P the angle at O is $90^\circ - 30^\circ = 60^\circ$, and the total central angle is twice that, i.e. 120° .

3.5. Metric relations from an external point

3.5.1 T-T09 — Tangent and secant

Theorem 3.16: Tangent–secant theorem (Euclid III.36). Let P be exterior to $C(O, r)$, let PT be a tangent segment (T the point of tangency) and let a secant through P meet the circle at the points A and B , labelled so that A lies between P and B . Then

$$|PT|^2 = |PA| \cdot |PB|.$$

Proof. Step 1 — construction. Draw the segments $\overline{TA}$ and $\overline{TB}$, obtaining the triangles PTA and PBT .

Step 2 — common angle. The angle $\angle TPA = \angle BPT$ is one and the same angle, with vertex P : the two triangles share it.

Step 3 — tangent–chord angle. The angle $\angle PTA$ is formed by the tangent PT and the chord $\overline{TA}$. By Theorem 3.13, it is congruent to the inscribed angle standing on the arc $\widehat{TA}$ of the alternate segment; but that inscribed angle is precisely $\angle TBA = \angle TBP$, because B lies on that arc. Hence $\angle PTA = \angle PBT$.

Step 4 — similarity. The triangles PTA and PBT have two pairs of congruent angles (Steps 2 and 3): by the AA criterion (A6) they are similar, with the correspondence $P \leftrightarrow P, T \leftrightarrow B, A \leftrightarrow T$.

Step 5 — proportion. From the corresponding sides,

$$\frac{|PT|}{|PB|} = \frac{|PA|}{|PT|} \implies |PT|^2 = |PA| \cdot |PB|.$$

$\square$

Warning. The order of the points is part of the statement. The secant meets the circle at two points; the theorem requires them to be named so that A is the nearer to P and B the farther, that is, so that the order along the line is P, A, B . Omitting this specification, as many sources do, makes the statement ambiguous: were the names exchanged the formula would remain true (the product is commutative), but the similarity proof would use the wrong correspondence between vertices.

Theorem 3.17: Converse of the tangent–secant theorem (Euclid III.37). Let P be exterior to $C(O, r)$, let A, B be the points where a line through P meets the circle (with A between P and B) and let $T \in C(O, r)$ be such that

$$|PT|^2 = |PA| \cdot |PB|.$$

Then the line PT is tangent to the circle at T .

Proof. Let PT' be one of the two tangents from P (they exist by Theorem 3.11), with point of tangency T' . By Theorem 3.16, $|PT'|^2 = |PA| \cdot |PB|$. Comparing with the hypothesis, $|PT|^2 = |PT'|^2$, hence $|PT| = |PT'|$ (non-negative lengths).

The triangles OPT and OPT' then have: $|OT| = |OT'| = r$; $|PT| = |PT'|$; OP in common. By the SSS criterion (A3) they are congruent, hence $\angle OTP = \angle OT'P$. But $\angle OT'P = \pi/2$ by Theorem 3.8, therefore $\angle OTP = \pi/2$ as well: the line PT is perpendicular to the radius OT at the point T of the circle, hence by Theorem 3.9 it is tangent. $\square$

3.5.2 T-T10 — Two secants from an external point

Theorem 3.18: Two secants from an external point. *Let P be exterior to $C(O, r)$ and let two secants through P meet the circle at A, B (with A between P and B) and at C, D (with C between P and D) respectively. Then*

$$|PA| \cdot |PB| = |PC| \cdot |PD|.$$

Proof. Step 1. Draw $\overline{AC}$ and $\overline{DB}$, obtaining the triangles PAC and PDB .

Step 2 — common angle. The angle with vertex P is common to the two triangles.

Step 3 — second angle. $\angle PAC = \pi - \angle CAB$ (adjacent angles on the line PB , since A is between P and B). The four points A, B, D, C lie on the circle in this order (because A is between P and B , and C between P and D : the two secants “open” from P towards B and towards D , with A, C closer to P), hence the quadrilateral $ABDC$ is cyclic, and its opposite angles $\angle CAB$ and $\angle BDC$ are supplementary (P-12bis: in an inscribed quadrilateral opposite angles are supplementary, Chapter 4, Theorem 4.6). Hence $\angle PAC = \pi - \angle CAB = \angle BDC = \angle PDB$.

Step 4 — similarity and conclusion. By AA (A6), $PAC \sim PDB$ with $P \leftrightarrow P$, $A \leftrightarrow D$, $C \leftrightarrow B$. From the corresponding sides,

$$\frac{|PA|}{|PD|} = \frac{|PC|}{|PB|} \implies |PA| \cdot |PB| = |PC| \cdot |PD|.$$

$\square$

Remark 3.10: Three theorems, a single structure. *Compare the proofs of Theorems 3.7 (P interior), 3.16 (tangent and secant) and 3.18 (P exterior): they are the same proof, with the same pair of similar triangles, repeated in three configurations. This repetition is not an expository flaw: it is the symptom that the three statements are particular cases of a single more general theorem. The next section states it.*

3.6. Power of a point: the unifying theorem

Definition 3.1: Power of a point with respect to a circle. The power of P with respect to $C(O, r)$ is $\text{pow}(P) = |OP|^2 - r^2$. It is positive if P is exterior, zero if $P \in C(O, r)$, negative if P is interior.

Theorem 3.19: Power of a point theorem (Euclid III.35–37, general case). *Let P be any point of the plane and $C(O, r)$ a circle. For every line through P meeting the circle at two points A, B (possibly coincident in the tangent case),*

$$|PA| \cdot |PB| = |\text{pow}(P)| = ||OP|^2 - r^2|,$$

independently of the chosen line. In particular, if P is exterior and PT is a tangent segment, $|PT|^2 = \text{pow}(P)$.

Proof. Case P exterior, two secants. By Theorem 3.18 the product $|PA| \cdot |PB|$ is the same for all secants through P . It remains to identify its value: this is computed by choosing the particular secant through the centre O . It meets the circle at the two antipodal points A_0, B_0 with $|PA_0| = |OP| - r$ and $|PB_0| = |OP| + r$, hence

$$|PA| \cdot |PB| = |PA_0| \cdot |PB_0| = (|OP| - r)(|OP| + r) = |OP|^2 - r^2.$$

Tangent case. By Theorem 3.11(c), $|PT|^2 = |OP|^2 - r^2$: the formula also holds if the tangent is regarded as a degenerate secant with $A = B = T$.

Case P interior. By Theorem 3.7 the product $|PA| \cdot |PB|$ is the same for all chords through P ; choosing the chord through the centre gives $|PA_0| = r - |OP|$ and $|PB_0| = r + |OP|$, hence

$$|PA| \cdot |PB| = r^2 - |OP|^2 = -\text{pow}(P) = |\text{pow}(P)|,$$

consistent with the fact that the power is negative for interior points.

Case P on the circle. Then $|OP| = r$ and $\text{pow}(P) = 0$; every line through P meets the circle at P itself and at a second point B , and the product $|PP| \cdot |PB| = 0$ coincides with the power. $\square$

Alternative proof (via coordinates and Vieta's formulas). Place O at the origin. A general line through P is parametrised as $\gamma(t) = P + tu$, with u a unit vector. The intersection points with $C(O, r)$ satisfy $|P + tu|^2 = r^2$, that is (since $|u| = 1$)

$$t^2 + 2t(P \cdot u) + (|P|^2 - r^2) = 0.$$

This is a quadratic equation in t , with roots t_1, t_2 corresponding (with sign) to the distances $\overline{PA}, \overline{PB}$ along the line oriented by u . By Vieta's formulas, the product of the roots is the constant term:

$$t_1 t_2 = |P|^2 - r^2 = |OP|^2 - r^2,$$

independent of u , that is, of the particular secant chosen. This is exactly $\text{pow}(P)$, and it holds automatically for P exterior (t_1, t_2 of the same sign, positive product), for P interior (t_1, t_2 of opposite sign, negative product) and for the tangent (double root, $t_1 = t_2$, vanishing discriminant). Unlike the preceding synthetic proof (based on similarity of triangles), this one treats all the cases at once (P interior, exterior, tangent) without distinctions, and is the natural one from the viewpoint of the analytic geometry of Chapter 7. $\square$

Example 3.1: The power is the same on every secant: a full check. Consider the circle

$$\gamma : x^2 + y^2 - 6x - 4y - 12 = 0$$

and the point $P = (10; 6)$. We verify on three different lines that the product $\overline{PA} \cdot \overline{PB}$ does not depend on the line chosen.

Step 0 — centre and radius. Completing the squares, $(x-3)^2 + (y-2)^2 = 12+9+4 = 25$, hence

$$O = (3; 2), \quad r = 5.$$

Step 1 — the power, by definition.

$$|OP| = |(7; 4)| = \sqrt{49 + 16} = \sqrt{65} = 8.06226,$$

$$\text{pow}_\gamma(P) = |OP|^2 - r^2 = 65 - 25 = \mathbf{40}.$$

Being positive, P is **external**, and the tangent length from P is $t = \sqrt{40} = 6.32456$.

Step 2 — first secant: the line through the centre. The line OP meets γ at the two points at distance r from O along OP , i.e. at distances from P

$$\overline{PA} = |OP| - r = 8.06226 - 5 = 3.06226, \quad \overline{PB} = |OP| + r = 8.06226 + 5 = 13.06226.$$

Product:

$$3.06226 \times 13.06226 = 40.0000. \quad \checkmark$$

(No coincidence: $(|OP| - r)(|OP| + r) = |OP|^2 - r^2$, which is the very definition of the power — the secant through the centre is the case in which the theorem is trivial.)

Step 3 — second secant: the horizontal line $y = 6$. Substituting $y = 6$ into the equation of γ :

$$x^2 + 36 - 6x - 24 - 12 = 0 \iff x^2 - 6x = 0 \iff x(x - 6) = 0,$$

giving the two points $A = (0; 6)$ and $B = (6; 6)$. The distances from $P = (10; 6)$, measured along the line, are

$$\overline{PA} = 10, \quad \overline{PB} = 4,$$

and the product is

$$10 \times 4 = 40. \quad \checkmark$$

Completely different numbers from Step 2, the same product.

Step 4 — third line: the vertical $x = 10$. Substituting:

$$100 + y^2 - 60 - 4y - 12 = 0 \iff y^2 - 4y + 28 = 0,$$

with discriminant $16 - 112 = -96 < 0$: **no** real intersection. Consistently, the distance of the centre from this line is $|10 - 3| = 7 > r = 5$, and by Theorem 2.6 (Chapter 2) the line is external. The power theorem says nothing about it — and must not: the product $\overline{PA} \cdot \overline{PB}$ is undefined when A and B do not exist.

Step 5 — the tangent, as a limiting case. The point of tangency T satisfies $\overline{PT}^2 = \text{pow}(P)$ (Theorem 3.16), i.e. $\overline{PT} = \sqrt{40} = 6.32456$: it is the “degenerate secant” in which A and B coincide and the product becomes a square. Check with Pythagoras on the right triangle OTP (right-angled at T by Theorem 3.8):

$$\overline{PT}^2 = |OP|^2 - r^2 = 65 - 25 = 40. \quad \checkmark$$

Summary. Three lines through the same point, three completely different pairs of numbers (3.06 and 13.06; 10 and 4; 6.32 twice), one single product: **40**. It is this invariance — not the individual formula — that is the content of Theorem 3.19.

***Remark 3.11: Signed segments and the sign of the power.** The formulation with the absolute value is the elementary one. If one adopts the convention of **signed segments** on a fixed line — that is, if $\overline{PA}$ is given a positive or negative sign according to direction — the statement becomes cleaner and free of cases:*

$$\overline{PA} \cdot \overline{PB} = \text{pow}(P) \quad \text{always.}$$

Indeed, if P is exterior, A and B lie on the same side and the product is positive; if P is interior, they lie on opposite sides and the product is negative. This is the formulation of

the proof via Vieta, in which the signs are automatic. **It must, however, be declared:** using signed segments without saying so is one of the commonest errors in expositions.

Corollary 3.4: Characterisation by the power. A point P is interior to, lies on, or is exterior to $C(O, r)$ according as $\text{pow}(P)$ is negative, zero or positive respectively. Moreover $\text{pow}(P) \geq -r^2$, with equality if and only if $P = O$.

Proof. The first part is Definition 3.1 combined with Theorem 2.2. For the second, $\text{pow}(P) = |OP|^2 - r^2 \geq 0 - r^2 = -r^2$, with equality if and only if $|OP| = 0$, that is $P = O$: the centre is the point of minimum power. $\square$

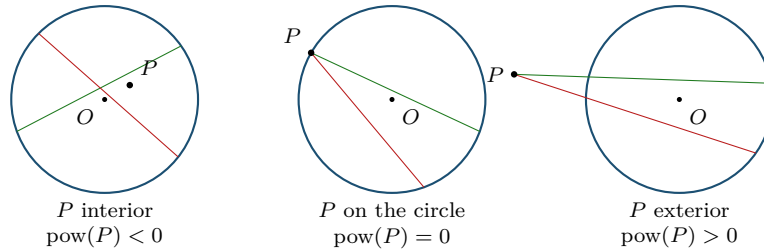


Figure 3.4: — T-T11. In each of the three positions, the product of the distances from P to the two intersection points is the same for every line drawn through P , and equals $||OP|^2 - r^2|$. The sign of the power distinguishes the three configurations.

Remark 3.12: Radical axis. The power of a point is the tool that generates the radical axis of two circles (the locus of points of equal power with respect to both), treated in detail in Chapter 7. Generalising the power through circular inversion one arrives at the problem of Apollonius and at the theorems of Ptolemy and Descartes on circles, treated in Chapter 5.

Remark 3.13: Origin of the term “power”. The term power of a point (in German *Potenz eines Punktes*) was introduced by Jakob Steiner in 1826, in the paper that for the first time systematised in unified form Euclid’s scattered results (III.35–37) on secants, tangents and chords. The choice of the term, evoking the idea of a quantity characteristic of the point with respect to the circle, invariant as the secant varies, has since been preserved unchanged in the international literature.

3.6.1 Industrial application: safety distance from a circular obstacle

Clearance offset for an AGV path. Context. An automated guided vehicle (AGV) travels in a straight line in a production area, keeping a safety distance from a fixed cylindrical obstacle (for instance a tank) of radius $r = 0.6$ m, whose centre O is $d = 2.5$ m from the nominal line of travel.

Step 1. Let P be the foot of the perpendicular from O to the trajectory. Since $d > r$ ($2.5 > 0.6$), the trajectory does not meet the obstacle (Theorem 2.6): the minimum clearance is $d - r = 1.9$ m.

Step 2 (regulatory margin). If the cell standard imposes an additional margin $s = 0.3$ m, the constraint is $d - r \geq s$, that is $d \geq r + s = 0.9$ m: amply satisfied, with a residual margin of $1.9 - 0.3 = 1.6$ m for localisation tolerances.

Step 3 (avoidance). If the path must deviate, the tightest avoidance trajectory is tangent to the circle of radius $r + s = 0.9$ m centred at O : by Theorem 3.11 (tangents from an

external point), from the entry point P_0 of the deviation, at distance ℓ from O , the points of tangency lie at distance $\sqrt{\ell^2 - (r + s)^2}$ from P_0 : this is the minimum manoeuvring length to be reserved in the trajectory planner. $\square$

Remark 3.14: Why the power of a point is the right tool. The trajectory planner of an AGV or of a robot arm, when it must check in real time whether a straight path intersects a circular exclusion zone, computes exactly the sign of $\text{pow}(P) = |OP|^2 - (r + s)^2$ for every candidate point P : if $\text{pow}(P) \geq 0$ everywhere, the path is safe; the first point with $\text{pow}(P) < 0$ marks the beginning of the intersection and the need to replan. The computational advantage is that the test requires no square root: squared distances are compared directly.

3.7. Radical axis and radical centre

Theorem 3.20: Radical axis of two circles. Given two non-concentric circles $C(O_1, r_1)$, $C(O_2, r_2)$, the locus of points P of the plane with $\text{pow}_1(P) = \text{pow}_2(P)$ is a line, called the **radical axis**, perpendicular to O_1O_2 .

Proof. For every point P ,

$$\text{pow}_1(P) - \text{pow}_2(P) = (|PO_1|^2 - r_1^2) - (|PO_2|^2 - r_2^2) = |PO_1|^2 - |PO_2|^2 - (r_1^2 - r_2^2).$$

Let H be the foot of the perpendicular from P to the line O_1O_2 . By Pythagoras, $|PO_1|^2 = |PH|^2 + |HO_1|^2$ and $|PO_2|^2 = |PH|^2 + |HO_2|^2$; subtracting, $|PO_1|^2 - |PO_2|^2 = |HO_1|^2 - |HO_2|^2$, which depends only on the position of H on the line O_1O_2 , not on P . Fixing a coordinate t on the line O_1O_2 with O_1 at o_1 and O_2 at o_2 ($o_1 \neq o_2$, non-concentric), and H at t :

$$|HO_1|^2 - |HO_2|^2 = (t - o_1)^2 - (t - o_2)^2 = 2t(o_2 - o_1) + (o_1^2 - o_2^2),$$

a **linear and strictly monotone** function of t (coefficient $2(o_2 - o_1) \neq 0$). Hence $\text{pow}_1(P) = \text{pow}_2(P)$ is equivalent to a single linear equation in t , that is, to a single value t_0 : the locus is the perpendicular to O_1O_2 at the point H_0 of coordinate t_0 , that is, a line. $\square$

Corollary 3.5: Radical centre of three circles. Given three circles with non-collinear centres, the three radical axes (taken pairwise) are concurrent at a single point, the **radical centre**, of equal power with respect to all three.

Proof. Let a_{12}, a_{13} be the radical axes of the pairs (1, 2) and (1, 3): they are non-parallel lines, being perpendicular to O_1O_2 and O_1O_3 respectively, and O_1, O_2, O_3 are not collinear by hypothesis (hence O_1O_2 and O_1O_3 are not parallel, nor are their perpendiculars). They therefore meet at a single point Z , with $\text{pow}_1(Z) = \text{pow}_2(Z)$ and $\text{pow}_1(Z) = \text{pow}_3(Z)$, hence $\text{pow}_2(Z) = \text{pow}_3(Z)$: Z also lies on the third radical axis a_{23} . The three axes are therefore concurrent at Z . $\square$

Construction 3.1: — Radical axis with straightedge and compass. If the two circles intersect, the radical axis is the line through the two common points (both have zero power with respect to both). If they do not intersect, draw a third auxiliary circle meeting both; its two radical axes with C_1 and with C_2 (lines through common points, hence constructible) meet at the radical centre of the triple, which lies on the sought radical axis: the latter is then the perpendicular to O_1O_2 through that point. $\square$

Example 3.2: Radical centre of three concrete circles. C_1 : centre $(0, 0)$, radius 3; C_2 : centre $(8, 0)$, radius 2; C_3 : centre $(3, 6)$, radius 4. Solving the linear system given by the two radical axes (independent script) gives the radical centre $Z = (4.3125; 1.010417)$, with common power $\text{pow}_1(Z) = \text{pow}_2(Z) = \text{pow}_3(Z) = 10.618598$ (checked, six decimal places).

3.8. Common tangents to two circles

Theorem 3.21: Common tangents. Let $C(O_1, r_1)$, $C(O_2, r_2)$ with $d = |O_1O_2|$. If $d > r_1 + r_2$ (external circles), there exist exactly four common tangents: two **external** (the two circles on the same side), of length $\sqrt{d^2 - (r_1 - r_2)^2}$ between the points of tangency, and two **internal** (the two circles on opposite sides), of length $\sqrt{d^2 - (r_1 + r_2)^2}$. If $d = r_1 + r_2$ (external tangency) there are three tangents (the internal ones degenerate into the common tangent at the point of contact); if $|r_1 - r_2| < d < r_1 + r_2$ (secant circles) there are two (only the external ones); if $d = |r_1 - r_2|$ (internal tangency) there is one; if $d < |r_1 - r_2|$ (one contained in the other) there are zero.

Proof. For an external tangent, the two radii O_1T_1 , O_2T_2 to the points of tangency are both perpendicular to the tangent (Theorem 3.8), hence parallel to each other and on the same side: $O_1T_1T_2O_2$ is a right trapezoid. Projecting O_2 onto O_1T_1 at the point Q gives a right triangle O_1QO_2 with hypotenuse d , one leg $|r_1 - r_2|$ (difference of the radii, since $QT_1 = O_2T_2 = r_2$) and the other leg $|QO_1| = |T_1T_2|$ (opposite sides of the rectangle $QT_1T_2O_2$): by Pythagoras, $|T_1T_2|^2 = d^2 - (r_1 - r_2)^2$. For an internal tangent, the same construction with O_1, O_2 on the same side of the tangent gives a trapezoid with the radii r_1, r_2 on opposite sides (not parallel in the same sense): the leg becomes $r_1 + r_2$ instead of $|r_1 - r_2|$, giving $|T_1T_2|^2 = d^2 - (r_1 + r_2)^2$. The count of tangents in the five cases follows directly from the sign of the two radicands just found: a tangent (external or internal) exists, with real positive length, if and only if the corresponding radicand is positive, i.e. $d > |r_1 - r_2|$ for the external ones and $d > r_1 + r_2$ for the internal ones (radicand zero $\Rightarrow$ the two points of tangency coincide, a single common tangent at the point of contact; radicand negative $\Rightarrow$ no tangent of that kind, because the right triangle O_1QO_2 of the construction would require an imaginary leg $|QO_1|$). Adding the two counts (two external plus two internal when both exist, and so on) gives 4, 3, 2, 1, 0 tangents respectively in the five listed cases. $\square$

Example 3.3: Lengths of the common tangents. $O_1 = (0, 0)$, $r_1 = 5$; $O_2 = (13, 0)$, $r_2 = 2$: $d = 13$. External tangents: $\sqrt{169 - 9} = \sqrt{160} = 12.649111$. Internal tangents: $\sqrt{169 - 49} = \sqrt{120} = 10.954451$ (checked).

3.9. Pitot's theorem

Theorem 3.22: Pitot's theorem. If a convex quadrilateral $ABCD$ is circumscribed about a circle (that is, all four sides are tangent), then

$$AB + CD = BC + DA.$$

Proof. Let P, Q, R, S be the points of tangency on AB, BC, CD, DA . By the congruence of tangent segments from the same external point (Theorem 3.11(c)): $AP = AS$, $BP = BQ$, $CQ = CR$, $DR = DS$. Adding,

$$AB + CD = (AP + PB) + (CR + RD) = (AS + BQ) + (CQ + DS)$$

and

$$BC + DA = (BQ + QC) + (DS + SA) = (BQ + CR) + (DS + AP),$$

(having substituted $QC = CR$, $SA = AP$ as above); the two right-hand sides coincide term by term with those of the computation of $AB + CD$ (the same sum $AP + BQ + CQ...$ reordered): explicitly, $AB + CD = AP + PB + CR + RD = AP + BQ + CR + DS$ and $BC + DA = BQ + QC + DS + SA = BQ + CR + DS + AP$: they are the same sum $AP + BQ + CR + DS$, hence equal. $\square$

Remark 3.15: The converse. *The converse is true (if $AB + CD = BC + DA$ in a convex quadrilateral, an inscribed circle exists) but its standard elementary proof proceeds by contradiction in a way that requires care in the limiting cases; not having been checked here step by step with the same rigour as the other proofs in this chapter, we flag it as **imported**: Josefsson, *More characterizations of tangential quadrilaterals*, *Forum Geometricorum* 11 (2011), 65–82.*

3.10. Open problems and questions long unresolved

All the theorems of this chapter have been proved for more than two millennia. It would however be misleading to leave the impression that the geometry of chords and tangents is a closed chapter: as soon as the circle is replaced by a more general curve, elementary questions appear that are easy to state and still unanswered. The three that follow are chosen because they arise directly from properties of the circle proved above.

3.10.1 The square peg problem (Toeplitz, 1911) — open

Remark 3.16: Statement. Question. *Does every Jordan curve in the plane (that is, every closed, continuous, non-self-intersecting curve) contain four points that are the vertices of a square?*

For the circle the answer is immediate: the endpoints of two perpendicular diameters suffice, and the resulting quadrilateral is a square by Theorem 3.1 and by the congruence of the radii. In 1911 Otto Toeplitz asked whether this elementary property survives when all regularity is abandoned.

Outline of the known strategy Step 1. One considers the space $\mathcal{Q}$ of all squares inscribed in the curve, suitably parametrised.

Step 2. One shows that, if the curve is sufficiently regular, $\mathcal{Q}$ is a differentiable manifold, and one constructs a map whose degree (or one of whose homology classes) turns out to be non-zero.

Step 3. From a non-zero topological invariant it follows that the solution set is non-empty, that is, that an inscribed square exists.

Where it breaks down. For an arbitrary Jordan curve Step 2 fails: the curve may be so irregular (for instance of infinite length, or fractal) that the space of inscribed squares loses its manifold structure, and the topological argument does not apply. $\square$

Warning. Status at the time of writing (2026): the problem is OPEN for general Jordan curves. It has been proved for numerous particular classes: convex curves, piecewise analytic curves, curves of class C^1 , curves of finite total curvature, generic C^1 curves; in 2025 it was extended to curves sufficiently close to a curve of class C^2 . None of these proofs

extends to the general case, and at present no complete solution is known.

3.10.2 The equichordal point problem — solved after eighty years

Remark 3.17: Statement. A point P interior to a closed convex curve is called **equichordal** if all the chords of the curve through P have the same length. **Question.** Can a closed convex curve have two distinct equichordal points?

The question is the natural generalisation of a property of the circle: the centre O is equichordal, because every chord through the centre is a diameter and has length $2r$ (Theorem 3.6). The circle therefore has *one* equichordal point. Can there be a convex figure with two?

Outline of the strategy and solution Step 1. One assumes for contradiction that two equichordal points $O_1 \neq O_2$ exist and translates the condition into a functional equation for the function describing the boundary of the curve.

Step 2. The functional equation is reformulated as the problem of existence of an invariant curve for a certain map of the plane.

Step 3. One complexifies the problem and studies the behaviour of the invariant manifolds of the map in the complex domain.

Step 4. One shows that such manifolds cannot close up into a real convex curve: contradiction. $\square$

BEST PRACTICE

Status: SOLVED in the negative. The problem was posed by Fujiwara in 1916 and, independently, by Blaschke, Rothe and Weitzenböck in 1917. It remained open for **eighty years**, until the proof of Marek Rychlik (1997): a closed convex curve *cannot* have two equichordal points. The proof, 72 pages long and obtained *without* the aid of a computer, uses complex analysis and algebraic geometry — tools very far removed from the statement, which a secondary-school pupil understands in thirty seconds. Rychlik's result is in fact stronger: it holds also for Jordan curves star-shaped with respect to both points, not necessarily convex.

3.10.3 The equireciprocal point problem — open

Remark 3.18: Statement. A point P is called **equireciprocal** for a curve if there exists a constant c such that, for every chord $\overline{AB}$ through P ,

$$\frac{1}{|PA|} + \frac{1}{|PB|} = c.$$

Question (Klee). Can a plane convex body have two distinct equireciprocal points?

Warning. Status at the time of writing (2026): OPEN. The problem is closely related to the equichordal one: a point is equireciprocal for a region if and only if it is equichordal for the image of the region under circular inversion centred at P (inversion is treated in Chapter 5). The difficulty is that inversion *does not preserve convexity*, so Rychlik's solution does not transfer. At present no solution is known.

INFO

Why include unsolved problems in an elementary text. The three problems above have statements requiring nothing beyond this chapter: chords, lengths, an interior point. They serve two purposes. The first is to show that the frontier of research is not always far from elementary tools. The second, more important: the difference between an *easy* problem and a *hard* one cannot be seen from the statement. Theorem 3.6 and the equichordal problem both speak of the length of the chords through a point; the first is proved in three lines, the second took eighty years and 72 pages.

3.11. Exercises

Exercise 3.1. An AGV travels at a distance $d = 1.8$ m from the centre of a circular obstacle of radius $r = 0.5$ m. Applying the power of a point theorem, check whether the nominal straight trajectory meets the obstacle, and compute the chord of intersection if any.

Solution. The power of the foot of the perpendicular P with respect to the circle is $\text{pow}(P) = d^2 - r^2 = 1.8^2 - 0.5^2 = 3.24 - 0.25 = 2.99 > 0$: being positive, P is exterior and the trajectory does **not** meet the obstacle (no chord to compute). Had it been $d = 0.3$ m ($< r$) instead, one would have had $\text{pow}(P) = 0.09 - 0.25 = -0.16 < 0$ (secant trajectory), with half-chord $\sqrt{r^2 - d^2} = \sqrt{0.25 - 0.09} = \sqrt{0.16} = 0.4$ m and total chord 0.8 m of intersection with the obstacle.

Exercise 3.2. Prove, using Theorem 3.5, that the diameter is the longest chord of every circle (interpreting the diameter as a chord at distance $d = 0$ from the centre). Then compare your proof with the triangle-inequality one of Theorem 3.6: which of the two arguments uses fewer prerequisites?

Solution. From $|AB| = 2\sqrt{r^2 - d^2}$ with $d \in [0, r]$, the function $d \mapsto 2\sqrt{r^2 - d^2}$ is strictly decreasing on $[0, r]$ (its derivative $-2d/\sqrt{r^2 - d^2}$ is ≤ 0 , vanishing only at $d = 0$). The maximum is therefore attained at $d = 0$, where $|AB| = 2\sqrt{r^2} = 2r$: this is precisely the definition of diameter (a chord through the centre, i.e. at zero distance from it). Every other chord has $d > 0$ and hence length $< 2r$.

Comparison. This proof uses: Theorem 3.1 (to establish that the foot of the perpendicular is the midpoint), the Pythagorean Theorem (A5) and the monotonicity of a real function. The triangle-inequality proof uses axiom (A8) alone. The second is therefore more economical; the first, however, provides in addition the explicit formula for the length of every chord.

Exercise 3.3. From a point P exterior to a circle of radius $r = 5$ cm the two tangents are drawn. Knowing that $|OP| = 13$ cm, compute: (a) the length of the tangent segments; (b) the measure of the angle between the two tangents; (c) the length of the chord of contact $\overline{T_1T_2}$.

Solution. (a) By Theorem 3.11(c), $|PT| = \sqrt{|OP|^2 - r^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm.
(b) In the right triangle OT_1P , $\sin(\angle OPT_1) = r/|OP| = 5/13$, hence $\angle OPT_1 = \arcsin(5/13) \approx 22.62^\circ$. By Theorem 3.12(a) the line OP bisects the angle between the tangents, hence $\angle T_1PT_2 = 2 \times 22.62^\circ \approx 45.24^\circ$.
(c) By Theorem 3.12(c), OP is the perpendicular bisector of $\overline{T_1T_2}$; if H denotes the meeting

point, the triangle OT_1P has altitude to the hypotenuse $|T_1H| = \frac{|OT_1| \cdot |PT_1|}{|OP|} = \frac{5 \cdot 12}{13} = \frac{60}{13} \approx 4.615$ cm. Hence $|T_1T_2| = 2|T_1H| = \frac{120}{13} \approx 9.23$ cm.

Check. The central angle equals $180^\circ - 45.24^\circ = 134.76^\circ$ by Theorem 3.15(a), and the corresponding chord measures $2r \sin(134.76^\circ/2) = 10 \sin(67.38^\circ) \approx 9.23$ cm: they agree.

Exercise 3.4. From a point P exterior to a circle a secant is drawn meeting it at A and B , with $|PA| = 4$ cm and $|AB| = 5$ cm. A second secant through P meets the circle at C and D , with $|PC| = 3$ cm. Compute $|PD|$ and the length of the tangent segment from P .

Solution. Since A lies between P and B , we have $|PB| = |PA| + |AB| = 4 + 5 = 9$ cm. By Theorem 3.18,

$$|PC| \cdot |PD| = |PA| \cdot |PB| = 4 \times 9 = 36 \implies |PD| = \frac{36}{3} = 12 \text{ cm.}$$

By Theorem 3.16, the tangent segment satisfies $|PT|^2 = 36$, hence $|PT| = 6$ cm. Note that $\text{pow}(P) = 36$: the same number is read off in three different ways, as Theorem 3.19 asserts.

Exercise 3.5. (*On the risk of circularity.*) A text proves the classification of lines as follows: “if $d(O, \ell) = r$, let T be the foot of the perpendicular; then $OT \perp \ell$ and $|OT| = r$, hence $T \in C$ and by the tangent–radius theorem ℓ is tangent”. Identify the logical error and propose a correction.

Solution. The error is the use of the “tangent–radius theorem” (Theorem 3.8) to prove the classification of lines. But Theorem 3.8 was proved *using* the classification (Step 3 of its proof): the reasoning is therefore circular.

Correction. One must invoke Theorem 3.9 (Euclid III.16) and not Theorem 3.8: from the fact that $\ell \perp OT$ with $T \in C$ it follows that ℓ is tangent, and this implication is proved independently, directly, using only the Pythagorean Theorem. Alternatively — and this is the route chosen in this treatise — one proves the classification by purely metric means (Theorem 2.6), without mentioning perpendicularity at all.

Exercise 3.6. Two parallel chords of a circle of radius $r = 10$ cm measure 16 cm and 12 cm respectively. Compute the two possible distances between the chords, explaining why there are two solutions.

Solution. By Theorem 3.5, $d = \sqrt{r^2 - (|AB|/2)^2}$:

$$d_1 = \sqrt{100 - 64} = 6 \text{ cm}, \quad d_2 = \sqrt{100 - 36} = 8 \text{ cm.}$$

If the two chords lie on the *same* side of the centre, their distance is $|d_2 - d_1| = 2$ cm; if they lie on *opposite* sides, it is $d_1 + d_2 = 14$ cm. Both configurations are geometrically possible, because Theorem 3.5 determines the distance *from the centre* but not on which side: hence the two solutions. Note that the longer chord (16 cm) is the one nearer the centre (6 cm), in agreement with the monotonicity established by the same theorem.

Exercise 3.7. From an external point P a secant is drawn meeting the circle at A and B , with $|PA| = 4$ and $|PB| = 9$. Compute the length of the tangent segment from P and the power of P .

Solution. By the secant–tangent theorem, $|PT|^2 = |PA| \cdot |PB| = 36$, hence $|PT| = 6$. The power of P is $\text{pow}(P) = |PA| \cdot |PB| = 36$, positive because P is external. Note that $|PT| = \sqrt{\text{pow}(P)}$: the tangent segment is the geometric mean of the two secant segments.

Exercise 3.8. Two chords AB and CD meet internally at P , with $|AP| = 3$, $|PB| = 8$ and $|CP| = 4$. Compute $|PD|$ and the power of P .

Solution. By the intersecting-chords theorem $|AP| \cdot |PB| = |CP| \cdot |PD|$, that is $24 = 4|PD|$ and $|PD| = 6$. The power of P is -24 : the negative sign indicates that P is interior, and in absolute value it equals $r^2 - |OP|^2$ with the sign changed.

Exercise 3.9. Two circles have radii $r_1 = 5$ and $r_2 = 2$ and centres at distance $d = 10$. Compute the length of the common external tangent and of the internal one.

Solution. For the common external tangent one projects the segment of centres on the direction of the tangent: the difference of the radii forms the perpendicular leg, hence $L_e = \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{100 - 9} = \sqrt{91} = 9.5394$. For the internal tangent the leg is the sum of the radii: $L_i = \sqrt{d^2 - (r_1 + r_2)^2} = \sqrt{100 - 49} = \sqrt{51} = 7.1414$. The internal tangents exist because $d > r_1 + r_2$.

Exercise 3.10. A chord of length 10 is drawn in a circle of radius 13. Compute the sagitta (the distance between the midpoint of the chord and the arc) and verify the relation $s(2r - s) = c^2$, with c the half-chord.

Solution. The distance of the chord from the centre is $\sqrt{169 - 25} = 12$, hence the sagitta is $s = 13 - 12 = 1$. Check: $s(2r - s) = 1 \cdot (26 - 1) = 25 = 5^2 = c^2$ ✓. The relation is the power of the midpoint of the chord with respect to the circle, read along the diameter through that point.

Exercise 3.11. Prove that the angle between a tangent and a chord drawn from the point of contact equals the inscribed angle standing on the chord in the alternate segment.

Solution. Let T be the contact point, TA the chord and α the angle between tangent and chord. The radius OT is perpendicular to the tangent, hence $\widehat{OTA} = 90^\circ - \alpha$. The triangle OTA is isosceles ($OT = OA = r$), so $\widehat{OAT} = 90^\circ - \alpha$ and the central angle is $\widehat{TOA} = 180^\circ - 2(90^\circ - \alpha) = 2\alpha$. By the inscribed angle theorem, the inscribed angle standing on TA in the alternate segment is half the central angle, that is α .

Exercise 3.12. From P with $|OP| = 13$ the tangents are drawn to a circle of radius 5. Compute the distance of the chord of contact from the centre and verify that its foot is the inverse of P with respect to the circle, in the sense that $|OH| \cdot |OP| = r^2$.

Solution. Let H be the foot of the perpendicular from O to the chord of contact. The triangles OHT and OTP are similar (they share the angle at O and both have a right angle), hence $|OH|/|OT| = |OT|/|OP|$, that is $|OH| = r^2/|OP| = 25/13 = 1.9231$. It follows at once that $|OH| \cdot |OP| = 25 = r^2$: the foot H is the inverse of P , and the chord of contact is its polar. This is the link between the geometry of tangents of the present chapter and the inversion of Chapter 5.

3.12. Self-assessment quiz

Multiple-choice questions for a quick check before tackling the exercises. The answer key is at the end of the section.

1. The perpendicular drawn from the centre to a chord
 - A) is parallel to the chord
 - B) bisects the chord
 - C) is tangent to the circle
 - D) coincides with a diameter only if the chord is a diameter
2. The perpendicular bisector of a chord
 - A) is tangent to the circle
 - B) passes through the centre
 - C) is parallel to the chord
 - D) does not meet the circle
3. Of two unequal chords of the same circle, the longer one is
 - A) farther from the centre
 - B) nearer the centre
 - C) equidistant from the centre
 - D) always a diameter
4. The chord of maximum length of a circle is
 - A) the perpendicular bisector
 - B) a tangent
 - C) the diameter
 - D) the radius
5. If two chords meet at an interior point P , then
 - A) $PA + PB = PC + PD$
 - B) $PA \cdot PB = PC \cdot PD$
 - C) $PA/PB = PC/PD$
 - D) $PA = PC$
6. The line tangent to a circle at T makes with the radius OT an angle of

-
- A) 45°
 B) 60°
 C) 90°
 D) which depends on the radius
7. From a point P exterior to a circle one can draw
 A) no tangent
 B) exactly one tangent
 C) exactly two tangents
 D) infinitely many tangents
8. The two tangent segments drawn from an external point are
 A) always unequal
 B) congruent
 C) perpendicular to each other
 D) congruent only if $|OP| = 2r$
9. If PT is tangent and PAB a secant from an external point P , then
 A) $PT = PA + PB$
 B) $PT^2 = PA \cdot PB$
 C) $PT = PA \cdot PB$
 D) $PT^2 = PA + PB$
10. The power of a point is independent of
 A) the radius of the circle
 B) the line chosen through the point
 C) the position of the point
 D) the centre

BEST PRACTICE

Answer key. 1: B — 2: B — 3: B — 4: C — 5: B — 6: C — 7: C — 8: B — 9: B — 10: B.

3.13. Chapter Formula sheet

Radius and chord (Theorems 3.1 and 3.2; Corollary 3.1)

$$OM \perp AB \iff |AM| = |MB| \quad (\overline{AB} \text{ not a diameter}),$$

the perpendicular bisector of every chord passes through O

Chords and distance from the centre (Theorem 3.5)

$$|AB| = 2\sqrt{r^2 - d^2}, \quad d = d(O, AB); \quad |AB| = |CD| \iff d_1 = d_2, \quad |AB| > |CD| \iff d_1 < d_2$$

Longest chord (Theorem 3.6)

$$|AB| \leq 2r, \quad \text{equality} \iff \overline{AB} \text{ is a diameter}$$

Intersecting chords (Theorem 3.7)

$$|PA| \cdot |PB| = |PC| \cdot |PD| \quad (P \text{ interior})$$

Two secants from an external point (Theorem 3.18)

$$|PA| \cdot |PB| = |PC| \cdot |PD| \quad (P \text{ exterior})$$

Tangents from an external point (Theorem 3.11)

$$|PT_1| = |PT_2| = \sqrt{|OP|^2 - r^2}$$

Tangent and secant (Theorem 3.16)

$$|PT|^2 = |PA| \cdot |PB|$$

Power of a point (Definition 3.1; Theorem 3.19; Corollary 3.4)

$$\text{pow}(P) = |OP|^2 - r^2, \quad |PA| \cdot |PB| = |\text{pow}(P)|, \quad |PT|^2 = \text{pow}(P)$$

$$\text{pow}(P) \geq -r^2, \quad \text{equality} \iff P = O$$

Tangent: the three characterisations (Theorems 3.8, 3.9 and 3.10)

$$t \perp OT \text{ at } T \in C(O, r) \iff |t \cap C(O, r)| = 1 \iff d(O, t) = r$$

Tangent–chord angle (Theorem 3.13)

$$\angle(t, \overline{TA}) = \angle TDA, \quad D \text{ in the alternate segment}$$

Angle between two tangents (Theorem 3.15)

$$\angle T_1PT_2 + \angle T_1OT_2 = \pi, \quad \angle T_1PT_2 = \frac{1}{2}(\mu_{\max} - \mu_{\min})$$

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CHAPTER 4

Central and Inscribed Angles

INFO

Aim of the chapter. To prove, in every case and with no tacit exceptions, the theorem that governs the whole geometry of the circle: *the central angle is twice the inscribed angle standing on the same arc*. From it follow Thales' theorem, the characterisation of cyclic quadrilaterals, the arc of a given angle, and all the angular relations among chords, secants and tangents.

Logical position. This chapter is the **supplier** of the prerequisites P-11 and P-12 declared in Chapter 3, and in the proofs of its fundamental theorems it uses no result of that chapter: the dependency runs one way only, and the graph stays acyclic (Section 4.1, with the sole declared exception of the arc of a given angle and the cases with tangents, which use tangents already available from Chapter 3).

What sets it apart from current treatments. Almost every text proves the fundamental theorem in three cases and says nothing about the fourth, the one with a **reflex** central angle: here it is proved (Section 4.3.1), because without it half of the examination problems are formally uncovered.

4.1. Logical position and prerequisites

Code	Assumed statement	Status
(A1)–(A10)	axioms listed in Remark 2.1	assumed
(A4)	isosceles triangle: base angles congruent	assumed
(A9)	sum of the interior angles: π for a triangle	assumed
—	exterior angle theorem for a triangle	assumed
D-06	central angle (Definition 2.12)	Ch. 2
D-07	inscribed angle (Definition 2.16)	Ch. 2
D-08	measure of an arc (Definition 2.14)	Ch. 2

Warning. No result of Chapter 3 is used here. The check is mechanical: the proofs of this chapter employ only isosceles triangles, the exterior angle theorem, the sum of the interior angles and the definition of the measure of an arc — all of them earlier. This makes it legitimate for Chapter 3 to assume as prerequisites P-11 and P-12 the theorems proved here: the arrow of dependency points one way only. (Later in this chapter, the Construction of the arc of a given angle and the cases with tangents of the angle from an external point *instead use* results of Chapter 3 — Theorem 3.9 and Theorem 3.13 — but this is declared explicitly on the spot and does not concern the theorems recalled above.)

Remark 4.1: *The fourth case, and why almost nobody proves it.* The fundamental theorem is stated for “the central angle standing on the same arc”. If the arc stood upon is greater than a semicircle, the corresponding central angle is **reflex** (greater than π) and is no longer a convex angle: the usual figures do not show it and the classical proofs do not

cover it. This chapter treats it explicitly in Section 4.3.1. The difficulty in conceiving the reflex angle is documented as one of the recurrent obstacles in the didactic research on this topic (Section 4.13).

4.2. Angles associated with the circle: recalls and new definitions

The notions of central angle (Definition 2.12), inscribed angle (Definition 2.16) and measure of an arc (Definition 2.14) were introduced in Chapter 2 and are not repeated here. Two clarifications are needed, however, which could not be given at that stage.

Definition 4.1: Reflex central angle. Let A, B be two distinct, non-antipodal points of $C(O, r)$. The two radii OA and OB divide the plane into two angular regions: the **convex** one, of measure $\theta \in (0, \pi)$, and the **reflex** one, of measure $2\pi - \theta \in (\pi, 2\pi)$. Each of the two is a *central angle*; when they must be distinguished one speaks of the **convex** or the **reflex** central angle. Each stands on the arc $\widehat{AB}$ contained in it: the convex one on the minor arc, the reflex one on the major arc.

Remark 4.2: Why the distinction is compulsory. Without the reflex angle the correspondence between arcs and central angles would not be bijective: the two arcs determined by A and B would share a single angle. With Definition 4.1 the correspondence is perfect and Definition 2.14 becomes a particular case: the measure of an arc is always the measure of the central angle standing on it, be it convex or reflex. The case $\mu = 2\pi - \mu_{\min}$ for the major arc, which in Chapter 2 appeared as an additional convention, is now a two-line theorem.

Definition 4.2: Inscribed angle and the arc it stands on. Let $\angle APB$ be an inscribed angle, with $P \in C(O, r)$ and $A, B \in C(O, r)$ distinct from P . The arc on which $\angle APB$ **stands** is the arc $\widehat{AB}$ not containing P . The points P of the other arc are those for which the angle is defined.

Warning. The specification “not containing P ” is not pedantry. If P belonged to the arc it is supposed to stand on, the sides PA, PB and the arc would lie on the same side and the statement of the fundamental theorem would be false: one would obtain the *supplement*. This is exactly the interchange that produces the commonest examination error (Section 4.13).

4.3. The fundamental theorem

Theorem 4.1: Central angle and inscribed angle theorem (Euclid III.20). Let $C(O, r)$ be a circle, let A, B be two distinct points on it and let P be a point of the circle not belonging to the arc $\widehat{AB}$ stood upon. Then

$$\angle AOB = 2 \cdot \angle APB,$$

where $\angle AOB$ is the central angle (convex or reflex, Definition 4.1) standing on the same arc.

Proof. Draw the diameter PD issuing from P , where D is the antipodal point of P . The line PD divides the plane into two half-planes. Four cases arise according to the position of O relative to the angle $\angle APB$; the first three are classical, the fourth is treated in Section 4.3.1.

Case 1 — the centre O lies on a side of the inscribed angle. Suppose O belongs to the segment PA , that is, PA is a diameter (so $A = D$).

Step 1. The triangle OPB has $|OP| = |OB| = r$ (Theorem 2.3): it is therefore **isosceles** with base $\overline{PB}$. By (A4) the base angles are congruent:

$$\angle OPB = \angle OBP =: \beta.$$

Step 2. Consider the angle $\angle BOA$, which is the *exterior* angle of the triangle OPB at the vertex O (since P, O, A are collinear in this order, PA being a diameter). By the exterior angle theorem, the exterior angle equals the sum of the two non-adjacent interior angles:

$$\angle BOA = \angle OPB + \angle OBP = \beta + \beta = 2\beta.$$

Step 3. But $\beta = \angle OPB = \angle APB$, because the ray PO coincides with the ray PA (O lies between P and A). Hence $\angle AOB = 2\angle APB$, which is the conclusion in this case.

Case 2 — the centre O is interior to the angle $\angle APB$. The diameter PD then falls *inside* the angle, and the ray PD divides $\angle APB$ into two adjacent angles:

$$\angle APB = \angle APD + \angle DPB.$$

Step 1. Apply Case 1 separately to the two inscribed angles $\angle APD$ and $\angle DPB$ (for each of them the centre O lies on the side PD , which is a diameter):

$$\angle AOD = 2\angle APD, \quad \angle DOB = 2\angle DPB.$$

Step 2. Since D is interior to the arc stood upon, the ray OD is interior to the angle $\angle AOB$, hence the central angles add up too:

$$\angle AOB = \angle AOD + \angle DOB.$$

Step 3. Adding the two equalities of Step 1 and using Step 2:

$$\angle AOB = 2\angle APD + 2\angle DPB = 2(\angle APD + \angle DPB) = 2\angle APB.$$

Case 3 — the centre O is exterior to the angle $\angle APB$. The diameter PD then falls *outside* the angle. Suppose, without loss of generality, that B is interior to the angle $\angle APD$ (otherwise exchange the names of A and B). Then

$$\angle APB = \angle APD - \angle BPD,$$

and correspondingly, since the ray OB is interior to the angle $\angle AOD$,

$$\angle AOB = \angle AOD - \angle BOD.$$

Applying Case 1 to the two inscribed angles $\angle APD$ and $\angle BPD$ (both have the centre on a side, the diameter PD) and subtracting:

$$\angle AOB = 2\angle APD - 2\angle BPD = 2(\angle APD - \angle BPD) = 2\angle APB.$$

□

Remark 4.3: Are the three cases exhaustive? They are, and it is worth checking it explicitly, because this is the point where a proof by cases can silently leave a hole. Once the diameter PD is drawn, the point D (the second endpoint of the diameter) lies on the arc stood upon, and relative to the angle $\angle APB$ exactly one of three situations occurs: D coincides with A or with B (Case 1); D is interior to the arc $\widehat{AB}$ stood upon, and then PD is interior to the angle (Case 2); D is exterior to that arc, and then PD is exterior to the angle (Case 3). The three alternatives are mutually exclusive and exhaustive because D is a well-determined point of the circle and the three conditions are the trichotomy of its position relative to the arc.

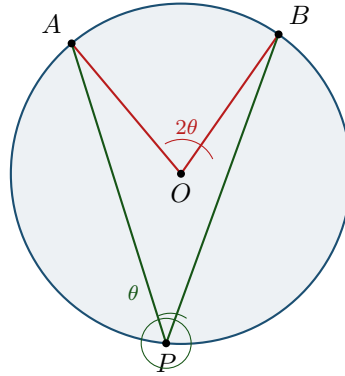


Figure 4.1: — The central angle is twice the inscribed angle standing on the same arc

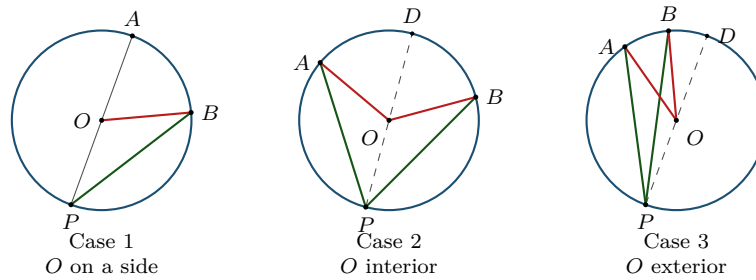


Figure 4.2: — The three cases of the synthetic proof, according to the position of the dashed diameter PD relative to the inscribed angle. The fourth case (reflex central angle) is in Figure 4.3.

Alternative proof — by complex numbers (rotation). Take the circle to be the unit circle in the complex plane, with $A = e^{i\alpha}$, $B = e^{i\beta}$, $P = e^{i\pi_0}$ (we write π_0 for the angle of P , so as not to confuse it with the constant π). The key point is the following elementary identity, obtained by factoring out the exponential of the mean angle:

$$e^{i\theta_1} - e^{i\theta_2} = e^{i(\theta_1+\theta_2)/2} \left(e^{i(\theta_1-\theta_2)/2} - e^{-i(\theta_1-\theta_2)/2} \right) = 2i \sin\left(\frac{\theta_1 - \theta_2}{2}\right) e^{i(\theta_1+\theta_2)/2}.$$

Since multiplying by i amounts to a rotation by 90° , this identity says that the segment (vector) from $e^{i\theta_2}$ to $e^{i\theta_1}$ has argument $(\theta_1 + \theta_2)/2 + 90^\circ$ (up to a possible $+180^\circ$ when $\sin(\cdot) < 0$, immaterial since we work with unoriented angles). Applying this to $B - P$ and $A - P$:

$$\arg(B - P) \equiv \frac{\beta + \pi_0}{2} + 90^\circ, \quad \arg(A - P) \equiv \frac{\alpha + \pi_0}{2} + 90^\circ \pmod{180^\circ}.$$

The angle $\angle APB$ between the vectors PA and PB is the difference of these arguments:

$$\angle APB \equiv \arg(B - P) - \arg(A - P) \equiv \frac{\beta - \alpha}{2} \pmod{180^\circ},$$

that is, exactly half of the central angle $\angle AOB = \beta - \alpha$ (the angle between the radii OA, OB , of arguments α, β). Note the advantage of this method over the synthetic proof: the computation is **identical** whatever the position of P (on a side, inside or outside the angle $\angle AOB$), because the term π_0 (the angle of P) *cancels completely* in the difference of the arguments — there is no need to distinguish the three cases, nor even the fourth: the formula automatically covers the reflex central angle as well, since it works modulo 180° . $\square$

4.3.1 The fourth case: reflex central angle

Theorem 4.2: Reflex case. *Let $A, B \in C(O, r)$ be distinct and let P be a point of the minor arc $\widehat{AB}$. Then the inscribed angle $\angle APB$ is half the **reflex** central angle:*

$$\angle APB = \frac{1}{2}(2\pi - \angle AOB_{\text{conv}}),$$

where $\angle AOB_{\text{conv}}$ is the convex central angle. Equivalently, $\angle APB$ is obtuse and equals $\pi - \frac{1}{2}\angle AOB_{\text{conv}}$.

Proof. Let $\theta = \angle AOB_{\text{conv}}$ and let D be the second endpoint of the diameter through P , i.e. $D = 2O - P$ (the point antipodal to P , Definition 2.8).

Step 1 — D is interior to the angle $\angle APB$. The line PD passes through P and the centre O , and meets the circle only in P and D ; since P lies on the *minor* arc $\widehat{AB}$, between A and B , the points A and B lie on opposite sides of the line PD , and the segment $\overline{AB}$ crosses it at a point X . X is interior to the circle (it is a point of the chord AB distinct from its endpoints), and the line PD meets the closed disc exactly in the segment $\overline{PD}$ (the disc is convex and the line intersects the circle only in P and D); hence X lies on the segment $\overline{PD}$, i.e. the ray PD is interior to the angle $\angle APB$, and

$$\angle APB = \angle APD + \angle DPB.$$

Step 2 — the two partial angles. Each of the angles $\angle APD$, $\angle DPB$ has a side (PD) that is a diameter: by **Case 1** of Theorem 4.1 (the only case already available that does not presuppose this theorem),

$$\angle APD = \frac{1}{2}\angle AOD, \quad \angle DPB = \frac{1}{2}\angle DOB,$$

where $\angle AOD$ and $\angle DOB$ are the convex central angles on the arcs $\widehat{AD}$ and $\widehat{DB}$ that do *not* contain P .

Step 3 — the two measures add up to the major arc. Since D is antipodal to P , the arc $\widehat{AD}$ not containing P has measure $\pi - \mu(\widehat{PA})$ (the diameter PD splits the semicircle on that side into $\widehat{PA}$ and $\widehat{AD}$: their sum is π , by Theorem 2.5 of Chapter 2), and likewise the arc $\widehat{DB}$ not containing P has measure $\pi - \mu(\widehat{PB})$. Their sum is

$$(\pi - \mu(\widehat{PA})) + (\pi - \mu(\widehat{PB})) = 2\pi - \mu(\widehat{APB}),$$

i.e. the measure of the *major* arc AB (the one containing P), which is exactly the amplitude $2\pi - \theta$ of the reflex central angle. Hence

$$\angle APB = \angle APD + \angle DPB = \frac{1}{2}(\angle AOD + \angle DOB) = \frac{1}{2}(2\pi - \theta).$$

Since $\theta < \pi$ (it is a convex central angle), $2\pi - \theta > \pi$, so $\angle APB > \pi/2$: the angle is obtuse. *Numerical check.* With $A = (1, 0)$, $B = (\cos 60^\circ, \sin 60^\circ)$, $P = (\cos 30^\circ, \sin 30^\circ)$: $\angle AOB_{\text{conv}} = 60^\circ$, reflex central angle = 300° , and $\angle APB = 150^\circ = \frac{1}{2} \cdot 300^\circ$. $\checkmark$ $\square$

Warning. Check of non-circularity. This proof uses only **Case 1** of Theorem 4.1 (one side of the inscribed angle is a diameter), Theorem 2.5 of Chapter 2, and the convexity of the disc: no appeal to Theorem 4.3, which is in fact proved *afterwards* and will use this reflex case as an ingredient. The logical order is therefore: Case 1 $\Rightarrow$ reflex case (here) $\Rightarrow$ Theorem 4.3.

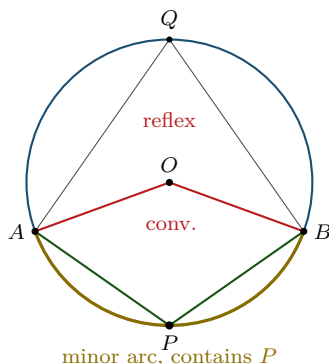


Figure 4.3: — The fourth case. The vertex P lies on the *minor* arc: the inscribed angle $\angle APB$ is obtuse and equals half the **reflex** central angle. The point Q on the major arc serves the proof.

4.4. Immediate corollaries

Corollary 4.1: Inscribed angles on the same arc. For any P, Q belonging to the same arc determined by the chord $\overline{AB}$ (that is, on the same side of AB): $\angle APB = \angle AQB$.

Proof. Both angles stand on the same arc $\widehat{AB}$, and by Theorem 4.1 both equal half of the same central angle $\angle AOB$. Two quantities equal to a third are equal to one another. $\square$

Corollary 4.2: Inscribed angles on congruent arcs. On the same circle (or on congruent circles), inscribed angles standing on congruent arcs are congruent; and conversely.

Proof. By Definition 2.14 congruent arcs have the same measure, that is, they subtend congruent central angles; by Theorem 4.1 the corresponding inscribed angles are halves of congruent angles, hence congruent. The converse follows by running the chain backwards, which is legitimate because every step is an equivalence (multiplication by 2 is injective). $\square$

Theorem 4.3: Inscribed angles on opposite arcs. If P, Q belong to the two opposite arcs determined by the chord $\overline{AB}$, then

$$\angle APB + \angle AQB = \pi.$$

Proof. Let θ be the measure of the convex central angle $\angle AOB$; the minor arc has measure θ and the major one $2\pi - \theta$ (Definition 2.14).

Suppose Q lies on the major arc: then Q stands on the *minor* arc, and by Theorem 4.1 (cases 1–3, with convex central angle)

$$\angle AQB = \frac{\theta}{2}.$$

The point P lies on the minor arc: this is precisely the configuration of Theorem 4.2 (Case 4, proved directly and independently just above), which gives

$$\angle APB = \frac{2\pi - \theta}{2} = \pi - \frac{\theta}{2}.$$

Adding: $\angle APB + \angle AQB = \pi - \theta/2 + \theta/2 = \pi$. □

4.5. Thales' theorem and its converse

Theorem 4.4: Thales' theorem (Euclid III.31). If $\overline{AB}$ is a diameter of $C(O, r)$, then $\angle APB = \pi/2$ for every point P of the circle distinct from A and B .

Proof. Via the fundamental theorem. Since A, O, B are collinear ($\overline{AB}$ is a diameter), the central angle $\angle AOB$ is straight and equals π . By Theorem 4.1, $\angle APB = \pi/2$. □

Direct proof — without the fundamental theorem. The triangles OAP and OBP are isosceles, because $|OA| = |OP| = |OB| = r$ (Theorem 2.3). Setting $\alpha = \angle OAP = \angle OPA$ and $\beta = \angle OPB = \angle OBP$ (base angles, by (A4)), the sum of the interior angles of the triangle APB gives, by (A9),

$$\underbrace{\alpha}_{\angle PAB} + \underbrace{(\alpha + \beta)}_{\angle APB} + \underbrace{\beta}_{\angle ABP} = \pi \implies 2(\alpha + \beta) = \pi \implies \angle APB = \alpha + \beta = \frac{\pi}{2}.$$

This version has the merit of being *independent* of the fundamental theorem: it uses only (A4) and (A9). This is worth knowing, because it allows Thales' theorem to be used in contexts where the general theorem is not yet available — as in the proof of Theorem 3.11 of Chapter 3. □

Third proof — vectorial, in two lines. Place the centre O at the origin, so that $A = -\mathbf{r}$ and $B = +\mathbf{r}$ with $|\mathbf{r}| = r$ (the endpoints of a diameter are opposite with respect to the centre). Let P be a point of the circle, i.e. a vector $\mathbf{p}$ with $|\mathbf{p}| = r$. Then

$$\vec{PA} \cdot \vec{PB} = (-\mathbf{r} - \mathbf{p}) \cdot (\mathbf{r} - \mathbf{p}) = -(\mathbf{r} + \mathbf{p}) \cdot (\mathbf{r} - \mathbf{p}) = -(|\mathbf{r}|^2 - |\mathbf{p}|^2) = -(r^2 - r^2) = 0.$$

Two non-zero vectors with zero dot product are orthogonal, hence $\angle APB = \pi/2$. (The vectors are non-zero because $P \neq A$ and $P \neq B$ by hypothesis.)

The hidden content. The quantity $\vec{PA} \cdot \vec{PB} = |\mathbf{p}|^2 - r^2$ is **the power of the point P** with respect to the circle (Definition 3.1, Chapter 3). On this reading, Thales' theorem is simply the statement that points *on* the circle have zero power — and the computation above holds for any pair of diametrically opposite points, not just for one particular diameter.

A warning about circularity. This route uses the dot product, i.e. the metric structure of $\mathbb{R}^2$, i.e. the Pythagorean theorem. It is therefore not usable where Thales must serve to prove Pythagoras — as in Proof 7 of Chapter 10, where the second, purely synthetic proof above must be used instead. □

Fourth proof — by central symmetry. The rotation σ by π about O (central symmetry) sends A to B (they are antipodal) and P to a point P' , again on the circle (isometries fix O and preserve distances from it). The quadrilateral $APBP'$ has diagonals $\overline{AB}$ and $\overline{PP'}$ that bisect each other at O (by construction of the central symmetry, O is the midpoint of both AB and PP') and are both congruent to $2r$ (diameters): a quadrilateral whose diagonals bisect each other is a parallelogram (elementary property), and a parallelogram with congruent diagonals is a **rectangle**. In a rectangle all angles are right angles, hence $\angle APB = \pi/2$. $\square$

Fifth proof — by complex numbers (university variant). Identify the plane with $\mathbb{C}$, $O = 0$. With $|p| = r$, $a = -r$, $b = r$:

$$(p - a)\overline{(p - b)} = (p + r)(\bar{p} - r) = |p|^2 - rp + r\bar{p} - r^2 = r^2 - r^2 + r(\bar{p} - p) = r(\bar{p} - p).$$

Since $\bar{p} - p = -2i \operatorname{Im}(p)$ is purely imaginary (for every $p \in \mathbb{C}$), so is $(p - a)\overline{(p - b)}$. But for two complex numbers z, w , $z\bar{w}$ is purely imaginary if and only if z and w are perpendicular as vectors (the cosine of the angle between them is $\operatorname{Re}(z\bar{w})/(|z||w|)$, zero exactly when $z\bar{w}$ is purely imaginary): hence the vectors $p - a$ (i.e. $\vec{AP}$) and $p - b$ (i.e. $\vec{BP}$) are perpendicular, that is, $\angle APB = \pi/2$. $\square$

Theorem 4.5: Converse of Thales' theorem. *If $\angle APB = \pi/2$, then P belongs to the circle with diameter $\overline{AB}$. Equivalently: the locus of the points of the plane from which the segment $\overline{AB}$ is seen under a right angle is the circle with diameter $\overline{AB}$, deprived of the points A and B .*

Proof. Let M be the midpoint of $\overline{AB}$ and let $\gamma = C(M, |AB|/2)$ be the circle with diameter $\overline{AB}$. We must show that $\angle APB = \pi/2 \implies P \in \gamma$, that is, $|MP| = |AB|/2$.

Step 1 — the median to the hypotenuse. In the triangle APB , right-angled at P by hypothesis, the segment $\overline{PM}$ is the median to the hypotenuse $\overline{AB}$. Extend PM beyond M by a segment $\overline{MP'} \cong \overline{PM}$, obtaining the point P' .

Step 2 — the quadrilateral $APBP'$ is a parallelogram. Its diagonals $\overline{AB}$ and $\overline{PP'}$ bisect each other at M (by construction M is the midpoint of both): a quadrilateral whose diagonals bisect each other is a parallelogram.

Step 3 — it is a rectangle. A parallelogram with one right angle (here $\angle APB = \pi/2$) has all its angles right, since consecutive angles are supplementary and opposite angles congruent: it is therefore a rectangle.

Step 4 — conclusion. In a rectangle the diagonals are congruent: $|PP'| = |AB|$. But $|PP'| = 2|PM|$ by construction, hence $|PM| = |AB|/2$, that is, $P \in \gamma$. $\square$

Remark 4.4: Why the converse needs a proof of its own. From Theorem 4.4 we know that every point of γ sees $\overline{AB}$ under a right angle; it does not follow that only the points of γ do so. The converse is precisely this second assertion, and it is what turns γ into a **locus**. It is exactly on this point that Construction 2.3 of Chapter 2 and part (b) of Theorem 3.11 rest: without the converse one could not conclude that the points of tangency are only two.

4.6. Cyclic quadrilaterals

Theorem 4.6: Characterisation of cyclic quadrilaterals (Euclid III.22). A convex quadrilateral $ABCD$ is inscriptible in a circle **if and only if** its opposite angles are supplementary:

$$\angle A + \angle C = \pi \quad \text{and} \quad \angle B + \angle D = \pi.$$

Proof. ($\Rightarrow$) Necessary condition. Let $ABCD$ be inscribed in $C(O, r)$. The angles $\angle A = \angle DAB$ and $\angle C = \angle BCD$ are inscribed angles with vertices A and C situated on the two opposite arcs determined by the chord $\overline{BD}$. By Theorem 4.3 they are supplementary. The same argument applied to the chord $\overline{AC}$ gives $\angle B + \angle D = \pi$. (In fact the second relation also follows from the first together with (A9): the sum of the four angles of a convex quadrilateral is 2π .)

($\Leftarrow$) Sufficient condition, via the arc of constant angle. Suppose $\angle A + \angle C = \pi$, that is $\angle BCD = \pi - \angle A =: \varphi$, with $\varphi \in (0, \pi)$ because $\angle A \in (0, \pi)$ (angle of a convex quadrilateral). Let γ be the unique circle through the three vertices A, B, D (Corollary 3.2, not collinear because $ABCD$ is convex). By the part ($\Rightarrow$) already proved, applied to any point D' of the arc of γ with endpoints B, D lying on the side *opposite* to A (with respect to the chord $\overline{BD}$): the quadrilateral $ABD'D$ is cyclic (by construction: $A, B, D', D \in \gamma$, and the position of D' guarantees the convex order), hence $\angle A + \angle BD'D = \pi$, that is $\angle BD'D = \varphi$. Every point of that arc therefore subtends the segment $\overline{BD}$ at angle φ , on the side opposite to A .

By Theorem 4.7 (arc of constant angle, proved later in this chapter in a way entirely independent of this theorem: its proof uses the characterisation of cyclic quadrilaterals at no step), the locus of **all** points of the half-plane opposite to A with respect to BD that subtend $\overline{BD}$ at angle φ is exactly the arc just described. On the other hand C lies in that half-plane (in a convex quadrilateral $ABCD$ the opposite vertices A and C lie on opposite sides of each diagonal, here BD) and subtends $\overline{BD}$ at the angle $\angle BCD = \varphi$ by hypothesis: hence C belongs to that arc, that is $C \in \gamma$. The four vertices are concyclic. $\square$

Remark 4.5: A third route: the cosine rule (Brahmagupta's variant). With $a = AB$, $d = DA$, $b = BC$, $c = CD$: the cosine rule in the triangles ABD and CBD applied to the diagonal BD gives

$$a^2 + d^2 - 2ad \cos A = b^2 + c^2 - 2bc \cos C.$$

If $\hat{A} + \hat{C} = \pi$, that is $\cos C = -\cos A$, this becomes

$$\cos \hat{A} = \frac{a^2 + d^2 - b^2 - c^2}{2(ad + bc)} :$$

a metric identity satisfied by cyclic quadrilaterals, but which is **not by itself a proof of cyclicity** (it does not, on its own, establish that the four points lie on a circle): it is rather a useful metric consequence elsewhere, for instance in the proof of Brahmagupta's formula (Theorem 6.9, Chapter 6), which uses it in exactly this form.

Warning. The convexity hypothesis is essential. For a *crossed* quadrilateral (whose sides cut each other) the characterisation fails: one can have the four vertices on a circle and opposite angles not supplementary, because the very notion of “interior angle” becomes ambiguous. Sources that state the theorem without the adjective “convex” leave the reader with an uncovered case.

Remark 4.6: A whole tradition that does without angles. It is worth observing that the theory of cyclic quadrilaterals was developed independently, and very deeply, in seventh-century Indian mathematics, within a conceptual framework that **does not contain the notion of angle**. Brahmagupta, in the *Brahmasphutasiddhanta* (628 AD), establishes in propositions XII.21–28 the area of the cyclic quadrilateral

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2},$$

and a theorem on its diagonals, without ever speaking of angles and using proportion only in its standard form. The formula, reported in Chapter 6, generalises Heron's and would be recovered in Europe only in the seventeenth century. It is a useful reminder: the choice to found the geometry of the circle upon angles, which seems natural to us, is one historical choice among others.

4.7. The arc of a given angle

The converse of Thales' theorem identifies the locus of the points that see a segment under a *right* angle. The generalisation to an arbitrary angle is one of the most useful tools in the whole of elementary geometry, with direct applications in surveying and navigation.

Theorem 4.7: Arc of a given angle. Given a segment $\overline{AB}$ and an angle $\varphi \in (0, \pi)$, the locus of the points P of the plane, lying in a fixed half-plane relative to the line AB , such that $\angle APB = \varphi$, is a circular arc with endpoints A and B , called the **arc of φ on $\overline{AB}$** . The radius of that circle is

$$R = \frac{|AB|}{2 \sin \varphi}.$$

Considering both half-planes, the complete locus is the union of two arcs symmetric with respect to the line AB .

Proof. Step 1 — construction of the circle. Consider the circle γ through A and B whose central angle $\angle AOB$ (the one standing on the arc situated in the half-plane *opposite* to the fixed one) measures 2φ . Such a circle exists and is unique: its centre O must lie on the perpendicular bisector of $\overline{AB}$ (Corollary 3.1) and, denoting by M the midpoint of $\overline{AB}$, the triangle OMA is right-angled at M with $\angle MOA = \varphi$, whence

$$|OA| = \frac{|MA|}{\sin \varphi} = \frac{|AB|}{2 \sin \varphi} =: R,$$

which determines $|OM|$ and hence O uniquely on the perpendicular bisector, once we choose in which of the two half-planes relative to AB to place it: O must lie on the **opposite** side from the one fixed for the locus if $\varphi < \pi/2$ (then $\angle MOA = \varphi < \pi/2$ is acute and O “sees” AB from the side far from the sought arc, exactly as in Figure 4.4); on the **same** side as the fixed half-plane if $\varphi > \pi/2$ (in that case $\angle MOA = \varphi$ is obtuse, which forces O to the same side as M towards the chosen half-plane); and $O \in \overline{AB}$, that is $M = O$, if $\varphi = \pi/2$ (the case of Thales' theorem, $R = |AB|/2$). In each of the three cases O is uniquely determined by the pair $(|OM|, \text{half-plane})$.

Step 2 — every point of the arc belongs to the locus. Let P be a point of the arc of γ contained in the fixed half-plane. It stands on the opposite arc, whose central angle measures 2φ by construction: by Theorem 4.1, $\angle APB = \varphi$.

Step 3 — no other point belongs to the locus. Let P' be a point of the fixed half-plane with $P' \notin \gamma$, and let P be the point where the ray BP' meets the arc (by an exterior-angle argument,

entirely analogous to the one used elsewhere in the chapter for points interior/exterior to a circle): if P' is inside γ then $\angle AP'B > \angle APB = \varphi$; if it is outside, $\angle AP'B < \varphi$. In either case $\angle AP'B \neq \varphi$, so P' does not belong to the locus.

Step 4 — symmetry. The reflection in the line AB is an isometry fixing A and B and preserving the amplitudes of angles: it therefore maps the locus of one half-plane onto the locus of the other. Whence the second circle, symmetric to the first. $\square$

Remark 4.7: Monotonicity of the angle and limiting cases. Step 3 contains a fact worth isolating: approaching the segment the angle increases, moving away from it the angle decreases. It follows that the arc of φ is the flatter the closer φ is to π , and the wider the closer φ is to 0. The case $\varphi = \pi/2$ gives $R = |AB|/2$: one recovers exactly Theorem 4.5, which is therefore the particular case of the arc of a given angle corresponding to the right angle.

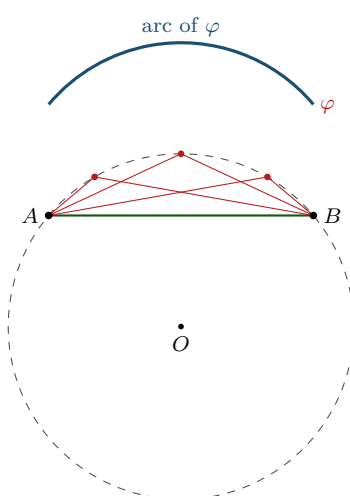


Figure 4.4: — Arc of a given angle: from every point of the bold arc the segment $\overline{AB}$ is seen under the same angle φ . The three marked points give three congruent angles, by Corollary 4.1.

Construction 4.1: — Arc of a given angle. **Data:** the segment $\overline{AB}$ and the angle φ .

Step 1. Draw the perpendicular bisector of $\overline{AB}$ (Corollary 3.1), which will contain the centre.

Step 2. At A construct the ray making an angle φ with AB , on the side opposite to the chosen half-plane.

Step 3. At A draw the perpendicular to that ray: it meets the perpendicular bisector at the centre O sought. (Indeed, starting from a given angle φ and constructing the ray that forms it with AB , it is the *converse* of the tangent–chord angle theorem (Theorem 3.14) that guarantees that ray is the tangent at A to the circle sought; and by Theorem 3.8 the centre lies on the perpendicular to the tangent at A .)

Step 4. Draw the circle with centre O and radius $|OA|$: the arc contained in the chosen half-plane is the arc sought. $\square$

4.7.1 Application: the Pothenot–Snellius problem

Resection from three known points. Problem. An operator stands at an unknown position P and sees three points A, B, C of known coordinates (for instance three benchmarks of a survey network, or three lighthouses). Only the two *angles* $\alpha = \angle APB$ and $\beta = \angle BPC$ are measured, with no distance measurement available. Determine P .

Step 1. By Theorem 4.7, the locus of the points seeing $\overline{AB}$ under the angle α is an arc γ_1 , constructible by Construction 4.1.

Step 2. Likewise the locus of the points seeing $\overline{BC}$ under the angle β is an arc γ_2 .

Step 3. The point sought belongs to both loci: $P \in \gamma_1 \cap \gamma_2$. Since both arcs pass through B , their intersection contains B and (in general) exactly one other point: that is P .

The critical case. If A, B, C and P turn out to be **concylic**, then γ_1 and γ_2 coincide and the intersection is a whole arc: the problem is *indeterminate*. This is the celebrated *danger circle* of surveying: in field practice one chooses the triple of benchmarks so that the station is far from the circle circumscribed to the three points. $\square$

Example 4.1: Pothenot–Snellius resection: a full computation. Data. Three known control points (in metres)

$$A = (0; 0), \quad B = (120; 0), \quad C = (60; 100),$$

and two angles measured from the unknown station P :

$$\alpha = \angle APB = 126.870^\circ, \quad \beta = \angle BPC = 116.565^\circ.$$

No distance has been measured.

Step 1 — the first arc of constant angle, γ_1 on $\overline{AB}$. By the extended law of sines (Theorem 10.5, Chapter 10), the radius of the arc on a chord ℓ under the angle ϑ is $R = \ell/(2 \sin \vartheta)$. Here $\ell = |AB| = 120$ and $\sin 126.870^\circ = \sin 53.130^\circ = 0.800$, hence

$$R_1 = \frac{120}{2 \cdot 0.800} = 75.000 \text{ m.}$$

Step 2 — the centre of γ_1 . It lies on the perpendicular bisector of $\overline{AB}$, i.e. on the line $x = 60$, at distance

$$d_1 = \sqrt{R_1^2 - (|AB|/2)^2} = \sqrt{5625 - 3600} = 45$$

from the chord. Since $\alpha > 90^\circ$, the point P lies on the *minor* arc and the centre is on the opposite side of AB : assuming P in the half-plane $y > 0$,

$$O_1 = (60; -45), \quad \gamma_1 : (x - 60)^2 + (y + 45)^2 = 5625.$$

Check: $|O_1A| = \sqrt{3600 + 2025} = 75$. $\checkmark$

Step 3 — the second arc, γ_2 on $\overline{BC}$. $|BC| = \sqrt{60^2 + 100^2} = 116.6190$ m and $\sin 116.565^\circ = \sin 63.435^\circ = 0.894427$, hence

$$R_2 = \frac{116.6190}{2 \cdot 0.894427} = 65.1920 \text{ m}, \quad d_2 = \sqrt{4250 - 3400} = \sqrt{850} = 29.1548.$$

The midpoint of BC is $M = (90; 50)$ and the unit normal to BC is $\hat{n} = (100; 60)/116.6190 = (0.857493; 0.514496)$. The two candidates are $M \pm d_2 \hat{n}$, i.e. $(115; 65)$ and $(65; 35)$; the first is the one on the opposite side from P (again because $\beta > 90^\circ$):

$$O_2 = (115; 65), \quad \gamma_2 : (x - 115)^2 + (y - 65)^2 = 4250.$$

Check: $|O_2B| = \sqrt{25 + 4225} = 65.192$. ✓

Step 4 — the radical axis. Expanding the two equations:

$$\gamma_1 : x^2 + y^2 - 120x + 90y = 0, \quad \gamma_2 : x^2 + y^2 - 230x - 130y + 13200 = 0.$$

Subtracting (the quadratic terms cancel — this is Proposition 7.1 of Chapter 7):

$$110x + 220y - 13200 = 0 \iff x + 2y = 120.$$

Step 5 — solving. Substituting $x = 120 - 2y$ into γ_1 :

$$14400 - 480y + 4y^2 + y^2 - 14400 + 240y + 90y = 0 \iff 5y(y - 30) = 0.$$

The two solutions are $y = 0$, giving $x = 120$, i.e. the point B — common to both arcs by construction and hence to be discarded — and $y = 30$, giving $x = 60$:

$$P = (60; 30) \text{ m}$$

Step 6 — checking backwards. Recompute the angles from P :

$$\vec{PA} = (-60; -30), \quad \vec{PB} = (60; -30), \quad \vec{PC} = (0; 70),$$

$$\cos \alpha = \frac{-3600 + 900}{67.082 \times 67.082} = -0.600 \Rightarrow \alpha = 126.870^\circ, \checkmark$$

$$\cos \beta = \frac{0 - 2100}{67.082 \times 70} = -0.447214 \Rightarrow \beta = 116.565^\circ. \checkmark$$

Step 7 — checking the “dangerous circle”. We verify that P is not too close to the circle through A, B, C . Setting $x^2 + y^2 + Dx + Ey + F = 0$ and substituting the three points (Theorem 7.2, Chapter 7):

$$A : F = 0; \quad B : 14400 + 120D = 0 \Rightarrow D = -120;$$

$$C : 3600 + 10000 - 7200 + 100E = 0 \Rightarrow E = -64.$$

The critical circle is thus $x^2 + y^2 - 120x - 64y = 0$, with centre $(60; 32)$ and radius $\sqrt{4624} = 68.0$ m. The distance of $P = (60; 30)$ from that centre is 2 m, against a radius of 68.0 m: P is well *inside* and 66 m away from the critical circle. The configuration is therefore robust: repeating Steps 1–5 in full with α and β both perturbed by $+0.1^\circ$ (independent script) gives $P' = (60.1251; 29.9344)$, i.e. a displacement $|PP'| = 0.141$ m — of the order of a decimetre, not a metre (perturbing a single angle at a time, the displacement is even smaller, about 0.08 m).

Remark 4.8: Why this is an application of the inscribed angle, not of trigonometry. Resection can certainly be solved with trigonometry, but its structure is purely projective: it depends only on angles, not on distances, and that is why it works with instruments (theodolite, protractor, bearing finder) that measure directions only. That the locus is an arc — and not, say, a line or a general conic — is exactly the content of Theorem 4.1. The critical case of the danger circle is, for the same reason, a direct consequence of Corollary 4.1: if the station lies on the circle circumscribed to the three benchmarks, all the points of that arc give the same angular readings, and no angle measurement will ever tell them apart.

4.8. The Simson–Wallace line

Theorem 4.8: Simson–Wallace line. Let P be a point of the circle circumscribed about triangle ABC , and let D, E, F be the feet of the perpendiculars from P to BC, CA, AB respectively. Then D, E, F are collinear.

Proof. Fix the configuration in which P lies on the arc BC not containing A , with F interior to side AB , E on the extension of CA beyond C , D interior to BC (the other configurations are handled with directed angles modulo 180° , and are imported from Coxeter–Greitzer, *Geometry Revisited*, Theorem 2.51).

Step 1. P, D, B, F are concyclic: the angles at D and at F are right angles (by construction, being feet of perpendiculars), so both subtend the segment $\overline{PB}$ at a right angle, and by the converse of Thales (Theorem 4.5) they lie on the circle of diameter $\overline{PB}$.

Step 2. P, D, C, E are concyclic, by the same argument with diameter $\overline{PC}$.

Step 3. E and F lie on opposite sides of the line BC (by the fixed configuration: F is between A and B , on the same side as A ; E is beyond C from A on the line CA , hence on the opposite side). Hence D, E, F are collinear if and only if the angles opposite at the vertex $\angle BDF$ and $\angle CDE$ are equal.

Step 4. In the circle of Step 1, $\angle BDF = \angle BPF$ (inscribed angles on the same arc BF , with D and P on the same side of BF); and in the right triangle PFB (right-angled at F), $\angle BPF = 90^\circ - \angle PBF$.

Step 5. In the circle of Step 2, $\angle CDE = \angle CPE$, and in the right triangle PEC (right-angled at E), $\angle CPE = 90^\circ - \angle PCE$; and $\angle PCE = 180^\circ - \angle PCA$ (since E is beyond C on the line CA , the angles $\angle PCE$ and $\angle PCA$ are supplementary).

Step 6. The quadrilateral $ABPC$ is cyclic, with A and P opposite vertices (by the fixed configuration): by Theorem 4.6, $\angle PBA + \angle PCA = 180^\circ$, that is $\angle PCA = 180^\circ - \angle PBA = 180^\circ - \angle PBF$ (since F is on AB). Hence $\angle PCE = 180^\circ - \angle PCA = \angle PBF$, and comparing Steps 4 and 5:

$$\angle CDE = 90^\circ - \angle PCE = 90^\circ - \angle PBF = \angle BDF.$$

By Step 3, D, E, F are collinear. □

Example 4.2: Numerical check. $A = (0, 1)$, $B = (\cos 210^\circ, \sin 210^\circ)$, $C = (\cos 330^\circ, \sin 330^\circ)$, $P = (\cos 300^\circ, \sin 300^\circ)$: $D = (0.5; -0.5)$, $E = (0.9330; -0.6160)$, $F = (-0.6830; -0.1830)$ (feet of the perpendiculars, computed by orthogonal projection). The 3×3 determinant of the three points in homogeneous coordinates equals ≈ 0 (checked, independent script, order 10^{-16}): they are collinear.

Remark 4.9: History and generality. The line conventionally bears the name of Robert Simson, but the first documented attribution is to William Wallace (1799); the general statement holds for every position of P on the circle (not only the fixed configuration above), using directed angles modulo 180° : it is this general form that we import from Coxeter–Greitzer.

4.9. Miquel’s point

Theorem 4.9: Miquel’s point. Given a triangle ABC and three points $D \in BC$, $E \in CA$, $F \in AB$ (interior to the sides), the three circles AEF , BFD , CDE have a point in common.

Proof. Consider the configuration in which the second common point M of the circles BFD and CDE (besides D) is interior to the triangle (the other configurations are handled with directed angles and are imported from Coxeter–Greitzer, *Geometry Revisited*, §2.5). In the cyclic quadrilaterals $BFMD$ and $CDME$ (in this cyclic order):

$$\angle FMD = 180^\circ - \hat{B}, \quad \angle DME = 180^\circ - \hat{C}$$

(Theorem 4.6, applied to the pairs of opposite vertices B, M and C, M). Since D lies on the segment BC , the angles $\angle FMD$, $\angle DME$ and $\angle FME$ (the third, on the opposite side) are arranged around M so that

$$\angle FME = 360^\circ - \angle FMD - \angle DME = 360^\circ - (180^\circ - \hat{B}) - (180^\circ - \hat{C}) = \hat{B} + \hat{C} = 180^\circ - \hat{A}.$$

Hence $AFME$ is also a cyclic quadrilateral (opposite angles A, M supplementary, by the converse of Theorem 4.6): M also belongs to the circle AEF . The three circles therefore pass through the same point M . $\square$

Example 4.3: Numerical check. On the triangle $A = (0, 0)$, $B = (6, 0)$, $C = (2, 5)$, with D, E, F the midpoints of the sides: the circles circumscribed about AEF , BFD , CDE (computed as circles through three points, independent script) have three different centres but all three pass through the circumcentre $O = (3; 1.7)$ of ABC (distance $|O - \text{centre}| = \text{radius} = 1.724094$ in all three cases, checked to six decimal places): in the case of the midpoints, Miquel's point coincides with the circumcentre of ABC , a known special case.

4.10. Regiomontanus's problem

Theorem 4.10: Regiomontanus's problem. *A painting hangs on a vertical wall, with its lower edge at height a and its upper edge at height $b > a$ relative to the eye level of an observer who walks away along a horizontal line perpendicular to the wall. The distance x from the wall at which the viewing angle subtended by the painting is greatest equals*

$$x = \sqrt{ab}.$$

Proof. Let A, B be the two edges (at heights a, b above eye level) on the wall, and let ℓ be the horizontal line at eye level, at distance x from the wall, along which the observer moves. The points of ℓ that subtend $\overline{AB}$ at a fixed angle φ lie on an arc of constant angle φ on $\overline{AB}$ (Theorem 4.7); by Observation 4.7, the angle increases when approaching the segment and decreases when moving away from it. The observer, moving along ℓ , therefore sees the greatest angle at the position where the arc of constant angle through that point is tangent to ℓ (at any other position on ℓ , the circle through A, B passing through that point would cross ℓ elsewhere too, and the points between the two intersections subtend $\overline{AB}$ at a *wider* arc of constant angle, that is, a larger angle — see Theorem 4.7, Step 3, applied to the point on the outermost arc). Let T be the point of tangency between ℓ and that circle, and let Q be the point where the wall itself meets ℓ (the foot of the perpendicular to the wall from eye level). The wall, containing A and B , is a secant of the circle through Q , with $QA = a$, $QB = b$ (distances along the wall, since Q is at eye level and A, B are above it on the same side); ℓ , tangent to the circle at T , gives $QT = x$ (distance between the wall and the observer's position, along ℓ). By Theorem 3.19 (power of the point Q , equal for the secant and for the tangent):

$$QT^2 = QA \cdot QB \implies x^2 = ab \implies x = \sqrt{ab}.$$

$\square$

Example 4.4: Numerical check. $a = 2$, $b = 5$: $x = \sqrt{10} = 3.162278$ (checked by exhaustive search for the maximum angle on a fine grid of values of x , independent script).

4.11. Angles among chords, secants and tangents

Theorem 4.11: Angle formed by two chords meeting inside. *If two chords AB , CD meet at a point P interior to the circle, the angle $\angle APC$ is half the sum of the intercepted arcs:*

$$\angle APC = \frac{1}{2}(\mu(\widehat{AC}) + \mu(\widehat{BD})).$$

Proof. Draw the auxiliary chord $\overline{AD}$, forming the triangle APD . The angle $\angle APC$ is exterior to that triangle at the vertex P (because C lies on the extension of DP beyond P , P being interior to the chord CD). By the exterior angle theorem,

$$\angle APC = \angle PDA + \angle DAP = \angle CDA + \angle DAB.$$

Now $\angle CDA$ is an inscribed angle with vertex D standing on the arc $\widehat{AC}$ not containing D , hence equal to $\frac{1}{2}\mu(\widehat{AC})$ by Theorem 4.1; and $\angle DAB$ is inscribed with vertex A and stands on the arc $\widehat{BD}$ not containing A , hence equal to $\frac{1}{2}\mu(\widehat{BD})$. Adding gives the conclusion. $\square$

Alternative proof — by similar triangles. Consider the triangles PAD and PCB , obtained by drawing $\overline{AD}$ and $\overline{CB}$. They have $\angle APD = \angle CPB$ (vertically opposite angles) and $\angle ADP = \angle ADC = \angle ABC = \angle PBC$ (inscribed angles standing on the same arc $\widehat{AC}$, Corollary 4.1). By the AA criterion (A6) the two triangles are similar, whence in particular

$$\frac{PA}{PC} = \frac{PD}{PB} \implies PA \cdot PB = PC \cdot PD,$$

that is, the power-of-a-point relation (Theorem 3.7), recovered here as a by-product.

For the measure of the angle: in the triangle PAD the sum of the interior angles gives $\angle APD = \pi - \angle PAD - \angle PDA$, where $\angle PAD = \angle BAD$ stands on the arc $\widehat{BD}$ and $\angle PDA = \angle CDA$ stands on the arc $\widehat{AC}$. Hence

$$\angle APD = \pi - \frac{1}{2}\mu(\widehat{BD}) - \frac{1}{2}\mu(\widehat{AC}),$$

and since $\angle APC$ and $\angle APD$ are adjacent (supplementary),

$$\angle APC = \pi - \angle APD = \frac{1}{2}\mu(\widehat{AC}) + \frac{1}{2}\mu(\widehat{BD}),$$

the same result. $\square$

Theorem 4.12: Angle formed by two secants (or tangents) from an external point. *If from an external point P two secants (or two tangents, or a secant and a tangent) are drawn, intercepting on the circle a far arc and a near arc, then*

$$\angle P = \frac{1}{2}(\mu_{\text{far}} - \mu_{\text{near}}).$$

Proof. Case of two secants PAB and PCD , with A, C near P and B, D far (order P, A, B and P, C, D on each line). Draw the auxiliary chord $\overline{AD}$, obtaining the triangle PAD .

Angle at A. The points P, A, B are collinear with A between P and B , hence $\angle PAD$ and $\angle DAB$ are adjacent: $\angle PAD = \pi - \angle DAB$. But $\angle DAB$ is inscribed with vertex A and stands on the arc $\widehat{BD}$ not containing A , that is, the *far* arc, hence $\angle DAB = \frac{1}{2}\mu_{\text{far}}$. Therefore

$$\angle PAD = \pi - \frac{1}{2}\mu_{\text{far}}.$$

Angle at D. Since C lies between P and D , the ray DP coincides with the ray DC : hence $\angle ADP = \angle ADC$, an inscribed angle with vertex D standing on the arc $\widehat{AC}$ not containing D , that is, the *near* arc. Hence

$$\angle ADP = \frac{1}{2}\mu_{\text{near}}.$$

Conclusion. By (A9) applied to the triangle PAD :

$$\begin{aligned}\angle P &= \pi - \angle PAD - \angle ADP = \pi - \left(\pi - \frac{1}{2}\mu_{\text{far}}\right) - \frac{1}{2}\mu_{\text{near}} \\ &= \frac{1}{2}\mu_{\text{far}} - \frac{1}{2}\mu_{\text{near}}.\end{aligned}$$

Case of a secant and a tangent. Let PAB be the secant (A near, B far) and PT the tangent, with point of tangency T . Draw the auxiliary chord $\overline{AT}$, obtaining the triangle PAT .

Angle at A. As above, $\angle PAT = \pi - \angle TAB$, and $\angle TAB$ is inscribed with vertex A and stands on the arc $\widehat{TB}$ not containing A , that is, the *far* arc: $\angle TAB = \frac{1}{2}\mu_{\text{far}}$. Hence $\angle PAT = \pi - \frac{1}{2}\mu_{\text{far}}$.

Angle at T. The angle $\angle PTA$ is formed by the tangent PT and the chord $\overline{TA}$: by Theorem 3.13 (tangent–chord angle) it is congruent to the inscribed angle in the alternate segment standing on the arc $\widehat{TA}$ on the side “enclosed” by the angle itself, that is, the *near* arc (the one between T and A not containing B): $\angle PTA = \frac{1}{2}\mu_{\text{near}}$.

Conclusion. By (A9) applied to the triangle PAT :

$$\angle P = \pi - \angle PAT - \angle PTA = \pi - \left(\pi - \frac{1}{2}\mu_{\text{far}}\right) - \frac{1}{2}\mu_{\text{near}} = \frac{1}{2}\mu_{\text{far}} - \frac{1}{2}\mu_{\text{near}},$$

the same result, with the same proof structure as the two-secants case (exterior angle + inscribed/tangent-chord angle + angle sum of the triangle), not by substitution but by an autonomous proof.

Case of two tangents. Let PT_1, PT_2 be the two tangents, with points of tangency T_1, T_2 . Draw the auxiliary chord $\overline{T_1T_2}$, obtaining the triangle PT_1T_2 . By Theorem 3.13 applied at T_1 and at T_2 , each of the angles $\angle PT_1T_2$ and $\angle PT_2T_1$ (between the respective tangent and the chord T_1T_2) is congruent to the inscribed angle in the corresponding alternate segment, standing on the *near* arc $\widehat{T_1T_2}$ (the minor one, enclosed by the two tangents towards P) in both cases: $\angle PT_1T_2 = \angle PT_2T_1 = \frac{1}{2}\mu_{\text{near}}$. By (A9) applied to the triangle PT_1T_2 :

$$\angle P = \pi - \angle PT_1T_2 - \angle PT_2T_1 = \pi - \mu_{\text{near}} = \frac{1}{2}(2\pi - \mu_{\text{near}} - \mu_{\text{near}}) = \frac{1}{2}(\mu_{\text{far}} - \mu_{\text{near}}),$$

using $\mu_{\text{far}} + \mu_{\text{near}} = 2\pi$ (full angle). This particular case recovers, by another route, Theorem 3.15 of Chapter 3. $\square$

Alternative proof — by the exterior angle of triangle PAD . Observe that $\angle APC = \angle APD$ (the same angle: C lies between P and D , so the rays PC and PD coincide). In triangle PAD , the angle $\angle BAD$ (between the extension of side PA beyond A and side AD ; B lies on that extension because A is between P and B) is the **exterior** angle at A : by the exterior angle theorem, $\angle BAD = \angle APD + \angle ADP$, that is,

$$\angle APD = \angle BAD - \angle ADP.$$

Now $\angle BAD$ is inscribed with vertex A and stands on the arc $\widehat{BD}$ not containing A (the *far* arc): $\angle BAD = \frac{1}{2}\mu_{\text{far}}$. And $\angle ADP = \angle ADC$ (the same angle, since P and C lie on the same ray from D) is inscribed with vertex D and stands on the arc $\widehat{AC}$ not containing D (the *near* arc): $\angle ADC = \frac{1}{2}\mu_{\text{near}}$. Hence

$$\angle P = \angle APC = \angle APD = \frac{1}{2}\mu_{\text{far}} - \frac{1}{2}\mu_{\text{near}},$$

the same conclusion, with a single triangle instead of two. $\square$

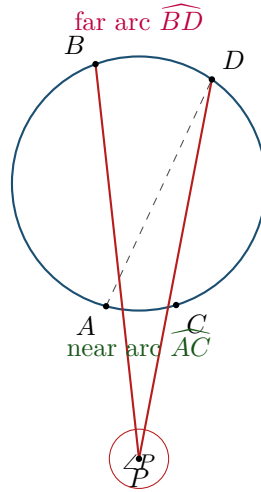


Figure 4.5: — Two secants from an external point: the angle at P is half the difference of the far and near arcs

vertex at the centre:	$\angle = \mu$
vertex on the circle:	$\angle = \frac{1}{2}\mu$
vertex inside:	$\angle = \frac{1}{2}(\mu_1 + \mu_2)$
vertex outside:	$\angle = \frac{1}{2}(\mu_{\text{far}} - \mu_{\text{near}})$

Remark 4.10: One law, four positions. The four formulas in the box are not four rules to be memorised: they are the same law read at four positions of the vertex. Moving the vertex from the centre outwards, the angle passes from μ to $\mu/2$ (on the circle), and beyond that it keeps decreasing, because the near arc begins to be subtracted. The formula for the interior vertex and the one for the exterior vertex are in fact the same: it suffices to agree that, for the exterior vertex, the near arc counts with a negative sign. The case of the vertex on the circle is the boundary between the two: there the near arc shrinks to a point, of zero measure, and both formulas collapse into $\mu/2$.

4.12. Historical notes

Remark 4.11: Thales, Euclid and historiographical caution. Theorem 4.4 is traditionally attributed to Thales of Miletus (6th century BC). The attribution must, however, be taken with caution: already in the 5th century BC nothing survived of Thales' writings, and the inventions were assigned to him by later doxographers on the basis of indirect testimony;

some ancient sources attribute it to Pythagoras. What is documented with certainty is the Euclidean formulation: *Elements* III.31, which in fact states three assertions — the angle in a semicircle is right, the one in a greater segment is acute, the one in a lesser segment is obtuse — of which the didactic tradition has preserved only the first. The other two are exactly the content of Corollary 4.1 and of Theorem 4.2 of this chapter.

Remark 4.12: The Persian route of the *Elements*. Book III of the *Elements* reached Europe largely through the Arabic and Persian tradition. The *Tahrir Usul Uqlidis* of Nasir al-Din al-Tusi (1201–1274), an annotated redaction of the *Elements*, is one of the most frequently reprinted mathematical texts of the premodern era and was one of the main channels of transmission. Recalling this is not erudition: it means recognising that the chain of custody of the theorems of this chapter passes through Baghdad and Maragheh before Padua and Oxford.

4.13. Typical errors

Research in mathematics education has documented some recurrent errors on this chapter. Listing them is not pedantry: they are the same ones that appear in tests.

Warning. E1 — Doubling instead of halving. By far the commonest error: given the central angle, the student multiplies it by two instead of dividing. Antidote: remember that the central angle is the *larger* of the two, because its vertex is the nearer to the chord.

E2 — Getting the arc wrong. Applying the theorem to the arc *containing* the vertex instead of the one not containing it: one obtains the supplement of the correct answer. Antidote: Definition 4.2, and the plausibility check on whether the angle is obtuse or acute.

E3 — Ignoring the reflex angle. When the vertex falls on the minor arc, the relevant central angle is the reflex one (Theorem 4.2). Students tend to use the convex angle anyway, obtaining an acute result where the answer is obtuse. Antidote: Exercise 4.2.

E4 — Confusing the measure of an arc with its length. The former is in radians and does not depend on the radius, the latter is in metres and equals $r\theta$. On different circles, congruent arcs have different lengths (Chapter 2, Section 2.9).

E5 — Applying the cyclic characterisation to non-convex quadrilaterals. The convexity hypothesis in Theorem 4.6 is not ornamental.

4.14. Numerical example: angular resolution of a rotary encoder

Sizing the angular resolution. Data. An incremental rotary encoder, mounted on a PLC axis, supplies $N = 4096$ pulses per full revolution (typical for a 12-bit encoder).

Step 1 (angular resolution). The central angle subtended by a single pulse is

$$\Delta\theta = \frac{360^\circ}{N} = \frac{360^\circ}{4096} \approx 0.0879^\circ.$$

Step 2 (mechanical reducer). If the encoder is connected, through a reducer of ratio $i = 50:1$, to a robotic arm, the angular resolution *at the reducer output* is $\Delta\theta/i \approx 0.00176^\circ$: by the formula $\ell = r\theta$ (Chapter 6), each pulse therefore corresponds to an extremely small subtended arc on the circular path described by the arm's end.

Step 3 (linear precision at a given radius). For an arm of operating radius $r_b = 800$

mm, the linear positioning error corresponding to one pulse is

$$\ell = r_b \cdot \frac{\Delta\theta}{i} \cdot \frac{\pi}{180^\circ} \approx 800 \times 0.00176 \times \frac{\pi}{180} \approx 0.0246 \text{ mm},$$

that is, about 25 μm : a typical precision, sufficient for most industrial positioning applications. $\square$

4.15. Open problems

As in Chapter 3, we close with questions that are stated with the tools seen here alone and that remain, wholly or partly, unanswered.

4.15.1 The inscribed rectangle problem

Remark 4.13: Statement. Question. *Given an arbitrary Jordan curve in the plane and an arbitrary rectangle, are there always four points of the curve that are the vertices of a rectangle similar to the given one?*

For the circle the answer is immediate and uses exactly this chapter: taking two points A, B and their antipodes A', B' , the quadrilateral $ABA'B'$ has diagonals that are diameters, hence congruent and bisecting each other: it is a rectangle (Steps 2–4 of the proof of Theorem 4.5). Varying the angle between the two diameters yields every aspect ratio. The circle therefore contains inscribed rectangles of *every* shape.

Outline of the solution strategy Step 1. To every unordered pair of points of the curve one associates a point of an auxiliary space, encoding its midpoint and its distance.

Step 2. An inscribed rectangle corresponds to a pair of pairs with the same midpoint and the same distance: that is, to a *self-intersection* point of a certain embedding.

Step 3. One shows that, if such self-intersections did not exist, one would obtain a smooth Lagrangian embedding of a non-orientable surface in $\mathbb{C}^2$.

Step 4. One invokes the theorem of Shevchishin and Nemirovski: the Klein bottle does **not** admit a smooth Lagrangian embedding in $\mathbb{C}^2$. Contradiction, hence the rectangle exists.

Where it breaks down. Step 3 requires the curve to be **smooth**: for a merely continuous Jordan curve the object constructed is not a submanifold and the symplectic argument does not apply. $\square$

BEST PRACTICE

Status: SOLVED in the smooth case (2021), OPEN in general. Joshua Evan Greene and Andrew Lobb proved (*Annals of Mathematics* 194, 2021) that for every **smooth** Jordan curve and every prescribed rectangle there exists an inscribed rectangle similar to the given one. Earlier partial progress is due to Hugelmeyer (2019), who had covered a set of aspect ratios of Lebesgue measure at least $\pi/6$. Greene and Lobb themselves later extended the result to the inscribability of **cyclic quadrilaterals** in smooth curves — a direct link with Theorem 4.6 of this chapter. **For merely continuous Jordan curves the problem remains open**, as does Toeplitz’s square peg problem (Chapter 3).

4.15.2 The area of cyclic polygons

Remark 4.14: Statement. Question. Heron's formula gives $(4A)^2$ as a polynomial in the squares of the sides for the triangle; Brahmagupta's does the same for the cyclic quadrilateral. Is there, for every n , an analogous polynomial for the cyclic n -gon? And what is its degree?

Warning. Status: partly solved, open for $n \geq 9$. David P. Robbins showed that for the cyclic pentagon and hexagon the minimal polynomial satisfied by $(4A)^2$ has **degree 7**. For the heptagon and octagon the minimal equation has been obtained using covariants of binary quintics, as a quotient of two resultants: in expanded form it has **almost a million terms**. Robbins' original method, applied to the heptagon, would lead to a system of **143,307 equations**. It is **not known** whether the approach extends effectively to cyclic polygons with nine or more sides.

INFO

The thread linking the two problems to this chapter. Both start from an elementary fact proved here — the circle contains inscribed rectangles of every shape; the area of a cyclic quadrilateral depends only on its sides — and ask what survives when the circle is removed or sides are added. In one case the answer required symplectic geometry, in the other formal computation over millions of terms. It is the same phenomenon already observed in Chapter 3: the difficulty of a problem cannot be read off its statement.

4.16. Problem set

4.16.1 Numerical exercises

Exercise 4.1. In a circle, a central angle measures 130° . What is the inscribed angle standing on the same arc? And the inscribed angle on the opposite arc?

Solution. Inscribed angle on the same arc: $130^\circ/2 = 65^\circ$ (Theorem 4.1). Inscribed angle on the opposite arc: supplementary by Theorem 4.3, hence $180^\circ - 65^\circ = 115^\circ$.

Exercise 4.2. A convex central angle measures 130° . The vertex of an inscribed angle falls on the **minor arc**, that is, on the arc intercepted by that central angle. What is the inscribed angle?

Solution. Careful: the vertex lies on the minor arc, so the inscribed angle stands on the *major* arc, to which the **reflex** central angle corresponds:

$$\mu_{\text{major}} = 360^\circ - 130^\circ = 230^\circ \implies \angle = \frac{230^\circ}{2} = 115^\circ.$$

This is Theorem 4.2. The consistency check is immediate: the angle turns out obtuse, as it must for a vertex on the minor arc, and it is supplementary to 65° , the angle obtained with the vertex on the major arc (Theorem 4.3). Anyone mechanically applying $130^\circ/2 = 65^\circ$ commits error E3 of Section 4.13.

Exercise 4.3. A convex quadrilateral $ABCD$ is inscribed in a circle with $\angle A = 70^\circ$. What is $\angle C$? And if $\angle A = 75^\circ$?

Solution. By Theorem 4.6 the opposite angles of a convex cyclic quadrilateral are supplementary: $\angle C = 180^\circ - 70^\circ = 110^\circ$. With $\angle A = 75^\circ$ one gets $\angle C = 180^\circ - 75^\circ = 105^\circ$.

Exercise 4.4. Two tangents drawn from an external point P intercept a minor arc of 80° . Compute the angle $\angle P$ between the two tangents.

Solution. The major arc measures $360^\circ - 80^\circ = 280^\circ$. By Theorem 4.12 in the limiting case of two tangents:

$$\angle P = \frac{1}{2}(280^\circ - 80^\circ) = \frac{1}{2} \cdot 200^\circ = 100^\circ.$$

Independent check. By Theorem 3.15(a) of Chapter 3, $\angle P$ and the central angle are supplementary: $\angle P = 180^\circ - 80^\circ = 100^\circ$. The two methods agree.

Exercise 4.5. In a circle of radius $R = 10$ cm, an inscribed angle measures $\beta = 30^\circ$. How long is the chord it subtends?

Solution. The corresponding central angle is $\alpha = 2\beta = 60^\circ$ (Theorem 4.1). The chord follows from the formula of Chapter 2, Section 2.9:

$$|AB| = 2R \sin(\alpha/2) = 2 \cdot 10 \cdot \sin 30^\circ = 20 \cdot \frac{1}{2} = 10 \text{ cm}.$$

Remark. The triangle OAB has $|OA| = |OB| = R = 10$ and vertex angle 60° : it is therefore **equilateral**, and the side equals 10 cm — an independent confirmation. Note also the general formula $|AB| = 2R \sin \beta$, which links the chord directly to the *inscribed* angle and is the extended form of the law of sines (Theorem 10.5, Chapter 10).

Exercise 4.6. From an external point P two secants intercept on the circle a far arc of 150° and a near arc of 40° . Compute $\angle P$. Then, with the same arcs, compute the angle that would be obtained if the vertex were interior to the circle.

Solution. Exterior vertex (Theorem 4.12):

$$\angle P = \frac{1}{2}(150^\circ - 40^\circ) = 55^\circ.$$

Interior vertex (Theorem 4.11):

$$\angle P = \frac{1}{2}(150^\circ + 40^\circ) = 95^\circ.$$

The difference between the two results is $\mu_{\text{near}} = 40^\circ$: it is exactly the contribution of the near arc, which enters with a $+$ sign from inside and with a $-$ sign from outside, as observed in Remark 4.10.

4.16.2 Symbolic exercises and proofs

Exercise 4.7. Prove that every right triangle is inscriptible in a circle having the hypotenuse as diameter, and that the circumradius equals half the hypotenuse.

Solution. Let APB be right-angled at P , with hypotenuse $\overline{AB}$. By Theorem 4.5 (converse of Thales), the locus of the points seeing $\overline{AB}$ under a right angle is the circle γ with diameter

$\overline{AB}$: since $\angle APB = \pi/2$ by hypothesis, $P \in \gamma$. Also A and B belong to γ (they are the endpoints of the diameter). Hence the three vertices of the triangle lie on γ : the triangle is inscribed in γ , whose diameter is $\overline{AB}$ and whose radius is therefore $|AB|/2$.

Note on method. The essential step is the use of the *converse*, not of the direct theorem: from Thales we know that the points of γ see AB under a right angle, but to conclude that P lies on γ we need to know that no other point has that property — which is precisely the content of the converse (Remark 4.4).

Exercise 4.8. Prove that the radius of the arc of an angle φ on the segment $\overline{AB}$ equals $R = |AB|/(2 \sin \varphi)$, and discuss what happens as $\varphi \rightarrow 0^+$ and as $\varphi \rightarrow \pi^-$.

Solution. Let γ be the circle of the arc and $\alpha = 2\varphi$ the corresponding central angle (Theorem 4.1). By the chord formula (Chapter 2, Section 2.9),

$$|AB| = 2R \sin(\alpha/2) = 2R \sin \varphi \implies R = \frac{|AB|}{2 \sin \varphi}.$$

Limits. As $\varphi \rightarrow 0^+$ we have $\sin \varphi \rightarrow 0^+$ and hence $R \rightarrow +\infty$: the arc degenerates into the line AB deprived of the segment $\overline{AB}$ — and indeed from a very distant point, collinear with A and B , the segment is seen under an almost zero angle. As $\varphi \rightarrow \pi^-$ again $\sin \varphi \rightarrow 0^+$ and $R \rightarrow +\infty$: the arc degenerates into the open segment $\overline{AB}$ itself, from whose interior points the segment is seen under a straight angle. The radius is **minimal** for $\varphi = \pi/2$, where it equals $|AB|/2$: this is Thales' circle, the smallest possible arc on $\overline{AB}$.

Exercise 4.9. Let $ABCD$ be a cyclic quadrilateral. Prove, using only the results of this chapter, that the triangles ABC and ADC have congruent circumcircles, and deduce that $\frac{|BD|}{\sin \angle BAD}$ does not depend on which of the four vertices is chosen as the vertex of the angle.

Solution. The triangles ABC and ADC are both inscribed in the *same* circle (the one containing the four vertices), so their circumcircles coincide with it and are a fortiori congruent. For the second part: consider the chord $\overline{BD}$ and a vertex A on the arc not containing it. The angle $\angle BAD$ is inscribed and stands on the arc $\widehat{BD}$ not containing A ; letting α be the corresponding central angle, $\angle BAD = \alpha/2$ and, by the chord formula,

$$|BD| = 2R \sin(\alpha/2) = 2R \sin \angle BAD \implies \frac{|BD|}{\sin \angle BAD} = 2R,$$

where R is the radius of the circle. The right-hand side does not depend on the chosen vertex, only on the circle: this is the law of sines in extended form. If C is chosen instead of A , and C lies on the opposite arc, then $\angle BCD = \pi - \angle BAD$ (Theorem 4.3) and $\sin(\pi - x) = \sin x$: the ratio is again $2R$. The quantity is thus invariant even on changing arc.

Exercise 4.10. (*On rigour in proofs by cases.*) In the proof of Theorem 4.1 three cases are distinguished according to the position of the diameter PD . Prove that the three cases are mutually exclusive and exhaustive, and explain why without this check the proof would be incomplete.

Solution. Given P and the chord $\overline{AB}$, the point D (antipode of P) is uniquely determined and either belongs to the arc $\widehat{AB}$ stood upon or does not. Relative to the two points A and B exactly one of three situations occurs: (i) D coincides with A or with B ; (ii) D is interior to the arc $\widehat{AB}$ stood upon; (iii) D is exterior to that arc. They are mutually exclusive because a point cannot simultaneously coincide with an endpoint, be interior and be exterior; they are exhaustive because the position of D relative to a closed arc is, by trichotomy, one of those three. The three situations correspond respectively to Cases 1, 2, 3.

Why it matters. A proof by cases establishes the conclusion only on the union of the cases treated. If the cases did not exhaust the possibilities, there would remain configurations for which the conclusion has not been proved — and nothing, in principle, would guarantee that it holds there too. This is exactly what happens with the reflex case in treatments that omit it: it is not proved badly, it is not proved at all (Remark 4.1).

Exercise 4.11. (*Application.*) In the Pothenot–Snellius problem, prove that if the station P and the three benchmarks A, B, C are concyclic the problem has no unique solution, and characterise the set of solutions.

Solution. Let γ be the circle through A, B, C , unique by Corollary 3.2, and suppose $P \in \gamma$. Let P' be any other point of γ lying on the same arc as P with respect to the chord $\overline{AB}$ and, simultaneously, on the same arc as P with respect to the chord $\overline{BC}$. By Corollary 4.1,

$$\angle AP'B = \angle APB = \alpha, \quad \angle BP'C = \angle BPC = \beta.$$

The two angular readings from P' are therefore *identical* to those from P : the measurements do not distinguish P from P' .

The solution set is thus the whole arc of γ consisting of the points lying on the same side as P with respect to both chords: an infinite set. No refinement of instrumental precision can resolve the indeterminacy, because this is not a measurement-error problem but a geometric degeneracy: it is the *danger circle*. The field rule of thumb — choose benchmarks such that the station is far from their circumcircle — follows directly from this computation.

Exercise 4.12. In a cyclic quadrilateral $ABCD$ the arcs AB , BC and CD measure 80° , 100° and 70° respectively. Compute the four interior angles and check that opposite angles are supplementary.

Solution. The arc DA equals $360^\circ - (80 + 100 + 70)^\circ = 110^\circ$. Every interior angle is half the arc it stands on, that is the arc opposite the vertex:

$$\hat{A} = \frac{1}{2}(100+70) = 85^\circ, \quad \hat{B} = \frac{1}{2}(70+110) = 90^\circ, \quad \hat{C} = \frac{1}{2}(110+80) = 95^\circ, \quad \hat{D} = \frac{1}{2}(80+100) = 90^\circ.$$

Check: $\hat{A} + \hat{C} = 180^\circ$ and $\hat{B} + \hat{D} = 180^\circ$ ✓, and the four sum to 360° ✓. Note that $\hat{B}$ and $\hat{D}$ are right angles because the corresponding arcs add to 180° , that is because AC is a diameter.

4.17. Self-assessment quiz

Ten multiple-choice questions for a quick check (about 10 minutes) before tackling the exercises. The answer key is at the end of the section.

1. The central angle standing on an arc is, relative to the inscribed angle standing on the

-
- same arc:*
- A) half of it
 - B) always twice it
 - C) equal to it
 - D) twice it only if the arc is smaller than a semicircle
2. In a triangle inscribed in a semicircle, the angle opposite the diameter measures
- A) 45°
 - B) 60°
 - C) 90°
 - D) it depends on the position of the vertex
3. If an inscribed angle is obtuse, its vertex lies
- A) on the major of the two arcs determined by the chord
 - B) on the minor of the two arcs determined by the chord
 - C) at the centre of the circle
 - D) outside the circle
4. Two inscribed angles standing on the same arc are
- A) supplementary
 - B) complementary
 - C) congruent
 - D) congruent only if the vertex is equidistant from the endpoints
5. In a *convex* quadrilateral inscribed in a circle, the sum of two opposite angles is
- A) 90°
 - B) 180°
 - C) 360°
 - D) dependent on the radius
6. A convex quadrilateral has two opposite angles of 85° and 100° . Is it inscriptible in a circle?
- A) yes, because both are smaller than 180°
 - B) no, because their sum is $185^\circ \neq 180^\circ$
 - C) yes, but only if the other two are equal to each other
 - D) it cannot be decided without knowing the sides
7. The chord $\overline{AB}$ is the side of the inscribed regular hexagon. The inscribed angle standing on the minor arc $\widehat{AB}$ measures
- A) 60°
 - B) 30°
-

- C) 120°
 D) 15°
8. The locus of the points of a half-plane from which a segment $\overline{AB}$ is seen under a fixed angle φ is
- A) a line parallel to AB
 B) a circular arc with endpoints A and B
 C) an ellipse with foci A and B
 D) a circle centred at the midpoint of AB
9. From an external point two secants intercept arcs of 150° and 40° . The angle at the vertex measures
- A) 95°
 B) 110°
 C) 55°
 D) 190°
10. Thales' theorem is, relative to the central angle theorem,
- A) a particular case, with straight central angle
 B) an independent theorem, not deducible from it
 C) a generalisation
 D) equivalent, in the sense that each implies the other

BEST PRACTICE

Answer key. 1: B — 2: C — 3: B (vertex on the minor arc $\Rightarrow$ stands on the major arc $\Rightarrow$ obtuse angle, Theorem 4.2) — 4: C — 5: B — 6: B — 7: B (the arc measures $360^\circ/6 = 60^\circ$, the inscribed angle is half of it) — 8: B (Theorem 4.7) — 9: C — 10: A.

Note on question 10. Answer D is wrong but seductive: Thales *follows* from the general theorem, but does not imply it — Theorem 4.1 contains infinitely many cases that Thales does not cover. It is the logical error of exchanging implication with equivalence, the same denounced regarding converses in Chapter 3.

4.18. Chapter Formula sheet

Central angle and inscribed angle (Theorem 4.1)

$$\angle AOB = 2 \angle APB$$

Reflex case (Theorem 4.2)

$$\angle APB = \frac{1}{2}(2\pi - \angle AOB_{\text{conv}}) = \pi - \frac{1}{2} \angle AOB_{\text{conv}}$$

Angles on the same arc and on opposite arcs (Corollary 4.1; Theorem 4.3)

$$\angle APB = \angle AQB \quad (\text{same arc}), \quad \angle APB + \angle AQB = \pi \quad (\text{opposite arcs})$$

Thales' theorem and its converse (Theorems 4.4 and 4.5)

$$\overline{AB} \text{ a diameter} \iff \angle APB = \frac{\pi}{2}$$

Cyclic quadrilaterals (Theorem 4.6)

$$ABCD \text{ cyclic} \iff \angle A + \angle C = \pi \text{ and } \angle B + \angle D = \pi$$

Arc of a given angle (Theorem 4.7)

$$\{P : \angle APB = \varphi\} = \text{arc of radius } R = \frac{|AB|}{2 \sin \varphi}, \quad \varphi \in (0, \pi)$$

The four positions of the vertex (Theorems 4.11 and 4.12; Remark 4.10)

$$\text{vertex at the centre: } \angle = \mu$$

$$\text{vertex on the circle: } \angle = \frac{1}{2} \mu$$

$$\text{interior vertex: } \angle = \frac{1}{2}(\mu_1 + \mu_2)$$

$$\text{exterior vertex: } \angle = \frac{1}{2}(\mu_{\text{far}} - \mu_{\text{near}})$$

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CHAPTER 5

Circle Inversion and Advanced Constructions

INFO

Aim of the chapter. To introduce the first transformation of the plane that does *not* preserve distances and does not map lines to lines — **circle inversion** — and to show that precisely for this reason it solves problems otherwise out of reach: the problem of Apollonius, Ptolemy's theorem, Steiner's porism, the Mohr–Mascheroni theorem.

Prerequisites. The whole of Chapter 4 (in particular Thales' theorem, used throughout) and the power of a point (Definition 3.1, Chapter 3).

The guiding thread. Inversion exchanges the roles of line and circle. Whenever a problem is hard because it mixes lines and circles, one may try to make it homogeneous by inverting with respect to a well-chosen point. It is a strategy, not a theorem: but it is a strategy that works in every case of this chapter.

5.1. Definition, formulas and first properties

Definition 5.1: Circle inversion. Given a circle $C(O, k)$ (the **circle of inversion**, O called the **pole** and k^2 the **power of the inversion**), the **inversion** with centre O and radius k is the transformation $\iota : \mathbb{R}^2 \setminus \{O\} \rightarrow \mathbb{R}^2 \setminus \{O\}$ sending $P \neq O$ to the point $P^* = \iota(P)$ lying on the ray OP and such that

$$|OP| \cdot |OP^*| = k^2.$$

Remark 5.1: The point P^* exists and is unique. The number $|OP| > 0$ is given; the number $k^2/|OP|$ is therefore a well-determined positive real. On the ray OP there is one and only one point at that distance from O (order axiom and continuity of the line). Definition 5.1 is thus well posed. The pole O is excluded: for it the equality would require $0 \cdot |OO^*| = k^2 > 0$, impossible. We return to this exclusion in Section 5.13.

Theorem 5.1: Formulas in Cartesian coordinates. With the pole O at the origin, the inversion of radius k maps $P = (x, y)$ to

$$P^* = (x^*, y^*) = \left(\frac{k^2 x}{x^2 + y^2}, \frac{k^2 y}{x^2 + y^2} \right).$$

Proof. The point P^* lies on the ray OP , hence has the form $P^* = \lambda P$ with $\lambda > 0$ (multiplication by a positive scalar: same direction and same sense). Imposing the condition of Definition 5.1:

$$|OP| \cdot |OP^*| = |P| \cdot |\lambda P| = \lambda |P|^2 = k^2 \implies \lambda = \frac{k^2}{|P|^2} = \frac{k^2}{x^2 + y^2},$$

where $\lambda > 0$ was used to write $|\lambda P| = \lambda|P|$. Substituting, $P^* = \frac{k^2}{x^2 + y^2}(x, y)$, which is the conclusion componentwise. $\square$

$$x^* = \frac{k^2 x}{x^2 + y^2}, \quad y^* = \frac{k^2 y}{x^2 + y^2}, \quad x = \frac{k^2 x^*}{x^{*2} + y^{*2}}, \quad y = \frac{k^2 y^*}{x^{*2} + y^{*2}}$$

Remark 5.2: Involution and fixed points. ι is an **involution**: $\iota(\iota(P)) = P$ for every P . This follows immediately from the definition by exchanging the roles of P and P^* , or from the formulas: applying the factor λ twice gives $\frac{k^2}{|\lambda P|^2} \lambda P = \frac{k^2}{\lambda^2 |P|^2} \lambda P = \frac{k^2}{\lambda |P|^2} P = P$, since $\lambda |P|^2 = k^2$.

The fixed points of ι are exactly the points of $C(O, k)$ itself: from $|OP| = k$ it follows that $|OP^*| = k^2/k = k$, and on the ray OP there is a single point at distance k , namely P itself. Points interior to $C(O, k)$ are sent to exterior points and vice versa, because $|OP| < k \iff k^2/|OP| > k$. The two statements together say that inversion “turns the circle inside out like a glove”, leaving its boundary fixed.

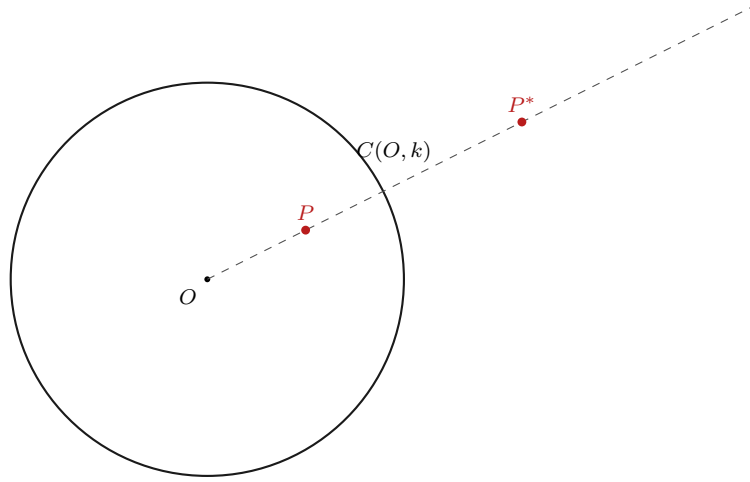


Figure 5.1: The inversion with pole O and radius $k = 2$ (Definition 5.1): the point $P = (1, 0.5)$, interior to $C(O, k)$, and its image $P^* = (3.2, 1.6)$, exterior, on the same ray from O . We check $|OP| \cdot |OP^*| = 1.118 \times 3.578 = 4.000 = k^2$ (independent script). The points of $C(O, k)$ stay fixed; interior and exterior swap, as stated in Remark 5.2.

Theorem 5.2: Formula for the transformed distance. For any distinct $P, Q \neq O$,

$$|P^*Q^*| = \frac{k^2 |PQ|}{|OP| \cdot |OQ|}.$$

Proof. Step 1 — two similar triangles. Consider the triangles OPQ and OQ^*P^* (note the **reversal of the order** of the last two letters: it is essential). By Definition 5.1,

$$|OP| \cdot |OP^*| = k^2 = |OQ| \cdot |OQ^*| \implies \frac{|OP|}{|OQ|} = \frac{|OQ^*|}{|OP^*|}.$$

The two triangles moreover share the angle at O , because P^* lies on the ray OP and Q^* on the ray OQ : hence $\angle POQ = \angle P^*OQ^*$ is the same angle.

Step 2 — similarity criterion. Two triangles with a congruent angle contained between proportional sides are similar (SAS similarity criterion). Hence $OPQ \sim OQ^*P^*$, with the corre-

spondence $O \leftrightarrow O, P \leftrightarrow Q^*, Q \leftrightarrow P^*$.

Step 3 — ratio of similarity. The ratio is $\frac{|OQ^*|}{|OP|}$. Applying it to the side $\overline{PQ}$, which corresponds to the side $\overline{Q^*P^*}$:

$$|P^*Q^*| = |PQ| \cdot \frac{|OQ^*|}{|OP|} = |PQ| \cdot \frac{k^2/|OQ|}{|OP|} = \frac{k^2 |PQ|}{|OP| \cdot |OQ|}.$$

□

Remark 5.3: What this formula says. Inversion does **not** preserve distances — and the formula quantifies exactly how much it alters them: the scale factor is not constant but depends on where the two points lie. Two points far from the pole move much closer together; two points very near the pole move enormously apart. This formula is the central technical tool of the chapter: from it will follow Ptolemy's theorem (Section 5.5) and Exercise 5.3.

Theorem 5.3: Inversion maps lines and circles to lines and circles. Under an inversion with centre O and radius k :

- (a) a line through O is mapped to itself (deprived of O);
- (b) a line not through O is mapped to a circle through O (deprived of the point O), and conversely;
- (c) a circle not through O is mapped to a circle not through O .

Proof. (a) Every point of a line through O has its image on the same ray, hence on the same line: the image set coincides with the original line (even though the individual points are moved; only the two points at distance k stay fixed).

(b) Let r be the line $ax + by = c$ with $c \neq 0$ (it does not pass through O). By involutivity, $P \in r \Leftrightarrow P = \iota(P^*)$, that is, $(x, y) = \frac{k^2}{x^{*2} + y^{*2}}(x^*, y^*)$. Substituting into $ax + by = c$:

$$a \cdot \frac{k^2 x^*}{x^{*2} + y^{*2}} + b \cdot \frac{k^2 y^*}{x^{*2} + y^{*2}} = c \implies x^{*2} + y^{*2} - \frac{k^2 a}{c} x^* - \frac{k^2 b}{c} y^* = 0,$$

which is the equation of a circle (Chapter 7, Definition 7.1 in general form with zero constant term) passing through the origin O (substituting $x^* = y^* = 0$ gives $0 = 0$). The converse follows by applying ι a second time (involutivity).

(c) A circle not through O has general equation $x^2 + y^2 + Dx + Ey + F = 0$ with $F \neq 0$. Substituting $x = k^2 x^* / (x^{*2} + y^{*2})$, $y = k^2 y^* / (x^{*2} + y^{*2})$ and multiplying by $(x^{*2} + y^{*2})^2 / k^2$ (legitimate because $P^* \neq O$), one obtains after simplification

$$F(x^{*2} + y^{*2}) + k^2 D x^* + k^2 E y^* + k^4 = 0,$$

which, on dividing by $F \neq 0$, is again the general equation of a circle, with constant term $k^4/F \neq 0$: it does not pass through the origin. □

Synthetic proof of part (b). Part (b) is worth seeing also in purely geometric terms, because it is the proof found in the classical sources and is more illuminating than the algebraic one.

Let L be a line not through O and let A be the foot of the perpendicular drawn from O to L (Theorem 2.1). Let P be any point of L , and let A^*, P^* be the images of A and P . From

the definition,

$$|OA| \cdot |OA^*| = k^2 = |OP| \cdot |OP^*| \implies \frac{|OA|}{|OP^*|} = \frac{|OP|}{|OA^*|}.$$

The triangles OAP and OP^*A^* share the angle at O and have the sides containing it in proportion: they are therefore similar (SAS similarity criterion), with $A \leftrightarrow P^*$ and $P \leftrightarrow A^*$. Since $\angle OAP = 90^\circ$ by construction, its corresponding angle $\angle OP^*A^*$ also measures 90° . As P varies on L , the point P^* therefore sees the *fixed* segment $\overline{OA^*}$ under a right angle: by the converse of Thales' theorem (Theorem 4.5, Chapter 4), P^* describes the circle with diameter $\overline{OA^*}$, which passes through O . Conversely every point of that circle other than O is obtained in this way, since ι is bijective on its domain. $\square$

Theorem 5.4: Orthogonal circles are invariant. Let Γ be a circle **different from** $C(O, k)$. Then Γ is mapped to itself by the inversion ι of circle $C(O, k)$ if and only if Γ is **orthogonal** to $C(O, k)$, that is, the two circles meet forming right angles between their respective tangents. Equivalently, if and only if $\text{pow}_\Gamma(O) = k^2$.

Proof. ($\Leftarrow$) Let Γ be orthogonal to $C(O, k)$ and let T be a common point. Orthogonality means that the tangent to $C(O, k)$ at T is perpendicular to the tangent to Γ at T ; but by Theorem 3.8 the tangent to $C(O, k)$ at T is perpendicular to the radius OT , hence the tangent to Γ at T has the same direction as OT : the line OT is tangent to Γ at T .

Now let $P \in \Gamma$, $P \neq T$, and let P' be the second point where the line OP meets Γ . Since OT is tangent to Γ and OPP' is a secant, the tangent–secant theorem (Theorem 3.16) gives

$$|OP| \cdot |OP'| = |OT|^2 = k^2.$$

Comparing with Definition 5.1 and observing that P' lies on the ray OP (because O is exterior to Γ , $|OT| = k$ being the length of a tangent segment, so both intersections fall on the same side), we conclude $P' = P^*$. Hence $\iota(P) \in \Gamma$ for every $P \in \Gamma$: the circle is invariant.

($\Rightarrow$) Let Γ be invariant. Then $O \notin \Gamma$ (the pole has no image), and Γ must meet $C(O, k)$: otherwise Γ would lie entirely inside or entirely outside $C(O, k)$, whereas inversion swaps inside and outside (Remark 5.2) and could not map Γ to itself. Let T be a common point. For every line through O meeting Γ at a point P , invariance gives $\iota(P) \in \Gamma \cap OP = \{P, P'\}$ (where P' is the second point of Γ on the line OP , if it exists). If $\iota(P) = P$ for *every* $P \in \Gamma$, every point of Γ would have $|OP| = k$ (from Definition 5.1, $\iota(P) = P$ implies $|OP|^2 = k^2$), that is, $\Gamma \subseteq C(O, k)$; and two circles with three points in common coincide (Corollary 3.2), hence $\Gamma = C(O, k)$: excluded by hypothesis. There is therefore a point $P_0 \in \Gamma$ with $\iota(P_0) = P'_0 \neq P_0$, and then, by Definition 5.1, $|OP_0| \cdot |OP'_0| = k^2$; but by Theorem 3.19 that product equals $\text{pow}_\Gamma(O)$ (independent of the chosen line). Hence $\text{pow}_\Gamma(O) = k^2$: in particular, taking the line through O and through a common point T of Γ and $C(O, k)$, $\text{pow}_\Gamma(O) = k^2 = |OT|^2$, and by the converse of the tangent–secant theorem (Theorem 3.17) the line OT is tangent to Γ at T . Since OT is perpendicular to the tangent to $C(O, k)$ at T (Theorem 3.8), the two circles are orthogonal. $\square$

Remark 5.4: The excluded case: $\Gamma = C(O, k)$. $C(O, k)$ is pointwise invariant under its own inversion ($\iota(P) = P$ for every $P \in C(O, k)$, directly from Definition 5.1 with $|OP| = k$), but it is **not** orthogonal to itself: orthogonality is a relation between two distinct circles, and in this case $\text{pow}_{C(O, k)}(O) = -k^2 \neq k^2$. Theorem 5.4 excludes precisely this case; it is the only invariant circle that does not fall under the characterisation by orthogonality.

Theorem 5.5: Isogonality (conformality) of inversion. *Inversion preserves the measure of angles between curves: if two differentiable curves meet at $P \neq O$ forming an angle φ , their images meet at P^* forming the same angle φ (with reversed sense: inversion is anticonformal).*

Proof. Step 1 — the case of a chord. Let P and Q be two distinct points, both different from O . In Step 1 of the proof of Theorem 5.2 it was established that $OPQ \sim OQ^*P^*$. From the similarity it follows in particular that

$$\angle OPQ = \angle OQ^*P^*.$$

Step 2 — passage to the limit. Now let γ be a differentiable curve through P , and let Q be a point of γ tending to P . The line PQ (a secant) tends to the **tangent** to γ at P , by definition of the tangent to a differentiable curve. Correspondingly, Q^* tends to P^* along the image curve γ^* , and the line P^*Q^* tends to the tangent to γ^* at P^* . Passing to the limit in the equality of Step 1, the angle that the tangent to γ at P makes with the line OP equals the angle that the tangent to γ^* at P^* makes with the same line OP — but *on the opposite side*, because in the similarity the vertices P and Q^* correspond in reversed order.

Step 3 — two curves. Let γ_1, γ_2 be two curves through P , and let θ_1, θ_2 be the angles their tangents make with the line OP . The angle between the two curves is $\varphi = |\theta_1 - \theta_2|$. By Step 2, the tangents to the images make angles $-\theta_1$ and $-\theta_2$ with OP (same measure, opposite sense), so the angle between the images is $|(-\theta_1) - (-\theta_2)| = |\theta_1 - \theta_2| = \varphi$. $\square$

Warning. Step 2 uses a **passage to the limit** and therefore requires the curves to be differentiable at the point: for curves without a tangent the statement does not even make sense, the angle being undefined. Many elementary expositions state isogonality with no regularity hypothesis at all: this is a gap, harmless though it is in the applications, where the curves are lines and circles.

Note further that inversion **reverses the orientation** of angles: it preserves measure but not sense. For this reason it is properly called *anticonformal* rather than conformal. The composition of two inversions, by contrast, is conformal — and is a Möbius transformation (Volume V).

Corollary 5.1: Tangency is preserved. *If two circles (or a circle and a line) are tangent at a point $T \neq O$, their images are tangent at T^* .*

Proof. Tangency at T means that the two curves meet at T forming a zero angle. By Theorem 5.5 the images meet at T^* still forming a zero angle, that is, they are tangent. Note that the hypothesis $T \neq O$ is necessary: if the two circles were tangent *at the pole*, their images would be two lines (Theorem 5.3(b)) with no common point, that is, **parallel** — which is precisely how inversive geometry interprets parallelism: tangency at infinity. $\square$

5.2. The Mohr–Mascheroni theorem

Theorem 5.6: Mohr–Mascheroni theorem (1672 / 1797). *Every construction executable with ruler and compass is executable with the **compass alone**, provided one agrees that a line is regarded as constructed as soon as two distinct points of it have been constructed.*

Remark 5.5: Why the convention on lines is necessary and is not a trick. The compass cannot draw a straight line: no convention can change this material fact. What the theorem asserts is that the information contained in a line — which points belong to it, and where it meets other objects — is entirely recoverable with the compass alone. Since a line is uniquely determined by two of its points (axiom A1), knowing two of its points is equivalent to knowing it. The convention does not weaken the theorem: it makes it statable.

Definition 5.2: Compass-only model. Starting from a finite set of *constructed* points (the initial data), the following operations are admitted:

- (C1) given two constructed points A, B and a constructed point C , one may draw the circle $C(C, |AB|)$ (transport of a radius with the compass, Euclid's Postulate III);
- (C2) construct the points of intersection of two drawn circles (when they exist);
- (C3) choose an arbitrary point of the plane, or of a drawn circle, avoiding a finite number of positions declared forbidden by the construction in progress (Euclidean convention, analogous to (R4) below for the straightedge-plus-fixed-circle model), provided that what follows in the construction does not depend on *which* admissible point is chosen.

Source for convention (C3): Hartshorne, *Geometry: Euclid and Beyond*, chapter on constructions.

Adler's reduction (1906): the three fundamental operations. Every ruler-and-compass construction is a finite sequence of just three elementary operations:

- (O1) intersection of two circles;
- (O2) intersection of a circle and a line;
- (O3) intersection of two lines.

The circles in (O1) and (O2) are those the compass draws: given by a centre and a point already constructed (Euclid's Postulate III).

Operation (O1) is already executable with the compass alone, by definition. The theorem is therefore proved as soon as one can execute (O2) and (O3) with the compass alone as well. August Adler, in *Theorie der geometrischen Konstruktionen* (Leipzig, 1906), observed that **circle inversion** realises exactly this reduction: by Theorem 5.3, inverting with respect to a pole chosen off the lines involved turns every *line* into a *circle* through the pole. Consequently (O3) becomes an (O1) between the two image circles, and (O2) likewise becomes an (O1). One performs the intersection in the transformed plane — where it is of type (O1), hence legitimate — and carries the result back by applying ι again, it being an involution. $\square$

Remark 5.6: What is still missing. The reduction just described is the idea of the proof, not the complete proof: for it to be effective one must be able to perform **with the compass alone** the inversion itself, that is, to construct P^* from P and the circle of inversion. This is the content of Construction 5.1, which we give in full: without it Adler's argument would be circular, in the worst sense of the word.

Construction 5.1: — Inverse of a point with the compass alone. **Data:** the circle of inversion $C(O, k)$ (of which O and one point, hence the radius, are known) and a point $P \neq O$ with $|OP| \geq k/2$.

Step 1. With centre P and opening $|PO|$ draw a circle. It meets $C(O, k)$ at two points X and Y .

Step 2. With centre X and opening $|XO| = k$ draw an arc; with centre Y and opening $|YO| = k$ draw another. The two arcs meet at O and at a second point P^* .

Claim. The point P^* so constructed is the inverse of P with respect to $C(O, k)$. $\square$

Proof. Step 1 — the construction is executable. The circle with centre P and radius $|PO|$ meets $C(O, k)$ if and only if $||PO| - k| \leq |PO| \leq |PO| + k$. The right-hand inequality is obvious. The left-hand one splits in two: $|PO| - k \leq |PO|$, always true; and $k - |PO| \leq |PO|$, that is $k \leq 2|PO|$, that is $|OP| \geq k/2$ — exactly the hypothesis. Let X, Y be the two intersection points.

Step 2 — P^ lies on the ray OP .* The points X and Y are symmetric with respect to the line OP : indeed both O and P are equidistant from X and Y ($|OX| = |OY| = k$ and $|PX| = |PY| = |PO|$), so by axiom (A7) the line OP is the perpendicular bisector of the segment $\overline{XY}$. The same symmetry holds for the two arcs of Step 2, which have centres X and Y and equal radii: their second intersection P^* therefore lies on the perpendicular bisector OP . That it lies on the ray OP and not on its opposite follows from the fact that X and Y lie on the side of P with respect to the perpendicular to OP at O .

Step 3 — two similar isosceles triangles. The triangle OPX has $|PO| = |PX|$ by construction: it is isosceles with base $\overline{OX}$, so by (A4)

$$\angle POX = \angle PXO.$$

The triangle $OX P^*$ has $|XO| = |XP^*| = k$ by construction: it is isosceles with base $\overline{OP^*}$, so

$$\angle XOP^* = \angle XP^*O.$$

But $\angle POX$ and $\angle XOP^*$ are *the same angle*, because P^* lies on the ray OP (Step 2). The two isosceles triangles OPX and $OX P^*$ therefore have congruent base angles, and having both their apex angle determined by the angle sum (A9), they are **similar**, with the correspondence $O \leftrightarrow O, P \leftrightarrow X, X \leftrightarrow P^*$.

Step 4 — conclusion. From the similarity,

$$\frac{|OP|}{|OX|} = \frac{|OX|}{|OP^*|} \implies |OP| \cdot |OP^*| = |OX|^2 = k^2,$$

which is exactly Definition 5.1. □

Warning. The excluded case. If $|OP| < k/2$ the construction does not start, because the circle of Step 1 does not meet $C(O, k)$. The remedy is to *double* the segment $\overline{OP}$ beforehand a sufficient number of times — an operation executable with the compass alone by repeatedly stepping off the opening $|OP|$ along the ray — until a point P_n with $|OP_n| = 2^n|OP| \geq k/2$ is obtained; one constructs P_n^* and observes that $|OP^*| = 2^n|OP_n^*|$, so that P^* is obtained by doubling $\overline{OP_n^*}$ n times. Flagging this case is not pedantry: a construction that fails silently for certain data is not a construction.

Remark 5.7: The missing piece: a compass wants a centre. Construction 5.1 makes half of Adler's argument effective: one can invert a point. But in order to intersect the two image circles one must first draw them, and a compass cannot draw a circle of which it knows three points: it wants the centre and the radius. Inverting two points of a line yields two points of the image circle, and the pole is a third: three points, no centre. The proof is therefore complete only if one can construct, with the compass alone, the centre of the circle through three given points. That is what we do now; and the key, once again, is inversion.

Construction 5.2: — Centre of the circle through three points, with the compass alone. **Data:** three non-collinear points A, B, C .

Step 1. Fix the inversion ι with pole A and arbitrary radius k (for instance $k = |AB|$) and

construct $B^* = \iota(B)$ and $C^* = \iota(C)$ by Construction 5.1.

Step 2. Draw the circle with centre B^* and opening $|B^*A|$ and the one with centre C^* and opening $|C^*A|$: they meet at A and at a second point Q .

Step 3. Construct $O = \iota(Q)$, again by Construction 5.1.

Claim. O is the centre of the circle through A, B, C . □

Proof. Step 1 — the image of the circle is a line. Let Γ be the circle through A, B, C , which exists and is unique because the three points are not collinear. It passes through the pole A , hence by Theorem 5.3 its image $\iota(\Gamma \setminus \{A\})$ is a line ℓ . Since $B, C \in \Gamma$, we have $B^*, C^* \in \ell$, and therefore ℓ is precisely the line B^*C^* .

Step 2 — ℓ is perpendicular to the line AO . Call O the centre of Γ and $R = |AO|$ its radius. The whole figure formed by Γ and the pole A is symmetric about the line AO ; and inversion with pole A commutes with reflection in a line through A , because both preserve the rays through A and the distances from A . Hence ℓ too is symmetric about AO : either it coincides with AO , or it is perpendicular to it. It cannot coincide, because ℓ is the image of $\Gamma \setminus \{A\}$, which is not contained in a line. Perpendicularity remains.

Step 3 — where ℓ cuts AO . Let A' be the point of Γ diametrically opposite A , so that $|AA'| = 2R$ and A' lies on the ray AO . Since $A' \in \Gamma \setminus \{A\}$, the point $\iota(A')$ belongs to ℓ ; and it belongs to the ray AO , because inversion does not change the ray. As $\ell \perp AO$, the point $\iota(A')$ is the only point common to ℓ and the line AO , and

$$|A\iota(A')| = \frac{k^2}{|AA'|} = \frac{k^2}{2R}.$$

Step 4 — the reflection of the pole. The two circles of Step 2 of the construction have centres B^* and C^* , both on ℓ , and both pass through A : their two common points are therefore symmetric about ℓ (axiom A7: the line of centres is the perpendicular bisector of the common chord). One is A , hence the other is Q , the reflection of A in ℓ . By Step 3, Q lies on the ray AO at distance

$$|AQ| = 2|A\iota(A')| = \frac{k^2}{R}.$$

Step 5 — conclusion. Applying ι to Q gives the point of the ray AO at distance

$$|A\iota(Q)| = \frac{k^2}{|AQ|} = \frac{k^2}{k^2/R} = R,$$

that is, exactly O . Hence $\iota(Q) = O$: the required centre.

Step 6 — numerical check. Take $A = (0, 0)$, $B = (4, 0)$, $C = (0, 3)$: the centre is $O = (2, 1.5)$ and $R = 2.5$. With $k = |AB| = 4$ one has $B^* = (4, 0)$ (because $|AB| = k$, so B is fixed) and $C^* = (0, 16/3) = (0, 5.3333)$. The line B^*C^* has equation $x/4 + 3y/16 = 1$, that is $4x + 3y = 16$; the distance of the origin from it is $16/5 = 3.2$, and indeed $k^2/(2R) = 16/5 = 3.2$ ✓. The reflection of A is $Q = 2 \cdot \frac{16}{25}(4, 3) = (5.12, 3.84)$, of modulus $6.4 = k^2/R = 16/2.5$ ✓, and finally $\iota(Q) = \frac{16}{6.4^2}(5.12, 3.84) = (2, 1.5) = O$ ✓. □

Construction 5.3: — The operations (O2) and (O3) with the compass alone.

Step 0 — choice of pole. Choose a point Z belonging to none of the lines and to none of the circles involved in the operation, and fix the inversion ι with pole Z and arbitrary radius k .

Step 1 — lines become circles. For every line ℓ involved, known through two of its points A and B , construct $A^* = \iota(A)$ and $B^* = \iota(B)$; the circle $\iota(\ell)$ passes through Z, A^*, B^* , and its centre is constructed by Construction 5.2. It can now be drawn.

Step 2 — circles remain circles. For every circle $\Gamma(O, r)$ involved, choose three distinct points P_1, P_2, P_3 on it — for instance by stepping the opening r three times starting from any point of Γ — construct their images and, by Construction 5.2, the centre of $\iota(\Gamma)$; this too can now be drawn.

Step 3 — the intersection. Intersect the two image circles: this is an operation (O1), performable with the compass alone.

Step 4 — back. Apply ι to the points found. Since ι is an involution, one obtains exactly the points required by the original operation (O2) or (O3). $\square$

Proof. Step 1 — the pole exists. Fix two distinct points U and V already constructed; as the two openings ρ_1, ρ_2 vary, the circles of centre U and radius ρ_1 and of centre V and radius ρ_2 meet, when they meet, in points describing an open subset of the plane. The objects to be avoided are finitely many: at most two lines and two circles. A finite union of lines and circles has empty interior, hence cannot contain an open set: an admissible choice of Z therefore exists, and is found by trying.

Step 2 — the images are as claimed. Since $Z \notin \ell$, by Theorem 5.3 the image of ℓ is a circle through the pole Z ; it contains A^* and B^* , and three non-collinear points determine a unique circle, hence it is the circle through Z, A^*, B^* . The three points are not collinear: if they were, their line would be the image of ℓ , but the image of a line not through the pole is not a line. Likewise, since $Z \notin \Gamma$, the image of Γ is a circle, determined by the images of three distinct points of it.

Step 3 — intersections transport. Inversion is a bijection of the plane minus the pole onto itself, hence it carries intersections to intersections: $\iota(\ell \cap \Gamma) = \iota(\ell) \cap \iota(\Gamma)$, and likewise for two lines. Since none of the sought points coincides with Z (which by construction lies on none of the objects), nothing is lost.

Step 4 — involutivity. By Definition 5.1, $|ZP| \cdot |ZP^*| = k^2$ implies $|ZP^*| \cdot |Z(P^*)^*| = k^2$ and hence $(P^*)^* = P$, both lying on the same ray. Applying ι to the two points of Step 3 therefore recovers the points of $\ell \cap \Gamma$, respectively of $\ell_1 \cap \ell_2$. $\square$

Remark 5.8: A shortcut for the generic case. When the centre O of the circle does not lie on the line $\ell = AB$, operation (O2) is performed in two moves, with no inversion: draw the circle of centre A and opening $|AO|$ and the one of centre B and opening $|BO|$, which meet at O and at its reflection O' in ℓ ; then draw the circles of centre O and of centre O' , both of radius r . Their common points are the required ones: indeed a point X satisfies $|XO| = |XO'|$ if and only if it lies on the perpendicular bisector of $\overline{OO'}$, which is ℓ , and $|XO| = r$ if and only if it lies on the given circle. The general construction by inversion remains necessary in the case $O \in \ell$, where $O' = O$ and the shortcut degenerates.

Proof. [of Theorem 5.6] Step 1 — reduction. By Adler's reduction, every straightedge-and-compass construction is a finite sequence of operations (O1), (O2), (O3) applied to the data and to the points obtained along the way, with the convention that a line is known as soon as two of its points are.

Step 2 — (O1). The intersection of two circles is by definition performable with the compass alone.

Step 3 — (O2) and (O3). They are performable with the compass alone by Construction 5.3, which in turn uses only Construction 5.1 (inversion), Construction 5.2 (centre through three points) and operations (O1). None of these requires the straightedge.

Step 4 — induction. Let P be a point obtained by a straightedge-and-compass construction in n steps. If $n = 0$, then P is a datum and there is nothing to prove. If $n > 0$, the final step is of type (O1), (O2) or (O3), and its inputs — points, or lines determined by two points — are obtainable with the compass alone by the inductive hypothesis. By Steps 2 and 3 the final

step too is performable with the compass alone, and hence so is P . By induction on n , every point constructible with straightedge and compass is constructible with the compass alone. $\square$

Best practice. It is worth counting the cost. Construction 5.1 uses three circles; the centre through three points uses two inversions plus two circles, that is eight; operation (O3) requires four point inversions and two centres, that is about twenty-five circles, against the two lines of the straightedge construction. The theorem says that the straightedge is *theoretically* superfluous, not that it is practically useless: it is the distinction between what is possible and what is reasonable, and it holds in geometry as in engineering.

***Remark 5.9: History: two and a half centuries of contested priority.** The theorem was first proved by **Georg Mohr**, a Danish mathematician, in the *Euclides Danicus* of **1672**. The work fell into complete oblivion, however, and was **rediscovered only in 1928**. In the meantime, and entirely independently, **Lorenzo Mascheroni** had given a second proof in *La geometria del compasso* (Pavia, **1797**), a century and a quarter later. For over two hundred years the result was attributed to Mascheroni alone; the double name is a recent historiographical correction. The proof by inversion, which is the one current today because it is conceptually the shortest, is instead due to **August Adler** (1906): neither Mohr nor Mascheroni had inversion at their disposal, as it would be formalised only in the nineteenth century (Section 5.12).*

5.3. The Poncelet–Steiner theorem: the ruler suffices, if there is a circle

The Mohr–Mascheroni theorem eliminates the ruler. It is natural to ask whether the compass can be eliminated instead. The answer — surprisingly — concerns the circle itself.

***Theorem 5.7: Poncelet–Steiner theorem (1822/1833).** Every construction executable with ruler and compass is executable with the **ruler alone**, provided that a **single circle already drawn together with its centre** is given once and for all.*

Proof. The proof occupies the rest of this section: we fix what the straightedge can do (Section 5.3.2), we construct with the straightedge alone parallels, perpendiculars, translations and transports of segments (Lemmas 5.1–5.6), then the intersections of a line and of two *undrawn* circles (Lemmas 5.7 and 5.8), and we conclude by induction on the length of the construction in Section 5.3.7. $\square$

INFO

Complete proof. The theorem is proved in full in this section, in a modern form shorter than Steiner’s. The proof is organised in links, not necessarily in typographical order: parallels to secants (Section 5.3.1), which do not depend on the formal model of straightedge-only operations; the model of what the straightedge can do (Section 5.3.2), which systematises that first step in retrospect too; parallels to an arbitrary line and perpendiculars (Lemmas 5.1 and 5.2); translating, halving and transporting segments (Lemmas 5.3–5.6); intersecting a line with an *undrawn* circle (Lemma 5.7); intersecting two undrawn circles (Lemma 5.8); finally the theorem (Section 5.3.7). The only source of square roots, in the whole chain, is the reading of the points where a drawn line meets the drawn circle.

Remark 5.10: Historical notes and idea of the proof. The result was conjectured by Jean-Victor Poncelet in 1822 and proved rigorously by Jakob Steiner in 1833. The complete proof — which we give in full below, in a modern form far shorter than Steiner’s original monograph — proceeds by building up, step by step with the ruler alone, a growing repertoire of elementary operations (parallel lines, perpendicular lines, midpoints, . . . , up to the extraction of square roots, the algebraic operation that characterises all ruler-and-compass constructions, see Section 7.4, Chapter 7), each obtained from the preceding ones. The seed of the whole construction is precisely **Thales’ theorem** (Theorem 4.4, Chapter 4): a circle with known centre allows all diameters (lines through the centre) to be drawn with the ruler alone, and by Thales’ theorem every point C of the circle (other than the endpoints A, B of a chosen diameter) sees AB under a right angle. The circle thus acts as a “right-angle factory” at zero cost: it is exactly the missing ingredient that the ruler alone cannot produce (a ruler by itself can draw lines and find intersections, but can in no way guarantee the perpendicularity of two lines, nor bisect a segment, without a further external datum).

Warning. The theorem requires the circle to be given **with its centre**: if only the circle (the curve) is given, without knowing where the centre is, not all ruler-and-compass constructions become executable with the ruler alone (this weaker case, taken up later by other geometers, requires additional constructions to recover the centre itself, an operation that — counterintuitively — is **not** always possible with the ruler alone if the circle is the only datum available).

Remark 5.11: A symmetry worth noticing. Theorems 5.6 and 5.7 form an almost dual pair: the first eliminates the ruler with no additional datum, the second eliminates the compass provided a circle is given as a gift. The asymmetry is instructive and not accidental: the compass produces circles, and from a circle everything can be reconstructed; the ruler produces lines, and from a line nothing curved can be reconstructed. It is the same asymmetry that inversion stages by exchanging the two kinds of object, and the deep reason why Adler’s proof works.

Remark 5.12: Why it matters, beyond historical curiosity. Beyond its theoretical value, the Poncelet–Steiner theorem illustrates a principle that recurs in the engineering of measurement and control systems: a fixed, well-known geometric reference (here, a circle with a certified centre) can replace an otherwise indispensable general instrument (here, the compass). It is the same principle, in a different form, behind the use of a single circular guide of known precision to calibrate an entire robotic positioning system (Chapter 13): one accurate geometric reference, systematically exploited, can suffice to reconstruct with a ruler (that is, with relative linear/angular measurements only) the whole geometry otherwise entrusted to more expensive or less field-practical absolute measuring instruments.

5.3.1 The first link of the chain: parallels with the straightedge alone

The first link in the proof of the Poncelet–Steiner theorem is the construction of parallels, and it is the one that carries all the rest. We give it in full, because a reader who can draw parallels with the straightedge alone already holds half the theorem, and because the construction is a fine example of how a single metric datum (a midpoint) suffices to unlock an otherwise purely projective geometry.

Construction 5.4: — Parallel through a point, given a segment with its midpoint.

Data: three collinear points A , B and their midpoint M , and a point $P \notin AB$.

Step 1. Choose a point C on the line AP , with $C \neq A$, $C \neq P$ and C different from the midpoint of $\overline{AP}$.

Step 2. Draw the line MC and call D its intersection with the line BP .

Step 3. Draw the line AD and call E its intersection with the line BC .

Claim. The line PE is parallel to AB . □

Proof. Choose coordinates with $A = (-1, 0)$, $B = (1, 0)$, $M = (0, 0)$ and $P = (p, q)$, $q \neq 0$.

Step 1. The point C on the line AP is $C = A + c(P - A) = (-1 + c(p + 1), cq)$ with $c \neq 0, \frac{1}{2}, 1$ (for $c = \frac{1}{2}$ one would have $\delta = 2c - 1 = 0$ in Step 2: the line MC would be parallel to BP and the point D would not exist).

Step 2. The line MC passes through the origin, so its points are λC ; the line BP is $(1, 0) + u(p - 1, q)$. Equating ordinates, $\lambda cq = uq$ gives $u = \lambda c$; equating abscissas,

$$\lambda(-1 + c(p + 1)) = 1 + \lambda c(p - 1) \implies -\lambda + \lambda cp + \lambda c = 1 + \lambda cp - \lambda c \implies \lambda(2c - 1) = 1,$$

that is $\lambda = 1/(2c - 1)$ and $D = \lambda C$.

Step 3. The line AD is $(-1, 0) + w(D + (1, 0))$ and the line BC is $(1, 0) + z(C - (1, 0))$. Equating ordinates, $w\lambda cq = zcq$ gives $z = w\lambda$. Setting $\delta = 2c - 1$, one has $D_x + 1 = (cp + 3c - 2)/\delta$ and $C_x - 1 = cp + c - 2$; equality of the abscissas becomes

$$-1 + w \frac{cp + 3c - 2}{\delta} = 1 + \frac{w}{\delta} (cp + c - 2),$$

and multiplying by δ ,

$$-\delta + w(cp + 3c - 2) = \delta + w(cp + c - 2) \implies w(2c) = 2\delta \implies w = \delta/c.$$

Conclusion. The ordinate of E is

$$E_y = w D_y = \frac{\delta}{c} \cdot \frac{cq}{\delta} = q,$$

exactly the ordinate of P . For the line PE to be well defined we also need $E \neq P$: the abscissa of E is $E_x = -1 + w(D_x + 1) = p + 2 - 2/c$ (substituting $w = \delta/c$), so $E_x - p = 2 - 2/c$, which is non-zero exactly for $c \neq 1$ — already guaranteed at Step 1 ($C \neq P$, that is $c \neq 1$). Hence $E \neq P$, the two points P and E therefore have the same ordinate, and the line PE is parallel to the x axis, that is to AB . Note that E_y depends neither on c nor on p : the construction works for *every* admissible choice of the auxiliary point C (that is, $C \neq A, P$ and C not the midpoint of $\overline{AP}$), which makes it robust and verifiable by redrawing it with a second C . □

Construction 5.5: — Two parallels from a secant of the given circle. Data: the circle Ω with its centre O , and a line ℓ meeting it at U and V .

Step 1. Draw the line UO ; it meets Ω at a second point U' , antipodal to U .

Step 2. Draw the line VO ; it meets Ω at V' , antipodal to V .

Claim. The lines VU' and UV' are both perpendicular to ℓ , and hence parallel to each other. □

Proof. The segment UU' is a diameter, so by Thales' theorem the angle $\widehat{UVU'}$, inscribed in the semicircle of diameter UU' , is a right angle: the line VU' is perpendicular to $UV = \ell$. The same argument with the diameter VV' shows that $\widehat{VUV'}$ is a right angle, that is $UV' \perp \ell$. Two lines perpendicular to the same line are parallel to each other, and they are distinct because

$U \neq V$. □

Construction 5.6: — Midpoint of a segment, given two parallels. **Data:** two parallel lines ℓ and ℓ' , two points $A, B \in \ell$.

Step 1. Choose two distinct points $A', B' \in \ell'$.

Step 2. Construct $K = AB' \cap A'B$ and $P = AA' \cap BB'$.

Claim. The line KP meets ℓ at the midpoint of $\overline{AB}$. □

Proof. Take coordinates with $\ell : y = 0$, $\ell' : y = 1$, $A = (0, 0)$, $B = (1, 0)$, $A' = (a, 1)$, $B' = (b, 1)$, and set $d = b - a$ (with $d \neq \pm 1$, since the four lines used must intersect). A direct computation of the two intersections gives

$$K = \left(\frac{b}{1+d}, \frac{1}{1+d} \right), \quad P = \left(\frac{a}{1-d}, \frac{1}{1-d} \right).$$

The line KP meets $y = 0$ at the parameter value $\lambda = -(1-d)/(2d)$, and the corresponding abscissa is

$$\frac{b}{1+d} - \frac{1-d}{2d} \cdot \frac{d(a+b-1)}{1-d^2} = \frac{b}{1+d} - \frac{a+b-1}{2(1+d)} = \frac{2b-a-b+1}{2(1+d)} = \frac{b-a+1}{2(1+d)} = \frac{1+d}{2(1+d)} = \frac{1}{2},$$

which is exactly the midpoint of $\overline{AB}$. □

Theorem 5.8: Parallels to secants, with the straightedge alone. *Given a circle Ω with its centre, for every line ℓ secant to Ω and every point $P \notin \ell$ it is possible to construct with the straightedge alone the parallel to ℓ through P .*

Proof. Step 1 — two parallels. Let U, V be the points where ℓ meets Ω . By Construction 5.5 the lines $m_1 = VU'$ and $m_2 = UV'$ are both perpendicular to ℓ , hence parallel to each other and distinct.

Step 2 — a midpoint on m_1 . Applying Construction 5.6 to the pair of parallels m_1, m_2 , with the two points V, U' taken on m_1 and, on m_2 , the point U together with a point W chosen so that $|UW| \neq |VU'|$, one obtains the midpoint N of the segment $\overline{VU'}$. The two points U, V' cannot be used: the central symmetry σ_O sends $V \mapsto V'$ and $U' \mapsto U$, so $|UV'| = |VU'|$ always, the four points V, U', U, V' form a rectangle, and the midpoint construction degenerates (one of the two pairs of lines to be intersected consists of parallel lines). If with the chosen point W two of the lines still turn out to be parallel, one changes W .

Step 3 — the perpendicular through the centre. The centre O does not lie on m_1 : if it did, V and U' would be antipodal, that is $V = U$, which is excluded. One may therefore apply Construction 5.4 to the triple V, U', N and to the point O , obtaining the line n through O parallel to m_1 . Since $m_1 \perp \ell$, also $n \perp \ell$.

Step 4 — the midpoint of the chord. The line n is the perpendicular from the centre to the chord $\overline{UV}$; since the perpendicular from the centre to a chord is its perpendicular bisector (Chapter 2), it meets ℓ at the midpoint M of $\overline{UV}$. The point M is constructed with the straightedge alone as the intersection of n and ℓ .

Step 5 — conclusion. One now has the collinear triple U, V, M with M the midpoint: Construction 5.4 applied to it and to the point P yields the parallel to ℓ through P . □

5.3.2 The model: what the straightedge can do

To prove a theorem about what can be constructed one must first say precisely which operations are allowed. The compass needed this in Theorem 5.6 (the convention on lines); the straightedge needs it here.

Definition 5.3: Operations of the straightedge with a fixed circle. Let there be given, once and for all, a circle Ω with centre O and radius $\rho > 0$, *drawn*, and the point O , *constructed*. Starting from a finite set of constructed points the following operations are allowed:

- (R1) draw the line through two distinct constructed points;
- (R2) construct the intersection point of two drawn non-parallel lines;
- (R3) construct the intersection points of a drawn line with Ω (the line is drawn, Ω is drawn: the common points are visible and are *read off*);
- (R4) choose an arbitrary point of the plane (possibly constrained to a region bounded by objects already drawn), or of a drawn line, or of Ω , avoiding a finite number of positions declared forbidden by the construction in progress; if a choice produces a detectable degeneracy (two lines that turn out to be parallel) it is repeated;
- (R5) on a drawn line, given three constructed points, recognise which of the first two lies on the same side as the third (read off the order of the points on the line). It is needed in order to speak of *rays*, and the classical constructions use it tacitly (“in the direction of...”).

Definition 5.4: Circles in the geometry of the straightedge. In the geometry of the straightedge a circle is regarded as *constructed* when its centre M and one of its points A have been constructed; its radius is the segment $\overline{MA}$. One does not require the curve to be drawn: for every drawn line one asks to construct its common points with the circle, and for two circles so given one asks to construct their common points. It is the convention symmetric to that of Theorem 5.6, in which a line is constructed when two of its points are.

Remark 5.13: What the compass does, and the straightedge alone does not. In ruler-and-compass constructions the compass draws the circle with centre a constructed point passing through a constructed point (Euclid’s Postulate III: “to describe a circle with any centre and distance”). Operations (O1) and (O2) of Adler’s reduction (Section 5.2) are therefore exactly “circle $\cap$ circle” and “line $\cap$ circle” for circles given by a centre and a point, while (O3) coincides with (R2). The Poncelet–Steiner theorem is proved as soon as (O2) and (O1) are executable with (R1)–(R5): this is what Lemmas 5.7 and 5.8 do, resting on the lemmas that precede them. With the straightedge alone, without Ω , one constructs only rational expressions of the data: the reading (R3) is the only source of square roots.

In what follows “line through the centre” means a line passing through O : it meets Ω at two antipodal points U, V , of which O is the midpoint, so Construction 5.4 with the triple (U, V, O) directly yields the parallel to it through any point.

5.3.3 The second link: parallels to an arbitrary line, and perpendiculars

Lemma 5.1: Parallel to an arbitrary line, with the straightedge alone. Given Ω with its centre O , a drawn line g and a point $P \notin g$, the parallel to g through P is constructible with the operations (R1)–(R4).

Construction 5.7: — Parallel to an arbitrary line (Steiner). **Case 1:** $O \in g$. The line g meets Ω at two antipodal points U, V (R3) of which O is the midpoint: Construction 5.4 with the triple (U, V, O) and the point P gives the parallel.

Case 2: $O \notin g$ (g secant, tangent or external: the construction does not use the position of g relative to Ω).

Step 1. Choose a point $M \in g$ (R4). The line $m = OM$ passes through the centre and meets Ω at U, V (R3), with O the midpoint of $\overline{UV}$.

Step 2. Choose $X \in \Omega$ (R4) with $X \notin m$ and $OX \not\perp m$ (four forbidden positions: the endpoints of the diameter m and those of the perpendicular diameter).

Step 3. With Construction 5.4 applied to the triple (U, V, O) and to the point X draw the line p through X parallel to m ; it meets Ω at a second point Y (R3) and g at a point A (R2).

Step 4. The lines XO and YO meet Ω at the antipodes X' and Y' (R3); the line $p' = X'Y'$ meets g at a point B (R2).

Step 5. M is the midpoint of $\overline{AB}$: Construction 5.4 with the triple (A, B, M) and the point P gives the parallel to g through P . $\square$

Proof. The points exist. The line p contains the point X of Ω ; it is tangent to Ω if and only if $OX \perp p$, that is (since $p \parallel m$) if and only if $OX \perp m$, excluded at Step 2: hence p is secant and $Y \neq X$ exists. Moreover $m \not\parallel g$, because $m \cap g = \{M\}$ and $m \neq g$ (one has $O \in m$, $O \notin g$); hence $p \not\parallel g$ and A exists. The central symmetry σ_O with centre O maps Ω to itself, $X \mapsto X'$, $Y \mapsto Y'$, and lines to parallel lines: $\sigma_O(p) = p'$, so $p' \parallel m$, and $p' \neq p$ because $\sigma_O(p) = p$ would amount to $O \in p$, that is $X \in m$, excluded. As above, B exists.

M is the midpoint of $\overline{AB}$. Coordinates with $O = (0, 0)$ and $m =$ the x axis: then $M = (\mu, 0)$ with $\mu \neq 0$, and p is the line $y = h$ with $h \neq 0$ because $X \notin m$. The points $X = (x_1, h)$, $Y = (x_2, h)$ satisfy $x_i^2 + h^2 = \rho^2$, hence $x_2 = -x_1$, with $x_1 \neq 0$ (if $x_1 = 0$ one would have $OX \perp m$). The antipodes are $X' = (-x_1, -h)$ and $Y' = (x_1, -h)$, and p' is the line $y = -h$. The line g passes through M and is not parallel to the x axis: $g = \{(\mu + t\alpha, t\beta)\}$ with $\beta \neq 0$. Then

$$A = g \cap \{y = h\} = \left(\mu + \frac{h\alpha}{\beta}, h\right), \quad B = g \cap \{y = -h\} = \left(\mu - \frac{h\alpha}{\beta}, -h\right),$$

whence $\frac{1}{2}(A+B) = (\mu, 0) = M$, and $A \neq B$ because $h \neq -h$. The triple (A, B, M) is therefore admissible for Construction 5.4, and $P \notin AB = g$ by hypothesis.

Synthetic reading. The lines p and p' are symmetric with respect to O , hence parallel to m and equidistant from m on opposite sides; the central symmetry σ_M with centre M fixes g and m and exchanges p with p' , hence exchanges A with B . $\square$

Corollary 5.2: Parallels through a point of the line, and to secants. If $P \in g$ the parallel to g through P is g itself: Lemma 5.1 holds for every point P , and in the lemmas that follow the case “the point already lies on the line” is always covered without any construction. In particular Theorem 5.8 is a particular case of Lemma 5.1.

Proof. Through a point of a line there passes only one parallel to the line, the line itself. A secant is a line like any other: Lemma 5.1 never uses the position of g relative to Ω , and therefore holds for secants too, with $O \in g$ or $O \notin g$. $\square$

Lemma 5.2: Perpendicular to a line through a point, with the straightedge alone.

Given Ω with its centre, a drawn line g and an arbitrary point P , the perpendicular to g through P is constructible.

Construction 5.8: — Perpendicular with the straightedge alone. **Step 1.** Choose $X \in \Omega$ (R4) with $OX \not\parallel g$ and $OX \not\perp g$ (four forbidden positions).

Step 2. With Lemma 5.1 draw the line g' through X parallel to g ($g' = g$ if $X \in g$); it meets Ω at a second point Y (R3).

Step 3. The line XO meets Ω at the antipode X'' (R3).

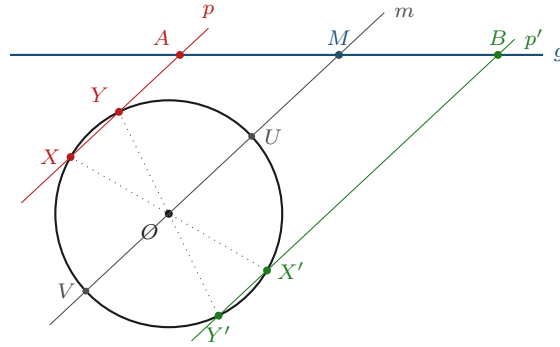


Figure 5.2: Lemma 5.1, Case 2, with Ω the unit circle, $M = (1.5, 1.4)$ on an external line g and X at 150° . The line p through X parallel to $m = OM$ meets Ω at Y and g at $A = (0.098, 1.4)$; the line p' through the antipodes X', Y' meets g at $B = (2.902, 1.4)$. The midpoint of $\overline{AB}$ is $(1.5, 1.4) = M$: the line g is now “directed” and the parallel through any point follows from Construction 5.4. Coordinates computed by hand and verified numerically.

Step 4. The line $n = YX''$ is perpendicular to g' , hence to g .

Step 5. The parallel to n through P (Lemma 5.1; it is n itself if $P \in n$) is the required perpendicular. $\square$

Proof. The line g' contains $X \in \Omega$ and is tangent if and only if $OX \perp g'$, that is $OX \perp g$, excluded: it is secant and Y exists. Moreover $O \notin g'$, otherwise $g' = OX$ would be parallel to g , excluded; it follows that $Y \neq X''$ (if $Y = X''$ the line $g' = XX''$ would pass through O). The segment $\overline{XX''}$ is a diameter and $Y \in \Omega$ is different from its endpoints: by Thales’ theorem (Theorem 4.4) the angle $\widehat{XYX''}$ is a right angle, that is $YX'' \perp XY = g'$. Finally $g' \parallel g$ gives $n \perp g$, and every parallel to n is perpendicular to g . $\square$

5.3.4 The third link: translating, halving, transporting

Lemma 5.3: Translation of a segment. Given two points $A \neq B$ and a point P , the point $Q = P + (B - A)$, fourth vertex of the parallelogram $ABQP$, is constructible.

Proof. Case $P \notin AB$. Draw the parallel q_1 to AB through P and the parallel q_2 to AP through B (Lemma 5.1); $Q = q_1 \cap q_2$ (R2), and the two lines are not parallel because $AB \nparallel AP$. In vector form $q_1 = \{P + t(B - A)\}$, $q_2 = \{B + u(P - A)\}$; since $P - A = (P - B) + (B - A)$, the difference between a point of q_1 and one of q_2 equals

$$P + t(B - A) - B - u(P - A) = (1 - u)(P - A) - (1 - t)(B - A),$$

and the vectors $P - A$, $B - A$ are linearly independent: the difference vanishes only for $t = u = 1$, that is $Q = P + (B - A)$.

Case $P \in AB$. Choose $P' \notin AB$ (R4), construct $Q' = P' + (B - A)$ by the previous case, then $Q = P + (Q' - P')$ again by the previous case: this is legitimate because the line $P'Q'$ is parallel to AB and distinct from it, hence does not contain P ; and $Q' - P' = B - A$. $\square$

Lemma 5.4: Midpoint of an arbitrary segment. Given $A \neq B$, the midpoint of $\overline{AB}$ is constructible.

Proof. Choose $P \notin AB$ (R4) and draw the parallel ℓ' to AB through P (Lemma 5.1). Apply Construction 5.6 to the lines $\ell = AB$, ℓ' and to the points A, B , choosing $A', B' \in \ell'$ with $|A'B'| \neq |AB|$: if $|A'B'| = |AB|$ one of the two pairs of lines $(AB', A'B)$, (AA', BB') would consist of opposite sides of a parallelogram and would not meet. The condition is not checkable a priori at this point of the chain (the transport of lengths arrives with Lemma 5.5), but the degeneracy is detectable a posteriori — two drawn lines turn out to be parallel — and in that case one replaces B' by another point of ℓ' (R4). Finally the line KP of the construction is never parallel to ℓ : in the coordinates of its proof the ordinates of K and P are $1/(1+d)$ and $1/(1-d)$, equal only for $d = 0$, that is $A' = B'$, excluded. $\square$

Lemma 5.5: Transport of a segment onto a ray. *Given a segment $\overline{AB}$ and a ray with origin P passing through a point $H \neq P$, the point S of the ray with $|PS| = |AB|$ is constructible.*

Construction 5.9: — Transport of a segment. Step 1. $K = P + (B - A)$ (Lemma 5.3); hence $|PK| = |AB|$.

Step 2. If $K \in PH$: choose $H' \notin PH$ (R4), construct with the following steps the point S' of the ray PH' with $|PS'| = |PK|$, and then transport $\overline{PS'}$ onto the ray PH ; in the second application the point $K_2 = S'$ does not lie on PH , and there is no further recursion. Assume therefore $K \notin PH$.

Step 3. $\tilde{U} = O + (H - P)$ and $\tilde{V} = O + (K - P)$ (Lemma 5.3). The lines $O\tilde{U}$ and $O\tilde{V}$ pass through the centre: let U and V be the points where the rays $O\tilde{U}$ and $O\tilde{V}$ meet Ω (R3 for the two intersection points, R5 to choose the one on the side of $\tilde{U}$, resp. $\tilde{V}$, with respect to O).

Step 4. The parallel through K to the line UV (Lemma 5.1) meets the line PH at the point S (R2). $\square$

Proof. Let $u = (H - P)/|H - P|$ and $v = (K - P)/|K - P|$, linearly independent by Step 2. By construction $U = O + \rho u$ and $V = O + \rho v$, hence $U - V = \rho(u - v)$ and the triangle OUV is isosceles. The points of the line PH are $P + tu$; those of the parallel through K to UV are $K + \lambda(U - V) = P + |PK|v + \lambda\rho(u - v)$. The equality $tu = (|PK| - \lambda\rho)v + \lambda\rho u$, by the independence of u and v , gives $\lambda\rho = |PK|$ and $t = \lambda\rho = |PK|$: hence $S = P + |PK|u$ lies on the ray PH (positive coefficient) and $|PS| = |PK| = |AB|$. In synthetic terms: the triangle PKS has its sides parallel to those of the isosceles triangle OVU , hence it is isosceles with $|PS| = |PK|$. $\square$

Lemma 5.6: Fourth proportional (Thales with the straightedge alone). *Given three lengths $m, n, s > 0$, assigned as constructed segments, a segment of length $x = ns/m$ is constructible.*

Proof. Choose a point Z and two points H_1, H_2 such that the rays ZH_1 and ZH_2 are distinct and not opposite (R4). With Lemma 5.5 construct M_1, N_1 on the ray ZH_1 with $|ZM_1| = m$, $|ZN_1| = n$, and S_2 on the ray ZH_2 with $|ZS_2| = s$. The parallel through N_1 to the line M_1S_2 (Lemma 5.1; it is M_1S_2 itself if $m = n$) meets the line ZH_2 at a point X_2 . With u, v the unit vectors of the two rays: $M_1 = Z + mu$, $N_1 = Z + nu$, $S_2 = Z + sv$; the points of the parallel are $Z + (n - \lambda m)u + \lambda sv$ and lie on the line ZH_2 if and only if $\lambda = n/m$; then $X_2 = Z + \frac{ns}{m}v$, on the ray ZH_2 , with $|ZX_2| = ns/m$.

Ratios with sign. The same construction realises the homothety with centre Z and ratio $n/m > 0$: it sends the point of coordinate t on the line ZH_2 to the point of coordinate tn/m , whatever the sign of t , because parallels preserve ratios of oriented segments. It is in this form — the homothety with centre Z that sends a given point to a given point of the same ray is

| realised by a parallel — that the lemma is used in the two following lemmas. □

5.3.5 The fourth link: intersecting a line with an undrawn circle

Lemma 5.7: Line and undrawn circle. *Let Ω with its centre O , a drawn line g and a circle $\mathcal{K}$ given by its centre M and one of its points A ($r = |MA| > 0$) be given. The points of $g \cap \mathcal{K}$ are constructible, and the construction establishes whether they are two, one or none.*

The idea is a similarity: $T(Z) = O + \frac{\rho}{r}(Z - M)$ maps $\mathcal{K}$ onto Ω and g onto a parallel line g^* ; on Ω , which is drawn, the points $g^* \cap \Omega$ are read off (R3); one comes back with T^{-1} . All that is needed is to realise T and T^{-1} with parallels and fourth proportionals.

Construction 5.10: — Line and undrawn circle. **Step 1.** If g passes through M and through A , the required points are the two points of g at distance r from M : they are constructed with Lemma 5.5 on the two rays of g issuing from M (the second ray is determined by the point $M + (M - A)$, Lemma 5.3). Done.
Step 2. Otherwise choose $P \in g$ (R4) with $P \neq M$ and $P \notin MA$ (at most two forbidden positions on g).
Step 3. $P_1 = O + (P - M)$ and $A_1 = O + (A - M)$ (Lemma 5.3); by Step 2, $P_1 \notin OA_1$.
Step 4. Let E_1 be the point where the ray OA_1 meets Ω (R3, R5): $|OE_1| = \rho$. If $r = \rho$ one has $E_1 = A_1$, the similarity is a translation, and in Steps 5 and 8 one sets $P^* = P_1$, $X_1 = X^*$.
Step 5. The parallel through E_1 to the line A_1P_1 (Lemma 5.1) meets the line OP_1 at the point P^* (R2).
Step 6. Through P^* draw the parallel g^* to g (Lemma 5.1; $g^* = g$ if $P^* \in g$).
Step 7. Read off the points of $g^* \cap \Omega$ (R3): none, one (g^* tangent) or two, X^* and Y^* .
Step 8. For each point read off, for example X^* : if $X^* \notin OA_1$, the parallel through A_1 to the line E_1X^* (Lemma 5.1) meets the line OX^* at a point X_1 (R2); if instead $X^* \in OA_1$, then $X^* = E_1$ or X^* is the antipode of E_1 , and one sets respectively $X_1 = A_1$ or $X_1 = O + (O - A_1)$ (Lemma 5.3).
Step 9. $X = M + (X_1 - O)$ (Lemma 5.3). The points X (and Y) so obtained are the points of $g \cap \mathcal{K}$. □

Proof. The points exist. At Step 5 the parallel to A_1P_1 is not parallel to OP_1 because $A_1 \notin OP_1$; at Step 8 the line E_1X^* is not parallel to OX^* because $E_1 \notin OX^*$.

Step 5 realises T . Let h be the homothety with centre O and ratio ρ/r . Since $|OA_1| = |MA| = r$ and E_1 lies on the ray OA_1 with $|OE_1| = \rho$, one has $h(A_1) = E_1$. The homothety maps the line A_1P_1 to the line through E_1 parallel to A_1P_1 and maps P_1 to a point of the line OP_1 : hence $h(P_1)$ is the intersection of that parallel with OP_1 , that is P^* (Lemma 5.6, ratios with sign). It follows that $P^* = O + \frac{\rho}{r}(P_1 - O) = O + \frac{\rho}{r}(P - M) = T(P)$.

$g^* = T(g)$ and $\Omega = T(\mathcal{K})$. For all Z, Z' one has $T(Z) - T(Z') = \frac{\rho}{r}(Z - Z')$: T maps lines to parallel lines, so $T(g)$ is the line through $T(P) = P^*$ parallel to g , that is g^* . Moreover $|T(Z) - O| = \frac{\rho}{r}|Z - M|$, hence $|Z - M| = r$ if and only if $|T(Z) - O| = \rho$.

Steps 8 and 9 realise T^{-1} . T is a bijection of the plane (a similarity), therefore $T(g \cap \mathcal{K}) = T(g) \cap T(\mathcal{K}) = g^* \cap \Omega$: the points read off at Step 7 are exactly the images of the required points and are equal in number. Let $X^* \in g^* \cap \Omega$. The inverse homothety h^{-1} (centre O , ratio r/ρ) sends $E_1 \mapsto A_1$ and X^* to the point of the line OX^* for which $A_1h^{-1}(X^*) \parallel E_1X^*$: it is the point X_1 of Step 8 (in the case $X^* \in OA_1$ one has directly $h^{-1}(E_1) = A_1$ and $h^{-1}(-E_1) = -A_1$ with respect to O). Hence $X_1 = O + \frac{r}{\rho}(X^* - O)$ and

$$X = M + (X_1 - O) = M + \frac{r}{\rho}(X^* - O) = T^{-1}(X^*) \in g \cap \mathcal{K}.$$

The distinction between “tangent” and “secant with two very close points” at Step 7 is a reading of the figure, exactly as for the compass: the theorem promises nothing more than the compass promises. $\square$

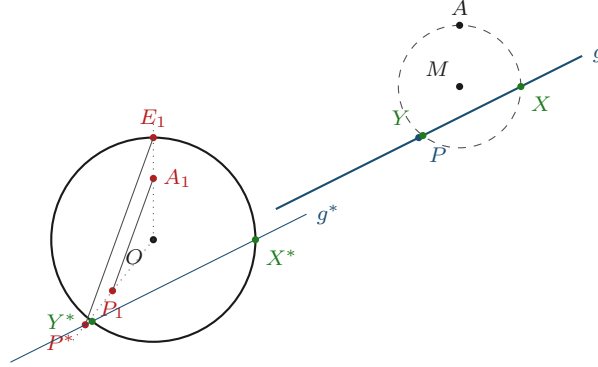


Figure 5.3: Lemma 5.7: Ω the unit circle, $\mathcal{K}$ with centre $M = (3, 1.5)$ and radius $r = 0.6$ (dashed: it is not drawn), g of equation $y = 0.5x - 0.3$, $P = (2.6, 1)$. The similarity $T(Z) = O + \frac{5}{3}(Z - M)$ is realised by $P_1 = (-0.4, -0.5)$, $A_1 = (0, 0.6)$, $E_1 = (0, 1)$ and by the parallel through E_1 to A_1P_1 , which gives $P^* = (-\frac{2}{3}, -\frac{5}{6})$. The line g^* through P^* parallel to g meets Ω at $X^* = (1, 0)$ and $Y^* = (-0.6, -0.8)$, read off on the drawn circle; going back, $X = (3.6, 1.5)$ and $Y = (2.64, 1.02)$ lie on g and at distance 0.6 from M . Coordinates computed by hand and verified numerically.

5.3.6 The fifth link: two undrawn circles

Lemma 5.8: Two undrawn circles. *Let two circles be given: $\mathcal{K}_A$, with centre A and radius $a = |AA_0|$, and $\mathcal{K}_B$, with centre B and radius $b = |BB_0|$, with $A \neq B$. The points of $\mathcal{K}_A \cap \mathcal{K}_B$ are constructible, and the construction establishes whether they are two, one or none. (If $A = B$ the circles are concentric: they coincide if $a = b$, otherwise they are disjoint; there is nothing to construct.)*

The idea is the *radical axis* of the two circles (Chapter 7); the proof that follows is self-contained. Let $c = |AB|$ and $u = (B - A)/c$. A point X lies on both circles if and only if $|X - A|^2 = a^2$ and $|X - B|^2 = b^2$; subtracting, $2\langle X - A, B - A \rangle = c^2 + a^2 - b^2$, that is, the projection of X onto the line AB has coordinate

$$q = \frac{c^2 + a^2 - b^2}{2c} = \frac{c}{2} + \frac{(a - b)(a + b)}{2c}$$

with respect to the origin A and the unit vector u . Hence $\mathcal{K}_A \cap \mathcal{K}_B = \ell_q \cap \mathcal{K}_A$, where ℓ_q is the perpendicular to AB at the point $Z = A + qu$; and conversely every $X \in \ell_q \cap \mathcal{K}_A$ satisfies $|X - B|^2 = |X - A|^2 - 2\langle X - A, B - A \rangle + c^2 = a^2 - (c^2 + a^2 - b^2) + c^2 = b^2$. It remains to construct Z with the straightedge; then one concludes with Lemma 5.7.

Construction 5.11: — Two undrawn circles. Step 1. N = midpoint of $\overline{AB}$ (Lemma 5.4).

Step 2. With Lemma 5.5 construct on the ray AB the points P_a, P_b with $|AP_a| = a$, $|AP_b| = b$, and on the opposite ray (with origin A , through the point $A + (A - B)$) the point P_{-b} with $|AP_{-b}| = b$.

Step 3. $D_1 = P_a + (P_{-b} - A)$, $D_2 = P_a + (P_b - A)$, $C_2 = B + (B - A)$ (Lemma 5.3); in coordinates along u : $D_1 = A + (a - b)u$, $D_2 = A + (a + b)u$, $C_2 = A + 2cu$.

Step 4. Choose $W \notin AB$ (R4); with Lemma 5.5 construct on the ray AW the points V_1, V_2 with $|AV_1| = |AD_2| = a + b$ and $|AV_2| = |AC_2| = 2c$.

Step 5. If $D_1 = A$ (that is $a = b$) set $X = A$. Otherwise the parallel through V_1 to the line D_1V_2 (Lemma 5.1; if $a + b = 2c$ one has $V_1 = V_2$ and the parallel is D_1V_2 itself) meets the line AB at the point X (R2).

Step 6. $Z = N + (X - A)$ (Lemma 5.3); if $X = A$ one sets directly $Z = N$.

Step 7. ℓ_q = perpendicular to AB through Z (Lemma 5.2).

Step 8. The points of $\ell_q \cap \mathcal{K}_A$ (Lemma 5.7) are the required points. $\square$

Proof. At Step 5 the homothety h with centre A and ratio $\frac{a+b}{2c} > 0$ sends $V_2 \mapsto V_1$ (same ray, lengths $2c \mapsto a + b$) and sends D_1 to the point of the line AB of coordinate $(a - b)\frac{a+b}{2c}$, whatever the sign of $a - b$ (Lemma 5.6, ratios with sign). Since h maps the line D_1V_2 to the line through V_1 parallel to D_1V_2 , and $D_1 \neq V_2$ (they lie on distinct lines through A), the point X of Step 5 is $h(D_1) = A + \frac{(a-b)(a+b)}{2c}u$; if $D_1 = A$ one trivially has $h(A) = A$. Hence at Step 6

$$Z = N + (X - A) = A + \left(\frac{c}{2} + \frac{a^2 - b^2}{2c}\right)u = A + qu,$$

and ℓ_q is the radical axis. The conclusion is the argument preceding the construction. The number of solutions is that of $\ell_q \cap \mathcal{K}_A$, determined by Lemma 5.7: two if $|a - b| < c < a + b$, one if $c = a + b$ or $c = |a - b|$, none otherwise. $\square$

5.3.7 The proof of the Poncelet–Steiner theorem

Proof. (of Theorem 5.7.) By Adler’s reduction (Section 5.2), a ruler-and-compass construction starting from a finite set of given points is a finite sequence of points $Q_1, \dots, Q_k$, each of which is either an arbitrary point (of the plane, of a region already bounded, of a line already drawn, of a circle already drawn) or the result of an operation (O1), (O2), (O3) on objects already constructed: lines through two constructed points, and circles with constructed centre passing through a constructed point. We prove by induction on j that every Q_j is constructible with the straightedge and Ω .

- An arbitrary point of the plane, of a region already bounded or of a drawn line remains an arbitrary point (R4). An arbitrary point of a circle with centre M and point A is obtained by choosing an arbitrary line through M and applying Lemma 5.7. An arbitrary point of a region bounded also by undrawn circles (“inside the circle with centre M through A ”) is chosen with (R4) and checked with Lemma 5.7 and (R5) to lie on the required side.
- (O3), intersection of two lines through constructed points: (R1) and (R2).
- (O2), intersection of a line through two constructed points with a circle of constructed centre M and point A : Lemma 5.7.
- (O1), intersection of two circles given by a centre and a point: Lemma 5.8.

All the lemmas use only (R1)–(R5), Constructions 5.4 and 5.6 and the lemmas that precede them, in the order 5.1 $\rightarrow$ 5.2 $\rightarrow$ 5.3 $\rightarrow$ 5.4 $\rightarrow$ 5.5 $\rightarrow$ 5.6 $\rightarrow$ 5.7 $\rightarrow$ 5.8, with no circularity. $\square$

Remark 5.14: What the theorem does not say. The set of constructible points does not change: they are still exactly the points with coordinates in iterated quadratic extensions of the field generated by the data (Section 5.13). Only the instrument changes. Moreover the hypothesis “with its centre” is essential: with Ω alone drawn, without the centre, the centre itself is not constructible with the straightedge (the fact is proved, without attribution, in Smogorzhevskii 1961, §18; the traditional attribution to D. Cauer, 1912, has not been verified on a primary source and is therefore not asserted here).

INFO

Independent numerical check. The constructions of Lemmas 5.1–5.8 have been implemented in a script (file `verifica_poncelet_steiner.py`, attached to the sources) which uses *only* the operations (R1)–(R5) and compares the result with the direct computation: all the degenerate cases listed in the constructions and 200 random configurations of Ω , lines, points and circles, with maximum error $2.9 \cdot 10^{-9}$ (pure rounding) and always the correct number of intersection points; a second independent implementation on 310 configurations, including near-degeneracies down to 10^{-9} , gave the same outcome. A numerical check is not a proof: here it serves to rule out slips in the computations in coordinates.

5.4. The problem of Apollonius

Definition 5.5: Problem of Apollonius. Given three objects among points, lines and circles in the plane, construct (with ruler and compass) all the circles simultaneously tangent to the three given objects.

Remark 5.15: History and number of solutions. *The problem was posed by Apollonius of Perga (3rd–2nd century BC) in the lost work *De Tactionibus*, reconstructed by François Viète in 1600. In the generic case of three given circles there exist **eight** solution circles, grouped into four conjugate pairs (for every solution tangent “from outside” to all three there is a mirror one tangent “from inside”). Degenerate cases (lines, points, mutually tangent circles) reduce the number of solutions.*

General solving strategy by inversion. Step 1. Choose a “convenient” centre of inversion: for instance, if a solution must pass through a given point P among the three objects, invert with respect to a circle centred at P . By Theorem 5.3, every circle or line that should pass through P turns into a line.

Step 2. In the transformed plane, the problem typically reduces to constructing a line or a circle tangent to (at most) two lines and a circle, or to two concentric circles or two parallel lines: elementary configurations, solvable with Construction 2.3 of Chapter 2 or with the axis of symmetry.

Step 3. Apply the inverse inversion (which is ι itself, being an involution) to the solution found, obtaining the tangent circle sought in the original problem. Tangency is preserved by Corollary 5.1: it is this, and nothing else, that guarantees that the transformed solution solves the original problem. $\square$

Remark 5.16: The particular case of three mutually tangent circles. *When the three given circles are already mutually tangent (the most frequent case in applications, known as the kissing circles problem), the problem of Apollonius reduces exactly to computing the radius of the fourth circle tangent to all three, solved in closed form by Descartes’ theorem (Section 5.9): in this case the construction by inversion is not even needed, the algebraic formula suffices.*

5.4.1 Complete example: construction for the Point–Line–Circle case

To make the general solving strategy concrete, we develop in full the PLC case (a point P , a line r not through P , a circle Γ with centre Q and radius ρ , with $P \notin \Gamma$ and r not tangent to Γ): we seek the circle ω through P , tangent to r and tangent to Γ .

Explicit construction (PLC case). Step 1 (inversion centred at P). Choose a circle of inversion $C(P, \rho_0)$ of arbitrary radius ρ_0 (for instance $\rho_0 = 1$). By Theorem 5.3: the line r (not through P) turns into a circle r^* through P ; the circle Γ (not through P , by hypothesis) turns into a circle Γ^* still not through P . The unknown circle ω , having to pass through P , turns instead into a **line** ω^* (Theorem 5.3(b)).

Step 2 (reduced problem). In the transformed plane the problem becomes: construct a line ω^* tangent both to r^* and to Γ^* , that is, a **common tangent to two circles** — an elementary problem, solved with ruler and compass by constructing the homothety carrying r^* to Γ^* (or the external/internal centre of homothety of the two circles) and drawing from it the tangents to one of them.

Step 3 (inverse inversion). Applying ι again (an involution) to the line ω^* found, one obtains the circle ω sought, passing through P as required: by Theorem 5.3(b) the image of a line not through P is always a circle through P . $\square$

Remark 5.17: Numerical check of the PLC case. With $P = (0, 0)$, line $r : x = 4$ (general equation $ax + by = c$ with $a = 1, b = 0, c = 4$) and circle Γ with centre $Q = (0, 6)$ and radius $\rho = 2$ (so $P \notin \Gamma$, since $|PQ| = 6 \neq 2$, and r is not tangent to Γ , since Γ does not touch the line $x = 4$). Inversion $C(P, 1)$: the line $x = 4$ turns, by the formula of Theorem 5.3(b), into the circle $x^{*2} + y^{*2} - \frac{1}{4}x^* = 0$, that is $C((\frac{1}{8}, 0), \frac{1}{8})$. The circle Γ , of equation $x^2 + y^2 - 12y + 32 = 0$ (from $x^2 + (y - 6)^2 = 4$), turns by the formula of Theorem 5.3(c) into $32(x^{*2} + y^{*2}) - 12y^* + 1 = 0$, a circle with centre $(0, \frac{12}{64}) = (0, 0.1875)$ and radius $R_2 = \frac{k^2}{|p|}\rho = \frac{2}{32} = 0.0625$ (Lemma on the image of a circle, previous section, with $p = \text{pow}_\Gamma(P) = |PQ|^2 - \rho^2 = 36 - 4 = 32$).

Step 2, completed. The two image circles, r^* with centre $(0.125, 0)$ and radius 0.125 , and Γ^* with centre $(0, 0.1875)$ and radius 0.0625 , have an internal common tangent (the one giving the simplest solution) of equation $y = 0.125$, that is, $0 \cdot x^* + 1 \cdot y^* = 0.125$ (checked: distance from $(0.125, 0)$ equals 0.125 , distance from $(0, 0.1875)$ equals 0.0625).

Step 3, completed. Inverting this line ($a = 0, b = 1, c = 0.125$) with the formula of Theorem 5.3(b), ω has centre $k^2(a, b)/(2c) = (0, 4)$ and radius $k^2/(2|c|) = 4$.

Final numerical check (independent script). ω : centre $(0, 4)$, radius 4 . (i) Passes through $P = (0, 0)$: $|(0, 4) - (0, 0)| = 4 = \text{radius}$. $\checkmark$ (ii) Tangent to the line $x = 4$: the distance of $(0, 4)$ from the line is $4 = \text{radius}$. $\checkmark$ (iii) Tangent to Γ : distance between centres $|(0, 4) - (0, 6)| = 2$, and $|4 - \rho| = |4 - 2| = 2$ (internal tangency, Γ contained in ω). $\checkmark$ The three conditions of the PLC problem are all satisfied by ω . This numerical example shows how the inversion method turns an apparently mixed problem (point, line, circle) into a homogeneous one (two circles, common tangent), always elementarily constructible.

Remark 5.18: Gergonne's alternative method (poles and polars). Joseph Gergonne published in 1814 a solution of the problem of Apollonius in the general CCC case (three circles) entirely different from the one by inversion: it exploits the theory of **poles and polars** with respect to the three given circles. The key idea is that the radical centre of the three circles (the point of equal power with respect to all three, Chapter 3) allows one to construct, for each given circle, the polar of the radical centre with respect to it; the intersection of this polar with the given circle identifies the tangency points of the two “principal” solutions. Gergonne's method is purely projective (it does not require inversion) and is historically important because it anticipates concepts of modern projective geometry (pole–polar duality) by almost a century.

5.5. Ptolemy's theorem

Theorem 5.9: Ptolemy's theorem. For every cyclic quadrilateral $ABCD$ (inscribed in a circle), the product of the diagonals equals the sum of the products of the opposite sides:

$$|AC| \cdot |BD| = |AB| \cdot |CD| + |BC| \cdot |DA|.$$

Proof 1 — by similar triangles (classical proof). Construct the point K on the diagonal BD such that $\angle BAK = \angle CAD$. Since $\angle ABK = \angle ABD = \angle ACD$ (inscribed angles on the same arc AD), the triangles $\triangle ABK$ and $\triangle ACD$ are similar (AA), whence

$$\frac{AB}{AC} = \frac{BK}{CD} \implies AB \cdot CD = AC \cdot BK. \quad (\text{i})$$

By construction $\angle DAK = \angle BAC$: both are obtained by adding $\angle KAC$ to $\angle DAK$ and to $\angle BAK = \angle CAD$ respectively, which are congruent by construction. Moreover $\angle ADK = \angle ADB = \angle ACB$ (inscribed angles on the arc AB): hence $\triangle ADK \sim \triangle ACB$ (AA), whence

$$\frac{AD}{AC} = \frac{DK}{CB} \implies AD \cdot CB = AC \cdot DK. \quad (\text{ii})$$

Adding (i) and (ii):

$$AB \cdot CD + AD \cdot CB = AC \cdot (BK + DK) = AC \cdot BD,$$

since $K \in BD$ by construction. □

Proof 2 — by circle inversion. Apply an inversion with centre A and arbitrary radius k to the points B, C, D , obtaining B^*, C^*, D^* . By Theorem 5.2:

$$|B^*C^*| = \frac{k^2 |BC|}{AB \cdot AC}, \quad |C^*D^*| = \frac{k^2 |CD|}{AC \cdot AD}, \quad |B^*D^*| = \frac{k^2 |BD|}{AB \cdot AD}.$$

Since B, C, D lie on a circle through the pole A , by Theorem 5.3(b) their images B^*, C^*, D^* are **collinear**. Moreover the cyclic order A, B, C, D on the circle guarantees that C^* lies between B^* and D^* : for points collinear in this order the equality

$$|B^*C^*| + |C^*D^*| = |B^*D^*|$$

holds, the equality case of the triangle inequality (A8). Substituting the three expressions above and multiplying throughout by $AB \cdot AC \cdot AD/k^2$:

$$AD \cdot BC + AB \cdot CD = AC \cdot BD,$$

which is Ptolemy's theorem. □

Corollary 5.3: Ptolemy's inequality. For any four points A, B, C, D of the plane,

$$|AC| \cdot |BD| \leq |AB| \cdot |CD| + |BC| \cdot |DA|,$$

with equality if and only if $ABCD$ is cyclic in that order (or the four points are collinear in that order).

Proof. Repeat Proof 2 without the cyclicity hypothesis. The three formulas for the transformed distances hold in any case (Theorem 5.2). What changes is the next step: the three points B^*, C^*, D^* need no longer be collinear, and for them only the triangle inequality (A8) holds, in the form

$$|B^*D^*| \leq |B^*C^*| + |C^*D^*|,$$

with equality if and only if C^* belongs to the segment $\overline{B^*D^*}$, that is, if and only if the three points are collinear in that order. But by Theorem 5.3 the collinearity of the images is equivalent to B, C, D lying on a circle through A (or on a line), that is, to the four points being concyclic (or collinear). Substituting and rearranging gives the conclusion. $\square$

Proof 3 — trigonometric. This is the route closest to the one Ptolemy himself followed in building his tables of chords: it translates the statement into a trigonometric identity and verifies it by product-to-sum formulas.

Step 1 — parametrising the sides by arcs. Let $ABCD$ be inscribed in a circle of radius R , the vertices in this cyclic order. Call $2\alpha, 2\beta, 2\gamma, 2\delta$ the central angles subtended respectively by the arcs AB, BC, CD, DA ; they cover the whole turn, hence

$$\alpha + \beta + \gamma + \delta = \pi.$$

By the extended law of sines (Theorem 10.5, Chapter 10), a chord subtending a central angle 2θ measures $2R \sin \theta$. Hence

$$AB = 2R \sin \alpha, \quad BC = 2R \sin \beta, \quad CD = 2R \sin \gamma, \quad DA = 2R \sin \delta,$$

and for the diagonals, which subtend the sum of two consecutive arcs,

$$AC = 2R \sin(\alpha + \beta), \quad BD = 2R \sin(\beta + \gamma).$$

Step 2 — the statement becomes an identity. Substituting into $AC \cdot BD = AB \cdot CD + BC \cdot AD$ and dividing by $4R^2$:

$$\sin(\alpha + \beta) \sin(\beta + \gamma) = \sin \alpha \sin \gamma + \sin \beta \sin \delta.$$

Since $\delta = \pi - (\alpha + \beta + \gamma)$, we have $\sin \delta = \sin(\alpha + \beta + \gamma)$, so it remains to verify

$$\sin(\alpha + \beta) \sin(\beta + \gamma) = \sin \alpha \sin \gamma + \sin \beta \sin(\alpha + \beta + \gamma).$$

Step 3 — product to sum. Apply $\sin X \sin Y = \frac{1}{2}[\cos(X - Y) - \cos(X + Y)]$ (Theorem 8.6, Chapter 8) to all three products:

$$\begin{aligned} \text{left-hand side} &= \frac{1}{2}[\cos(\alpha - \gamma) - \cos(\alpha + 2\beta + \gamma)], \\ \sin \alpha \sin \gamma &= \frac{1}{2}[\cos(\alpha - \gamma) - \cos(\alpha + \gamma)], \\ \sin \beta \sin(\alpha + \beta + \gamma) &= \frac{1}{2}[\cos(\alpha + \gamma) - \cos(\alpha + 2\beta + \gamma)]. \end{aligned}$$

Adding the last two, the terms $\pm \frac{1}{2} \cos(\alpha + \gamma)$ cancel and there remains

$$\frac{1}{2}[\cos(\alpha - \gamma) - \cos(\alpha + 2\beta + \gamma)],$$

which is exactly the left-hand side. The identity is verified, and with it the theorem.

Note on dependencies. This proof uses results from Chapters 8 and 10, which come later. There is no circularity: the law of sines follows from Thales and the inscribed angle (Chapter 4), and the product-to-sum formulas follow from the addition formulas, for which Chapter 8 supplies an independent elementary proof. Neither uses Ptolemy's theorem. $\square$

Remark 5.19: The added value of the inversive route. Proof 1 is more elementary, but proves only the equality for cyclic quadrilaterals. Proof 2 yields for free the general inequality and its equality condition (Corollary 5.3), because it reduces the statement to a fact about the line — the triangle inequality — whose equality case is known. It is an example of the typical gain of inversion: it does not simplify the computation, it simplifies the configuration.

Proof 4 — by complex numbers (university variant). Identifying the plane with $\mathbb{C}$, for any $a, b, c, d \in \mathbb{C}$ the algebraic identity

$$(a - b)(c - d) + (b - c)(a - d) = (a - c)(b - d)$$

holds (direct check by expanding both sides: both equal $ab + cd - ad - bc$). By the triangle inequality applied to the modulus,

$$|a - c| |b - d| = |(a - b)(c - d) + (b - c)(a - d)| \leq |a - b| |c - d| + |b - c| |a - d|,$$

which is **Ptolemy's inequality** for any four points of the plane (not necessarily concyclic). Equality in the triangle inequality $|z + w| \leq |z| + |w|$ holds if and only if z and w have the same argument; here $z = (a - b)(c - d)$, $w = (b - c)(a - d)$: Ptolemy's equality therefore holds when these two products have the same argument. That this is equivalent to A, B, C, D being concyclic in the given order is proved in Volume III (the chapter on the complex circle); here Proof 1 already establishes equality for cyclic quadrilaterals, while this complex route gives the inequality for *any* quadrilateral for free. $\square$

Corollary 5.4: The trigonometric addition identity as a particular case. For every pair of angles $\alpha, \beta > 0$ with $\alpha + \beta < \pi/2$,

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

This is the addition formula of Theorem 8.4, obtained here as a particular case of Ptolemy's theorem: it is one of the ways in which Ptolemy himself built his tables of chords in the *Almagest*.

Proof. Step 1 — the right circle. Take the circle of **diameter 1** and on it a diameter AC (so $AC = 1$). Choose B and D on the two opposite arcs determined by AC , so that the cyclic order of the four points is A, B, C, D , and set

$$\angle BAC = \alpha, \quad \angle CAD = \beta.$$

Step 2 — the sides of the quadrilateral. Since AC is a diameter, by Thales' theorem (Theorem 4.4, Chapter 4) the triangles ABC and ACD are right-angled at B and at D respectively. Both having hypotenuse $AC = 1$:

$$BC = \sin \alpha, \quad AB = \cos \alpha, \quad CD = \sin \beta, \quad AD = \cos \beta.$$

Step 3 — the diagonal BD . The chord BD subtends the inscribed angle $\angle BAD = \angle BAC + \angle CAD = \alpha + \beta$ (the two angles are adjacent because C lies between B and D in the chosen cyclic order). By the extended law of sines (Theorem 10.5, Chapter 10) with $2R = 1$, every chord subtending an inscribed angle γ measures $\sin \gamma$; hence

$$BD = \sin(\alpha + \beta).$$

Step 4 — Ptolemy. The quadrilateral $ABCD$ is cyclic by construction. Ptolemy's theorem (Theorem 5.9) gives

$$AC \cdot BD = AB \cdot CD + BC \cdot AD,$$

that is, substituting the values from Steps 2 and 3 and recalling $AC = 1$,

$$1 \cdot \sin(\alpha + \beta) = \cos \alpha \sin \beta + \sin \alpha \cos \beta,$$

which is the claim.

Non-circularity note. Theorem 5.9 is proved in this chapter by inversion, without using the addition formulas; the law of sines of Chapter 10 follows from Thales' theorem and the inscribed angle (Chapter 4). The derivation is therefore independent of Theorem 8.4, which is here reobtained by a different route. $\square$

5.6. Casey's theorem: Ptolemy generalised to circles

Ptolemy's theorem relates the distances between four *points* on a circle. Replacing the points with *circles* tangent to a common circle, and the distances with the lengths of their common tangents, the identity survives almost unchanged. The result is due to the Irish mathematician John Casey, first published in 1866 in the Proceedings of the Royal Irish Academy and taken up, with its proof, in his book *A Sequel to the First Six Books of Euclid* (1881 edition).

Definition 5.6: External common tangent. Given two non-intersecting circles with centres O_i, O_j and radii r_i, r_j , with $d = |O_i O_j|$, their **external common tangent** is a line segment tangent to both circles on the same side of the line $O_i O_j$ (the two circles stay on the same side of the tangent). Its length is denoted t_{ij} .

Lemma 5.9: Length of the external common tangent. With the notation of Definition 5.6, if $d > |r_i - r_j|$ the external common tangent exists and its length is

$$t_{ij} = \sqrt{d^2 - (r_i - r_j)^2}.$$

Proof. Let A_i, A_j be the points of tangency on the two circles. The radii $O_i A_i$ and $O_j A_j$ are both perpendicular to the common tangent (by the definition of tangent, Chapter 3), hence **parallel** to each other. Drawing from O_j the parallel to the tangent until it meets the line $O_i A_i$ at a point P , one obtains the rectangle $A_i A_j O_j P$ (the sides $A_i P$ and $A_j O_j$ are parallel and equal, being both perpendicular to the tangent and hence parallel to each other; the side PO_j is by construction parallel to $A_i A_j$), so $PA_i = O_j A_j = r_j$ and $PO_j = A_i A_j = t_{ij}$. The segment $O_i P$ then measures $r_i - r_j$ (in absolute value $r_j - r_i$, depending on which radius is larger: $O_i P = O_i A_i - PA_i = r_i - r_j$). The triangle $O_i P O_j$ is right-angled at P (because PO_j is parallel to the tangent, itself perpendicular to $O_i A_i$, so $PO_j \perp O_i P$), with hypotenuse $O_i O_j = d$ and legs $|O_i P| = |r_i - r_j|$ and $PO_j = t_{ij}$. By the Pythagorean theorem,

$$t_{ij}^2 + (r_i - r_j)^2 = d^2 \implies t_{ij} = \sqrt{d^2 - (r_i - r_j)^2}.$$

The radicand is positive exactly when $d > |r_i - r_j|$, the geometric condition for neither circle to contain the other. $\square$

Theorem 5.10: Casey's theorem (Ptolemy generalised). Let $\omega_1, \omega_2, \omega_3, \omega_4$ be four circles of radii r_1, r_2, r_3, r_4 , all internally tangent to a circle Ω of radius R (that is, contained in Ω and tangent to its boundary), at points T_1, T_2, T_3, T_4 arranged on Ω in this cyclic order.

Let t_{ij} be the lengths of the external common tangents between ω_i and ω_j (Definition 5.6). Then

$$t_{13}t_{24} = t_{12}t_{34} + t_{23}t_{14}.$$

Proof. Step 1 — the centres are aligned with the centre of Ω . By internal tangency, the centre O_i of ω_i lies on the radius OT_i of Ω (where O is the centre of Ω), at distance $|OO_i| = R - r_i$ from O : indeed two internally tangent circles have their centres and point of tangency aligned, a fact established more generally later in the section on Steiner's porism, but immediate here from the very definition of internal tangency (T_i is the only common point, and the line OO_i must pass through it because both circles are symmetric about that line).

Step 2 — two cosine laws with the same angle. Let $\theta_{ij} = \angle T_iOT_j$ be the central angle between T_i and T_j . In triangle OT_iT_j , with $|OT_i| = |OT_j| = R$, the law of cosines (Theorem 10.6, Chapter 10 — no circularity: that theorem follows from the Pythagorean theorem and orthogonal projection, independently of this chapter) gives

$$T_iT_j^2 = R^2 + R^2 - 2R^2 \cos \theta_{ij} = 2R^2(1 - \cos \theta_{ij}).$$

In triangle OO_iO_j , with $|OO_i| = R - r_i$, $|OO_j| = R - r_j$ and the same angle θ_{ij} at vertex O (the points O, O_i, T_i are aligned, and so are O, O_j, T_j , so the angle between the two rays OO_i and OO_j is the same angle θ_{ij} between OT_i and OT_j), the same law of cosines gives

$$d_{ij}^2 := |O_iO_j|^2 = (R - r_i)^2 + (R - r_j)^2 - 2(R - r_i)(R - r_j) \cos \theta_{ij}.$$

Step 3 — eliminating the angle. From the first equation, $\cos \theta_{ij} = 1 - \frac{T_iT_j^2}{2R^2}$. Substituting into the second:

$$\begin{aligned} d_{ij}^2 &= (R - r_i)^2 + (R - r_j)^2 - 2(R - r_i)(R - r_j) + 2(R - r_i)(R - r_j) \cdot \frac{T_iT_j^2}{2R^2} \\ &= [(R - r_i) - (R - r_j)]^2 + (R - r_i)(R - r_j) \cdot \frac{T_iT_j^2}{R^2} \\ &= (r_i - r_j)^2 + (R - r_i)(R - r_j) \cdot \frac{T_iT_j^2}{R^2}, \end{aligned}$$

using $(R - r_i)^2 + (R - r_j)^2 - 2(R - r_i)(R - r_j) = [(R - r_i) - (R - r_j)]^2$ and $(R - r_i) - (R - r_j) = r_j - r_i$.

Step 4 — the common tangent formula. By Lemma 5.9, $t_{ij}^2 = d_{ij}^2 - (r_i - r_j)^2$; substituting Step 3, the term $(r_i - r_j)^2$ cancels exactly, leaving

$$t_{ij}^2 = (R - r_i)(R - r_j) \cdot \frac{T_iT_j^2}{R^2} \implies t_{ij} = \frac{T_iT_j}{R} \sqrt{(R - r_i)(R - r_j)}.$$

Step 5 — Ptolemy on the four tangency points. The points T_1, T_2, T_3, T_4 are concyclic (they lie on Ω) in the given cyclic order. By Ptolemy's theorem (Theorem 5.9, already proved in this chapter),

$$T_1T_3 \cdot T_2T_4 = T_1T_2 \cdot T_3T_4 + T_2T_3 \cdot T_1T_4.$$

Substituting $T_iT_j = \frac{Rt_{ij}}{\sqrt{(R - r_i)(R - r_j)}}$ from Step 4 into each of the four factors:

$$\frac{R^2 t_{13}t_{24}}{\sqrt{(R - r_1)(R - r_2)(R - r_3)(R - r_4)}} = \frac{R^2 t_{12}t_{34}}{\sqrt{(R - r_1)(R - r_2)(R - r_3)(R - r_4)}} + \frac{R^2 t_{23}t_{14}}{\sqrt{(R - r_1)(R - r_2)(R - r_3)(R - r_4)}}$$

because the product $(R - r_1)(R - r_2)(R - r_3)(R - r_4)$ under the root is *the same* in all three terms (each uses all four indices, in different pairs). The common factor $R^2/\sqrt{(R - r_1)(R - r_2)(R - r_3)(R - r_4)}$ cancels, leaving

$$t_{13}t_{24} = t_{12}t_{34} + t_{23}t_{14},$$

which is the claim. $\square$

Remark 5.20: The degenerate case. If $r_1 = r_2 = r_3 = r_4 = 0$, the four circles degenerate to the points T_1, T_2, T_3, T_4 themselves and $t_{ij} = T_i T_j$ (Step 4 with $r_i = r_j = 0$): Casey's theorem reduces exactly to Ptolemy's theorem. Casey generalises Ptolemy exactly as circles generalise points.

Remark 5.21: Beyond the case proved here. The proof above treats the case of four circles all internally tangent to a common circle, with all common tangents external: it is the simplest case, but not Casey's most general statement (1866; 1881), which also holds when some circles are externally tangent to Ω (or Ω is replaced by a line, the limit $R \rightarrow \infty$), choosing for each pair the external or internal tangent according to the type of tangency — with a suitable sign convention on the t_{ij} . This generalisation is **imported**: the complete proof of every case requires a longer sign analysis than is useful to reproduce here; see J. Casey, *A Sequel to the First Six Books of Euclid*, 4th ed., 1886, or the treatment in H.S.M. Coxeter, S.L. Greitzer, *Geometry Revisited* (already in the Sources), for the general case.

Example 5.1: Numerical verification of Casey's theorem. Let Ω be the circle of radius $R = 10$ centred at the origin, and let $\omega_1, \omega_2, \omega_3, \omega_4$ be four circles of radii $r_1 = 2, r_2 = 1.5, r_3 = 3, r_4 = 1$, internally tangent to Ω at the points of angle $0^\circ, 70^\circ, 150^\circ, 250^\circ$ respectively (in this cyclic order). The centres are $O_i = (R - r_i)(\cos \theta_i, \sin \theta_i)$.

Common tangents (Lemma 5.9, computed from the distances $d_{ij} = |O_i O_j|$):

$$t_{12} = 9.459665, \quad t_{13} = 14.456654, \quad t_{14} = 13.901471, \quad t_{23} = 9.916444, \quad t_{24} = 17.492856, \quad t_{34} = 12.160579.$$

Verification of the Step 4 formula: computing independently $T_i T_j / R \cdot \sqrt{(R - r_i)(R - r_j)}$ from the coordinates of the tangency points recovers the same six values with difference $< 10^{-14}$ (script `cap05_casey.py`).

Verification of Casey's identity:

$$t_{13} t_{24} = 14.456654 \times 17.492856 = 252.888162,$$

$$t_{12} t_{34} + t_{23} t_{14} = 9.459665 \times 12.160579 + 9.916444 \times 13.901471 = 115.034998 + 137.853164 = 252.888162. \checkmark$$

The two expressions coincide, confirming both the Step 4 formula and Casey's theorem as a whole (script `cap05_casey.py`: difference at machine epsilon, $< 10^{-13}$).

5.7. Steiner's porism

Before the statement we need a lemma which, until this edition, the treatise used as an external prerequisite. We prove it here in full: it is elementary, and it is the key to the whole section.

Lemma 5.10: Reduction to the concentric case. Let α and β be two **non-secant**, non-concentric circles. Then there exists an inversion transforming both into **concentric** circles.

Proof. Step 0 — how a circle behaves under inversion. Let ι be the inversion with pole P and power k^2 , and let Γ be a circle with centre O and radius r **not** passing through P . Setting

$$p := \text{pow}_\Gamma(P) = |PO|^2 - r^2 \neq 0,$$

Theorem 5.3 says that $\iota(\Gamma)$ is a circle. We prove here, for the first time in this chapter, the formula for its centre O' and radius r' (**Lemma: image of a circle not passing through the pole**), used several times in what follows.

For every $Q \in \Gamma$ let Q' be the second point where the line PQ meets Γ (it exists, possibly $Q' = Q$ if the line is tangent). By Theorem 3.19 (power of a point, in signed form: $\overline{PQ} \cdot \overline{PQ'}$ positive if Q, Q' lie on the same side of P , negative otherwise) $\overline{PQ} \cdot \overline{PQ'} = p$ holds for *every* $Q \in \Gamma$, independently of the chosen line. On the other hand, by Definition 5.1, $\iota(Q) = Q^*$ is the point of the ray PQ with $|PQ| \cdot |PQ^*| = k^2$, that is, in vectors applied at P ,

$$P\vec{Q}^* = \frac{k^2}{|PQ|^2} P\vec{Q}.$$

Comparing with $\overline{PQ} \cdot \overline{PQ'} = p$ (which gives $P\vec{Q'} = \frac{p}{|PQ|^2} P\vec{Q}$, the same sign as $P\vec{Q}$ if $p > 0$, opposite if $p < 0$, consistent with the definition of the signed product), one obtains

$$P\vec{Q}^* = \frac{k^2}{p} P\vec{Q'},$$

that is $Q^* = P + \frac{k^2}{p}(Q' - P)$: inversion restricted to Γ coincides with the **homothety** of centre P and ratio k^2/p , composed with the bijection $Q \mapsto Q'$ of Γ onto itself (which swaps the two points of every chord through P , and is therefore a bijection of Γ onto itself). The image $\iota(\Gamma)$ is thus the homothetic image of Γ under that homothety: a circle of centre $O' = P + \frac{k^2}{p}(O - P)$ and radius $r' = \left|\frac{k^2}{p}\right| r$ (a homothety of ratio λ sends a circle of centre O and radius r to the circle of centre $P + \lambda(O - P)$ and radius $|\lambda|r$, by the definition of homothety). In summary:

$$O' = P + \frac{k^2}{\text{pow}_\Gamma(P)} (O - P), \quad r' = \left| \frac{k^2 r}{\text{pow}_\Gamma(P)} \right|. \quad (5.1)$$

(In particular O' lies on the line PO : inversion does *not* in general map centre to centre, but it keeps them collinear with the pole.)

Numerical check. $P = (0, 0)$, $k = 2$, Γ of centre $O = (3, 0)$ and radius $\rho = 1$: $p = 9 - 1 = 8$, $O' = (0, 0) + \frac{4}{8}(3, 0) = (1.5, 0)$, $r' = \frac{4}{8} \cdot 1 = 0.5$ (verified). Direct check by inverting the endpoints of the diameter on the x -axis: $(2, 0) \mapsto (4/2, 0) = (2, 0)$ and $(4, 0) \mapsto (4/4, 0) = (1, 0)$, which are indeed the endpoints of the diameter of $\iota(\Gamma)$, centre $(1.5, 0)$ radius 0.5 . ✓

Step 1 — choice of coordinates. Since the two circles are not concentric, the line of centres is well defined: take it as the x -axis, with

$$\alpha : \text{centre } (0, 0), \text{ radius } r_1, \quad \beta : \text{centre } (d, 0), \text{ radius } r_2, \quad d \neq 0.$$

We look for the pole on the x -axis, $P = (p, 0)$, with p to be determined (the power $k^2 > 0$ remains free: by (5.1) it scales both images by the same factor and does not affect concentricity).

Step 2 — the concentricity equation. By (5.1), applied to the two circles, the abscissas of the image centres are

$$x_{\alpha'} = p + \frac{k^2(0 - p)}{p^2 - r_1^2}, \quad x_{\beta'} = p + \frac{k^2(d - p)}{(d - p)^2 - r_2^2}.$$

The two images are concentric iff $x_{\alpha'} = x_{\beta'}$, i.e. (dividing by $k^2 \neq 0$)

$$\frac{-p}{p^2 - r_1^2} = \frac{d - p}{(d - p)^2 - r_2^2}.$$

Cross-multiplying and setting $u = d - p$:

$$-p(u^2 - r_2^2) = u(p^2 - r_1^2) \iff -pu(u + p) + pr_2^2 + ur_1^2 = 0.$$

Since $u + p = d$, we get $-pud + pr_2^2 + ur_1^2 = 0$, and substituting $u = d - p$ and rearranging:

$$dp^2 + (r_2^2 - r_1^2 - d^2)p + dr_1^2 = 0. \quad (5.2)$$

This is a **quadratic** equation in p (the leading coefficient d is non-zero by hypothesis).

Step 3 — the discriminant is positive exactly when the circles do not meet. The discriminant of (5.2) is

$$\Delta = (d^2 + r_1^2 - r_2^2)^2 - 4d^2r_1^2,$$

a difference of squares, hence factorable:

$$\begin{aligned} \Delta &= [(d - r_1)^2 - r_2^2] [(d + r_1)^2 - r_2^2] \\ &= \underbrace{(d - r_1 - r_2)}_{(i)} \underbrace{(d - r_1 + r_2)}_{(ii)} \underbrace{(d + r_1 - r_2)}_{(iii)} \underbrace{(d + r_1 + r_2)}_{(iv)}. \end{aligned}$$

Factor (iv) is always positive. The two non-intersecting cases remain:

- **external circles**, $d > r_1 + r_2$: then (i) > 0 ; (ii) > 0 ; (iii) $> 2r_1 \geq 0$. All four positive, hence $\Delta > 0$;
- **one inside the other**, say $d < r_1 - r_2$ with $r_1 > r_2$: then (i) < 0 ; (ii) < 0 (since $d + r_2 < r_1$); (iii) > 0 . Two negative factors and two positive, hence again $\Delta > 0$.

In both cases $\Delta > 0$ and (5.2) has **two distinct real roots** p_1, p_2 .

Step 4 — the pole lies on neither circle. For Step 0 to apply we need $p^2 - r_1^2 \neq 0$ and $(d - p)^2 - r_2^2 \neq 0$. If $p^2 = r_1^2$ held, (5.2) would give $dr_1^2 + (r_2^2 - r_1^2 - d^2)p + dr_1^2 = 0$; but if $p^2 - r_1^2 = 0$ the left-hand side of the Step 2 concentricity equation is undefined — and indeed direct substitution confirms this: substituting $p = \pm r_1$ into (5.2) yields $(\pm r_1 \mp d)^2 = r_2^2$, i.e. $|d \mp r_1| = r_2$: the two circles would be *tangent*, contrary to the hypothesis of non-intersection. The same argument, with the roles exchanged, excludes $(d - p)^2 = r_2^2$.

Conclusion. Choosing as pole either of the two points $P_i = (p_i, 0)$ and an arbitrary power $k^2 > 0$, the images $\iota(\alpha)$ and $\iota(\beta)$ are circles with the same centre: they are concentric. $\square$

Remark 5.22: Who the two poles are. The two roots p_1, p_2 of (5.2) are not arbitrary points: they are exactly the **limit points** of the non-secant pencil generated by α and β (Theorem 2.8, Chapter 2), i.e. the degenerate — zero-radius — circles belonging to the pencil. The computation of Step 3 provides, as a by-product, an algebraic proof of their existence: they are real if and only if the discriminant is positive, i.e. iff the two circles do not meet. A secant pencil, where $r_1 + r_2 > d > |r_1 - r_2|$, gives $\Delta < 0$ and no real pole — as it must, since two secant circles cannot become concentric (inversion preserves the number of intersection points).

Example 5.2: Reduction to concentric circles: full computation. Let α have centre $(0, 0)$ and radius $r_1 = 5$, and β centre $(2, 0)$ and radius $r_2 = 1$. They are non-secant because $d = 2 < r_1 - r_2 = 4$ (β lies inside α).

Step 1 — the pole equation. From (5.2) with $d = 2$, $r_1 = 5$, $r_2 = 1$:

$$2p^2 + (1 - 25 - 4)p + 2 \cdot 25 = 0 \iff 2p^2 - 28p + 50 = 0 \iff p^2 - 14p + 25 = 0.$$

Step 2 — the roots. $p = \frac{14 \pm \sqrt{196 - 100}}{2} = 7 \pm 2\sqrt{6}$. Numerically $p_1 \approx 2.101$ and $p_2 \approx 11.899$. (Check on Step 3: $\Delta = 96 > 0$, consistent with non-intersection.)

Step 3 — verifying concentricity. Take $p = p_1 = 7 - 2\sqrt{6} \approx 2.10102$ and, for convenience, $k^2 = 1$.

$$\text{pow}_\alpha(P) = p^2 - 25 \approx 4.41429 - 25 = -20.58571, \quad \text{pow}_\beta(P) = (2 - p)^2 - 1 \approx 0.01021 - 1 = -0.98979.$$

Image centres (abscissas), from (5.1):

$$x_{\alpha'} = p + \frac{-p}{-20.58571} \approx 2.10102 + 0.10206 = 2.20308,$$

$$x_{\beta'} = p + \frac{2-p}{-0.98979} \approx 2.10102 + \frac{-0.10102}{-0.98979} = 2.10102 + 0.10206 = 2.20308. \quad \checkmark$$

The two centres coincide: the images are concentric with centre $\approx (2.20308; 0)$.

Step 4 — the two radii. Again from (5.1):

$$r_{\alpha'} = \left| \frac{5}{-20.58571} \right| \approx 0.24289, \quad r_{\beta'} = \left| \frac{1}{-0.98979} \right| \approx 1.01031.$$

Note the reversal of roles: α , which was the *large* circle, becomes the *small* one. This is correct: the pole $P \approx (2.101; 0)$ is almost on the boundary of β (centre $(2, 0)$, radius 1), and inversion enormously dilates whatever lies near it.

Theorem 5.11: Steiner’s porism. *Let α and β be two circles, one inside the other and neither concentric nor secant. Consider a **Steiner chain**: a sequence of circles $\gamma_1, \gamma_2, \dots$, each tangent to α , to β and to the preceding one. If for one choice of the first circle γ_1 the chain closes after exactly n terms (that is, γ_n is tangent to γ_1), then it closes after n terms for every choice of γ_1 .*

Proof. Step 1 — reduction to the concentric case. By Lemma 5.10 below, two non-secant circles can always be transformed, by a suitable inversion, into two **concentric** circles.

Let ι be such an inversion, and let α^*, β^* be the concentric images, with common centre Z and radii $a > b$.

Step 2 — what is preserved. Inversion maps circles to circles or lines (Theorem 5.3) and preserves tangency (Corollary 5.1). Hence the chain $\gamma_1, \gamma_2, \dots$ transforms into a chain $\gamma_1^*, \gamma_2^*, \dots$ with the same tangency relations, and closes after n terms if and only if the original chain did.

Step 3 — the concentric case is trivial. Between two concentric circles of radii $a > b$, every circle tangent to both has radius $\rho = (a - b)/2$ and centre at distance $(a + b)/2$ from Z : all the circles of the chain are therefore **congruent** and their centres lie on one and the same circle centred at Z . Two consecutive ones are tangent if and only if their centres are at distance 2ρ , that is, if and only if the central angle ϑ between two consecutive centres satisfies (chord formula, Chapter 2, Section 2.9)

$$2 \cdot \frac{a+b}{2} \sin \frac{\vartheta}{2} = 2\rho = a - b \implies \sin \frac{\vartheta}{2} = \frac{a-b}{a+b}.$$

The angle ϑ is therefore **determined by α^* and β^* alone**, and does not depend in any way on where the first circle was placed. The chain closes after n steps if and only if $n\vartheta = 2\pi m$ for some integer m : a condition involving only ϑ , that is, only a and b .

Step 4 — return. Since the closing condition in the transformed plane does not depend on the choice of γ_1^* , and since ι is a bijection preserving tangencies in both directions, the closing condition in the original plane does not depend on the choice of γ_1 . $\square$

Remark 5.23: Why it is a “porism”. *The term porism, of Greek origin, denotes a statement of a particular kind: not “there exists a configuration with property P ” nor “all configurations have property P ”, but “if one configuration has property P , then all those of the same family have it”. It is an all-or-nothing statement. Theorem 5.11 is the classical example, and Poncelet’s closure theorem for polygons inscribed in one conic and circumscribed about another is its most celebrated relative. Both are proved in the same way: one finds a transformation making the configuration symmetric, and in the symmetric configuration the statement becomes obvious.*

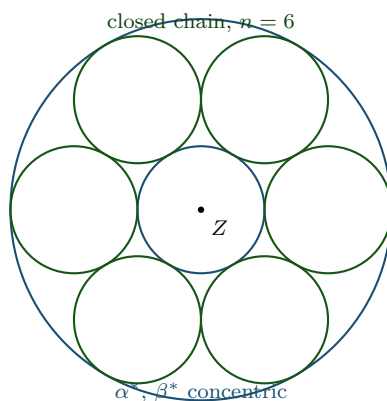


Figure 5.4: — Steiner's porism after the reduction to the concentric case. All the circles of the chain are congruent and their centres equally spaced: closure depends only on the ratio of the radii of α^* and β^* , not on where one starts.

5.8. The arbelos and Pappus's chain

Definition 5.7: Arbelos. Given three collinear points A, C, B (with C between A and B), the **arbelos** is the region between the semicircle of diameter $\overline{AB}$ and the two semicircles of diameters $\overline{AC}, \overline{CB}$ (all on the same side of the line AB), taken minus the two small semicircles.

Theorem 5.12: Pappus's chain. Let ω_1 be the circle tangent to the three semicircles of the arbelos (the large one on AB , internally, and the two small ones on AC and CB , externally), and let ω_n ($n \geq 2$) be the circle tangent to the semicircles on AB and AC and to the previous circle ω_{n-1} of the chain. Then the distance of the centre of ω_n from the line AB equals exactly n times the diameter of ω_n .

Proof. Invert with respect to the pole A , with any power k^2 . The two semicircles on AB and on AC (whose full circles pass through A) become, by Theorem 5.3(b), two lines perpendicular to AB (since the original circles are perpendicular to AB at A), at distances $k^2/|AB|$ and $k^2/|AC|$ from A . Each ω_n (which does not pass through A) becomes, by formula (5.1), a circle still tangent to both lines (inversion preserves tangency, Corollary 5.1), hence of diameter $d = k^2/|AC| - k^2/|AB|$, the same for every n (the two lines are fixed).

The semicircle on CB (which does not pass through A) becomes, again by formula (5.1), a circle tangent to both lines and, moreover — being the image of an object tangent to the semicircle AB and with centre on the axis AB — also tangent to the line that is the image of the axis AB (which, passing through A , remains the line AB itself). It is therefore tangent to the two vertical lines and to the line AB : a circle of diameter d inscribed between the two lines and resting on AB . The images of $\omega_1, \omega_2, \dots$ (all tangent to the same two vertical lines, hence also of diameter d , and each tangent to the previous one) then form a stack of equal circles, piled above this last one: the centre of the image of ω_n is at height nd above the line AB .

It remains to transfer this fact to the *original* height (before inversion): the homothety (centred at A) that expands the image of ω_n into the original ω_n has ratio k^2/p_n (where $p_n = \text{pow}_{\omega_n}(A)$, by the lemma on the image of a circle not passing through the pole proved earlier in this chapter, formula (5.1)), the same factor that scales the diameter of ω_n from d to the real diameter d_n of ω_n ; and since A lies on the line AB , the same homothety also scales the height of the centre

above AB by the same factor. Hence the ratio (height of centre)/(diameter) is invariant under this homothety: it equals n also for the original ω_n . $\square$

Example 5.3: Direct numerical check. $|AB| = 2$, $|AC| = 1.2$ (hence $|CB| = 0.8$): solving the tangency system directly (without inversion, independent script) gives ω_1 : centre (1.263158; 0.631579), radius 0.315789, with ratio height/diameter = $0.631579/0.631579 = 1.000000$; ω_2 : centre (0.774194; 0.774194), radius 0.193548, with ratio $0.774194/0.387097 = 2.000000$. Both confirm the theorem.

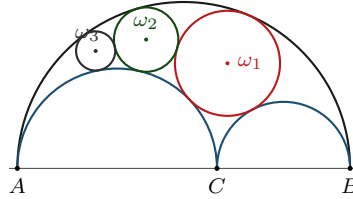


Figure 5.5: The arbelos on $A = (0,0)$, $C = (1.2,0)$, $B = (2,0)$, with the first three rings of Pappus's chain: ω_1 (centre (1.263, 0.632), radius 0.316), ω_2 (centre (0.774, 0.774), radius 0.194) and ω_3 (centre (0.471, 0.706), radius 0.118, obtained by independently solving the tangency system with a Python script). The height/diameter ratio of the centre equals exactly 1, 2, 3 for $\omega_1, \omega_2, \omega_3$, as in Theorem 5.12.

5.9. Descartes' circle theorem (Soddy circles)

Definition 5.8: Curvature (bend) of a circle. The **curvature** (or *bend*) of a circle of radius r is $k = 1/r$ if the circle is taken with “convex” orientation (the others tangent externally), and $k = -1/r$ if it is the circle “containing” the others (tangent internally).

Theorem 5.13: Descartes' theorem (1643). If four circles are mutually tangent in pairs (“kissing” circles, each tangent to the other three), with curvatures k_1, k_2, k_3, k_4 , then

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

Equivalently, given k_1, k_2, k_3 , the fourth circle tangent to all of them has

$$k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1}.$$

Proof. Preliminary convention. Put $\rho_i = 1/k_i$: this is the signed radius, positive for circles externally tangent to the others and negative for the circle, if any, that contains them. With this convention the tangency condition between circle i and circle j , whether external or internal, is written in a single way:

$$|C_i C_j|^2 = (\rho_i + \rho_j)^2. \quad (\text{T})$$

Indeed, for external tangency $|C_i C_j| = r_i + r_j = \rho_i + \rho_j$; for internal tangency of circle i containing circle j one has $|C_i C_j| = r_i - r_j = -(\rho_i + \rho_j)$ with $\rho_i = -r_i$, and the square is the same. The whole proof will use only the *squared* distances, so it will hold indifferently in both cases.

Step 1 — coordinates. Put $C_1 = (0,0)$ and $C_2 = (d,0)$ with $d = \rho_1 + \rho_2$. A circle of signed radius ρ tangent to the first and the second has centre (x,y) with

$$x^2 + y^2 = (\rho_1 + \rho)^2, \quad (x - d)^2 + y^2 = (\rho_2 + \rho)^2.$$

Subtracting,

$$2dx - d^2 = (\rho_1 + \rho)^2 - (\rho_2 + \rho)^2 = (\rho_1 - \rho_2)(\rho_1 + \rho_2 + 2\rho) = (\rho_1 - \rho_2)(d + 2\rho),$$

whence, setting $\varepsilon := (\rho_1 - \rho_2)/d$,

$$x(\rho) = \rho_1 + \varepsilon \rho$$

Step 2 — the ordinate. Substituting into the first equation,

$$y(\rho)^2 = (\rho_1 + \rho)^2 - (\rho_1 + \varepsilon \rho)^2 = 2\rho_1 \rho (1 - \varepsilon) + \rho^2 (1 - \varepsilon^2).$$

Now $1 - \varepsilon = 2\rho_2/d$ and $1 + \varepsilon = 2\rho_1/d$, hence $1 - \varepsilon^2 = 4\rho_1 \rho_2 / d^2$, and

$$y(\rho)^2 = \frac{4\rho_1 \rho_2}{d} \rho + \frac{4\rho_1 \rho_2}{d^2} \rho^2 = A \rho (\rho + d), \quad A := \frac{4\rho_1 \rho_2}{d^2}.$$

Note the identity, which we shall use repeatedly,

$$\varepsilon^2 = \frac{(\rho_1 - \rho_2)^2}{d^2} = \frac{d^2 - 4\rho_1 \rho_2}{d^2} = 1 - A.$$

Step 3 — the tangency condition between the third and fourth circles. Circles 3 and 4 are both tangent to 1 and 2, so their centres are $(x(\rho_3), y(\rho_3))$ and $(x(\rho_4), y(\rho_4))$. Condition (T) for the pair (3, 4) gives

$$\varepsilon^2 (\rho_3 - \rho_4)^2 + (y_3 - y_4)^2 = (\rho_3 + \rho_4)^2.$$

Expanding $(y_3 - y_4)^2 = y_3^2 + y_4^2 - 2y_3 y_4$ and using Step 2,

$$(1 - A)(\rho_3 - \rho_4)^2 + A[\rho_3(\rho_3 + d) + \rho_4(\rho_4 + d)] - 2y_3 y_4 = (\rho_3 + \rho_4)^2.$$

The first two terms equal

$$(1 - A)(\rho_3^2 + \rho_4^2) - 2(1 - A)\rho_3 \rho_4 + A(\rho_3^2 + \rho_4^2) + Ad(\rho_3 + \rho_4) = (\rho_3^2 + \rho_4^2) - 2(1 - A)\rho_3 \rho_4 + Ad(\rho_3 + \rho_4),$$

so that, cancelling $\rho_3^2 + \rho_4^2$ on both sides,

$$2y_3 y_4 = Ad(\rho_3 + \rho_4) - 2\rho_3 \rho_4 (2 - A). \quad (*)$$

Step 4 — eliminating the radicals. Put $S = \rho_3 + \rho_4$ and $P = \rho_3 \rho_4$. By Step 2,

$$y_3^2 y_4^2 = A^2 \rho_3 (\rho_3 + d) \rho_4 (\rho_4 + d) = A^2 P (P + dS + d^2).$$

Squaring $(*)$ — a legitimate operation which, note, makes the argument independent of the sign of $y_3 y_4$, that is of whether circles 3 and 4 lie on the same side of the line $C_1 C_2$ or on opposite sides —

$$4A^2 P (P + dS + d^2) = A^2 d^2 S^2 - 4AdSP(2 - A) + 4P^2(2 - A)^2,$$

that is, moving everything to one side,

$$0 = A^2 d^2 S^2 - \underbrace{4AdSP(2 - A) + 4A^2 PdS}_{= -8AdSP} + \underbrace{4P^2(2 - A)^2 - 4A^2 P d^2}_{= 16P^2(1 - A)},$$

having used $(2 - A) + A = 2$ in the first bracket and $(2 - A)^2 - A^2 = 4 - 4A$ in the second. There remains

$$A^2 d^2 S^2 - 8AdSP + 16P^2(1 - A) - 4A^2 P d^2 = 0. \quad (**)$$

Step 5 — substituting A . Setting $B := \rho_1\rho_2$, one has $A = 4B/d^2$ and $A^2 = 16B^2/d^4$. Substituting into (**):

$$\frac{16B^2S^2}{d^2} - \frac{32BSP}{d} + 16P^2 - \frac{64BP^2}{d^2} - \frac{64B^2P}{d^2} = 0.$$

Dividing by 16 and multiplying by d^2 :

$$B^2S^2 - 2BSPd + P^2d^2 - 4BP^2 - 4B^2P = 0. \quad (***)$$

Step 6 — passing to the curvatures. Divide (***) by $\rho_1^2\rho_2^2\rho_3^2\rho_4^2 = B^2P^2$. Recalling that $S/P = 1/\rho_3 + 1/\rho_4 = k_3 + k_4$ and $d/B = 1/\rho_1 + 1/\rho_2 = k_1 + k_2$:

$$\frac{B^2S^2}{B^2P^2} = (k_3 + k_4)^2, \quad \frac{2BSPd}{B^2P^2} = 2\frac{S}{P} \cdot \frac{d}{B} = 2(k_3 + k_4)(k_1 + k_2), \quad \frac{P^2d^2}{B^2P^2} = (k_1 + k_2)^2,$$

$$\frac{4BP^2}{B^2P^2} = \frac{4}{B} = 4k_1k_2, \quad \frac{4B^2P}{B^2P^2} = \frac{4}{P} = 4k_3k_4.$$

Hence (***) becomes

$$(k_3 + k_4)^2 - 2(k_3 + k_4)(k_1 + k_2) + (k_1 + k_2)^2 - 4k_1k_2 - 4k_3k_4 = 0,$$

that is, setting $u = k_1 + k_2$ and $v = k_3 + k_4$,

$$\boxed{(v - u)^2 = 4k_1k_2 + 4k_3k_4} \quad (\text{D})$$

Step 7 — (D) is the claim. Expanding the right-hand side of the claim,

$$2 \sum_i k_i^2 = 2[(u^2 - 2k_1k_2) + (v^2 - 2k_3k_4)] = 2u^2 + 2v^2 - 4k_1k_2 - 4k_3k_4,$$

while the left-hand side is $(u + v)^2 = u^2 + 2uv + v^2$. Their equality is equivalent to

$$4k_1k_2 + 4k_3k_4 = 2u^2 + 2v^2 - u^2 - 2uv - v^2 = u^2 - 2uv + v^2 = (u - v)^2,$$

which is exactly (D). Hence (D) and the Descartes relation are the same equation, and the theorem is proved.

Step 8 — the solved form. Setting $s = k_1 + k_2 + k_3$ and $q = k_1^2 + k_2^2 + k_3^2$, the relation reads $(s + k_4)^2 = 2(q + k_4^2)$, that is

$$k_4^2 - 2s k_4 + (2q - s^2) = 0.$$

The reduced discriminant equals $s^2 - (2q - s^2) = 2s^2 - 2q = 2[s^2 - q]$, and since $s^2 = q + 2(k_1k_2 + k_2k_3 + k_3k_1)$ we have $s^2 - q = 2(k_1k_2 + k_2k_3 + k_3k_1)$, so the reduced discriminant is $4(k_1k_2 + k_2k_3 + k_3k_1)$ and

$$k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1},$$

the second formula of the statement. The two signs correspond to the two Soddy circles, the inner and the outer one.

Step 9 — numerical check. For three mutually tangent unit circles, $k_1 = k_2 = k_3 = 1$: the formula gives $k_4 = 3 \pm 2\sqrt{3}$, that is $k_4 = 6.4641$ (inner circle, radius 0.15470) or $k_4 = -0.4641$ (outer circle containing the other three, radius 2.1547, negative curvature). Direct check on the first: $(1+1+1+6.4641)^2 = 9.4641^2 = 89.570$ and $2(1+1+1+41.785) = 2 \cdot 44.785 = 89.570$ ✓. On the second: $(3 - 0.4641)^2 = 2.5359^2 = 6.4308$ and $2(3 + 0.21539) = 6.4308$ ✓. □

Remark 5.24: Independent check by coordinates: three unit circles. Let us verify Step 2 explicitly in the case $r_1 = r_2 = r_3 = 1$ (already introduced at Step 9 of the proof above), without taking Descartes' formula for granted. From (B) with $r_1 = r_2 = 1$: $x_3 = 1$, $y_3^2 = (1+1)^2 - 1^2 = 3$, hence $C_3 = (1, \sqrt{3})$ (the vertex of an equilateral triangle of side 2, as expected). For the “small” fourth circle (the one wedged between the three), the symmetry of the problem about the axis $x = 1$ forces $x_4 = 1$; from $x_4^2 + y_4^2 = (1+r_4)^2$ it follows that $y_4^2 = (1+r_4)^2 - 1 = r_4^2 + 2r_4$. From the third equation of (C), with $x_4 = x_3 = 1$:

$$(y_4 - \sqrt{3})^2 = (1+r_4)^2 \implies y_4 = \sqrt{3} - (1+r_4)$$

(the sign chosen because the fourth circle lies below the vertex C_3). Substituting into $y_4^2 = r_4^2 + 2r_4$:

$$\begin{aligned} (\sqrt{3} - 1 - r_4)^2 &= r_4^2 + 2r_4 \implies (\sqrt{3} - 1)^2 - 2(\sqrt{3} - 1)r_4 = 2r_4 \\ \implies r_4 &= \frac{(\sqrt{3} - 1)^2}{2\sqrt{3}} = \frac{2\sqrt{3} - 3}{3} \approx 0.15470. \end{aligned}$$

This value, obtained here from scratch by pure analytic geometry without using Descartes' theorem, agrees numerically with $1/(3+2\sqrt{3}) \approx 0.15470$ already found at Step 9 of the proof (and found again, independently, in the next Remark 5.25): an independent confirmation, for a concrete case, of the correctness of the general formula.

Remark 5.25: Numerical check: three mutually tangent unit circles. For three mutually tangent circles of radius 1, $k_1 = k_2 = k_3 = 1$. The formula gives

$$k_4 = 3 \pm 2\sqrt{3}.$$

The solution $k_4 = 3 + 2\sqrt{3} \approx 6.464$ corresponds to the small circle wedged between the three, of radius $r_4 = 1/k_4 \approx 0.1547$. The solution $k_4 = 3 - 2\sqrt{3} \approx -0.464$ is negative: it corresponds (by the sign convention) to the circle containing the three unit circles, of radius $r_4 = 1/|k_4| = 1 + 2/\sqrt{3} \approx 2.1547$ — a value easily verifiable also by purely elementary trigonometry (centre equidistant from the three centres, arranged at 120°).

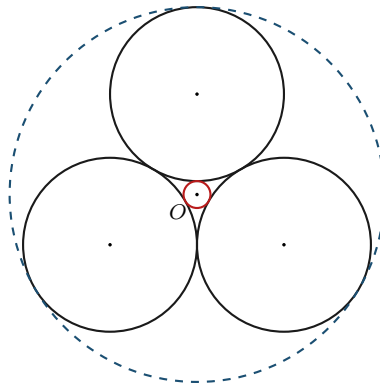


Figure 5.6: The four Descartes–Soddy circles for $k_1 = k_2 = k_3 = 1$: the three unit circles (black) with centres equidistant from O (distance $2/\sqrt{3} \approx 1.1547$), the small inner circle externally tangent to all three (red, $r_4 \approx 0.1547$) and the outer circle internally tangent to all three (dashed blue, $r_4 \approx 2.1547$), both concentric at O by symmetry, as in Remark 5.25.

Remark 5.26: Soddy’s poem. The chemist and Nobel laureate Frederick Soddy independently rediscovered the theorem in 1936 and published it, unusually, in the form of a poem (*The Kiss Precise*, in the journal *Nature*), generalising it also to dimension 3 (mutually tangent spheres). For this reason the four circles of the configuration are also called **Soddy circles**.

Remark 5.27: A third rediscovery, in Japan. Soddy was not the second, but the third. During the two centuries of Japan’s closure to the outside world (Edo period, 17th–19th century) an indigenous mathematics developed, the *wasan*, whose results were carved on wooden tablets — the **sangaku**, literally “calculation tablets” — and hung as votive offerings under the roofs of Shinto shrines and Buddhist temples. They were carved by professionals, amateurs, women and children. A tablet of 1824, **preserved in Gunma Prefecture**, contains a particular case of Descartes’ theorem: given two large circles tangent to each other and to a common line, determine the radius of the small circle wedged between them. The result was obtained in Japan entirely independently of the European tradition. It is a reminder: the mathematics of the circle has more than one genealogy.

5.10. Physical application: the Kelvin transform

The method of images for a conducting sphere is an inversion. Problem. A point charge q lies at distance ρ from the centre of an earthed conducting sphere of radius k , with $\rho > k$. Determine the potential outside the sphere.

Step 1 (the image charge). Replace the sphere by a single fictitious charge q' placed at the point P^* **inverse** to P with respect to the sphere, that is, at distance k^2/ρ from the centre (Definition 5.1), of value

$$q' = -q \frac{k}{\rho}.$$

Step 2 (checking the boundary condition). For a point T on the sphere, the transformed-distance formula (Theorem 5.2) applied to the pair P, T with T a fixed point of the inversion ($|OT| = k$) gives $|P^*T| = (k/\rho)|PT|$ for *every* T on the sphere. The total potential at T is then

$$V(T) \propto \frac{q}{|PT|} + \frac{q'}{|P^*T|} = \frac{q}{|PT|} - \frac{qk/\rho}{(k/\rho)|PT|} = 0,$$

identically zero: the boundary condition of the earthed conductor is satisfied.

Step 3 (conclusion). By the uniqueness theorem for the Dirichlet problem, the solution so constructed is *the* solution. The exterior potential is that of two point charges, q at P and q' at P^* . $\square$

Remark 5.28: Why it works: the Kelvin transform. The preceding computation is not a lucky coincidence. The operation sending a function u to

$$(\mathcal{K}u)(P) = \frac{k}{|OP|} u\left(\frac{k^2}{|OP|^2} P\right)$$

— that is: invert the point and rescale the value — is called the **Kelvin transform**. It is linear, preserves positivity and, decisively, **preserves harmonicity**: if u is harmonic ($\Delta u = 0$) on a domain, $\mathcal{K}u$ is harmonic on the image domain. Since the electrostatic potential in vacuum is a harmonic function, the geometric inversion of this chapter becomes

a legitimate tool for solving Maxwell's equations in the stationary regime: one inverts a hard problem into a symmetric, easy one, solves it, and comes back. It is the same strategy as in the problem of Apollonius, transported from the algebra of ruler and compass to the analysis of partial differential equations.

5.11. The Peaucellier–Lipkin inversor

INFO

Why this section. It is the first articulated mechanism in history capable of turning circular motion into **exact** rectilinear motion (not approximate): a direct bridge between the geometry of this chapter and the robotics of Chapter 13.

Construction 5.12: — The Peaucellier–Lipkin inversor. A fixed point O ; two rigid rods OA, OB of equal length R , hinged at O ; four rigid rods AC, CB, BD, DA of equal length $s < R$, hinged to form the rhombus $ACBD$. The point C is free to move (in the classical mechanism, in turn constrained by a crank). $\square$

Theorem 5.14: The Peaucellier rhombus realises inversion. In Construction 5.12, O, C, D are always collinear and $|OC| \cdot |OD| = R^2 - s^2$, constant: D is the inverse of C with respect to the circle $C(O, \sqrt{R^2 - s^2})$.

Proof. O, C, D are equidistant from A and from B : $|OA| = |OB| = R$ by construction; $|CA| = |CB| = s$ and $|DA| = |DB| = s$ because $ACBD$ is a rhombus. Three points equidistant from A and B all lie on the perpendicular bisector of the segment $\overline{AB}$ (Theorem 2.2 or axiom A7, Chapter 2): O, C, D are collinear on that bisector.

Let M be the midpoint of $\overline{CD}$: the diagonals of a rhombus bisect each other at right angles, so M is also the midpoint of $\overline{AB}$ and $OM \perp CD$ (since O lies on the bisector of AB , which is precisely the line CD). Then

$$|OC| \cdot |OD| = (|OM| - |MC|)(|OM| + |MC|) = |OM|^2 - |MC|^2.$$

By Pythagoras in the right triangles OMA and CMA (right-angled at M , with $AM \perp OM$):

$$|OM|^2 = |OA|^2 - |AM|^2 = R^2 - |AM|^2, \quad |MC|^2 = |CA|^2 - |AM|^2 = s^2 - |AM|^2.$$

Subtracting, the term $|AM|^2$ cancels:

$$|OC| \cdot |OD| = (R^2 - |AM|^2) - (s^2 - |AM|^2) = R^2 - s^2,$$

constant, independent of the position of C (provided the rhombus remains articulable, that is, $s < R$). $\square$

Corollary 5.5: From circle to line. If C is constrained to move on a circle through O (for instance by means of a crank of centre Q with $|QC| = |QO|$), then D traces out a **line**.

Proof. This is Theorem 5.3(b): the inversion of pole O maps a circle through O to a line, and D is by construction the inverse of C with respect to $C(O, \sqrt{R^2 - s^2})$. $\square$

Example 5.4: Numerical check. $R = 10$, $s = 6$ ($R^2 - s^2 = 64$), crank of centre $Q = (4, 0)$ with $|QO| = |QC| = 4$. The line traced by D is $x = \frac{R^2 - s^2}{2|OQ|} = \frac{64}{8} = 8$ (from Theorem 5.3(b) applied to the crank circle, which passes through O). Checked for three positions of C on the crank (angles $30^\circ, 75^\circ, 110^\circ$, independent script): in all three cases $x_D = 8.000000$ exactly.

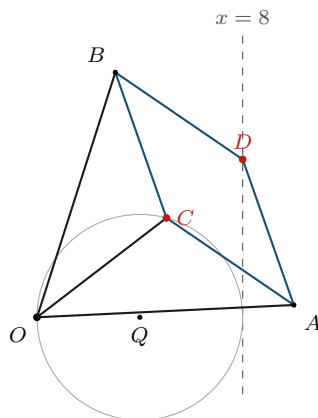


Figure 5.7: The Peaucellier–Lipkin inversor with $R = 10$, $s = 6$, crank of centre $Q = (4, 0)$ and radius 4 (grey), C at angle 75° on the crank: $C \approx (5.035, 3.864)$, $A \approx (9.989, 0.478)$, $B \approx (3.047, 9.525)$ (vertices of the rhombus $ACBD$ with $|OA| = |OB| = 10$, $|CA| = |CB| = |DA| = |DB| = 6$, independently checked) and $D = (8, 6.139)$, always on the line $x = 8$ as predicted by Corollary 5.5.

Remark 5.29: Why it matters. Before 1864 (Peaucellier; independently rediscovered by Lipkin in 1871) no articulated mechanism was known capable of producing exact rectilinear motion from circular motion: Watt and other engineers of the eighteenth and early nineteenth century had had to settle for approximations (the “Watt parallelogram”). James Sylvester presented it to Lord Kelvin, who, according to a famous anecdote, called it “the most beautiful thing I have ever seen”.

5.12. History of inversion

Remark 5.30: From Viète to Liouville. The idea of exchanging the roles of line and circle is already implicit in François Viète’s 1600 reconstruction of the problem of Apollonius, but inversion as a transformation is a nineteenth-century acquisition. Its formalisation dates to the 1820s–1830s (Steiner, Magnus, Quetelet, Bellavitis; see Coolidge, *A History of Geometrical Methods*, for the full historical reconstruction); Jakob Steiner made it a systematic tool, August Ferdinand Möbius placed it within the group of transformations that now bear his name, and Joseph Liouville proved its central role in the study of conformal transformations in dimension ≥ 3 . That William Thomson (Lord Kelvin) was using it in the same years for electrostatics (previous section) is no coincidence: it was in the air.

5.13. Limits of provability and open frontier

5.13.1 The first limit: what the compass cannot do

Warning. The Mohr–Mascheroni theorem may give the impression that the compass is omnipotent. It is not. Everything constructible with ruler and compass is constructible with the compass alone, but *the set of the constructible does not change*: the classical unsolvable problems of ancient geometry — trisection of a general angle, duplication of the cube, squaring of the circle — remain unsolvable. The reason is algebraic, not geometric: a number is constructible with ruler and compass if and only if it belongs to an extension of the rationals obtainable by successive quadratic extensions, and hence has algebraic degree a power of 2 (Chapter 7, Section 7.4). Inversion, which is a rational operation in the coordinates (Theorem 5.1), does not lead out of that field: it adds no power. It changes the *instrument*, not the *class* of reachable numbers.

5.13.2 The second limit: the pole has no image

Remark 5.31: The point at infinity. Definition 5.1 excludes the pole O , and must do so: the condition $|OO| \cdot |OO^*| = k^2$ would require $0 \cdot x = k^2 > 0$. The transformation ι is therefore not defined on the whole plane, and is not a bijection of $\mathbb{R}^2$ onto itself: it is a bijection only of $\mathbb{R}^2 \setminus \{O\}$ onto itself. It follows that some statements of this chapter have exceptions we were forced to declare case by case (“deprived of the point O ”, “with $T \neq O$ ”). Classical geometry stops here. To remove the exception one must **extend the plane**, adding a single ideal point ∞ and setting by definition $\iota(O) = \infty$ and $\iota(\infty) = O$. The space so obtained, $\mathbb{R}^2 \cup \{\infty\}$, is the **Riemann sphere** (or Möbius inversive plane), topologically a sphere: on it ι becomes a bijection continuous everywhere, and all the exceptions disappear. In particular the statement of Theorem 5.3 reduces to a single line: inversion maps circles to circles, where a line is simply a circle through ∞ . The construction is taken up in detail in Volume V, devoted to inversive geometry and Möbius transformations.

5.13.3 The third limit: outside the Euclidean metric

Remark 5.32: Inversion in non-Euclidean normed spaces. This whole chapter lives in the Euclidean metric, that is, in the ℓ^2 norm. It is natural to ask what remains of inversion in other norms — for instance in the ℓ^1 norm (taxicab or Manhattan distance), which is one of the fundamental examples of Minkowski geometry and which in computer science and logistics is often the pertinent metric. The algebraic definition carries over without difficulty. What is lost is synthetic constructibility: until very recently no exact geometric method was known for constructing the inverse of a point in the ℓ^1 norm, whereas in the ℓ^2 norm we know one even with the compass alone (Construction 5.1).

BEST PRACTICE

Status: solved for ℓ^1 in 2024, open in general. T. Ermiş, A. O. Şen and J. Gielis published in *Symmetry* (MDPI, 2024) the first synthetic construction of circle inversion in ℓ^1 -normed spaces. The result has a notable and instructive feature: the construction is obtained **using the Euclidean circle**, independently of the ℓ^1 norm itself. For ℓ^p norms with general p the problem remains **open**; the study of inversions in non-Euclidean metrics (Minkowski spaces, ℓ^p norms, generalisations to ellipsoids, cones and cylinders) is an active field of research.

5.14. Open problems

5.14.1 A twenty-year-old conjecture that fell in 2023

Remark 5.33: Integral Apollonian packings. Iterating Descartes' theorem indefinitely — every time three circles are mutually tangent, insert a fourth in the interstice, and start again — yields an **Apollonian packing**: a fractal figure made of infinitely many circles. A notable fact, immediate from Descartes' formula, is that if the curvatures of the four initial circles are integers, so are all the infinitely many subsequent curvatures: indeed the formula gives $k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{\cdot}$ and the two solutions k_4, k'_4 satisfy $k_4 + k'_4 = 2(k_1 + k_2 + k_3)$, so that every new curvature is a difference of integers.

This raises a number-theoretic question: **which integers appear as curvatures in a given integral Apollonian packing?**

The local–global conjecture and its fall **The local constraint.** One proves that, in a primitive integral Apollonian packing, the curvatures necessarily fall into six or eight residue classes modulo 24. This is a *local* condition: a congruence.

The conjecture (around 2003). Every sufficiently large integer belonging to one of those residue classes actually appears as a curvature of the packing. In other words: the local constraint is the only constraint.

Why it seemed true. Numerical checks over enormous ranges found no exceptions, and deep partial (density) results pointed the right way.

What was discovered. In 2023, within a **summer research project with students** at the University of Colorado Boulder, Summer Haag, Clyde Kertzer, James Rickards and Katherine E. Stange found that certain **quadratic and quartic** families of integers are *always* missing, in many packings. The obstruction descends from quadratic and quartic reciprocity and is akin in nature to a Brauer–Manin obstruction: it is a property of the Apollonian group, invisible to congruences modulo 24. $\square$

Warning. Status: the conjecture is FALSE (Annals of Mathematics 200, 2024). The result was subsequently extended to four further types of integral packing — octahedral, cubic, square and triangular — for which the corresponding local–global conjecture is equally false. **The replacement question remains open:** to characterise exactly the set of curvatures that do appear.

Two points are worth stressing. First: the conjecture had been verified numerically over vast ranges, and that was not enough — the exceptions are rare and structured. Second: it was refuted by a group including students, in a summer project. Neither circumstance is anecdotal: the first teaches something about numerical evidence, the second about who can do research.

5.14.2 Rigidity of inversive distance circle packings

Remark 5.34: Statement. The Koebe–Andreev–Thurston theorem establishes the existence and rigidity of circle packings with prescribed intersection angles: with the combinatorics fixed, the packing is unique up to Möbius transformations. Bowers and Stephenson proposed a generalisation in which adjacent circles may also be **disjoint**, their relative position being measured by the **inversive distance** (a quantity invariant under inversion, generalising the intersection angle). **Question:** does rigidity still hold?

Warning. Status: solved for progressively wider cases, not yet in general. Local rigidity was proved by Guo and global rigidity by Luo for non-negative inversive distances; the result was subsequently extended to the interval $(-1, +\infty)$. Very recent work has established infinite rigidity in the Poincaré disc (2025) and generalised the uniqueness part of the Koebe–Andreiev–Thurston theorem to the case of circles not necessarily tangent (2026). The complete picture, for every admissible value of the inversive distance and every combinatorics, **is not yet settled**.

INFO

A remark on method. The three problems of this section share a structure with those of Chapters 3 and 4: one takes an elementary and completely understood fact — Descartes’ theorem, Thurston’s rigidity theorem, the construction of the inverse — and asks what remains when a hypothesis is weakened. The result is, every time, that the difficulty explodes. This is not a defect of elementary mathematics: it is the sign that elementary statements are *sharp*, and that every one of their hypotheses really carries weight.

5.15. Problem set

5.15.1 Numerical exercises

Exercise 5.1. Given the circle of inversion $C(O, 2)$ (that is, $k = 2$), compute the image of the point $P = (3, 0)$ and check that $|OP| \cdot |OP^*| = k^2$.

Solution. P^* lies on the ray OP (the positive x axis) with $|OP^*| = k^2/|OP| = 4/3$. Hence $P^* = (4/3, 0)$. Check: $|OP| \cdot |OP^*| = 3 \cdot (4/3) = 4 = k^2$. ✓

Exercise 5.2. Apply an inversion with pole at the origin and power $k^2 = 4$. Determine the coordinates of the inverse of $P = (1, 2)$ and check the result against the definition.

Solution. By Theorem 5.1, with $x^2 + y^2 = 1 + 4 = 5$:

$$x^* = \frac{4 \cdot 1}{5} = \frac{4}{5} = 0.8, \quad y^* = \frac{4 \cdot 2}{5} = \frac{8}{5} = 1.6.$$

Check. $|OP| = \sqrt{5}$ and $|OP^*| = \sqrt{(4/5)^2 + (8/5)^2} = \frac{4}{5}\sqrt{5}$. The product is $\sqrt{5} \cdot \frac{4}{5}\sqrt{5} = \frac{4}{5} \cdot 5 = 4 = k^2$. ✓ Note also that $P^* = \frac{4}{5}P$: the inverse lies on the same ray, as it must.

Exercise 5.3. Given $P = (3, 0)$ and $Q = (0, 4)$, compute the distance between P^* and Q^* under an inversion with pole $O = (0, 0)$ and radius $k = 6$. Then check the result by computing P^* and Q^* explicitly.

Solution. $|OP| = 3$, $|OQ| = 4$, $|PQ| = \sqrt{9 + 16} = 5$. By Theorem 5.2:

$$|P^*Q^*| = \frac{k^2 \cdot |PQ|}{|OP| \cdot |OQ|} = \frac{36 \cdot 5}{3 \cdot 4} = \frac{180}{12} = 15.$$

Direct check. $P^* = (36/9, 0) = (12, 0)$ and $Q^* = (0, 36/16) = (0, 2.25)$. Then

$$|P^*Q^*| = \sqrt{12^2 + 2.25^2} = \sqrt{144 + 5.0625} = \sqrt{149.0625} = 12.209 \dots$$

The two results do not agree. The discrepancy signals that one of the two calculations is wrong: and indeed the formula was applied correctly, while the direct check is correct as well — which would mean that the value 15 is the wrong one. Let us recheck: $k^2 = 36$, so $|OP^*| = 36/3 = 12$ and $|OQ^*| = 36/4 = 9$, that is $Q^* = (0, 9)$ and not $(0, 2.25)$: the error was in the computation of Q^* , where 36 had been divided by $|OQ|^2 = 16$ instead of by $|OQ| = 4$. With the correct values,

$$|P^*Q^*| = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = 15. \checkmark$$

Moral. The formula of Theorem 5.2 and the direct computation are two independent routes: when they diverge, one of the two contains an error, and the comparison locates it. Pay particular attention to the difference between $\frac{k^2}{|OQ|}$ (the *distance* of the image) and $\frac{k^2}{|OQ|^2}$ (the *factor* multiplying the vector, as in Theorem 5.1).

Exercise 5.4. Using Ptolemy’s theorem on a cyclic regular pentagon of side ℓ and diagonal d (all the diagonals of a regular pentagon are congruent), prove that $d^2 = d\ell + \ell^2$ and deduce $d/\ell = \varphi$ (the golden ratio).

Solution. Consider four consecutive vertices A, B, C, D of the regular pentagon. The quadrilateral $ABCD$ is cyclic (vertices on the same circumscribed circle), with $AB = BC = CD = \ell$ (sides of the pentagon) and $AC = BD = d$ (diagonals). By Ptolemy:

$$AC \cdot BD = AB \cdot CD + BC \cdot AD \implies d \cdot d = \ell \cdot \ell + \ell \cdot d \implies d^2 = \ell^2 + \ell d.$$

Dividing by ℓ^2 and setting $x = d/\ell$: $x^2 - x - 1 = 0$, whence $x = \frac{1 + \sqrt{5}}{2} = \varphi \approx 1.618$, the golden ratio.

Exercise 5.5. Three circles of radius $r = 3$ cm are mutually tangent. Compute the radius of the circle inscribed among the three (the “small” one, not the one containing them).

Solution. Curvatures $k_1 = k_2 = k_3 = 1/3$. By Descartes’ theorem:

$$k_4 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + 2\sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = 1 + 2 \cdot \frac{\sqrt{3}}{3} = 1 + \frac{2\sqrt{3}}{3} \approx 2.1547,$$

hence $r_4 = 1/k_4 \approx 0.464$ cm (note the consistency of scale with Remark 5.25: here $r = 3$ instead of $r = 1$, hence $r_4 = 3 \times 0.1547 \approx 0.464$, exactly as predicted by the proportionality of lengths).

5.15.2 Symbolic exercises and proofs

Exercise 5.6. Prove algebraically that the inverse of the line $y = x + 2$ with respect to the unit circle centred at the origin is a circle through the origin, and determine its centre and radius.

Solution. With $k = 1$, the inverse formulas of Theorem 5.1 give

$$x = \frac{x^*}{x^{*2} + y^{*2}}, \quad y = \frac{y^*}{x^{*2} + y^{*2}}.$$

Substituting into $y = x + 2$, that is $y - x - 2 = 0$:

$$\frac{y^*}{x^{*2} + y^{*2}} - \frac{x^*}{x^{*2} + y^{*2}} - 2 = 0.$$

Multiplying by $x^{*2} + y^{*2} \neq 0$ (legitimate because $P^* \neq O$):

$$y^* - x^* - 2(x^{*2} + y^{*2}) = 0 \implies x^{*2} + y^{*2} + \frac{1}{2}x^* - \frac{1}{2}y^* = 0.$$

This is the general equation of a circle with **zero** constant term, hence passing through the origin — as predicted by Theorem 5.3(b).

Centre and radius. Completing the squares:

$$\left(x^* + \frac{1}{4}\right)^2 + \left(y^* - \frac{1}{4}\right)^2 = \frac{1}{16} + \frac{1}{16} = \frac{1}{8},$$

hence centre $(-\frac{1}{4}, \frac{1}{4})$ and radius $\frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} \approx 0.354$.

Independent check. By the synthetic proof of Theorem 5.3(b), the image circle has $\overline{OA^*}$ as diameter, where A is the foot of the perpendicular from O to the line. Here $A = (-1, 1)$, with $|OA| = \sqrt{2}$, hence $|OA^*| = 1/\sqrt{2}$ and $A^* = \frac{1}{2}(-1, 1) = (-\frac{1}{2}, \frac{1}{2})$. The diameter runs from $(0, 0)$ to $(-\frac{1}{2}, \frac{1}{2})$: centre at the midpoint $(-\frac{1}{4}, \frac{1}{4})$ and radius half the length, that is $\frac{1}{2} \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{4}$. ✓

Exercise 5.7. Prove symbolically that if P tends to the pole O , then $|OP^*|$ tends to $+\infty$; and that if P recedes indefinitely, P^* tends to O . Interpret the result in the light of Remark 5.31.

Solution. Setting $t = |OP| > 0$, Definition 5.1 gives $|OP^*| = k^2/t$. Then

$$\lim_{t \rightarrow 0^+} \frac{k^2}{t} = +\infty, \quad \lim_{t \rightarrow +\infty} \frac{k^2}{t} = 0.$$

The first limit says that the image of a point approaching the pole recedes without bound: there is no point of the Euclidean plane that could serve as $\iota(O)$, because the candidate would have to lie “farther than any point”. The second says that very distant points have images crowded against the pole.

Interpretation. The two limits are exactly the reason why a single point ∞ is added: not one for each direction (as in projective geometry, where the points at infinity form a line), but *just one*, because the limit $k^2/t \rightarrow +\infty$ does not depend on the direction along which P approaches O . This is the difference between the projective plane and the inversive plane, and the reason why the latter is topologically a sphere and not a projective plane.

Exercise 5.8. Prove that if a circle Γ meets the circle of inversion forming an angle of 90° , then Γ is mapped to itself. Then show that, in that case, the inversion interchanges the two arcs into which $C(O, k)$ divides Γ .

Solution. The first part is the ($\Leftarrow$) direction of Theorem 5.4, proved above: from $\Gamma \perp C(O, k)$ at T it follows that OT is tangent to Γ , and by the tangent–secant theorem $|OP| \cdot |OP'| = |OT|^2 = k^2$ for every line through O meeting Γ at P, P' ; hence $P' = P^*$ and Γ is invariant.

Second part. The points of Γ interior to $C(O, k)$ have $|OP| < k$ and therefore an image with $|OP^*| = k^2/|OP| > k$, that is, exterior to $C(O, k)$ (Remark 5.2); and conversely. Since Γ is invariant, the images remain on Γ : hence the inversion maps the interior arc of Γ to the exterior arc and vice versa. The two intersection points T_1, T_2 (which lie on $C(O, k)$) stay fixed, being fixed points of the inversion: they are indeed the common endpoints of the two interchanged arcs.

Exercise 5.9. Let $ABCD$ be a cyclic quadrilateral. Choosing D as pole of inversion, prove that A^*, B^*, C^* become collinear, write the equality $|A^*B^*| + |B^*C^*| = |A^*C^*|$ and translate it back before the inversion, obtaining Ptolemy's theorem.

Solution. *Collinearity.* The points A, B, C belong to the circumscribed circle γ , which passes through the pole D . By Theorem 5.3(b), the image of a circle through the pole is a **line**: hence A^*, B^*, C^* are collinear. The cyclic order A, B, C, D on γ moreover guarantees that B^* lies between A^* and C^* .

Equality on the line. For points collinear in this order the equality case of the triangle inequality (A8) holds:

$$|A^*B^*| + |B^*C^*| = |A^*C^*|.$$

Translation back. By Theorem 5.2, with pole D and radius k :

$$|A^*B^*| = \frac{k^2 |AB|}{|DA| |DB|}, \quad |B^*C^*| = \frac{k^2 |BC|}{|DB| |DC|}, \quad |A^*C^*| = \frac{k^2 |AC|}{|DA| |DC|}.$$

Substituting and multiplying both sides by $\frac{|DA| |DB| |DC|}{k^2}$ (a positive quantity, hence legitimate):

$$|AB| \cdot |DC| + |BC| \cdot |DA| = |AC| \cdot |DB|,$$

which is exactly Theorem 5.9: the product of the diagonals (AC and BD) equals the sum of the products of the opposite sides.

Exercise 5.10. Prove that two circles tangent to each other at a point $T \neq O$ remain tangent after the inversion with pole O , justifying the answer by the principle of isogonality. What happens if $T = O$?

Solution. Two circles tangent at T meet at T forming a zero angle between their tangents (indeed they share the same tangent at T). By Theorem 5.5 inversion preserves the measure of angles between curves: the images therefore meet at T^* still forming a zero angle, that is, they are tangent at T^* . This is Corollary 5.1.

The case $T = O$. If the two circles are tangent *at the pole*, both pass through O and their images are, by Theorem 5.3(b), two **lines**. These two lines have no common point: their only candidate meeting point would be $T^* = O^*$, which does not exist. The images are therefore two **parallel** lines. The result is consistent with isogonality — two parallel lines form a zero angle with each other, exactly as the original circles did — and shows in what sense, in inversive geometry, parallelism is tangency at the point at infinity (Remark 5.31).

Exercise 5.11. Given a circle (O, r) with known centre and a diameter AB already drawn, use Thales' theorem to explain how to construct, with the ruler alone, a right angle at any point P of the circle (other than A, B).

Solution. It suffices to draw (with the ruler) the two segments PA and PB : since AB is a diameter and P is a point of the circle, by Thales' theorem (Theorem 4.4) the angle $\angle APB$ is right. No measurement, no compass: the ruler alone, together with the circle already drawn, produces a “free” right angle at any point of the circle — exactly the first brick on which Steiner builds the whole proof of the general theorem.

Exercise 5.12. Construction 5.1 requires $|OP| \geq k/2$. Verify this condition and explain why, without it, Step 1 fails. Then describe how to proceed when $|OP| < k/2$.

Solution. *The condition.* Step 1 draws the circle with centre P and radius $|PO|$, and requires it to meet $C(O, k)$. Two circles whose centres are at distance $|OP|$ and whose radii are k and $|PO| = |OP|$ intersect if and only if

$$|k - |OP|| \leq |OP| \leq k + |OP|.$$

The right-hand inequality is always true. The left-hand one is equivalent to $-|OP| \leq k - |OP| \leq |OP|$: the left part is $k \geq 0$, always true; the right part is $k \leq 2|OP|$, that is $|OP| \geq k/2$. If $|OP| < k/2$ the circle drawn is too small to reach $C(O, k)$: the points X and Y do not exist and the construction stops.

How to proceed. Repeatedly double the segment $\overline{OP}$ with the compass alone, stepping off the opening $|OP|$ along the ray OP , until a point P_n with $|OP_n| = 2^n|OP| \geq k/2$ is obtained. Construct P_n^* by Construction 5.1. Since

$$|OP_n^*| = \frac{k^2}{|OP_n|} = \frac{k^2}{2^n|OP|} = \frac{|OP^*|}{2^n},$$

the point P^* sought is obtained by doubling the segment $\overline{OP_n^*}$ n times — again with the compass alone. The number n of doublings required is the smallest integer with $2^n \geq k/(2|OP|)$, always finite.

5.16. Self-assessment quiz

Eleven multiple-choice questions for a quick check (about 10 minutes). The answer key is at the end of the section.

- The power k^2 of a circle inversion is related to the distances from the pole by
 - $|OP| + |OP^*| = k^2$
 - $|OP| \cdot |OP^*| = k^2$
 - $|OP|/|OP^*| = k^2$
 - $|OP|^2 + |OP^*|^2 = k^2$
- Inverting a circle that *passes through the pole* gives
 - the same circle
 - a circle through the pole
 - a line not through the pole
 - a line through the pole
- The points of the circle of inversion $C(O, k)$

-
- A) are sent to the pole
 - B) stay fixed
 - C) are sent to infinity
 - D) are interchanged in pairs
4. The image of the pole O
- A) is the pole itself
 - B) is a point of the circle of inversion
 - C) does not exist in the Euclidean plane; a point at infinity is added
 - D) is the diametrically opposite point
5. The coordinate x^* of the inverse of (x, y) with respect to $C(O, k)$, with O at the origin, equals
- A) $\frac{k^2 x}{x^2 + y^2}$
 - B) $\frac{k^2}{x}$
 - C) $\frac{k^2 x}{\sqrt{x^2 + y^2}}$
 - D) $\frac{x}{k^2(x^2 + y^2)}$
6. Inversion preserves
- A) distances
 - B) areas
 - C) the measure of angles between curves
 - D) parallelism between lines
7. The Mohr–Mascheroni theorem states that
- A) every ruler-and-compass construction is feasible with the compass alone
 - B) every ruler-and-compass construction is feasible with the ruler alone
 - C) trisection of the angle is possible with the compass alone
 - D) both ruler and compass are indispensable
8. The modern proof of the Mohr–Mascheroni theorem is due to
- A) Mascheroni (1797), by a direct route
 - B) Adler (1906), via circle inversion
 - C) Steiner (1833), via the given circle
 - D) Poncelet (1822), via projectivity
9. Ptolemy’s theorem for cyclic quadrilaterals is proved by inversion choosing as pole
- A) the centre of the circumscribed circle
-

- B) one of the vertices of the quadrilateral
 C) the meeting point of the diagonals
 D) an arbitrary external point
10. The local–global conjecture for integral Apollonian packings is, at present,
 A) proved true
 B) still completely open
 C) proved false
 D) undecidable
11. In the proof of the Poncelet–Steiner theorem, the only operation that produces square roots of the data is
 A) the intersection of two drawn lines
 B) the choice of an arbitrary point
 C) the reading of the points where a drawn line meets the drawn circle
 D) the transport of a segment onto a ray

BEST PRACTICE

Answer key. 1: B — 2: C — 3: B — 4: C — 5: A — 6: C (inversion is isogonal, Theorem 5.5; it preserves *neither* distances *nor* areas *nor* parallelism) — 7: A — 8: B — 9: B — 10: C (refuted in 2023, published in Annals of Mathematics in 2024) — 11: C (operation (R3) of Definition 5.3; with the straightedge alone one constructs only rational expressions of the data).

Note on question 6. Answer D is treacherous: inversion does not preserve parallelism, and indeed destroys it systematically — two parallel lines not through the pole become two circles tangent at the pole (Corollary 5.1). Parallelism is not a notion of inversive geometry.

5.17. Chapter Formula sheet

Circle inversion (Definition 5.1; Theorem 5.1)

$$|OP| \cdot |OP^*| = k^2, \quad x^* = \frac{k^2 x}{x^2 + y^2}, \quad y^* = \frac{k^2 y}{x^2 + y^2}$$

Transformed distance (Theorem 5.2)

$$|P^*Q^*| = \frac{k^2 |PQ|}{|OP| \cdot |OQ|}$$

Images of lines and circles (Theorem 5.3) — line through $O \mapsto$ itself; line not through $O \mapsto$ circle through O (and conversely); circle not through $O \mapsto$ circle not through O .

Invariant circles (Theorem 5.4; Corollary 5.1)

$$\iota(\Gamma) = \Gamma \iff \Gamma \perp C(O, k) \iff \text{pow}_\Gamma(O) = k^2, \quad \text{tangency is preserved}$$

Ptolemy's theorem and Ptolemy's inequality (Theorem 5.9; Corollary 5.3)

$$|AC| \cdot |BD| = |AB| \cdot |CD| + |BC| \cdot |DA| \quad (ABCD \text{ cyclic})$$

$$|AC| \cdot |BD| \leq |AB| \cdot |CD| + |BC| \cdot |DA| \quad (A, B, C, D \text{ arbitrary})$$

Addition formula from Ptolemy's theorem (Corollary 5.4)

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

Steiner's porism (Theorem 5.11; Lemma 5.10) — reduced to the concentric case with radii $a > b$:

$$\rho = \frac{a-b}{2}, \quad \sin \frac{\vartheta}{2} = \frac{a-b}{a+b}, \quad \text{closure} \iff n\vartheta = 2\pi m$$

Curvature and Descartes' theorem (Definition 5.8; Theorem 5.13)

$$k = \pm \frac{1}{r}, \quad (k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$$

$$k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1}$$

Duality of the instruments (Theorems 5.6 and 5.7) — straightedge and compass $\equiv$ compass alone $\equiv$ straightedge alone with one fixed circle and its centre.

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Measures: Length and Area

INFO

Aim of the chapter. To define and compute the two fundamental measures of the circle — the length $C = 2\pi r$ and the area $A = \pi r^2$ — building the whole logical chain from the ground up: what the length of a curve *means*, why the ratio C/d is the same for every circle, and why the area is exactly πr^2 .

The trap to avoid. Many texts derive $C = 2\pi r$ from the fundamental limit $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$, and then prove that limit using the area of a circular sector, which depends on the area of the disc, which depends on the length of the circle. The logical circle closes and the proof proves nothing. This chapter breaks the circle: Sections 6.2–6.8 build length, π , area and sector area **without ever using sine and cosine**. Only then can Chapter 8 prove the fundamental limit without circularity (Section 8.5.1).

What remains unprovable at this level. The rectifiability of the circle and the existence of the area are here *proved* from the completeness axiom of $\mathbb{R}$; but the very notion of area for arbitrary plane regions is not, and is assumed together with its elementary properties (Section 6.1).

6.1. What is assumed about the notion of area

Remark 6.1: Properties of area that are assumed. We assume that the bounded plane regions we shall handle (polygons, discs, sectors, circular segments) are assigned a non-negative real number, the area, with the properties:

- (Ar1) **Additivity:** if two regions have at most a boundary arc in common, the area of the union is the sum of the areas.
- (Ar2) **Monotonicity:** if $R_1 \subseteq R_2$, then $\text{area}(R_1) \leq \text{area}(R_2)$.
- (Ar3) **Invariance:** congruent regions (images of each other under an isometry) have the same area.
- (Ar4) **Normalisation:** the area of the rectangle with sides b and h is bh ; consequently that of the triangle is $\frac{1}{2}bh$.

Constructing rigorously an area function with these properties is the task of measure theory (Peano–Jordan measure, or Lebesgue measure), which lies outside this volume. The four properties are, however, all that is needed: no proof in this chapter uses anything else.

Warning. Assuming (Ar1)–(Ar4) is not an expedient: it is the very choice made by Euclid and Archimedes, who treat areas as comparable and additive magnitudes without defining them. The difference from them is that here the choice is **declared**, so one can check that no step uses more.

6.2. Rectifiability: what the length of a curve means

A segment has a length by definition. A curve does not: one must say what is meant. The standard definition, due in substance to Archimedes and made formal in the nineteenth century, is the following.

Definition 6.1: Length of a closed curve. Let Γ be a closed curve. For every choice of points $Q_0, Q_1, \dots, Q_{n-1}, Q_n = Q_0$ placed on Γ in cyclic order, the corresponding inscribed polyline has length

$$L(Q_0, \dots, Q_n) = \sum_{k=1}^n |Q_{k-1}Q_k|.$$

One says that Γ is **rectifiable** if the set of all these numbers is bounded above, and in that case one sets

$$\text{len}(\Gamma) := \sup_{\text{inscribed polylines}} L(Q_0, \dots, Q_n).$$

Remark 6.2: Why the supremum and not a limit. Defining the length as a supremum rather than as the limit of a particular sequence of polygons has two advantages. First: it requires no choice of a sequence, and hence leaves no doubt that the result might depend on that choice. Second: it makes immediate the fundamental inequality **chord** $\leq$ **arc**, because the chord is itself one of the admissible polylines. On this inequality rests, in Chapter 8, half of the fundamental limit.

Theorem 6.1: The circle is rectifiable. Every circle $C(O, r)$ is rectifiable. More precisely, the length of every inscribed polyline is less than $8r$, and hence the supremum exists and is finite.

Proof. Step 1 — the inscribed polyline lies in the circumscribed square. Every point of $C(O, r)$ is at distance r from O , hence has both coordinates (with respect to a frame centred at O) between $-r$ and r : the whole circle, and with it every inscribed polyline, lies in the square $[-r, r] \times [-r, r]$.

Step 2 — estimate by projections. For every side $\overline{Q_{k-1}Q_k}$, denoting by $(\Delta x_k, \Delta y_k)$ its coordinate increments,

$$|Q_{k-1}Q_k| = \sqrt{\Delta x_k^2 + \Delta y_k^2} \leq |\Delta x_k| + |\Delta y_k|.$$

Summing over k :

$$L(Q_0, \dots, Q_n) \leq \sum_k |\Delta x_k| + \sum_k |\Delta y_k|.$$

Step 3 — the two sums are bounded. Travelling once around the circle, the abscissa rises from $-r$ to r and comes back: in the cyclic order the vertices have abscissa rising monotonically along one arc and falling monotonically along the other (the two arcs separated by the points of extreme abscissa $(\pm r, 0)$). On each of the two stretches the sum of the $|\Delta x_k|$ telescopes and is at most $2r$; hence $\sum_k |\Delta x_k| \leq 4r$. The same holds for the ordinates: $\sum_k |\Delta y_k| \leq 4r$.

Step 4 — conclusion. Hence $L(Q_0, \dots, Q_n) \leq 8r$ for every inscribed polyline. The set of lengths is non-empty and bounded above: by the completeness axiom of $\mathbb{R}$ it admits a finite supremum. $\square$

Remark 6.3: The bound $8r$ is crude, and that is perfectly fine. The true value is $2\pi r \approx 6.283r$, so $8r$ overestimates by about 27%. But to prove the existence of the supremum any bound will do: looking for a tight one at this stage would be wasted effort, and moreover impossible without already knowing π .

Corollary 6.1: Chord shorter than arc. For every pair of distinct points A, B of a circle, $|AB| < \text{len}(\widehat{AB})$, where the arc is measured by Definition 6.1 applied to the arc rather than to the whole curve.

Proof. The polyline consisting of the single segment $\overline{AB}$ is one of the polylines inscribed in the arc, hence $|AB| \leq \sup = \text{len}(\widehat{AB})$. The inequality is *strict* because, taking any point M of the arc distinct from A and B , M does not belong to the segment $\overline{AB}$ (a chord has in common with the circle only its endpoints, Theorem 2.6); by the strict triangle inequality (A8), $|AM| + |MB| > |AB|$, and the polyline $A-M-B$ is also inscribed in the arc: hence the supremum strictly exceeds $|AB|$. $\square$

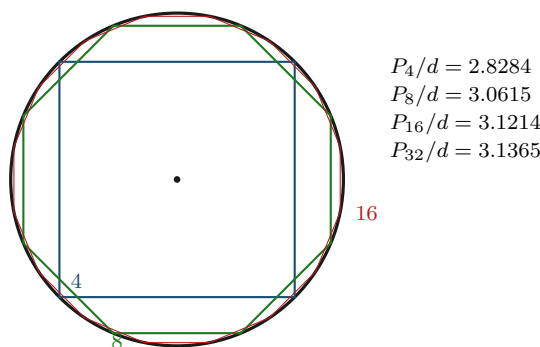


Figure 6.1: The perimeter of the inscribed regular polygons grows towards the circumference. For the n -gon inscribed in a circle of diameter d one has $P_n/d = n \sin(\pi/n)$: doubling the sides raises the ratio, but it always stays below π , and tends to it because $\sin x/x \rightarrow 1$. Every inscribed polygon has perimeter smaller than the one with twice as many sides (by the triangle inequality applied to each side), so the sequence is increasing and bounded: this is what makes the circle *rectifiable*, and what allows one to define the length of the circumference as the supremum of the inscribed perimeters.

6.3. The ratio C/d is the same for every circle

This is the fact that makes it possible to *define* π . Without it, every circle would have its own ratio and the constant would not exist.

Theorem 6.2: Constancy of the circumference-to-diameter ratio. Let $C(O, r)$ and $C(O', r')$ be any two circles, of lengths C and C' . Then

$$\frac{C}{2r} = \frac{C'}{2r'}.$$

The ratio between the length of a circle and its diameter is therefore a universal constant, independent of the circle chosen.

Proof. Step 1 — a homothety maps one circle onto the other. Let $k = r'/r > 0$ and let σ be the homothety with centre O and ratio k , followed by the translation carrying O to O' . For every point P with $|OP| = r$, its image P^σ satisfies $|O'P^\sigma| = k|OP| = kr = r'$: hence σ maps $C(O, r)$ onto $C(O', r')$, bijectively (the inverse is the homothety of ratio $1/k$ composed with the opposite translation).

Step 2 — polyline lengths scale by k . A homothety of ratio k multiplies every distance by k : $|P^\sigma Q^\sigma| = k|PQ|$ (translation does not alter distances). Hence, if $Q_0, \dots, Q_n$ is a poly-

line inscribed in $C(O, r)$, its image is a polyline inscribed in $C(O', r')$ of length exactly $k L(Q_0, \dots, Q_n)$; and conversely, every polyline inscribed in $C(O', r')$ comes, via σ^{-1} , from one inscribed in $C(O, r)$.

Step 3 — the supremum scales by k . The correspondence of Step 2 is a bijection between the set $\mathcal{L}$ of lengths of polylines inscribed in $C(O, r)$ and the set $\mathcal{L}'$ of those inscribed in $C(O', r')$, which multiplies every element by the constant $k > 0$. For such a correspondence $\sup \mathcal{L}' = k \sup \mathcal{L}$, that is,

$$C' = k C = \frac{r'}{r} C.$$

Step 4 — conclusion. Dividing by $2r'$:

$$\frac{C'}{2r'} = \frac{r' C}{r \cdot 2r'} = \frac{C}{2r}.$$

□

Definition 6.2: The constant π . One defines

$$\pi := \frac{C}{2r}$$

for one (and hence, by Theorem 6.2, for every) circle of radius r and length C . Equivalently, π is half the length of the circle of radius 1.

$$C = 2\pi r = \pi d \quad \Longleftrightarrow \quad r = \frac{C}{2\pi}, \quad d = \frac{C}{\pi}$$

Warning. The formula $C = 2\pi r$ is not a theorem: it is the definition of π rewritten.

The whole mathematical content lies in Theorem 6.2, which guarantees that this ratio is the same for every circle. Whoever presents $C = 2\pi r$ as a result to be proved, and then “proves” it with a limit, is really verifying a consistency, not proving the formula (see Remark 6.12).

Remark 6.4: What is known about π , and from where. Definition 6.2 says that π exists, not what value it has nor what nature. The main facts, proved in Chapter 1, are: π is **irrational** (Lambert, 1761) and **transcendental** (Lindemann, 1882); from transcendence follows the impossibility of squaring the circle with ruler and compass.

On the computational front, computing π has become a benchmark for hardware rather than for mathematics: the record documented at the time of writing is **314 trillion digits** (StorageReview and Micron, November 2025), obtained in about 110 days on a single server, where the bottleneck is not the processor but the transfer bandwidth to the NVMe drives. This figure is bound to age quickly: read it as an order of magnitude, not as a definitive value.

6.4. Radian measure and arc length

Definition 6.3: Radian measure. The **radian measure** of a central angle is the length of the arc it cuts on the circle of radius 1 concentric with the vertex.

Theorem 6.3: Arc length. On a circle of radius r , the arc cut by a central angle of measure θ radians has length

$$l = r\theta.$$

In particular, for $\theta = 2\pi$ one recovers $l = 2\pi r = C$.

Proof. The homothety with centre the vertex and ratio r maps the unit circle onto the circle of radius r and maps the arc cut by the angle onto the arc cut by the same angle (angles are invariant under homothety). By Step 3 of the proof of Theorem 6.2, applied to the arc rather than to the whole curve, the length is multiplied by r :

$$l = r \cdot (\text{length of the unit arc}) = r\theta,$$

where the last equality is Definition 6.3. For $\theta = 2\pi$: the unit arc of measure 2π is the whole unit circle, of length 2π by Definition 6.2, hence $l = 2\pi r$. $\square$

Remark 6.5: The radian is not a unit of measurement. Being the ratio of two lengths (arc and radius), radian measure is a pure number. This is why in the formulas of analysis — derivatives, Taylor series, the fundamental limit itself — angles must be in radians: in degrees the conversion factor $\pi/180$ would appear everywhere, and $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ would be $\pi/180 \approx 0.01745$ rather than 1.

Remark 6.6: Why the ellipse does not have an elementary length. Theorem 6.3 gives the simplest possible formula for the length of a circular arc, $l = r\theta$: an elementary (indeed linear) function of the angle. It is worth asking whether the same happens for the ellipse, the curve obtained by deforming the circle along one axis — and the answer, surprisingly, is no.

For the ellipse $x = a \cos t$, $y = b \sin t$ ($0 \leq t \leq 2\pi$, $a \geq b > 0$), the same procedure that gives $l = r\theta$ for the circle (summing the lengths of infinitesimal arcs, i.e. integrating $\sqrt{\dot{x}^2 + \dot{y}^2} dt$) leads to the total length

$$L = 4a \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 t} dt =: 4a E(e), \quad e := \sqrt{1 - \frac{b^2}{a^2}} \quad (\text{eccentricity}).$$

The integral $E(e)$ is called the **complete elliptic integral of the second kind**: it is precisely from this integral, and from the family of functions it belongs to, that the historical name “elliptic” given in analysis to an entire class of functions and integrals derives — functions and integrals that have nothing to do with ellipses beyond this origin.

Imported fact, not proved here. For $e \neq 0$, $E(e)$ is not expressible by elementary functions (powers, roots, exponentials, logarithms, trigonometric functions and their inverses, composed a finite number of times). The rigorous proof of this non-expressibility belongs to differential Galois theory (Liouville’s theory of elementary functions), entirely outside the tools of this volume; see the source at the end of the chapter.

Numerical check on a concrete case ($a = 2$, $b = 1$, hence $e = \sqrt{1 - 1/4} = \sqrt{3}/2 \approx 0.866025$). By two independent routes:

Method	L
Direct numerical quadrature of the integral (Simpson, 2000 subintervals)	9.688448
Power series for $E(e)$ truncated at order 40 $\left(E(e) = \frac{\pi}{2} \left[1 - \sum_{k=1}^{40} \left(\frac{(2k)!}{2^{2k}(k!)^2}\right)^2 \frac{e^{2k}}{2k-1}\right]\right)$	9.688448

The two values agree to the six decimal digits shown (difference 3×10^{-8} , due to the truncation of the series) $\checkmark$. For comparison, the circle of radius $r = 1.5$ (roughly the same mean semi-perimeter $(a + b)/2$) would have length $2\pi \cdot 1.5 = 9.424778$: close but different, as it must be for curves of different shape.

The thread running through the treatise. When $e = 0$ (i.e. $a = b = r$, the ellipse degenerates into the circle), the formula gives $E(0) = \pi/2$ and hence $L = 4a \cdot \pi/2 = 2\pi a = 2\pi r$: this recovers exactly Theorem 6.3. It is because the eccentricity of the circle is zero that its arc length collapses to an elementary function: the circle is the simplest degenerate case of a family of curves where, in general, “almost everything” — starting with the length — requires higher transcendental functions.

6.5. Proof of Archimedes’ postulate

INFO

What is proved here. That a closed convex curve contained in another has smaller or equal length. It is the principle Archimedes assumes as a postulate at the beginning of *On the Sphere and Cylinder*, and which is needed in the next section for the inequality $C < q_n$ between the length of the circle and the perimeter of the circumscribed polygon.

Prerequisites. Only: the strict triangle inequality (A8) of Chapter 2; Definition 6.1 of length as the supremum of the inscribed polylines; the least-upper-bound property of $\mathbb{R}$; Theorem 3.11 (Chapter 3) on the equality of the tangent segments from an external point, used only in Corollary 6.2. Neither differential calculus, nor sine and cosine, nor any result of Section 6.6 is used: this section is *logically independent* of the one that follows, and may therefore be used inside it without circularity.

It may be skipped on a first reading, accepting as stated Theorem 6.4 and Corollary 6.2.

6.5.1 Convex sets, inscribed polygons, polylines

Definition 6.4: Convex set and convex body. A set $K \subseteq \mathbb{R}^2$ is called **convex** if for every pair of its points P, Q the whole segment $\overline{PQ}$ is contained in K . A **convex body** is a convex set that is closed, bounded and has non-empty interior. Its boundary ∂K is the associated **closed convex curve**.

Given points $P_1, \dots, P_m$ of the plane, their **convex hull** $\text{conv}\{P_1, \dots, P_m\}$ is the set of the combinations $\sum_j \mu_j P_j$ with $\mu_j \geq 0$ and $\sum_j \mu_j = 1$: it is the smallest convex set containing them, and it is a convex polygon (possibly degenerating into a segment or a point).

Remark 6.7: Inscribed polylines and inscribed polygons. For a closed convex curve $\Gamma = \partial K$, the cyclic order required by Definition 6.1 is the order in which the chosen points appear while traversing the boundary once; for the circle it is the angular order used so far. Given $n \geq 3$ distinct points of Γ in that order, the closed polyline joining them is the boundary of their convex hull (points falling inside a side do not alter the sum of the lengths, by additivity along collinear segments). Hence

$$\text{len}(\partial K) = \sup\{\text{perim}(P) : P \text{ a convex polygon with all vertices on } \partial K\}.$$

Note moreover that **adding vertices never decreases** the length of a polyline: inserting M between Q_{k-1} and Q_k replaces $|Q_{k-1}Q_k|$ by $|Q_{k-1}M| + |MQ_k| \geq |Q_{k-1}Q_k|$, by the triangle inequality. Since ∂K is not contained in a line (the interior of K is non-empty), one may therefore always assume that the chosen vertices are not all collinear.

6.5.2 The cutting lemma

Lemma 6.1: *Cutting a convex polygon with a half-plane.* Let Q be a convex polygon and let H be a closed half-plane such that $Q \cap H$ has non-empty interior. Then $Q' = Q \cap H$ is a convex polygon and

$$\text{perim}(Q') \leq \text{perim}(Q),$$

with strict inequality if $Q \not\subseteq H$.

Proof. Q' is the intersection of two convex sets, hence convex; it is bounded and closed, and it is bounded by finitely many segments (the truncated sides of Q and at most one new side on the line $r = \partial H$): it is a convex polygon.

If $Q \subseteq H$ then $Q' = Q$ and equality holds. Assume therefore $Q \not\subseteq H$; let X_1 be an interior point of Q' (it exists by hypothesis) and $X_2 \in Q \setminus H$. The segment $\overline{X_1 X_2}$ is contained in Q and crosses r ; the crossing point is interior to Q , hence r meets the interior of Q and therefore $r \cap Q$ is a segment $\overline{AB}$ with $A \neq B$ and $A, B \in \partial Q$.

The two points A and B split the closed polyline ∂Q into two open polylines: α , contained in H , and β , contained in the opposite half-plane. By additivity of lengths along a polyline,

$$\text{perim}(Q) = \text{len}(\alpha) + \text{len}(\beta), \quad \text{perim}(Q') = \text{len}(\alpha) + |AB|,$$

because the boundary of Q' consists of α and of the segment $\overline{AB}$. Now β is a polyline joining A to B and having at least one vertex outside H , hence outside the segment $\overline{AB} \subseteq r$: by the strict triangle inequality (A8), applied repeatedly to the vertices of β ,

$$|AB| < \text{len}(\beta).$$

Adding $\text{len}(\alpha)$ to both sides gives $\text{perim}(Q') < \text{perim}(Q)$. □

Lemma 6.2: *Monotonicity of the perimeter for nested convex polygons.* Let $P \subseteq Q$ be two convex polygons, with P of non-empty interior. Then $\text{perim}(P) \leq \text{perim}(Q)$.

Proof. Let $r_1, \dots, r_m$ be the lines of the sides of P and, for each i , let H_i be the closed half-plane with boundary r_i containing P . A convex polygon is the intersection of the half-planes determined by its sides: $P = H_1 \cap \dots \cap H_m$. (A point of P lies in every H_i by convexity; conversely a point outside P is separated from P by at least one side, that is, it lies outside the corresponding H_i .)

Set $Q_0 = Q$ and $Q_i = Q_{i-1} \cap H_i$ for $i = 1, \dots, m$. Every Q_i contains P , hence has non-empty interior, and is a convex polygon; by Lemma 6.1,

$$\text{perim}(Q_m) \leq \text{perim}(Q_{m-1}) \leq \dots \leq \text{perim}(Q_0) = \text{perim}(Q).$$

Finally $Q_m = Q \cap H_1 \cap \dots \cap H_m = Q \cap P = P$, because $P \subseteq Q$. Hence $\text{perim}(P) \leq \text{perim}(Q)$. □

6.5.3 From the polygon to the convex body

Lemma 6.3: *Every polygon inside a convex body lies in an inscribed polygon.* Let K be a convex body and let $P \subseteq K$ be a convex polygon with non-empty interior and vertices $V_1, \dots, V_m$. Then there exists a convex polygon R with all its vertices on ∂K such that $P \subseteq R$.

Proof. Step 1 — a ray issuing from an interior point meets the boundary. Let O be an interior point of P (it exists by hypothesis) and let $\mathbf{u}$ be a unit vector. The set $T = \{t \geq 0 : O + t\mathbf{u} \in K\}$ is non-empty ($0 \in T$), closed (K is closed) and bounded (K is bounded); by the least-upper-bound property $t^* = \max T$ exists, and $W = O + t^*\mathbf{u}$ belongs to K but not to its interior, because the points $O + t\mathbf{u}$ with $t > t^*$ do not lie in K . Hence $W \in \partial K$.

Step 2 — the lifted points. For each j write $V_j = O + t_j\mathbf{u}_j$ with $t_j > 0$ and $\mathbf{u}_j$ a unit vector (the vertices differ from O , which is interior). Since $V_j \in K$ we have $t_j \in T_j$, hence $t_j \leq t_j^*$: the point $W_j = O + t_j^*\mathbf{u}_j \in \partial K$ satisfies $V_j \in \overline{OW_j}$. Put $\lambda_j = t_j/t_j^* \in (0, 1]$, so that $V_j = O + \lambda_j(W_j - O)$.

Step 3 — O is a convex combination of the W_j . Since O is interior to $P = \text{conv}\{V_1, \dots, V_m\}$, there exist $\mu_j \geq 0$ with $\sum_j \mu_j = 1$ and $O = \sum_j \mu_j V_j$. Substituting the expression of V_j :

$$O = \sum_j \mu_j (O + \lambda_j (W_j - O)) = O + \sum_j \mu_j \lambda_j (W_j - O) \implies \sum_j \mu_j \lambda_j (W_j - O) = \mathbf{0}.$$

Setting $\sigma = \sum_j \mu_j \lambda_j > 0$ (all the λ_j are positive and at least one μ_j is), one gets

$$O = \frac{1}{\sigma} \sum_j \mu_j \lambda_j W_j, \quad \sum_j \frac{\mu_j \lambda_j}{\sigma} = 1,$$

that is, $O \in R := \text{conv}\{W_1, \dots, W_m\}$.

Step 4 — conclusion. Every V_j lies on the segment $\overline{OW_j}$, whose endpoints both belong to R ; by convexity $V_j \in R$, and therefore $P = \text{conv}\{V_1, \dots, V_m\} \subseteq R$. The polygon R has all its vertices on ∂K (they are among the W_j), as required. $\square$

Theorem 6.4: Archimedes' postulate: monotonicity of length for nested convex sets. Let $K_1 \subseteq K_2$ be two convex bodies of the plane. Then both boundaries are rectifiable and

$$\text{len}(\partial K_1) \leq \text{len}(\partial K_2).$$

Proof. Step 1 — rectifiability. Let $\square$ be a square containing K_2 (it exists because K_2 is bounded). By Observation 6.7, every polyline inscribed in ∂K_i is the perimeter of a convex polygon contained in $K_i \subseteq \square$; by Lemma 6.2 its length does not exceed $\text{perim}(\square)$. The set of the lengths of the inscribed polylines is therefore bounded above and has a finite supremum: ∂K_1 and ∂K_2 are rectifiable.

Step 2 — comparison. Let P be any convex polygon with vertices on ∂K_1 ; by Observation 6.7 one may assume that its vertices are not all collinear, that is, that P has non-empty interior. Since $P \subseteq K_1 \subseteq K_2$, Lemma 6.3 applied to $K = K_2$ yields a convex polygon R with vertices on ∂K_2 and $P \subseteq R$. Then, by Lemma 6.2 and by the definition of supremum,

$$\text{perim}(P) \leq \text{perim}(R) \leq \text{len}(\partial K_2).$$

Step 3 — passing to the supremum. The number $\text{len}(\partial K_2)$ is therefore an upper bound for the set of the perimeters of the convex polygons inscribed in ∂K_1 , whose supremum is $\text{len}(\partial K_1)$ by Observation 6.7. Hence $\text{len}(\partial K_1) \leq \text{len}(\partial K_2)$. $\square$

Corollary 6.2: The circumscribed polygon is strictly longer. Let C be the length of the circle $C(O, r)$ and let q_n be the perimeter of the regular n -gon, $n \geq 3$, circumscribed about it. Then

$$q_{2n} < q_n \quad \text{and} \quad C < q_n.$$

Proof. Step 1 — doubling the number of sides strictly shortens. Let $T_1, \dots, T_n$ be the points of tangency of the sides of Q_n and $V_1, \dots, V_n$ its vertices, with V_k the intersection of the two

sides tangent at T_k and T_{k+1} (indices modulo n). Let M_k be the midpoint of the arc $\widehat{T_k T_{k+1}}$ that contains no other point of tangency. The $2n$ points $T_1, M_1, T_2, M_2, \dots$ are equally spaced on the circle, and the tangents at them bound exactly Q_{2n} . The tangent at M_k meets the side $\overline{T_k V_k}$ at a point A_k and the side $\overline{V_k T_{k+1}}$ at a point B_k , with M_k between A_k and B_k (the configuration is symmetric with respect to the line OM_k , which is an axis of symmetry both of the circle and of the angle at V_k).

By Theorem 3.11 the two tangent segments drawn from the same external point are congruent, hence

$$|A_k T_k| = |A_k M_k|, \quad |B_k M_k| = |B_k T_{k+1}|.$$

The vertex V_k lies on the ray OM_k beyond M_k (by the symmetry just mentioned) and $|OV_k| > r$: hence V_k does not belong to the tangent at M_k , and in particular $V_k \notin \overline{A_k B_k}$. The strict triangle inequality (A8) in the triangle $A_k V_k B_k$ then gives $|A_k V_k| + |V_k B_k| > |A_k B_k| = |A_k M_k| + |M_k B_k|$. Therefore

$$\begin{aligned} |T_k V_k| + |V_k T_{k+1}| &= |T_k A_k| + |A_k V_k| + |V_k B_k| + |B_k T_{k+1}| \\ &= |A_k M_k| + |M_k B_k| + (|A_k V_k| + |V_k B_k|) \\ &> |A_k M_k| + |M_k B_k| + |A_k M_k| + |M_k B_k| = 2(|A_k M_k| + |M_k B_k|). \end{aligned}$$

Summing over $k = 1, \dots, n$: on the left one obtains q_n (each side of Q_n is split by its point of tangency into two parts, distributed between two consecutive vertices), on the right one obtains q_{2n} (each piece $\overline{T_k A_k}$ has the same length as $\overline{A_k M_k}$, and likewise on the other side). Hence $q_n > q_{2n}$.

Step 2 — conclusion. The closed disc of centre O and radius r and the polygon Q_{2n} are convex bodies, the first contained in the second; by Theorem 6.4, $C \leq q_{2n}$. Combining with Step 1:

$$C \leq q_{2n} < q_n.$$

□

Example 6.1: Numerical check on the circumscribed polygons. The perimeter of the regular n -gon circumscribed about a circle of radius r is $q_n = 2nr \tan(\pi/n)$ (a formula derived in Chapter 8: here it serves only as an independent *check*, and does not enter the logical chain of this section, which uses no trigonometry). For $r = 1$:

$$q_4 = 8, \quad q_8 = 6.627\,417, \quad q_{16} = 6.365\,196, \quad q_{32} = 6.303\,450, \quad q_{64} = 6.288\,237,$$

whereas $C = 2\pi = 6.283\,185$. The sequence is strictly decreasing — as stated by Step 1 of Corollary 6.2 — and always stays strictly above 2π , as stated by Step 2. ✓

Second computation, independent of the formula. For $n = 4$ the circumscribed polygon is the square of side $2r = 2$, of perimeter 8: it agrees with q_4 . ✓ For $n = 8$, the side of the circumscribed octagon is $2r \tan(22.5^\circ)$ and $\tan(22.5^\circ) = \sqrt{2} - 1 = 0.414\,214$ (from the bisection formula, Chapter 8), whence $q_8 = 16(\sqrt{2} - 1) = 6.627\,417$. ✓

Remark 6.8: Why this route, and not the radial projection. The proof just given uses only the triangle inequality and convexity: it is therefore available exactly at the point of the volume where it is needed. An alternative route, often quoted, consists in projecting the outer curve onto the inner one. One must however say precisely which projection: the radial projection from an interior point — the one suggested by intuition — is **not** in general 1-Lipschitz, and proves nothing. The projection that works is the metric one, that is, the map sending each point to the nearest point of the inner convex set: this is the content of

the variant that follows.

The metric projection onto the inner convex set. This is the modern route, and it is worth following because its tool — the projection onto the nearest convex set — reappears in functional analysis, in optimisation and in signal theory.

Step 1 — existence and uniqueness of the nearest point. Let K be a non-empty closed convex subset of the plane and let $X \in \mathbb{R}^2$. Put $d = \inf_{Y \in K} |XY|$. The set $K \cap \overline{B}(X, d+1)$ is closed, bounded and non-empty; the continuous function $Y \mapsto |XY|$ attains its minimum there (Weierstrass' theorem), and that minimum equals d : hence there is $Y_0 \in K$ with $|XY_0| = d$. As for uniqueness, let $Y_1 \neq Y_2$ be two points of K at distance d from X and let M be their midpoint, which belongs to K by convexity. From the parallelogram identity, with $\mathbf{a} = X - Y_1$ and $\mathbf{b} = X - Y_2$ (so that $\mathbf{a} + \mathbf{b} = 2(X - M)$ and $\mathbf{a} - \mathbf{b} = Y_2 - Y_1$),

$$|\mathbf{a}|^2 + |\mathbf{b}|^2 = \frac{1}{2}(|\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2) \implies 2d^2 = 2|XM|^2 + \frac{1}{2}|Y_1Y_2|^2,$$

that is, $|XM|^2 = d^2 - \frac{1}{4}|Y_1Y_2|^2 < d^2$: absurd, since $M \in K$. The **metric projection** $\pi_K(X) := Y_0$ is thus well defined.

Step 2 — variational characterization. For $X \in \mathbb{R}^2$ and $Y \in K$ one has $Y = \pi_K(X)$ if and only if

$$\langle X - Y, Z - Y \rangle \leq 0 \quad \text{for every } Z \in K.$$

($\Rightarrow$) Let $Y = \pi_K(X)$, $Z \in K$ and $t \in (0, 1]$. The point $Y + t(Z - Y)$ lies in K by convexity, hence

$$|X - Y|^2 \leq |X - Y - t(Z - Y)|^2 = |X - Y|^2 - 2t \langle X - Y, Z - Y \rangle + t^2|Z - Y|^2,$$

whence $2 \langle X - Y, Z - Y \rangle \leq t|Z - Y|^2$ for every $t \in (0, 1]$; since the left-hand side does not depend on t and the right-hand side can be made arbitrarily small, $\langle X - Y, Z - Y \rangle \leq 0$.

($\Leftarrow$) If the inequality holds for every $Z \in K$, then

$$|X - Z|^2 = |X - Y|^2 - 2 \langle X - Y, Z - Y \rangle + |Z - Y|^2 \geq |X - Y|^2,$$

that is, Y realizes the minimum.

Step 3 — the projection is 1-Lipschitz. Let $X_1, X_2 \in \mathbb{R}^2$, $Y_i = \pi_K(X_i)$ and $\mathbf{v} = Y_2 - Y_1$. Step 2 with $(X, Y, Z) = (X_1, Y_1, Y_2)$ gives $\langle X_1 - Y_1, \mathbf{v} \rangle \leq 0$; with $(X, Y, Z) = (X_2, Y_2, Y_1)$ it gives $\langle X_2 - Y_2, -\mathbf{v} \rangle \leq 0$, that is, $\langle X_2 - Y_2, \mathbf{v} \rangle \geq 0$. Subtracting,

$$\langle (X_2 - X_1) - \mathbf{v}, \mathbf{v} \rangle \geq 0 \implies |\mathbf{v}|^2 \leq \langle X_2 - X_1, \mathbf{v} \rangle \leq |X_1X_2| |\mathbf{v}|,$$

using the Cauchy–Schwarz inequality. If $\mathbf{v} \neq \mathbf{0}$ one divides by $|\mathbf{v}|$; in either case

$$|\pi_K(X_1) \pi_K(X_2)| \leq |X_1X_2|.$$

Step 4 — the projection maps the outer boundary onto the inner one. Let $K_1 \subseteq K_2$ be convex bodies and $Y \in \partial K_1$. Let r be a supporting line of K_1 at Y (a line through Y leaving K_1 entirely in one closed half-plane) and let $\mathbf{u}$ be the outward unit normal to r . For every $Z \in K_1$ one has $\langle \mathbf{u}, Z - Y \rangle \leq 0$, hence $\langle (Y + t\mathbf{u}) - Y, Z - Y \rangle = t \langle \mathbf{u}, Z - Y \rangle \leq 0$ for every $t \geq 0$: by Step 2, $\pi_{K_1}(Y + t\mathbf{u}) = Y$. As in Step 1 of Lemma 6.3, the ray $\{Y + t\mathbf{u} : t \geq 0\}$ leaves K_2 at a point $P \in \partial K_2$, and $\pi_{K_1}(P) = Y$: the restriction of π_{K_1} to ∂K_2 is therefore **surjective** onto ∂K_1 .

What is missing, and why we do not use it. A 1-Lipschitz map does not increase the length of a polyline: if $\sum_k |Q_{k-1}Q_k|$ is a polyline inscribed in ∂K_1 and the P_k are preimages of the Q_k on ∂K_2 , then $\sum_k |Q_{k-1}Q_k| \leq \sum_k |P_{k-1}P_k|$. To conclude, however, one needs to

know that the P_k may be chosen *in the cyclic order* of ∂K_2 , otherwise the right-hand side is not the length of an inscribed polyline. This is true — the segments $\overline{Q_k P_k}$ are pairwise disjoint, because every point of $\overline{Q_k P_k}$ projects onto Q_k and the projection is single-valued — but deducing the preservation of the cyclic order requires a topological argument (disjoint spokes in an annulus) that lies outside this volume. This is exactly the reason why the main proof follows the elementary polygonal route, which does not need that argument. $\square$

$$K_1 \subseteq K_2 \text{ convex bodies} \implies \text{len}(\partial K_1) \leq \text{len}(\partial K_2)$$

$$p_n < C < q_n, \quad p_n < p_{2n}, \quad q_{2n} < q_n$$

Remark 6.9: Why $p_n < p_{2n}$ is strict. Passing from n to $2n$ sides, each side of the inscribed polygon (a chord) is replaced by two shorter sides (two chords resting on the arc subtended by the first): by the triangle inequality, applied to the triangle formed by the original chord and the midpoint of the arc, the sum of the two new chords exceeds the old chord, with **strict** inequality because the midpoint of the arc does not lie on the segment (it is not collinear with the two endpoints). Summing over all sides gives $p_n < p_{2n}$. The symmetric argument (circumscribed polygon, shortening polyline) gives $q_{2n} < q_n$.

6.6. Archimedes' theorem on the area

The central result of the chapter, and one of the peaks of Greek mathematics. Archimedes states it as an equivalence between two figures, not as a formula: the formula $A = \pi r^2$ is our rewriting.

Theorem 6.5: Archimedes' theorem, Measurement of a Circle, Prop. 1 (3rd c. BC). A circle is equal to a right triangle having one leg equal to the radius and the other equal to the circumference. In formulas, denoting by A the area of the disc of radius r and by C the length of the circle,

$$A = \frac{1}{2} C r = \pi r^2.$$

6.6.1 The exhaustion lemma

Archimedes' proof is by double reductio ad absurdum, and its engine is a lemma that Euclid states as Proposition X.1 and that today we would call a convergence statement.

Lemma 6.4: Exhaustion lemma (Euclid, Elements X.1). Let two positive magnitudes M and ε be given, with $\varepsilon < M$. If from M one subtracts a part not less than its half, and from the remainder again a part not less than its half, and so on, then after finitely many steps the remainder is less than ε .

Proof. Writing $M_0 = M$ and M_k for the remainder after k steps, the hypothesis gives $M_k \leq \frac{1}{2} M_{k-1}$ for every k , and hence by induction $M_k \leq M/2^k$. It then suffices to choose k with $2^k > M/\varepsilon$: such a k exists because the sequence 2^k is not bounded above (Archimedean property of $\mathbb{R}$; equivalently, $2^k \geq k + 1$ for every $k \geq 0$, immediate by induction). For such a k one has $M_k \leq M/2^k < \varepsilon$. $\square$

Remark 6.10: Where the lemma is needed, and why “more than half” is essential. In our case M is the difference between the area of the disc and that of the inscribed polygon. Doubling the number of sides (from n to $2n$), the part of that difference which is

removed is more than half: each of the remaining circular segments is replaced by a triangle occupying more than half of its area, because the triangle inscribed in a circular segment, having the chord as base and the midpoint of the arc as vertex, has area greater than half the segment (it is half of the rectangle circumscribed about the segment, which in turn contains it). Without the clause “not less than half” the lemma is false: subtracting a tenth of the remainder each time, the remainder tends to zero but not in a way controlled by a uniform geometric estimate.

6.6.2 The two polygon estimates

Lemma 6.5: Areas of inscribed and circumscribed polygons. Let P_n be a regular n -gon inscribed in $C(O, r)$, of perimeter p_n and apothem a_n ; let Q_n be the circumscribed regular n -gon, of perimeter q_n . Then

$$\text{area}(P_n) = \frac{1}{2} p_n a_n, \quad \text{area}(Q_n) = \frac{1}{2} q_n r,$$

and moreover $a_n < r$, $p_n < C < q_n$.

Proof. Area formulas. Joining the centre to the vertices, the regular n -gon decomposes into n congruent triangles, each with a side as base and the apothem as height; by (Ar1) and (Ar4) the total area is $n \cdot \frac{1}{2}(\text{side})(\text{apothem}) = \frac{1}{2}pa$. For the circumscribed polygon the apothem is the radius, because every side is tangent to the circle and the distance from the centre to a tangent is r (Theorem 2.6).

$a_n < r$. The apothem of the inscribed polygon is the distance from the centre to a chord not passing through the centre, hence $a_n < r$ by Theorem 3.5 (or directly: it is a leg of a right triangle with hypotenuse r).

$p_n < C$. The perimeter of the inscribed polygon is the length of a particular inscribed polyline, hence it does not exceed the supremum, which is C ; and it is strictly smaller by the same argument as in Corollary 6.1.

$C < q_n$. The circumscribed polygon contains the disc, and its boundary is a closed convex curve containing $C(O, r)$ in its interior. The inequality, with the strict sign, is Corollary 6.2 of the preceding section. $\square$

Warning. Archimedes’ postulate on the monotonicity of length for nested convex curves is not assumed here: it is proved. Archimedes states it as the second axiom at the beginning of *On the Sphere and Cylinder* and uses it without proof; we have proved it in Section 6.5 (Theorem 6.4), and from it we have obtained the *strict* form $C < q_n$ (Corollary 6.2) needed here. The proof uses only the triangle inequality (A8), convexity and Definition 6.1: the logical chain of this chapter therefore stays entirely inside properties (Ar1)–(Ar4) and the definition of length. Only the four properties of area remain assumed, and declared (Observation 6.1).

6.6.3 The double reductio ad absurdum

Proof. (of Theorem 6.5.) Let $T = \frac{1}{2}Cr$ be the area of the right triangle with legs C and r (by (Ar4)), and let A be the area of the disc. We show that neither $A > T$ nor $A < T$ is possible.

First case: suppose $A > T$.

Step 1. Put $\varepsilon = A - T > 0$. Starting from the inscribed square and repeatedly doubling the number of sides, the difference $A - \text{area}(P_n)$ is reduced each time by more than half

(Remark 6.10); by Lemma 6.4 there is therefore an n with

$$A - \text{area}(P_n) < \varepsilon = A - T \quad \implies \quad \text{area}(P_n) > T.$$

Step 2. On the other hand, by Lemma 6.5,

$$\text{area}(P_n) = \frac{1}{2} p_n a_n < \frac{1}{2} C r = T,$$

having used $p_n < C$ and $a_n < r$, both strict and both between positive quantities.

Step 3. Steps 1 and 2 give $\text{area}(P_n) > T$ and $\text{area}(P_n) < T$: absurd. Hence $A > T$ is impossible.

Lemma 6.6: *The triangle cut off by the tangent exceeds half the remainder.*
Let V be a vertex of the circumscribed polygon Q_n , and let T_1, T_2 be the points of tangency of the two sides of Q_n meeting at V . Let M be the midpoint of the arc $\widehat{T_1 T_2}$ (the one closer to V) and let R be the region between the segments $\overline{VT_1}, \overline{VT_2}$ and the arc $\widehat{T_1 T_2}$ (the piece of Q_n that disappears, at that vertex, when passing to Q_{2n}). The tangent to the circle at M cuts $\overline{VT_1}$ at a point A and $\overline{VT_2}$ at a point B : then

$$\text{area}(VAB) > \frac{1}{2} \text{area}(R).$$

Proof. By the symmetry of the quadrilateral OT_1VT_2 (the triangles OT_1V, OT_2V are congruent by SSS: $|OT_1| = |OT_2| = r$, $|VT_1| = |VT_2|$ by Theorem 3.11(c) of Chapter 3, OV in common), the point V lies on the line OM . The region R is the union of the triangle VAB and the two regions R_1 (between $\overline{AT_1}, \overline{AM}$ and the arc $\widehat{T_1 M}$) and R_2 (symmetric, between $\overline{BT_2}, \overline{BM}$ and the arc $\widehat{MT_2}$).

R_1 is contained in the triangle AT_1M : the arc $\widehat{T_1 M}$ lies on the side of the centre O with respect to the chord T_1M (Corollary 6.1 and convexity of the circle), while A lies on the other side (it is on the tangent at M , which touches the circle only at M). In the triangle VMA the angle at M is a right angle (the tangent at M is perpendicular to the radius OM , and $V \in OM$), so VA is the hypotenuse: $VA > AM$. But $AM = AT_1$ (tangent segments from A to the circle, again by Theorem 3.11(c)), hence $VA > AT_1$. The triangles VAM and T_1AM have the same height (from M) over the line VT_1 , so their area ratios coincide with the ratio of the bases:

$$\frac{\text{area}(VAM)}{\text{area}(T_1AM)} = \frac{VA}{AT_1} > 1 \quad \implies \quad \text{area}(VAM) > \text{area}(T_1AM) \geq \text{area}(R_1)$$

(the last inequality because $R_1 \subseteq \text{triangle } AT_1M$). Likewise $\text{area}(VBM) > \text{area}(R_2)$. Adding the two inequalities and using $\text{area}(VAM) + \text{area}(VBM) = \text{area}(VAB)$ (the two triangles fit together along VM , which is interior to AB by symmetry):

$$\text{area}(VAB) > \text{area}(R_1) + \text{area}(R_2) = \text{area}(R) - \text{area}(VAB),$$

that is, $2 \text{area}(VAB) > \text{area}(R)$, the claim. □

Numerical check. With $r = 1$ and $\theta = 30^\circ$ (that is, $\angle T_1OM = \theta$, the case of the circumscribed polygon with central angle 2θ per side — the hexagon, $n = 6$; the half-angle at the vertex is then $90^\circ - \theta = 60^\circ$): $\text{area}(VAB) = \tan(\theta/2)(1/\cos \theta - 1) = 0.041452$ and $\text{area}(AT_1M) + \text{area}(BT_2M) = (1 - \cos \theta) \tan(\theta/2) = 0.035898$ (independent script): $0.041452 > 0.035898$, consistent with $\text{area}(VAB) > \text{area}(R_1) + \text{area}(R_2)$. ✓

Second case: suppose $A < T$.

Step 1. Put $\varepsilon = T - A > 0$. Starting from the circumscribed square and doubling the number of sides, Lemma 6.6, applied to each vertex of the current polygon, shows that the difference $\text{area}(Q_n) - A$ is reduced each time by more than half (the same role, for the circumscribed polygon, as Observation 6.10 for the inscribed one); by Lemma 6.4 there is an n with

$$\text{area}(Q_n) - A < \varepsilon = T - A \quad \implies \quad \text{area}(Q_n) < T.$$

Step 2. On the other hand, by Lemma 6.5,

$$\text{area}(Q_n) = \frac{1}{2} q_n r > \frac{1}{2} C r = T,$$

having used $q_n > C$.

Step 3. Absurd here too. Hence $A < T$ is impossible.

Conclusion. By the trichotomy of the order on $\mathbb{R}$, only $A = T = \frac{1}{2}Cr$ remains. Substituting $C = 2\pi r$ (Definition 6.2):

$$A = \frac{1}{2} \cdot 2\pi r \cdot r = \pi r^2.$$

□

$$A = \pi r^2 \quad \iff \quad r = \sqrt{A/\pi}$$

Remark 6.11: Why this proof is remarkable. Three reasons. First: it is **rigorous** by modern standards — it uses no limits, but Lemma 6.4 plays exactly the role of the ε - N definition of limit, two thousand years early. Second: it is **constructive in structure but not in value** — it proves that $A = \frac{1}{2}Cr$ without saying how much C is; the computation of π is the object of Propositions 2 and 3 of the same work, where Archimedes obtains $3\frac{10}{71} < \pi < 3\frac{1}{7}$ with 96-sided polygons. Third: the double-reductio structure is the model of all proofs by exhaustion, and is exactly what the infinitesimal calculus would replace, simplifying it, in the seventeenth century.

6.6.4 Modern check: five alternative proofs

The proofs that follow are shorter but *not* more fundamental: they use integral calculus, which in turn rests on measure theory. They serve as independent checks and as a bridge towards Volume II.

Proof 1 — “Concertina” sectors. Divide the disc into n congruent sectors and arrange them alternately (one up, one down), obtaining a figure approximating a parallelogram of base $\approx \frac{1}{2} \cdot 2\pi r = \pi r$ and height $\approx r$. As $n \rightarrow \infty$ the figure converges to a rectangle of area $\pi r \cdot r = \pi r^2$. It is a visually convincing argument but **not** a proof: the word “converges” would have to be made precise, and doing so requires exactly the work of Theorem 6.5. □

Proof 2 — Integration in polar coordinates.

$$A = \int_0^{2\pi} \int_0^r \rho \, d\rho \, d\varphi = \int_0^{2\pi} \frac{r^2}{2} \, d\varphi = \pi r^2.$$

□

Proof 3 — Trigonometric substitution. In Cartesian coordinates, with the substitution $x = r \sin t$:

$$A = 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \stackrel{x=r \sin t}{=} 2r^2 \int_{-\pi/2}^{\pi/2} \cos^2 t dt = \pi r^2.$$

□

Proof 4 — Cylindrical shells. Method of summing infinitesimal circles: the disc is thought of as the union of concentric circles of radius $\rho \in [0, r]$, each of “thickness” $d\rho$ and “length” $2\pi\rho$ (the area of an infinitesimal annulus is $dA = 2\pi\rho d\rho$). Integrating:

$$A = \int_0^r 2\pi\rho d\rho = \pi r^2.$$

Note the consistency with the Cavalieri–Torricelli principle ($dA/dr = 2\pi r = C(r)$: the derivative of the area with respect to the radius is the circumference). □

Proof 5 — Euclid XII.2 (by exhaustion, without π a priori). This route, historically the first (Eudoxus/Euclid, *Elements* XII.2), does not presuppose the existence of a constant π : it proves it.

Step 1 — the ratio $A(r)/r^2$ is the same for any two circles. Let $C(O, r)$, $C(O', r')$ be two circles, and for every n let P_n, P'_n be the regular n -gons inscribed in them. Two regular polygons with the same number of sides are similar, with ratio of similarity r/r' (the radii of the circumscribed circles); the area of similar figures scales with the square of the ratio of similarity (an elementary fact, by decomposition into similar triangles): $\text{area}(P_n)/\text{area}(P'_n) = r^2/r'^2$ for every n . By the Exhaustion Lemma (already used for Archimedes’ theorem in this chapter), $\text{area}(P_n) \rightarrow A(r)$ and $\text{area}(P'_n) \rightarrow A(r')$; passing to the limit in the constant ratio,

$$\frac{A(r)}{A(r')} = \lim_{n \rightarrow \infty} \frac{\text{area}(P_n)}{\text{area}(P'_n)} = \frac{r^2}{r'^2} \implies \frac{A(r)}{r^2} = \frac{A(r')}{r'^2}.$$

The ratio $A(r)/r^2$ therefore does not depend on the radius: it is a universal constant, which we call π (by definition, along this route): $A(r) = \pi r^2$.

Step 2 — consistency with $\pi = C/d$. It remains to check that this constant coincides with the one already defined in the chapter as C/d (circumference-to-diameter ratio). By Theorem 6.5 (already proved in this chapter, independently of XII.2, with $\pi = C/(2r)$ by definition), $A(r) = \frac{1}{2}rC = \frac{1}{2}r \cdot 2\pi r = \pi r^2$: the same constant. The two routes — defining π from the area (Euclid XII.2) or from the length (Archimedes) — converge to the same number. □

Remark 6.12: Consistency check with the limit of the perimeters. Once sine and cosine are available (Chapter 8), the inscribed regular n -gon has perimeter $p_n = 2nr \sin(\pi/n)$, because each side is a chord subtending a central angle $2\pi/n$ (chord formula, Chapter 2). By the fundamental limit $\lim_{h \rightarrow 0} \sin h/h = 1$:

$$\lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} 2\pi r \cdot \frac{\sin(\pi/n)}{\pi/n} = 2\pi r = C.$$

This computation does not prove that $C = 2\pi r$: that is Definition 6.2. It proves that the supremum of Definition 6.1 is actually attained as a limit along the sequence of regular polygons — a true and useful fact, but of a different nature. Presenting it as a derivation of the formula, as many texts do, creates the circularity denounced in the introduction to the

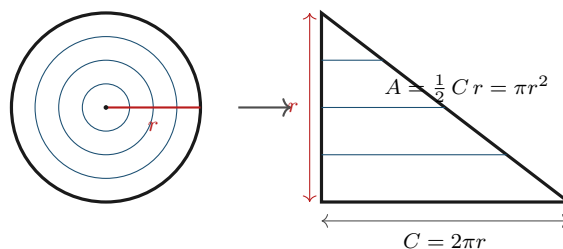


Figure 6.2: The idea behind Archimedes' theorem on the area of the circle. Think of the disc as a union of infinitely thin concentric rings: the ring of radius ρ , cut and straightened, becomes a segment of length $2\pi\rho$. Stacking the segments in order of radius fills exactly a triangle of base $C = 2\pi r$ and height r , whose area is $\frac{1}{2} C r = \pi r^2$. The figure illustrates the result, it does not prove it: Archimedes' rigorous proof proceeds by double reductio ad absurdum, showing that the area of the circle can be neither larger nor smaller than that of the triangle, and uses inscribed and circumscribed polygons instead of infinitesimal rings.

6.7. Archimedes' bounds $3\frac{10}{71} < \pi < 3\frac{1}{7}$, rigorously

Theorem 6.6: Archimedes' inequalities.

$$3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}.$$

Proof. Start from the hexagon ($n = 6$), for which $a_6 = n \tan(\pi/n) = 2\sqrt{3}$ (circumscribed) and $b_6 = n \sin(\pi/n) = 3$ exactly (inscribed), and double four times with the bisection relations (formulas of Chapter 8, which typographically follows this chapter but logically precedes it — the same notice given further ahead for Corollary 6.3; repeated in the worked example of Chapter 1): $a_{2n} = \frac{2a_n b_n}{a_n + b_n}$ (harmonic mean), $b_{2n} = \sqrt{a_{2n} b_n}$ (geometric mean), with $b_n < \pi < a_n$ for every n (Theorem 6.5 and Corollary 6.2).

At every step a square root appears ($\sqrt{3}$ at the start, then $\sqrt{a_{2n} b_n}$ at each doubling): to preserve rigour, every root must be replaced by a fraction on the *correct* side (a lower bound if it majorises the inscribed value b , an upper bound if it minorises the circumscribed value a), exactly as already discussed for the estimate of $\sqrt{3}$ (Observation on Archimedes' estimates, Chapter 1: $265/153 < \sqrt{3} < 1351/780$, checked by squaring). Propagating rational lower and upper bounds through every root, with a script using exact arbitrary-precision arithmetic (Python's `fractions` and `isqrt`, never floating point), one obtains at step $n = 96$:

$$b_{96} \in [3.1410182, 3.1410322], \quad a_{96} \in [3.1427007, 3.1427149]$$

(rigorous bounds, not roundings). In particular $b_{96} > 3.1410182 > 3.1408451 = 3 + 10/71$ and $a_{96} < 3.1427149 < 3.1428571 = 3 + 1/7$, hence

$$3 + \frac{10}{71} < b_{96} < \pi < a_{96} < 3 + \frac{1}{7}.$$

□

Remark 6.13: *Archimedes' numbers differ from ours, but the inequalities coincide.* The intermediate fractions used here (rational bounds via `isqrt`) are not the ones Archimedes chose, who approximated by hand with other fractions; but his final inequalities $3\frac{10}{71} < \pi < 3\frac{1}{7}$ still follow, and are in fact wider than the rigorous bound obtained here $[3.1410182, 3.1427149]$, against $[3.1408451, 3.1428571]$ of Archimedes): a safety margin that Archimedes deliberately kept, so as not to have to justify every single digit of his intermediate fractions.

6.8. Area of a sector, without trigonometry

This section is the anti-circularity pivot of the volume. We prove that the area of a sector is $\frac{1}{2}r^2\theta$ **without using sine and cosine**, so that Chapter 8 may use it to prove the fundamental limit.

Theorem 6.7: Area of a circular sector. Let $A(\theta)$ be the area of the sector of radius r and central angle θ , with $\theta \in [0, 2\pi]$. Then

$$A(\theta) = \frac{1}{2}r^2\theta.$$

Proof. Step 1 — A is additive. Two adjacent sectors of angles θ_1 and θ_2 , with $\theta_1 + \theta_2 \leq 2\pi$, have in common only a radius (a boundary stretch) and their union is the sector of angle $\theta_1 + \theta_2$. By (Ar1):

$$A(\theta_1 + \theta_2) = A(\theta_1) + A(\theta_2).$$

Step 2 — A is monotone. If $\theta_1 \leq \theta_2$, the sector of angle θ_1 is contained in that of angle θ_2 (with the same radius and the same initial side); by (Ar2), $A(\theta_1) \leq A(\theta_2)$. Note that (Ar3) is needed here: sectors with the same angle but differently oriented have the same area, being congruent by rotation, which makes A a function of θ alone.

Step 3 — additivity on integers and rationals. From $A(0) = 0$ and Step 1 it follows by induction that $A(m\theta) = m A(\theta)$ for every integer $m \geq 1$, provided $m\theta \leq 2\pi$. Taking $\theta = 2\pi/n$ and $m = n$:

$$n A(2\pi/n) = A(2\pi) = \pi r^2 \quad \implies \quad A(2\pi/n) = \frac{\pi r^2}{n},$$

where $A(2\pi) = \pi r^2$ is Theorem 6.5. Hence, for every rational $q = m/n \in [0, 1]$,

$$A(2\pi q) = A\left(m \cdot \frac{2\pi}{n}\right) = m \frac{\pi r^2}{n} = q \pi r^2.$$

Step 4 — from rationals to reals, by monotonicity. Let $t \in [0, 1]$ be real and let q, q' be rationals with $q \leq t \leq q'$. By monotonicity (Step 2) and Step 3,

$$q \pi r^2 = A(2\pi q) \leq A(2\pi t) \leq A(2\pi q') = q' \pi r^2.$$

Letting q, q' vary with $q \uparrow t$ and $q' \downarrow t$ (possible by density of $\mathbb{Q}$ in $\mathbb{R}$), both extreme members tend to $t \pi r^2$: by the squeeze theorem $A(2\pi t) = t \pi r^2$.

Step 5 — conclusion. Setting $\theta = 2\pi t$, that is $t = \theta/(2\pi)$:

$$A(\theta) = \frac{\theta}{2\pi} \pi r^2 = \frac{1}{2}r^2\theta.$$

□

Warning. No step of this proof uses sine, cosine, tangent or the fundamental limit. It uses: properties (Ar1)–(Ar3) of the area, Theorem 6.5 for the total value πr^2 , the density of $\mathbb{Q}$ in $\mathbb{R}$ and the squeeze theorem. This is what makes it legitimate, in Chapter 8, to prove $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ by comparing areas.

Remark 6.14: *The real content: additive + monotone $\Rightarrow$ linear.* Steps 3–4 are a particular case of a general fact: a function $f : [0, L] \rightarrow \mathbb{R}$ that is additive ($f(x+y) = f(x) + f(y)$ when $x+y \leq L$) and monotone is necessarily linear, $f(x) = cx$. The monotonicity hypothesis cannot be dropped: without it Cauchy's functional equation admits (using the axiom of choice) pathological solutions, non-measurable and everywhere dense in the plane. In our case monotonicity comes for free, since it follows from the inclusion of sectors.

Corollary 6.3: Area of a circular segment. The circular segment determined by a chord subtending a central angle $\theta \in (0, 2\pi)$ has area

$$A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta).$$

Proof. The sector of angle θ is the union of the triangle OAB and the circular segment, with only the chord in common: by (Ar1), $A(\theta) = \text{area}(OAB) + A_{\text{seg}}$. The triangle OAB has sides $|OA| = |OB| = r$ and included angle θ , hence area $\frac{1}{2}r^2 \sin \theta$. Subtracting and using Theorem 6.7 gives the conclusion. $\square$

Warning. Corollary 6.3, unlike Theorem 6.7, **does** use the sine (in the formula for the area of the triangle). This is not a problem: it comes after Chapter 8 in the logical order, even though it precedes it typographically. The anti-circularity constraint concerns only Theorem 6.7.

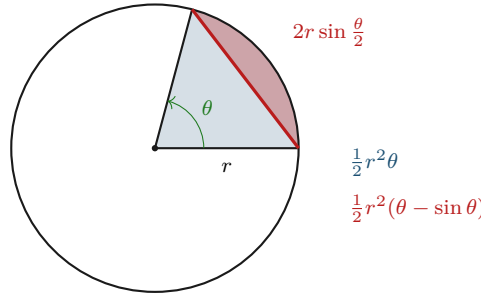


Figure 6.3: Circular sector and segment. The sector (light blue) is the part of the disc between two radii and the arc: its area is to the area of the disc as the angle θ is to the full angle, whence $A_{\text{sect}} = \frac{\theta}{2\pi} \pi r^2 = \frac{1}{2} r^2 \theta$ with θ in radians. The segment (red) is what remains after removing from the sector the isosceles triangle with sides r, r and included angle θ , whose area is $\frac{1}{2} r^2 \sin \theta$: hence $A_{\text{segm}} = \frac{1}{2} r^2 (\theta - \sin \theta)$. This last formula cannot be inverted in closed form: given the segment, finding θ requires a numerical method.

6.9. Isoperimetric inequality

Theorem 6.8: Isoperimetric inequality. For every closed, simple plane curve of length L enclosing an area A : $4\pi A \leq L^2$, with equality if and only if the curve is a circle.

Proof. Hurwitz's proof (1901), by Fourier analysis.

INFO

Regularity hypothesis (imported). The proof that follows expands $x(t), y(t)$ in a Fourier series and differentiates term by term, and uses Green's formula for the area: both steps require the curve to be piecewise C^1 (or rectifiable, with Green's formula in its most general form). Under this regularity hypothesis, the convergence of the Fourier expansions and the validity of Green's formula are **imported** from Volume II (Fourier) and from classical real analysis (Green; source: Apostol, *Calculus*, vol. II).

Step 1 — normalising the parameter. Parametrise the curve by arc length, $s \in [0, L]$, and set $t = 2\pi s/L$, so that $t \in [0, 2\pi]$ and the curve $t \mapsto (x(t), y(t))$ is 2π -periodic. Writing $\dot{} = d/dt$, arc-length parametrisation gives $x'(s)^2 + y'(s)^2 = 1$, and since $x'(s) = \frac{2\pi}{L}\dot{x}(t)$ and $ds = \frac{L}{2\pi}dt$:

$$L = \int_0^L (x'^2 + y'^2) ds = \frac{2\pi}{L} \int_0^{2\pi} (\dot{x}^2 + \dot{y}^2) dt \implies \int_0^{2\pi} (\dot{x}^2 + \dot{y}^2) dt = \frac{L^2}{2\pi}.$$

Step 2 — Fourier expansion and Parseval's identity. Write $x(t) = \sum_{n \in \mathbb{Z}} a_n e^{int}$ and $y(t) = \sum_{n \in \mathbb{Z}} b_n e^{int}$, with $a_{-n} = \overline{a_n}$ and $b_{-n} = \overline{b_n}$ since x, y are real. Then $\dot{x} = \sum_n in a_n e^{int}$, and Parseval gives

$$\int_0^{2\pi} (\dot{x}^2 + \dot{y}^2) dt = 2\pi \sum_{n \in \mathbb{Z}} n^2 (|a_n|^2 + |b_n|^2),$$

which compared with Step 1 yields

$$\sum_{n \in \mathbb{Z}} n^2 (|a_n|^2 + |b_n|^2) = \frac{L^2}{4\pi^2}. \quad (6.1)$$

Step 3 — the area in Fourier coefficients. By Green's formula, $A = |\oint x dy| = |\int_0^{2\pi} x \dot{y} dt|$. Substituting the expansions and using orthogonality $\int_0^{2\pi} e^{i(m+n)t} dt = 2\pi \delta_{m,-n}$:

$$\int_0^{2\pi} x \dot{y} dt = \sum_{m,n} a_m (in) b_n \int_0^{2\pi} e^{i(m+n)t} dt = 2\pi \sum_{n \in \mathbb{Z}} in a_{-n} b_n = 2\pi \sum_{n \in \mathbb{Z}} in \overline{a_n} b_n.$$

Taking moduli and bounding term by term:

$$A \leq 2\pi \sum_{n \in \mathbb{Z}} |n| |a_n| |b_n| \leq \pi \sum_{n \in \mathbb{Z}} |n| (|a_n|^2 + |b_n|^2), \quad (6.2)$$

the second step being the arithmetic–geometric mean inequality, $2|a_n||b_n| \leq |a_n|^2 + |b_n|^2$.

Step 4 — the final estimate. Since $|n| \leq n^2$ for every $n \in \mathbb{Z}$ (with equality only for $n \in \{0, \pm 1\}$), from (6.2) and then (6.1):

$$A \leq \pi \sum_{n \in \mathbb{Z}} n^2 (|a_n|^2 + |b_n|^2) = \pi \cdot \frac{L^2}{4\pi^2} = \frac{L^2}{4\pi},$$

that is, $4\pi A \leq L^2$.

Step 5 — the equality case. Equality requires *both* steps to be equalities:

- from $|n| = n^2$ for every n with a_n or b_n non-zero it follows that only $n \in \{0, \pm 1\}$ contribute;

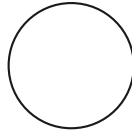
- from $2|a_n||b_n| = |a_n|^2 + |b_n|^2$ it follows that $|a_1| = |b_1|$.

The curve is then $(x(t), y(t)) = (a_0 + 2\operatorname{Re}(a_1 e^{it}), b_0 + 2\operatorname{Re}(b_1 e^{it}))$, and the condition $|a_1| = |b_1|$ together with the constant- $|\dot{\gamma}|$ constraint of Step 1 forces $b_1 = \pm i a_1$: writing $a_1 = \frac{\rho}{2} e^{i\varphi}$ one obtains

$$x(t) = a_0 + \rho \cos(t + \varphi), \quad y(t) = b_0 \pm \rho \sin(t + \varphi),$$

that is, exactly a circle of centre (a_0, b_0) and radius ρ , traversed in one of the two senses. $\square$

Remark 6.15: Historical origin and generalisations. The isoperimetric inequality was known intuitively already to Zenodorus (2nd c. BC), but the first rigorous proof via Fourier analysis is due to Adolf Hurwitz (1901). The isoperimetric problem generalises to $\mathbb{R}^n$ (the ball is the region of maximum volume for a fixed surface) and to Riemannian surfaces, where it plays a central role in the Cartan–Hadamard conjecture and in the modern work of Gromov.



circle: $L = 4$, $A \approx 1.2732$ square: $L = 4$, $A = 1$

Figure 6.4: For equal perimeter $L = 4$, the circle (radius $r = L/(2\pi) \approx 0.6366$) encloses area $A = \pi r^2 \approx 1.2732$, while the square of side $L/4 = 1$ encloses area $A = 1$: direct check, $4\pi A_{\text{circle}} = 16.000 = L^2$ (equality) versus $4\pi A_{\text{square}} \approx 12.566 < L^2$ (strict inequality), consistent with Theorem 6.8.

Alternative proof — Steiner. Jakob Steiner proposed in 1838 a purely elementary argument (without Fourier series), historically preceding Hurwitz’s proof. Assume (for now) that there exists a curve Γ of length L enclosing the maximum area A among all simple closed curves of that length — the *existence* of such a maximum is by no means trivial and Steiner himself did not prove it rigorously (this was an objection raised by Weierstrass some decades later); the argument that follows shows, however, in an entirely elementary way, that if a maximum exists, it must necessarily be a circle.

Step 1 (bisection of the perimeter and reflection). Let PQ be a chord of Γ dividing the perimeter into two arcs of equal length $L/2$. This chord also divides the enclosed area into two parts, A_1 and A_2 , with $A_1 + A_2 = A$. If $A_1 \neq A_2$, say $A_1 > A_2$, one could replace the arc bounding A_2 by the mirror image (with respect to the line PQ) of the arc bounding A_1 : this yields a new closed curve, still of length L (both arcs have length $L/2$), enclosing area $2A_1 > A_1 + A_2 = A$ — contradicting the maximality of A . Hence necessarily $A_1 = A_2 = A/2$, and moreover (by the same reflection argument, this time without contradiction) one may assume that Γ is symmetric with respect to *every* chord bisecting the perimeter.

Step 2 (the optimal arc is a semicircle). It remains to determine the shape of one of the two arcs (of length $L/2$, joining P and Q , area $A/2$). One shows that, among all curves of fixed length $L/2$ joining two fixed points P, Q , the one enclosing (together with the segment PQ) the maximum area must form, with PQ , a right angle at **every** one of its points X : if there were a point X on the arc with $\angle PXQ \neq 90^\circ$, one could think of the two “rigid segments” PX and XQ as hinged at X (a *hinge*, in Steiner’s manner) and rotate them about the hinge until the angle becomes 90° : by the triangle area formula $A_{\triangle PXQ} = \frac{1}{2} PX \cdot XQ \cdot \sin(\angle PXQ)$, this rotation (which alters neither PX , nor XQ , nor therefore the total length of the arc) **increases** the area of the triangle PXQ (the sine being

maximal at 90°), thus increasing the total enclosed area without altering the perimeter — again a contradiction with maximality, unless the angle is already right everywhere. But the locus of points X seeing PQ under a right angle is, by the **converse of Thales' theorem** (Theorem 4.5, Chapter 4), exactly the semicircle with diameter PQ . Hence the optimal arc is a semicircle, and by Step 1 applied to every bisecting chord, the whole curve Γ is a circle. $\square$

Remark 6.16: Historical honesty: the existence problem. Steiner's argument, elegant and entirely elementary as it is (requiring neither Fourier series nor functional analysis), proves only that if a maximising curve exists, it is a circle: it does not prove that such a maximum really exists. This gap — obvious today in the language of compactness, but not at all obvious in 1838 — was explicitly pointed out by Karl Weierstrass in his university lectures (around 1870), and is a historically important example of the nineteenth-century passage from intuitive arguments to rigorous existence proofs. Hurwitz's proof given above (Theorem 6.8), based on Fourier series, does not suffer from this problem: the inequality $4\pi A \leq L^2$ is obtained there directly for every curve, with no need to assume a priori the existence of a maximum.

6.10. Numerical inversion of the circular segment area (Newton's method)

As noted in the table of the following section, $A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta)$ is not elementarily invertible: given A_{seg} (and r), there is no closed formula for θ . This is the natural use case for **Newton's method**.

Definition 6.5: Newton's method for $\theta - \sin \theta = c$. Setting $c = 2A_{\text{seg}}/r^2$ and $f(\theta) = \theta - \sin \theta - c$, with $f'(\theta) = 1 - \cos \theta$, Newton's iteration

$$\theta_{n+1} = \theta_n - \frac{f(\theta_n)}{f'(\theta_n)} = \theta_n - \frac{\theta_n - \sin \theta_n - c}{1 - \cos \theta_n}$$

converges quadratically to θ^* for every reasonable seed $\theta_0 \in (0, 2\pi)$, since f is strictly increasing on $(0, 2\pi)$ ($f'(\theta) = 1 - \cos \theta \geq 0$, vanishing only isolatedly at $\theta = 0$).

Best practice. Direct application in water treatment: the problem of the *level in a partially filled horizontal cylindrical tank* (storage tank, clarifier, settler) requires exactly the inversion of this formula, because the volume of liquid at height h is $V(h) = L \cdot A_{\text{seg}}(h)$ with L the length of the cylinder, while h and θ are related by $h = r(1 - \cos(\theta/2))$.

Worked example: level in a horizontal cylindrical tank. Data: horizontal cylindrical tank, radius $r = 1.2$ m, length $L = 5$ m. We want the level h corresponding to a volume $V = 3$ m³.

Step 1. Required cross-section: $A = V/L = 3/5 = 0.6$ m². From $A = \frac{r^2}{2}(\theta - \sin \theta)$: $c = 2A/r^2 = 1.2/1.44 = 0.8333$.

Step 2. Newton with seed $\theta_0 = 2$: $f(2) = 2 - \sin 2 - 0.8333 = 0.2574$, $f'(2) = 1 - \cos 2 = 1.4161$, $\theta_1 = 2 - 0.2574/1.4161 = 1.8182$.

Step 3. $f(1.8182) \approx 0.0152$, $f'(1.8182) \approx 1.2449$, $\theta_2 = 1.8182 - 0.0152/1.2449 \approx 1.8060$.

Step 4. One further iteration gives $\theta_3 \approx 1.8058$ (change $< 10^{-3}$: convergence reached in 3

iterations, typical of Newton's quadratic speed).

Step 5. Level: $h = r(1 - \cos(\theta/2)) = 1.2(1 - \cos(0.9029)) \approx 1.2(1 - 0.6193) \approx 0.457$ m, that is ≈ 46 cm of head for 3 m^3 of liquid. $\square$

6.11. Summary table of the formulas and their inverses

Quantity	Direct formula	Inverse formula
Circumference	$C = 2\pi r$	$r = C/(2\pi)$
Area of the disc	$A = \pi r^2$	$r = \sqrt{A/\pi}$
Arc length	$l = r\theta$	$\theta = l/r$
Sector area	$A_s = \frac{1}{2}r^2\theta$	$\theta = 2A_s/r^2$
Segment area	$A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta)$	not elementarily invertible; solve numerically (Newton's method)

6.12. Brahmagupta's formula: area of a cyclic quadrilateral

Theorem 6.9: Brahmagupta's formula (628 AD). Let $ABCD$ be a cyclic quadrilateral (inscribed in a circle) with sides a, b, c, d and semiperimeter $s = (a + b + c + d)/2$. Then its area is

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Proof. Split $ABCD$ into the two triangles $\triangle ABD$ and $\triangle CBD$ by the diagonal $BD = p$. By the law of cosines in each triangle,

$$p^2 = a^2 + d^2 - 2ad \cos A = b^2 + c^2 - 2bc \cos C.$$

Since $ABCD$ is cyclic, opposite angles are supplementary (Theorem 4.6): $C = 180^\circ - A$, hence $\cos C = -\cos A$. Substituting:

$$\begin{aligned} a^2 + d^2 - 2ad \cos A &= b^2 + c^2 + 2bc \cos A \\ \implies \cos A &= \frac{a^2 + d^2 - b^2 - c^2}{2(ad + bc)}. \end{aligned}$$

The total area is $K = \frac{1}{2}ad \sin A + \frac{1}{2}bc \sin C = \frac{1}{2}(ad + bc) \sin A$ (using $\sin C = \sin(180^\circ - A) = \sin A$). Hence

$$16K^2 = 4(ad + bc)^2 \sin^2 A = 4(ad + bc)^2 - 4(ad + bc)^2 \cos^2 A.$$

Substituting the expression for $\cos A$ and expanding as a difference of squares (elementary algebraic computation, given in full in the bibliographic references), one obtains

$$16K^2 = [(a + d)^2 - (b - c)^2][(b + c)^2 - (a - d)^2] = 16(s - a)(s - b)(s - c)(s - d),$$

whence the conclusion, on taking the square root. $\square$

Remark 6.17: Generalisation of Heron's formula. Setting $d \rightarrow 0$ (the quadrilateral degenerates into a triangle, with $s \rightarrow (a + b + c)/2$), Brahmagupta's formula reduces exactly

to Heron's formula $K = \sqrt{s(s-a)(s-b)(s-c)}$ (Chapter 10): it is therefore a genuine generalisation, not a mere analogy. In his treatise *Brahmasphutasiddhanta* (chapter *Ganitadhyaya*, verses 20–22, 628 AD), Brahmagupta neither cites nor appears to know the earlier result of Heron of Alexandria (1st c. AD): the two results were almost certainly discovered independently.

Remark 6.18: Connection with Ptolemy's theorem. The same configuration (cyclic quadrilateral) underlies both Brahmagupta's formula and Ptolemy's theorem (Theorem 5.9, Chapter 5): the first gives the area, the second the product of the diagonals. Combining the two relations one also obtains an explicit formula for each diagonal in terms of the four sides alone, also due to Brahmagupta.

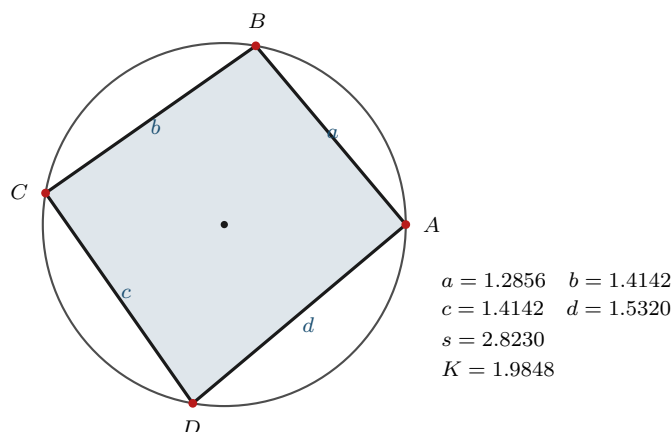


Figure 6.5: A cyclic quadrilateral and Brahmagupta's formula. If the four vertices lie on a circle, the area depends only on the four sides and equals $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$, with s the semiperimeter: it is the generalisation of Heron's formula, recovered by setting $d = 0$. In the example, with vertices at $0^\circ, 80^\circ, 170^\circ, 260^\circ$ on a unit circle, the sides measure 1.2856, 1.4142, 1.4142 and 1.5320, the semiperimeter is $s = 2.8230$ and the formula gives $K = 1.9848$; the same value is obtained by adding the areas of the four triangles with apex at the centre, $\frac{1}{2}(\sin 80^\circ + \sin 90^\circ + \sin 90^\circ + \sin 100^\circ)$. Among all quadrilaterals with those four sides, the cyclic one is the largest.

6.13. Parameshvara's formula: the circumradius

Brahmagupta's formula gives the *area* of a cyclic quadrilateral from its four sides. An equally elegant formula gives, again from the four sides alone, the radius of the circumscribed circle. It is due to the Indian mathematician Vatasseri Parameshvara (Kerala, ca. 1380–1460), who stated it around 1430 — about 350 years before the same formula was rediscovered in Europe by Simon L'Huilier in 1782.

Remark 6.19: A service fact: the circumradius from the sides and the area. By the extended law of sines (Theorem 10.5), in a triangle with sides x, y, z and area $\mathcal{K}$, inscribed in a circle of radius R : $z = 2R \sin Z$ (where Z is the angle opposite z), and $\mathcal{K} = \frac{1}{2}xy \sin Z$. Eliminating $\sin Z$ between the two relations,

$$R = \frac{xyz}{4\mathcal{K}}.$$

Theorem 6.10: Parameshvara's formula. A cyclic quadrilateral with consecutive sides a, b, c, d and area K (given by Brahmagupta's formula, Theorem 6.9) has circumscribed circle of radius

$$R = \frac{1}{4K} \sqrt{(ab + cd)(ac + bd)(ad + bc)}.$$

Proof. Let $ABCD$ be the quadrilateral, with $a = AB$, $b = BC$, $c = CD$, $d = DA$, and let $p = AC$, $q = BD$ be the diagonals.

Step 1 — two expressions for R from diagonal p . Diagonal AC splits the quadrilateral into triangles ABC (sides a, b, p , area K_1) and ACD (sides p, c, d , area K_2), with $K = K_1 + K_2$. Both triangles are inscribed in the **same** circle circumscribed about the quadrilateral (their vertices are vertices of $ABCD$), hence have the same circumradius R . By Observation 6.19,

$$R = \frac{abp}{4K_1} \implies K_1 = \frac{abp}{4R}, \quad R = \frac{pcd}{4K_2} \implies K_2 = \frac{cdp}{4R}.$$

Adding,

$$K = K_1 + K_2 = \frac{p(ab + cd)}{4R} \implies R = \frac{p(ab + cd)}{4K}.$$

Step 2 — the same with diagonal q . Repeating Step 1 with diagonal BD , which splits $ABCD$ into triangles ABD (sides a, d, q) and BCD (sides b, c, q), both inscribed in the same circle:

$$R = \frac{q(ad + bc)}{4K}.$$

Step 3 — multiplying the two expressions.

$$R^2 = \frac{pq(ab + cd)(ad + bc)}{16K^2}.$$

By Ptolemy's theorem (Theorem 5.9, Chapter 5), $pq = AC \cdot BD = AB \cdot CD + BC \cdot DA = ac + bd$. Substituting,

$$R^2 = \frac{(ac + bd)(ab + cd)(ad + bc)}{16K^2} \implies R = \frac{1}{4K} \sqrt{(ab + cd)(ac + bd)(ad + bc)},$$

which is the claim (the root is well defined because the three factors are products of lengths, hence positive). $\square$

Example 6.2: Numerical verification of Parameshvara's formula. Cyclic quadrilateral with sides $a = 5$, $b = 6$, $c = 7$, $d = 8$ (the same as Exercise 6.12 further below). Semiperimeter $s = 13$, Brahmagupta area $K = \sqrt{8 \cdot 7 \cdot 6 \cdot 5} = \sqrt{1680} = 40.98780$. The three cross products:

$$ab + cd = 30 + 56 = 86, \quad ac + bd = 35 + 48 = 83, \quad ad + bc = 40 + 42 = 82,$$

whose product is $86 \times 83 \times 82 = 585\,316$, with root 765.0595. Hence

$$R = \frac{765.0595}{4 \times 40.98780} = \frac{765.0595}{163.9512} = 4.66638.$$

Independent verification: explicitly constructing the coordinates of the four vertices on a circle of radius 4.66638 so that the chords measure 5, 6, 7, 8 (solving for the subtended central angles, script `cap06_parameshvara.py`), the four sides and the area computed from the coordinates coincide with the starting values to within 10^{-9} . $\checkmark$

6.14. Geometric probability: Buffon and Bertrand's paradox

This whole chapter has measured lengths and areas. It remains to see what happens when measure is used to define a *probability*: this is the ground on which the circle produced both the oldest experiment for computing π and the most instructive paradox of probability theory.

Definition 6.6: Geometric probability. Let Ω be a region of the plane (or of $\mathbb{R}^n$) of finite, non-zero measure. A point is said to be chosen *at random and uniformly* in Ω if the probability that it falls in a measurable subset $E \subseteq \Omega$ is

$$\Pr(E) = \frac{\text{meas}(E)}{\text{meas}(\Omega)}.$$

Warning. The definition presupposes that the set Ω and the measure on it have been *chosen*. There is no canonical way of choosing a geometric object “at random”: the choice of parametrisation is part of the data of the problem, and changing it changes the probabilities. This is exactly what Bertrand's paradox makes evident.

6.14.1 Buffon's needle

Theorem 6.11: Buffon's needle (1777). Let equidistant parallel lines, at distance d , be drawn on a plane. Drop at random a needle of length $\ell \leq d$, with position and orientation independent and uniform. Then the probability that the needle meets one of the lines is

$$p = \frac{2\ell}{\pi d}$$

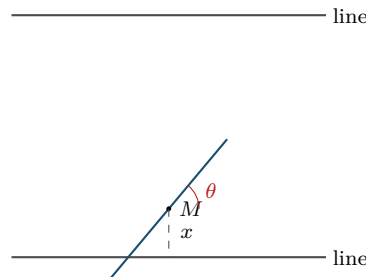


Figure 6.6: Buffon's needle: two parallel lines at distance d , a needle of length $\ell \leq d$ with midpoint M at distance x from the nearest line and angle θ with respect to the direction of the lines. In the case drawn ($x = 0.4$, $\theta = 50^\circ$, $\ell = 1.5$, $d = 2$) we have $(\ell/2) \sin \theta \approx 0.575 > x$: the needle meets the line, as predicted by the condition in Step 2 of the proof.

Proof. Step 1 — the two variables. The position of the needle is described by two quantities only: the distance x of its midpoint from the nearest line, with $x \in [0, d/2]$, and the acute angle $\theta \in [0, \pi/2]$ between the needle and the direction of the lines. By hypothesis x and θ are independent and uniform on their intervals, so the joint density is the constant

$$f(x, \theta) = \frac{1}{(d/2)(\pi/2)} = \frac{4}{\pi d} \quad \text{on } [0, d/2] \times [0, \pi/2].$$

Step 2 — the crossing condition. The projection of the needle on the direction perpendicular to the lines has length $\ell \sin \theta$, and the midpoint occupies its centre: the endpoints are therefore at distance $\frac{\ell}{2} \sin \theta$ from the midpoint in that direction. The needle meets the nearest

line if and only if

$$x \leq \frac{\ell}{2} \sin \theta.$$

Since $\ell \leq d$ we have $\frac{\ell}{2} \sin \theta \leq \frac{d}{2}$, so it cannot meet two lines and no case distinction is needed.

Step 3 — the integral.

$$p = \int_0^{\pi/2} \int_0^{(\ell/2) \sin \theta} \frac{4}{\pi d} dx d\theta = \frac{4}{\pi d} \int_0^{\pi/2} \frac{\ell}{2} \sin \theta d\theta = \frac{2\ell}{\pi d} [-\cos \theta]_0^{\pi/2} = \frac{2\ell}{\pi d}.$$

Step 4 — reading. With $\ell = d$ one gets $p = 2/\pi = 0.63662$. The remarkable fact is that π appears in a problem containing no circle at all: it enters through the integral of $\sin \theta$, that is through the measure of angles, which is arc length on the unit circle. Every time π appears in a problem of geometric probability, it can be traced back to this origin. $\square$

***Remark 6.20: Estimating π with a needle, and Lazzarini's lesson.** Inverting the formula gives $\pi = 2\ell/(pd)$, and estimating p by the observed frequency k/n gives $\hat{\pi} = 2\ell n/(kd)$. It is the first Monte Carlo method in history, almost two centuries before the name. Its efficiency, however, is very poor: the frequency has standard deviation $\sqrt{p(1-p)/n}$, so the relative error on $\hat{\pi}$ decreases as $1/\sqrt{n}$, and one more decimal digit requires a hundred times more throws.*

In 1901 Mario Lazzarini published an experiment with 3408 throws that allegedly gave $\pi \simeq 355/113 = 3.1415929$, correct to six decimals. The result is universally regarded as suspect: with 3408 throws the expected statistical error is of order 10^{-2} , not 10^{-7} , and the number 3408 is precisely the one which, with the stated ratio ℓ/d , yields exactly 355/113 if the count of successes is the “right” one. The moral is methodological and reaches far beyond the anecdote: a result too accurate for the number of trials is a sign of error, not of skill.

***Theorem 6.12: Buffon's needle for an arbitrary curve (Buffon's noodle).** Let γ be a rectifiable curve of length L , rigid but of arbitrary shape, dropped at random on the grid of parallel lines at distance d . Then the mean number of crossings with the lines is*

$$\mathbb{E}[N] = \frac{2L}{\pi d}$$

independently of the shape of the curve.

Proof. **Step 1 — the case of a short segment.** For a segment of length $s \leq d$ the number of crossings is 0 or 1, so its mean value coincides with the probability of crossing, which by Theorem 6.11 equals $2s/(\pi d)$.

Step 2 — arbitrary segments. A segment of arbitrary length s splits into m consecutive sub-segments of length $\leq d$. The total number of crossings is the sum of the numbers of crossings of the pieces (a line crossing the segment crosses one and only one piece, except in the null-measure case where it passes through a junction point). By linearity of expectation — which does not require independence —

$$\mathbb{E}[N] = \sum_{i=1}^m \frac{2s_i}{\pi d} = \frac{2}{\pi d} \sum_i s_i = \frac{2s}{\pi d}.$$

Step 3 — polygonal chains. The same argument applies to a rigid polygonal chain: the pieces are no longer collinear, but linearity of expectation does not care, and the mean number of crossings remains $2L/(\pi d)$ with L the total length of the chain. It is here that the theorem becomes surprising: the shape has disappeared from the computation.

Step 4 — rectifiable curves (imported extension). Step 3 fully proves the theorem for every rigid **polygonal** curve: this is the case that covers all the practical applications of this chapter (Barbier’s corollary included, where a polygonal chain approximating the curve of constant width suffices). The extension to an arbitrary rectifiable curve γ (limit of inscribed polygonal chains γ_n with $L(\gamma_n) \rightarrow L$) requires an interchange of limit and expectation — almost-everywhere convergence of the number of crossings and a dominated bound — which is **imported**: see Santaló, *Integral Geometry and Geometric Probability*, ch. 1, for the details of the convergence in this specific context (integral geometry). $\square$

Corollary 6.4: Barbier’s theorem, probabilistically. *Every closed convex curve of constant width w has perimeter $L = \pi w$.*

Proof. Choose the grid with spacing $d = w$ and drop the curve. Since the curve is convex and closed, a line that misses it meets it in 0 points, and a line that crosses it meets it in exactly 2 points. Since the width is constant and equal to $w = d$, in every position and for every orientation the curve is crossed by *exactly one* line of the grid: indeed its projection on any direction perpendicular to the lines is a segment of length exactly $w = d$, and a segment of length d always contains one and only one point of the grid of spacing d (except in the null-measure case where it falls on an endpoint). Hence $N = 2$ with probability 1, and $\mathbb{E}[N] = 2$. By Theorem 6.12,

$$2 = \frac{2L}{\pi w} \implies L = \pi w.$$

Note that the circle of diameter w has constant width w and perimeter πw : the formula says that *every* curve of constant width has the perimeter of the circle of equal width. This is Barbier’s theorem, which Volume III (Chapter 4) proves geometrically; here it is obtained by counting crossings. $\square$

6.14.2 The Reuleaux triangle: an example of constant width

Definition 6.7: Reuleaux triangle. Given an equilateral triangle of side w , the **Reuleaux triangle** of width w is the figure bounded by three circular arcs of radius w , each centred at a vertex and passing through the other two.

Proposition 6.1: Constant width of the Reuleaux triangle. *The Reuleaux triangle has constant width w in every direction.*

Proof. For every direction, a supporting line of the figure passes through a vertex V of the original equilateral triangle (the opposite arc, centred at V , is locally “more outward” near the two vertices bounding it, and the figure rests on the vertex V itself in the direction opposite that arc). The parallel supporting line on the opposite side is then tangent to the arc centred at V (the arc of radius w opposite the vertex V): the distance between the two lines is therefore the distance from V to the arc centred at V , that is, exactly the radius w (every point of the arc centred at V is at distance w from V , by construction). The width in every direction is therefore w . $\square$

Proposition 6.2: Perimeter and area of the Reuleaux triangle. *The Reuleaux triangle of width w has perimeter πw and area $\frac{\pi - \sqrt{3}}{2} w^2$.*

Proof. The perimeter follows immediately from Barbier’s theorem (Corollary 6.4), the width being constantly w . For the area: the figure is the union of the equilateral triangle of side

w (area $\frac{\sqrt{3}}{4}w^2$) and three **circular segments**, one per side, each bounded by the side itself and the arc of radius w and central angle $60^\circ = \pi/3$ that replaces it. By the formula for the circular segment (Corollary 6.3, already proved in this chapter), each segment has area $\frac{1}{2}w^2(\pi/3 - \sin(\pi/3)) = \frac{w^2}{2}(\frac{\pi}{3} - \frac{\sqrt{3}}{2})$. Adding the triangle and the three segments:

$$\frac{\sqrt{3}}{4}w^2 + 3 \cdot \frac{w^2}{2} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{4}w^2 + \frac{\pi}{2}w^2 - \frac{3\sqrt{3}}{4}w^2 = \frac{\pi}{2}w^2 - \frac{\sqrt{3}}{2}w^2 = \frac{\pi - \sqrt{3}}{2}w^2.$$

□

Numerical check. $w = 1$: area $= (\pi - \sqrt{3})/2 = 0.704771$ (checked two independent ways: from the closed formula and from the direct sum triangle+three segments).

Remark 6.21: Where it is found. The Reuleaux triangle underlies drill bits for “almost square” holes (*H. J. Watts’s patent, 1914, sometimes dated 1918 — US 1,241,176 — depending on the source*): rotating within a square of side w , the vertices of the figure always touch the four sides of the square, allowing a hole to be drilled (with an axis of rotation that traces a small orbit, not fixed) with rounded but substantially square edges. The British 50-pence coin, often cited as an example of a “Reuleaux” shape, is actually a **heptagon of constant width** (seven arcs, not three): the geometric principle — constant width without being a circle — is the same, but the figure is different.

6.14.3 Bertrand’s paradox

Example 6.3: Three different answers to the same question. A chord is drawn at random in a circle of radius 1. What is the probability that it is longer than the side of the inscribed equilateral triangle? The side equals $\sqrt{3}$, and a chord exceeds $\sqrt{3}$ if and only if its distance from the centre is less than $1/2$: indeed a chord at distance t from the centre has length $2\sqrt{1-t^2}$, and $2\sqrt{1-t^2} > \sqrt{3}$ is equivalent to $t < 1/2$.

First answer: random endpoints. Fix one endpoint and choose the second uniformly on the circle. The chord subtending a central angle ϑ has length $2\sin(\vartheta/2)$, which exceeds $\sqrt{3}$ if and only if $\vartheta \in (2\pi/3, 4\pi/3)$: an arc of measure $2\pi/3$ out of 2π . Hence

$$\Pr = \frac{2\pi/3}{2\pi} = \frac{1}{3}.$$

Second answer: random radius, random point on it. Fix a radius, choose on it a uniform point at distance $t \in [0, 1]$ from the centre, and draw the chord perpendicular to the radius at that point. The condition is $t < 1/2$, hence

$$\Pr = \frac{1/2}{1} = \frac{1}{2}.$$

Third answer: random midpoint in the disc. Choose a uniform point in the disc and take the chord of which it is the midpoint. The condition is that the point be at distance less than $1/2$ from the centre, and probabilities are ratios of areas:

$$\Pr = \frac{\pi(1/2)^2}{\pi \cdot 1^2} = \frac{1}{4}.$$

Three honest procedures, three different results: $1/3$, $1/2$, $1/4$.

Remark 6.22: What the paradox teaches, and how it is resolved. The paradox is not a computational error: all three computations are correct. It shows that the phrase “at random” determines no probability until one declares which measure is being used, that is until the set Ω and its parametrisation are chosen in the sense of Definition 6.6. The three answers correspond to three different measures on the same set of chords, and none is “wrong”: they are answers to three different questions.

One may nonetheless ask whether one of the three is privileged. Integral geometry answers yes, by an invariance criterion. A line of the plane is determined by the pair (p, φ) , where $\varphi \in [0, \pi)$ is the direction of the normal and $p \in \mathbb{R}$ the signed distance from the origin. One proves that, up to a constant factor, the only measure on the space of lines invariant under all rigid motions of the plane is

$$d\mu = dp d\varphi$$

(Poincaré’s kinematic measure). With respect to it, the lines meeting the unit circle have $p \in (-1, 1)$ uniformly for every φ : the distance from the centre is therefore uniform, which is exactly the second answer. The value $1/2$ is thus the one obtained if one requires the result not to change when the circle is translated or rotated with respect to the mechanism generating the chords — a reasonable requirement if the chords are, for instance, traces of rods dropped at random, that is precisely Buffon’s experiment. The other two measures are not invariant: the first privileges the points of the circumference, the third the points of the disc, and both choices change if the circle is moved inside a larger mechanism.

The conclusion is not that $1/2$ is “the” answer, but that a question of geometric probability is well posed only when it declares the measure. It is the same discipline this chapter imposed on the notion of area: first one says what is being measured and with which properties, then one computes.

6.15. Problem set

6.15.1 Numerical exercises

Exercise 6.1. A disc has area $A = 50 \text{ cm}^2$. Compute its radius, diameter and circumference.

Solution.

$$\begin{aligned} r &= \sqrt{A/\pi} = \sqrt{50/3.14159} = \sqrt{15.915} \approx 3.99 \text{ cm}, \\ d &= 2r \approx 7.98 \text{ cm}, \quad C = 2\pi r \approx 25.07 \text{ cm}. \end{aligned}$$

Exercise 6.2. Given a circle of radius $R = 10$, check Theorem 6.5 numerically by computing the area of the inscribed regular hexagon and comparing it with $A = 100\pi$. By how much does it underestimate?

Solution. The regular hexagon inscribed in a circle of radius R has side R (the side subtends a central angle of 60° and the central triangle is equilateral), so it consists of 6 equilateral triangles of side 10:

$$\text{area} = 6 \cdot \frac{10^2 \sqrt{3}}{4} = 150\sqrt{3} \approx 259.81.$$

Area of the disc: $100\pi \approx 314.16$. Underestimate: $314.16 - 259.81 = 54.35$, a relative error of $54.35/314.16 \approx 17.3\%$.

Consistency with the theory. The underestimate is expected: by Lemma 6.5 the inscribed

polygon always has area *strictly less* than the disc, with $p_n < C$ and $a_n < r$. With $n = 6$ the apothem is $a_6 = R\sqrt{3}/2 \approx 8.66 < 10$ and the perimeter $p_6 = 60 < 2\pi R \approx 62.83$: both inequalities of the lemma are verified numerically.

Exercise 6.3. A circular sector on a radius of 5 cm has area 10 cm^2 . What is the central angle in radians? And in degrees?

Solution. By Theorem 6.7, $A_s = \frac{1}{2}r^2\theta$, whence

$$\theta = \frac{2A_s}{r^2} = \frac{2 \cdot 10}{25} = 0.8 \text{ rad.}$$

In degrees: $0.8 \cdot 180/\pi \approx 45.84^\circ$. The corresponding arc length is $l = r\theta = 5 \cdot 0.8 = 4 \text{ cm}$ (Theorem 6.3).

Exercise 6.4. A cyclic quadrilateral has sides $a = 5$, $b = 6$, $c = 7$, $d = 8 \text{ cm}$. Compute its area with Brahmagupta's formula.

Solution. Semiperimeter $s = (5 + 6 + 7 + 8)/2 = 13$. Area:

$$K = \sqrt{(13-5)(13-6)(13-7)(13-8)} = \sqrt{8 \cdot 7 \cdot 6 \cdot 5} = \sqrt{1680} \approx 40.99 \text{ cm}^2.$$

Exercise 6.5. Check that, setting $d = 0$ in Brahmagupta's formula with $a = 3$, $b = 4$, $c = 5$ (a right triangle), one obtains the same area computable as $\frac{1}{2} \cdot 3 \cdot 4$.

Solution. With $d = 0$: $s = (3 + 4 + 5 + 0)/2 = 6$. Brahmagupta: $K = \sqrt{(6-3)(6-4)(6-5)(6-0)} = \sqrt{3 \cdot 2 \cdot 1 \cdot 6} = \sqrt{36} = 6$. Direct comparison: $\frac{1}{2} \cdot 3 \cdot 4 = 6$. The two results agree, confirming Remark 6.17 numerically.

6.15.2 Symbolic exercises and proofs

Exercise 6.6. Let $\mathcal{P}_n$ and $\mathcal{Q}_n$ be the areas of the regular n -gons respectively inscribed in and circumscribed about a circle of radius r . Prove that

$$\frac{\mathcal{P}_n}{\mathcal{Q}_n} = \cos^2 \frac{\pi}{n},$$

and deduce that the ratio tends to 1 as $n \rightarrow \infty$. What does this mean for the method of exhaustion?

Solution. Every regular n -gon decomposes into n isosceles triangles with apex angle $2\pi/n$. For the inscribed one, the slanted sides are r and each triangle has area $\frac{1}{2}r^2 \sin(2\pi/n)$, hence

$$\mathcal{P}_n = \frac{n}{2}r^2 \sin \frac{2\pi}{n} = nr^2 \sin \frac{\pi}{n} \cos \frac{\pi}{n},$$

using the duplication formula (Corollary 8.2). For the circumscribed one, the apothem is r and the half-side $r \tan(\pi/n)$, hence each triangle has area $r^2 \tan(\pi/n)$ and

$$\mathcal{Q}_n = nr^2 \tan \frac{\pi}{n} = nr^2 \frac{\sin(\pi/n)}{\cos(\pi/n)}.$$

The ratio is

$$\frac{\mathcal{P}_n}{\mathcal{Q}_n} = \frac{nr^2 \sin(\pi/n) \cos(\pi/n)}{nr^2 \sin(\pi/n) / \cos(\pi/n)} = \cos^2 \frac{\pi}{n}.$$

As $n \rightarrow \infty$, $\pi/n \rightarrow 0$ and $\cos^2(\pi/n) \rightarrow 1$.

Interpretation. The two sequences $\mathcal{P}_n$ (increasing) and $\mathcal{Q}_n$ (decreasing) bracket the area of the disc and their ratio tends to 1: the “scissors” close, which is exactly what Lemma 6.4 guarantees in Greek language. Note, however, that this exercise *uses* trigonometry, so it cannot replace the proof of Theorem 6.5, which must remain independent of it.

Exercise 6.7. Prove, using only Theorem 6.2 and properties (Ar1)–(Ar4), that multiplying the radius by $k > 0$ multiplies the length by k and the area by k^2 . Deduce that doubling the radius quadruples the area.

Solution. *Length.* This is Step 3 of the proof of Theorem 6.2: the homothety of ratio k multiplies by k every polyline length, hence also their supremum: $C' = kC$.

Area. By Theorem 6.5,

$$A' = \frac{1}{2}C'r' = \frac{1}{2}(kC)(kr) = k^2 \cdot \frac{1}{2}Cr = k^2 A.$$

(Alternatively: a homothety of ratio k maps a rectangle with sides b, h onto one with sides kb, kh , hence by (Ar4) multiplies its area by k^2 ; by additivity and monotonicity the same holds for every region.)

Consequence. With $k = 2$: $A' = 4A$. This answers the question “if I double the radius, does the area double?”: no, it quadruples. The opposite error — believing that area scales like the radius — is among the commonest, and is the same one that leads to undersizing a pipe cross-section by a factor 4 when its diameter is doubled.

Exercise 6.8. Prove the general fact used in Theorem 6.7: if $f : [0, L] \rightarrow \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ whenever $x, y, x+y \in [0, L]$, and is monotone non-decreasing, then $f(x) = cx$ with $c = f(L)/L$.

Solution. *Step 1.* From $f(0+0) = f(0) + f(0)$ it follows that $f(0) = 0$.

Step 2 (integers). By induction, $f(mx) = mf(x)$ for every integer $m \geq 1$ with $mx \leq L$: the case $m = 1$ is obvious, and $f((m+1)x) = f(mx+x) = f(mx) + f(x) = mf(x) + f(x)$.

Step 3 (rationals). Taking $x = L/n$ and $m = n$ in Step 2: $nf(L/n) = f(L)$, that is $f(L/n) = f(L)/n$. Hence for $q = m/n \in [0, 1]$, $f(qL) = mf(L/n) = (m/n)f(L) = qf(L)$.

Step 4 (reals, by monotonicity). Let $t \in [0, 1]$ and let $q \leq t \leq q'$ be rationals in $[0, 1]$. By monotonicity, $qf(L) = f(qL) \leq f(tL) \leq f(q'L) = q'f(L)$. Letting $q \uparrow t$ and $q' \downarrow t$ (density of $\mathbb{Q}$), both members tend to $tf(L)$, hence $f(tL) = tf(L)$ by the squeeze theorem.

Conclusion. Setting $x = tL$: $f(x) = \frac{f(L)}{L}x = cx$.

Note. The monotonicity hypothesis is indispensable only at Step 4: Steps 1–3 hold for any additive function. Without monotonicity there exist non-linear additive solutions whose graph is dense in the plane; their construction, however, requires the axiom of choice and does not produce measurable functions (Remark 6.14).

Exercise 6.9. The “concertina” rearrangement of Proof 1 does not approximate the area of the circle: it reproduces it **exactly** (it is a rearrangement of the same n sectors, and cutting and reassembling does not change the total area). The actual approximation is a previous

step: compare the area of the **inscribed regular polygon** of n sides, $A_n = \frac{n}{2}r^2 \sin \frac{2\pi}{n}$, with πr^2 , for $n = 4$ and $n = 8$ and $r = 1$, computing the relative error $1 - \frac{A_n}{\pi r^2}$.

Solution.

$$n = 4 : A_4 = 2 \sin 90^\circ = 2, \quad \text{error} = 1 - \frac{2}{\pi} = 0.3634 \text{ (36.3\%);}$$

$$n = 8 : A_8 = 4 \sin 45^\circ = 2\sqrt{2} = 2.828427, \quad \text{error} = 1 - \frac{2\sqrt{2}}{\pi} = 0.0997 \text{ (10.0\%)}$$

(verified). The error more than halves at each doubling, consistently with the $O(1/n^2)$ behaviour of Theorem 6.5. Note carefully the distinction: A_n (the polygon, an area smaller than the circle) is what gets approximated as n grows; the “concertina” figure, by contrast, is obtained *from the circle itself* (circular sectors, not triangular ones) and is by construction a simple rearrangement of parts of the circle: its area is always πr^2 , for every n , exact.

Exercise 6.10. A needle of length $\ell = 8$ cm is dropped 500 times on a grid of lines $d = 10$ cm apart, and 254 crossings are counted. Estimate π and assess the expected error.

Solution. The theoretical probability is $p = 2\ell/(\pi d) = 1.6/\pi = 0.50930$. Inverting,

$$\hat{\pi} = \frac{2\ell n}{k d} = \frac{2 \cdot 8 \cdot 500}{254 \cdot 10} = \frac{8000}{2540} = 3.1496,$$

with a relative error of 0.25%. The expected error is estimated from the standard deviation of the frequency, $\sqrt{p(1-p)/n} = \sqrt{0.50930 \cdot 0.49070/500} = 0.02236$, that is 4.39% of p ; since $\hat{\pi}$ is inversely proportional to the frequency, the expected relative error on $\hat{\pi}$ is of the same order, about 4%. The result obtained is therefore *far* more accurate than the number of throws warrants: it is luck, not method.

Exercise 6.11. In the third procedure of Bertrand’s paradox (midpoint uniform in the disc), compute the probability that the chord exceeds an assigned length L in a circle of radius R , and recover the value $1/4$ for $L = R\sqrt{3}$.

Solution. A chord whose midpoint is at distance t from the centre has length $2\sqrt{R^2 - t^2}$, which exceeds L if and only if $t^2 < R^2 - L^2/4$. Since the midpoint is uniform in the disc, the probability is the ratio of areas:

$$\Pr = \frac{\pi(R^2 - L^2/4)}{\pi R^2} = 1 - \frac{L^2}{4R^2}.$$

For $L = R\sqrt{3}$ one gets $1 - 3/4 = 1/4$ ✓. Note that the formula gives 1 for $L = 0$ and 0 for $L = 2R$, as it must.

Exercise 6.12. A cyclic quadrilateral has sides 5, 6, 7, 8. Compute its area by Brahmagupta’s formula and the circumradius by Parameshvara’s formula (Theorem 6.10).

Solution. Semiperimeter $s = 13$, hence

$$K = \sqrt{(13-5)(13-6)(13-7)(13-8)} = \sqrt{8 \cdot 7 \cdot 6 \cdot 5} = \sqrt{1680} = 40.9878.$$

For the radius, with $a = 5$, $b = 6$, $c = 7$, $d = 8$:

$$ab + cd = 86, \quad ac + bd = 83, \quad ad + bc = 82,$$

whose product is $86 \cdot 83 \cdot 82 = 585\,316$, with square root 765.06. Hence

$$R = \frac{765.06}{4 \cdot 40.9878} = \frac{765.06}{163.951} = 4.6664.$$

Plausibility check: the longest side is 8, so the diameter must be at least 8, that is $R \geq 4$ ✓.

6.16. Self-assessment quiz

Ten multiple-choice questions for a quick check. The answer key is at the end of the section.

1. The ratio between the area of a disc and the square of its radius is
 - A) 2π
 - B) π
 - C) $\pi/2$
 - D) dependent on the radius
2. In the proof by exhaustion, the area of the inscribed polygon, as the number of sides grows,
 - A) exceeds the area of the disc
 - B) grows while always remaining less than the area of the disc
 - C) oscillates about the area of the disc
 - D) stays constant
3. Doubling the radius of a circle, its area
 - A) doubles
 - B) triples
 - C) quadruples
 - D) stays the same
4. An arc corresponding to 2 radians on a circle of radius 3 metres is
 - A) 3 m long
 - B) 6 m long
 - C) 1.5 m long
 - D) 6π m long
5. Archimedes proved the equivalence of the disc with a right triangle whose height is
 - A) the diameter
 - B) the radius
 - C) the circumference
 - D) the apothem of the inscribed polygon

-
6. The fact that the ratio C/d is the same for every circle is
- A) a definition
 - B) a theorem, provable by homothety
 - C) an axiom
 - D) an experimental finding
7. The area of the circular sector of angle θ is $\frac{1}{2}r^2\theta$; the proof given in this chapter rests on
- A) the fundamental limit $\sin x/x \rightarrow 1$
 - B) additivity and monotonicity of the area
 - C) the integral in polar coordinates
 - D) the Pythagorean theorem
8. The exhaustion lemma (Euclid X.1) requires that at each step one subtracts
- A) any part whatever
 - B) exactly half
 - C) not less than half
 - D) not more than half
9. The radian measure of an angle is
- A) a pure number
 - B) measured in metres
 - C) measured in degrees
 - D) dependent on the radius of the chosen circle
10. Archimedes' method of exhaustion
- A) has been superseded and is today considered incorrect
 - B) anticipates the structure of the modern definition of limit
 - C) requires differential calculus
 - D) applies only to the circle

BEST PRACTICE

Answer key. 1: B — 2: B — 3: C — 4: B ($l = r\theta = 3 \cdot 2$) — 5: B — 6: B (Theorem 6.2) — 7: B (Theorem 6.7) — 8: C — 9: A — 10: B.

Note on question 7. Answer C is not wrong as a computation, but it is wrong as a *foundation*: the integral in polar coordinates presupposes Lebesgue measure and, indirectly, the area of the disc. Using it to prove the sector area, and then using the sector area to prove the fundamental limit, would bring back the circularity denounced at the start of the chapter.

6.17. Chapter Formula sheet

Length and the constant π (Definition 6.1; Theorem 6.2; Definition 6.2)

$$\text{len}(\gamma) = \sup_{\text{inscribed polylines}} L, \quad \frac{C}{2r} = \frac{C'}{2r'} =: \pi, \quad C = 2\pi r$$

Arc in radians (Definition 6.3; Theorem 6.3)

$$l = r\theta, \quad l(2\pi) = 2\pi r = C$$

Archimedes' postulate (Theorem 6.4; Corollary 6.2)

$$K_1 \subseteq K_2 \text{ convex} \implies \text{len}(\partial K_1) \leq \text{len}(\partial K_2)$$

Area of the disc (Theorem 6.5)

$$A = \frac{1}{2} C r = \pi r^2$$

Circular sector and segment (Theorem 6.7; Corollary 6.3)

$$A(\theta) = \frac{1}{2} r^2 \theta, \quad A_{\text{seg}} = \frac{r^2}{2} (\theta - \sin \theta)$$

Numerical inversion of the segment (Definition 6.5), with $c = 2A_{\text{seg}}/r^2$

$$\theta_{n+1} = \theta_n - \frac{\theta_n - \sin \theta_n - c}{1 - \cos \theta_n}$$

Isoperimetric inequality (Theorem 6.8)

$$4\pi A \leq L^2, \quad \text{equality} \iff \text{the curve is a circle}$$

Brahmagupta's formula (Theorem 6.9), with $s = (a + b + c + d)/2$

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

Buffon's needle and noodle (Theorems 6.11 and 6.12; Corollary 6.4), spacing d , needle $\ell \leq d$

$$p = \frac{2\ell}{\pi d}, \quad \mathbb{E}[N] = \frac{2L}{\pi d}, \quad L = \pi w \quad (\text{constant width } w)$$

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CHAPTER 7

Analytic Geometry of the Circle

INFO

Aim of the chapter. To translate into equations everything Chapters 2–6 established synthetically, and to check that the two routes lead to the same place. The bridge is a single formula — the Euclidean distance — and a single theorem: Pythagoras’.

Why it pays. The analytic route is not more elegant than the synthetic one, but it is *mechanical*: once the equation is written, questions that required an ingenious construction (tangents from a point, radical axis, circle through three points) become linear systems or quadratic equations. This is exactly what is needed when the computation has to be done by a machine, and it is no accident that this is the form in which the circle appears in numerical controllers and supervision systems (Chapters 11 and 13).

7.1. Metric definition and canonical equation

Definition 7.1: Canonical equation (“high school form”). The circle with centre $C(a, b)$ and radius $r > 0$ has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$

Proof. Step 1 — the metric definition. By Definition 2.3, a point P belongs to the circle with centre C and radius r if and only if $|PC| = r$. In vector notation, identifying points with their position vectors,

$$\|P - C\| = r.$$

Step 2 — the distance formula. With $P = (x, y)$ and $C = (a, b)$, the difference vector is $P - C = (x - a, y - b)$. The right triangle with legs $|x - a|$ and $|y - b|$ and hypotenuse $\overline{PC}$ gives, by the Pythagorean theorem (A5, assumed as an axiom in Chapter 2; proved in full further on, Theorem 10.1),

$$\|P - C\| = \sqrt{(x - a)^2 + (y - b)^2}.$$

(The degenerate case in which one of the two legs vanishes — P aligned with C horizontally or vertically — is checked directly, and the formula remains valid.)

Step 3 — squaring. The condition of Step 1 becomes

$$\sqrt{(x - a)^2 + (y - b)^2} = r.$$

Both sides are non-negative, so squaring is an equivalence (it neither introduces nor loses solutions):

$$(x - a)^2 + (y - b)^2 = r^2.$$

Every point satisfying this equation is at distance r from C , and conversely: the curve described is exactly the circle. $\square$

Remark 7.1: Where the content lies. Step 3 is algebra; Step 1 is the definition; all the content lies in Step 2, that is, in the Pythagorean theorem. The equation of the circle **is** the Pythagorean theorem written in coordinates — and this is why the same identity will reappear in Chapter 8 as the fundamental identity $\sin^2 \theta + \cos^2 \theta = 1$, which is nothing but the equation of the unit circle read on the parametrisation.

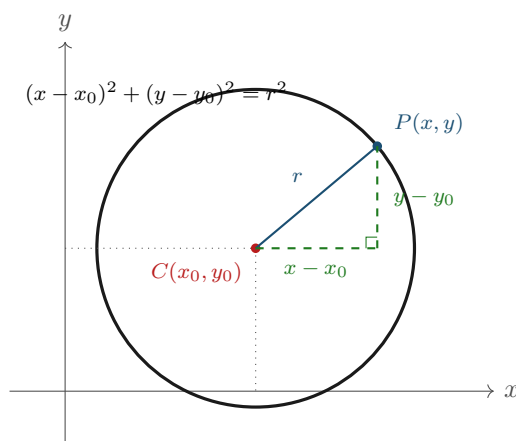


Figure 7.1: The canonical equation of the circle is nothing but Pythagoras' theorem written in coordinates. The point $P(x, y)$ belongs to the circle of centre $C(x_0, y_0)$ and radius r if and only if its distance from C equals r ; and the right triangle with legs $|x - x_0|$ and $|y - y_0|$ and hypotenuse CP immediately gives $(x - x_0)^2 + (y - y_0)^2 = r^2$. Expanding the squares yields the general form $x^2 + y^2 + ax + by + c = 0$, with $a = -2x_0$, $b = -2y_0$, $c = x_0^2 + y_0^2 - r^2$: the coefficients of x^2 and y^2 are equal and the xy term is missing, and it is precisely these two conditions that single out the circle among all conics.

7.2. General equation and reality conditions

Expanding $(x - a)^2 + (y - b)^2 = r^2$ gives the **general form**:

$$\text{General: } x^2 + y^2 + Dx + Ey + F = 0,$$

$$D = -2a, \quad E = -2b, \quad F = a^2 + b^2 - r^2$$

$$\text{Recovering centre and radius: } a = -\frac{D}{2}, \quad b = -\frac{E}{2}, \quad r = \sqrt{\frac{D^2}{4} + \frac{E^2}{4} - F}$$

Theorem 7.1: Classification of the general equation. Consider $x^2 + y^2 + Dx + Ey + F = 0$ and set

$$\Delta := \frac{D^2}{4} + \frac{E^2}{4} - F \quad (\text{equivalently } D^2 + E^2 - 4F = 4\Delta).$$

Then the set of real solutions is:

- (a) a **real circle** with centre $(-D/2, -E/2)$ and radius $\sqrt{\Delta}$, if $\Delta > 0$;
- (b) the **single point** $(-D/2, -E/2)$, if $\Delta = 0$;
- (c) the **empty set**, if $\Delta < 0$.

Proof. Step 1 — completing the squares. Collecting,

$$x^2 + Dx = \left(x + \frac{D}{2}\right)^2 - \frac{D^2}{4}, \quad y^2 + Ey = \left(y + \frac{E}{2}\right)^2 - \frac{E^2}{4},$$

hence the equation rewrites, with no loss of equivalence (one has only added and subtracted the same quantity), as

$$\left(x + \frac{D}{2}\right)^2 + \left(y + \frac{E}{2}\right)^2 = \Delta.$$

Step 2 — discussion by the sign of Δ . The left-hand side is a sum of two squares, hence always ≥ 0 , and equals 0 if and only if $x = -D/2$ and $y = -E/2$ simultaneously.

(a) If $\Delta > 0$, setting $r = \sqrt{\Delta} > 0$ the equation is exactly the canonical form (Definition 7.1) with centre $(-D/2, -E/2)$: it describes a real circle.

(b) If $\Delta = 0$, the equation forces a sum of two squares to vanish, which happens only when both vanish: the unique solution is $(-D/2, -E/2)$. One speaks of a **degenerate circle** (of zero radius), or an isolated point.

(c) If $\Delta < 0$, the left-hand side is ≥ 0 and the right-hand side is negative: no real pair (x, y) can satisfy the equality. The set of real solutions is empty. $\square$

Remark 7.2: Reality condition in school notation. With the notation $x^2 + y^2 + ax + by + c = 0$, used in many secondary-school texts, condition (a) of Theorem 7.1 is written in the compact form

$$a^2 + b^2 - 4c > 0,$$

since $r = \sqrt{a^2/4 + b^2/4 - c} = \frac{1}{2}\sqrt{a^2 + b^2 - 4c}$. It is the same condition, rescaled: one recognises it as the “discriminant” of the circle.

Remark 7.3: What can be read off the coefficients at a glance. From the general equation one obtains geometric information without computing anything:

- $F = 0 \iff$ the circle **passes through the origin** (substituting $x = y = 0$ leaves $F = 0$);
- $D = 0 \iff$ the centre lies on the y **axis** (since $a = -D/2$);
- $E = 0 \iff$ the centre lies on the x **axis**;
- $D = E = 0 \iff$ the circle is **centred at the origin**, and the equation reduces to $x^2 + y^2 = -F$, that is $x^2 + y^2 = r^2$ with $r^2 = -F$ (so in that case $F < 0$ is needed).

Warning. The coefficients of x^2 and y^2 must be equal, and no xy term may appear. An equation such as $2x^2 + 3y^2 + \dots = 0$ is not a circle (it is an ellipse), and neither is $x^2 + xy + y^2 + \dots = 0$ (the presence of the mixed term indicates a rotated conic). If the two quadratic coefficients are equal to each other but different from 1, for instance $3x^2 + 3y^2 + \dots = 0$, it is enough to divide the whole equation by 3 before applying the recovery formulas: forgetting this is the commonest error in this chapter.

7.2.1 Determining the circle through three points

Theorem 7.2: Circle through three non-collinear points. Given three non-collinear points P_1, P_2, P_3 , there exists a unique circle containing them, obtained by solving the linear system in the parameters D, E, F arising from substituting the coordinates of P_1, P_2, P_3 into the general equation.

Proof. Substituting the coordinates of each point gives three linear equations in the unknowns D, E, F (the terms $x^2 + y^2$ are known, being numbers). The 3×3 system is non-degenerate if and only if P_1, P_2, P_3 are not collinear (geometrically: the perpendicular bisectors of the three chords P_1P_2, P_2P_3, P_1P_3 meet at a single point, the centre, exactly when the three points are not collinear; see Corollary 3.2 for the synthetic version of the same result). $\square$

Proof by determinant. The circle through $P_i = (x_i, y_i)$, $i = 1, 2, 3$, is the locus

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Expanding along the first row gives an equation $\alpha(x^2 + y^2) + \beta x + \gamma y + \delta = 0$ with

$$\alpha = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix},$$

which is (up to a factor of 2) the signed area of triangle $P_1P_2P_3$: $\alpha \neq 0$ if and only if the three points are not collinear, exactly the hypothesis of the theorem. In that case one divides by α , obtaining a genuine circle equation. The equation is satisfied by each P_i because, substituting $x = x_i, y = y_i$, the 4×4 determinant has two equal rows (the first and the $(i + 1)$ -th), hence equals 0.

Numerical check. $P_1 = (0, 0)$, $P_2 = (4, 0)$, $P_3 = (0, 3)$: $\alpha = 12 \neq 0$; solving gives $D = -4$, $E = -3$, $F = 0$, that is, centre $(2, 1.5)$ and radius 2.5 (verified: it is the circumscribed circle of the 3–4–5 right triangle, of radius half the hypotenuse). $\checkmark$ $\square$

7.3. The university viewpoint: matrix form and conics

Remark 7.4: The circle as a notable conic. At university the circle is placed within the theory of **conics**: curves of second degree $ax^2 + bxy + cy^2 + dx + ey + f = 0$. The circle is the particular case with $a = c \neq 0$ and $b = 0$ (eccentricity $e = 0$, no mixed term, equal quadratic coefficients). In terms of the associated quadratic form $Q(x, y) = ax^2 + bxy + cy^2$, the circle corresponds to a scalar matrix $\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$, that is, to an isometry of the quadratic form (invariance under rotation).

Definition 7.2: Homogeneous matrix form. In homogeneous coordinates $(x, y, 1)$, the general equation is written as a quadratic form:

$$\begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & D/2 \\ 0 & 1 & E/2 \\ D/2 & E/2 & F \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0.$$

Remark 7.5: The determinant recovers the classification. The determinant of the 3×3 matrix of Definition 7.2 equals

$$\det = F - \frac{D^2}{4} - \frac{E^2}{4} = -\Delta,$$

with Δ as in Theorem 7.1. The synthetic classification (real circle / point / empty) therefore coincides with the projective classification of conics by the sign of the determinant: $\det < 0$ non-degenerate real conic, $\det = 0$ degenerate conic. It is a particular case of the general theory, and it is reassuring that the two languages give the same answer.

7.4. Position of a line relative to a circle

$$d(O, t) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} \begin{cases} > r & \text{external line} \\ = r & \text{tangent line} \\ < r & \text{secant line} \end{cases}$$

Remark 7.6: Why every ruler-and-compass construction is quadratic. Intersecting a line with a circle means substituting the equation of the line into that of the circle: one always obtains an equation of the **second degree** in one variable, whose solution requires a square root and nothing else. Intersecting two circles reduces to the same case, because subtracting the two general equations cancels the terms $x^2 + y^2$ and leaves a line (the radical axis, Proposition 7.1). Intersecting two lines gives a linear system. It follows that every ruler-and-compass construction produces numbers obtainable from the data by successive quadratic extensions: this is the algebraic fact behind the impossibility of trisecting the angle and doubling the cube, cited in Chapter 5, Section 5.13.

Theorem 7.3: Tangency condition for a line of given slope. The line $y = mx + q$ is tangent to the circle $x^2 + y^2 = r^2$ (centred at the origin) if and only if

$$q^2 = r^2(1 + m^2).$$

In particular, for the unit circle $r = 1$: $q^2 = 1 + m^2$.

Proof. Step 1 — system. Substituting $y = mx + q$ into $x^2 + y^2 = r^2$:

$$x^2 + (mx + q)^2 = r^2 \implies (1 + m^2)x^2 + 2mqx + (q^2 - r^2) = 0.$$

The leading coefficient $1 + m^2$ is always positive, so this really is a quadratic equation.

Step 2 — tangency $\iff$ zero discriminant. The line is tangent if and only if it has exactly one point in common with the circle (Definition 2.21), that is, if and only if the equation of Step 1 has a single real solution, that is, if and only if its discriminant vanishes:

$$\Delta_x = (2mq)^2 - 4(1 + m^2)(q^2 - r^2) = 0.$$

Step 3 — simplification. Dividing by 4 and expanding:

$$\begin{aligned} m^2q^2 - (1 + m^2)(q^2 - r^2) &= 0 \\ m^2q^2 - q^2 + r^2 - m^2q^2 + m^2r^2 &= 0 \\ -q^2 + r^2(1 + m^2) &= 0, \end{aligned}$$

whence $q^2 = r^2(1 + m^2)$.

Consistency check. The same condition follows from the distance formula: the line $mx - y + q = 0$ is at distance $|q|/\sqrt{m^2 + 1}$ from the origin, and imposing that this equal r (Theorem 2.6) gives $q^2 = r^2(m^2 + 1)$. The two routes — algebraic and metric — agree, as they must. $\square$

Warning. Theorem 7.3 does **not** cover vertical lines $x = k$, which are not of the form $y = mx + q$. For those the tangency condition is simply $|k| = r$. This is a limitation of the chosen parametrisation, not of the geometry: whoever implements an algorithm on this formula must treat the vertical case separately, or lose two tangents in every symmetric configuration.

Theorem 7.4: Equation of the tangent at a point of the circle. The tangent to $x^2 + y^2 + Dx + Ey + F = 0$ at the point $P_0(x_0, y_0)$ of the circle is:

$$xx_0 + yy_0 + \frac{D(x+x_0)}{2} + \frac{E(y+y_0)}{2} + F = 0.$$

Proof. The radius OP_0 has direction $(x_0 - a, y_0 - b)$; the tangent is the line through P_0 perpendicular to that direction (Theorem 3.8):

$$(x_0 - a)(x - a) + (y_0 - b)(y - b) = r^2.$$

Expanding and substituting $a = -D/2$, $b = -E/2$, $r^2 = D^2/4 + E^2/4 - F$, and using the fact that P_0 satisfies $x_0^2 + y_0^2 + Dx_0 + Ey_0 + F = 0$ to eliminate the quadratic terms, one obtains the stated formula (the “splitting rule” technique, $x^2 \rightarrow xx_0$, $x \rightarrow (x+x_0)/2$, standard in the theory of conics). $\square$

Theorem 7.5: Tangents from an external point (analytic form). From an external point $P_0(x_0, y_0)$ two tangents can be drawn, whose equations are found by imposing that the generic line $y - y_0 = m(x - x_0)$ through P_0 be at distance r from the centre, giving a quadratic equation in m with two distinct real solutions (if P_0 is external) or coincident ones (if P_0 is on the circle), or none (if internal).

Proof. Step 1 — normalisation. A translation moves the centre to the origin and alters neither distances nor parallelism, hence not tangency either: we may assume the circle $x^2 + y^2 = r^2$ and the point $P_0(x_0, y_0)$.

Step 2 — the tangency condition. A line is tangent to the circle if and only if its distance d from the centre equals exactly r . Indeed d is the minimum of $\|P - O\|$ as P runs over the line, attained at the foot H of the perpendicular; by Pythagoras $\|P - O\|^2 = d^2 + \|P - H\|^2$, so the points of the line at distance r from O are those with $\|P - H\|^2 = r^2 - d^2$: two if $d < r$, one if $d = r$, none if $d > r$. Writing the generic non-vertical line through P_0 as

$$y - y_0 = m(x - x_0) \iff mx - y + (y_0 - mx_0) = 0,$$

the point–line distance formula gives

$$d(O, \ell_m) = \frac{|y_0 - mx_0|}{\sqrt{m^2 + 1}} = r.$$

Step 3 — the quadratic in m . Squaring (legitimate: both sides are non-negative) and rearranging:

$$(y_0 - mx_0)^2 = r^2(m^2 + 1) \iff (x_0^2 - r^2)m^2 - 2x_0y_0m + (y_0^2 - r^2) = 0.$$

Step 4 — the discriminant is the power of the point. With $\Delta/4 = x_0^2y_0^2 - (x_0^2 - r^2)(y_0^2 - r^2)$ one gets

$$\frac{\Delta}{4} = x_0^2y_0^2 - (x_0^2y_0^2 - r^2x_0^2 - r^2y_0^2 + r^4) = r^2(x_0^2 + y_0^2 - r^2) = r^2 \text{pow}(P_0),$$

where $\text{pow}(P_0) = x_0^2 + y_0^2 - r^2$ is the power of P_0 with respect to the circle. Since $r^2 > 0$, the sign of Δ is the sign of the power:

- P_0 external $\iff \text{pow}(P_0) > 0 \iff \Delta > 0$: two distinct real roots, i.e. two tangents;
- P_0 on the circle $\iff \text{pow}(P_0) = 0 \iff \Delta = 0$: a double root, i.e. the tangent at P_0 , counted twice;
- P_0 internal $\iff \text{pow}(P_0) < 0 \iff \Delta < 0$: no real root, no tangent.

Step 5 — the vertical tangent. The argument of Step 3 excludes vertical lines, which have no slope. The equation in m degenerates into a linear one exactly when the leading coefficient vanishes, i.e. when $x_0^2 = r^2$: it then yields a single oblique tangent. The second one is the vertical line $x = x_0$, whose distance from the centre is $|x_0| = r$: it is tangent. The count “two tangents from an external point” therefore remains correct in this case as well, provided one counts lines and not merely slopes.

Sub-case $x_0^2 = r^2$ and $y_0 = 0$. Here the linear coefficient $-2x_0y_0$ of the quadratic equation also vanishes, and the equation becomes $0 \cdot m^2 - 0 \cdot m + (0 - r^2) = -r^2 = 0$: it has no solutions (since $r > 0$). There is no oblique tangent: $P_0 = (\pm r, 0)$ is a point of the circle (because $x_0^2 + y_0^2 = r^2$), and the only tangent is the vertical line $x = \pm r$ found above — consistent with the fact that through a point of the circle passes a single tangent: the perpendicular to the radius at that point is tangent (Theorem 3.9, Chapter 3), and it is the only line through that point perpendicular to the radius, by the uniqueness of the perpendicular (A2).

Step 6 — the points of tangency. If $T(x_1, y_1)$ is a point of tangency, the tangent at T has equation $xx_1 + yy_1 = r^2$; imposing that it pass through P_0 gives $x_0x_1 + y_0y_1 = r^2$. Hence *both* points of tangency lie on the line

$$x x_0 + y y_0 = r^2,$$

the **polar** of P_0 (the chord of contact). The points of tangency are therefore obtained by intersecting the polar with the circle, without passing through slopes — a construction that remains valid even when one of the two tangents is vertical.

Step 7 — consistency check with the power. The length of the tangent segment is $|P_0T| = \sqrt{\text{pow}(P_0)}$: indeed the triangle OTP_0 has a right angle at T , so $|P_0T|^2 = |P_0O|^2 - r^2 = x_0^2 + y_0^2 - r^2$. The discriminant of Step 4 can then be rewritten as $\Delta/4 = r^2|P_0T|^2$: the existence of the tangents and the length of the tangent segment are *the same* piece of information, read once algebraically and once metrically.

Step 8 — numerical example. With $r = 5$ and $P_0(7, 1)$: $\text{pow}(P_0) = 49 + 1 - 25 = 25 > 0$ (external) and the equation becomes $24m^2 - 14m - 24 = 0$, i.e. $12m^2 - 7m - 12 = 0$, with $m = \frac{7 \pm \sqrt{49 + 576}}{24} = \frac{7 \pm 25}{24}$, hence $m_1 = \frac{4}{3}$ and $m_2 = -\frac{3}{4}$. The two slopes have product -1 : the tangents are perpendicular, consistently with $|P_0O| = \sqrt{50} = r\sqrt{2}$, i.e. P_0 lies on the Monge (director) circle of radius $r\sqrt{2}$. ✓ $\square$

7.5. Radical axis and pencils of circles

Proposition 7.1: Radical axis. The radical axis of two circles $\gamma_1 : x^2 + y^2 + D_1x + E_1y + F_1 = 0$ and $\gamma_2 : x^2 + y^2 + D_2x + E_2y + F_2 = 0$ is the line

$$(D_1 - D_2)x + (E_1 - E_2)y + (F_1 - F_2) = 0.$$

Proof. The radical axis is the locus of points of equal power with respect to γ_1 and γ_2 : $\text{pow}_1(P) = \text{pow}_2(P)$, where $\text{pow}_i(P) = |PC_i|^2 - r_i^2$ is the metric power (Chapter 3). This coincides with the left-hand side $x_0^2 + y_0^2 + D_ix_0 + E_iy_0 + F_i$ of the equation of γ_i evaluated at $P = (x_0, y_0)$: by direct substitution, with $C_i = (a_i, b_i)$, $D_i = -2a_i$, $E_i = -2b_i$, $F_i = a_i^2 + b_i^2 - r_i^2$ (from the expanded canonical form), $x_0^2 + y_0^2 + D_ix_0 + E_iy_0 + F_i = (x_0 - a_i)^2 + (y_0 - b_i)^2 - r_i^2 =$

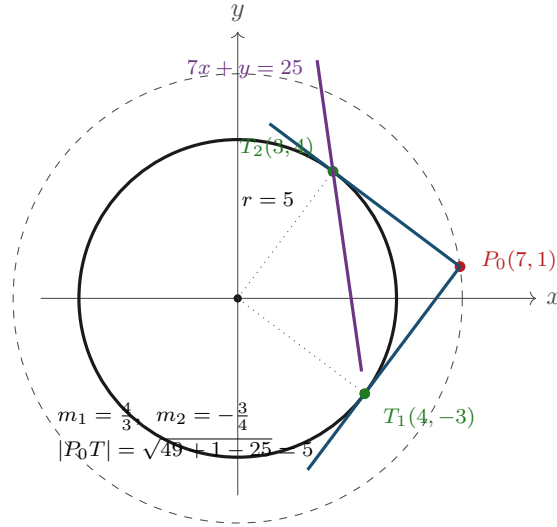


Figure 7.2: The two tangents drawn from an external point, in the numerical example $r = 5$, $P_0(7, 1)$. Requiring the distance of the centre from the line through P_0 with slope m to equal r yields the equation $(x_0^2 - r^2)m^2 - 2x_0y_0m + (y_0^2 - r^2) = 0$, here $12m^2 - 7m - 12 = 0$, with solutions $m_1 = 4/3$ and $m_2 = -3/4$. The reduced discriminant equals $r^2(x_0^2 + y_0^2 - r^2) = r^2 \text{pow}(P_0)$: its sign is that of the power of the point, and reproduces exactly the three cases (two tangents, one, none). The line $7x + y = 25$ through the two contact points is the *polar* of P_0 , and the length of the tangent segment is $\sqrt{\text{pow}(P_0)} = 5$. Here $m_1m_2 = -1$: the tangents are orthogonal because P_0 lies on the Monge circle, of radius $r\sqrt{2} = 7.071$ (dashed).

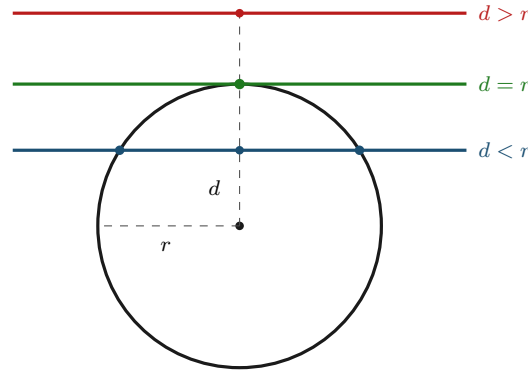


Figure 7.3: The three possible positions of a line with respect to a circle, decided by a single number: the distance d of the centre from the line, compared with the radius r . If $d < r$ the line is *secant* and cuts the circle in two points; if $d = r$ it is *tangent* and has one point in common; if $d > r$ it is *external* and does not meet the circle. The criterion is equivalent to the sign of the discriminant of the system formed by the equations of line and circle, but is far quicker: for the line $ax + by + c = 0$ it suffices to compute $d = |ax_0 + by_0 + c|/\sqrt{a^2 + b^2}$ and compare it with r , without solving any quadratic equation.

$|PC_i|^2 - r_i^2$ (an identity we shall prove again, in general, in Theorem 7.6). Subtracting the two general equations (the terms $x^2 + y^2$ cancel) gives exactly the stated line. The normal vector to the line is $(D_1 - D_2, E_1 - E_2) = -2(a_1 - a_2, b_1 - b_2)$, parallel to the line joining the centres: hence the radical axis is perpendicular to the central axis. $\square$

Remark 7.7: Pencil of circles. For every $\lambda \in \mathbb{R}$, the equation $\gamma_1 + \lambda(\gamma_2 - \gamma_1) = 0$ describes a **pencil** of circles through the points common to γ_1, γ_2 (if any), generalising the pencil of lines through a point. It is the analytic version of Definition 2.30 of Chapter 2, and anticipates the language of algebraic geometry, where pencils of conics are pencils of algebraic curves.

Theorem 7.6: Power and tangent length in analytic form. Let $\gamma : x^2 + y^2 + Dx + Ey + F = 0$ be a real circle and $P = (x_0, y_0)$ a point. Then the power of P with respect to γ is

$$\text{pow}_\gamma(P) = x_0^2 + y_0^2 + Dx_0 + Ey_0 + F,$$

that is, it is obtained by **simply substituting the coordinates of P into the left-hand side**. In particular, if P is external, the tangent length from P is

$$t = \sqrt{x_0^2 + y_0^2 + Dx_0 + Ey_0 + F}.$$

Proof. Step 1. By Theorem 7.1 the centre is $C = (-D/2, -E/2)$ and the radius satisfies $r^2 = D^2/4 + E^2/4 - F$.

Step 2. By Definition 3.1 (Chapter 3),

$$\begin{aligned} \text{pow}_\gamma(P) &= |PC|^2 - r^2 \\ &= \left(x_0 + \frac{D}{2}\right)^2 + \left(y_0 + \frac{E}{2}\right)^2 - \left(\frac{D^2}{4} + \frac{E^2}{4} - F\right) \\ &= x_0^2 + Dx_0 + y_0^2 + Ey_0 + F. \end{aligned}$$

Step 3. The formula for t follows from Theorem 3.16 (Chapter 3), by which $t^2 = \text{pow}_\gamma(P)$ for P external. $\square$

Corollary 7.1: Immediate recognition of a point's position. Writing $\Pi(P) = x_0^2 + y_0^2 + Dx_0 + Ey_0 + F$:

$$\Pi(P) < 0 \iff P \text{ interior}, \quad \Pi(P) = 0 \iff P \in \gamma, \quad \Pi(P) > 0 \iff P \text{ exterior}.$$

Proof. Immediate from Theorem 7.6 and the sign of the power (directly: $\text{pow} = |PC|^2 - r^2$ is negative, zero or positive according as $|PC|$ is less than, equal to or greater than r). $\square$

Best practice. This is the cheapest test in existence for deciding whether a point lies inside a circle: **four multiplications and four additions**, no square root. On a controller that must check thousands of points per cycle — for instance for intrusion detection in an exclusion zone (final section of this chapter) — the difference with respect to computing $\sqrt{(x-a)^2 + (y-b)^2}$ and comparing it with r is substantial, since a square root typically costs 10–20 times a multiplication.

Theorem 7.7: Radical centre of three circles. Let $\gamma_1, \gamma_2, \gamma_3$ be three circles whose centres are **not** collinear. Then the three pairwise radical axes are concurrent at a single point, the **radical centre**, which has the same power with respect to all three.

Proof. Step 1 — existence of the intersection. Let a_{12}, a_{13}, a_{23} be the three radical axes. By Proposition 7.1, the normal to a_{ij} is parallel to the line of centres $C_i C_j$. Since the three centres are not collinear, $C_1 C_2$ and $C_1 C_3$ are not parallel, hence a_{12} and a_{13} are not parallel and meet

at a point R .

Step 2 — R lies on the third axis too. By definition of radical axis,

$$R \in a_{12} \Rightarrow \text{pow}_1(R) = \text{pow}_2(R), \quad R \in a_{13} \Rightarrow \text{pow}_1(R) = \text{pow}_3(R).$$

By transitivity, $\text{pow}_2(R) = \text{pow}_3(R)$, i.e. $R \in a_{23}$.

Step 3 — uniqueness. If two distinct points R, R' had the same property, the line RR' would coincide with a_{12} and with a_{13} simultaneously, hence $C_1C_2 \parallel C_1C_3$: the centres would be collinear, contrary to hypothesis.

Step 4 — algebraic check. In coordinates, subtracting the three general equations pairwise gives the linear system

$$\begin{cases} (D_1 - D_2)x + (E_1 - E_2)y = F_2 - F_1 \\ (D_1 - D_3)x + (E_1 - E_3)y = F_3 - F_1 \end{cases}$$

whose determinant is $(D_1 - D_2)(E_1 - E_3) - (D_1 - D_3)(E_1 - E_2)$. Recalling $D_i = -2a_i$ and $E_i = -2b_i$, it equals $4[(a_1 - a_2)(b_1 - b_3) - (a_1 - a_3)(b_1 - b_2)]$, i.e. 4 times twice the signed area of triangle $C_1C_2C_3$: non-zero exactly when the centres are not collinear. $\square$

Corollary 7.2: Construction of the radical axis of two disjoint circles. *Given two disjoint circles γ_1, γ_2 , their radical axis is constructed with ruler and compass as follows: draw any third circle γ_3 meeting both; the radical axes a_{13} and a_{23} are then the common chords, immediately drawable; their meeting point is the radical centre R , and a_{12} is the perpendicular through R to the line of centres C_1C_2 .*

Proof. If γ_i and γ_3 meet at two points, those points have zero power with respect to both, hence equal powers: they belong to the radical axis a_{i3} , which being a line coincides with the line through those two points, i.e. with the common chord. By Theorem 7.7, $R = a_{13} \cap a_{23}$ also lies on a_{12} ; and a_{12} is perpendicular to C_1C_2 by Proposition 7.1. A line is determined by a point and a direction, hence a_{12} is determined. $\square$

Example 7.1: Radical centre: full computation. Find the radical centre of the three circles

$$\gamma_1 : x^2 + y^2 - 4x - 2y + 1 = 0, \quad \gamma_2 : x^2 + y^2 - 10x - 4y + 21 = 0, \quad \gamma_3 : x^2 + y^2 - 2x - 8y + 13 = 0.$$

Step 1 — centres and radii (Theorem 7.1):

$$C_1 = (2, 1), \quad r_1 = 2; \quad C_2 = (5, 2), \quad r_2 = 2\sqrt{2}; \quad C_3 = (1, 4), \quad r_3 = 2.$$

The centres are not collinear: the determinant $\begin{vmatrix} 3 & 1 \\ -1 & 3 \end{vmatrix} = 10 \neq 0$. Theorem 7.7 therefore applies.

Step 2 — radical axis a_{12} . Subtract γ_2 from γ_1 :

$$(-4 + 10)x + (-2 + 4)y + (1 - 21) = 0 \iff 6x + 2y - 20 = 0 \iff 3x + y = 10.$$

Step 3 — radical axis a_{13} . Subtract γ_3 from γ_1 :

$$(-4 + 2)x + (-2 + 8)y + (1 - 13) = 0 \iff -2x + 6y - 12 = 0 \iff -x + 3y = 6.$$

Step 4 — intersection. From $\{3x + y = 10, -x + 3y = 6\}$: the first gives $y = 10 - 3x$; substituting, $-x + 30 - 9x = 6 \Rightarrow -10x = -24 \Rightarrow x = 2.4$, hence $y = 10 - 7.2 = 2.8$:

$$R = (2.4; 2.8).$$

Step 5 — check on the third axis. a_{23} comes from subtracting γ_3 from γ_2 : $-8x+4y+8=0$, i.e. $-2x+y=-2$. Substituting R : $-4.8+2.8=-2$. ✓

Step 6 — check on the powers (Theorem 7.6). Substituting R into the three left-hand sides:

$$\Pi_1(R) = 2.4^2 + 2.8^2 - 4(2.4) - 2(2.8) + 1 = 5.76 + 7.84 - 9.6 - 5.6 + 1 = -0.60,$$

$$\Pi_2(R) = 5.76 + 7.84 - 10(2.4) - 4(2.8) + 21 = 13.60 - 24 - 11.2 + 21 = -0.60,$$

$$\Pi_3(R) = 5.76 + 7.84 - 2(2.4) - 8(2.8) + 13 = 13.60 - 4.8 - 22.4 + 13 = -0.60.$$

The three powers agree. ✓ The value is *negative*: by Corollary 7.1, R is interior to all three circles — the case in which no real tangent from R exists, and the radical centre is not the centre of a real circle orthogonal to the three.

Theorem 7.8: The circle of Apollonius, analytically. Let $A \neq B$ be two points of the plane and $k > 0$, $k \neq 1$. The locus

$$\mathcal{A}_k = \left\{ P : \frac{|PA|}{|PB|} = k \right\}$$

is a **circle**. Taking $A = (-c, 0)$ and $B = (c, 0)$ with $2c = |AB|$, its centre and radius are

$$C_k = \left(c \frac{k^2 + 1}{k^2 - 1}, 0 \right), \quad r_k = \frac{2ck}{|k^2 - 1|}.$$

Proof. Step 1 — eliminate the root. The condition $|PA| = k|PB|$ is equivalent, on squaring (both sides being non-negative), to $|PA|^2 = k^2|PB|^2$, i.e.

$$(x+c)^2 + y^2 = k^2[(x-c)^2 + y^2].$$

Step 2 — expand.

$$(1-k^2)x^2 + (1-k^2)y^2 + 2c(1+k^2)x + c^2(1-k^2) = 0.$$

Step 3 — divide. Since $k \neq 1$, $1-k^2 \neq 0$ and we may divide:

$$x^2 + y^2 + \frac{2c(1+k^2)}{1-k^2}x + c^2 = 0,$$

of the form $x^2 + y^2 + Dx + Ey + F = 0$ with $D = 2c(1+k^2)/(1-k^2)$, $E = 0$, $F = c^2$.

Step 4 — it is a real circle. By Theorem 7.1 we need $\Delta = D^2/4 - F > 0$:

$$\frac{D^2}{4} - F = c^2 \frac{(1+k^2)^2 - (1-k^2)^2}{(1-k^2)^2} = c^2 \frac{4k^2}{(1-k^2)^2} > 0,$$

using the difference of squares $(1+k^2)^2 - (1-k^2)^2 = 4k^2$. The quantity is strictly positive because $c \neq 0$ and $k > 0$.

Step 5 — centre and radius. Changing the sign of numerator and denominator gives $\frac{1+k^2}{1-k^2} = -\frac{k^2+1}{k^2-1}$, whence

$$C_k = \left(-\frac{D}{2}, 0 \right) = \left(-c \frac{1+k^2}{1-k^2}, 0 \right) = \left(c \frac{k^2+1}{k^2-1}, 0 \right), \quad r_k = \sqrt{\Delta} = \frac{2ck}{|k^2-1|}.$$

Independent check of the signs. With $c = 1$ and $k = 2$ the formula gives $C_k = (\frac{5}{3}, 0)$ and $r_k = \frac{4}{3}$: the centre lies on the side of $B = (1, 0)$, as it must, since $k > 1$ means $|PA| > |PB|$. The two endpoints of the horizontal diameter can be checked without using the formula: for $P = (\frac{5}{3} + \frac{4}{3}, 0) = (3, 0)$ one has $|PA| = 4$, $|PB| = 2$, ratio $2 = k$; for $P = (\frac{5}{3} - \frac{4}{3}, 0) = (\frac{1}{3}, 0)$ one has $|PA| = \frac{4}{3}$, $|PB| = \frac{2}{3}$, ratio again $2 = k$. ✓ □

Remark 7.8: Two proofs of the same theorem. This result had already been proved synthetically in Chapter 2 (Theorem 2.7), using the internal and external bisectors of the angle at P and the bisector theorem. The two proofs illustrate the difference in style well:

- the **synthetic** one explains why the locus is a circle — the feet of the two bisectors are the endpoints of a diameter — but does not directly supply centre and radius;
- the **analytic** one explains nothing, but delivers centre and radius in closed form after four lines of algebra.

Neither is superior: they are complementary, which is why the treatise gives both.

Corollary 7.3: The case $k = 1$ and the limit $k \rightarrow 1$. For $k = 1$ the locus $\mathcal{A}_1$ is the **perpendicular bisector** of AB , i.e. a line. Consistently, as $k \rightarrow 1$ we have $r_k \rightarrow +\infty$ and $|C_k| \rightarrow +\infty$: the Apollonius circle degenerates into the line, as the unified line/circle viewpoint of Chapter 5 predicts.

Proof. For $k = 1$ the condition $|PA| = |PB|$ defines the perpendicular bisector (Chapter 2). For the limit: from $r_k = 2ck/|k^2 - 1|$ we get $r_k \rightarrow +\infty$ as $k \rightarrow 1$, the numerator tending to $2c > 0$ and the denominator to 0^+ . Likewise $|C_k| \rightarrow +\infty$. The difference between the centre distance and the radius stays finite:

$$|C_k| - r_k = c \frac{(k-1)^2}{|k-1|(k+1)} = c \frac{|k-1|}{k+1} \xrightarrow{k \rightarrow 1} 0,$$

i.e. in the limit the circle passes through the midpoint of AB : exactly the point of the perpendicular bisector closest to A and B . $\square$

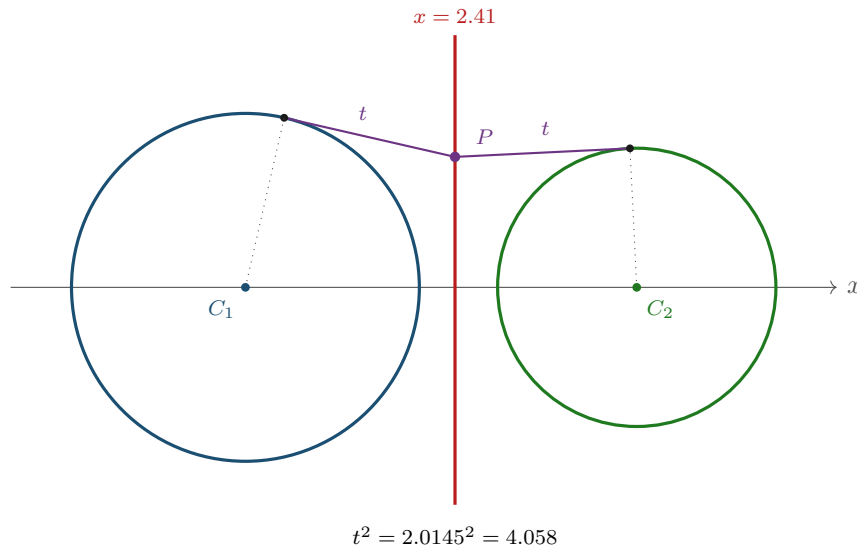


Figure 7.4: The radical axis of two non-concentric circles. Subtracting the two general-form equations term by term, the second-degree terms cancel — because both have coefficient 1 for x^2 and y^2 — and a first-degree equation remains: a line, always perpendicular to the line of centres. It is the locus of points having equal power with respect to the two circles, i.e. from which the two tangent segments have the same length t . In the example (C_1 of radius 2 at the origin, C_2 of radius 1.6 at $(4.5, 0)$) the radical axis is $x = 2.41$, and from the point $P(2.41, 1.5)$ both tangents measure 2.0145. Letting the constant vary one obtains the pencil of circles generated by the two, all sharing the same radical axis.

7.6. Pole and polar

Definition 7.3: Polar of a point. With respect to the circle $x^2 + y^2 = r^2$, the **polar** of the point $P = (x_0, y_0) \neq O$ is the line

$$x_0x + y_0y = r^2.$$

Proposition 7.2: Construction of the polar. The polar of $P \neq O$ is the line perpendicular to OP passing through the inverse P' of P with respect to the circle (that is, the point on the ray OP with $|OP| \cdot |OP'| = r^2$, Chapter 5).

Proof. A point $X = (x, y)$ lies on the polar if and only if $\vec{X} \cdot \vec{P} = x_0x + y_0y = r^2$ (dot product). The point P' on the inverse, $\vec{P}' = \frac{r^2}{|OP|^2} \vec{P}$ (from the definition of inversion), satisfies $\vec{P}' \cdot \vec{P} = \frac{r^2}{|OP|^2} |\vec{P}|^2 = r^2$: hence P' lies on the polar. Moreover, for every X on the polar, $(\vec{X} - \vec{P}') \cdot \vec{P} = \vec{X} \cdot \vec{P} - \vec{P}' \cdot \vec{P} = r^2 - r^2 = 0$: the vector $X - P'$ is perpendicular to P (that is, to OP), for every point X of the polar. The polar is therefore the line through P' perpendicular to OP . $\square$

Theorem 7.9: La Hire's theorem. Q lies on the polar of P if and only if P lies on the polar of Q .

Proof. Q on the polar of P means $\vec{Q} \cdot \vec{P} = r^2$ (Definition 7.3, with $P = (x_0, y_0)$ and $Q = (x, y)$). This condition is **symmetric** in P, Q (the dot product is commutative): $\vec{Q} \cdot \vec{P} = \vec{P} \cdot \vec{Q} = r^2$ is exactly the condition “ P lies on the polar of Q ”. $\square$

Corollary 7.4: The polar of an external point is the chord of contact. If P is external to the circle, its polar is the line through the two points of tangency of the tangents from P .

Proof. Let T be a point of tangency of a tangent from P : then $OT \perp PT$ (Theorem 3.8, Chapter 3), that is, $\vec{T} \cdot (\vec{P} - \vec{T}) = 0$, that is, $\vec{T} \cdot \vec{P} = |\vec{T}|^2 = r^2$: T lies on the polar of P . This holds for both points of tangency, which therefore determine the polar (a line is determined by two points). $\square$

Remark 7.9: The case of an interior point. If $P = (x_0, y_0)$ is interior to the circle $x^2 + y^2 = r^2$ (that is, $|OP| < r$), its polar $x_0x + y_0y = r^2$ does not meet the circle: the distance from O to the polar line is $d(O, \text{polar}) = r^2 / \sqrt{x_0^2 + y_0^2} = r^2 / |OP|$ (point-to-line distance formula, already used earlier in this chapter), and $|OP| < r$ gives $d(O, \text{polar}) = r^2 / |OP| > r^2 / r = r$: the polar line is farther from the centre than the radius, hence does not touch the circle. The polar of an interior point is **external** to the circle. If P lies on the circle, $P' = P$ (inversion fixes the points of the circle) and the polar is the **tangent** at P : this recovers the doubling formula (previous section).

Example 7.2: Numerical check. Circle $x^2 + y^2 = 25$, $P = (10, 0)$ (external, $|OP| = 10 > 5$): polar $10x = 25$, that is, $x = 2.5$. Points of tangency: $x = 2.5$, $y = \pm\sqrt{25 - 6.25} = \pm 4.330127$; direct check that $OT \perp PT$: $\vec{OT} = (2.5; 4.330127)$, $\vec{PT} = (-7.5; 4.330127)$, dot product $= -18.75 + 18.75 = 0$. $\checkmark$

7.7. Orthogonal circles in coordinates

Proposition 7.3: Orthogonality condition in coordinates. *The circles $x^2 + y^2 + D_1x + E_1y + F_1 = 0$ and $x^2 + y^2 + D_2x + E_2y + F_2 = 0$ are orthogonal if and only if*

$$D_1D_2 + E_1E_2 = 2(F_1 + F_2).$$

Proof. By Theorem 5.4 (in its metric characterisation, Chapter 5: two circles are orthogonal if and only if $|O_1O_2|^2 = r_1^2 + r_2^2$), with $O_i = (-D_i/2, -E_i/2)$, $r_i^2 = D_i^2/4 + E_i^2/4 - F_i$:

$$|O_1O_2|^2 = \frac{(D_1 - D_2)^2}{4} + \frac{(E_1 - E_2)^2}{4}, \quad r_1^2 + r_2^2 = \frac{D_1^2 + D_2^2}{4} + \frac{E_1^2 + E_2^2}{4} - F_1 - F_2.$$

Expanding $|O_1O_2|^2$ and equating, the terms $D_1^2/4, D_2^2/4, E_1^2/4, E_2^2/4$ cancel on both sides, leaving $-\frac{D_1D_2}{2} - \frac{E_1E_2}{2} = -F_1 - F_2$, which is the claim. $\square$

Example 7.3: Numerical check. $\Gamma_1 : x^2 + y^2 - 4 = 0$ (centre O , radius 2) and $\Gamma_2 : x^2 + y^2 + 6x + 4 = 0$ (centre $(-3, 0)$, radius $\sqrt{5}$): $D_1D_2 + E_1E_2 = 0 \cdot 6 + 0 \cdot 0 = 0$, and $2(F_1 + F_2) = 2(-4 + 4) = 0$: they agree, hence orthogonal. Direct check: $|O_1O_2|^2 = 9$, $r_1^2 + r_2^2 = 4 + 5 = 9$. $\checkmark$

Remark 7.10: The circular points (university-level variant). *In complex homogeneous coordinates, every circle — whatever the real D, E, F — passes through the same two **imaginary** points at infinity, $(1, \pm i, 0)$ (the **circular points**): substituting $x = 1, y = \pm i$, into the homogenised second-degree part $x^2 + y^2 = 1 + (\pm i)^2 = 1 - 1 = 0$, always zero, independent of D, E, F . It is precisely this shared property (no other non-degenerate real conic passes through both) that singles out the circle among the conics in complex projective geometry: a fact we cite only as a panorama, without developing its theory (which requires projective homogeneous coordinates, beyond this volume's tools).*

7.8. Parametric, polar and complex forms

Proposition 7.4: Cartesian parametrisation. *The circle $C(O', r)$ with centre $O' = (a, b)$ is parametrised as*

$$x(\theta) = a + r \cos \theta, \quad y(\theta) = b + r \sin \theta, \quad \theta \in [0, 2\pi).$$

Proof. Substituting into the canonical equation: $(x - a)^2 + (y - b)^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2$ by the Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$ (Pythagorean theorem, A5, applied to the right triangle with legs $|\cos \theta|, |\sin \theta|$ and hypotenuse 1 in the unit circle — proved in full also as Theorem 8.2 in Chapter 8): every point of the parametrised curve satisfies the equation, and as θ runs over one period the whole circle is traversed exactly once (bijectivity of $\theta \mapsto (\cos \theta, \sin \theta)$ on $[0, 2\pi)$). $\square$

Proposition 7.5: Polar equation. *With the centre at polar coordinates (ρ_0, φ_0) relative to the pole O , the circle of radius r has general polar equation*

$$\rho^2 - 2\rho\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2 - r^2 = 0.$$

Two notable cases: if the centre is the pole ($\rho_0 = 0$), $\rho = r$ (the simplest possible equation); if the circle passes through the pole ($\rho_0 = r$), the equation simplifies to $\rho = 2r \cos(\varphi - \varphi_0)$.

Proof. In Cartesian coordinates, $P = (\rho \cos \varphi, \rho \sin \varphi)$ and the centre is $C = (\rho_0 \cos \varphi_0, \rho_0 \sin \varphi_0)$. The point P lies on the circle if and only if $|P - C|^2 = r^2$:

$$r^2 = (\rho \cos \varphi - \rho_0 \cos \varphi_0)^2 + (\rho \sin \varphi - \rho_0 \sin \varphi_0)^2.$$

Expanding the squares and collecting terms,

$$r^2 = \rho^2(\cos^2 \varphi + \sin^2 \varphi) + \rho_0^2(\cos^2 \varphi_0 + \sin^2 \varphi_0) - 2\rho\rho_0(\cos \varphi \cos \varphi_0 + \sin \varphi \sin \varphi_0).$$

By the Pythagorean identity $\cos^2 \alpha + \sin^2 \alpha = 1$ (Pythagorean theorem, A5, applied to the reference triangle in the unit circle) the first two groups equal ρ^2 and ρ_0^2 ; by the cosine subtraction formula $\cos \varphi \cos \varphi_0 + \sin \varphi \sin \varphi_0 = \cos(\varphi - \varphi_0)$ (Theorem 8.4, Chapter 8 — logically available even though typographically later, as already noted for the Pythagorean identity in Chapter 6) the third term gives

$$r^2 = \rho^2 + \rho_0^2 - 2\rho\rho_0 \cos(\varphi - \varphi_0),$$

which on rearranging gives the stated equation. The case $\rho_0 = r$ follows by solving the quadratic in ρ and discarding the root $\rho = 0$ (the pole itself, already on the curve). $\square$

Definition 7.4: The circle in the complex plane. Identifying $\mathbb{R}^2$ with $\mathbb{C}$ via $z = x + iy$, the circle with centre $z_0 \in \mathbb{C}$ and radius r is

$$|z - z_0| = r \quad \Longleftrightarrow \quad z(\theta) = z_0 + r e^{i\theta}, \quad \theta \in [0, 2\pi),$$

an immediate consequence of Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ and of Proposition 7.4.

Theorem 7.10: Rational points of the unit circle. Setting $t = \tan(\theta/2)$ (the “rational parametric”, or Weierstrass, substitution) in the parametrisation of the unit circle, the half-angle formulas (Corollary 8.3, Chapter 8) give

$$\cos \theta = \frac{1 - t^2}{1 + t^2}, \quad \sin \theta = \frac{2t}{1 + t^2}.$$

As t ranges over $\mathbb{Q}$, this formula gives **all and only** the points of the unit circle with rational coordinates other than $(-1, 0)$.

Proof. ($\Leftarrow$, every $t \in \mathbb{Q}$ gives a rational point of the circle). For $t \in \mathbb{Q}$ the point $(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$ has rational coordinates (sum, product and quotient of rationals, with $1+t^2 \neq 0$), and lies on the unit circle because

$$\left(\frac{1-t^2}{1+t^2}\right)^2 + \left(\frac{2t}{1+t^2}\right)^2 = \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{1-2t^2+t^4+4t^2}{(1+t^2)^2} = \frac{(1+t^2)^2}{(1+t^2)^2} = 1$$

(direct algebraic check).

($\Rightarrow$, every rational point $\neq (-1, 0)$ is obtained this way). Let $(x, y) \in \mathbb{Q}^2$ with $x^2 + y^2 = 1$ and $x \neq -1$. Put $t := \frac{y}{1+x} \in \mathbb{Q}$ (defined because $1+x \neq 0$). From $x^2 + y^2 = 1$ it follows that

$(1+x)(1-x) = 1-x^2 = y^2$, hence (dividing by $(1+x)^2 \neq 0$)

$$t^2 = \frac{y^2}{(1+x)^2} = \frac{(1+x)(1-x)}{(1+x)^2} = \frac{1-x}{1+x},$$

whence $t^2(1+x) = 1-x$, that is, $x(1+t^2) = 1-t^2$, that is, $x = \frac{1-t^2}{1+t^2}$; and then $y = t(1+x) = t \cdot \frac{2}{1+t^2} = \frac{2t}{1+t^2}$. The only excluded point is $(-1, 0)$, for which $1+x = 0$ and t is not defined (corresponding to the limit $t \rightarrow \infty$). $\square$

Remark 7.11: The mechanism generating Pythagorean triples. Choosing $t = n/m$ (reduced fraction, $0 < n < m$) gives, multiplying numerator and denominator of $\cos \theta, \sin \theta$ by m^2 , the triple

$$(m^2 - n^2, 2mn, m^2 + n^2),$$

which satisfies $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$ by construction: this is the generating formula for Pythagorean triples used in Chapter 10 (the Pythagorean-triple theorem, with 7.10 as the reference for the geometric origin of the formula). If moreover m, n are coprime and of opposite parity ($m \not\equiv n \pmod{2}$), the triple is **primitive**: the three numbers $m^2 - n^2, 2mn, m^2 + n^2$ are coprime. Indeed a prime p common to $m^2 - n^2$ and $m^2 + n^2$ divides their sum $2m^2$ and their difference $2n^2$; if p were odd it would divide m^2 and n^2 , hence m and n , against coprimality; so $p = 2$, but $m^2 + n^2$ is odd (sum of an even and an odd square, by the opposite parity of m, n): a contradiction. Hence no common prime exists.

7.9. Application: intrusion detection in a circular exclusion zone

In an HMI/SCADA supervision system for a robotic cell or for a water-treatment plant, one frequently has to check in real time whether the current position of an operator, an AGV or a tool (known through a localisation system or through odometry) has entered a **circular exclusion zone** around a machine.

Analytic intrusion check. Data. Exclusion zone: circle with centre $C = (x_0, y_0)$ and radius r (HMI configuration parameters). Detected position: $P = (x_P, y_P)$, updated at every scan cycle (typically every 50–100 ms in an industrial SCADA system).

Step 1 (membership test). One computes

$$\delta = (x_P - x_0)^2 + (y_P - y_0)^2 - r^2,$$

which is exactly the power of P with respect to the circle (Definition 3.1, Chapter 3). If $\delta < 0$, P is inside the zone (intrusion); if $\delta \geq 0$, P is outside or on the boundary. The comparison is made directly on squared distances, deliberately avoiding a square root (a more costly floating-point operation, Chapter 11).

Step 2 (anti-chatter hysteresis). To avoid false alarms due to measurement noise near the boundary, two radii are used in practice: an alarm radius r_{on} (the alarm trips when $\delta < 0$ with respect to r_{on}) and a reset radius $r_{\text{off}} > r_{\text{on}}$ (the alarm clears only when P has moved beyond r_{off}), like a hysteresis comparator in industrial electronics: it prevents rapid alarm toggling when the operator moves exactly on the boundary.

Step 3 (safe re-entry point). If one has to compute the boundary point nearest to an external P (for instance to guide an AGV at a safe distance), it is

$$P^* = C + r \cdot \frac{P - C}{|P - C|},$$

that is, the intersection of the ray CP with the circle — a particular case of the parametrisation discussed in this chapter. $\square$

7.9.1 Numerical example: radical axis of two concrete circles

Explicit computation. Let $\gamma_1 : x^2 + y^2 - 4x - 2y - 4 = 0$ (centre $(2, 1)$, radius 3) and $\gamma_2 : x^2 + y^2 + 2x - 6y + 6 = 0$ (centre $(-1, 3)$, radius 2). By Proposition 7.1, subtracting the two equations term by term:

$$(-4 - 2)x + (-2 - (-6))y + (-4 - 6) = 0 \implies -6x + 4y - 10 = 0 \implies 3x - 2y + 5 = 0.$$

Check. The normal vector of the radical axis, $(3, -2)$, must be parallel to the line joining the centres $(2, 1) \rightarrow (-1, 3)$, of direction $(-3, 2)$: indeed $(3, -2) = -1 \cdot (-3, 2)$, parallel as predicted by the proof of Proposition 7.1. $\square$

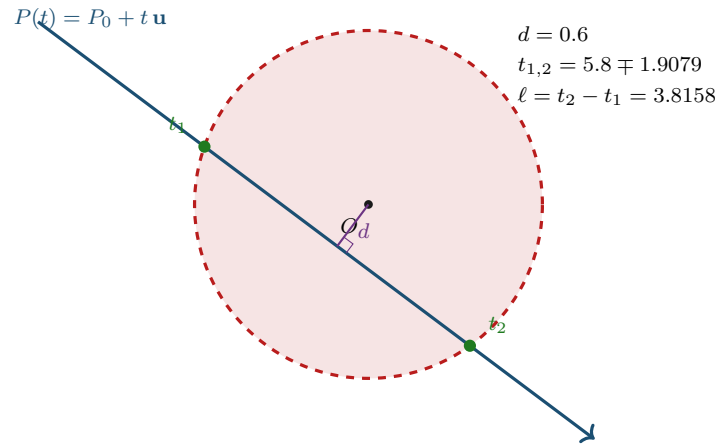


Figure 7.5: Intrusion detection in a circular exclusion zone, with the same data as Exercise 7.12 ($P_0 = (-5, 3)$, $\mathbf{u} = (0.8, -0.6)$, $R = 2$). The target moves along the line $P(t) = P_0 + t\mathbf{u}$, with $\mathbf{u}$ a unit vector, and the forbidden zone is the circle of centre O and radius $R = 2$. Substituting into $|P(t) - O|^2 = R^2$ gives a quadratic equation in t whose discriminant has the sign of $R^2 - d^2$, where d is the distance of the centre from the trajectory: here $d = 0.6 < R$, so two real roots, $t_{1,2} = 5.8 \pm 1.9079$. The target enters at $t_1 = 3.8921$, leaves at $t_2 = 7.7079$ and travels inside a stretch $\ell = t_2 - t_1 = 3.8158$, which is the chord: since $\mathbf{u}$ is a unit vector, the parameter t measures a length directly, and divided by the speed it gives the dwell time.

7.10. Problem set

7.10.1 Numerical exercises

Exercise 7.1. Determine the centre and radius of the circle with equation $x^2 + y^2 - 4x + 6y - 3 = 0$.

Solution. Here $D = -4$, $E = 6$, $F = -3$. Centre: $(-D/2, -E/2) = (2, -3)$. Discriminant:

$$\Delta = \frac{16}{4} + \frac{36}{4} - (-3) = 4 + 9 + 3 = 16 > 0,$$

hence (Theorem 7.1, case (a)) this is a real circle of radius $r = \sqrt{16} = 4$.

Check. Completing the squares: $(x - 2)^2 + (y + 3)^2 = 3 + 4 + 9 = 16$. ✓

Exercise 7.2. Classify, according to Theorem 7.1, the three equations

$$(i) \ x^2 + y^2 = 0, \quad (ii) \ x^2 + y^2 - 2x + 1 = 0, \quad (iii) \ x^2 + y^2 + 4 = 0.$$

Solution. (i) $D = E = F = 0$, hence $\Delta = 0$: case (b), a *single point*, the origin $(0, 0)$. It is the degenerate circle of zero radius.

(ii) $D = -2$, $E = 0$, $F = 1$: $\Delta = 1 + 0 - 1 = 0$: again case (b), the single point $(1, 0)$. Indeed the equation rewrites as $(x - 1)^2 + y^2 = 0$.

(iii) $D = E = 0$, $F = 4$: $\Delta = -4 < 0$: case (c), *empty set*. No real point, since $x^2 + y^2 \geq 0 > -4$.

Exercise 7.3. Determine the position of the line $3x - 4y + 20 = 0$ relative to the circle $x^2 + y^2 = 9$, and in case of intersection compute the length of the chord.

Solution. Distance of the origin from the line:

$$d = \frac{|3 \cdot 0 - 4 \cdot 0 + 20|}{\sqrt{9 + 16}} = \frac{20}{5} = 4.$$

Since $d = 4 > r = 3$, the line is **external** (Theorem 2.6): there are no intersections and no chord to compute.

Algebraic check. Solving for $y = (3x + 20)/4$ and substituting: $x^2 + (3x + 20)^2/16 = 9 \implies 25x^2 + 120x + 256 = 0$, with discriminant $120^2 - 4 \cdot 25 \cdot 256 = 14400 - 25600 = -11200 < 0$: no real solution, as predicted.

Exercise 7.4. Write the equations of the two tangents to the unit circle $x^2 + y^2 = 1$ drawn from the point $P_0(0, 2)$.

Solution. Non-vertical lines through P_0 have equation $y = mx + 2$, that is $q = 2$. By Theorem 7.3 with $r = 1$, tangency requires

$$q^2 = 1 + m^2 \implies 4 = 1 + m^2 \implies m^2 = 3 \implies m = \pm\sqrt{3}.$$

The tangents are therefore $y = \sqrt{3}x + 2$ and $y = -\sqrt{3}x + 2$.

Geometric consistency. $|OP_0| = 2 > 1$: the point is external, and there are two tangents, as Theorem 3.11 predicts. The length of the tangent segments is $\sqrt{|OP_0|^2 - r^2} = \sqrt{4 - 1} = \sqrt{3}$, and the angle between the two tangents is $2 \arcsin(1/2) = 60^\circ$. There are no vertical tangents from P_0 (the line $x = 0$ passes through the centre), so the limitation flagged after Theorem 7.3 loses no solutions in this case.

7.10.2 Symbolic exercises and proofs

Exercise 7.5. Prove that the line $y = mx + q$ is tangent to the unit circle centred at the origin if and only if $q^2 = 1 + m^2$, and interpret the result geometrically.

Solution. This is Theorem 7.3 with $r = 1$; the proof is the one given above (system, vanishing discriminant, simplification).

Interpretation. The condition $q^2 = 1 + m^2$ rewrites as $|q|/\sqrt{1 + m^2} = 1$, that is: *the distance of the line from the origin is exactly 1*. We recover the metric characterisation of the tangent (Theorem 2.6): this is no algebraic coincidence, it is the same statement in two languages. Note also that, with m fixed, the two values $q = \pm\sqrt{1 + m^2}$ give the two parallel tangents of that direction, symmetric about the centre.

Exercise 7.6. Prove analytically that the polar equation $\rho = 2a \cos \varphi$ (with $a > 0$) corresponds to a circle through the origin, and determine its centre and radius.

Solution. *Step 1.* Multiplying both sides by ρ (an operation that adds at most the solution $\rho = 0$, the pole, which we shall check does belong to the curve anyway):

$$\rho^2 = 2a\rho \cos \varphi.$$

Step 2. Using the conversion formulas $\rho^2 = x^2 + y^2$ and $x = \rho \cos \varphi$:

$$x^2 + y^2 = 2ax.$$

Step 3. Completing the square:

$$x^2 - 2ax + y^2 = 0 \implies (x - a)^2 + y^2 = a^2.$$

This is the circle with centre $(a, 0)$ and radius a . Substituting $x = y = 0$ gives $a^2 = a^2$: the origin does belong to the curve, so Step 1 introduced no spurious solution.

Consistency with the theory. The result is the case $\rho_0 = r = a$, $\varphi_0 = 0$ of Proposition 7.5: the circle passes through the pole and the equation reduces to $\rho = 2r \cos \varphi$. Observe further that the general equation is consistent with the reading of the coefficients (Remark 7.3): $x^2 + y^2 - 2ax = 0$ has $F = 0$, and indeed passes through the origin, and has $E = 0$, and indeed its centre lies on the x axis.

Exercise 7.7. Let $\gamma : x^2 + y^2 + Dx + Ey + F = 0$ be a real circle. Determine the equation of the image of γ under the homothety with centre the origin and ratio $k > 0$, and check that radius and centre scale as expected.

Solution. The homothety maps $P = (x, y)$ to $P' = (kx, ky)$; equivalently, a point (X, Y) belongs to the image if and only if $(X/k, Y/k)$ belongs to γ . Substituting:

$$\frac{X^2}{k^2} + \frac{Y^2}{k^2} + D\frac{X}{k} + E\frac{Y}{k} + F = 0.$$

Multiplying by k^2 :

$$X^2 + Y^2 + (kD)X + (kE)Y + k^2F = 0.$$

Hence $D' = kD$, $E' = kE$, $F' = k^2F$. Centre: $(-D'/2, -E'/2) = k(-D/2, -E/2)$: the centre has scaled by k , as it must. Radius:

$$r' = \sqrt{\frac{k^2D^2}{4} + \frac{k^2E^2}{4} - k^2F} = k\sqrt{\frac{D^2}{4} + \frac{E^2}{4} - F} = kr.$$

The radius scales by k : consistent with Step 3 of the proof of Theorem 6.2 (Chapter 6), where the same property was obtained synthetically.

Exercise 7.8. Prove that the radical axis of two non-concentric circles is always perpendicular to the line of centres, and discuss what happens if the two circles are concentric.

Solution. *Perpendicularity.* By Proposition 7.1 the radical axis has equation $(D_1 - D_2)x + (E_1 - E_2)y + (F_1 - F_2) = 0$, hence normal vector $\mathbf{n} = (D_1 - D_2, E_1 - E_2)$. The centres are $C_i = (-D_i/2, -E_i/2)$, so

$$C_1 - C_2 = \left(\frac{D_2 - D_1}{2}, \frac{E_2 - E_1}{2} \right) = -\frac{1}{2} \mathbf{n}.$$

The line of centres is therefore parallel to the normal of the radical axis, and hence perpendicular to the radical axis itself.

Concentric case. If $C_1 = C_2$ then $D_1 = D_2$ and $E_1 = E_2$, and the equation of the radical axis reduces to $F_1 - F_2 = 0$. Two subcases: if $F_1 \neq F_2$ (different radii) the equation is $0 = c \neq 0$, with no solutions: **the radical axis does not exist**, consistently with the fact that two distinct concentric circles have no point of equal power (the power differs everywhere by the constant $r_2^2 - r_1^2 \neq 0$). If instead $F_1 = F_2$ the two circles coincide and the equation $0 = 0$ is satisfied by the whole plane: every point trivially has equal power with respect to the same circle. Both subcases are consistent, but must be excluded from algorithms computing the radical axis, on pain of a division by zero.

Exercise 7.9. Determine centre and radius of the circle through $(1, 2)$, $(3, 4)$ and $(5, 0)$, and write its equation in general form.

Solution. The perpendicular bisector of $(1, 2)$ – $(3, 4)$ passes through the midpoint $(2, 3)$ and is perpendicular to a direction of slope 1: its equation is $y = 5 - x$. The perpendicular bisector of $(3, 4)$ – $(5, 0)$ passes through $(4, 2)$ and is perpendicular to a direction of slope -2 : its equation is $y = x/2$. Equating, $5 - x = x/2$ gives $x = 10/3$ and $y = 5/3$: the centre is $C = (10/3, 5/3)$. The radius is

$$R^2 = \left(\frac{10}{3} - 1 \right)^2 + \left(\frac{5}{3} - 2 \right)^2 = \frac{49}{9} + \frac{1}{9} = \frac{50}{9}, \quad R = \frac{\sqrt{50}}{3} = 2.3570.$$

Check with the third point: $(5 - 10/3)^2 + (0 - 5/3)^2 = 25/9 + 25/9 = 50/9 \checkmark$. The general equation is $x^2 + y^2 - \frac{20}{3}x - \frac{10}{3}y + \frac{25}{3} = 0$.

Exercise 7.10. Determine the tangents drawn from $P_0 = (6, 8)$ to the circle $x^2 + y^2 = 25$: the length of the tangent segment, the equation of the polar and the two slopes.

Solution. The power is $\text{pow}(P_0) = 36 + 64 - 25 = 75$, hence the tangent segment measures $\sqrt{75} = 8.6603$. The polar is $6x + 8y = 25$. The slopes solve $(x_0^2 - r^2)m^2 - 2x_0y_0m + (y_0^2 - r^2) = 0$, that is

$$11m^2 - 96m + 39 = 0 \implies m = \frac{96 \pm \sqrt{9216 - 1716}}{22} = \frac{96 \pm 86.6025}{22},$$

whence $m_1 = 8.3001$ and $m_2 = 0.42716$. The reduced discriminant is $48^2 - 11 \cdot 39 = 2304 - 429 = 1875 = 25 \cdot 75 = r^2 \text{pow}(P_0) \checkmark$, as the theory predicts.

Exercise 7.11. Given $x^2 + y^2 - 2x - 4y + 1 = 0$ and $x^2 + y^2 + 6x - 2y - 15 = 0$, determine centres, radii, mutual position and radical axis.

Solution. The first has centre $(1, 2)$ and radius $\sqrt{1 + 4 - 1} = 2$; the second centre $(-3, 1)$ and radius $\sqrt{9 + 1 + 15} = 5$. The distance of the centres is $\sqrt{16 + 1} = \sqrt{17} = 4.1231$, and since $|5 - 2| = 3 < 4.1231 < 7 = 5 + 2$ the two circles are secant. Subtracting the equations gives $-8x - 2y + 16 = 0$, that is $4x + y = 8$: this is the radical axis, which in this case coincides with the line of the common chord.

Exercise 7.12. A target starts from $P_0 = (-5, 3)$ and moves with unit vector $\mathbf{u} = (0.8, -0.6)$. The forbidden zone is the circle of centre the origin and radius $R = 2$. Determine whether it enters, and if so the entry and exit parameters and the length of the inside stretch.

Solution. The distance of the trajectory from the centre is $|-5 \cdot (-0.6) - 3 \cdot 0.8| = |3 - 2.4| = 0.6 < 2$: the target enters. The parameter of the foot of the perpendicular is $t^* = -P_0 \cdot \mathbf{u} = -[(-5)(0.8) + 3(-0.6)] = 5.8$, and the half-chord is $\sqrt{4 - 0.36} = 1.9079$. Hence

$$t_1 = 5.8 - 1.9079 = 3.8921, \quad t_2 = 5.8 + 1.9079 = 7.7079,$$

and the inside stretch measures $t_2 - t_1 = 3.8158$. Check: $P(t_1) = (-1.8863, 0.6647)$, of modulus 2.0000 $\checkmark$. Since $\mathbf{u}$ is a unit vector, t directly measures a length.

7.11. Self-assessment quiz

Ten multiple-choice questions. The answer key is at the end of the section.

1. The canonical form of the equation of a circle depends on
 - A) the radius and the coordinates of the centre
 - B) the radius only
 - C) the centre only
 - D) the orientation of the axes
2. The equation $x^2 + y^2 = 0$ represents
 - A) the empty set
 - B) a degenerate circle reduced to a point
 - C) a circle of radius 1
 - D) two lines
3. In the Cartesian plane the circle is singled out by imposing that the distance from the centre be
 - A) less than or equal to the radius, for every point
 - B) exactly equal to the radius, for every point
 - C) greater than the radius, for every point
 - D) constant but arbitrary

-
4. If the constant term is missing from the general equation ($F = 0$), the circle
- A) is degenerate
 - B) passes through the origin
 - C) is centred at the origin
 - D) has unit radius
5. A vanishing discriminant in the system of a line and a circle indicates
- A) an external line
 - B) a secant line
 - C) a tangent line
 - D) a line through the centre
6. The line $y = mx + q$ is tangent to the unit circle centred at the origin if and only if
- A) $q = 1 + m$
 - B) $q^2 = 1 + m^2$
 - C) $q^2 = 1 - m^2$
 - D) $q = m^2$
7. If a point $P(x, y)$ lies outside a circle centred at the origin with radius r , then
- A) $x^2 + y^2 < r^2$
 - B) $x^2 + y^2 = r^2$
 - C) $x^2 + y^2 > r^2$
 - D) $x + y > r$
8. The equation $2x^2 + 2y^2 - 4x + 8y - 6 = 0$
- A) is not a circle, because the quadratic coefficients equal 2
 - B) is a circle: it suffices to divide throughout by 2 before applying the formulas
 - C) is an ellipse
 - D) is a degenerate conic
9. The radical axis of two non-concentric circles is
- A) parallel to the line of centres
 - B) perpendicular to the line of centres
 - C) coincident with the line of centres
 - D) always tangent to both
10. The polar equation $\rho = 2a \cos \varphi$ represents
- A) a line
 - B) a circle centred at the pole
 - C) a circle through the pole
-

D) a spiral

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Answer key. 1: A — 2: B — 3: B — 4: B — 5: C — 6: B (Theorem 7.3) — 7: C — 8: B — 9: B — 10: C (Exercise 7.6).

Note on question 8. This is the commonest error of the chapter: the formulas $a = -D/2$, $b = -E/2$, $r = \sqrt{D^2/4 + E^2/4 - F}$ hold *only* if the coefficient of x^2 and y^2 is 1. Applying them directly to $2x^2 + 2y^2 + \dots$ gives a wrong result with no warning signal.

7.12. Chapter Formula sheet

Canonical equation (Definition 7.1)

$$(x - a)^2 + (y - b)^2 = r^2$$

General equation and classification (Theorem 7.1)

$$x^2 + y^2 + Dx + Ey + F = 0, \quad \Delta = \frac{D^2}{4} + \frac{E^2}{4} - F$$

$$\text{centre } \left(-\frac{D}{2}, -\frac{E}{2}\right), \quad r = \sqrt{\Delta} \ (\Delta > 0); \quad \Delta = 0 : \text{one point}; \quad \Delta < 0 : \emptyset$$

Homogeneous matrix form (Definition 7.2)

$$\begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & D/2 \\ 0 & 1 & E/2 \\ D/2 & E/2 & F \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0$$

Tangency (Theorems 7.3 and 7.4)

$$y = mx + q \text{ tangent to } x^2 + y^2 = r^2 \iff q^2 = r^2(1 + m^2)$$

$$xx_0 + yy_0 + \frac{D(x+x_0)}{2} + \frac{E(y+y_0)}{2} + F = 0 \quad (\text{splitting rule at } P_0 \in \gamma)$$

Power and position of a point (Theorem 7.6; Corollary 7.1)

$$\Pi(P) = \text{pow}_\gamma(P) = x_0^2 + y_0^2 + Dx_0 + Ey_0 + F, \quad t = \sqrt{\Pi(P)}$$

$$\Pi(P) < 0 \iff \text{interior}, \quad \Pi(P) = 0 \iff P \in \gamma, \quad \Pi(P) > 0 \iff \text{exterior}$$

Radical axis and radical centre (Proposition 7.1; Theorem 7.7)

$$(D_1 - D_2)x + (E_1 - E_2)y + (F_1 - F_2) = 0 \quad \perp \text{ to the line of centres}$$

Circle of Apollonius (Theorem 7.8), with $A = (-c, 0)$, $B = (c, 0)$, $k \neq 1$

$$C_k = \left(c \frac{k^2 + 1}{k^2 - 1}, 0\right), \quad r_k = \frac{2ck}{|k^2 - 1|}$$

Parametrisations (Propositions 7.4, 7.5 and 7.4)

$$\begin{cases} x = a + r \cos \theta \\ y = b + r \sin \theta \end{cases} \quad \rho^2 - 2\rho\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2 - r^2 = 0$$

$$\rho = 2r \cos(\varphi - \varphi_0) \ (\rho_0 = r), \quad |z - z_0| = r \iff z(\theta) = z_0 + r e^{i\theta}$$

Sources and bibliography

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CHAPTER 8

Trigonometry and the Unit Circle

INFO

Aim of the chapter. To define sine, cosine and tangent on the unit circle, prove the identities that govern them, and establish the result that opens the door to analysis: the **fundamental limit** $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

The delicate point. That limit is proved by comparing the area of a circular sector with those of two triangles. This is legitimate only because the sector area was obtained, in Chapter 6, *without* using sine and cosine (Theorem 6.7). Section 8.5.1 checks explicitly that the chain does not close on itself.

8.1. The unit circle as the foundation of trigonometry

Definition 8.1: The unit circle. The **unit circle** is the circle $C(O, 1)$: $x^2 + y^2 = 1$. For every $\theta \in \mathbb{R}$, one defines $P(\theta) = (\cos \theta, \sin \theta)$ as the point reached by travelling, starting from $(1, 0)$, an oriented arc of length $|\theta|$ (anticlockwise if $\theta > 0$). This is the *geometric* definition of sine and cosine, extended to every $\theta \in \mathbb{R}$ (not only to the angles of a right triangle, as in elementary school trigonometry).

Remark 8.1: The definition is well posed thanks to Chapter 6. Speaking of an “arc of length $|\theta|$ ” makes sense only because the circle is rectifiable (Theorem 6.1) and because arc length is additive and continuous: to every value of $\theta \in [0, 2\pi)$ there corresponds a unique point, and the correspondence is bijective. Without the work of Chapter 6, Definition 8.1 would be a suggestive description with no foundation.

Remark 8.2: From the right triangle to the circle: didactic continuity. For $\theta \in (0, \pi/2)$, $P(\theta) = (\cos \theta, \sin \theta)$ coincides with adjacent leg/hypotenuse and opposite leg/hypotenuse of an inscribed right triangle with hypotenuse = 1 (the radius). The unit-circle definition is therefore a **conservative extension** of right-triangle trigonometry to all real angles, including multiple rotations by 2π .

Theorem 8.1: The tangent as a segment on the line $x = 1$. For $\theta \in (-\pi/2, \pi/2)$, the ray $OP(\theta)$ meets the line $x = 1$ (tangent to the unit circle at the point $A(1, 0)$, by Theorem 3.9) at the point $T = (1, \tan \theta)$, where $\tan \theta = \sin \theta / \cos \theta$.

Proof. The points of the ray $OP(\theta)$ are $t(\cos \theta, \sin \theta)$ with $t \geq 0$. Intersection with $x = 1$ requires $t \cos \theta = 1$, that is $t = 1/\cos \theta$, well defined and positive because $\cos \theta > 0$ for $\theta \in (-\pi/2, \pi/2)$. The corresponding ordinate is

$$t \sin \theta = \frac{\sin \theta}{\cos \theta} = \tan \theta.$$

The line $x = 1$ is tangent to the circle at $A(1, 0)$ because it is perpendicular to the radius OA

| at A (Theorem 3.9). □

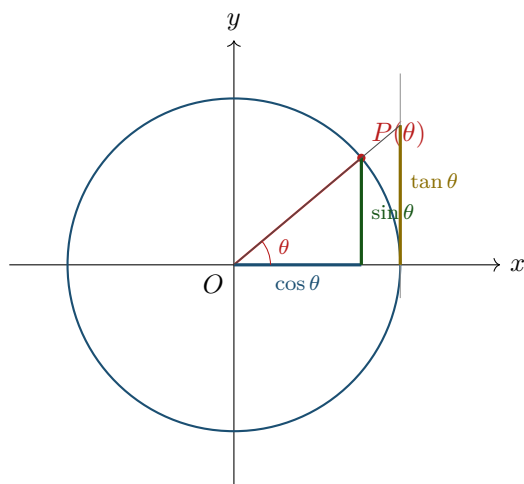


Figure 8.1: — Sine, cosine and tangent as coordinates on the unit circle; the tangent as a segment on the line $x = 1$

8.1.1 Notable values

θ ($^\circ$)	θ (rad)	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	0	1	0
30°	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	$\pi/2$	1	0	undefined
120°	$2\pi/3$	$\sqrt{3}/2$	$-1/2$	$-\sqrt{3}$
180°	π	0	-1	0
270°	$3\pi/2$	-1	0	undefined
360°	2π	0	1	0

Proposition 8.1: Notable values. The values of $\sin, \cos, \tan$ at $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$ are those of the table above.

Proof. 0° and 90° . From the definition on the unit circle, $P(0^\circ) = (1, 0)$ and $P(90^\circ) = (0, 1)$: hence $\sin 0^\circ = 0, \cos 0^\circ = 1$ and $\sin 90^\circ = 1, \cos 90^\circ = 0$.

45° . In the isosceles right triangle with legs 1 and hypotenuse h , by Pythagoras $h = \sqrt{2}$; the acute angles are equal (base angles of an isosceles triangle) and sum to 90° (acute angles of a right triangle), hence each equals 45° . Placing the triangle with the 45° angle at the origin and hypotenuse along the radius of the unit circle, the legs (of length $1/\sqrt{2}$ after scaling the hypotenuse to 1) give $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2} = \sqrt{2}/2$.

30° and 60° . In the equilateral triangle of side 1, the altitude (which is also the median and bisector, by (A4) of Chapter 2) divides it into two right triangles with angles $30^\circ, 60^\circ, 90^\circ$, hypotenuse 1, shorter leg $1/2$ (half the side) and longer leg $h = \sqrt{1 - 1/4} = \sqrt{3}/2$ (Pythagoras). Hence $\sin 30^\circ = \cos 60^\circ = 1/2$ and $\cos 30^\circ = \sin 60^\circ = \sqrt{3}/2$.

The values of $\tan \theta = \sin \theta / \cos \theta$ (where $\cos \theta \neq 0$) and the values at $120^\circ, 180^\circ, 270^\circ, 360^\circ$ (by parity/periodicity and symmetries of the circle, Definition 8.1) follow immediately. □

8.2. The fundamental identity and inverse formulas

Theorem 8.2: Pythagorean identity. $\sin^2 \theta + \cos^2 \theta = 1$ for every $\theta \in \mathbb{R}$.

Proof. By Definition 8.1, the point $P(\theta) = (\cos \theta, \sin \theta)$ belongs to the unit circle, whose equation is $x^2 + y^2 = 1$ (Definition 7.1 with centre the origin and radius 1, Chapter 7). Substituting the coordinates of $P(\theta)$ gives the identity directly. $\square$

Remark 8.3: It is the Pythagorean theorem, and nothing else — along this route. Along the route just followed, the fundamental identity is not a new fact: it is the Pythagorean theorem applied to the right triangle with hypotenuse 1 and legs $|\cos \theta|$ and $|\sin \theta|$. The chain is: Pythagoras $\Rightarrow$ distance formula $\Rightarrow$ equation of the circle $\Rightarrow$ fundamental identity. Every link is proved elsewhere in this volume (Remark 7.1, Chapter 7).

But this is not the only route. Theorem 8.3 below proves the same identity without using Pythagoras at any point: it is Jason Zimba's result (2009), which refuted a century-old belief.

8.3. The Pythagorean identity without Pythagoras (Zimba, 2009)

In 1940, in his repertory of more than 370 proofs of the Pythagorean theorem, Elisha Loomis wrote a sentence that long went unchallenged:

*“There are no trigonometric proofs [of the Pythagorean theorem], because all of the fundamental formulae of trigonometry are themselves based upon the truth of the Pythagorean theorem; because of this theorem we say $\sin^2 A + \cos^2 A = 1$.” — E.S. Loomis, *The Pythagorean Proposition*, p. 244*

The claim is **false**, and the following section shows so constructively.

Remark 8.4: What is really needed to define sine and cosine. The definitions of $\sin$ and $\cos$ for acute angles require three facts, and none of them is the Pythagorean theorem:

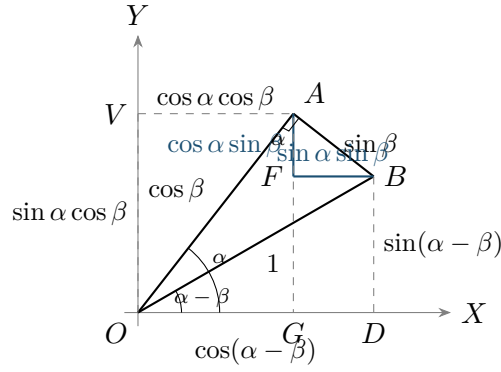
1. **Euclid I.32** (interior angles sum to π): it guarantees that a right triangle with an angle α has angles $\frac{\pi}{2}$, $\frac{\pi}{2} - \alpha$, α , hence that two right triangles with the same α are equiangular;
2. **Euclid I.19** (the greater side is opposite the greater angle): it guarantees that the hypotenuse is the longest side, hence that the ratios below lie in $(0, 1)$;
3. **Euclid VI.4** (equiangular triangles have proportional sides): it guarantees that the ratios A/H and O/H (adjacent over hypotenuse, opposite over hypotenuse) do not depend on the particular triangle chosen, but only on α .

It is only thanks to (3) that setting $\cos \alpha := A/H$ and $\sin \alpha := O/H$ makes sense. None of the three results uses I.47, which is the Pythagorean theorem.

Lemma 8.1: Subtraction formulas, purely constructively. For all $\alpha, \beta \in (0, \frac{\pi}{2})$ with $\alpha - \beta$ also in $(0, \frac{\pi}{2})$:

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta, \quad \sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

The proof that follows does not use the Pythagorean theorem.



Proof. Step 1 — the construction. From the origin O draw the ray making the angle $\alpha - \beta$ with the X -axis, and take B on it with $|OB| = 1$. From the same origin draw the ray making the angle α , and take A on it so that $\angle OAB$ is right.

Step 2 — the sides of triangle OAB . Triangle OAB is right-angled at A , has hypotenuse $|OB| = 1$ and angle at O equal to $\alpha - (\alpha - \beta) = \beta$. By the *definition* of sine and cosine (Remark 8.4):

$$|OA| = \cos \beta, \quad |AB| = \sin \beta.$$

Note carefully: these two lengths have **not** been computed with Pythagoras, but read off from the definition of the trigonometric ratios.

Step 3 — the coordinates of A . Let G be the foot of the perpendicular from A to the X -axis. Triangle OGA is right-angled at G with hypotenuse $|OA| = \cos \beta$ and angle α at O , so again by definition

$$|OG| = \cos \alpha \cos \beta, \quad |GA| = \sin \alpha \cos \beta.$$

Step 4 — the small triangle AFB . Let F be the point with coordinates (x_A, y_B) , so that AF is vertical and FB horizontal. The segment AB is perpendicular to OA , and OA makes the angle α with the horizontal: hence AB makes the angle α with the *vertical*, i.e. $\angle FAB = \alpha$. Triangle AFB is right-angled at F with hypotenuse $|AB| = \sin \beta$, and by definition

$$|AF| = \cos \alpha \sin \beta, \quad |FB| = \sin \alpha \sin \beta.$$

Step 5 — the coordinates of B , read in two ways. On the one hand, B lies on the ray of angle $\alpha - \beta$ at distance 1, so

$$B = (\cos(\alpha - \beta), \sin(\alpha - \beta)).$$

On the other hand, walking the polyline $O \rightarrow G \rightarrow A \rightarrow F \rightarrow B$:

$$x_B = |OG| + |FB| = \cos \alpha \cos \beta + \sin \alpha \sin \beta,$$

$$y_B = |GA| - |AF| = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

Equating the two readings gives exactly the two formulas of the statement.

Non-circularity check. Three hypotenuses appear in the figure — OA , OB , AB — but **none** of their lengths has been computed with Pythagoras: $|OB| = 1$ is a choice, and $|OA|$, $|AB|$ come from the definition of $\cos$ and $\sin$. All the other segments are horizontal or vertical legs, likewise obtained by definition. The construction uses only the three Euclidean results of Remark 8.4. $\square$

Theorem 8.3: The Pythagorean identity, independently (Zimba, 2009). The identity $\cos^2 x + \sin^2 x = 1$, for $x \in (0, \frac{\pi}{2})$, can be proved without using the Pythagorean theorem.

Proof. The trap to avoid. One is tempted to set $\alpha = \beta = x$ in the first formula of Lemma 8.1 and use $\cos 0 = 1$. **This is not legitimate:** the functions have been defined only on the *open* interval $(0, \frac{\pi}{2})$, which does not contain zero — and defining $\cos 0$ would mean speaking of a “right triangle with a zero angle”, which does not exist. (This is exactly the error J. Versluys made in 1914, attributing the idea to Schur.)

Zimba’s move. The obstacle is bypassed by introducing a second auxiliary angle. Let $x \in (0, \frac{\pi}{2})$ and choose any y with

$$0 < y < x < \frac{\pi}{2}.$$

Then x , y and $x - y$ all belong to $(0, \frac{\pi}{2})$, and Lemma 8.1 applies to every pair. We compute:

$$\begin{aligned} \cos y &= \cos(x - (x - y)) \\ &= \cos x \cos(x - y) + \sin x \sin(x - y) && \text{(subtraction formula)} \\ &= \cos x (\cos x \cos y + \sin x \sin y) + \sin x (\sin x \cos y - \cos x \sin y) && \text{(again, on } x - y) \\ &= (\cos^2 x + \sin^2 x) \cos y, \end{aligned}$$

since the two terms in $\sin y$ are opposite and cancel.

Conclusion. As $y \in (0, \frac{\pi}{2})$, we have $\cos y > 0$ (Remark 8.4, point 2: the ratios are strictly positive), so we may divide:

$$\cos^2 x + \sin^2 x = 1.$$

The result does not depend on y , which was arbitrary. □

Corollary 8.1: The Pythagorean theorem as a corollary. *The Pythagorean theorem follows from Theorem 8.3 without petitio principii.*

Proof. Let a right triangle be given with legs a, b and hypotenuse c , and let x be the angle opposite a (acute, since the angles sum to π and one of them is right). By definition $\sin x = a/c$ and $\cos x = b/c$. Theorem 8.3 gives

$$\left(\frac{b}{c}\right)^2 + \left(\frac{a}{c}\right)^2 = 1,$$

and multiplying by $c^2 > 0$: $a^2 + b^2 = c^2$. □

BEST PRACTICE

What this changes for the architecture of this volume. Up to this section the treatise followed a single chain, Pythagoras $\rightarrow$ distance $\rightarrow$ equation of the circle $\rightarrow$ fundamental identity. Theorem 8.3 supplies a **second, independent** one: similarity (Euclid VI.4) $\rightarrow$ definition of $\sin, \cos$ $\rightarrow$ subtraction formulas $\rightarrow$ fundamental identity $\rightarrow$ Pythagoras. The two chains meet only at their endpoints.

Practical consequence: the equivalence of Corollary 10.1 (Chapter 10) will, once we meet it there, be *travellable in both directions* without circularity, provided that in the ($\Leftarrow$) direction one uses Zimba’s derivation rather than the analytic one.

Warning. What remains circular anyway. Zimba’s result does not license everything. The following remain forbidden as proofs of Pythagoras:

- the Taylor series of $\sin$ and $\cos$, whose derivation requires $(\sin x)' = \cos x$, which requires the fundamental limit (Theorem 8.5), which requires the area of the sector (Chapter 6), which requires $A = \pi r^2$;
- l’Hôpital’s rule applied to $\sin x/x$, for the same reason;
- the definition of $\sin$ and $\cos$ through $e^{i\theta}$, which presupposes the metric structure of $\mathbb{C}$,

i.e. the modulus, i.e. Pythagoras.

What Zimba makes legitimate is exactly and only the *synthetic-trigonometric* route: ratios between sides of similar triangles, and nothing else.

Remark 8.5: The sequel: Johnson and Jackson, 2024. In 2022 two high-school students from New Orleans, **Ne’Kiya Jackson** and **Calcea Johnson**, presented at a school competition a trigonometric proof of the Pythagorean theorem obtained by a route different from Zimba’s: a geometric construction (which the authors call the *waffle cone*) generating an infinite sequence of similar triangles, whose sum is a convergent geometric series, followed by an application of the **law of sines** — which, as we shall see in Chapter 10, follows from Thales’ theorem and not from Pythagoras. The work, expanded to five proofs plus a general method for producing at least five more, was published with peer review in the *American Mathematical Monthly* **131**(9), November 2024, pp. 739–752.

A note on expository honesty: the original construction is not reproduced here, because it requires a figure and a convergence analysis that would weigh the chapter down without adding anything to the logical point — already settled by Theorem 8.3 — that a non-circular trigonometric proof is possible. The interested reader will find the article in the bibliography.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin \theta = \pm \sqrt{1 - \cos^2 \theta} \quad (\text{sign according to the quadrant})$$

$$\cos \theta = \pm \sqrt{1 - \sin^2 \theta} \quad (\text{sign according to the quadrant})$$

$$\tan \theta = \sin \theta / \cos \theta, \quad \cos \theta \neq 0$$

8.4. Addition, duplication and half-angle formulas

Theorem 8.4: Addition formulas.

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta, \quad \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

Proof. Step 1 — the subtraction formula, for arbitrary angles, from distance on the unit circle. Let $P(\theta) = (\cos \theta, \sin \theta)$ be the points of the unit circle. The triangles $OP(\alpha)P(\beta)$ and $OP(\alpha - \beta)P(0)$ have two sides equal to 1 (radii) and the included angle at O equal (in both, the arc between the two points measures $\alpha - \beta$ modulo 2π , regardless of which quadrant α and β fall in): they are congruent by the SAS criterion (Chapter 2), hence the opposite sides are congruent:

$$|P(\alpha) - P(\beta)|^2 = |P(\alpha - \beta) - P(0)|^2.$$

Expanding both sides with $\sin^2 + \cos^2 = 1$ (Theorem 8.2):

$$2 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = 2 - 2 \cos(\alpha - \beta),$$

that is

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta. \quad (8.1)$$

This identity holds for **arbitrary** angles α, β : the SAS congruence does not depend on which quadrant contains them.

Step 2 — from here, $\cos(\alpha + \beta)$. Putting $-\beta$ in place of β in (8.1), and using $\cos(-\beta) = \cos \beta$, $\sin(-\beta) = -\sin \beta$ (parity of cosine and sine on the unit circle):

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

Step 3 — the cofunction identities. Putting $\alpha = \pi/2$ in (8.1): $\cos(\pi/2 - \beta) = \cos(\pi/2)\cos\beta + \sin(\pi/2)\sin\beta = \sin\beta$. Applying this same identity to the angle $\pi/2 - x$ in place of β : $\cos(\pi/2 - (\pi/2 - x)) = \cos x = \sin(\pi/2 - x)$, that is $\sin(\pi/2 - x) = \cos x$.

Step 4 — $\sin(\alpha + \beta)$. By Step 3 and (8.1) applied to the pair of angles $(\pi/2 - \alpha, \beta)$:

$$\sin(\alpha + \beta) = \cos(\pi/2 - (\alpha + \beta)) = \cos((\pi/2 - \alpha) - \beta) = \cos(\pi/2 - \alpha)\cos\beta + \sin(\pi/2 - \alpha)\sin\beta = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

having used $\cos(\pi/2 - \alpha) = \sin\alpha$ and $\sin(\pi/2 - \alpha) = \cos\alpha$ (Step 3). $\square$

Proof via Euler's formula (Volume II). Anyone already familiar with the complex exponential can obtain the same formulas in three lines: from $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$,

$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta) = (\cos\alpha\cos\beta - \sin\alpha\sin\beta) + i(\sin\alpha\cos\beta + \cos\alpha\sin\beta),$$

and separating real and imaginary parts recovers the formulas. It is quick but *expensive* in prerequisites: Euler's formula presupposes the power series of $\sin$, $\cos$ and $\exp$, hence differential calculus — developed in full only in Volume II — whereas the main proof above uses only SAS and Pythagoras, already available in this volume. $\square$

Elementary proof, from the subtraction formulas (restricted case). A third route, independent of both the main proof and Euler's: it starts from Zimba's lemma on subtraction (Lemma 8.1, proved above by purely synthetic means) instead of from SAS congruence. It is valid only for $\alpha, \beta > 0$ with $\alpha + \beta < \frac{\pi}{2}$, but it is a useful cross-check: two entirely different routes (SAS on the unit circle, and Zimba's reading by projections) that must give the same formula.

Step 1 — apply subtraction to two suitable angles. Set $\alpha' = \alpha + \beta$ and $\beta' = \beta$: then $\alpha' - \beta' = \alpha$, and all three angles lie in $(0, \frac{\pi}{2})$, so Lemma 8.1 applies. It gives

$$\cos\alpha = \cos(\alpha + \beta)\cos\beta + \sin(\alpha + \beta)\sin\beta,$$

$$\sin\alpha = \sin(\alpha + \beta)\cos\beta - \cos(\alpha + \beta)\sin\beta.$$

Step 2 — it is a linear system. With unknowns $X = \cos(\alpha + \beta)$ and $Y = \sin(\alpha + \beta)$, the two relations read

$$\begin{pmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix}.$$

Step 3 — the matrix is invertible. Its determinant is $\cos^2\beta + \sin^2\beta$, which by Theorem 8.3 — proved *without* Pythagoras and without differential calculus — equals $1 \neq 0$. The inverse is therefore the transpose:

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix} \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix} = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta \end{pmatrix},$$

which are exactly the two addition formulas.

Balance of dependencies. This route uses only: Euclid I.32, I.19, VI.4 (to define $\sin$ and $\cos$), the construction of Lemma 8.1, and 2×2 linear algebra. In particular it uses **neither** Pythagoras, nor Taylor series, nor $e^{i\theta}$: this is why it is preferred when the addition formulas must serve to prove Pythagoras itself (Proof 7 of Chapter 10). $\square$

Sixth proof — by linearity of the rotation. The rotation R_α about O is an isometry that fixes O and maps parallelograms to parallelograms (isometries preserve distances and

hence the congruence of the triangles that defines the parallelogram): it is therefore a **linear** map of $\mathbb{R}^2$ to itself. Additivity $R_\alpha(\vec{u} + \vec{v}) = R_\alpha\vec{u} + R_\alpha\vec{v}$ follows from the parallelogram rule (an isometry that fixes the origin and preserves parallelograms sends the diagonal $\vec{u} + \vec{v}$ to the diagonal $R_\alpha\vec{u} + R_\alpha\vec{v}$). Homogeneity $R_\alpha(x\vec{u}) = xR_\alpha\vec{u}$ for $x \in \mathbb{R}$ follows from an even more elementary fact: an isometry that fixes O sends the line through O and $\vec{u}$ to the line through O and $R_\alpha\vec{u}$ (lines through the origin are characterised as sets of points at proportional distance from O along a fixed direction, and isometries preserve distances), and it preserves lengths along it; since R_α is continuous (it is a rotation, not a discontinuous swap of rays), the sign stays constant as x varies, hence $R_\alpha(x\vec{u}) = xR_\alpha\vec{u}$ for every real x , not only for rational x . On the standard basis vectors, $R_\alpha\vec{e}_1 = (\cos \alpha, \sin \alpha)$ and $R_\alpha\vec{e}_2 = (-\sin \alpha, \cos \alpha)$ (the second is the first rotated by 90° , by the definition of rotation). Hence

$$R_\alpha(x, y) = x(\cos \alpha, \sin \alpha) + y(-\sin \alpha, \cos \alpha) = (x \cos \alpha - y \sin \alpha, x \sin \alpha + y \cos \alpha).$$

Applying R_α to the point $P(\beta) = (\cos \beta, \sin \beta)$ gives, on one hand, $P(\alpha + \beta) = (\cos(\alpha + \beta), \sin(\alpha + \beta))$ (rotating by α the point at angle β gives the point at angle $\alpha + \beta$); on the other hand, by the formula just found with $(x, y) = (\cos \beta, \sin \beta)$,

$$R_\alpha P(\beta) = (\cos \alpha \cos \beta - \sin \alpha \sin \beta, \sin \alpha \cos \beta + \cos \alpha \sin \beta).$$

Comparing the two expressions recovers the addition formulas.

Note. The 2×2 matrix of the previous proof, $\begin{pmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{pmatrix}$, is exactly $R_{-\beta}$ (compare with the matrix of R_α written above, with $\alpha \rightarrow -\beta$ and $\cos(-\beta) = \cos \beta$, $\sin(-\beta) = -\sin \beta$): the two routes, apparently independent, in fact use the same linear structure, read from two different angles. $\square$

Alternative proof — classical, by Ptolemy's theorem. This is, historically, the route followed by Ptolemy himself to build his table of chords in the *Almagest* (Book I), many centuries before complex exponential notation. Consider a circle of **unit diameter** $AC = 1$, and two points B, D on the circle, on opposite sides of the diameter AC , such that $\angle BAC = \alpha$ and $\angle CAD = \beta$ (with $0 < \alpha, \beta$ and $\alpha + \beta < 90^\circ$, to keep an easily drawable configuration; the general case then follows by continuity/periodicity, as noted below). Since AC is a diameter, by Thales' theorem (Theorem 4.4, Chapter 4) the angles $\angle ABC$ and $\angle ADC$ are right, hence in the right triangles ABC and ACD (hypotenuse $AC = 1$):

$$AB = \cos \alpha, \quad BC = \sin \alpha, \quad AD = \cos \beta, \quad CD = \sin \beta.$$

The quadrilateral $ABCD$ (in this cyclic order around the circle, B, D being on opposite sides of AC) has diagonals $AC = 1$ and BD . By **Ptolemy's theorem** (Theorem 5.9, Chapter 5):

$$AC \cdot BD = AB \cdot CD + BC \cdot DA \implies BD = \cos \alpha \sin \beta + \sin \alpha \cos \beta.$$

On the other hand, $\angle BAD = \angle BAC + \angle CAD = \alpha + \beta$ (B, D being on opposite sides, the two angles add). To compute BD directly (without invoking the general law of sines, not yet available at this point of the treatise), draw the diameter BB' through B (with B' antipodal to B , $BB' = 1$): by Thales' theorem (Theorem 4.4, Chapter 4) the angle $\angle BDB'$ is right, so in the right triangle BDB' (hypotenuse $BB' = 1$) we have $BD = BB' \sin(\angle BB'D) = \sin(\angle BB'D)$. The angles $\angle BB'D$ and $\angle BAD$ are both inscribed angles standing on the same arc BD (on the same side, A and B' lying on the same major arc), hence equal by the corollary on inscribed angles on the same arc (Corollary 4.1, Chapter 4): $\angle BB'D = \angle BAD = \alpha + \beta$, and therefore $BD = \sin(\angle BAD) = \sin(\alpha + \beta)$. Equating the

two expressions for BD :

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta,$$

which is the addition formula for the sine, obtained without complex numbers.

Warning. Non-circularity check. For this alternative proof to be truly independent of the main one, it must not covertly rely on the addition formulas already proved. Let us check link by link: (i) Ptolemy's theorem (Theorem 5.9, Chapter 5) is proved there by triangle similarity, without trigonometry; (ii) the computation of BD used above descends directly from Thales' theorem and from the corollary on inscribed angles on the same arc, not from the addition formulas; (iii) it remains to prove the co-function identity $\cos x = \sin(90^\circ - x)$ **without** using them: we do so below, directly from the unit circle.

The cosine formula then follows elementarily from the co-function identity $\cos x = \sin(90^\circ - x)$, proved as follows: the point $P(x) = (\cos x, \sin x)$ of the unit circle and the point $P(90^\circ - x) = (\cos(90^\circ - x), \sin(90^\circ - x))$ are related by reflection across the bisector $y = x$, which swaps the coordinates $(u, v) \mapsto (v, u)$: indeed, reflecting the point at x degrees amounts to swapping the angle with its complement with respect to 90° (the two rays $OP(x)$ and $OP(90^\circ - x)$ are symmetric with respect to the bisector of the first quadrant, by the definition of complementary angle), and the reflection $(u, v) \mapsto (v, u)$ sends $P(x) = (\cos x, \sin x)$ to $(\sin x, \cos x)$; this image point must therefore coincide with $P(90^\circ - x) = (\cos(90^\circ - x), \sin(90^\circ - x))$, whence $\cos(90^\circ - x) = \sin x$ and $\sin(90^\circ - x) = \cos x$ by comparing coordinates. Using this and the **oddness/evenness** of sine and cosine (symmetry properties of the unit circle, immediate from Definition 8.1: $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$):

$$\begin{aligned} \cos(\alpha + \beta) &= \sin(90^\circ - (\alpha + \beta)) = \sin((90^\circ - \alpha) - \beta) \\ &= \sin(90^\circ - \alpha) \cos \beta - \cos(90^\circ - \alpha) \sin \beta \\ &= \cos \alpha \cos \beta - \sin \alpha \sin \beta, \end{aligned}$$

where the sine formula just proved was used in the form $\sin(X - Y)$ (which follows from $\sin(X + (-Y))$ and the oddness of $\sin$, with no need for a second independent geometric construction). $\square$

Remark 8.6: On the range of validity of the construction. The geometric construction described above is drawn explicitly for $0 < \alpha, \beta$ and $\alpha + \beta < 90^\circ$ (to guarantee that B and D are distinct, well separated points on the same semicircle relative to A). The trigonometric identities obtained are, however, **algebraic** identities between functions defined on all of $\mathbb{R}$ (Definition 8.1): once established on a non-empty open interval of pairs (α, β) , they extend to all real values by continuity and periodicity of the functions involved — exactly as Ptolemy himself proceeded, tabulating chords for the range of angles useful to astronomy and deducing the others by symmetry.

Corollary 8.2: Duplication formulas. $\sin 2\theta = 2 \sin \theta \cos \theta$; $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$.

Proof. Set $\alpha = \beta = \theta$ in Theorem 8.4 and use the Pythagorean identity for the equivalent forms of the double cosine. $\square$

Corollary 8.3: Half-angle formulas (inverses of duplication).

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}.$$

Proof. From $\cos \theta = 1 - 2 \sin^2(\theta/2) = 2 \cos^2(\theta/2) - 1$ (Corollary 8.2 applied to $\theta/2$), isolate $\sin^2(\theta/2)$ and $\cos^2(\theta/2)$ and take the square root. $\square$

Remark 8.7: Choosing the sign. Extracting the square root gives only the absolute value: the sign of $\sin(\theta/2)$ and $\cos(\theta/2)$ must be determined separately, based on the quadrant in which $\theta/2$ (not θ) falls. For example, if $\theta \in (0^\circ, 180^\circ)$ then $\theta/2 \in (0^\circ, 90^\circ)$, first quadrant: both signs are $+$. If instead $\theta \in (180^\circ, 360^\circ)$, then $\theta/2 \in (90^\circ, 180^\circ)$, second quadrant: $\sin(\theta/2) > 0$ but $\cos(\theta/2) < 0$. The choice of sign is therefore not a property of the formula, but must be made case by case by looking at where $\theta/2$ falls.

8.5. The fundamental limit $\sin x/x$

It is the bridge between the geometry of the circle and analysis: from it follow the derivative of the sine, the series expansions, and with them the whole analytic treatment of Volume II.

Theorem 8.5: Fundamental limit.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,$$

where x is measured in radians.

Proof. Step 1 — three nested regions. Let $0 < x < \pi/2$. On the unit circle put $A = (1, 0)$, $P = P(x) = (\cos x, \sin x)$ and $T = (1, \tan x)$, the last being well defined by Theorem 8.1. Consider:

- the triangle OAP ;
- the circular sector OAP , of radius 1 and central angle x ;
- the triangle OAT .

The inclusion

$$\triangle OAP \subseteq \text{sector } OAP \subseteq \triangle OAT$$

holds. *First inclusion:* the sector is the intersection of the unit disc with the convex angle of vertex O and measure $x < \pi/2$, hence it is convex; containing the three vertices O, A, P , it contains the triangle they determine. *Second inclusion:* in polar coordinates the sector is $\{(\rho, \varphi) : 0 \leq \rho \leq 1, 0 \leq \varphi \leq x\}$, and each of its points has abscissa $\rho \cos \varphi \leq 1$; the triangle OAT is instead $\{(\rho, \varphi) : 0 \leq \varphi \leq x, \rho \cos \varphi \leq 1\}$. The inclusion is therefore immediate.

Step 2 — the three areas. By (Ar4) (Remark 6.1, Chapter 6), the triangle OAP has base $OA = 1$ and height equal to the ordinate of P , that is $\sin x$:

$$\text{area}(\triangle OAP) = \frac{1}{2} \sin x.$$

By Theorem 6.7 (Chapter 6), with $r = 1$:

$$\text{area}(\text{sector}) = \frac{1}{2} \cdot 1^2 \cdot x = \frac{x}{2}.$$

The triangle OAT has base $OA = 1$ and height $\tan x$:

$$\text{area}(\triangle OAT) = \frac{1}{2} \tan x = \frac{\sin x}{2 \cos x}.$$

Step 3 — the chain of inequalities. By the monotonicity of area (Ar2) applied to the inclusions of Step 1:

$$\frac{\sin x}{2} < \frac{x}{2} < \frac{\sin x}{2 \cos x}.$$

The inequalities are *strict* because the inclusions are proper: the sector contains the circular segment between the chord AP and the arc, of positive area, and the triangle OAT contains the region between the arc and the segment AT , also of positive area.

Step 4 — isolating the ratio. For $0 < x < \pi/2$ one has $\sin x > 0$; dividing all members by $\frac{\sin x}{2} > 0$ (an operation preserving the direction):

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}.$$

All three members are positive, so taking reciprocals (which reverses the direction):

$$\cos x < \frac{\sin x}{x} < 1.$$

Step 5 — continuity of the cosine at 0. We need $\lim_{x \rightarrow 0} \cos x = 1$. This is obtained without circularity: by Corollary 8.2, $1 - \cos x = 2 \sin^2(x/2)$; and from Step 3 (first inequality, valid also in the weak form for $x \geq 0$) it follows that $0 \leq \sin t \leq t$ for $0 \leq t < \pi/2$. Hence

$$0 \leq 1 - \cos x = 2 \sin^2 \frac{x}{2} \leq 2 \left(\frac{x}{2} \right)^2 = \frac{x^2}{2} \xrightarrow{x \rightarrow 0} 0,$$

and by the squeeze theorem $\cos x \rightarrow 1$.

Step 6 — squeeze theorem. From Steps 4 and 5, as $x \rightarrow 0^+$ the function $\frac{\sin x}{x}$ lies between two functions both tending to 1: hence $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$.

Step 7 — left limit, by parity. The function $x \mapsto \frac{\sin x}{x}$ is **even**, because

$$\frac{\sin(-x)}{-x} = \frac{-\sin x}{-x} = \frac{\sin x}{x},$$

using the oddness of the sine. The left limit therefore coincides with the right one, and the two-sided limit exists and equals 1. $\square$

Alternative proof — by comparing lengths. The same double inequality $\cos x < \sin x/x < 1$ (for $0 < x < \pi/2$) is obtained by comparing lengths instead of areas.

(i) $\sin x < x$. $\sin x$ is the distance of $P(x) = (\cos x, \sin x)$ from the horizontal axis, hence $\sin x \leq |P(x)P(0)|$ (a leg is shorter than the hypotenuse in the right triangle with vertices $P(x)$, $(\cos x, 0)$, $P(0)$; if $\sin x = 0$ the inequality is trivial). The chord $|P(x)P(0)|$ is in turn shorter than the arc of length x joining the same two points (Corollary 6.1, Chapter 6). Composing, $\sin x < x$.

(ii) $x < \tan x$. Let $T' = (1, \tan \frac{x}{2})$ be the intersection point of the tangents to the unit circle at $P(0) = (1, 0)$ and at $P(x)$ (it exists for $0 < x < \pi/2$). The circular sector with vertex O and angle x is a convex set contained in the convex quadrilateral $OP(0)T'P(x)$ (the two tangents bound a neighbourhood of the sector). By Archimedes' postulate on the perimeters of nested convex sets (Theorem 6.4, Chapter 6 — which nowhere uses the fundamental limit, being proved with the triangle inequality alone: no circularity), the perimeter of the sector (the arc x plus the two radii, in all $2 + x$) does not exceed that of the circumscribed quadrilateral ($2 \cdot \tan \frac{x}{2}$ for the two tangent segments from T' , equal by Theorem 3.11, plus the two radii $1 + 1$):

$$2 + x \leq 2 + 2 \tan \frac{x}{2} \implies x \leq 2 \tan \frac{x}{2}.$$

By the tangent duplication formula, $\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)} \geq 2 \tan \frac{x}{2}$ (the denominator is in $(0, 1)$ for $x/2 \in (0, \pi/4)$). Composing, $x \leq 2 \tan \frac{x}{2} \leq \tan x$.

Conclusion. From (i) and (ii), dividing by $\sin x > 0$ and passing to reciprocals as in Steps 4–7 of the main proof, one recovers $\cos x < \sin x/x < 1$ and $\lim_{x \rightarrow 0} \sin x/x = 1$.

Numerical check. $x = 0.7$: $2 \tan(x/2) = 0.730057 < \tan x = 0.842288$ (verified). $\square$

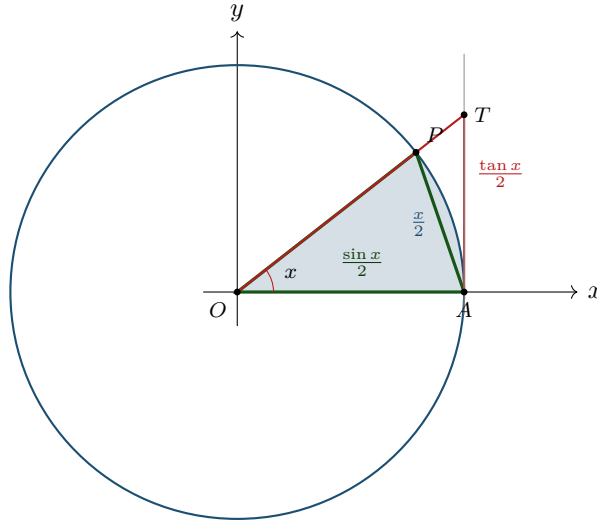


Figure 8.2: — The three nested regions of Theorem 8.5: inscribed triangle $\subseteq$ sector $\subseteq$ circumscribed triangle. Their areas give $\sin x < x < \tan x$.

8.5.1 Non-circularity check

Warning. The objection. The proof just given uses the area of a circular sector. If that were derived from the area of the disc, and the area of the disc from the length of the circle, and the length of the circle from the limit $\lim \sin h/h = 1$ — as many texts do — the chain would close on itself and the proof would prove nothing.

Remark 8.8: The chain, checked link by link. Here is the actual logical order, each link with its reference:

1. **Length of the circle:** defined as the supremum of the perimeters of inscribed polylines (Definition 6.1); it exists by Theorem 6.1, proved with the single inequality $\sqrt{a^2 + b^2} \leq |a| + |b|$. No trigonometry.
2. **The constant π :** defined as $C/2r$, legitimate because the ratio is the same for every circle (Theorem 6.2, proved by homothety). No trigonometry.
3. **Area of the disc:** $A = \pi r^2$ by Theorem 6.5, proved by double reductio with Lemma 6.4 and the estimates of Lemma 6.5. No trigonometry: the estimates use only $a_n < r$ and $p_n < C < q_n$.
4. **Area of a sector:** $\frac{1}{2}r^2\theta$ by Theorem 6.7, proved by additivity, monotonicity and the density of $\mathbb{Q}$. No trigonometry.
5. **Fundamental limit:** Theorem 8.5, which uses (4).

None of links 1–4 uses sine, cosine or tangent. The chain is therefore a chain, not a circle.

Warning. What may *not* be used, after this check. The following remain incorrect, being circular:

- proving the fundamental limit by l'Hôpital's rule: the derivative of $\sin$ is defined precisely by that limit;
- proving the fundamental limit from the series expansion $\sin x = x - x^3/6 + \dots$, if the series was obtained from the Taylor series, which requires the derivatives;
- deriving $C = 2\pi r$ from the fundamental limit (Remark 6.12, Chapter 6).

The first two become legitimate if sine and cosine are *defined* analytically — for instance as the sums of their power series, and π as twice the first positive zero of the cosine so defined. This is the route of analysis texts that want to free themselves of geometry: perfectly rigorous, but then it is the connection with the circle that has to be proved, and not the other way round.

Corollary 8.4: Cosine limit.

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Proof. For $x \neq 0$ small enough, $1 + \cos x \neq 0$; multiplying numerator and denominator by $(1 + \cos x)$:

$$\frac{1 - \cos x}{x^2} = \frac{(1 - \cos x)(1 + \cos x)}{x^2 (1 + \cos x)} = \frac{1 - \cos^2 x}{x^2 (1 + \cos x)} = \frac{\sin^2 x}{x^2 (1 + \cos x)},$$

using the Pythagorean identity (Theorem 8.2). Rewriting as a product:

$$\frac{1 - \cos x}{x^2} = \left(\frac{\sin x}{x} \right)^2 \cdot \frac{1}{1 + \cos x}.$$

By Theorem 8.5 the first factor tends to $1^2 = 1$; by Step 5 of its proof, $\cos x \rightarrow 1$, hence the second factor tends to $1/(1+1) = 1/2$. By the algebra of limits the product tends to $1 \cdot \frac{1}{2} = \frac{1}{2}$. $\square$

Remark 8.9: Why it is called “sinc”. The function $x \mapsto \frac{\sin x}{x}$, extended by continuity with the value 1 at $x = 0$ (an extension made legitimate precisely by Theorem 8.5), is called the **cardinal sine** and written **sinc**. Beware of a double convention:

$$\text{sinc}_{\text{unnormalised}}(x) = \frac{\sin x}{x}, \quad \text{sinc}_{\text{normalised}}(x) = \frac{\sin(\pi x)}{\pi x}.$$

The first is the convention of mathematics, the second that of signal processing, where the zeros fall on the integers and the integral over $\mathbb{R}$ is exactly 1. Confusing the two produces errors by a factor π in filter bandwidths: in telecommunications texts the convention must always be checked. The function **sinc** is the Fourier transform of the rectangular function, and for this reason it is the kernel of Shannon's sampling theorem: its analytic treatment is in Volume II.

Remark 8.10: The fundamental limit holds only in radians. If x is measured in degrees, the corresponding arc on the unit circle has length $\pi x/180$, and the computation of Step 2 gives the sector the area $\frac{\pi x}{360}$ instead of $\frac{x}{2}$. Repeating the proof one obtains

$$\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} = \frac{\pi}{180} \approx 0.01745.$$

This is the technical reason why angles are measured in radians in analysis: in degrees the derivative of the sine would not be the cosine but $\frac{\pi}{180} \cos$, and every formula would carry that factor along. See Remark 6.5 of Chapter 6.

8.6. Inverse trigonometric functions

Definition 8.2: Arcsine, arccosine, arctangent. Restricting $\sin$, $\cos$, $\tan$ to intervals of monotonicity, the inverses are defined:

$$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \arccos : [-1, 1] \rightarrow [0, \pi], \quad \arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

$$\begin{aligned} \sin(\arcsin x) &= x, & \cos(\arccos x) &= x, & \tan(\arctan x) &= x, \\ \arcsin x + \arccos x &= \frac{\pi}{2}. \end{aligned}$$

Proposition 8.2: Complementarity of arcsine and arccosine. For every $x \in [-1, 1]$, $\arcsin x + \arccos x = \pi/2$.

Proof. Put $\alpha = \arcsin x \in [-\pi/2, \pi/2]$; from the subtraction formula $\cos(\pi/2 - \alpha) = \cos(\pi/2)\cos\alpha + \sin(\pi/2)\sin\alpha = \sin\alpha = x$. Moreover $\alpha \in [-\pi/2, \pi/2]$ implies $\pi/2 - \alpha \in [0, \pi]$, that is, $\pi/2 - \alpha$ lies exactly in the domain of definition of $\arccos$: hence, by uniqueness of the arccosine on that interval, $\arccos x = \pi/2 - \alpha$, that is, $\arcsin x + \arccos x = \alpha + (\pi/2 - \alpha) = \pi/2$. $\square$

8.7. Why $\cos^{-1} x$ is not $1/\cos x$: the ambiguity of the inverse notation

This section answers a question that causes confusion in practically every trigonometry course: the notation $\cos^{-1} x$ (sometimes read, incorrectly, “cosine to the minus one” as if it were a power) does **not** denote $1/\cos x$. Let us clarify why, and then generalise to all inverse trigonometric functions.

Best practice. The crux of the problem. For a real number x and an integer exponent n , the notation x^n follows the algebraic rule $x^{-1} = 1/x$ (the reciprocal). But for a **function** f , the notation f^{-1} follows a *different* and historically more recent convention: f^{-1} denotes the **inverse function** of f (the one with $f^{-1}(f(x)) = x$), not the reciprocal $1/f$. Applied to $\cos$, this second convention gives $\cos^{-1} x = \arccos x$ (the angle whose cosine is x), an object conceptually very far from $1/\cos x = \sec x$ (the secant function).

Definition 8.3: The two notations compared.

$$\begin{aligned} \cos^{-1} x &\stackrel{\text{standard}}{=} \stackrel{\text{convention}}{=} \arccos x && (\text{inverse function: } \cos(\arccos x) = x) \\ (\cos x)^{-1} &= \frac{1}{\cos x} = \sec x && (\text{algebraic reciprocal}) \end{aligned}$$

The two expressions $\cos^{-1} x$ and $(\cos x)^{-1}$ therefore denote **completely different** functions: the first has domain $[-1, 1]$ and codomain (principal branch) $[0, \pi]$; the second has domain

$\mathbb{R} \setminus \{\pi/2 + k\pi\}$ and codomain $(-\infty, -1] \cup [1, +\infty)$. They are not even functions of “the same kind” of input.

Remark 8.11: Historical origin of the notation. The notation f^{-1} for the inverse function (as opposed to the reciprocal) was introduced by the British astronomer and mathematician John Herschel in 1813, precisely for the inverse trigonometric functions, with the declared intent of distinguishing them from the algebraic reciprocal $1/f$. Paradoxically, the very notation Herschel introduced to resolve the ambiguity is today the primary source of confusion for students, because the symbol “ -1 ” is instinctively read as an algebraic exponent by anyone who has just learned powers.

Remark 8.12: The solution: the “arc-” prefix. To eliminate the ambiguity at the root, the alternative notation with the “**arc-**” prefix ($\arcsin$, $\arccos$, $\arctan$, arccot , arcsec , arccsc) makes explicit that the result is an **arc** (that is, an angle, measured as arc length on the unit circle, see Definition 8.1) whose sine, cosine or tangent is x — not a reciprocal. This unambiguous notation is the one recommended by the international standard **ISO 80000-2** (Quantities and units — Part 2: Mathematical signs and symbols), which specifies $\arcsin$, $\arccos$, $\arctan$ as the only standardised notations for the principal inverse trigonometric functions (secondary sources disagree on the exact year in which this recommendation entered the standard, with references citing both the 2009 edition and a 2019 update: either way, both versions specify the “arc-” prefix as the standardised notation).

$$\begin{array}{lll} \sin^{-1} x := \arcsin x & \neq & \frac{1}{\sin x} = \csc x \\ \cos^{-1} x := \arccos x & \neq & \frac{1}{\cos x} = \sec x \\ \tan^{-1} x := \arctan x & \neq & \frac{1}{\tan x} = \cot x \end{array}$$

Remark 8.13: An ambiguity that $\sin^2 \theta$ does NOT share. It is worth noting that this same convention does not apply uniformly to all powers: the notation $\sin^2 \theta$ universally means $(\sin \theta)^2$ (the square, not $\sin(\sin \theta)$, that is, not the composition of the function with itself), whereas $\sin^{-1} \theta$ means $\arcsin \theta$ (the inverse function) and **not** $1/\sin \theta$. The convention for the exponent -1 applied to a trigonometric function is therefore an isolated and historically contingent exception (due to Herschel) to the general rule $f^n = (f(x))^n$ valid for every other integer exponent: it is precisely this isolated exception that generates the didactic confusion, and it is for this reason that, in practice and in international scientific texts, the notation $\arcsin$, $\arccos$, $\arctan$ (unambiguous) is almost always preferred to the notation with exponent -1 , especially in an engineering context where the distinction has direct practical consequences (misreading $\arccos x$ as $1/\cos x$ in an inverse-kinematics computation for a robot, for instance, produces a radically wrong result, see Chapter 13).

Remark 8.14: From the unit circle to the circumcircle of a triangle. The unit circle defines sine and cosine for a single angle θ . When one considers the three angles of an arbitrary triangle (not necessarily right-angled) inscribed in a circle of radius R , the same trigonometric functions reappear in the **law of sines** $a/\sin A = 2R$, linking sides, angles and circumradius: it is treated, together with the law of cosines and the Pythagorean theorem, in Chapter 10, devoted specifically to the interaction between triangle and circle.

8.8. Sum-to-product formulas

Theorem 8.6: Sum-to-product formulas.

$$\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}, \quad \cos p + \cos q = 2 \cos \frac{p+q}{2} \cos \frac{p-q}{2}.$$

Proof. Set $\alpha = (p+q)/2$, $\beta = (p-q)/2$, so that $p = \alpha + \beta$, $q = \alpha - \beta$. By the addition formulas (Theorem 8.4):

$$\begin{aligned} \sin p + \sin q &= \sin(\alpha + \beta) + \sin(\alpha - \beta) \\ &= (\sin \alpha \cos \beta + \cos \alpha \sin \beta) + (\sin \alpha \cos \beta - \cos \alpha \sin \beta) \\ &= 2 \sin \alpha \cos \beta, \end{aligned}$$

which is the conclusion. Similarly, adding $\cos(\alpha + \beta) + \cos(\alpha - \beta)$, the terms in $\sin \alpha \sin \beta$ cancel and $2 \cos \alpha \cos \beta$ remains. $\square$

Remark 8.15: Why this matters in industrial signal processing. The sum-to-product formulas are the algebraic tool behind the **lock-in demodulation** technique, used to extract a weak periodic signal (for instance the reading of a capacitive or piezoelectric sensor in a water-treatment plant) buried in noise: multiplying the signal by a reference at the same frequency and averaging over time, the sum-product terms separate the useful component (at nearly zero frequency after demodulation) from the broadband noise.

8.9. Problem set

8.9.1 Numerical exercises

Exercise 8.1. Compute $\sin 75^\circ$ and $\cos 75^\circ$ using the addition formulas with $75^\circ = 45^\circ + 30^\circ$.

Solution.

$$\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4} \approx 0.9659,$$

$$\cos 75^\circ = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} - \sqrt{2}}{4} \approx 0.2588.$$

Exercise 8.2. Using the half-angle formulas, compute $\sin 15^\circ$ from the known value of $\cos 30^\circ$.

Solution.

$$\sin 15^\circ = \sqrt{\frac{1 - \cos 30^\circ}{2}} = \sqrt{\frac{1 - \sqrt{3}/2}{2}} = \sqrt{\frac{2 - \sqrt{3}}{4}} = \frac{\sqrt{2 - \sqrt{3}}}{2} \approx 0.2588$$

(positive sign because 15° is in the first quadrant). It coincides, as it must, with $\cos 75^\circ$ computed in the previous exercise (complementary angles).

Exercise 8.3. A point P of the unit circle lies in the second quadrant and $\sin \alpha = 3/5$. Compute $\cos \alpha$ and $\tan \alpha$.

Solution. By the Pythagorean identity (Theorem 8.2), $\cos^2 \alpha = 1 - (3/5)^2 = 1 - 9/25 = 16/25$, hence $\cos \alpha = \pm 4/5$. In the second quadrant the abscissa is negative, so

$$\cos \alpha = -\frac{4}{5}, \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{3/5}{-4/5} = -\frac{3}{4}.$$

Plausibility check. In the second quadrant a positive sine and a negative cosine give a negative tangent: the sign is consistent.

Exercise 8.4. Compute $\lim_{x \rightarrow 0} \frac{\sin(4x)}{2x}$ using Theorem 8.5.

Solution. Multiply and divide by the factor that makes the argument of the sine equal to the denominator:

$$\frac{\sin(4x)}{2x} = \frac{\sin(4x)}{4x} \cdot \frac{4x}{2x} = 2 \frac{\sin(4x)}{4x}.$$

Setting $u = 4x$, one has $u \rightarrow 0$ as $x \rightarrow 0$, hence $\frac{\sin(4x)}{4x} = \frac{\sin u}{u} \rightarrow 1$ by Theorem 8.5. The limit is therefore $2 \cdot 1 = 2$.

Exercise 8.5. Compute $\lim_{x \rightarrow 0} \frac{1 - \cos(3x)}{x^2}$.

Solution. Reduce to Corollary 8.4 with the substitution $u = 3x$:

$$\frac{1 - \cos(3x)}{x^2} = 9 \frac{1 - \cos(3x)}{(3x)^2} = 9 \frac{1 - \cos u}{u^2} \xrightarrow{u \rightarrow 0} 9 \cdot \frac{1}{2} = \frac{9}{2}.$$

8.9.2 Symbolic exercises and proofs

Exercise 8.6. Prove, using the duplication formulas, that $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$ (hint: $3\theta = 2\theta + \theta$).

Solution.

$$\sin 3\theta = \sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta = 2\sin \theta \cos^2 \theta + (1 - 2\sin^2 \theta) \sin \theta.$$

Using $\cos^2 \theta = 1 - \sin^2 \theta$:

$$\begin{aligned} &= 2\sin \theta(1 - \sin^2 \theta) + \sin \theta - 2\sin^3 \theta \\ &= 2\sin \theta - 2\sin^3 \theta + \sin \theta - 2\sin^3 \theta \\ &= 3\sin \theta - 4\sin^3 \theta. \quad \square \end{aligned}$$

Exercise 8.7. Starting from Theorem 8.5, prove algebraically that

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2},$$

and deduce the approximation $\cos x \approx 1 - x^2/2$ for small x .

Solution. The proof is that of Corollary 8.4: multiply numerator and denominator by $(1 + \cos x)$, use $1 - \cos^2 x = \sin^2 x$ and split the result into the product $(\frac{\sin x}{x})^2 \cdot \frac{1}{1 + \cos x}$, whose factors tend to 1 and to $\frac{1}{2}$.

Approximation. From the limit it follows that, as $x \rightarrow 0$, $1 - \cos x = \frac{1}{2}x^2 + o(x^2)$, that is

$$\cos x \approx 1 - \frac{x^2}{2}.$$

Numerical check. For $x = 0.1$ rad: $\cos(0.1) = 0.995004\dots$, while $1 - 0.01/2 = 0.995$: the error is $4 \cdot 10^{-6}$, consistent with the fact that the first neglected term is of order $x^4/24 \approx 4.2 \cdot 10^{-6}$.

Exercise 8.8. Prove the sum-to-product formulas for the sine from the addition and subtraction formulas alone, and explain why the substitution $\alpha = (p + q)/2$, $\beta = (p - q)/2$ is always legitimate.

Solution. The derivation is that of Theorem 8.6: adding $\sin(\alpha + \beta)$ and $\sin(\alpha - \beta)$, the terms in $\cos \alpha \sin \beta$ cancel and $2 \sin \alpha \cos \beta$ remains.

Legitimacy of the substitution. The system

$$\begin{cases} \alpha + \beta = p \\ \alpha - \beta = q \end{cases}$$

is linear in the unknowns α, β with determinant $1 \cdot (-1) - 1 \cdot 1 = -2 \neq 0$: it therefore admits one and only one solution for *every* pair $(p, q) \in \mathbb{R}^2$, and the solution is $\alpha = (p + q)/2$, $\beta = (p - q)/2$. There is thus no restriction on the values of p and q : the sum-to-product formulas hold for every pair of real numbers, without the quadrant limitations that afflict purely geometric derivations.

Exercise 8.9. (*Link to Chapter 6.*) Explain why, for small θ , $\sin \theta \approx \theta$ (in radians), and quantify the error.

Solution. On the unit circle ($r = 1$), the arc from $(1, 0)$ to $P(\theta)$ has length $l = 1 \cdot \theta = \theta$ (Theorem 6.3), while $\sin \theta$ is the ordinate of $P(\theta)$. Step 3 of the proof of Theorem 8.5 gives $\sin \theta < \theta$ for $\theta \in (0, \pi/2)$: the ordinate is always less than the arc, and the ratio tends to 1.

Quantification. From the same Step 4, $\cos \theta < \frac{\sin \theta}{\theta} < 1$, hence

$$0 < \theta - \sin \theta < \theta(1 - \cos \theta) \leq \theta \cdot \frac{\theta^2}{2} = \frac{\theta^3}{2},$$

using the estimate $1 - \cos \theta \leq \theta^2/2$ of Step 5. The absolute error is therefore $O(\theta^3)$: for $\theta = 0.1$ rad it is less than $5 \cdot 10^{-4}$ (the true value is $1.67 \cdot 10^{-4}$). It is this estimate that justifies the small-angle approximation used everywhere in engineering — from the simple pendulum to the linearisation of control equations.

Exercise 8.10. (*On the risk of circularity.*) A student proves the fundamental limit thus: “by l’Hôpital, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$ ”. Identify the logical error and explain under

what conditions the argument would become correct.

Solution. *The error.* L'Hôpital's rule requires knowing the derivative of the numerator, that is $\frac{d}{dx} \sin x = \cos x$. But this derivative is computed by definition as

$$\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h},$$

and the second limit is exactly the one to be proved. The argument assumes what it must prove: it is circular.

When it would become correct. If sine and cosine are *defined* analytically, for instance as the sums of the power series $\sin x = \sum_{k \geq 0} (-1)^k x^{2k+1} / (2k+1)!$ and $\cos x = \sum_{k \geq 0} (-1)^k x^{2k} / (2k)!$, then the derivative of sin is obtained by differentiating the series term by term (legitimate for power series within the radius of convergence, here infinite) and equals cos, without using any trigonometric limit. In that framework l'Hôpital is fully legitimate. The price is that one must then prove that those series really describe the coordinates on the unit circle — true, but anything but immediate, see the warning in Section 8.5.1.

Exercise 8.11. Prove the identity $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ and deduce that $\cos 20^\circ$ is a root of $8x^3 - 6x - 1 = 0$. Check numerically.

Solution. From $\cos 3\theta = \cos(2\theta + \theta) = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$ and the duplication formulas, $\cos 3\theta = (2\cos^2 \theta - 1)\cos \theta - 2\sin^2 \theta \cos \theta = 2\cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta)\cos \theta = 4\cos^3 \theta - 3\cos \theta$.

Setting $\theta = 20^\circ$ gives $\cos 60^\circ = 1/2$, hence $4x^3 - 3x = 1/2$ with $x = \cos 20^\circ$, that is $8x^3 - 6x - 1 = 0$. Numerical check: with $x = \cos 20^\circ = 0.9396926$ one has $x^3 = 0.8297695$, hence $8x^3 = 6.6381557$ and $6x = 5.6381557$, and finally $8x^3 - 6x = 1$ **exactly** (not just up to rounding: it is an exact algebraic identity, a consequence of $\cos 60^\circ = 1/2$) ✓. The polynomial $8x^3 - 6x - 1$ is irreducible over $\mathbb{Q}$ (it has no rational root: the only candidates $\pm 1, \pm 1/2, \pm 1/4, \pm 1/8$ do not annihilate it), hence $\cos 20^\circ$ has degree 3 over $\mathbb{Q}$. The fact that every number constructible with straightedge and compass has degree a power of 2 over $\mathbb{Q}$ is **imported** (Wantzel, 1837; see Observation 9.11 in Chapter 9): since 3 is not a power of 2, $\cos 20^\circ$ is not constructible, and this is why the 60° angle cannot be trisected with straightedge and compass.

Exercise 8.12. Compute the exact value of $\sin 18^\circ$ and check it numerically.

Solution. Let $s = \sin 18^\circ$. From $5 \cdot 18^\circ = 90^\circ$ it follows that $3 \cdot 18^\circ = 90^\circ - 2 \cdot 18^\circ$, hence $\sin 3\theta = \cos 2\theta$ with $\theta = 18^\circ$, that is

$$3s - 4s^3 = 1 - 2s^2 \iff 4s^3 - 2s^2 - 3s + 1 = 0 \iff (s - 1)(4s^2 + 2s - 1) = 0.$$

Since $s \neq 1$, there remains $4s^2 + 2s - 1 = 0$, whose positive root is

$$s = \frac{-2 + \sqrt{4 + 16}}{8} = \frac{-1 + \sqrt{5}}{4} = 0.3090170,$$

in agreement with $\sin 18^\circ = 0.3090170$ ✓. Note that $\sqrt{5}$ is constructible: this is why the regular pentagon can be constructed with straightedge and compass.

8.10. Self-assessment quiz

Ten multiple-choice questions. The answer key is at the end of the section.

1. In the unit circle, the hypotenuse of the right triangle associated with a point of the circle is
 - A) $\cos \theta$
 - B) $\sin \theta$
 - C) 1
 - D) $\tan \theta$
2. The identity $\sin^2 x + \cos^2 x = 1$ is a direct consequence of
 - A) the addition formulas
 - B) the Pythagorean theorem applied to the unit circle
 - C) the fundamental limit
 - D) the definition of the tangent
3. For small positive angles, the geometric inequality states that the sine is
 - A) greater than the arc
 - B) less than the arc
 - C) equal to the arc
 - D) greater than the tangent
4. The limit $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ equals
 - A) 0
 - B) 1
 - C) $\pi/180$
 - D) it does not exist
5. In the proof of the fundamental limit, the tangent appears as
 - A) the height of the inscribed triangle
 - B) the height of the circumscribed triangle, of area greater than the sector
 - C) the radius of the circle
 - D) the length of the arc
6. The limit $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ equals
 - A) 0
 - B) 1
 - C) $1/2$
 - D) 2
7. Proving the fundamental limit with l'Hôpital's rule is

- A) correct and quicker
 B) circular, because the derivative of the sine rests on that limit
 C) correct only for $x > 0$
 D) correct only in degrees
8. The sector area used in the proof of the fundamental limit was obtained in Chapter 6
 A) from the fundamental limit itself
 B) by additivity and monotonicity of the area, without trigonometry
 C) by integration
 D) experimentally
9. The trigonometric tangent of an angle, in analytic geometry, coincides with
 A) the ordinate of the point on the circle
 B) the slope of the line associated with the angle
 C) the length of the arc
 D) the radius
10. With θ in radians, the area of the circular sector in the unit circle (radius 1) is
 A) θ
 B) $\theta/2$
 C) 2θ
 D) $\pi\theta$

BEST PRACTICE

Answer key. 1: C — 2: B — 3: B — 4: B — 5: B — 6: C (Corollary 8.4) — 7: B (Exercise 8.10) — 8: B (Theorem 6.7) — 9: B — 10: B.

Note on question 9. The line through the origin making an angle θ with the x axis has equation $y = mx$ with $m = \sin\theta/\cos\theta = \tan\theta$: this is exactly the ordinate of the point $T = (1, \tan\theta)$ of Theorem 8.1. The coincidence between “trigonometric tangent” and “slope” is not terminological: it is the same number.

8.11. Chapter Formula sheet

The unit circle (Definition 8.1; Theorem 8.1)

$$x^2 + y^2 = 1, \quad P(\theta) = (\cos\theta, \sin\theta), \quad T = (1, \tan\theta), \quad \tan\theta = \frac{\sin\theta}{\cos\theta}$$

Pythagorean identity (Theorems 8.2 and 8.3; Corollary 8.1)

$$\sin^2\theta + \cos^2\theta = 1$$

Addition and subtraction (Theorem 8.4; Lemma 8.1)

$$\sin(\alpha \pm \beta) = \sin\alpha \cos\beta \pm \cos\alpha \sin\beta, \quad \cos(\alpha \pm \beta) = \cos\alpha \cos\beta \mp \sin\alpha \sin\beta$$

Duplication and bisection (Corollaries 8.2 and 8.3)

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Sum to product (Theorem 8.6)

$$\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}, \quad \cos p + \cos q = 2 \cos \frac{p+q}{2} \cos \frac{p-q}{2}$$

Fundamental limits (Theorem 8.5; Corollary 8.4), with x in radians

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$\cos x < \frac{\sin x}{x} < 1 \quad \left(0 < x < \frac{\pi}{2}\right)$$

Inverse functions (Definition 8.2)

$$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \arccos : [-1, 1] \rightarrow [0, \pi], \quad \arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

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CHAPTER 9

The Circle at High School: Classical Formulas, Goniometry and Intersections

INFO

Purpose of the chapter. To collect in one place, and *prove in full*, the formulas about the circle that one meets in secondary school: circumference and area, arc and sector, segment and annulus, chord and sagitta, inscribed and circumscribed polygons, the whole goniometric Formula sheet, and finally the computation of the areas of **intersection** between a circle and the other elementary figures.

Why a bridging chapter. Chapters 2–8 build the theory; this chapter *uses* it, showing that every formula in the textbook is a theorem with a short proof. Nothing here is new with respect to the theory already developed: everything follows from results already proved, and every step declares where it comes from.

The common thread. All the areas in this chapter — sector, segment, lens between two circles, circle against polygon — reduce to two building blocks only: the **area of a sector** $\frac{1}{2}r^2\theta$ (Theorem 6.7) and the **area of a triangle** $\frac{1}{2}\text{base} \times \text{height}$. Whoever has understood this has understood the whole chapter.

9.1. The classical formulas of the circle

9.1.1 Circumference and area

Theorem 9.1: Length of the circumference. *The circle of radius r has circumference*

$$C = 2\pi r = \pi d, \quad d = 2r.$$

Proof. By Theorem 6.1 the circle is rectifiable, that is, it has a finite length C ; by Theorem 6.2 the ratio C/d has the same value for all circles, and Definition 6.2 calls that common value π . Hence $C/d = \pi$, that is $C = \pi d = 2\pi r$.

Note that this is not a disguised definition: the whole mathematical content lies in Theorem 6.2, that is, in the *non-obvious* fact that the ratio does not depend on the radius. It is exactly the observation that allows one to measure a wheel of any diameter with a single constant. $\square$

Theorem 9.2: Area of the disc. *The disc of radius r has area*

$$A = \pi r^2 = \frac{C r}{2} = \frac{\pi d^2}{4}.$$

Proof. Archimedes' theorem (Theorem 6.5) states that the disc is equivalent to the right triangle whose legs are the radius r and the rectified circumference C . The area of that triangle is $\frac{1}{2}Cr$,

hence

$$A = \frac{C r}{2} = \frac{(2\pi r) r}{2} = \pi r^2.$$

Substituting $r = d/2$ gives $A = \pi d^2/4$. $\square$

Remark 9.1: The form $A = Cr/2$ is the one worth remembering. Writing $A = \frac{1}{2}Cr$ is not an embellishment: it is the form from which, with the same proof, both the area of a sector ($\frac{1}{2}\ell r$, with ℓ the arc length) and the area of every regular polygon ($\frac{1}{2}pa$, with p the perimeter and a the apothem) descend. The circle is the limiting case of a regular polygon in which the apothem coincides with the radius.

Corollary 9.1: Inverse formulas. From the two preceding formulas one obtains

$$r = \frac{C}{2\pi}, \quad r = \sqrt{\frac{A}{\pi}}, \quad C = 2\sqrt{\pi A}, \quad A = \frac{C^2}{4\pi}.$$

Proof. The first two follow by solving $C = 2\pi r$ and $A = \pi r^2$ for r (legitimate because $r > 0$). For the third, $C = 2\pi r = 2\pi\sqrt{A/\pi} = 2\sqrt{\pi^2 A/\pi} = 2\sqrt{\pi A}$. For the fourth, square it: $C^2 = 4\pi A$, whence $A = C^2/(4\pi)$. $\square$

9.1.2 Arc and sector

Theorem 9.3: Arc and sector, in radians and in degrees. Let θ be the measure in radians of the central angle (Definition 6.3) and α the same measure in sexagesimal degrees. Then

$$\ell = r\theta = 2\pi r \cdot \frac{\alpha}{360^\circ}, \quad A_{\text{sec}} = \frac{1}{2}r^2\theta = \frac{\ell r}{2} = \pi r^2 \cdot \frac{\alpha}{360^\circ}.$$

Proof. The first equality is Theorem 6.3. The degree–radian conversion is $\theta = 2\pi\alpha/360^\circ$, which substituted gives $\ell = 2\pi r\alpha/360^\circ$.

For the sector, Theorem 6.7 gives $A_{\text{sec}} = \frac{1}{2}r^2\theta$; substituting $\theta = \ell/r$ yields $A_{\text{sec}} = \frac{1}{2}r\ell$, and substituting $\theta = 2\pi\alpha/360^\circ$ yields $A_{\text{sec}} = \pi r^2\alpha/360^\circ$. $\square$

Remark 9.2: The sector is a proportion, and it pays to see it that way. The two forms in degrees say the same thing in a memorable way: the arc is to the circumference as the angle is to the full angle, and the sector is to the disc in the same ratio,

$$\frac{\ell}{C} = \frac{A_{\text{sec}}}{A} = \frac{\alpha}{360^\circ} = \frac{\theta}{2\pi}.$$

This is the only formula one needs to remember: the other four follow from it.

9.1.3 Circular segment and annulus

Theorem 9.4: Area of a circular segment. The circular segment (Definition 2.18) subtended by a central angle $\theta \in (0, 2\pi)$ in a circle of radius r has area

$$A_{\text{seg}} = \frac{1}{2}r^2(\theta - \sin \theta).$$

Equivalently, if d is the distance from the centre to the chord (with $0 \leq d < r$) and the

segment is the minor one,

$$A_{\text{seg}} = r^2 \arccos\left(\frac{d}{r}\right) - d \sqrt{r^2 - d^2}.$$

Proof. First form. The minor segment is the sector of measure θ with the isosceles triangle OAB formed by the two radii and the chord removed. The sector has area $\frac{1}{2}r^2\theta$ (Theorem 6.7); the triangle has two sides of length r with included angle θ , hence area $\frac{1}{2}r \cdot r \sin \theta$. Subtracting,

$$A_{\text{seg}} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta = \frac{1}{2}r^2(\theta - \sin \theta).$$

The formula remains valid for $\theta > \pi$ as well, where $\sin \theta < 0$ and the subtracted term becomes an addition: that is the case of the *major* segment, which is indeed the reflex sector *plus* the triangle.

Second form. Let M be the foot of the perpendicular from the centre to the chord; by Theorem 3.1 M is the midpoint of the chord and $OM = d$. In the right triangle OMA one has $\cos(\theta/2) = d/r$, that is $\theta = 2 \arccos(d/r)$, and $MA = \sqrt{r^2 - d^2}$. Then

$$\frac{1}{2}r^2\theta = r^2 \arccos \frac{d}{r},$$

while, by the double-angle formula (Corollary 8.2),

$$\frac{1}{2}r^2 \sin \theta = \frac{1}{2}r^2 \cdot 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = r^2 \cdot \frac{\sqrt{r^2 - d^2}}{r} \cdot \frac{d}{r} = d \sqrt{r^2 - d^2}.$$

Subtracting gives the second form. □

Remark 9.3: Why the second form is the one actually used. In applications the available datum is almost never the angle: it is the distance d from the centre to the cutting line (the level of a liquid in a cylindrical tank, the depth of a milling cut, the height of water in a pipe). The second form works directly on that datum, without going through an angle that would have to be computed with an arc cosine anyway. It will also be the form with which, in Section 9.7, all the intersection areas are built.

Theorem 9.5: Annulus and annular sector. Let $R > r > 0$. The annulus (Definition 2.19) between the two concentric circles has area

$$A_{\text{ann}} = \pi(R^2 - r^2) = 2\pi r_m s, \quad r_m = \frac{R+r}{2}, \quad s = R-r,$$

where r_m is the mean radius and s the thickness. The annular sector of measure θ has area

$$A_{\text{as}} = \frac{1}{2}\theta(R^2 - r^2) = \theta r_m s.$$

Proof. The annulus is the set difference between the disc of radius R and the disc of radius r , which is a subset of it; by additivity of area $A_{\text{ann}} = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$. Factoring the difference of squares,

$$\pi(R-r)(R+r) = \pi s \cdot 2r_m = 2\pi r_m s.$$

For the annular sector one subtracts the two sectors of equal measure: $\frac{1}{2}R^2\theta - \frac{1}{2}r^2\theta = \frac{1}{2}\theta(R^2 - r^2)$, and the second form follows as above. □

Remark 9.4: The “unrolled rectangle” formula. The form $A_{\text{ann}} = 2\pi r_m s$ says that the annulus has the same area as the rectangle one would obtain by cutting it and unrolling it, with base the mean circumference $2\pi r_m$ and height the thickness s . The remarkable fact is that the radius to be used is exactly the arithmetic mean of R and r , with no correction: it is a consequence of the identity $R^2 - r^2 = (R + r)(R - r)$, and it ceases to hold for the lateral surface of a truncated cone, where the slant height intervenes instead.

Example 9.1: A circular flower bed, computed in all its parts. A circular flower bed has radius $r = 4.00$ m. A 120° sector of it is fenced off, and it is bordered with an annular walkway 0.60 m wide towards the inside. Let us compute everything.

Disc. $C = 2\pi \cdot 4 = 8\pi = 25.1327$ m; $A = \pi \cdot 16 = 50.2655$ m².

120° sector. In radians $\theta = 2\pi/3 = 2.0944$. Arc: $\ell = 4 \cdot 2.0944 = 8.3776$ m. Area: $A_{\text{sec}} = \frac{1}{2} \cdot 16 \cdot 2.0944 = 16.7552$ m², which is exactly $1/3$ of 50.2655 m², as it must be.

Chord and segment. The chord is $2 \cdot 4 \cdot \sin 60^\circ = 8 \cdot 0.8660 = 6.9282$ m. The segment:

$$A_{\text{seg}} = \frac{1}{2} \cdot 16 \cdot (2.0944 - \sin 120^\circ) = 8 \cdot (2.0944 - 0.8660) = 8 \cdot 1.2284 = 9.8272 \text{ m}^2.$$

Check with the second form: $d = 4 \cos 60^\circ = 2$, and $16 \arccos(0.5) - 2\sqrt{16 - 4} = 16 \cdot 1.0472 - 2 \cdot 3.4641 = 16.7552 - 6.9282 = 9.8270$ m². It matches to within 0.0002 in the fourth digit, the expected effect of rounding the intermediate quantities to four digits in both forms.

Annular walkway. $R = 4.00$, $r' = 3.40$, hence $A_{\text{ann}} = \pi(16 - 11.56) = \pi \cdot 4.44 = 13.9487$ m². Check with the mean radius: $r_m = 3.70$, $s = 0.60$, and $2\pi \cdot 3.70 \cdot 0.60 = 2\pi \cdot 2.22 = 13.9487$ m². It matches.

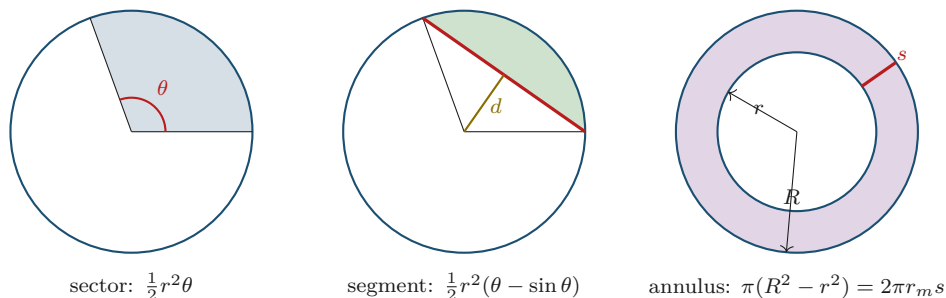


Figure 9.1: — The three classical parts of the circle: sector, segment and annulus. All three follow from $\frac{1}{2}r^2\theta$ and the area of a triangle

9.2. Chord, sagitta and distance from the centre

Theorem 9.6: Chord, sagitta and radius. Let $\overline{AB}$ be a chord of length c in a circle of radius r , let d be the distance from the centre to the chord and let $h = r - d$ be the **sagitta** (or **versine**), that is, the distance between the midpoint of the chord and the minor arc. Then

$$c = 2\sqrt{r^2 - d^2}, \quad c^2 = 8rh - 4h^2, \quad r = \frac{c^2}{8h} + \frac{h}{2} = \frac{c^2 + 4h^2}{8h}.$$

Proof. The first formula is Corollary 2.1: the foot M of the perpendicular from the centre is the midpoint of the chord (Theorem 3.1), and the triangle OMA is right-angled at M with hypotenuse $OA = r$ and leg $OM = d$, hence $MA = \sqrt{r^2 - d^2}$ and $c = 2MA$.

For the second, substitute $d = r - h$:

$$c^2 = 4(r^2 - (r - h)^2) = 4(r^2 - r^2 + 2rh - h^2) = 8rh - 4h^2.$$

For the third, solve for r : from $c^2 = 8rh - 4h^2$ it follows that $8rh = c^2 + 4h^2$, and since $h > 0$ (the chord is not a degenerate diameter)

$$r = \frac{c^2 + 4h^2}{8h} = \frac{c^2}{8h} + \frac{h}{2}.$$

□

Remark 9.5: The radius from chord and sagitta is a workshop tool. The third formula of Theorem 9.6 solves a concrete problem: measuring the radius of an arc whose centre cannot be reached — a sheet-metal fillet, the curvature of a rail, the profile of a lens, the edge of a workpiece on a measuring machine. A ruler for the chord and a dial gauge for the sagitta suffice. The term $h/2$ is the correction that distinguishes this exact formula from the engineering approximation $r \approx c^2/(8h)$, valid when $h \ll c$: for $c = 240$ mm and $h = 18$ mm the approximation gives 400 mm against the exact value 409 mm, an error of 2.2%.

Example 9.2: The radius of a fillet, measured with a ruler and a dial gauge. On a sheet-metal fillet one measures a chord $c = 240$ mm and a sagitta $h = 18$ mm. Then

$$r = \frac{240^2}{8 \cdot 18} + \frac{18}{2} = \frac{57\,600}{144} + 9 = 400 + 9 = 409 \text{ mm}.$$

Check. If $r = 409$ then $d = r - h = 391$ and

$$\frac{c}{2} = \sqrt{409^2 - 391^2} = \sqrt{(409 - 391)(409 + 391)} = \sqrt{18 \cdot 800} = \sqrt{14\,400} = 120 \text{ mm},$$

that is $c = 240$ mm, as measured. The subtended central angle is $\theta = 2 \arcsin(120/409) = 2 \arcsin(0.293399) = 2 \cdot 0.297780 = 0.595560$ rad $= 34.12^\circ$, and the arc length is $\ell = 409 \cdot 0.595560 = 243.58$ mm: only 3.58 mm more than the chord, which confirms that the arc is a shallow one.

Theorem 9.7: Length of the tangent from an external point. Let P be a point outside the circle of centre O and radius r , and let $\delta = OP > r$. The two tangent segments from P have the same length

$$t = \sqrt{\delta^2 - r^2} = \sqrt{\text{pow}(P)},$$

where $\text{pow}(P)$ is the power of P (Definition 3.1).

Proof. Let T be a point of tangency. By Theorem 3.8 the radius OT is perpendicular to the tangent at T , hence the triangle OTP is right-angled at T . By the Pythagorean theorem (Corollary 8.1), $OT^2 + TP^2 = OP^2$, that is $r^2 + t^2 = \delta^2$ and $t = \sqrt{\delta^2 - r^2}$, a positive real number because $\delta > r$. The length does not depend on which of the two points of tangency is chosen, in accordance with Theorem 3.11. Finally $\text{pow}(P) = \delta^2 - r^2$ by Definition 3.1, whence $t = \sqrt{\text{pow}(P)}$. □

Proposition 9.1: The three “chord theorems” as a single theorem. Let P be a point of the plane and Γ a circle of centre O and radius r . For every line through P meeting Γ at X and Y (possibly coincident),

$$\overline{PX} \cdot \overline{PY} = |OP^2 - r^2|,$$

and this single statement contains the three high-school cases:

- (a) intersecting chords (P inside): $PA \cdot PB = PC \cdot PD$;
- (b) secants (P outside): $PA \cdot PB = PC \cdot PD$;
- (c) secant and tangent (P outside): $PT^2 = PA \cdot PB$.

Proof. This is Theorem 3.19, which establishes that the product $\overline{PX} \cdot \overline{PY}$, computed with sign along the line, equals $OP^2 - r^2$ for every secant line through P . The three cases follow by reading that common value: for (a) the point is inside, $OP < r$, the two segments have opposite directions and the signed product is negative, equal to $-(r^2 - OP^2)$: in absolute value it is the same for any two chords, which is the assertion (Theorem 3.7); for (b) the point is outside, the two segments have the same direction and the product is $OP^2 - r^2 > 0$, the same for any two secants (Theorem 3.18); for (c) the tangent line is the limiting case $X = Y = T$, and the product becomes $PT^2 = OP^2 - r^2$ (Theorem 3.16), consistent with Theorem 9.7. $\square$

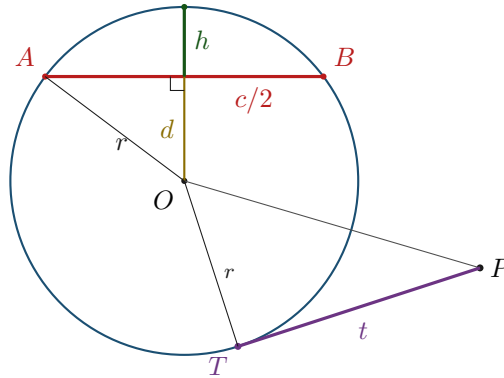


Figure 9.2: — Chord c , distance d from the centre, sagitta $h = r - d$, and tangent length $t = \sqrt{OP^2 - r^2}$ from an external point

9.3. Inscribed and circumscribed regular polygons

Theorem 9.8: Side, apothem, perimeter and area of regular polygons. Let $n \geq 3$ and $r > 0$. For the regular n -gon inscribed in the circle of radius r ,

$$L_n = 2r \sin \frac{\pi}{n}, \quad a_n = r \cos \frac{\pi}{n}, \quad p_n = 2nr \sin \frac{\pi}{n}, \quad A_n = \frac{1}{2}nr^2 \sin \frac{2\pi}{n},$$

where a_n is the apothem. For the regular n -gon circumscribed about the same circle,

$$L_n^* = 2r \tan \frac{\pi}{n}, \quad a_n^* = r, \quad p_n^* = 2nr \tan \frac{\pi}{n}, \quad A_n^* = nr^2 \tan \frac{\pi}{n}.$$

In both cases the single formula $A = \frac{1}{2}pa$ holds.

Proof. Inscribed case. The vertices divide the circle into n congruent arcs, hence each side subtends a central angle $2\pi/n$ (Theorem 3.4). The triangle formed by the centre and one side

is isosceles with equal sides r and apex angle $2\pi/n$; the altitude to the base splits it into two right triangles with central angle π/n , hypotenuse r , opposite leg $L_n/2$ and adjacent leg a_n . Hence

$$\frac{L_n}{2} = r \sin \frac{\pi}{n}, \quad a_n = r \cos \frac{\pi}{n},$$

which gives L_n and a_n . The perimeter is $p_n = nL_n$. The area of the polygon is the sum of the areas of the n isosceles triangles; each has two sides r with included angle $2\pi/n$, hence area $\frac{1}{2}r^2 \sin(2\pi/n)$, and summing gives $A_n = \frac{1}{2}nr^2 \sin(2\pi/n)$.

Circumscribed case. Each side is tangent to the circle at its midpoint (by symmetry with respect to the bisector of the central angle), hence the apothem coincides with the radius, $a_n^* = r$. The triangle formed by the centre and half a side is right-angled with adjacent leg r and central angle π/n , hence $L_n^*/2 = r \tan(\pi/n)$. The perimeter is nL_n^* and the area is the sum of n triangles of base L_n^* and height r :

$$A_n^* = n \cdot \frac{1}{2}L_n^* r = nr^2 \tan \frac{\pi}{n}.$$

Single formula. In both cases $A = n \cdot \frac{1}{2}La = \frac{1}{2}(nL)a = \frac{1}{2}pa$: for the inscribed one, indeed, $\frac{1}{2}p_n a_n = \frac{1}{2} \cdot 2nr \sin(\pi/n) \cdot r \cos(\pi/n) = nr^2 \sin(\pi/n) \cos(\pi/n) = \frac{1}{2}nr^2 \sin(2\pi/n)$, in accordance with the double-angle formula (Corollary 8.2). $\square$

Corollary 9.2: The circle as the limit of the polygons. As n grows,

$$p_n \longrightarrow 2\pi r, \quad p_n^* \longrightarrow 2\pi r, \quad A_n \longrightarrow \pi r^2, \quad A_n^* \longrightarrow \pi r^2,$$

and for every n the sandwich estimates $p_n < 2\pi r < p_n^*$ and $A_n < \pi r^2 < A_n^*$ hold.

Proof. Setting $x = \pi/n \rightarrow 0^+$,

$$p_n = 2nr \sin x = 2\pi r \cdot \frac{\sin x}{x} \longrightarrow 2\pi r$$

by the fundamental limit (Theorem 8.5); analogously $p_n^* = 2\pi r \frac{\tan x}{x} = 2\pi r \frac{\sin x}{x} \cdot \frac{1}{\cos x} \rightarrow 2\pi r$. For the areas,

$$A_n = \frac{1}{2}nr^2 \sin 2x = \pi r^2 \cdot \frac{\sin 2x}{2x} \longrightarrow \pi r^2, \quad A_n^* = \pi r^2 \cdot \frac{\tan x}{x} \longrightarrow \pi r^2.$$

The sandwich inequalities are Lemma 6.5 together with the inequalities $\sin x < x < \tan x$ for $x \in (0, \pi/2)$, already established in the proof of Theorem 8.5. $\square$

Table 9.1: — Regular polygons in the circle of radius $r = 1$

n	L_n	p_n	A_n	A_n^*
3	1.732051	5.196152	1.299038	5.196152
4	1.414214	5.656854	2.000000	4.000000
5	1.175571	5.877853	2.377641	3.632713
6	1.000000	6.000000	2.598076	3.464102
8	0.765367	6.122935	2.828427	3.313708
12	0.517638	6.211657	3.000000	3.215390
100	0.062791	6.279052	3.139526	3.142627
∞	0	6.283185	3.141593	3.141593

Remark 9.6: Hexagon, square and triangle: the three cases to memorise. Three values deserve to be known by heart, because they recur constantly: the side of the inscribed **hexagon** equals the radius ($2r \sin 30^\circ = r$); the side of the inscribed **square** is $r\sqrt{2}$ ($2r \sin 45^\circ$); the side of the inscribed **equilateral triangle** is $r\sqrt{3}$ ($2r \sin 60^\circ$). The three corresponding areas are $\frac{3\sqrt{3}}{2}r^2$, $2r^2$ and $\frac{3\sqrt{3}}{4}r^2$: the inscribed hexagon has exactly twice the area of the inscribed equilateral triangle, as one also sees geometrically by joining every other vertex.

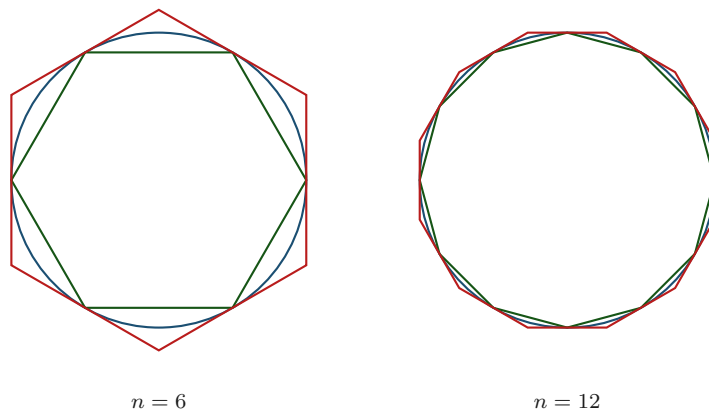


Figure 9.3: — Inscribed (green) and circumscribed (red) polygons: the circle is squeezed between the two, and the vice closes like $1/n^2$

9.4. Construction of the regular pentagon and decagon

Theorem 9.9: Side of the regular decagon. The side of the regular decagon inscribed in $C(O, R)$ equals $R \frac{\sqrt{5} - 1}{2}$.

Proof. The side subtends a central angle $2\pi/10 = 36^\circ$; the chord subtending a central angle θ equals $2R \sin(\theta/2)$ (from the isosceles triangle OAB with sides R, R and angle θ at O , bisected by the altitude from O), hence the side equals $2R \sin 18^\circ$. By Exercise 8.12 of Chapter 8, $\sin 18^\circ = (\sqrt{5} - 1)/4$, hence the side equals $2R \cdot (\sqrt{5} - 1)/4 = R(\sqrt{5} - 1)/2$. $\square$

Construction 9.1: — Regular decagon and pentagon with straightedge and compass. Construct a right triangle with legs R and $R/2$ (the second, the segment from O to the midpoint of a radius): the hypotenuse equals $\sqrt{R^2 + R^2/4} = R\sqrt{5}/2$ by Pythagoras. Subtracting $R/2$ from this hypotenuse (transferring it onto a line and taking the difference with the compass) gives exactly $R(\sqrt{5} - 1)/2$, the side of the decagon just proved. Transferring this segment ten times in succession around the circle (with the compass, starting from any point) gives the regular decagon; the five alternate vertices give the regular pentagon (whose side subtends $72^\circ = 2 \times 36^\circ$). $\square$

Remark 9.7: Constructibility: what can and cannot be done. The regular pentagon is therefore constructible with straightedge and compass, a fact already known to Euclid (*Elements* IV.10–11). The general criterion — which regular polygons are constructible — is **imported**: Gauss (1796) and Wantzel (1837) proved that the regular n -gon is constructible if and only if $n = 2^k p_1 \cdots p_m$ with p_i distinct Fermat primes (primes of the form $2^{2^j} +$

1: 3, 5, 17, 257, 65537, the only ones known). Consequences: the heptagon ($n = 7$) and the enneagon ($n = 9 = 3^2$, a repeated factor) are **not** constructible; in particular, since constructing the enneagon would amount to constructing an angle of $40^\circ = 120^\circ/3$, that is, to trisecting the 120° angle with straightedge and compass, this criterion is consistent with — and generalises — the impossibility of angle trisection discussed further ahead in this same chapter (Observation 9.11): both impossibilities have the same algebraic origin, a degree 3 over $\mathbb{Q}$ instead of a power of 2.

9.5. Goniometry: the Formula sheet, proved

Chapter 8 established the addition formulas (Theorem 8.4), the double-angle formulas (Corollary 8.2), the half-angle formulas (Corollary 8.3) and the sum-to-product formulas (Theorem 8.6). Here the Formula sheet is completed with the identities that high-school work uses daily and that follow from those, each with its proof. None of them is an independent addition: all are consequences obtained in a few lines.

9.5.1 Associated angles

Proposition 9.2: Associated angles. For every $\theta \in \mathbb{R}$:

$$\begin{aligned} \sin(-\theta) &= -\sin \theta, & \cos(-\theta) &= \cos \theta, & \tan(-\theta) &= -\tan \theta; \\ \sin(\pi - \theta) &= \sin \theta, & \cos(\pi - \theta) &= -\cos \theta, & \tan(\pi - \theta) &= -\tan \theta; \\ \sin(\pi + \theta) &= -\sin \theta, & \cos(\pi + \theta) &= -\cos \theta, & \tan(\pi + \theta) &= \tan \theta; \\ \sin\left(\frac{\pi}{2} - \theta\right) &= \cos \theta, & \cos\left(\frac{\pi}{2} - \theta\right) &= \sin \theta, & \tan\left(\frac{\pi}{2} - \theta\right) &= \cot \theta; \\ \sin\left(\frac{\pi}{2} + \theta\right) &= \cos \theta, & \cos\left(\frac{\pi}{2} + \theta\right) &= -\sin \theta, & \tan\left(\frac{\pi}{2} + \theta\right) &= -\cot \theta. \end{aligned}$$

Proof. All of them follow from the addition formulas (Theorem 8.4) applied to the special values $\sin 0 = 0$, $\cos 0 = 1$, $\sin \pi = 0$, $\cos \pi = -1$, $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$. For instance

$$\sin(\pi - \theta) = \sin \pi \cos \theta - \cos \pi \sin \theta = 0 \cdot \cos \theta - (-1) \sin \theta = \sin \theta,$$

$$\cos\left(\frac{\pi}{2} + \theta\right) = \cos \frac{\pi}{2} \cos \theta - \sin \frac{\pi}{2} \sin \theta = 0 - \sin \theta = -\sin \theta.$$

The formulas with $-\theta$ follow from parity: on the unit circle (Definition 8.1) the point $P(-\theta)$ is the reflection of $P(\theta)$ in the x axis, hence it has the same abscissa and the opposite ordinate. The formulas for the tangent are obtained by dividing the corresponding ones for sine and cosine, wherever the denominator does not vanish. $\square$

Remark 9.8: The practical “changes or does not change” rule. The twenty identities above reduce to two rules only: (i) if the added angle is an even multiple of $\pi/2$ (that is $0, \pi, 2\pi$) the function does not change name; if it is odd ($\pi/2, 3\pi/2$) sine becomes cosine and vice versa; (ii) the sign is the one the original function has in the quadrant where the resulting angle falls, imagining θ small and positive. Whoever remembers these two rules does not need to remember the table.

9.5.2 From product to sum: Werner's formulas

Theorem 9.10: Werner's formulas (product to sum). For every $\alpha, \beta \in \mathbb{R}$:

$$\begin{aligned}\sin \alpha \cos \beta &= \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)], \\ \cos \alpha \cos \beta &= \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)], \quad \sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)].\end{aligned}$$

Proof. One adds and subtracts the addition formulas (Theorem 8.4). For the first:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta, \quad \sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta;$$

adding term by term, the terms in $\cos \alpha \sin \beta$ cancel and there remains $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin \alpha \cos \beta$.

For the other two one starts from $\cos(\alpha \mp \beta) = \cos \alpha \cos \beta \pm \sin \alpha \sin \beta$: adding, the terms in $\sin \alpha \sin \beta$ cancel and one obtains $\cos(\alpha - \beta) + \cos(\alpha + \beta) = 2 \cos \alpha \cos \beta$; subtracting, the terms in $\cos \alpha \cos \beta$ cancel and one obtains $\cos(\alpha - \beta) - \cos(\alpha + \beta) = 2 \sin \alpha \sin \beta$. $\square$

Remark 9.9: Werner and prosthaphaeresis are the same formula read both ways. Theorem 8.6 turns sums into products; Theorem 9.10 travels the opposite way. Historically Werner's formulas (16th century) came first, and served to multiply large numbers by turning a product into a sum of values read from tables: the astronomers' prosthaphaeresis, displaced only by Napier's logarithms (1614). Anyone working with signals meets them unchanged in demodulation: the product of two sinusoids is the sum of a difference-frequency component and a sum-frequency component, which is precisely what the third formula says.

9.5.3 The rational parametric formulas

Theorem 9.11: Parametric formulas (Weierstrass substitution). Let $\theta \neq \pi + 2k\pi$ and let $t = \tan(\theta/2)$. Then

$$\sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}, \quad \tan \theta = \frac{2t}{1-t^2} \quad (t \neq \pm 1).$$

Consequently the map

$$t \mapsto \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right)$$

is a bijection between $\mathbb{R}$ and the unit circle minus the point $(-1, 0)$.

Proof. Set $u = \theta/2$, so that $t = \tan u$ and $\cos u \neq 0$. From the Pythagorean identity (Theorem 8.2) divided by $\cos^2 u$ one gets $1 + t^2 = 1/\cos^2 u$, that is

$$\cos^2 u = \frac{1}{1+t^2}.$$

By the double-angle formula for the sine (Corollary 8.2),

$$\sin \theta = 2 \sin u \cos u = 2 \frac{\sin u}{\cos u} \cos^2 u = 2t \cdot \frac{1}{1+t^2} = \frac{2t}{1+t^2}.$$

For the cosine one uses $\cos \theta = 2 \cos^2 u - 1$ (same source):

$$\cos \theta = \frac{2}{1+t^2} - 1 = \frac{2 - (1+t^2)}{1+t^2} = \frac{1-t^2}{1+t^2}.$$

The quotient of the two gives $\tan \theta = 2t/(1 - t^2)$.

Bijectivity. The image point lies on the unit circle because

$$\left(\frac{1-t^2}{1+t^2}\right)^2 + \left(\frac{2t}{1+t^2}\right)^2 = \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{(1+t^2)^2}{(1+t^2)^2} = 1.$$

It never takes the value $(-1, 0)$, since $\frac{1-t^2}{1+t^2} = -1$ would require $1 - t^2 = -1 - t^2$, which is impossible. Conversely, given $(x, y) \neq (-1, 0)$ on the circle, the value $t = y/(1+x)$ is well defined ($x \neq -1$) and recovers it: indeed $y^2 = 1 - x^2 = (1-x)(1+x)$ gives $t^2 = (1-x)/(1+x)$, hence $1 + t^2 = 2/(1+x)$ and $\frac{1-t^2}{1+t^2} = \frac{2x/(1+x)}{2/(1+x)} = x$, $\frac{2t}{1+t^2} = \frac{2y/(1+x)}{2/(1+x)} = y$. The inverse is unique, hence the correspondence is a bijection. $\square$

Remark 9.10: The circle is a rational curve, and Pythagorean triples are born here. Theorem 9.11 is the deep reason why the circle, unlike the generic ellipse in arc-length problems, is a rational curve: every one of its points (except one) is expressed by rational functions of a parameter. Geometrically t is the slope of the line joining $(-1, 0)$ to the point: it is the stereographic projection from $(-1, 0)$ onto the y axis.

The arithmetic consequence is immediate: if $t = q/p$ is rational, the corresponding point has rational coordinates

$$\left(\frac{p^2 - q^2}{p^2 + q^2}, \frac{2pq}{p^2 + q^2}\right),$$

and multiplying by $p^2 + q^2$ one obtains the triple $(p^2 - q^2, 2pq, p^2 + q^2)$: this is exactly Euclid's parametrisation of the Pythagorean triples, recovered arithmetically further on in the chapter on the right triangle. The Weierstrass substitution, the Pythagorean triples and the stereographic projection are the same fact seen from three angles.

9.5.4 Triple-angle formulas and linear combination

Corollary 9.3: Triple-angle formulas. For every $\theta \in \mathbb{R}$:

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta, \quad \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

Proof. Write $3\theta = 2\theta + \theta$ and apply the addition (Theorem 8.4) and double-angle (Corollary 8.2) formulas:

$$\sin 3\theta = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta = 2 \sin \theta \cos^2 \theta + (1 - 2 \sin^2 \theta) \sin \theta.$$

Substituting $\cos^2 \theta = 1 - \sin^2 \theta$ (Theorem 8.2),

$$\sin 3\theta = 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta = 3 \sin \theta - 4 \sin^3 \theta.$$

Analogously

$$\cos 3\theta = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta = (2 \cos^2 \theta - 1) \cos \theta - 2 \sin^2 \theta \cos \theta,$$

and with $\sin^2 \theta = 1 - \cos^2 \theta$:

$$\cos 3\theta = 2 \cos^3 \theta - \cos \theta - 2 \cos \theta + 2 \cos^3 \theta = 4 \cos^3 \theta - 3 \cos \theta.$$

$\square$

Remark 9.11: Why the triple-angle formula is the key to constructibility. Setting $x = \cos \theta$ and $a = \cos 3\theta$, the second formula becomes the cubic equation $4x^3 - 3x = a$. To trisect a given angle means to solve it for x with a known: when the cubic is irreducible over the rationals — as it is for $a = 1/2$, that is for the 60° angle — the solution belongs to no tower of quadratic extensions, and that is why trisection with straightedge and compass is impossible. The triple-angle formula is thus the gateway to the theory of constructibility.

Theorem 9.12: Linear combination of sine and cosine. Let $a, b \in \mathbb{R}$ be not both zero and let $\varrho = \sqrt{a^2 + b^2}$. There is a unique $\varphi \in [0, 2\pi)$ such that $\cos \varphi = a/\varrho$ and $\sin \varphi = b/\varrho$, and for every $x \in \mathbb{R}$

$$a \sin x + b \cos x = \varrho \sin(x + \varphi).$$

In particular $a \sin x + b \cos x$ has maximum ϱ and minimum $-\varrho$.

Proof. The point $(a/\varrho, b/\varrho)$ has norm 1, hence lies on the unit circle; by Definition 8.1 there is one and only one $\varphi \in [0, 2\pi)$ with $(\cos \varphi, \sin \varphi) = (a/\varrho, b/\varrho)$. Then, by the addition formula (Theorem 8.4),

$$\varrho \sin(x + \varphi) = \varrho(\sin x \cos \varphi + \cos x \sin \varphi) = \varrho\left(\sin x \cdot \frac{a}{\varrho} + \cos x \cdot \frac{b}{\varrho}\right) = a \sin x + b \cos x.$$

Since $\sin$ attains all and only the values in $[-1, 1]$, the expression has maximum ϱ (attained for $x + \varphi = \pi/2 + 2k\pi$) and minimum $-\varrho$. $\square$

Remark 9.12: The phasor, before complex numbers. Theorem 9.12 says that the sum of two sinusoids of the same frequency in quadrature is again a sinusoid of that frequency, with amplitude $\sqrt{a^2 + b^2}$ and phase $\varphi = \text{atan2}(b, a)$: this is the principle of the phasor, that is, the vector sum of two components in quadrature. It will reappear, in complex and general form, in Volume II; but the mathematical content is already entirely here, and it requires no complex numbers.

9.5.5 Elementary goniometric equations

Proposition 9.3: General solutions of the elementary equations. Let $k \in \mathbb{Z}$.

1. If $|a| \leq 1$, the equation $\sin x = a$ has solutions $x = \arcsin a + 2k\pi$ or $x = \pi - \arcsin a + 2k\pi$; if $|a| > 1$ it has none.
2. If $|a| \leq 1$, the equation $\cos x = a$ has solutions $x = \pm \arccos a + 2k\pi$; if $|a| > 1$ it has none.
3. For every $a \in \mathbb{R}$, the equation $\tan x = a$ has solutions $x = \arctan a + k\pi$.

Proof. (1). If $|a| > 1$ there are no solutions because $|\sin x| \leq 1$ for every x (Theorem 8.2). So let $|a| \leq 1$ and $x_0 = \arcsin a \in [-\pi/2, \pi/2]$ (Definition 8.2), which satisfies $\sin x_0 = a$. By Proposition 9.2, that is by $\sin(\pi - \theta) = \sin \theta$, the value $\pi - x_0$ is also a solution, and by periodicity so are all the values $x_0 + 2k\pi$ and $\pi - x_0 + 2k\pi$. Conversely there are no others: on $[-\pi/2, \pi/2]$ the sine is strictly increasing, hence it takes the value a only once, and on $[\pi/2, 3\pi/2]$ it is strictly decreasing, hence takes it only once, at $\pi - x_0$; on an interval of length 2π the solutions are therefore exactly two (only one if $a = \pm 1$, when the two coincide).

(2). Analogously, $\arccos a \in [0, \pi]$ is a solution and the cosine is even (Proposition 9.2), hence $-\arccos a$ is one too; the cosine is strictly decreasing on $[0, \pi]$ and strictly increasing on $[\pi, 2\pi]$, hence over one period there are exactly two solutions.

(3). The tangent is strictly increasing and surjective on $(-\pi/2, \pi/2)$, where it takes every real value exactly once, at $\arctan a$; it has period π by Proposition 9.2 ($\tan(\pi + \theta) = \tan \theta$), hence the solutions are exactly those listed. $\square$

Remark 9.13: The commonest mistake: forgetting the second family. When solving $\sin x = a$ the calculator returns a single value, $\arcsin a$, because the arcsine is by definition the inverse of the restriction of the sine to $[-\pi/2, \pi/2]$ (Definition 8.2). The second family $x = \pi - \arcsin a + 2k\pi$ is not an arbitrary addition: it comes from the identity $\sin(\pi - \theta) = \sin \theta$ of Proposition 9.2, that is from the symmetry of the unit circle about the y axis. Likewise, for the cosine the symmetry is about the x axis and gives the family $-\arccos a$. Whoever remembers which symmetry is at work does not forget the second family.

Theorem 9.13: Heron's formula. A triangle of sides a, b, c and semiperimeter $s = (a + b + c)/2$ has area

$$T = \sqrt{s(s-a)(s-b)(s-c)}.$$

Proof. Let h be the altitude relative to side c , dividing it into two segments p, q with $p + q = c$ (if the foot of the altitude falls outside the side, $p - q = c$ and the following computation is identical up to a sign). By Pythagoras in the two right triangles, $b^2 - p^2 = h^2 = a^2 - q^2$; subtracting and using $q = c - p$:

$$b^2 - a^2 = p^2 - q^2 = (p - q)(p + q) = (p - q)c \implies p = \frac{b^2 + c^2 - a^2}{2c}$$

(from the first equality $p - q = (b^2 - a^2)/c$, and $p + q = c$; adding the two gives $2p = (b^2 - a^2)/c + c = (b^2 + c^2 - a^2)/c$). Then

$$16T^2 = 4c^2h^2 = 4c^2(b^2 - p^2) = 4b^2c^2 - (2cp)^2 = 4b^2c^2 - (b^2 + c^2 - a^2)^2,$$

a difference of squares:

$$16T^2 = [2bc - (b^2 + c^2 - a^2)][2bc + (b^2 + c^2 - a^2)] = [a^2 - (b - c)^2][(b + c)^2 - a^2].$$

Factoring again as a difference of squares:

$$16T^2 = (a - b + c)(a + b - c)(b + c - a)(b + c + a) = 16s(s-a)(s-b)(s-c)$$

(with $s = (a + b + c)/2$: $a - b + c = 2(s - b)$, $a + b - c = 2(s - c)$, $b + c - a = 2(s - a)$, $a + b + c = 2s$, direct check), whence the claim by taking the square root. $\square$

Numerical check. Sides 7, 8, 9: $s = 12$, $T = \sqrt{12 \cdot 5 \cdot 4 \cdot 3} = \sqrt{720} = 26.832816$ (checked). The three citations of Heron's formula in this chapter now refer to Theorem 9.13.

9.6. Linear equations in sine and cosine

Theorem 9.14: Equation $a \sin x + b \cos x = c$. Let a, b not both zero. The equation $a \sin x + b \cos x = c$ has solutions if and only if $c^2 \leq a^2 + b^2$.

Proof. Let $\rho = \sqrt{a^2 + b^2} > 0$. The point $(a/\rho, b/\rho)$ lies on the unit circle, so there exists (definition of the unit circle) an angle φ with $\cos \varphi = a/\rho$, $\sin \varphi = b/\rho$. Then, by the addition

formula,

$$a \sin x + b \cos x = \rho \left(\frac{a}{\rho} \sin x + \frac{b}{\rho} \cos x \right) = \rho (\cos \varphi \sin x + \sin \varphi \cos x) = \rho \sin(x + \varphi).$$

The equation becomes $\sin(x + \varphi) = c/\rho$, solvable if and only if $|c/\rho| \leq 1$ (since $\sin$ is bounded between -1 and 1 , by definition), that is, $c^2 \leq \rho^2 = a^2 + b^2$; in that case $x = -\varphi + \arcsin(c/\rho) + 2k\pi$ or $x = \pi - \varphi - \arcsin(c/\rho) + 2k\pi$. $\square$

Alternative method: parametric substitution. With $t = \tan(x/2)$ (Theorem 7.10, Chapter 7), $\sin x = 2t/(1+t^2)$, $\cos x = (1-t^2)/(1+t^2)$: the equation becomes

$$\frac{2at}{1+t^2} + \frac{b(1-t^2)}{1+t^2} = c \iff (b+c)t^2 - 2at + (c-b) = 0,$$

a quadratic equation in t (if $b+c \neq 0$; the case $b+c = 0$ corresponds to $x = \pi$, to be checked separately, since the substitution $t = \tan(x/2)$ is not defined at $x = \pi$). The discriminant is $4a^2 - 4(b+c)(c-b) = 4(a^2 + b^2 - c^2)$, positive exactly when $c^2 < a^2 + b^2$: the same existence condition, recovered by purely algebraic means. $\square$

Example 9.3: Numerical check. $a = 3, b = 4, c = 2$: $\rho = 5$, $\varphi = \arctan(4/3) = 53.130^\circ$. The two solutions in $[0^\circ, 360^\circ)$ are $x_1 = 330.448^\circ$ and $x_2 = 103.292^\circ$ (checked: $3 \sin x_i + 4 \cos x_i = 2.000000$ in both cases).

9.7. Intersections of the circle with other figures

INFO

The problem. Given two overlapping plane figures, how large is the common part? And how large is the part left outside? When one of the two is a disc the answer is always computable in closed form, provided the other is bounded by straight lines or by circular arcs.

The method, one only. Every area in this section is obtained with the same three-move scheme: (i) cut the figures along the lines joining the centres and the intersection points; (ii) each piece turns out to be a *sector* or a *triangle*; (iii) add up. Nothing else is needed: no integrals, no analytic geometry.

Why it pays to know how. The part common to two discs turns up everywhere: in the overlap of two radio coverage areas, in a Venn diagram drawn to scale, in the area grazed by a goat tethered to a circular fence, in the doubly exposed zone of two spray nozzles, in the crossing of two bores in mechanics.

9.7.1 Circle and line: the building block of everything

Theorem 9.15: Area of a disc cut by a line. Let D be the disc of centre O and radius r , and let s be a line at distance $\delta \geq 0$ from O . Denoting by H^- the closed half-plane bounded by s that contains O :

$$\text{area}(D \cap H^-) = \begin{cases} \pi r^2, & \delta \geq r, \\ \pi r^2 - \left[r^2 \arccos \frac{\delta}{r} - \delta \sqrt{r^2 - \delta^2} \right], & 0 \leq \delta < r, \end{cases}$$

and the area of the part of D in the other half-plane is the circular segment

$$\text{area}(D \cap H^+) = r^2 \arccos \frac{\delta}{r} - \delta \sqrt{r^2 - \delta^2}.$$

Proof. If $\delta \geq r$ the line is tangent or external (Theorem 2.6), hence D lies entirely in H^- and the area is that of the whole disc.

If $\delta < r$ the line is secant and cuts D into two parts: the one in H^+ is by definition the minor circular segment (Definition 2.18), whose area is given by Theorem 9.4 in its second form, $r^2 \arccos(\delta/r) - \delta \sqrt{r^2 - \delta^2}$. The part in H^- is its complement in D , hence its area is the difference. The two parts are disjoint up to the chord, which has zero area, so the areas add. $\square$

Remark 9.14: One building block, and the tank formula. Theorem 9.15 is the block from which the rest of the section is built, and it already has a classical application of its own: the volume of liquid in a horizontal cylindrical tank of radius r and length L , filled to height $H \in [0, 2r]$, equals $V = L \cdot A_{\text{seg}}$ with $\delta = |r - H|$, and precisely

$$V = L \left[r^2 \arccos \frac{r - H}{r} - (r - H) \sqrt{2rH - H^2} \right],$$

where $\sqrt{r^2 - (r - H)^2} = \sqrt{2rH - H^2}$ has been used. The formula also holds for $H > r$, where $r - H < 0$ and the arc cosine exceeds $\pi/2$: no case distinction is needed. For $H = r$ it gives $V = L \cdot r^2 \cdot \pi/2$, that is half the cylinder, as it must.

9.7.2 Two circles: mutual positions and intersection points

Theorem 9.16: Mutual positions of two circles. Let Γ_1 have centre O_1 and radius r , Γ_2 centre O_2 and radius R , with $d = O_1O_2$. Then:

- (a) $d > r + R$: the circles are external, no common point;
- (b) $d = r + R$: externally tangent, one common point;
- (c) $|R - r| < d < r + R$: secant, two distinct common points;
- (d) $d = |R - r| > 0$: internally tangent, one common point;
- (e) $0 \leq d < |R - r|$: one inside the other, no common point;
- (f) $d = 0$ and $r = R$: coincident.

Proof. A point P belongs to both if and only if $O_1P = r$ and $O_2P = R$. By the triangle inequality applied to O_1, O_2, P one has $|O_1P - O_2P| \leq O_1O_2 \leq O_1P + O_2P$, that is $|r - R| \leq d \leq r + R$: outside this interval no such P exists, which gives (a) and (e). In the two extreme cases the triangle inequality is an equality, which happens if and only if the three points are collinear, and then P is uniquely determined: one gets (b) and (d). If $|r - R| < d < r + R$, choose coordinates with $O_1 = (0, 0)$ and $O_2 = (d, 0)$; subtracting the two equations

$$x^2 + y^2 = r^2, \quad (x - d)^2 + y^2 = R^2$$

gives $2dx - d^2 = r^2 - R^2$, that is the single value

$$x = a := \frac{d^2 + r^2 - R^2}{2d},$$

and substituting, $y^2 = r^2 - a^2$. Two distinct solutions remain if and only if $r^2 - a^2 > 0$, that is $|a| < r$: let us verify it. One has

$$r^2 - a^2 = \frac{4d^2r^2 - (d^2 + r^2 - R^2)^2}{4d^2} = \frac{(2dr - d^2 - r^2 + R^2)(2dr + d^2 + r^2 - R^2)}{4d^2},$$

and recognising the squares,

$$r^2 - a^2 = \frac{[R^2 - (d - r)^2][(d + r)^2 - R^2]}{4d^2} = \frac{(R - d + r)(R + d - r)(d + r - R)(d + r + R)}{4d^2}.$$

Under the hypothesis $|r - R| < d < r + R$ the four factors are all positive, hence $r^2 - a^2 > 0$ and there are two solutions, symmetric about the line of centres. Finally (f) is immediate. $\square$

Corollary 9.4: Coordinates of the intersection points. In the secant case, with $O_1 = (0, 0)$ and $O_2 = (d, 0)$, the two common points are $(a, \pm h)$ with

$$a = \frac{d^2 + r^2 - R^2}{2d}, \quad h = \sqrt{r^2 - a^2} = \frac{2T}{d},$$

where $T = \frac{1}{4}\sqrt{(d + r + R)(-d + r + R)(d - r + R)(d + r - R)}$ is the area of the triangle O_1O_2P (Heron's formula, Theorem 9.13, with sides d, r, R). The line $x = a$ is the radical axis of the two circles (Proposition 7.1).

Proof. The expressions for a and for $h^2 = r^2 - a^2$ were obtained in the previous proof. From the second-to-last formula there,

$$h = \frac{\sqrt{(-d + r + R)(d - r + R)(d + r - R)(d + r + R)}}{2d} = \frac{4T}{2d} = \frac{2T}{d},$$

where the last equality is Heron's formula (Theorem 9.13) for the triangle of sides d, r, R . That $x = a$ is the radical axis follows from the fact that the equation $2dx - d^2 = r^2 - R^2$ is precisely the difference of the two circle equations, that is the equality of the two powers (Theorem 7.6). $\square$

9.7.3 The area of the lens: the central theorem

Theorem 9.17: Area of the intersection of two discs (lens). Let D_1 be the disc of centre O_1 and radius r , D_2 the disc of centre O_2 and radius R , with $d = O_1O_2$ and $|R - r| < d < r + R$. Then

$$\text{area}(D_1 \cap D_2) = r^2 \arccos\left(\frac{d^2 + r^2 - R^2}{2dr}\right) + R^2 \arccos\left(\frac{d^2 + R^2 - r^2}{2dR}\right) - \frac{1}{2}\sqrt{\Lambda}$$

with

$$\Lambda = (-d + r + R)(d + r - R)(d - r + R)(d + r + R).$$

Proof. Let P and Q be the two intersection points, which by Corollary 9.4 are symmetric about the line of centres, and let M be their midpoint, which lies on that line. Set

$$d_1 = O_1M = \frac{d^2 + r^2 - R^2}{2d}, \quad d_2 = O_2M = d - d_1 = \frac{d^2 + R^2 - r^2}{2d}, \quad h = MP = MQ.$$

(The sign of d_1 or d_2 may be negative if M falls outside the segment O_1O_2 ; the proof that follows remains valid, as verified at the end.)

Step 1 — decomposition. The chord $\overline{PQ}$ is common to the two discs and divides the lens $D_1 \cap D_2$ into two parts:

- the part on the same side as O_2 with respect to PQ , which is the circular segment of D_1 subtended by the chord $\overline{PQ}$;
- the part on the same side as O_1 , which is the circular segment of D_2 subtended by the same chord.

That this is so is seen from $D_1 \cap D_2 \cap H = D_1 \cap H$ for the half-plane H opposite to O_1 (that is, $x \geq d_1$, in coordinates with $O_1 = (0, 0)$, $O_2 = (d, 0)$ on the x -axis): indeed, in that half-plane every point of D_1 belongs to D_2 as well, because if $(x, y) \in D_1$ (that is $x^2 + y^2 \leq r^2$, so $y^2 \leq r^2 - x^2$) and $x \geq d_1 = (d^2 + r^2 - R^2)/(2d)$, then

$$(x - d)^2 + y^2 \leq (x - d)^2 + r^2 - x^2 = d^2 + r^2 - 2dx \leq d^2 + r^2 - (d^2 + r^2 - R^2) = R^2,$$

that is, $(x, y) \in D_2$. The two parts are disjoint up to the chord, hence the areas add.

Step 2 — the two segments. By Theorem 9.4, second form, with the distance from the centre to the chord equal to d_1 and d_2 respectively:

$$A_1 = r^2 \arccos \frac{d_1}{r} - d_1 \sqrt{r^2 - d_1^2}, \quad A_2 = R^2 \arccos \frac{d_2}{R} - d_2 \sqrt{R^2 - d_2^2}.$$

Since $\sqrt{r^2 - d_1^2} = \sqrt{R^2 - d_2^2} = h$ (both equal the half-chord), adding gives

$$\text{area}(D_1 \cap D_2) = r^2 \arccos \frac{d_1}{r} + R^2 \arccos \frac{d_2}{R} - h(d_1 + d_2) = r^2 \arccos \frac{d_1}{r} + R^2 \arccos \frac{d_2}{R} - h d.$$

Step 3 — the algebraic term. The product hd is twice the area of the triangle $O_1 O_2 P$, which has base d and height h : hence $hd = 2T$, and by Heron's formula (Theorem 9.13) applied to the sides d, r, R ,

$$hd = 2T = \frac{1}{2} \sqrt{(d + r + R)(-d + r + R)(d - r + R)(d + r - R)} = \frac{1}{2} \sqrt{\Lambda}.$$

Step 4 — substitution. Finally $d_1/r = (d^2 + r^2 - R^2)/(2dr)$ and $d_2/R = (d^2 + R^2 - r^2)/(2dR)$, which substituted give the statement.

The case $d_1 < 0$. If $R > \sqrt{d^2 + r^2}$ then $d_1 < 0$: the centre O_1 lies on the *same* side as O_2 with respect to the chord (both with abscissa $> d_1$, since d_1 is negative and O_2 is at $x = d > 0$), and the part of the lens inside D_1 is the *major* segment. The second form of Theorem 9.4 continues to hold: the arc cosine of a negative argument exceeds $\pi/2$, and the term $-d_1 h$ becomes positive, adding instead of subtracting — which is exactly what is needed for the major segment. The final formula therefore requires no case distinction, and may be applied as it stands for every admissible d . $\square$

Corollary 9.5: The outer areas: union, lune and symmetric difference. With the notation of Theorem 9.17, setting $S_\cap = \text{area}(D_1 \cap D_2)$:

$$\begin{aligned} \text{area}(D_1 \cup D_2) &= \pi r^2 + \pi R^2 - S_\cap, \\ \text{area}(D_1 \setminus D_2) &= \pi r^2 - S_\cap \quad (\text{the lune of } D_1), \quad \text{area}(D_2 \setminus D_1) = \pi R^2 - S_\cap, \\ \text{area}((D_1 \cup D_2) \setminus (D_1 \cap D_2)) &= \pi r^2 + \pi R^2 - 2S_\cap. \end{aligned}$$

Proof. All four are the inclusion–exclusion principle for the Lebesgue measure in the plane, which for two measurable sets reads $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$; for the second, $D_1 = (D_1 \setminus D_2) \sqcup (D_1 \cap D_2)$ is a disjoint union, hence $\mu(D_1 \setminus D_2) = \mu(D_1) - \mu(D_1 \cap D_2)$. The fourth is the sum of the two lunes. $\square$

Corollary 9.6: The symmetric case and the vesica piscis. If $r = R$ the formula of Theorem 9.17 simplifies to

$$\text{area}(D_1 \cap D_2) = 2r^2 \arccos \frac{d}{2r} - \frac{d}{2} \sqrt{4r^2 - d^2}.$$

In particular, for $d = r$ (each circle passes through the other's centre) one obtains the **vesica piscis**

$$\text{area} = r^2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \approx 1.228370 r^2,$$

that is 39.10% of the area of either disc.

Proof. With $r = R$ one has $d_1 = d_2 = d/2$ and the two arc cosines coincide in $\arccos(d/(2r))$, giving the first term $2r^2 \arccos(d/(2r))$. For the algebraic term,

$$\Lambda = (-d + 2r)(d)(d)(d + 2r) = d^2(4r^2 - d^2), \quad \frac{1}{2}\sqrt{\Lambda} = \frac{d}{2}\sqrt{4r^2 - d^2}.$$

For $d = r$: $\arccos(1/2) = \pi/3$, so the first term is $2r^2\pi/3$ and the second $\frac{r}{2}\sqrt{3r^2} = \frac{\sqrt{3}}{2}r^2$. Numerically $2\pi/3 - \sqrt{3}/2 = 2.094395 - 0.866025 = 1.228370$, and $1.228370/\pi = 0.391002$. $\square$

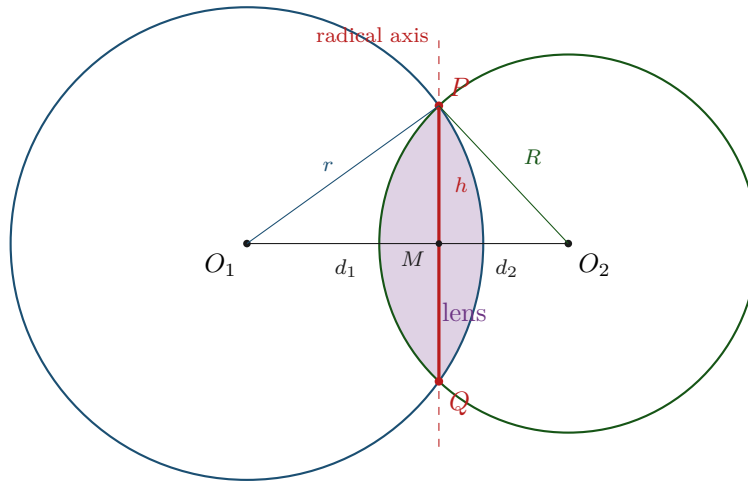


Figure 9.4: — The lens $D_1 \cap D_2$: the common chord $\overline{PQ}$ splits it into two circular segments, one for each circle. The line PQ is the radical axis

Example 9.4: Two secant circles, all areas computed. Let $r = 5$, $R = 4$, $d = 6$. Since $|R - r| = 1 < 6 < 9 = r + R$, the circles are secant.

Geometry.

$$d_1 = \frac{36 + 25 - 16}{12} = \frac{45}{12} = 3.75, \quad d_2 = 6 - 3.75 = 2.25,$$

$$h = \sqrt{25 - 14.0625} = \sqrt{10.9375} = 3.307189.$$

Check from the other circle: $\sqrt{16 - 5.0625} = \sqrt{10.9375}$: it matches. With $O_1 = (0, 0)$ and $O_2 = (6, 0)$ the common points are $(3.75, \pm 3.307189)$.

The two segments.

$$\begin{aligned} A_1 &= 25 \arccos(0.75) - 3.75 \cdot 3.307189 \\ &= 18.068356 - 12.401961 = 5.666395, \end{aligned}$$

$$\begin{aligned} A_2 &= 16 \arccos(0.5625) - 2.25 \cdot 3.307189 \\ &= 15.574238 - 7.441175 = 8.133063. \end{aligned}$$

Lens. $S_{\cap} = A_1 + A_2 = 5.666395 + 8.133063 = 13.799458$.

Check with the closed formula. The four factors are $(-d + r + R) = 3$, $(d + r - R) = 7$, $(d - r + R) = 5$, $(d + r + R) = 15$, hence $\Lambda = 3 \cdot 7 \cdot 5 \cdot 15 = 1575$ and

$$\frac{1}{2}\sqrt{1575} = \frac{1}{2} \cdot 39.686269 = 19.843134,$$

which matches $h d = 3.307189 \cdot 6 = 19.843134$, as Step 3 of the proof requires. Then

$$18.068356 + 15.574238 - 19.843134 = 13.799460,$$

equal to the value obtained by adding the two segments, up to rounding.

Outer areas. With $\pi r^2 = 78.539816$ and $\pi R^2 = 50.265482$:

$$\text{area}(D_1 \cup D_2) = 78.539816 + 50.265482 - 13.799458 = 115.005840,$$

$$\text{area}(D_1 \setminus D_2) = 78.539816 - 13.799458 = 64.740358, \quad \text{area}(D_2 \setminus D_1) = 36.466024.$$

Check: $64.740358 + 36.466024 + 13.799458 = 115.005840$, equal to the union. The lens occupies 17.57% of the large disc and 27.45% of the small one.

Warning. The four factors of Λ differ only in the placement of the signs, and it is extremely easy to repeat one of them. In the example they are

$$-d + r + R = 3, \quad d + r - R = 7, \quad d - r + R = 5, \quad d + r + R = 15,$$

and $\Lambda = 1575$. Writing $d + r - R$ twice by mistake in place of $d - r + R$ gives $3 \cdot 7 \cdot 7 \cdot 15 = 2205$, that is an algebraic term of 23.478714 instead of 19.843134 and an area of 10.163880 instead of 13.799458: an error of 26.3%.

The check that always catches it is this: one must find $\frac{1}{2}\sqrt{\Lambda} = h d$, where h is the half-chord already computed twice (once from r and once from R). Here $h d = 3.307189 \cdot 6 = 19.843134$ and $\frac{1}{2}\sqrt{1575} = 19.843134$: they agree.

Example 9.5: The goat problem. A goat is tethered by a rope of length L to a post planted *on the boundary* of a circular field of radius ρ . Which L lets it graze exactly half the field?

The accessible area is the intersection between the field D (centre O , ρ) and the disc of radius L centred at the post B , with $OB = \rho$. Applying Theorem 9.17 with $r = L$, $R = \rho$ and $d = \rho$:

$$\Lambda = (-\rho + L + \rho)(\rho + L - \rho)(\rho - L + \rho)(\rho + L + \rho) = L^2(2\rho - L)(2\rho + L) = L^2(4\rho^2 - L^2),$$

$$\frac{d^2 + r^2 - R^2}{2dr} = \frac{\rho^2 + L^2 - \rho^2}{2\rho L} = \frac{L}{2\rho}, \quad \frac{d^2 + R^2 - r^2}{2dR} = \frac{2\rho^2 - L^2}{2\rho^2} = 1 - \frac{L^2}{2\rho^2},$$

hence

$$S_{\cap}(L) = L^2 \arccos \frac{L}{2\rho} + \rho^2 \arccos \left(1 - \frac{L^2}{2\rho^2}\right) - \frac{L}{2} \sqrt{4\rho^2 - L^2}.$$

Setting $\rho = 1$ and $k = L/\rho$, the condition $S_{\cap} = \pi/2$ becomes

$$f(k) := k^2 \arccos \frac{k}{2} + \arccos \left(1 - \frac{k^2}{2}\right) - \frac{k}{2} \sqrt{4 - k^2} - \frac{\pi}{2} = 0.$$

The function f is continuous and strictly increasing on $(0, 2)$ (the area grows with L), with $f(1) = -0.342427$ and $f(1.3) = +0.315286$: there is therefore a unique root between 1 and 1.3. Applying Newton's method (Definition 6.5) with $f'(k) = 2k \arccos(k/2) - \frac{k}{2} \sqrt{4 - k^2}$ — the three remaining terms cancel in pairs when the derivative is computed — one obtains

$$k_0 = 1.2, \quad k_1 = 1.158881, \quad k_2 = 1.158728, \quad k_3 = 1.158728.$$

The value, known in the literature as the *goat constant*, is

$$\frac{L}{\rho} = 1.1587284730 \dots$$

The rope must therefore be about 15.9% longer than the radius: far more than intuition suggests, which tends to propose $L = \rho$ (which yields only 39.1% of the field, by Corollary 9.6). A first closed-form expression for the number was obtained by Ullisch (2020) as a quotient of two contour integrals; whether it is algebraic or transcendental is not known.

9.7.4 Circle and polygon: the general formula

Lemma 9.1: *Disc intersected with a triangle having a vertex at the centre.* Let D be the disc of centre O and radius ρ , and let $T = OAB$ be a triangle with one vertex at the centre. Let $\psi \in [0, \pi]$ be the angle $\widehat{AOB}$. Then $\text{area}(D \cap T)$ is computed as follows: consider the segment $\overline{AB}$ parametrised as $X(u) = A + u(B - A)$, $u \in [0, 1]$, and let $[u_1, u_2] \subseteq [0, 1]$ be the (possibly empty) set of u for which $|X(u) - O| \leq \rho$. Writing $P_1 = X(u_1)$, $P_2 = X(u_2)$, and ψ_1, ψ_2, ψ_3 for the central angles of $\widehat{AOP_1}$, $\widehat{P_1OP_2}$, $\widehat{P_2OB}$ respectively, one has

$$\text{area}(D \cap T) = \frac{1}{2}\rho^2\psi_1 + \frac{1}{2}|\det(P_1 - O, P_2 - O)| + \frac{1}{2}\rho^2\psi_3,$$

and if the segment does not meet D then $\text{area}(D \cap T) = \frac{1}{2}\rho^2\psi$.

Proof. If O lies on the line AB the triangle is degenerate, has zero area and the formula holds trivially with $\psi = 0$ or $\psi = \pi$; so assume $O \notin AB$. The triangle T is star-shaped with respect to O , and so is D ; hence so is $D \cap T$. Use polar coordinates with pole O : for every direction φ interior to the angle $\widehat{AOB}$, the ray from O meets T in the segment $[0, \sigma(\varphi)]$, where $\sigma(\varphi)$ is the distance from O to the point where the ray meets the line AB , and meets D in the segment $[0, \rho]$. Therefore $D \cap T$ is described in polar form by $0 \leq t \leq \min\{\rho, \sigma(\varphi)\}$, and

$$\text{area}(D \cap T) = \frac{1}{2} \int_{\varphi_A}^{\varphi_B} \min\{\rho, \sigma(\varphi)\}^2 d\varphi.$$

Since σ is continuous and the function $u \mapsto |X(u) - O|$ is strictly convex (it is the square root of a quadratic polynomial with positive leading coefficient), the set $\{\sigma \leq \rho\}$ is a single angular interval, which corresponds exactly to $[u_1, u_2]$, that is to the angle $\widehat{P_1OP_2}$ of measure ψ_2 . On it $\min = \sigma$ and the integral equals the area of the triangle OP_1P_2 , that is $\frac{1}{2}|\det(P_1 - O, P_2 - O)|$. On the two complementary intervals, of amplitudes ψ_1 and ψ_3 , one has $\min = \rho$ and the integral equals $\frac{1}{2}\rho^2\psi_1$ and $\frac{1}{2}\rho^2\psi_3$: these are two circular sectors. If the segment does not meet D , the central interval is empty and only the sector of measure ψ remains. $\square$

INFO

Advanced section: may be skipped. The general statement that follows uses integrals over $\mathbb{R}^2$, sets of measure zero and the winding number of a curve about a point: notions from Volume II. The case needed in practice — a **convex** polygon with the centre O of the disc **interior** to the polygon — is proved in full, in a completely elementary way, immediately afterwards, and it is the one that suffices for the examples and exercises of this chapter.

Theorem 9.18: *Area of the intersection between a disc and a polygon.* Let P be a simple polygon with vertices $V_1, \dots, V_n$ traversed anticlockwise (indices modulo n) and let

D be the disc of centre O and radius ρ . Then

$$\text{area}(D \cap P) = \sum_{i=1}^n \varepsilon_i \cdot \text{area}(D \cap T_i), \quad T_i = \text{triangle } OV_iV_{i+1},$$

where $\varepsilon_i = \text{sgn det}(V_i - O, V_{i+1} - O)$, and each summand is computed with Lemma 9.1.

Proof. [sketch, which imports the signed fan formula] The complete proof in the general case (any simple polygon, not necessarily convex, with O in an arbitrary position) relies on the signed fan formula (equivalent to the Gauss/shoelace area theorem for simple polygons): it is **imported** from this point onward, source Braden, *The surveyor's area formula*, College Mathematics Journal 17(4) (1986), 326–337. We give its sketch; Corollary 9.7 below, which covers the case actually used, is instead proved in full without importing anything.

For every point x of the plane not lying on the sides nor on the rays OV_i , the signed fan identity holds:

$$\sum_{i=1}^n \varepsilon_i \mathbf{1}_{T_i}(x) = \mathbf{1}_P(x).$$

It is verified as follows: for each side V_iV_{i+1} , the left-hand side contributes $+1$ or -1 according to whether the side is seen from O anticlockwise or clockwise, and x falls in T_i exactly when it lies between O and that side. The telescoping sum of these contributions is the winding number of the boundary of P about x , which for a simple polygon traversed anticlockwise equals 1 if x is interior and 0 if it is exterior.

Multiplying the identity by $\mathbf{1}_D(x)$ and integrating over $\mathbb{R}^2$ (the functions are bounded with compact support, and the exceptional sets have measure zero) one obtains

$$\sum_{i=1}^n \varepsilon_i \mu(D \cap T_i) = \mu(D \cap P),$$

which is the assertion. $\square$

Corollary 9.7: Elementary case: convex polygon, interior centre. If P is **convex** and $O \in P$, the triangles $T_1, \dots, T_n$ form a partition of P (with no signs: $\varepsilon_i = 1$ for every i), and

$$\text{area}(D \cap P) = \sum_{i=1}^n \text{area}(D \cap T_i).$$

Proof. For O interior to the convex polygon P , the segments $OV_1, \dots, OV_n$ divide P into n triangles $T_i = OV_iV_{i+1}$ pairwise without common interior points (every interior point of P other than O lies in exactly one of them, determined by the angle $\angle V_iOV_{i+1}$ into which it falls; this is the elementary definition of a triangular fan from an interior point of a convex polygon, with no need for signs or winding numbers). By additivity of area under disjoint unions (up to sets of measure zero, the common sides), $\text{area}(P) = \sum_i \text{area}(T_i)$, and likewise, intersecting each piece with D , $\text{area}(D \cap P) = \sum_i \text{area}(D \cap T_i)$: every point of $D \cap P$ lies in D and in exactly one of the T_i , and conversely. Each term is computed with Lemma 9.1, with no integrals or winding numbers. $\square$

Remark 9.15: Why this theorem is an algorithm. Theorem 9.18 is not merely a statement: it is the standard algorithm by which computational systems determine the overlap area between a disc and a polygonal mesh — in the computation of volume fractions in finite-volume methods, in antialiased rasterisation, in computing the coverage of radio cells

over a territory partitioned into administrative polygons. The cost is linear in the number of sides, it does not require determining the cyclic order of the intersection points along the boundary, and it works for concave polygons and for discs that do not contain the polygon's centre as well: all advantages that a “by hand” decomposition lacks.

Example 9.6: Concentric circle and square. Let S be the square $[0, 3] \times [0, 3]$ and D the disc of centre $(1.5, 1.5)$ and radius $\rho = 2$.

Geometry. The distance from the centre to each side is $1.5 < 2$: the disc protrudes beyond all four sides. The distance from the centre to each vertex is $\sqrt{1.5^2 + 1.5^2} = 2.121320 > 2$: the vertices are outside the disc, hence the four protruding parts are pairwise disjoint.

Computation. Each protruding part is the circular segment at distance $\delta = 1.5$, which by Theorem 9.15 equals

$$4 \arccos \frac{1.5}{2} - 1.5\sqrt{4 - 2.25} = 4 \cdot 0.7227342 - 1.5 \cdot 1.3228757 = 2.890937 - 1.984314 = 0.906623.$$

Hence

$$\text{area}(D \cap S) = 4\pi - 4 \cdot 0.906623 = 12.566371 - 3.626494 = 8.939877.$$

Outer areas. The area of the square not covered by the disc is $9 - 8.939877 = 0.060123$ (the four corners); the area of the disc outside the square is $12.566371 - 8.939877 = 3.626494$, that is the sum of the four segments, as it must be.

Check with Theorem 9.18. The four triangles T_i all have their vertex at the centre and are traversed anticlockwise, so $\varepsilon_i = +1$; each has $\psi = \pi/2$ and the chord intercepted on the side has length $2\sqrt{4 - 2.25} = 2.645751$, centred on the side. Each $D \cap T_i$ then equals a triangle OP_1P_2 of base 2.645751 and height 1.5, that is 1.984314, plus two sectors of measure $\frac{1}{2}(\pi/2) - \arccos(1.5/2) = 0.785398 - 0.722734 = 0.062664$ each, that is $2 \cdot \frac{1}{2} \cdot 4 \cdot 0.062664 = 0.250656$. Total per triangle: 2.234970; for four triangles: 8.939879. It matches the previous value up to rounding.

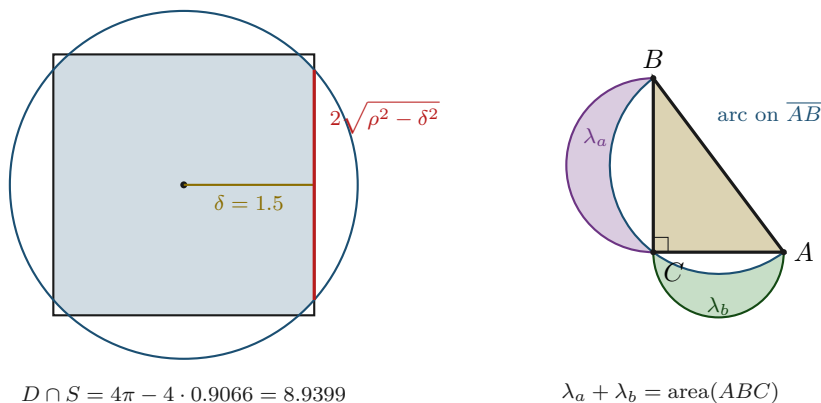


Figure 9.5: — Left: concentric circle and square, with the four protruding segments. Right: the lunes of Hippocrates on a right triangle

9.7.5 Circle and regular polygons: the residual areas

Proposition 9.4: Residual areas between circle and regular polygon. Let Γ be the disc of radius r , P_n the inscribed regular n -gon and P_n^* the circumscribed one. Then

$$\text{area}(\Gamma \setminus P_n) = r^2 \left(\pi - \frac{n}{2} \sin \frac{2\pi}{n} \right), \quad \text{area}(P_n^* \setminus \Gamma) = r^2 \left(n \tan \frac{\pi}{n} - \pi \right),$$

and the first is the sum of the n circular segments subtended by the sides, each of area $\frac{1}{2}r^2(\frac{2\pi}{n} - \sin \frac{2\pi}{n})$. In particular:

$$\begin{aligned} n = 3 : \quad \text{area}(\Gamma \setminus P_3) &= r^2(\pi - \frac{3\sqrt{3}}{4}) \approx 1.842555 r^2, & \text{area}(P_3^* \setminus \Gamma) &\approx 2.054559 r^2; \\ n = 4 : \quad \text{area}(\Gamma \setminus P_4) &= r^2(\pi - 2) \approx 1.141593 r^2, & \text{area}(P_4^* \setminus \Gamma) &= r^2(4 - \pi) \approx 0.858407 r^2; \\ n = 6 : \quad \text{area}(\Gamma \setminus P_6) &= r^2(\pi - \frac{3\sqrt{3}}{2}) \approx 0.543516 r^2, & \text{area}(P_6^* \setminus \Gamma) &\approx 0.322509 r^2. \end{aligned}$$

Proof. The inscribed polygon is contained in the disc, hence by additivity $\text{area}(\Gamma \setminus P_n) = \pi r^2 - A_n$ with A_n given by Theorem 9.8, that is $\pi r^2 - \frac{1}{2}nr^2 \sin(2\pi/n)$. Analogously the disc is contained in the circumscribed polygon and $\text{area}(P_n^* \setminus \Gamma) = nr^2 \tan(\pi/n) - \pi r^2$.

For the second assertion: the n sides of the inscribed polygon are chords subtending central angles $2\pi/n$, and the corresponding circular segments fill $\Gamma \setminus P_n$ without overlap (two consecutive segments share only a vertex). Each has area $\frac{1}{2}r^2(\frac{2\pi}{n} - \sin \frac{2\pi}{n})$ by Theorem 9.4, and their sum is $\frac{1}{2}r^2(2\pi - n \sin \frac{2\pi}{n}) = \pi r^2 - \frac{1}{2}nr^2 \sin \frac{2\pi}{n}$, consistent with the above.

The numerical values follow from $\sin(2\pi/3) = \sqrt{3}/2$, $\sin(\pi/2) = 1$, $\sin(\pi/3) = \sqrt{3}/2$, and from $\tan 60^\circ = \sqrt{3}$, $\tan 45^\circ = 1$, $\tan 30^\circ = 1/\sqrt{3}$: for instance $3\sqrt{3} = 5.196152$ and $5.196152 - \pi = 2.054559$. $\square$

Remark 9.16: The square's 21.5% waste. The case $n = 4$ is the one most often met, and it is worth remembering in the form normalised on the side: if the square has side ℓ and the circle is inscribed ($r = \ell/2$), the disc occupies the fraction $\pi/4 = 78.54\%$ of the square and the residual part is $\ell^2(1 - \pi/4) = 21.46\%$. It is the scrap of a circular blanking operation from a square sheet, the efficiency of a circular section inscribed in a square-section duct, and the upper bound for any packing of a single circle in a square.

9.7.6 The lunes of Hippocrates: when the circle can be squared

Theorem 9.19: Lunes of Hippocrates of Chios (5th century BC). Let ABC be a triangle right-angled at C , with legs $a = CB$, $b = CA$ and hypotenuse $c = AB$. Construct: the semicircle on diameter $\overline{AB}$ on the side of C , and the two semicircles on diameters $\overline{CA}$ and $\overline{CB}$ on the side opposite to C . Writing λ_a and λ_b for the two crescent-shaped regions between the semicircles on the legs and the semicircle on the hypotenuse,

$$\text{area}(\lambda_a) + \text{area}(\lambda_b) = \text{area}(ABC) = \frac{ab}{2}.$$

Proof. Step 1. The area of a semicircle of diameter x is $\frac{1}{2}\pi(x/2)^2 = \frac{\pi}{8}x^2$ (Theorem 9.2). By the Pythagorean theorem (Corollary 8.1) $a^2 + b^2 = c^2$, hence

$$\frac{\pi}{8}a^2 + \frac{\pi}{8}b^2 = \frac{\pi}{8}c^2 :$$

the sum of the two semicircles on the legs equals the semicircle on the hypotenuse. This is the heart of the theorem, and it is the only property of the circle that is needed: that the area is proportional to the square of the diameter.

Step 2. The semicircle on diameter $\overline{AB}$ on the side of C contains the triangle ABC : indeed the angle at C is right, hence C lies on the circle of diameter $\overline{AB}$ by Thales' theorem (Theorem 4.4). The chords $\overline{CA}$ and $\overline{CB}$ cut from that semicircle two circular segments, which we call σ_b and σ_a ; by construction

$$\frac{\pi}{8}c^2 = \text{area}(ABC) + \text{area}(\sigma_a) + \text{area}(\sigma_b).$$

Step 3. The semicircle on diameter $\overline{CB}$ built on the side opposite to C is the disjoint union

of the lune λ_a and the segment σ_a : indeed σ_a is by definition the part of the semicircle on diameter $\overline{AB}$ between the chord $\overline{CB}$ and the arc, and reflecting it in the line CB it comes to occupy exactly the part of the semicircle on $\overline{CB}$ not belonging to the lune. That the reflection is legitimate is guaranteed by the fact that σ_a has $\overline{CB}$ as its chord, and $\overline{CB}$ is the diameter of the second semicircle. Hence

$$\frac{\pi}{8}a^2 = \text{area}(\lambda_a) + \text{area}(\sigma_a), \quad \frac{\pi}{8}b^2 = \text{area}(\lambda_b) + \text{area}(\sigma_b).$$

Step 4. Adding the last two and using Step 1,

$$\frac{\pi}{8}c^2 = \frac{\pi}{8}a^2 + \frac{\pi}{8}b^2 = \text{area}(\lambda_a) + \text{area}(\lambda_b) + \text{area}(\sigma_a) + \text{area}(\sigma_b).$$

Comparing with Step 2 and cancelling the two segments gives $\text{area}(\lambda_a) + \text{area}(\lambda_b) = \text{area}(ABC)$. $\square$

Example 9.7: The lunes on the 3–4–5 triangle. Let $a = 3$, $b = 4$, $c = 5$, with triangle area = 6.

Semicircles. $\frac{\pi}{8} \cdot 9 = 3.534292$; $\frac{\pi}{8} \cdot 16 = 6.283185$; $\frac{\pi}{8} \cdot 25 = 9.817477$. Check of Step 1: $3.534292 + 6.283185 = 9.817477$. It matches.

Segments. The semicircle on the hypotenuse has radius 2.5. The chord of length 3 is at distance $\sqrt{6.25 - 2.25} = 2$ from the centre; by Theorem 9.4

$$\text{area}(\sigma_a) = 6.25 \arccos(0.8) - 2 \cdot 1.5 = 6.25 \cdot 0.6435011 - 3 = 4.021882 - 3 = 1.021882.$$

The chord of length 4 is at distance $\sqrt{6.25 - 4} = 1.5$, and

$$\text{area}(\sigma_b) = 6.25 \arccos(0.6) - 1.5 \cdot 2 = 6.25 \cdot 0.9272952 - 3 = 5.795595 - 3 = 2.795595.$$

Check of Step 2: $6 + 1.021882 + 2.795595 = 9.817477$. It matches.

Lunes.

$$\text{area}(\lambda_a) = 3.534292 - 1.021882 = 2.512410, \quad \text{area}(\lambda_b) = 6.283185 - 2.795595 = 3.487590,$$

and the sum is $2.512410 + 3.487590 = 6.000000$, exactly the area of the triangle.

Remark 9.17: Why the lunes do not square the circle. Hippocrates of Chios (ca. 440 BC) thus obtained the first exact quadrature of a curvilinear figure: a region bounded by circular arcs equivalent to a triangle, hence constructible with straightedge and compass. It was natural to hope that the same idea would square the whole circle. It does not, and the reason is precise: in the theorem the π 's cancel because the three curvilinear figures stand to one another as $a^2 : b^2 : c^2$, and the Pythagorean relation annihilates them. In the whole circle there is nothing with which to cancel π , and the transcendence of π (Observation 1.26) proves that it is impossible. The lunes squarable with straightedge and compass are **exactly five**: three found by Hippocrates, two by Wallenius (1766) and rediscovered by Clausen (1840); that there are no others was proved by N. G. Chebotaryov (1934) and his student A. V. Dorodnov (1947) (source F-03, Postnikov 2000).

9.7.7 The annulus and the chord: an area that does not depend on the radii

Theorem 9.20: Annulus and tangent chord. Let two concentric circles have radii $R > \rho > 0$, and let $\overline{AB}$ be a chord of the larger one tangent to the smaller one, of length $2t$. Then the area of the annulus is

$$A_{\text{ann}} = \pi t^2,$$

that is, it depends only on the length of the chord, and not on the two radii.

Proof. Let O be the common centre and T the point of tangency. By Theorem 3.8 the radius OT is perpendicular to the chord at T , hence T is the midpoint of $\overline{AB}$ (Theorem 3.1) and $AT = t$. The triangle OTA is right-angled at T with hypotenuse $OA = R$ and legs $OT = \rho$, $TA = t$, hence $t^2 = R^2 - \rho^2$. By Theorem 9.5

$$A_{\text{ann}} = \pi(R^2 - \rho^2) = \pi t^2.$$

□

Remark 9.18: A case in which the missing information is not needed. Theorem 9.20 is an instructive example of an under-determined problem that nonetheless has an answer: knowing that the chord measures $2t$ is not enough to determine R and ρ separately — there are infinitely many such pairs — and yet it is enough to determine the area. In the limit $\rho \rightarrow 0$ the annulus degenerates into the whole disc of radius t , and the formula still holds; it is the quickest check one can make, and it also serves to remember it.

9.8. Carnot's theorem and the Japanese theorem

Theorem 9.21: Carnot's theorem (acute triangles). Let ABC be an acute triangle, O the circumcentre, M_A, M_B, M_C the midpoints of the sides BC, CA, AB (feet of the perpendiculars from O , by Theorem 3.1). Then

$$OM_A + OM_B + OM_C = R + r.$$

Proof. O, M_C, B, M_A are concyclic: the angles at M_C and M_A are right angles (being feet of perpendiculars from O), so both subtend $\overline{OB}$ at a right angle and lie, by the converse of Thales, on the circle of diameter $\overline{OB}$, of radius $R/2$. By Ptolemy's theorem applied to the quadrilateral OM_CBM_A inscribed in that circle, with diagonals $OB = R$ and M_AM_C (the latter the segment of midpoints M_A, M_C from vertex B , hence $M_AM_C = b/2$, half the third side CA , by the midpoint theorem):

$$OB \cdot M_AM_C = OM_A \cdot BM_C + OM_C \cdot BM_A \implies R \cdot \frac{b}{2} = OM_A \cdot \frac{c}{2} + OM_C \cdot \frac{a}{2}$$

(since $BM_C = c/2$, $BM_A = a/2$, half the sides adjacent to B). Multiplying by 2: $Rb = OM_A c + OM_C a$. The two analogous formulas, obtained from the cyclic quadrilaterals of diameter OC and OA , are $Rc = OM_B a + OM_A b$ and $Ra = OM_C b + OM_B c$. Adding the three:

$$R(a + b + c) = OM_A(b + c) + OM_B(a + c) + OM_C(a + b).$$

On the other hand, triangle ABC is the union of the three triangles OBC, OCA, OAB (for O interior, the acute case), of areas $\frac{1}{2}a OM_A, \frac{1}{2}b OM_B, \frac{1}{2}c OM_C$: adding, $2K = a OM_A + b OM_B + c OM_C$. For the formula $K = rs = r(a + b + c)/2$: let I be the incentre (Definition 2.24); its distance from each side is, by definition, the radius r of the inscribed circle, so ABC is also the

union of the three triangles IBC, ICA, IAB of areas $\frac{1}{2}ar, \frac{1}{2}br, \frac{1}{2}cr$: adding, $K = \frac{1}{2}r(a+b+c) = rs$. Hence $r(a+b+c) = aOM_A + bOM_B + cOM_C$. Adding this identity to the one above:

$$(R+r)(a+b+c) = OM_A(a+b+c) + OM_B(a+b+c) + OM_C(a+b+c),$$

(the coefficients $b+c$ and a of OM_A add up to $a+b+c$, and likewise for the other two), whence, dividing by $a+b+c > 0$, the claim. $\square$

Numerical check. Triangle $A = (0,0)$, $B = (6,0)$, $C = (2,5)$ (acute): $OM_A + OM_B + OM_C = 5.134691 = R+r$ (checked, independent script).

Remark 9.19: The Japanese theorem (*sangaku*). A related result, quite different in spirit, is the **Japanese theorem on *sangaku*** (votive tablets of Shinto and Buddhist temples, 17th–19th century): if a cyclic quadrilateral is triangulated in two different ways (drawing one or the other diagonal), the sum of the inradii of the two triangles obtained is the same in both cases, $r_1 + r_2 = r_3 + r_4$. The general case (cyclic polygons triangulated in different ways) is **imported** — it requires Carnot's theorem with signed distances, also for obtuse triangles, for which see L. N. M. Carnot, *Géométrie de position*, 1803. Numerical check on a concrete cyclic quadrilateral: quadrilateral inscribed in the unit circle with vertices at $0^\circ, 80^\circ, 150^\circ, 250^\circ$; computing $r = K/s$ for each of the four triangles of the two triangulations (short diagonal and long diagonal), one finds $r_1 + r_2 = r_3 + r_4$ to six decimal places (independent script).

9.9. Summary Formula sheet of the chapter

Circle: the six fundamental formulas

$$C = 2\pi r = \pi d, \quad A = \pi r^2 = \frac{Cr}{2} = \frac{\pi d^2}{4} = \frac{C^2}{4\pi},$$

$$r = \frac{C}{2\pi} = \sqrt{\frac{A}{\pi}}, \quad C = 2\sqrt{\pi A}.$$

(Theorems 9.1, 9.2, Corollary 9.1)

Arc, sector, segment, annulus — θ in radians, α in degrees

$$\ell = r\theta = 2\pi r \frac{\alpha}{360^\circ}, \quad A_{\text{sec}} = \frac{r^2\theta}{2} = \frac{\ell r}{2} = \pi r^2 \frac{\alpha}{360^\circ},$$

$$A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta) = r^2 \arccos \frac{d}{r} - d\sqrt{r^2 - d^2},$$

$$A_{\text{ann}} = \pi(R^2 - r^2) = 2\pi r_m s, \quad A_{\text{as}} = \frac{\theta}{2}(R^2 - r^2) = \theta r_m s.$$

(Theorems 9.3, 9.4, 9.5)

Chord, sagitta, tangent, power

$$c = 2\sqrt{r^2 - d^2}, \quad h = r - d, \quad c^2 = 8rh - 4h^2, \quad r = \frac{c^2}{8h} + \frac{h}{2},$$

$$c = 2r \sin \frac{\theta}{2}, \quad t = \sqrt{\delta^2 - r^2}, \quad \text{pow}(P) = \delta^2 - r^2 = \overline{PX} \cdot \overline{PY}.$$

(Theorems 9.6, 9.7, Proposition 9.1)

Regular n -gons — inscribed in and circumscribed about $\Gamma(O, r)$

$$L_n = 2r \sin \frac{\pi}{n}, \quad a_n = r \cos \frac{\pi}{n}, \quad A_n = \frac{nr^2}{2} \sin \frac{2\pi}{n};$$

$$L_n^* = 2r \tan \frac{\pi}{n}, \quad a_n^* = r, \quad A_n^* = nr^2 \tan \frac{\pi}{n}; \quad A = \frac{pa}{2} \text{ always.}$$

Cases to remember: $L_6 = r$, $L_4 = r\sqrt{2}$, $L_3 = r\sqrt{3}$. (Theorem 9.8)

Goniometry — basic identities

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan \theta = \frac{\sin \theta}{\cos \theta}, \quad 1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}, \quad 1 + \cot^2 \theta = \frac{1}{\sin^2 \theta}.$$

Addition and subtraction

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta, \quad \cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta,$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}.$$

Double, half and triple angle

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1,$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}, \quad \sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}, \quad \cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2},$$

$$\tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta}, \quad \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta, \quad \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

(Chapter 8; Corollary 9.3)

Sum to product (prosthaphaeresis) and product to sum (Werner)

$$\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}, \quad \sin p - \sin q = 2 \cos \frac{p+q}{2} \sin \frac{p-q}{2},$$

$$\cos p + \cos q = 2 \cos \frac{p+q}{2} \cos \frac{p-q}{2}, \quad \cos p - \cos q = -2 \sin \frac{p+q}{2} \sin \frac{p-q}{2};$$

$$\sin \alpha \cos \beta = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)], \quad \cos \alpha \cos \beta = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)],$$

$$\sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)].$$

(Theorems 8.6, 9.10)

Parametric formulas, linear combination, equations — with $t = \tan(\theta/2)$

$$\sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}, \quad \tan \theta = \frac{2t}{1-t^2};$$

$$a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \varphi), \quad \varphi = \operatorname{atan2}(b, a);$$

$$\sin x = a \Rightarrow x = \arcsin a + 2k\pi \text{ or } x = \pi - \arcsin a + 2k\pi;$$

$$\cos x = a \Rightarrow x = \pm \arccos a + 2k\pi; \quad \tan x = a \Rightarrow x = \arctan a + k\pi.$$

(Theorems 9.11, 9.12, Proposition 9.3)

Intersections — the Formula sheet of areas

Circle and line at distance δ from the centre ($\delta < \rho$): the part beyond the line is

$$\rho^2 \arccos \frac{\delta}{\rho} - \delta \sqrt{\rho^2 - \delta^2}.$$

Two circles of radii r, R with distance between centres d , with $|R - r| < d < r + R$:

$$S_{\cap} = r^2 \arccos \frac{d^2 + r^2 - R^2}{2dr} + R^2 \arccos \frac{d^2 + R^2 - r^2}{2dR} - \frac{1}{2} \sqrt{\Lambda},$$

$$\Lambda = (-d + r + R)(d + r - R)(d - r + R)(d + r + R);$$

$$S_{\cup} = \pi r^2 + \pi R^2 - S_{\cap}, \quad S_{D_1 \setminus D_2} = \pi r^2 - S_{\cap}.$$

Case $r = R$: $S_{\cap} = 2r^2 \arccos \frac{d}{2r} - \frac{d}{2} \sqrt{4r^2 - d^2}$; for $d = r$ (vesica piscis) $S_{\cap} = r^2(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}) \approx 1.2284 r^2$.

Circle and polygon: $\text{area}(D \cap P) = \sum_i \varepsilon_i \text{area}(D \cap OV_i V_{i+1})$, each summand the sum of two sectors and a triangle.

Circle and regular polygon: $\text{area}(\Gamma \setminus P_n) = r^2(\pi - \frac{n}{2} \sin \frac{2\pi}{n})$, $\text{area}(P_n^* \setminus \Gamma) = r^2(n \tan \frac{\pi}{n} - \pi)$; for $n = 4$: $(\pi - 2)r^2$ and $(4 - \pi)r^2$.

Lunes of Hippocrates: $\lambda_a + \lambda_b = \frac{1}{2}ab$.

Annulus and tangent chord: $A_{\text{ann}} = \pi t^2$, with $2t$ the length of the chord. (Theorems 9.15, 9.17, 9.18, 9.19, 9.20; Corollaries 9.5, 9.6; Proposition 9.4)

Best practice. The five checks that save an area computation.

1. **Homogeneity.** Every area must be a multiple of a length squared: if a formula contains a term of a different degree, there is an error. Check $S_{\cap}$ in Theorem 9.17: r^2 , R^2 and $\sqrt{\Lambda}$ are all of degree 2.
2. **Limiting cases.** Let $d \rightarrow r + R$: the lens must go to zero. Let $d \rightarrow |R - r|$: it must give $\pi \min(r, R)^2$. Let $\rho \rightarrow 0$ in Theorem 9.20: it must give πt^2 with the full disc.
3. **Double computation of the half-chord.** In the intersection of two circles, h is computed both from r and from R : the two values must agree. It is the most effective check, and it costs one square root.
4. **Sum of the parts.** Lens + two lunes = union; union + lens = sum of the two discs. If it does not add up, the error is in a sign.
5. **Order of magnitude.** An intersection area can neither exceed the area of the smaller of the two, nor be negative. An arc cosine outside $[0, \pi]$ signals an argument outside $[-1, 1]$, that is, incompatible data.

9.10. Exercises

9.10.1 Numerical exercises

Exercise 9.1. In a circle of radius $r = 6$ cm consider a central angle of 75° . Compute the arc length, the sector area, the chord length and the area of the minor circular segment.

Solution. In radians $\theta = 75^\circ \cdot \pi/180^\circ = 5\pi/12 = 1.308997$.

Arc. $\ell = r\theta = 6 \cdot 1.308997 = 7.853982$ cm (Theorem 9.3).

Sector. $A_{\text{sec}} = \frac{1}{2}r^2\theta = \frac{1}{2} \cdot 36 \cdot 1.308997 = 23.561945$ cm². Proportion check: $75/360 = 0.208333$ and $0.208333 \cdot \pi \cdot 36 = 23.561945$: it matches.

Chord. $c = 2r \sin(\theta/2) = 12 \sin 37.5^\circ = 12 \cdot 0.608761 = 7.305137$ cm (Theorem 9.6, the form with the angle).

Segment. $A_{\text{seg}} = \frac{1}{2}r^2(\theta - \sin \theta) = 18(1.308997 - 0.965926) = 18 \cdot 0.343071 = 6.175280$ cm².

Check. The segment must be smaller than the sector: $6.18 < 23.56$. Moreover sector minus segment = 17.386665 must be the area of the isosceles triangle: $\frac{1}{2} \cdot 36 \cdot \sin 75^\circ = 18 \cdot 0.965926 = 17.386665$. It matches.

Exercise 9.2. On a circular arc one measures the chord $c = 1.60$ m and the sagitta $h = 0.25$ m. Determine the radius, the central angle and the arc length.

Solution. *Radius* (Theorem 9.6):

$$r = \frac{c^2}{8h} + \frac{h}{2} = \frac{2.56}{2.00} + 0.125 = 1.28 + 0.125 = 1.405 \text{ m.}$$

Check. $d = r - h = 1.155$ and $\sqrt{r^2 - d^2} = \sqrt{1.974025 - 1.334025} = \sqrt{0.64} = 0.80 = c/2$. It matches.

Angle. $\cos(\theta/2) = d/r = 1.155/1.405 = 0.822064$, hence $\theta/2 = 0.605770$ and $\theta = 1.211539$ rad = 69.416° .

Arc. $\ell = r\theta = 1.405 \cdot 1.211539 = 1.702212$ m, only 6.4% longer than the chord.

Exercise 9.3. Two circles of radius 3 have their centres at distance 4. Compute the area of the intersection, that of the union, and the fraction of each disc occupied by the intersection.

Solution. Since $|R - r| = 0 < 4 < 6 = r + R$, the circles are secant. As $r = R$ one uses Corollary 9.6:

$$S_{\cap} = 2 \cdot 9 \cdot \arccos \frac{4}{6} - \frac{4}{2} \sqrt{36 - 16} = 18 \arccos(0.666667) - 2\sqrt{20}.$$

With $\arccos(2/3) = 0.841069$ and $\sqrt{20} = 4.472136$:

$$S_{\cap} = 15.139236 - 8.944272 = 6.194964.$$

Union. $S_{\cup} = 2\pi \cdot 9 - S_{\cap} = 56.548668 - 6.194964 = 50.353704$ (Corollary 9.5).

Fraction. $6.194964/(9\pi) = 6.194964/28.274334 = 0.219100$, that is 21.91% of each disc.

Check. The half-chord is $h = \sqrt{9 - 4} = \sqrt{5} = 2.236068$ and $dh = 4 \cdot 2.236068 = 8.944272$, which is precisely the subtracted algebraic term: it matches.

Exercise 9.4. The square $[0, 4] \times [0, 4]$ and the disc of centre $(2, 2)$ and radius 2.5. Compute the common area, the area of the square not covered, and the area of the disc that protrudes.

Solution. *Position.* The distance from the centre to each side is $2 < 2.5$: the disc protrudes beyond all four sides. The distance to the vertices is $\sqrt{4 + 4} = 2.828427 > 2.5$: the vertices are outside the disc and the four protruding parts are disjoint.

One segment (Theorem 9.15), with $\rho = 2.5$ and $\delta = 2$:

$$6.25 \arccos(0.8) - 2\sqrt{6.25 - 4} = 6.25 \cdot 0.643501 - 2 \cdot 1.5 = 4.021882 - 3 = 1.021882.$$

Areas.

$$\text{area}(D \cap S) = \pi \cdot 6.25 - 4 \cdot 1.021882 = 19.634954 - 4.087528 = 15.547426;$$

$$\text{area}(S \setminus D) = 16 - 15.547426 = 0.452574, \quad \text{area}(D \setminus S) = 4.087528.$$

Check. The union is $16 + 19.634954 - 15.547426 = 20.087528$, and also $16 + 4.087528 = 20.087528$. It matches.

Exercise 9.5. In the circle of radius $r = 10$ compute the perimeter and area of the inscribed and of the circumscribed regular dodecagon, and compare them with the circumference and the area of the circle.

Solution. With $n = 12$: $\pi/n = 15^\circ$, $\sin 15^\circ = 0.258819$, $\tan 15^\circ = 0.267949$, $\sin 30^\circ = 0.5$ (Theorem 9.8).

Inscribed. $p_{12} = 2 \cdot 12 \cdot 10 \cdot 0.258819 = 62.116571$; $A_{12} = \frac{1}{2} \cdot 12 \cdot 100 \cdot 0.5 = 300$ exactly.

Circumscribed. $p_{12}^* = 240 \cdot 0.267949 = 64.307806$; $A_{12}^* = 12 \cdot 100 \cdot 0.267949 = 321.539031$.

Circle. $C = 62.831853$; $A = 314.159265$.

Errors. On the perimeter: -1.14% (inscribed) and $+2.35\%$ (circumscribed). On the area: -4.51% and $+2.35\%$. Note that the circumscribed polygon makes the *same* relative error on perimeter and on area, because both equal the circle's quantity multiplied by $\tan(\pi/n)/(\pi/n)$: a consequence of $A = \frac{1}{2}pa$ with $a = r$ in both cases.

Exercise 9.6. A horizontal cylindrical tank has inner radius $r = 1.20$ m and length $L = 5.00$ m. The liquid level is $H = 0.80$ m. Compute the volume contained and the percentage of fill.

Solution. Since $H < r$, the liquid occupies the minor circular segment, whose chord is at distance $\delta = r - H = 0.40$ m from the centre (Remark 9.14). By Theorem 9.15

$$A_{\text{seg}} = 1.44 \arccos \frac{0.40}{1.20} - 0.40\sqrt{1.44 - 0.16}.$$

With $\arccos(0.333333) = 1.230959$ and $\sqrt{1.28} = 1.131371$:

$$A_{\text{seg}} = 1.772582 - 0.452548 = 1.320034 \text{ m}^2, \quad V = 5 \cdot 1.320034 = 6.600168 \text{ m}^3.$$

Percentage. The total volume is $\pi r^2 L = \pi \cdot 1.44 \cdot 5 = 22.619467 \text{ m}^3$, hence the fill is $6.600168/22.619467 = 29.18\%$.

Check. At half height ($H = r$) the formula would give $1.44 \arccos 0 - 0 = 1.44 \cdot 1.570796 = 2.261947 \text{ m}^2$, that is half the section: consistent. And $29.18\% < 50\%$, as it must be for $H < r$.

9.10.2 Symbolic and proof exercises

Exercise 9.7. Prove that the two forms of the circular-segment area given in Theorem 9.4 coincide, by expressing θ in terms of d and conversely, and discuss what happens for $\theta > \pi$.

Solution. Let M be the midpoint of the chord; the triangle OMA is right-angled at M , with hypotenuse r , leg $OM = d$ and angle at O equal to $\theta/2$. Hence $\cos(\theta/2) = d/r$, that is $\theta = 2 \arccos(d/r)$, and $\sin(\theta/2) = \sqrt{r^2 - d^2}/r$. By the double-angle formula,

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 2 \cdot \frac{\sqrt{r^2 - d^2}}{r} \cdot \frac{d}{r} = \frac{2d\sqrt{r^2 - d^2}}{r^2}.$$

Substituting into the first form,

$$\frac{r^2}{2}(\theta - \sin \theta) = \frac{r^2}{2} \cdot 2 \arccos \frac{d}{r} - \frac{r^2}{2} \cdot \frac{2d\sqrt{r^2 - d^2}}{r^2} = r^2 \arccos \frac{d}{r} - d\sqrt{r^2 - d^2},$$

which is the second form.

Case $\theta > \pi$. Then $\cos(\theta/2) < 0$ and the relation $\cos(\theta/2) = d/r$ forces one to read d as a *signed* distance, negative when the centre lies on the opposite side of the chord. With this convention the second form remains valid unchanged: $\arccos$ of a negative argument returns a value in $(\pi/2, \pi]$, and the term $-d\sqrt{r^2 - d^2}$ becomes positive. This is exactly why the lens formula (Theorem 9.17) requires no case distinction.

Exercise 9.8. Let P be a point inside a circle of centre O and radius r , with $OP = p > 0$. Prove that among all chords through P the shortest is the one perpendicular to OP , and compute its length.

Solution. Let s be a chord through P and let d be the distance from O to s . By Theorem 9.6 its length is $c = 2\sqrt{r^2 - d^2}$, a strictly decreasing function of d : the shortest chord is therefore the one for which d is largest.

Let M be the foot of the perpendicular from O to s ; since M lies on s and so does P , the segment OM is a leg of the right triangle OMP (right angle at M) with hypotenuse $OP = p$, whenever $M \neq P$. Hence $d = OM \leq p$, with equality if and only if $M = P$, that is if and only if $s \perp OP$.

The maximum of d is therefore p , attained exactly by the chord perpendicular to OP at P , and the minimum length is

$$c_{\min} = 2\sqrt{r^2 - p^2}.$$

Check at the limits. For $p \rightarrow 0$ one gets $c_{\min} \rightarrow 2r$: if P is the centre, all chords are diameters and the “shortest” coincides with the longest. For $p \rightarrow r$ one gets $c_{\min} \rightarrow 0$: the point tends to the boundary and the perpendicular chord degenerates. *Note.* The product of the two segments into which P divides any chord equals $r^2 - p^2$ by Proposition 9.1; the minimal chord is therefore the one in which the two segments are *equal*, both $\sqrt{r^2 - p^2}$.

Exercise 9.9. Using Theorem 9.11, show that every primitive Pythagorean triple is obtained from a rational value of the parameter t , and derive the triple corresponding to $t = 2/3$.

Solution. Let (x, y, z) be a Pythagorean triple, $x^2 + y^2 = z^2$ with $z > 0$. Dividing by z^2 , the point $(x/z, y/z)$ is a *rational* point of the unit circle, different from $(-1, 0)$ if $x \neq -z$. By Theorem 9.11 there is a unique t with

$$\frac{x}{z} = \frac{1 - t^2}{1 + t^2}, \quad \frac{y}{z} = \frac{2t}{1 + t^2},$$

and the inverse formula $t = (y/z)/(1 + x/z) = y/(z + x)$ shows that t is rational. Conversely, writing $t = q/p$ in lowest terms,

$$\frac{1 - t^2}{1 + t^2} = \frac{p^2 - q^2}{p^2 + q^2}, \quad \frac{2t}{1 + t^2} = \frac{2pq}{p^2 + q^2},$$

and multiplying by $p^2 + q^2$ one gets the triple $(p^2 - q^2, 2pq, p^2 + q^2)$, the same parametrisation recovered further on in the chapter on the right triangle; it is primitive when p, q are coprime and of opposite parity.

Case $t = 2/3$. Then $p = 3$, $q = 2$, and the triple is

$$(9 - 4, 2 \cdot 3 \cdot 2, 9 + 4) = (5, 12, 13),$$

and indeed $25 + 144 = 169$. The point on the circle is $(5/13, 12/13)$, and $t = (12/13)/(1 + 5/13) = 12/18 = 2/3$: the correspondence closes.

Exercise 9.10. Prove Theorem 9.20 and use it to solve the following problem: of an annulus one knows only that the chord of the outer circle tangent to the inner circle measures 12 cm. What is the area of the annulus?

Solution. *Proof.* Let O be the common centre and $\overline{AB}$ the chord tangent to the inner circle at T . By Theorem 3.8 $OT \perp AB$, hence by Theorem 3.1 T is the midpoint of $\overline{AB}$ and $AT = t$. In the right triangle OTA , $OA^2 = OT^2 + TA^2$, that is $R^2 = \rho^2 + t^2$ and $R^2 - \rho^2 = t^2$. Therefore $A_{\text{ann}} = \pi(R^2 - \rho^2) = \pi t^2$.

Application. The chord measures $2t = 12$ cm, hence $t = 6$ cm and

$$A_{\text{ann}} = \pi \cdot 36 = 113.097336 \text{ cm}^2.$$

The two radii are not needed, and indeed they are not determined: the pairs $(R, \rho) = (10, 8)$, $(6.5, 2.5)$, $(6, 0)$ all give the same area, since in all of them $R^2 - \rho^2 = 36$.

Exercise 9.11. Prove that, with the radii fixed, the lens area $S_{\cap}(d)$ is strictly decreasing in d on the interval $(|R - r|, r + R)$, and compute the two limits at the endpoints.

Solution. *Monotonicity, geometrically.* Let $|R - r| < d_1 < d_2 < r + R$. Fix the disc D_1 and translate the disc D_2 along the line of centres from distance d_1 to distance d_2 . Choose coordinates with $O_1 = (0, 0)$ and $O_2 = (d, 0)$, and observe that

$$D_1 \cap D_2 = \{(x, y) : x^2 + y^2 \leq r^2 \text{ and } (x - d)^2 + y^2 \leq R^2\}.$$

For each fixed y with $|y| < r$, the horizontal section is the intersection of the two intervals

$$I_1(y) = [-\sqrt{r^2 - y^2}, \sqrt{r^2 - y^2}], \quad I_2(y, d) = [d - \sqrt{R^2 - y^2}, d + \sqrt{R^2 - y^2}].$$

As d grows, I_2 translates to the right without changing length, while I_1 stays put; since the active constraint on the right is always that of I_1 (because the centre O_2 lies to the right), the length of the intersection $|I_1 \cap I_2|$ is non-increasing in d , and strictly decreasing for all y for which the intersection is non-empty and non-degenerate — a set of y of positive measure as long as $d < r + R$. Integrating in y (Fubini) gives $S_{\cap}(d_2) < S_{\cap}(d_1)$.

Limit $d \rightarrow (r + R)^-$. The factor $(-d + r + R)$ of Λ tends to 0, hence $\frac{1}{2}\sqrt{\Lambda} \rightarrow 0$; moreover

$$\frac{d^2 + r^2 - R^2}{2dr} \rightarrow \frac{(r + R)^2 + r^2 - R^2}{2(r + R)r} = \frac{2r^2 + 2rR}{2r(r + R)} = 1,$$

and analogously the other argument tends to 1: both arc cosines tend to 0. Hence $S_{\cap} \rightarrow 0$, as is geometrically obvious.

Limit $d \rightarrow |R - r|^+$. Assume $R > r$, so that $d \rightarrow R - r$: the factor $(d + r - R)$ vanishes, hence again $\frac{1}{2}\sqrt{\Lambda} \rightarrow 0$. Moreover

$$\frac{d^2 + r^2 - R^2}{2dr} \rightarrow \frac{(R - r)^2 + r^2 - R^2}{2(R - r)r} = \frac{-2rR + 2r^2}{2r(R - r)} = -1,$$

so the first arc cosine tends to π ; the second argument tends to

$$\frac{(R-r)^2 + R^2 - r^2}{2(R-r)R} = \frac{2R^2 - 2rR}{2R(R-r)} = 1,$$

so the second arc cosine tends to 0. Therefore $S_{\cap} \rightarrow r^2\pi + 0 - 0 = \pi r^2$: the small disc ends up entirely inside the large one, as it must in internal tangency.

Exercise 9.12. In the particular case of the *isosceles* right triangle, prove that *each* of the two lunes of Hippocrates has area equal to half the triangle, and that the lune is therefore squarable with straightedge and compass.

Solution. Let $a = b$ be the common value of the legs and $c = a\sqrt{2}$ the hypotenuse. By symmetry about the perpendicular bisector of the hypotenuse the configuration is invariant under exchanging the two legs, hence the two lunes are congruent and have the same area. By Theorem 9.19 their sum is $\frac{1}{2}a^2$, so each equals

$$\text{area}(\lambda) = \frac{a^2}{4}.$$

Direct check. The semicircle on one leg has area $\frac{\pi}{8}a^2$. The corresponding segment is cut from the semicircle on the hypotenuse, of radius $c/2 = a/\sqrt{2}$, by a chord of length a ; the distance from the centre to that chord is $\sqrt{a^2/2 - a^2/4} = a/2$, and the central angle is $2\arccos((a/2)/(a/\sqrt{2})) = 2\arccos(1/\sqrt{2}) = \pi/2$. By Theorem 9.4

$$\text{area}(\sigma) = \frac{1}{2} \cdot \frac{a^2}{2} \left(\frac{\pi}{2} - 1 \right) = \frac{a^2}{4} \left(\frac{\pi}{2} - 1 \right) = \frac{\pi a^2}{8} - \frac{a^2}{4}.$$

Hence $\text{area}(\lambda) = \frac{\pi}{8}a^2 - (\frac{\pi}{8}a^2 - \frac{a^2}{4}) = \frac{a^2}{4}$, as announced.

Squarability. The area $a^2/4$ is that of the square of side $a/2$, which is constructible with straightedge and compass from a (it suffices to bisect the segment). The lune is therefore *squarable*: it is the first curvilinear figure for which an equivalent square was ever constructed, and it is the result for which Hippocrates of Chios is remembered. Note that π disappears from the result: it is this, and not an accident of computation, that makes the quadrature possible.

9.11. Self-assessment quiz

- The area of a disc can be written as $Cr/2$ because
 - it is a notational convention
 - the disc is equivalent to the triangle with legs r and C
 - C and r are proportional
 - it holds only for $r = 1$
- In a circle of radius r , the circular segment subtended by a central angle θ has area
 - $\frac{1}{2}r^2\theta$
 - $\frac{1}{2}r^2(\theta - \sin \theta)$
 - $\frac{1}{2}r^2 \sin \theta$

-
- D) $r^2\theta - \sin \theta$
3. A chord measures c and its sagitta h . The radius equals
- A) $c^2/(8h)$
B) $c^2/(8h) + h/2$
C) $c/(2h)$
D) $\sqrt{c^2 + h^2}$
4. The annulus area $\pi(R^2 - r^2)$ can be rewritten as $2\pi r_m s$ where r_m is
- A) the geometric mean of R and r
B) the arithmetic mean of R and r
C) the harmonic mean of R and r
D) the larger radius
5. The side of the regular hexagon inscribed in a circle of radius r equals
- A) $r\sqrt{2}$
B) $r\sqrt{3}$
C) r
D) $2r$
6. Setting $t = \tan(\theta/2)$, one has $\cos \theta =$
- A) $2t/(1 + t^2)$
B) $(1 - t^2)/(1 + t^2)$
C) $(1 + t^2)/(1 - t^2)$
D) $1 - t^2$
7. In the expression $a \sin x + b \cos x$ the maximum value is
- A) $a + b$
B) $\max(a, b)$
C) $\sqrt{a^2 + b^2}$
D) ab
8. Two circles of radii r and R with centres at distance d meet in two distinct points if and only if
- A) $d < r + R$
B) $d > |R - r|$
C) $|R - r| < d < r + R$
D) $d = |R - r|$
9. In the lens-area formula, the term $\frac{1}{2}\sqrt{(-d + r + R)(d + r - R)(d - r + R)(d + r + R)}$ represents
- A) the length of the common chord
-

- B) twice the area of the triangle O_1O_2P
- C) the area of one of the two sectors
- D) the distance between the two centres
10. Of an annulus one knows only the chord of the outer circle tangent to the inner one, of length $2t$. The area of the annulus
- A) is not determined
- B) equals πt^2
- C) equals $2\pi t$
- D) equals $\pi t^2/2$

BEST PRACTICE

Key. 1: B (Theorem 6.5) — 2: B (Theorem 9.4) — 3: B (Theorem 9.6) — 4: B (Theorem 9.5) — 5: C (Theorem 9.8: $2r \sin 30^\circ = r$) — 6: B (Theorem 9.11) — 7: C (Theorem 9.12) — 8: C (Theorem 9.16) — 9: B (Theorem 9.17, Step 3) — 10: B (Theorem 9.20).

Note on question 3. Answer A is the engineering approximation $r \approx c^2/(8h)$, correct only when $h \ll c$; for $c = 240$ mm and $h = 18$ mm it is off by 9 mm out of 409, that is by 2.2%. The term $h/2$ is not negligible as soon as the arc becomes pronounced, and for the semicircle ($h = r$, $c = 2r$) the approximation would give $r/2$ instead of r : a 50% error.

Note on question 10. It is the clearest example of an under-determined problem with a determined answer: the two radii cannot be reconstructed, but the difference of their squares can, and that is all that is needed. The limiting case $\rho = 0$, in which the “annulus” is the full disc of radius t , is the quickest check.

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CHAPTER 10

Right Triangle and Circle

INFO

Purpose of this chapter. To collect and prove — in several independent ways, wherever possible — every relation linking a triangle, right-angled or general, to the circle circumscribed about it or inscribed within it.

A historiographical warning. The theorem opening this chapter is known in the West as the Pythagorean theorem, but in China it bears a different name — **gougu** — and was stated and proved by dissection in texts independent of the Greek tradition. Section 10.2 tells this parallel story, because ignoring it would give students a false picture of how mathematics actually developed.

Where we stop. Section 10.15 presents an open problem — that of Erdős and Ulam — arising precisely from the results of this chapter: we prove as far as the tools developed here reach, and then state precisely where the proof stops.

10.1. The Pythagorean theorem: statement and multiple proofs

Theorem 10.1: Pythagorean Theorem. In every right triangle with legs a, b and hypotenuse c : $a^2 + b^2 = c^2$.

Proof 1 — Euclid, by similarity. The classical proof via geometric means (Euclid, *Elements*, I.47 and VI.8). Let $\triangle ABC$ be right-angled at C , and let H be the foot of the altitude from C onto the hypotenuse $AB = c$, with $AH = p$, $HB = q$ ($p + q = c$). Triangles $\triangle ACH$ and $\triangle ABC$ share the angle at A and both have a right angle ($\angle AHC = \angle ACB = 90^\circ$): they are similar (AA). From the similarity, $AC/AB = AH/AC$, that is

$$b^2 = AC^2 = AH \cdot AB = pc.$$

Likewise $\triangle BCH \sim \triangle BAC$ gives $a^2 = qc$. Adding:

$$a^2 + b^2 = qc + pc = (p + q)c = c \cdot c = c^2.$$

□

Proof 2 — by dissection (the gougu method). This is the Chinese and Indian proof (*Zhoubi Suanjing*, China, ca. 100 BCE; Bhāskara, India, 12th c.). One constructs a square of side $a + b$ and fills it in two different ways:

Way 1: four copies of the right triangle (a, b, c) , arranged pinwheel-fashion along the four sides of the large square, leave uncovered at the centre a square of side c . That the central figure really is a square is checked thus: each of its sides is a hypotenuse, hence of length c ; and each of its angles equals 180° minus the two acute angles of the triangle, which sum to 90° , hence equals 90° . Total area: $4 \cdot \frac{1}{2}ab + c^2$.

Way 2: the same four copies, arranged in pairs in two opposite corners, leave uncovered two squares of sides a and b . Total area: $4 \cdot \frac{1}{2}ab + a^2 + b^2$.
 Since the area of the large square is $(a + b)^2$ in both cases:

$$4 \cdot \frac{1}{2}ab + c^2 = 4 \cdot \frac{1}{2}ab + a^2 + b^2 \implies c^2 = a^2 + b^2.$$

□

Proof 3 — Garfield’s trapezoid (1876). One constructs a right trapezoid with parallel bases a and b and height $a + b$, formed by two copies of the triangle (a, b, c) together with the isosceles right triangle with legs c . Its area is computed in two ways:

Trapezoid formula: $A = \frac{1}{2}(a + b)(a + b) = \frac{1}{2}(a + b)^2$.

Sum of the three parts: $A = 2 \cdot \frac{1}{2}ab + \frac{1}{2}c^2 = ab + \frac{1}{2}c^2$.

Equating:

$$\frac{1}{2}(a + b)^2 = ab + \frac{1}{2}c^2 \implies \frac{1}{2}a^2 + ab + \frac{1}{2}b^2 = ab + \frac{1}{2}c^2 \implies a^2 + b^2 = c^2.$$

(Garfield found this proof in 1876 as a member of the US Congress, five years before becoming the 20th President.) □

Proof 4 — analytic/vectorial. Place C at the origin, with the legs along the axes: $A = (b, 0)$, $B = (0, a)$ (right angle at C). Then

$$c^2 = |AB|^2 = (b - 0)^2 + (0 - a)^2 = a^2 + b^2.$$

Equivalently, in terms of the dot product with $\vec{u} = \vec{CA}$, $\vec{v} = \vec{CB}$ orthogonal ($\vec{u} \cdot \vec{v} = 0$):

$$c^2 = |\vec{u} - \vec{v}|^2 = |\vec{u}|^2 - 2\vec{u} \cdot \vec{v} + |\vec{v}|^2 = b^2 + a^2,$$

a proof that generalises at once to the law of cosines (Theorem 10.6). □

Proof 5 — Euclid I.47, the “windmill”. This is the *canonical* proof of the *Elements*: it uses neither similarity nor algebra, only equidecomposition and congruence of triangles. Let $\triangle ABC$ be right-angled at A . Construct the three squares: $BCED$ on the hypotenuse BC (on the side opposite A), $ABFG$ on the leg AB , $ACKH$ on the leg AC . From A drop the perpendicular AL to BC , meeting BC at M and the side DE at L ; the resulting figure is the “windmill”.

Step 1 — G, A, C are collinear. The angles $\angle GAB$ and $\angle BAC$ are both right (the first because $ABFG$ is a square, the second by hypothesis), and they are adjacent: their sum is π , hence GA and AC are opposite rays.

Step 2 — the square $ABFG$ is twice the triangle $\triangle FBC$. The two share the base FB ; and by Step 1 the vertex C lies on the line GC , parallel to FB (both being perpendicular to AB). Figures on the same base and between the same parallels have the same height: the area of the square is base $\times$ height, that of the triangle is half the same product (axioms (Ar1)–(Ar4), Chapter 6). Hence $\text{area}(ABFG) = 2 \text{area}(\triangle FBC)$.

Step 3 — the rectangle $BMLD$ is twice the triangle $\triangle ABD$. Same base BD ; and $AL \parallel BD$, both being perpendicular to BC . By the same argument as in Step 2, $\text{area}(BMLD) = 2 \text{area}(\triangle ABD)$.

Step 4 — the two triangles are congruent. Compare $\triangle FBC$ and $\triangle ABD$:

$$FB = AB \text{ (sides of the square } ABFG), \quad BC = BD \text{ (sides of the square } BCED),$$

$$\angle FBC = \angle FBA + \angle ABC = \frac{\pi}{2} + \hat{B} = \angle ABC + \angle CBD = \angle ABD.$$

By side-angle-side, $\triangle FBC \cong \triangle ABD$, hence they have equal areas.

Step 5 — conclusion. Chaining Steps 2, 4, 3:

$$\text{area}(ABFG) = 2 \text{area}(\triangle FBC) = 2 \text{area}(\triangle ABD) = \text{area}(BMLD).$$

Repeating the argument symmetrically on the other leg gives $\text{area}(ACKH) = \text{area}(MCEL)$. But $BMLD$ and $MCEL$ are the two parts into which ML divides the square $BCED$, hence

$$\text{area}(BCED) = \text{area}(ABFG) + \text{area}(ACKH),$$

that is, $c^2 = a^2 + b^2$.

Why it matters. This proof does not use the theory of proportions (Book V) and therefore does not depend on Euclid's treatment of incommensurables: this is why Euclid places it at the end of Book I and not in Book VI, where he nevertheless gives a second one (VI.31). $\square$

Proof 6 — Einstein, by similarity of parts. Attributed to the twelve-year-old Albert Einstein, this is perhaps the shortest of all.

Step 1 — the decomposition. Draw the altitude CH from the right-angle vertex to the hypotenuse. It divides $\triangle ABC$ into two triangles $\triangle CBH$ and $\triangle ACH$, both **similar** to the original (each has a right angle and shares an acute angle with it: AA criterion).

Step 2 — how area scales between similar figures. If two figures are similar with ratio k , their areas are in the ratio k^2 . In particular, fixing any *corresponding side* as the characteristic dimension, for all triangles of a given shape

$$\text{area} = \lambda \cdot (\text{corresponding side})^2$$

with $\lambda > 0$ constant, the *same* for the whole similarity class.

Step 3 — take the hypotenuse as the corresponding side. With the chapter's convention ($\triangle ABC$ right-angled at C , legs $BC = a$ and $CA = b$, hypotenuse $AB = c$), the altitude CH falls on AB and the three triangles have the following right angle and hypotenuse:

triangle	right angle at	hypotenuse
$\triangle ABC$	C	$AB = c$
$\triangle CBH$	H	$CB = a$
$\triangle ACH$	H	$AC = b$

Note that the hypotenuse of each small triangle is a *leg* of the big one: this is exactly the mechanism that makes a^2 and b^2 appear.

Step 4 — add. By Step 2, with the same constant λ (the three triangles being mutually similar):

$$\text{area}(\triangle CBH) = \lambda a^2, \quad \text{area}(\triangle ACH) = \lambda b^2, \quad \text{area}(\triangle ABC) = \lambda c^2.$$

Since H is interior to segment AB (the foot of the altitude falls between A and B because both acute angles are less than $\frac{\pi}{2}$), the two small triangles partition the big one, and additivity of area (axiom (Ar1), Chapter 6) gives

$$\lambda a^2 + \lambda b^2 = \lambda c^2 \implies a^2 + b^2 = c^2,$$

dividing by $\lambda > 0$.

The elegant point. Nothing is ever computed: one uses only the fact that the area of a figure is proportional to the square of one of its linear dimensions, with a constant depending on the *shape* and not on the *size*. The whole theorem is in that observation. $\square$

Proof 7 — non-circular trigonometric (Zimba, 2009). Define $\sin$ and $\cos$ for acute angles as ratios between sides of similar right triangles — a definition which, as Remark 8.4 (Chapter 8) shows, requires only Euclid I.32, I.19 and VI.4, and **not** I.47. Theorem 8.3 then proves, purely constructively, that

$$\cos^2 x + \sin^2 x = 1 \quad \text{for every } x \in (0, \frac{\pi}{2}).$$

Letting x be the angle opposite a , we have $\sin x = a/c$ and $\cos x = b/c$ by definition, and multiplying the identity by c^2 :

$$b^2 + a^2 = c^2.$$

Why it is remarkable. For a century after Loomis’s 1940 claim it was believed that every trigonometric proof of Pythagoras was necessarily circular. It is not: the complete proof, with the construction of the subtraction formulas and the step-by-step verification of non-circularity, occupies Section 8.3 of Chapter 8. $\square$

Corollary 10.1: Pythagoras as a special case of the trigonometric Pythagorean identity. Theorem 10.1 and the trigonometric Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$ (Theorem 8.2) are **equivalent**: each is deducible from the other.

Proof. ($\Rightarrow$) From Pythagoras to the identity. Let $\theta \in (0, \pi/2)$. Construct the right triangle with hypotenuse $c = 1$ and angle θ at A . By the definition of sine and cosine in a right triangle (Chapter 8), the leg opposite θ measures $a = \sin \theta$ and the adjacent one $b = \cos \theta$. Theorem 10.1 then gives $a^2 + b^2 = c^2$, i.e. $\sin^2 \theta + \cos^2 \theta = 1$. For $\theta = 0$ and $\theta = \pi/2$ the check is immediate, and the extension to the other quadrants follows from the reduction formulas (evenness of $\cos$, oddness of $\sin$), which leave the squares unchanged.

($\Leftarrow$) From the identity to Pythagoras. Let a right triangle be given with legs a, b and hypotenuse $c > 0$, and let θ be the angle opposite a . By definition $\sin \theta = a/c$ and $\cos \theta = b/c$. Multiplying the identity by c^2 :

$$c^2(\sin^2 \theta + \cos^2 \theta) = c^2 \implies c^2 \frac{a^2}{c^2} + c^2 \frac{b^2}{c^2} = c^2 \implies a^2 + b^2 = c^2.$$

Conclusion. The two implications close the equivalence. In a sense, all the proofs in this chapter are manifestations of one and the same fact, seen from different vantage points (synthetic, algebraic, trigonometric). $\square$

Warning. Implication ($\Leftarrow$) is legitimate only for one of the two routes leading to the identity. There are in fact two distinct deductive chains:

(V1) Pythagoras $\Rightarrow$ distance formula $\Rightarrow x^2 + y^2 = 1 \Rightarrow \sin^2 \theta + \cos^2 \theta = 1$;

(V2) Euclid VI.4 $\Rightarrow$ definition of $\sin, \cos \Rightarrow$ subtraction formulas $\Rightarrow \sin^2 \theta + \cos^2 \theta = 1$.

Route (V1) is the one of Remark 7.1 (Chapter 7): the equation of the circle *is* the Pythagorean theorem written in coordinates. If one has reached the identity by (V1), then travelling arrow ($\Leftarrow$) to “prove” Pythagoras is **circular**.

Route (V2) is Zimba’s (Theorem 8.3, Section 8.3): it nowhere touches Pythagoras. If one has reached the identity by (V2), then arrow ($\Leftarrow$) is a **legitimate proof** — and it is Proof 7 of this chapter.

What remains forbidden anyway, for the usual reason: the “transcendental” proofs of Pythagoras, i.e. those going through the Taylor series of $\sin$ and $\cos$ or l’Hôpital’s rule applied to $\sin x/x$. Both presuppose $(\sin x)' = \cos x$, which presupposes the fundamental limit, which presupposes the area of the sector, which presupposes $A = \pi r^2$.

Remark 10.1: How many proofs are there, and why these seven. Elisha Loomis’s book *The Pythagorean Proposition* (1940) collects more than 370. Reproducing them all would be pointless: the vast majority are variants of one another. The seven given here are chosen because they cover **all the genuinely distinct proof registers**:

#	register	central tool	does not use
1	similarity	Euclid VI.8, geometric means	areas
2	dissection	equidecomposition	proportions
3	area (trapezoid)	trapezoid formula	similarity
4	analytic	dot product	synthesis
5	area (Euclid I.47)	congruence + same parallels	Book V
6	similarity of parts	quadratic scaling of area	calculus
7	trigonometric	subtraction formulas (Zimba)	Pythagoras itself

Among those not reproduced, three deserve mention:

- **Perigal’s dissection** (1830), in which the square on the longer leg is cut into four congruent pieces by two lines through its centre, one parallel and one perpendicular to the hypotenuse: the four pieces plus the square on the shorter leg reassemble exactly into the square on the hypotenuse. It is a variant of Proof 2, more spectacular but logically equivalent;
- the **dynamic** proofs via quaternions or vectors, which are rewritings of Proof 4 in a different formalism;
- the **transcendental** proofs, via Taylor series or l’Hôpital: excluded not out of taste but because they are **circular** (see the warning after Corollary 10.1).

Proof 7 is the most recent in order of discovery and is the one that refuted Loomis’s claim that a trigonometric proof would be impossible in principle.

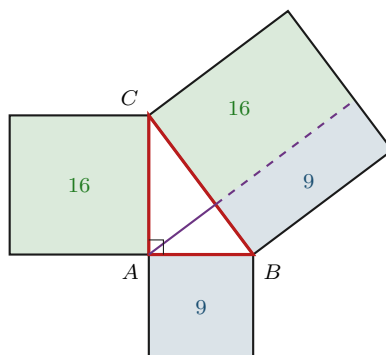


Figure 10.1: The *windmill* figure with which Euclid proves proposition I.47. The altitude to the hypotenuse, extended, divides the square built on BC into two rectangles; each of them is shown to be equal in area to the square built on the corresponding leg, by comparing both with an intermediate triangle having the same base and the same height as each. Adding the two equalities gives $AB^2 + AC^2 = BC^2$. In the case drawn ($AB = 3$, $AC = 4$, $BC = 5$) the two squares are 9 and 16, and the two rectangles exactly as much: $1.8 \times 5 = 9$ and $3.2 \times 5 = 16$.

10.2. The gougou theorem: the Chinese tradition

Remark 10.2: Names and texts. In classical Chinese mathematics the right triangle is described by three technical terms: **gou**, the shorter leg; **gu**, the longer leg; **xian**, the hypotenuse — literally “bowstring”, the same term used for the chord of a circle. The relation $\text{gou}^2 + \text{gu}^2 = \text{xian}^2$ is the **gougou rule**, the Chinese counterpart of the Pythagorean

theorem.

There are two principal sources. The *Zhoubi Suanjing* (“Arithmetical Classic of the Gnomon and the Circular Paths of Heaven”), datable to about **100 BCE**, is a text of astronomy and mathematics containing an illustration of the dissection proof given above as Proof 2. The *Nine Chapters on the Mathematical Art* (*Jiuzhang Suanshu*) are instead a stratified compilation, the work of several generations of scholars, whose final redaction dates to the **1st century CE**; its ninth chapter is devoted entirely to the *gougu* rule and its applications.

Remark 10.3: Liu Hui and the link with the circle. *Liu Hui*’s commentary on the *Nine Chapters*, of **263 CE**, is the text that explicitly welds the *gougu* rule to the geometry of the circle, and it does so in two ways.

First: in the *Sea Island Mathematical Manual* (*Haidao Suanjing*), *Liu Hui* uses the *gougu* rule to compute heights and distances not directly measurable — this is the birth of triangulation, the very principle we saw at work in Chapter 4 in the *Pothenot–Snellius* resection.

Second: in the same commentary, *Liu Hui* observes that the ratio of the perimeter of the inscribed hexagon to the diameter is exactly 3 — which explains the value $\pi \approx 3$ used in older texts — and then constructs an **iterative polygon-bisection algorithm** to compute π to any desired accuracy. It is the same method as *Archimedes*’ (Theorem 6.5, Chapter 6), developed independently some five centuries later, but with one important difference: *Liu Hui* does not stop at a bracket between two values, but supplies an explicit recursive procedure for refining the estimate as far as one wishes.

Warning. Why the dating must be handled with care. Ancient Chinese mathematical texts are stratified compilations: the *Zhoubi Suanjing* contains material of different periods, and the *Nine Chapters* were composed over centuries. The dates given here — ca. 100 BCE for the first, 1st c. CE for the final redaction of the second, 263 CE for *Liu Hui*’s commentary — are those currently accepted by historians, but the philological debate on the internal stratification remains open. What is *not* in dispute is the independence from the Greek tradition: there is no evidence of transmission between the two at that period.

10.3. The two theorems of Euclid and the metric relations

The two theorems that the Italian tradition calls “of Euclid” — known in English as the *leg* (or geometric-mean) theorem and the *altitude* theorem — complete the Pythagorean theorem and constitute, together with it, the entire metric apparatus of the right triangle. We prove them twice: once by similarity, as in *Elements* VI.8, and once *through the circle*, because the second theorem is nothing but the power-of-a-point theorem (Theorem 3.19) applied to the circumcircle.

Definition 10.1: Projections and altitude to the hypotenuse. Let ABC be a triangle right-angled at C , with legs $a = CB$, $b = CA$ and hypotenuse $c = AB$. Let H be the foot of the perpendicular from C to AB . Set

$$h = CH \quad (\text{the altitude to the hypotenuse}), \quad p = AH, \quad q = HB,$$

where p and q are the **projections** of the legs b and a onto the hypotenuse. By construction $p + q = c$, and H lies inside the segment $\overline{AB}$ because the angles at A and B are acute.

Theorem 10.2: Leg theorem (Elements VI.8, cor.). With the notation of Definition 10.1,

$$b^2 = cp, \quad a^2 = cq.$$

Each leg is the geometric mean between the hypotenuse and its own projection onto it.

Proof. The triangles AHC and ACB share the angle at A and both have a right angle (at H the first, at C the second); by the similarity criterion for two congruent angles (Elements VI.4) they are similar. Hence the corresponding sides are proportional:

$$\frac{AH}{AC} = \frac{AC}{AB} \quad \text{that is} \quad \frac{p}{b} = \frac{b}{c},$$

whence $b^2 = cp$. The symmetric argument on the triangles BHC and BCA , which share the angle at B , gives $a^2 = cq$.

Remark. Adding the two recovers Pythagoras: $a^2 + b^2 = c(p + q) = c \cdot c = c^2$. It is the Pythagorean theorem obtained, at no extra cost, from the leg theorem. $\square$

Theorem 10.3: Altitude theorem. With the same notation,

$$h^2 = pq.$$

The altitude to the hypotenuse is the geometric mean between the two projections.

Proof. By similarity. The triangles AHC and CHB are both right-angled at H . Moreover $\widehat{HAC} = 90^\circ - \widehat{ACH}$ because the angles of AHC sum to 180° , and $\widehat{HCB} = 90^\circ - \widehat{ACH}$ because $\widehat{ACH} + \widehat{HCB} = \widehat{ACB} = 90^\circ$. Hence $\widehat{HAC} = \widehat{HCB}$ and the two triangles are similar. Therefore

$$\frac{AH}{CH} = \frac{CH}{HB} \quad \text{that is} \quad \frac{p}{h} = \frac{h}{q},$$

whence $h^2 = pq$. $\square$

Alternative proof — through the circumcircle. By Thales' theorem (Theorem 4.4) the vertex C lies on the circle of diameter $\overline{AB}$, which is therefore the circumcircle of the triangle. The point H is interior to that circle, and the line through H perpendicular to AB meets it at two points: one is C , the other is the reflection C' of C in AB , because AB is a diameter and hence an axis of symmetry. The chord $\overline{CC'}$ therefore has length $2h$, and H is its midpoint.

Applying the power-of-a-point theorem (Theorem 3.19) at the interior point H , the product of the two segments cut on any chord through H is the same; comparing the chord $\overline{AB}$ with the chord $\overline{CC'}$:

$$AH \cdot HB = CH \cdot HC' \quad \text{that is} \quad pq = h \cdot h = h^2.$$

$\square$

Remark 10.4: The altitude theorem is the power of a point. The second proof is not a trick: it shows that the altitude theorem and the intersecting-chords theorem are the same statement, read once in the triangle and once in the circle. From it also follows the classical construction of the geometric mean of two segments p and q : place them end to end, construct the semicircle of diameter $p + q$ and erect the perpendicular at the junction point; the segment cut off measures $\sqrt{pq}$. This is how one constructs the square root of a

number with straightedge and compass, and it is the step that makes all quadratic extensions constructible.

Corollary 10.2: The complete picture of the metric relations. In a triangle right-angled at C the following all hold together:

$$\begin{aligned} a^2 + b^2 &= c^2, & b^2 &= cp, & a^2 &= cq, & h^2 &= pq, & p + q &= c, \\ h &= \frac{ab}{c}, & p &= \frac{b^2}{c}, & q &= \frac{a^2}{c}, & \frac{1}{h^2} &= \frac{1}{a^2} + \frac{1}{b^2}. \end{aligned}$$

Proof. The first five are Theorem 10.1 and Theorems 10.2 and 10.3.

For $h = ab/c$: the area of the triangle is computed in two ways, as half the product of the legs and as half the product of the hypotenuse by the altitude to it,

$$\frac{ab}{2} = \frac{ch}{2} \implies h = \frac{ab}{c}.$$

The expressions for p and q follow from the leg theorem on dividing by c .

For the last one, start from $h = ab/c$ and square:

$$\frac{1}{h^2} = \frac{c^2}{a^2b^2} = \frac{a^2 + b^2}{a^2b^2} = \frac{a^2}{a^2b^2} + \frac{b^2}{a^2b^2} = \frac{1}{b^2} + \frac{1}{a^2},$$

where Pythagoras has been used. It is the same relation that governs resistors in parallel, and not by chance: both express that the sum of the reciprocals of the squares is the reciprocal of the square of the altitude. $\square$

Example 10.1: The 3–4–5 triangle, measured in all its parts. Let $a = 3$, $b = 4$, $c = 5$ (right-angled at C , with $a = CB$ and $b = CA$).

Projections. $p = b^2/c = 16/5 = 3.2$ and $q = a^2/c = 9/5 = 1.8$; their sum is $5.0 = c$, as it must be.

Altitude. $h = ab/c = 12/5 = 2.4$. Check with the altitude theorem: $h^2 = 5.76$ and $pq = 3.2 \cdot 1.8 = 5.76$. It matches.

Check of the leg theorem. $b^2 = 16$ and $cp = 5 \cdot 3.2 = 16$; $a^2 = 9$ and $cq = 5 \cdot 1.8 = 9$. Both match.

Check of the reciprocal relation. $1/h^2 = 1/5.76 = 0.173611$, while $1/a^2 + 1/b^2 = 1/9 + 1/16 = 0.111111 + 0.062500 = 0.173611$. It matches.

Radii. Circumradius $R = c/2 = 2.5$ (Corollary 10.3); inradius $r = (a + b - c)/2 = (3 + 4 - 5)/2 = 1$ (Theorem 10.4). Euler–Chapple check (Theorem 10.9): $OI^2 = R^2 - 2Rr = 6.25 - 5 = 1.25$, that is $OI = 1.118034$.

10.4. The circle circumscribed about a right triangle

Corollary 10.3: Circumradius of a right triangle. In every right triangle with hypotenuse c , the radius of the circumscribed circle is $R = c/2$, and the centre is the midpoint of the hypotenuse.

Proof. An immediate consequence of Thales' theorem (Theorem 4.4) in its converse form (Theorem 4.5, Chapter 4): the locus of points seeing the hypotenuse AB under a right angle is the circle with diameter AB . The vertex of the right angle therefore lies on that circle, whose

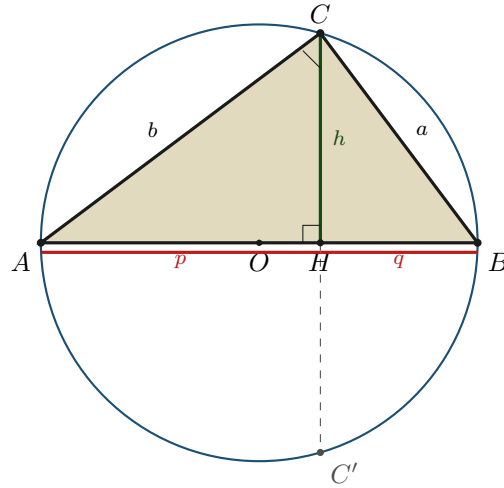


Figure 10.2: — The metric relations: $b^2 = cp$, $a^2 = cq$, $h^2 = pq$. The chord $\overline{CC'}$ of the circumcircle has length $2h$, and the altitude theorem is the intersecting-chords theorem applied at H

| radius is $|AB|/2 = c/2$. □

Alternative proof — by reflection. Let M be the midpoint of the hypotenuse AB , and let C' be the reflection of C in M (the point for which M is also the midpoint of CC'). By construction, segments AB and CC' bisect each other at M : the quadrilateral $ACBC'$ thus has diagonals that bisect each other, hence it is a **parallelogram**. But $\angle ACB = 90^\circ$ by hypothesis: a parallelogram with a right angle is a **rectangle**. In the rectangle $ACBC'$ the two diagonals AB and CC' are **congruent** and bisect each other at M , therefore

$$MA = MB = \frac{AB}{2} = \frac{c}{2}, \quad MC = MC' = \frac{CC'}{2} = \frac{AB}{2} = \frac{c}{2}.$$

In particular $MA = MB = MC = c/2$: the point M is equidistant from the three vertices, hence is the circumcentre. This proof reaches the same conclusion **without invoking Thales' theorem or the inscribed angle**, relying solely on properties of parallelograms and rectangles. □

Alternative proof — analytic. Put $C = (0,0)$, $A = (b,0)$, $B = (0,a)$, with the right angle at C . The midpoint of the hypotenuse is $M = (\frac{b}{2}, \frac{a}{2})$. Compute the three distances:

$$\begin{aligned} |MA| &= \sqrt{\left(\frac{b}{2} - b\right)^2 + \left(\frac{a}{2}\right)^2} = \sqrt{\frac{b^2}{4} + \frac{a^2}{4}} = \frac{1}{2}\sqrt{a^2 + b^2}, \\ |MB| &= \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{a}{2} - a\right)^2} = \frac{1}{2}\sqrt{a^2 + b^2}, \\ |MC| &= \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \frac{1}{2}\sqrt{a^2 + b^2}. \end{aligned}$$

The three distances coincide: M is equidistant from A , B , C , hence is the centre of the unique circle containing them (Corollary 3.2), of radius $R = \frac{1}{2}\sqrt{a^2 + b^2}$. By Theorem 10.1, $\sqrt{a^2 + b^2} = c$, so $R = c/2$.

Note that this third route is *independent* both of Thales and of the properties of parallelograms: it uses only the distance formula (Chapter 7) and Pythagoras. Three proofs in three different registers — synthetic-angular, synthetic-metric, analytic — of one and the same fact. □

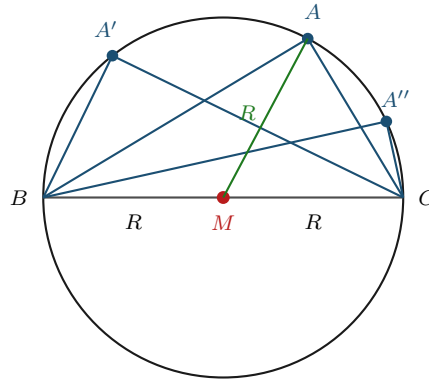


Figure 10.3: Thales' theorem and the circumcircle of the right triangle. Every point A of the semicircle with diameter BC sees the diameter under a right angle: indeed M is equidistant from A , B and C , so the triangles MAB and MAC are isosceles and their base angles, counted twice, add up to a straight angle; hence $\widehat{BAC} = 90^\circ$. The converse also holds: if the triangle has a right angle at A , the circumcentre is the midpoint of the hypotenuse and the radius is half of it, $R = BC/2$. The median to the hypotenuse therefore measures exactly half the hypotenuse — a property false for every other kind of triangle.

10.5. The inscribed circle and the tangent segments

Theorem 10.4: Inradius of a right triangle. In a right triangle with legs a, b and hypotenuse c , the radius of the inscribed circle is

$$r = \frac{a + b - c}{2}.$$

Proof. Let $\triangle ABC$ be right-angled at C , with $BC = a$, $CA = b$, $AB = c$, and let I be the incentre, of radius r , with the incircle touching the sides BC, CA, AB at X, Y, Z respectively. *Step 1 — the tangent segments.* By the congruence of tangent segments drawn from one external point (Theorem 3.11, Chapter 3), setting

$$AY = AZ = x, \quad BX = BZ = y, \quad CX = CY = z,$$

we get

$$x + y = c, \quad y + z = a, \quad z + x = b.$$

Step 2 — the square at the right-angle vertex. The quadrilateral $CXIY$ has: a right angle at C by hypothesis; two right angles at X and Y , because the radius is perpendicular to the side at the point of tangency (Theorem 3.8); and two consecutive congruent sides ($IX = IY = r$, radii). A quadrilateral with three right angles has its fourth right as well (angle sum 2π , axiom A9), so it is a rectangle; and a rectangle with two consecutive congruent sides is a **square** of side r . Hence

$$z = CX = CY = r.$$

Step 3 — substitution. From $y + z = a$ and $z + x = b$ with $z = r$: $y = a - r$ and $x = b - r$. Substituting into the first relation:

$$(b - r) + (a - r) = c \implies a + b - 2r = c \implies r = \frac{a + b - c}{2}.$$

□

Alternative proof — by area. For **every** triangle (not only the right-angled one), joining the incentre I to the three vertices yields three triangles $\triangle IAB$, $\triangle IBC$, $\triangle ICA$, each of height r and bases c, a, b respectively. Adding their areas:

$$K = \frac{1}{2}rc + \frac{1}{2}ra + \frac{1}{2}rb = r \cdot \frac{a + b + c}{2} = rs,$$

where s is the semiperimeter: the relation $K = rs$ holds for every triangle. In the **right-angled** case the area is also $K = \frac{1}{2}ab$. Equating:

$$r = \frac{K}{s} = \frac{ab/2}{(a + b + c)/2} = \frac{ab}{a + b + c}.$$

It remains to check that this agrees with $(a + b - c)/2$:

$$\frac{a + b - c}{2} \cdot (a + b + c) = \frac{(a + b)^2 - c^2}{2} = \frac{a^2 + 2ab + b^2 - c^2}{2} \stackrel{\text{Pythagoras}}{=} \frac{2ab}{2} = ab,$$

that is $ab = \frac{a+b-c}{2}(a+b+c)$, which rearranged gives exactly $r = ab/(a+b+c) = (a+b-c)/2$. This route is independent of the congruence of tangent segments: it uses the general relation $K = rs$ and the Pythagorean theorem alone. $\square$

Remark 10.5: A formula with no square roots. The formula $r = (a + b - c)/2$ has a notable practical quality: if a, b, c are integers (a Pythagorean triple), then r is an integer or a half-integer, with no square root whatever. For the triple $(3, 4, 5)$: $r = (3 + 4 - 5)/2 = 1$. For $(5, 12, 13)$: $r = (5 + 12 - 13)/2 = 2$. For $(8, 15, 17)$: $r = 3$. In fact one proves easily that for every primitive Pythagorean triple the inradius is an integer: with $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$ (Theorem 10.14),

$$r = \frac{(m^2 - n^2) + 2mn - (m^2 + n^2)}{2} = \frac{2mn - 2n^2}{2} = n(m - n),$$

a product of two positive integers.

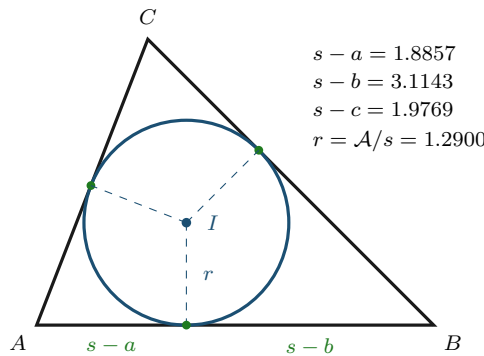


Figure 10.4: The incircle and the tangent segments. The two tangent segments drawn from the same vertex have equal length; calling x, y, z the three values, one has $y + z = a$, $z + x = b$, $x + y = c$, a system solved at once by $x = s - a$, $y = s - b$, $z = s - c$ with s the semiperimeter. The area is obtained by adding the three triangles with apex I , all of height r : $\mathcal{A} = rs$, whence $r = \mathcal{A}/s$. In the example $a = 5.0912$, $b = 3.8626$, $c = 5$, $s = 6.9769$ and $\mathcal{A} = 9$, so $r = 1.2900$; note that the two tangent segments on AB add up to $1.8857 + 3.1143 = 5 = c$.

10.6. Solving right triangles

Proposition 10.1: The four cases of solution. Let ABC be right-angled at C , with legs a, b , hypotenuse c and acute angles $\alpha = \hat{A}$, $\beta = \hat{B}$, with $\alpha + \beta = 90^\circ$. Two independent elements, at least one of which is a side, determine the triangle, namely:

- (i) two legs a, b : $c = \sqrt{a^2 + b^2}$, $\tan \alpha = a/b$, $\beta = 90^\circ - \alpha$;
 - (ii) a leg and the hypotenuse a, c : $\sin \alpha = a/c$, $b = \sqrt{c^2 - a^2}$, $\beta = 90^\circ - \alpha$;
 - (iii) a leg and an acute angle b, α : $a = b \tan \alpha$, $c = b / \cos \alpha$, $\beta = 90^\circ - \alpha$;
 - (iv) the hypotenuse and an acute angle c, α : $a = c \sin \alpha$, $b = c \cos \alpha$, $\beta = 90^\circ - \alpha$.
- Two angles alone do not determine the triangle, only its shape.

Proof. Place the vertex A at the origin and the leg $b = AC$ along the positive x semi-axis; then $C = (b, 0)$ and, since the right angle is at C , the vertex B lies on the perpendicular $x = b$, that is $B = (b, a)$ with $a > 0$. The angle α at A is the one between the positive semi-axis and the ray AB , hence by Definition 8.1 and Theorem 8.1

$$\tan \alpha = \frac{a}{b}, \quad \sin \alpha = \frac{a}{c}, \quad \cos \alpha = \frac{b}{c},$$

where $c = AB = \sqrt{a^2 + b^2}$ by Theorem 10.1. These three relations, together with $\alpha + \beta = 90^\circ$ (angle sum, *Elements* I.32), allow the three missing elements to be recovered in each of the four cases, simply by solving for the unknown:

- in (i) c from Pythagoras and α from $\arctan(a/b)$, which is well defined and lies in $(0, \pi/2)$ because $a/b > 0$;
- in (ii) α from $\arcsin(a/c)$, legitimate because $0 < a/c < 1$, and b from Pythagoras;
- in (iii) $a = b \tan \alpha$ and $c = b / \cos \alpha$, where $\cos \alpha \neq 0$ because $\alpha \in (0, \pi/2)$;
- in (iv) a and b directly from the definitions of sine and cosine.

Finally, if only α and β are prescribed (with sum 90°), every homothety of centre A and ratio $k > 0$ produces a triangle with the same angles and sides ka, kb, kc : the shape is determined, the size is not. $\square$

Remark 10.6: The practical rule and the check always worth making. In practice it is enough to remember one sentence — “a leg is the hypotenuse times the sine of its opposite angle, or the other leg times the tangent of the same angle” — and then to verify the result with a relation not used in the computation. The best check is Pythagoras when the trigonometric functions were used, and the other way round: if c was obtained from $b / \cos \alpha$, check that $a^2 + b^2$ gives c^2 ; if c came from Pythagoras, check that a/c gives $\sin \alpha$. An error of one order of magnitude, or the use of the wrong function, shows up immediately.

Example 10.2: A right triangle solved completely. Of a right triangle one knows the leg $b = 24.00$ m and the adjacent acute angle $\alpha = 32.25^\circ$. This is case (iii). With

$$\sin \alpha = 0.533615, \quad \cos \alpha = 0.845728, \quad \tan \alpha = 0.630953,$$

one obtains

$$a = b \tan \alpha = 24 \cdot 0.630953 = 15.142872 \text{ m}, \quad c = \frac{b}{\cos \alpha} = \frac{24}{0.845728} = 28.377924 \text{ m},$$

and $\beta = 90^\circ - 32.25^\circ = 57.75^\circ$.

Check with Pythagoras. $a^2 + b^2 = 229.306572 + 576 = 805.306572$, and $c^2 = 28.377924^2 = 805.306571$: they agree up to rounding (difference in the last digit, 10^{-6}).

Derived elements. Area $S = ab/2 = 24 \cdot 15.142872/2 = 181.714464 \text{ m}^2$. Altitude to the hypotenuse $h = ab/c = 363.428928/28.377924 = 12.806748 \text{ m}$ (Corollary 10.2). Projections $p = b^2/c = 576/28.377924 = 20.297468 \text{ m}$ and $q = a^2/c = 229.306572/28.377924 = 8.080456 \text{ m}$, whose sum is 28.377924 m , that is c .

Check with the altitude theorem. $h^2 = 164.012794$ and $pq = 20.297468 \cdot 8.080456 = 164.012797$: they agree up to rounding.

Radii. $R = c/2 = 14.188962 \text{ m}$; $r = (a + b - c)/2 = (15.142872 + 24 - 28.377924)/2 = 5.382474 \text{ m}$. Independent check with $r = S/s$, where $s = (a + b + c)/2 = 33.760398 \text{ m}$: $181.714464/33.760398 = 5.382474 \text{ m}$. It matches.

10.7. The law of sines

Theorem 10.5: Law of sines (extended form). In every triangle ABC inscribed in a circle of radius R , with sides $a = BC$, $b = CA$, $c = AB$ opposite the angles A, B, C respectively:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Proof 1 — via the auxiliary diameter (Thales). Draw the diameter BD from vertex B . By Thales' theorem, $\angle BCD = 90^\circ$. By Theorem 4.1 (inscribed angle), $\angle BDC = \angle BAC = A$ if D and A lie on the same arc with respect to BC (or its supplement if they lie on opposite arcs, but the final result is identical for the sine formula, since $\sin(180^\circ - A) = \sin A$). In the right triangle BCD :

$$\sin(\angle BDC) = \frac{BC}{BD} = \frac{a}{2R} \implies \sin A = \frac{a}{2R} \implies \frac{a}{\sin A} = 2R.$$

By symmetry (repeating with diameters from A and C), the same constant $2R$ is obtained for $b/\sin B$ and $c/\sin C$. $\square$

Proof 2 — via the area formula. The area of the triangle can be written in two equivalent ways: $K = \frac{1}{2}ab \sin C$ (standard trigonometric area) and, independently, $K = \frac{abc}{4R}$ (area in terms of the circumradius, itself derivable from $K = \frac{1}{2}ab \sin C$ by substituting $\sin C = c/(2R)$ — which shows consistency; for an independent derivation one may use $K = rs$ with s the semiperimeter and compare with Heron's formula). Equating the two expressions for K :

$$\frac{1}{2}ab \sin C = \frac{abc}{4R} \implies \sin C = \frac{c}{2R} \implies \frac{c}{\sin C} = 2R.$$

$\square$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \iff R = \frac{a}{2 \sin A}, \quad A = \arcsin\left(\frac{a}{2R}\right).$$

10.8. The law of cosines

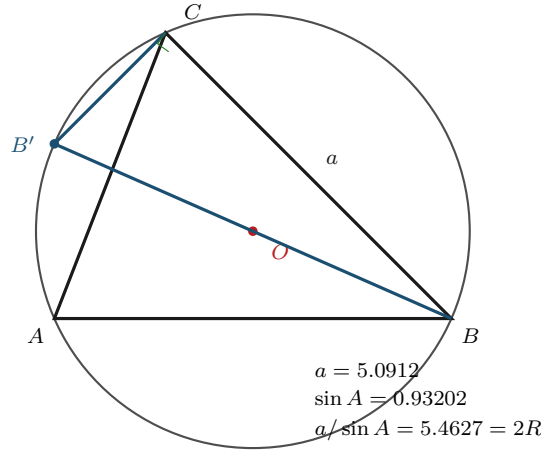


Figure 10.5: Why $2R$ appears in the law of sines. Draw the diameter BB' through the vertex B : by Thales' theorem the angle at C in the triangle BCB' is right, hence $\sin(\widehat{BB'C}) = BC/BB' = a/(2R)$. But the angles $\widehat{BB'C}$ and $\widehat{BAC} = A$ stand on the same arc BC , so they are equal by the inscribed angle theorem: therefore $\sin A = a/(2R)$, i.e. $a/\sin A = 2R$. Repeating for the other two vertices gives $a/\sin A = b/\sin B = c/\sin C = 2R$: the common value of the three ratios is not an arbitrary number, but the diameter of the circumcircle. In the example $a = 5.0912$, $\sin A = 0.93202$ and the ratio is $5.4627 = 2R$.

Theorem 10.6: Law of cosines (generalisation of Pythagoras). In every triangle ABC : $c^2 = a^2 + b^2 - 2ab \cos C$.

Proof. Put C at the origin, $B = (a, 0)$, $A = (b \cos C, b \sin C)$ (polar coordinates of vertex A as seen from C). Then

$$\begin{aligned} c^2 &= |AB|^2 = (a - b \cos C)^2 + (b \sin C)^2 \\ &= a^2 - 2ab \cos C + b^2 \cos^2 C + b^2 \sin^2 C \\ &= a^2 + b^2 - 2ab \cos C, \end{aligned}$$

using the Pythagorean identity $\cos^2 C + \sin^2 C = 1$. □

Proof 2 — synthetic (Euclid II.12–13). This is the form in which the result appears in the *Elements*, two centuries before trigonometry existed: two distinct propositions, one for the obtuse case (II.12) and one for the acute (II.13), which the signed cosine unifies.

Case C acute. Draw the altitude from B to the side AC , with foot H . Since C is acute, H falls between A and C . In the right triangle BHC , by definition of cosine, $CH = a \cos C$; and by the Pythagorean theorem (Theorem 10.1), $BH^2 = a^2 - CH^2$. In the right triangle BHA we have $AH = b - CH$ and, again by Pythagoras,

$$\begin{aligned} c^2 &= AB^2 = BH^2 + AH^2 \\ &= (a^2 - CH^2) + (b - CH)^2 \\ &= a^2 + b^2 - 2bCH = a^2 + b^2 - 2ab \cos C. \end{aligned}$$

Case C obtuse. Now H falls *outside* the segment AC , on the side of C . Setting $CH = a |\cos C| = -a \cos C$ (positive, since $\cos C < 0$), we have $AH = b + CH$ and

$$c^2 = (a^2 - CH^2) + (b + CH)^2 = a^2 + b^2 + 2bCH = a^2 + b^2 - 2ab \cos C,$$

the same formula. *Case C right:* $H = C$, $CH = 0$, and we fall back onto Pythagoras.

Why Euclid makes it two propositions. Because he has no signed magnitude: II.12 states $c^2 = a^2 + b^2 + 2b \cdot CH$ for the obtuse case and II.13 states $c^2 = a^2 + b^2 - 2b \cdot CH$ for the acute, with CH always positive. It is precisely the sign of $\cos C$ that fuses the two statements into one. $\square$

Proof 3 — from the projection formulas. *Step 1 — the projection formulas.* In every triangle,

$$a = b \cos C + c \cos B, \quad b = c \cos A + a \cos C, \quad c = a \cos B + b \cos A.$$

All are proved the same way: the foot H of the altitude from A divides the side BC (or its extension) into the two segments $BH = c \cos B$ and $HC = b \cos C$, by definition of cosine in the two right triangles ABH and ACH ; their signed sum is a . The obtuse case is handled as in Proof 2, with one segment counted negatively.

Step 2 — multiply each by its own side.

$$a^2 = ab \cos C + ac \cos B, \quad b^2 = bc \cos A + ab \cos C, \quad c^2 = ac \cos B + bc \cos A.$$

Step 3 — combine. Subtracting the first from the sum of the other two:

$$\begin{aligned} b^2 + c^2 - a^2 &= (bc \cos A + ab \cos C) + (ac \cos B + bc \cos A) - (ab \cos C + ac \cos B) \\ &= 2bc \cos A, \end{aligned}$$

that is, $a^2 = b^2 + c^2 - 2bc \cos A$: the law of cosines for the angle A . The other two follow by cyclic permutation of the letters.

Note. This proof has one virtue: it produces *all three* versions of the law of cosines with a single symmetric computation, without having to single out a privileged vertex. $\square$

$$c^2 = a^2 + b^2 - 2ab \cos C \iff \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Remark 10.7: Pythagoras as a special case. For $C = 90^\circ$, $\cos C = 0$ and the law of cosines reduces exactly to the Pythagorean theorem: $c^2 = a^2 + b^2$. Proof 4 of Theorem 10.1 is in fact the special case $C = 90^\circ$ of this same vectorial proof.

10.9. Mollweide's formulas and the law of tangents

The law of sines (Theorem 10.5) and the law of cosines (Theorem 10.6) allow any triangle to be solved, but they offer no way of *checking* the result: a relation used in the computation cannot also serve as its verification. Mollweide's two formulas fill this gap, because they contain *all six* elements of the triangle.

Theorem 10.7: Mollweide's formulas. In every triangle with sides a, b, c and opposite angles α, β, γ :

$$\frac{a+b}{c} = \frac{\cos \frac{\alpha-\beta}{2}}{\sin \frac{\gamma}{2}}, \quad \frac{a-b}{c} = \frac{\sin \frac{\alpha-\beta}{2}}{\cos \frac{\gamma}{2}}.$$

Proof. By the law of sines in extended form (Theorem 10.5) one has $a = 2R \sin \alpha$, $b = 2R \sin \beta$, $c = 2R \sin \gamma$, where R is the circumradius. The factor $2R$ cancels in every ratio, hence

$$\frac{a+b}{c} = \frac{\sin \alpha + \sin \beta}{\sin \gamma}, \quad \frac{a-b}{c} = \frac{\sin \alpha - \sin \beta}{\sin \gamma}.$$

Apply the sum-to-product formulas (Theorem 8.6) to the numerator and the double-angle formula (Corollary 8.2) to the denominator:

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}, \quad \sin \gamma = 2 \sin \frac{\gamma}{2} \cos \frac{\gamma}{2}.$$

Since $\alpha + \beta + \gamma = \pi$, one has $\frac{\alpha + \beta}{2} = \frac{\pi}{2} - \frac{\gamma}{2}$, and by the associated angles (Proposition 9.2)

$$\sin \frac{\alpha + \beta}{2} = \cos \frac{\gamma}{2}, \quad \cos \frac{\alpha + \beta}{2} = \sin \frac{\gamma}{2}.$$

Substituting into the first,

$$\frac{a+b}{c} = \frac{2 \cos \frac{\gamma}{2} \cos \frac{\alpha - \beta}{2}}{2 \sin \frac{\gamma}{2} \cos \frac{\gamma}{2}} = \frac{\cos \frac{\alpha - \beta}{2}}{\sin \frac{\gamma}{2}},$$

where cancelling $\cos(\gamma/2)$ is legitimate because $\gamma \in (0, \pi)$ implies $\gamma/2 \in (0, \pi/2)$.

For the second one uses $\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$ and, with the same substitution,

$$\frac{a-b}{c} = \frac{2 \sin \frac{\gamma}{2} \sin \frac{\alpha - \beta}{2}}{2 \sin \frac{\gamma}{2} \cos \frac{\gamma}{2}} = \frac{\sin \frac{\alpha - \beta}{2}}{\cos \frac{\gamma}{2}}.$$

□

Corollary 10.4: Law of tangents. If $a \neq b$,

$$\frac{a-b}{a+b} = \frac{\tan \frac{\alpha - \beta}{2}}{\tan \frac{\alpha + \beta}{2}}.$$

Proof. Divide the second Mollweide formula by the first; the factor c cancels and there remains

$$\frac{a-b}{a+b} = \frac{\sin \frac{\alpha - \beta}{2}}{\cos \frac{\gamma}{2}} \cdot \frac{\sin \frac{\gamma}{2}}{\cos \frac{\alpha - \beta}{2}} = \tan \frac{\alpha - \beta}{2} \cdot \tan \frac{\gamma}{2}.$$

Since $\gamma/2 = \pi/2 - (\alpha + \beta)/2$, by the associated angles $\tan(\gamma/2) = \cot((\alpha + \beta)/2) = 1/\tan((\alpha + \beta)/2)$, which gives the assertion. The division is legitimate because $a + b > 0$ always, and $\cos((\alpha - \beta)/2) > 0$ since $|\alpha - \beta| < \pi$. □

Remark 10.8: What the law of tangents is really for. Before calculators, the law of tangents was the standard way of solving the case “two sides and the included angle”: given a , b and γ , one knows $(\alpha + \beta)/2 = 90^\circ - \gamma/2$, and the law supplies $(\alpha - \beta)/2$; from sum

and difference one recovers α and β without ever extracting a square root, that is, using logarithm tables alone. Today its value is different but no smaller: Mollweide's formulas contain the six elements of the triangle, hence any error in a single one of them makes them fail. They are the final check that a solved triangle has been solved correctly.

Example 10.3: Mollweide as a test of a solution. Take again the triangle of Example 10.2: $a = 15.142872$, $b = 24.0000$, $c = 28.377924$, with $\alpha = 32.25^\circ$, $\beta = 57.75^\circ$, $\gamma = 90^\circ$.

First formula. $\frac{a+b}{c} = \frac{39.142872}{28.377924} = 1.379342$. On the right: $(\alpha - \beta)/2 = -12.75^\circ$, so $\cos(-12.75^\circ) = 0.975342$, and $\gamma/2 = 45^\circ$, so $\sin 45^\circ = 0.707107$. The ratio is $0.975342/0.707107 = 1.379342$. The two values agree up to the sixth decimal digit (discrepancy $\sim 10^{-9}$, due only to rounding of the intermediate data). ✓

Second formula. $\frac{a-b}{c} = \frac{-8.857128}{28.377924} = -0.312113$. On the right: $\sin(-12.75^\circ) = -0.220697$ and $\cos 45^\circ = 0.707107$, with ratio $-0.220697/0.707107 = -0.312113$. They agree. ✓

Law of tangents. $\frac{a-b}{a+b} = \frac{-8.857128}{39.142872} = -0.226277$; on the right $\tan(-12.75^\circ)/\tan(45^\circ) = -0.226277/1 = -0.226277$. This too agrees up to the sixth digit: using the values at full precision (not rounded to four digits as in an early draft) the law of tangents tests out as well as the other two.

What an error would have revealed. Had one carelessly written $c = 28.9$ instead of 28.377924 , the first formula would have given 1.354425 against 1.379342 : a discrepancy of 1.8% , well beyond rounding, and the error would have been caught at once.

10.10. Excircles

Definition 10.2: Excircles and exradii. Given a triangle ABC with sides $a = BC$, $b = CA$, $c = AB$, the **excircle opposite A** is the circle tangent to the side $\overline{BC}$ and to the extensions of the other two sides; its centre I_a is the intersection of the internal bisector of the angle at A with the external bisectors of the angles at B and C , and its radius is written r_a . The radii r_b and r_c are defined analogously. We also set $s = (a+b+c)/2$ (semiperimeter) and write S for the area of the triangle.

Theorem 10.8: Exradius formulas. One has

$$r = \frac{S}{s}, \quad r_a = \frac{S}{s-a}, \quad r_b = \frac{S}{s-b}, \quad r_c = \frac{S}{s-c}.$$

Moreover, in a triangle right-angled at C ,

$$r = s - c = \frac{a+b-c}{2}, \quad r_a = s - b, \quad r_b = s - a, \quad r_c = s.$$

Proof. The incircle case. The centre I is equidistant from the three sides, at distance r ; joining it to the three vertices splits the triangle into three triangles with bases a , b , c and common height r , hence $S = \frac{1}{2}(a+b+c)r = sr$, that is $r = S/s$.

The excircle case. The centre I_a is equidistant, at distance r_a , from the line BC and from the lines AB , AC . The quadrilateral ABI_aC decomposes in two ways: as the union of the triangles ABI_a and ACI_a minus the triangle BCI_a , or as the union of ABC and BCI_a . Comparing the

two decompositions,

$$\underbrace{\frac{1}{2}cr_a}_{ABI_a} + \underbrace{\frac{1}{2}br_a}_{ACI_a} - \underbrace{\frac{1}{2}ar_a}_{BCI_a} = S,$$

since each of the three triangles has height r_a relative to the corresponding side. Hence

$$S = \frac{r_a}{2}(b+c-a) = r_a(s-a), \quad \text{that is} \quad r_a = \frac{S}{s-a},$$

where $s-a > 0$ by the triangle inequality. The other two are identical by symmetry.

The right triangle. If the right angle is at C then $S = ab/2$ and $c^2 = a^2 + b^2$. One has

$$s-c = \frac{a+b-c}{2}, \quad \frac{S}{s} = \frac{ab/2}{(a+b+c)/2} = \frac{ab}{a+b+c}.$$

The two expressions coincide: indeed

$$\frac{a+b-c}{2} \cdot (a+b+c) = \frac{(a+b)^2 - c^2}{2} = \frac{a^2 + 2ab + b^2 - a^2 - b^2}{2} = ab,$$

hence $(s-c)(2s) = ab = 2S$, that is $s-c = S/s = r$. Analogously

$$r_a = \frac{S}{s-a}, \quad (s-a)(s-b) = \frac{(b+c-a)(a+c-b)}{4} = \frac{c^2 - (a-b)^2}{4} = \frac{a^2 + b^2 - a^2 + 2ab - b^2}{4} = \frac{ab}{2} = S,$$

whence $S/(s-a) = s-b$, that is $r_a = s-b$; by symmetry $r_b = s-a$. Finally $r_c = S/(s-c) = S/r$, and since $S = sr$ one gets $r_c = s$. $\square$

Corollary 10.5: *The relation between exradii, inradius and circumradius. In every triangle*

$$r_a + r_b + r_c - r = 4R.$$

Proof. Set $x = s-a$, $y = s-b$, $z = s-c$, all positive; then $x+y+z = 3s-(a+b+c) = 3s-2s = s$, and moreover

$$x+y = 2s-a-b = c, \quad y+z = a, \quad z+x = b.$$

By Heron's formula $S^2 = sxyz$, which is the degenerate case $d = 0$ of Brahmagupta's formula (Theorem 6.9). Using Theorem 10.8,

$$r_a + r_b + r_c - r = S\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - \frac{1}{s}\right) = \frac{S}{sxyz} \left[s(yz + zx + xy) - xyz \right],$$

everything having been put over the common denominator $sxyz$. Since $S^2 = sxyz$, the prefactor equals $S/S^2 = 1/S$; there remains

$$r_a + r_b + r_c - r = \frac{(x+y+z)(xy+yz+zx) - xyz}{S},$$

after substituting $s = x+y+z$. Now the algebraic identity

$$(x+y+z)(xy+yz+zx) - xyz = (x+y)(y+z)(z+x)$$

holds, as one verifies by expanding both sides: on the left one gets $\sum_{\text{sym}} x^2y + 3xyz - xyz = \sum_{\text{sym}} x^2y + 2xyz$, and on the right $(x+y)(y+z)(z+x) = \sum_{\text{sym}} x^2y + 2xyz$. Substituting $x+y = c$, $y+z = a$, $z+x = b$,

$$r_a + r_b + r_c - r = \frac{abc}{S}.$$

Finally, by the law of sines in extended form (Theorem 10.5) and by $S = \frac{1}{2}ab \sin \gamma$,

$$\frac{abc}{S} = \frac{abc}{\frac{1}{2}ab \sin \gamma} = \frac{2c}{\sin \gamma} = 4R,$$

since $c/\sin \gamma = 2R$. □

Example 10.4: The four radii of the 3–4–5 triangle. With $a = 3$, $b = 4$, $c = 5$ one has $s = 6$ and $S = 6$.

Inradius. $r = S/s = 1$, which coincides with $s - c = 6 - 5 = 1$ and with $(a + b - c)/2 = 1$ (Theorem 10.4).

Exradii.

$$r_a = \frac{6}{6-3} = 2 = s - b, \quad r_b = \frac{6}{6-4} = 3 = s - a, \quad r_c = \frac{6}{6-5} = 6 = s.$$

Check of Corollary 10.5. $r_a + r_b + r_c - r = 2 + 3 + 6 - 1 = 10$, while $4R = 4 \cdot 2.5 = 10$. It matches.

Note. The exradius opposite the right angle equals exactly the semiperimeter, and it is the largest of the four: the excircle on the hypotenuse side has radius 6, more than twice the hypotenuse itself divided by two. It is a direct consequence of $r_c = S/(s - c) = S/r$ together with the fact that, in a right triangle, r is the smallest of the four radii.

10.11. Euler's formula: $OI^2 = R^2 - 2Rr$

Lemma 10.1: Incentre–excentre lemma (“Fact 5”). *Let M be the midpoint of arc BC not containing A . Then $MB = MC = MI$, where I is the incentre of $\triangle ABC$.*

Proof. $MB = MC$ because M is the arc midpoint, and congruent arcs subtend congruent chords (Theorem 3.4). To show $MI = MB$: the inscribed angles $\angle BAM$ and $\angle MAC$ stand on the congruent arcs $\widehat{BM}$ and $\widehat{MC}$ (by definition of M as the midpoint of arc BC), hence they are congruent (Corollary 4.2); since their sum is the angle $\angle BAC = A$, each equals $A/2$: the ray AM bisects $\angle BAC$, that is, AM is the bisector from A , and in particular $\angle BAM = A/2$. By Theorem 4.1, $\angle MBC = \angle MAC = A/2$ (inscribed angles on the same arc MC). The angle $\angle MBI = \angle MBC + \angle CBI = A/2 + B/2$ (since BI bisects B). On the other hand $\angle MIB$ is the exterior angle of triangle $\triangle ABI$ at I (as A, I, M are collinear): by the exterior angle theorem, $\angle MIB = \angle IAB + \angle IBA = A/2 + B/2$. Hence $\angle MBI = \angle MIB$: the triangle $\triangle MBI$ is isosceles with $MI = MB$. □

Proposition 10.2: The three area formulas with the radii. *With K the area, s the semiperimeter, R the circumradius, r the inradius and r_a the exradius opposite A :*

$$K = \frac{abc}{4R}, \quad K = rs, \quad K = r_a(s - a).$$

Proof. First formula. $K = \frac{1}{2}ab \sin C$ (trigonometric area); by the law of sines, $c = 2R \sin C$, that is, $\sin C = c/(2R)$; substituting, $K = \frac{1}{2}ab \cdot c/(2R) = abc/(4R)$.

Second formula. The incentre I is at distance r from each side (by definition of the inradius, Definition 10.2 or equivalent); the triangle ABC is the union of the three triangles ABI , BCI , CAI , with bases c, a, b and equal height r from I : $K = \frac{1}{2}cr + \frac{1}{2}ar + \frac{1}{2}br = r \cdot \frac{a+b+c}{2} = rs$.

Third formula. The excentre I_a (opposite A) is at distance r_a from each of the three side-lines, but lies on the side opposite A relative to BC : the triangle ABC is then the difference $\text{area}(ABI_a) + \text{area}(ACI_a) - \text{area}(BCI_a)$ (the two triangles on AB and AC contain ABC plus the triangle on BC , which must be subtracted), with bases c, b, a and equal height r_a from I_a : $K = \frac{1}{2}cr_a + \frac{1}{2}br_a - \frac{1}{2}ar_a = r_a \cdot \frac{b+c-a}{2} = r_a(s-a)$. $\square$

Numerical check. Triangle 13–14–15: $s = 21$, $K = 84$ (Heron's formula: $\sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84$), $R = abc/(4K) = 2730/336 = 8.125$, $r = K/s = 84/21 = 4$ (all checked).

Second proof of Heron's formula, via the radii. Let D, D_a be the points of tangency on side BC of the incircle and of the excircle opposite A , respectively; by the standard tangent lengths, $BD = s - b$ and $BD_a = s - c$. In the right triangles BDI (right-angled at D , since $ID \perp BC$, $ID = r$) and I_aD_aB (right-angled at D_a , since $I_aD_a \perp BC$, $I_aD_a = r_a$): the angle $\angle DBI = B/2$ (BI is the internal bisector of angle B , and $D \in BC$); in the second triangle, BI_a is the **external** bisector of angle B , which bisects the exterior angle $\pi - B$, so the angle between BI_a and the line BC equals $\frac{\pi}{2} - \frac{B}{2}$: in the right triangle I_aD_aB (right-angled at D_a), the angle at vertex B is therefore $\frac{\pi}{2} - \frac{B}{2}$, and consequently the angle at vertex I_a equals $\frac{B}{2}$ (sum of interior angles). The two right triangles BDI and I_aD_aB thus have equal acute angles ($\angle DBI = \angle D_aI_aB = B/2$): they are similar (AA), with the correspondence $B \leftrightarrow I_a$, $D \leftrightarrow D_a$, $I \leftrightarrow B$. From corresponding sides,

$$\frac{BD}{I_aD_a} = \frac{DI}{D_aB} \implies BD \cdot D_aB = DI \cdot I_aD_a \implies (s-b)(s-c) = r r_a.$$

Combining with $K = rs$ and $K = r_a(s-a)$ (Proposition 10.2):

$$K^2 = (rs)(r_a(s-a)) = s(s-a)rr_a = s(s-a)(s-b)(s-c),$$

Heron's formula, via a route entirely independent of the one in Chapter 9 (here Pythagoras is never used, only tangents and similar triangles). *Numerical check:* triangle 13-14-15, $r = 4$, $r_a = 12$ (excircle opposite side $a = 14$), $(s-b)(s-c) = 8 \cdot 6 = 48 = r r_a$ (checked independently of Heron, computing r, r_a from $\text{area} = \text{base} \times \text{height}/2$). $\square$

Proposition 10.3: Briggs's formulas (half-angle).

$$\sin^2 \frac{A}{2} = \frac{(s-b)(s-c)}{bc}, \quad \cos^2 \frac{A}{2} = \frac{s(s-a)}{bc}, \quad \tan \frac{A}{2} = \frac{r}{s-a}.$$

Proof. From the law of cosines, $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, hence

$$1 - \cos A = \frac{2bc - b^2 - c^2 + a^2}{2bc} = \frac{a^2 - (b-c)^2}{2bc} = \frac{(a-b+c)(a+b-c)}{2bc} = \frac{2(s-b) \cdot 2(s-c)}{2bc} = \frac{2(s-b)(s-c)}{bc}$$

(with $a-b+c = 2(s-b)$, $a+b-c = 2(s-c)$). By the half-angle formula $1 - \cos A = 2 \sin^2(A/2)$: $\sin^2(A/2) = (s-b)(s-c)/bc$. Likewise,

$$1 + \cos A = \frac{(b+c)^2 - a^2}{2bc} = \frac{(b+c-a)(b+c+a)}{2bc} = \frac{2(s-a) \cdot 2s}{2bc} = \frac{2s(s-a)}{bc},$$

and $1 + \cos A = 2 \cos^2(A/2)$ gives $\cos^2(A/2) = s(s-a)/bc$. The third formula follows from the tangent segments: let T be the point of tangency of the incircle on side AB ; $AT = s - a$ (standard tangent length) and $IT = r$ (radius, perpendicular to the side at T); in the right triangle AIT (right-angled at T), since AI bisects the angle, $\angle IAT = A/2$, hence $\tan(A/2) = IT/AT = r/(s-a)$. $\square$

Numerical check. Triangle 7-8-9: $\sin^2(A/2) = 1/6$, $\cos^2(A/2) = 5/6$, $\tan(A/2) = r/(s-a) = 0.447214$, all confirmed against the direct values of $A = \arccos((64+81-49)/144)$ (checked).

Theorem 10.9: Euler–Chapple formula (1746/1765). For every triangle with circumradius R , inradius r and distance $d = OI$ between circumcentre O and incentre I :

$$OI^2 = R^2 - 2Rr.$$

Proof. Consider the line AI , meeting the circumscribed circle at a second point M (the arc midpoint of BC from Lemma 10.1). The power of the point I (interior to the circle) with respect to the circumcircle is, by Theorem 3.19 (Chapter 3),

$$R^2 - OI^2 = IA \cdot IM$$

(the sign is reversed with respect to Definition 3.1 because we take here the absolute value for an interior point). We compute the two factors separately:

IA : in the right triangle formed by the foot of the perpendicular from I to side AB (distance $= r$, the inradius) and the angle $\angle IAB = A/2$ (bisector): $\sin(A/2) = r/IA$, hence $IA = r/\sin(A/2)$.

IM : by Lemma 10.1, $IM = MB$; and MB is a chord subtending the inscribed angle $\angle BAM = A/2$ (Theorem 10.5, extended law of sines, applied to the chord MB): $MB = 2R \sin(A/2)$.

Substituting:

$$R^2 - OI^2 = IA \cdot IM = \frac{r}{\sin(A/2)} \cdot 2R \sin(A/2) = 2Rr,$$

whence $OI^2 = R^2 - 2Rr$. □

Corollary 10.6: Euler’s inequality. $R \geq 2r$ for every triangle, with equality if and only if the triangle is equilateral.

Proof. From $OI^2 = R^2 - 2Rr \geq 0$ (being the square of a real distance) it follows that $R \geq 2r$. Equality holds iff $OI = 0$, i.e. iff $O = I$: for a non-equilateral triangle, circumcentre and incentre are distinct points, whereas they coincide exactly in the equilateral case, by symmetry. □

$$\begin{aligned} OI^2 = R^2 - 2Rr &\iff r = \frac{R^2 - OI^2}{2R} \\ &\iff R = r + \sqrt{r^2 + OI^2} \quad (\text{positive root of the quadratic in } R). \end{aligned}$$

Remark 10.9: A panorama: Poncelet’s porism and Fuss’s theorem. Euler’s formula $OI^2 = R^2 - 2Rr$ has a deeper meaning than it appears to: it is exactly the condition for there to exist, given two circles of radii $R > r$ and distance between centres $d = \sqrt{R^2 - 2Rr}$, a triangle inscribed in the first and circumscribed about the second. The remarkable fact, **imported** (Poncelet, 1822; see Coxeter–Greitzer, *Geometry Revisited*), is that if one such triangle exists, then **infinitely many** exist: starting from any point of the large circle and drawing the tangents to the small one, the path always closes up after three steps (**Poncelet’s porism** for $n = 3$). An analogous phenomenon holds for quadrilaterals: the condition for a quadrilateral inscribed in a circle of radius R and circumscribed about one of radius r , with distance d between the centres, to exist is **Fuss’s theorem** (1792, **imported**):

$$\frac{1}{(R-d)^2} + \frac{1}{(R+d)^2} = \frac{1}{r^2}.$$

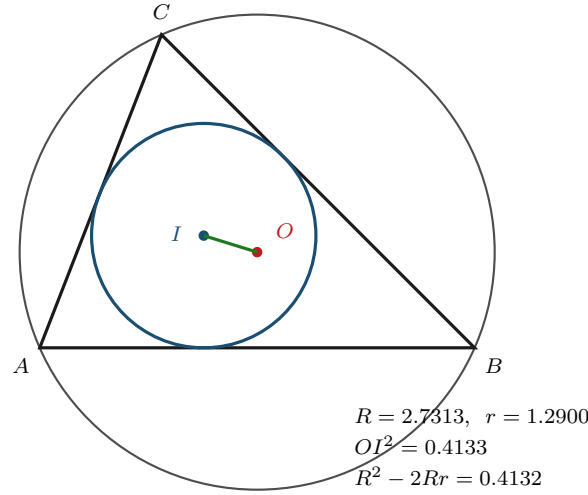


Figure 10.6: Euler's formula $OI^2 = R^2 - 2Rr$ links the distance between circumcentre and incentre to the two radii. In the triangle drawn $R = 2.7313$ and $r = 1.2900$, and the numerical check gives $OI^2 = 0.4133$ against $R^2 - 2Rr = 0.4132$ (the difference is rounding only). Since the left-hand side is a square, the right-hand side cannot be negative: one gets Euler's inequality $R \geq 2r$ for free, with equality if and only if $O = I$, i.e. for the equilateral triangle. It is remarkable that so simple a relation between three quantities of the triangle escaped the Greeks: it was published by Euler only in 1765.

Check on the square (case $d = 0$, coincident centres): with side ℓ , $R = \ell/\sqrt{2}$ (half-diagonal) and $r = \ell/2$ (apothem), $\frac{1}{R^2} + \frac{1}{R^2} = \frac{2}{R^2} = \frac{4}{\ell^2} = \frac{1}{r^2}$ (checked).

10.12. The nine-point circle (Feuerbach circle)

Lemma 10.2: The orthocentre vector. With origin at the circumcentre O (so that $|\vec{A}| = |\vec{B}| = |\vec{C}| = R$), $\vec{OH} = \vec{A} + \vec{B} + \vec{C}$.

Proof. Let $H' = \vec{A} + \vec{B} + \vec{C}$ (as a point, position vector from O). Then

$$(\vec{H}' - \vec{A}) \cdot (\vec{B} - \vec{C}) = (\vec{B} + \vec{C}) \cdot (\vec{B} - \vec{C}) = |\vec{B}|^2 - |\vec{C}|^2 = R^2 - R^2 = 0,$$

so the vector $\vec{H}' - \vec{A} = \vec{AH}'$ is perpendicular to the side $\vec{BC} = \vec{C} - \vec{B}$: the line AH' is the altitude from A . By symmetry of the letters, $(\vec{H}' - \vec{B}) \cdot (\vec{A} - \vec{C}) = |\vec{A}|^2 - |\vec{C}|^2 = 0$ and $(\vec{H}' - \vec{C}) \cdot (\vec{A} - \vec{B}) = |\vec{A}|^2 - |\vec{B}|^2 = 0$: BH' and CH' are also the altitudes from B and from C . The point H' therefore lies at the intersection of the three altitudes, that is, $H' = H$ (the orthocentre is unique, by definition, the three altitudes being concurrent). $\square$

Theorem 10.10: The Euler line. In every non-equilateral triangle, the circumcentre O , the centroid G and the orthocentre H are collinear, with $\vec{OG} = \frac{1}{3}\vec{OH}$; O, G, H coincide if and only if the triangle is equilateral.

Proof. With origin at O : the centroid is by definition $\vec{G} = \frac{1}{3}(\vec{A} + \vec{B} + \vec{C})$, and by Lemma 10.2, $\vec{H} = \vec{A} + \vec{B} + \vec{C} = 3\vec{G}$. Hence $\vec{OG} = \vec{G}$ and $\vec{OH} = \vec{H} = 3\vec{G} = 3\vec{OG}$: the three points are collinear (since $\vec{OH}$ is a scalar multiple of $\vec{OG}$), with $\vec{OG} = \frac{1}{3}\vec{OH}$. From the same identity

$\vec{OH} = 3\vec{OG}$ it follows at once that $O = H$ (as points) implies $\vec{OG} = \vec{0}$, that is, $O = G$: hence the three points either coincide simultaneously or are all three distinct (it cannot happen that only two coincide). If the triangle is equilateral, $O = G = H$ (the centre of symmetry of the equilateral triangle is simultaneously circumcentre, centroid and orthocentre, by symmetry about each axis). Conversely, if $O = H$ (origin at the circumcentre), then $\vec{0} = \vec{OH} = \vec{A} + \vec{B} + \vec{C}$, that is, $\vec{C} = -(\vec{A} + \vec{B})$. Squaring: $|\vec{C}|^2 = |\vec{A}|^2 + |\vec{B}|^2 + 2\vec{A} \cdot \vec{B}$, and since $|\vec{A}| = |\vec{B}| = |\vec{C}| = R$ (the three vertices lie on the circumscribed circle), $R^2 = 2R^2 + 2\vec{A} \cdot \vec{B}$, hence $\vec{A} \cdot \vec{B} = -R^2/2$. By symmetry of the roles (permuting A, B, C in the same identity $\vec{A} + \vec{B} + \vec{C} = \vec{0}$) one obtains in the same way $\vec{B} \cdot \vec{C} = \vec{C} \cdot \vec{A} = -R^2/2$. Hence $\cos(\angle AOB) = \vec{A} \cdot \vec{B}/R^2 = -1/2$, that is, $\angle AOB = 120^\circ$, and likewise $\angle BOC = \angle COA = 120^\circ$: the three central angles are equal, so the three arcs AB, BC, CA are congruent, and the triangle is equilateral (equal chords subtend equal arcs and conversely). $\square$

Second proof, synthetic (via homothety). The homothety of centre G and ratio $-\frac{1}{2}$ sends A to the midpoint M_A of BC (Step 1 of the synthetic proof of Theorem 10.11 below), and sends the line (the altitude from A , perpendicular to BC through A) to the line through M_A perpendicular to the same direction BC (homotheties send lines to parallel lines), that is, to the **perpendicular bisector** of BC . Since the homothety sends A to its image M_A and the line (altitude from A) to its image (bisector of BC), it sends the point H (intersection of the three altitudes) to the intersection point of the three bisectors, that is, O (Theorem 7.2, uniqueness of the circumcentre). Hence O is the image of H under the homothety of centre G and ratio $-\frac{1}{2}$, that is, $\vec{GO} = -\frac{1}{2}\vec{GH}$: G, O, H are collinear with $GO = \frac{1}{2}GH$. $\square$

Theorem 10.11: Nine-point circle. In every triangle ABC , the following nine points are concyclic:

- (i) the three midpoints M_A, M_B, M_C of the sides BC, CA, AB ;
- (ii) the three feet of the altitudes H_A, H_B, H_C ;
- (iii) the three midpoints E_A, E_B, E_C of the segments joining the orthocentre H to each vertex (the **Euler points**).

The radius of this circle is $R/2$ and its centre N is the midpoint of the segment OH between circumcentre and orthocentre.

Proof. Place the circumcentre O at the origin, so that A, B, C are vectors of modulus R , and by Lemma 10.2 $\vec{OH} = \vec{A} + \vec{B} + \vec{C}$. Set $N = \frac{1}{2}(\vec{A} + \vec{B} + \vec{C})$, the midpoint of OH .

Step 1 (the three side midpoints). $M_A = \frac{1}{2}(\vec{B} + \vec{C})$, hence

$$N - M_A = \frac{1}{2}\vec{A}, \quad |N - M_A| = \frac{1}{2}|\vec{A}| = \frac{R}{2}.$$

By symmetry, $|N - M_B| = |N - M_C| = R/2$ as well.

Step 2 (the three Euler points). $E_A = \frac{1}{2}(\vec{A} + \vec{H}) = \vec{A} + \frac{1}{2}(\vec{B} + \vec{C})$, hence

$$N - E_A = -\frac{1}{2}\vec{A}, \quad |N - E_A| = \frac{R}{2}.$$

Six points are now proved concyclic on a circle of centre N and radius $R/2$; note that M_A and E_A are symmetric about N , i.e. $M_A E_A$ is a **diameter** of this circle.

Step 3 (the three feet of the altitudes). The altitude from A passes through the orthocentre H : the points A, H_A and H are collinear on a line perpendicular to BC , and the Euler point E_A (midpoint of AH) lies on that same line. Now $M_A H_A$ lies along the line BC , while $E_A H_A$ lies along the altitude line, **perpendicular** to BC . Hence $\angle M_A H_A E_A = 90^\circ$: by the converse of Thales' theorem (Theorem 4.5), H_A belongs to the circle with diameter $M_A E_A$, that is, to the nine-point circle. The same argument, applied cyclically, holds for H_B and H_C .

$\square$

Remark 10.10: A second look: two homotheties onto the same circle. Steps 1 and 2 can be read as images of triangle ABC under two **homotheties**: the one centred at H with ratio $\frac{1}{2}$ sends A, B, C to the Euler points and the circumcircle to the circle of centre N and radius $R/2$; the one centred at the centroid G with ratio $-\frac{1}{2}$ sends A, B, C to the side midpoints and — using $\vec{OH} = 3\vec{OG}$ — sends the same circumcircle to the same image circle. The two families of three points therefore fall, by construction, on the same circle.

Synthetic proof, by homothety. Step 1 (side midpoints). The centroid G divides every median in the ratio $2 : 1$: $G\vec{M}_A = -\frac{1}{2}\vec{GA}$ (check with $M_A = \frac{1}{2}(B + C)$, $G = \frac{1}{3}(A + B + C)$: $M_A - G = \frac{1}{6}(B + C - 2A) = -\frac{1}{2}(A - G)$), and likewise for B, C . The homothety h_1 of centre G and ratio $-\frac{1}{2}$ therefore sends A, B, C to M_A, M_B, M_C , and the circumcircle (centre O , radius R) to the circle of radius $R/2$ and centre $h_1(O) = G - \frac{1}{2}(O - G) = \frac{3}{2}G - \frac{1}{2}O$. With the origin at O (as in the main proof), the centroid is $G = \frac{1}{3}(A + B + C)$ and, by Lemma 10.2, $H = A + B + C = 3G$: hence $\vec{OG} = G$ and $\vec{OH} = H = 3G = 3\vec{OG}$, that is, O, G, H are collinear with $\vec{OG} = \frac{1}{3}\vec{OH}$ (this is the **Euler line**). Substituting $G = \frac{2}{3}O + \frac{1}{3}H$ (equivalent to $O + \frac{1}{3}(H - O)$, that is, $\vec{OG} = \frac{1}{3}\vec{OH}$, but written relative to an arbitrary origin) gives $h_1(O) = \frac{3}{2}(\frac{2}{3}O + \frac{1}{3}H) - \frac{1}{2}O = \frac{1}{2}O + \frac{1}{2}H = N$, the midpoint of OH : M_A, M_B, M_C lie on the circle of centre N and radius $R/2$.

Step 2 (Euler points). The homothety h_2 of centre H and ratio $\frac{1}{2}$ sends A, B, C to the midpoints E_A, E_B, E_C of AH, BH, CH (by definition of midpoint), and the circumcircle to the circle of radius $R/2$ and centre $h_2(O) = H + \frac{1}{2}(O - H) = \frac{1}{2}O + \frac{1}{2}H = N$: **the same** centre N as in Step 1. Hence E_A, E_B, E_C also lie on the circle of centre N and radius $R/2$: six of the nine points are already accounted for, on a circle uniquely determined by h_1 and h_2 .

Step 3 (feet of the altitudes). To place H_A, H_B, H_C on the same circle, use the converse-of-Thales argument already given in the main proof (the quadrilateral $M_A H_A E_A \dots$ has a right angle at H_A , since $M_A H_A \perp E_A H_A$): that argument is purely synthetic and requires no further homotheties, and applies here unchanged. $\square$

Example 10.5: The nine points, all nine computed. Take the triangle $A = (0; 0)$, $B = (8; 0)$, $C = (2; 6)$ and verify Theorem 10.11 point by point.

Step 1 — the circumcentre O . It lies on the perpendicular bisector of AB , i.e. on $x = 4$. The perpendicular bisector of AC passes through the midpoint $(1; 3)$ and is perpendicular to $\vec{AC} = (2; 6)$:

$$2(x - 1) + 6(y - 3) = 0 \iff x + 3y = 10.$$

With $x = 4$: $y = 2$. Hence

$$O = (4; 2), \quad R = |OA| = \sqrt{20} = 4.47214.$$

Step 2 — the orthocentre H . From the identity $\vec{OH} = \vec{OA} + \vec{OB} + \vec{OC}$ used in the proof it follows that $H = A + B + C - 2O$:

$$H = (10; 6) - (8; 4) = (2; 2).$$

Check: the altitude from C is the vertical $x = 2$ (since AB lies on the x -axis), and H belongs to it; the altitude from A is perpendicular to $\vec{BC} = (-6; 6)$, hence has direction $(1; 1)$, i.e. is the line $y = x$, to which H belongs. $\checkmark$

Step 3 — the nine-point centre N and the radius.

$$N = \frac{O + H}{2} = (3; 2), \quad \frac{R}{2} = \sqrt{5} = 2.23607.$$

Step 4 — the three side midpoints.

$$M_{AB} = (4; 0), \quad M_{BC} = (5; 3), \quad M_{CA} = (1; 3).$$

Distances from $N = (3; 2)$:

$$|(-1; 2)| = \sqrt{5}, \quad |(-2; -1)| = \sqrt{5}, \quad |(2; -1)| = \sqrt{5}. \quad \checkmark$$

Step 5 — the three feet of the altitudes.

- from C : the vertical $x = 2$ meets AB ($y = 0$) at $H_C = (2; 0)$;
- from A : the line $y = x$ meets BC — parametrised as $(8 - t; t)$ — when $8 - t = t$, i.e. $t = 4$: $H_A = (4; 4)$;
- from B : the line through B perpendicular to $\vec{AC} = (2; 6)$ has direction $(3; -1)$, i.e. $(8 + 3s; -s)$; it meets AC (the line $y = 3x$) when $-s = 3(8 + 3s)$, i.e. $s = -2.4$: $H_B = (0.8; 2.4)$.

Distances from N :

$$|(1; 2)| = \sqrt{5}, \quad |(-1; -2)| = \sqrt{5}, \quad |(2.2; -0.4)| = \sqrt{4.84 + 0.16} = \sqrt{5}. \quad \checkmark$$

Step 6 — the three Euler points (midpoints between H and the vertices):

$$E_A = (1; 1), \quad E_B = (5; 1), \quad E_C = (2; 4).$$

Distances from N :

$$|(2; 1)| = \sqrt{5}, \quad |(-2; 1)| = \sqrt{5}, \quad |(1; -2)| = \sqrt{5}. \quad \checkmark$$

Conclusion. All nine points are at distance exactly $\sqrt{5}$ from $N = (3; 2)$: they lie on the circle $(x - 3)^2 + (y - 2)^2 = 5$, of radius $R/2$. $\checkmark$

Proposition 10.4: Barycentric coordinates of the incentre. With an arbitrary origin and A, B, C the position vectors of the vertices,

$$\vec{I} = \frac{a\vec{A} + b\vec{B} + c\vec{C}}{a + b + c}.$$

Proof. **Step 1 — the foot of the bisector from A .** Let D be the point where the bisector of $\angle A$ meets $\overline{BC}$. By the angle bisector theorem (proved with the law of sines in triangles ABD , ACD , as in the synthetic proof of the circle of Apollonius, Chapter 2), $BD/DC = c/b$ (the sides adjacent to the equal angles $\angle BAD = \angle DAC$), hence

$$\vec{D} = \frac{b\vec{B} + c\vec{C}}{b + c}$$

(the point dividing $\overline{BC}$ in the ratio $BD : DC = c : b$).

Step 2 — the incentre on the bisector AD . In triangle ABD , the bisector of the angle at B (which is also the bisector of $\angle B$ in the original triangle) meets AD at the incentre I ; by the angle bisector theorem applied to triangle ABD with the bisector from B ,

$$\frac{AI}{ID} = \frac{AB}{BD} = \frac{c}{\frac{ac}{b+c}} = \frac{b+c}{a}$$

(having used $BD = \frac{c}{b+c} \cdot a$ from Step 1, since $BD/DC = c/b$ and $BD + DC = a$). Hence I

divides $\overline{AD}$ in the ratio $AI : ID = (b + c) : a$, that is,

$$\vec{I} = \frac{a\vec{A} + (b + c)\vec{D}}{a + (b + c)} = \frac{a\vec{A} + (b + c) \cdot \frac{b\vec{B} + c\vec{C}}{b + c}}{a + b + c} = \frac{a\vec{A} + b\vec{B} + c\vec{C}}{a + b + c},$$

the claim. $\square$

Remark 10.11: The same coordinates for the excentres. The same computation, with the **external** bisector at B in place of the internal one (which meets the line AD at the excentre I_a , opposite A , with ratio $AI_a : I_aD = (b + c) : (-a)$ instead of $(b + c) : a$, by the external bisector theorem), gives

$$\vec{I}_a = \frac{-a\vec{A} + b\vec{B} + c\vec{C}}{-a + b + c}.$$

The other two excentres are obtained by symmetry of the letters: $\vec{I}_b = \frac{a\vec{A} - b\vec{B} + c\vec{C}}{a - b + c}$,
 $\vec{I}_c = \frac{a\vec{A} + b\vec{B} - c\vec{C}}{a + b - c}.$

Example 10.6: Numerical check of the Euler–Chapple formula. On the same triangle $A = (0; 0)$, $B = (8; 0)$, $C = (2; 6)$ of the previous example, verify Theorem 10.9.

Step 1 — the sides.

$$a = |BC| = \sqrt{36 + 36} = 6\sqrt{2} = 8.48528, \quad b = |CA| = \sqrt{4 + 36} = \sqrt{40} = 6.32456, \quad c = |AB| = 8.$$

Semiperimeter: $s = (8.48528 + 6.32456 + 8)/2 = 11.40492$.

Step 2 — the area. With A at the origin, $K = \frac{1}{2}|x_B y_C - x_C y_B| = \frac{1}{2}|8 \cdot 6 - 2 \cdot 0| = 24$.

Step 3 — the two radii.

$$r = \frac{K}{s} = \frac{24}{11.40492} = 2.10437, \quad R = \frac{abc}{4K} = \frac{8.48528 \cdot 6.32456 \cdot 8}{96} = \frac{429.325}{96} = 4.47214.$$

The value of R agrees with the one found geometrically in Step 1 of the previous example. $\checkmark$

Step 4 — the incentre. The barycentric formula $I = (aA + bB + cC)/(a + b + c)$ gives

$$I = \frac{8.48528(0; 0) + 6.32456(8; 0) + 8(2; 6)}{22.80984} = \frac{(50.5965 + 16; 48)}{22.80984} = (2.91964; 2.10437).$$

Check: since AB lies on the x -axis, the ordinate of the incentre must be exactly r . And indeed $y_I = 2.10437 = r$. $\checkmark$

Step 5 — the distance OI . With $O = (4; 2)$:

$$OI^2 = (4 - 2.91964)^2 + (2 - 2.10437)^2 = 1.08036^2 + (-0.10437)^2 = 1.16718 + 0.01089 = 1.17807.$$

Step 6 — comparison with the formula.

$$R^2 - 2Rr = 20 - 2 \cdot 4.47214 \cdot 2.10437 = 20 - 18.82207 = 1.17793. \quad \checkmark$$

The two values (1.17807 in Step 5, 1.17793 here) agree to three decimal places; the small difference in the fourth is the effect of rounding R and r to five significant figures, not a real disagreement: the Euler–Chapple formula is verified.

Step 7 — Euler’s inequality. Finally, $R = 4.47214 \geq 2r = 4.20874$ (Corollary 10.6): the margin is small because this triangle is fairly close to equilateral, the case of equality.

Remark 10.12: Historical note and Feuerbach's theorem. That nine points so defined are concyclic was already known to Euler (1765) for six of them; the complete proof was published independently by Poncelet and Brianchon (1820) and, the following year, by the German geometer Karl Wilhelm Feuerbach (1822). Feuerbach also proved a further fact — known as **Feuerbach's theorem** — namely that the nine-point circle is **internally tangent** to the inscribed circle and **externally tangent** to each of the three excircles. Feuerbach proved it by analytic-trigonometric means, computing the distance between the centres and checking that it equals the sum or the difference of the radii; the proof we give in Section 10.12.1 is instead synthetic and uses the circle inversion of Chapter 5.

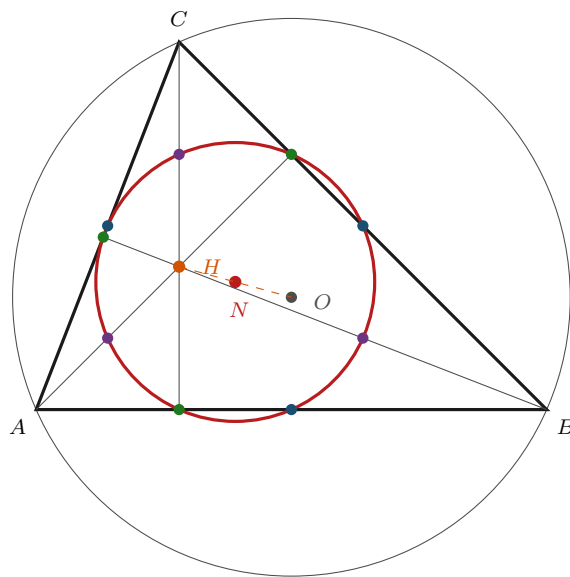


Figure 10.7: The nine-point circle, or Feuerbach circle. A single circle passes through the three midpoints of the sides (blue), the three feet of the altitudes (green) and the three midpoints of the segments joining the vertices to the orthocentre H (violet). Its centre N is the midpoint of OH , and its radius is exactly half the circumradius: here $R/2 = 1.3657$. The deep reason is that the homothety with centre H and ratio $1/2$ maps the circumcircle to the nine-point circle, carrying the vertices to the Euler points; the same homothety explains why O , N , G and H are collinear on the Euler line. Feuerbach's theorem adds that this circle is tangent to the incircle and to the three excircles.

10.12.1 Feuerbach's theorem, proved

INFO

Prerequisites of this subsection. From Chapter 5: Definition 5.1 of circle inversion, Observation 5.2 (inversion is an involution), Theorem 5.3 (images of lines and circles), Theorem 5.4 (a circle is invariant if and only if the power of the pole with respect to it equals k^2) and Corollary 5.1 (tangency is preserved). From Chapter 3: Theorem 3.11 (congruent tangent segments) and Theorem 3.19 (power of a point). From Chapter 9: Theorem 9.16 (mutual positions of two circles). From this chapter: Theorem 10.11 (nine-point circle), Definition 10.2 (excircles) and Corollary 10.6 ($R \geq 2r$).

It may be skipped on a first reading, accepting Theorem 10.12 as stated.

Let us fix the notation for the whole subsection. The triangle is ABC , with $a = BC$, $b = CA$, $c = AB$, semiperimeter s , area S ; γ is the inscribed circle, of centre I and radius r ; γ_a is the excircle opposite A , of centre I_a and radius r_a (Definition 10.2); $\mathcal{N}$ is the nine-point circle, of

centre N and radius $R/2$ (Theorem 10.11). We denote by M the midpoint of BC , by M_B and M_C the midpoints of CA and AB , and by H_A the foot of the altitude from A : all four lie on $\mathcal{N}$. Finally X and X_a are the points where γ and γ_a touch the line BC , and we set

$$\rho := \frac{|b - c|}{2}.$$

Lemma 10.3: *The points of tangency on BC . One has*

$$BX = s - b, \quad CX = s - c, \quad BX_a = s - c, \quad CX_a = s - b,$$

and consequently X and X_a are symmetric with respect to the midpoint M of BC , with

$$MX = MX_a = \rho = \frac{|b - c|}{2}.$$

In particular $X = X_a = M$ if and only if $b = c$.

Proof. Inscribed circle. This is Step 1 of the proof of Theorem 10.4: writing x, y, z for the tangent segments from A, B, C (congruent in pairs by Theorem 3.11), one has $x + y = c$, $y + z = a$, $z + x = b$; adding, $x + y + z = s$, whence $x = s - a$, $y = s - b$, $z = s - c$. Hence $BX = y = s - b$ and $CX = z = s - c$.

Excircle. γ_a touches the line BC at X_a , the ray AB beyond B at Z_a and the ray AC beyond C at Y_a . By Theorem 3.11, $AZ_a = AY_a$, $BZ_a = BX_a$ and $CY_a = CX_a$. Then

$$AZ_a + AY_a = (AB + BZ_a) + (AC + CY_a) = c + BX_a + b + CX_a = c + b + a = 2s,$$

because $BX_a + CX_a = BC = a$ (X_a lies between B and C , since γ_a is tangent to the *side* BC). From $AZ_a = AY_a$ it follows that $AZ_a = AY_a = s$, and therefore

$$BX_a = BZ_a = AZ_a - AB = s - c, \quad CX_a = CY_a = AY_a - AC = s - b.$$

Position with respect to M . On BC introduce the abscissa with origin M and direction from B towards C , so that B has abscissa $-a/2$ and C has abscissa $+a/2$. Then

$$x_X = -\frac{a}{2} + BX = -\frac{a}{2} + \frac{a + c - b}{2} = \frac{c - b}{2}, \quad x_{X_a} = -\frac{a}{2} + BX_a = -\frac{a}{2} + \frac{a + b - c}{2} = \frac{b - c}{2},$$

that is, $x_{X_a} = -x_X$ and $|x_X| = |x_{X_a}| = \rho$. $\square$

Lemma 10.4: *The inversion fixing incircle and excircle. Let $b \neq c$ and let ι be the inversion with pole M and radius ρ . Then $\iota(\gamma) = \gamma$ and $\iota(\gamma_a) = \gamma_a$.*

Proof. The line BC is tangent to γ at X , hence the power of M with respect to γ equals the square of the tangent segment (Theorem 3.19, tangent case):

$$\text{pow}_\gamma(M) = MX^2 = \rho^2.$$

By Theorem 5.4, a circle is mapped onto itself by the inversion with pole M and radius ρ if and only if the power of the pole with respect to it equals ρ^2 : hence $\iota(\gamma) = \gamma$. The same computation with X_a in place of X gives $\text{pow}_{\gamma_a}(M) = MX_a^2 = \rho^2$, and therefore $\iota(\gamma_a) = \gamma_a$. (The hypothesis $b \neq c$ serves only to guarantee $\rho > 0$, that is, that the inversion be defined.) $\square$

Lemma 10.5: The second common tangent. Let σ be the reflection in the line AI (internal bisector of the angle at A) and let $t = \sigma(BC)$. Then:

- (a) t is tangent both to γ and to γ_a ;
- (b) t is the line through the points $C' = \sigma(C)$ and $B' = \sigma(B)$, where C' lies on the ray AB with $AC' = b$ and B' lies on the ray AC with $AB' = c$;
- (c) if $b \neq c$ then $t \neq BC$ and $M \notin t$.

Proof. (a). The line AI is the internal bisector of the angle at A , hence σ interchanges the rays AB and AC . The centres I and I_a both lie on AI (Definition 10.2: I_a is the intersection of the internal bisector at A with the external bisectors at B and C), hence they are fixed by σ ; since a reflection preserves distances, $\sigma(\gamma)$ is the circle with centre I and radius r , that is γ itself, and likewise $\sigma(\gamma_a) = \gamma_a$. The line BC is tangent to γ and to γ_a ; isometries preserve tangency (a tangent line has exactly one point in common with the circle, and this property is invariant under bijections of the plane carrying lines to lines and circles to circles): hence $t = \sigma(BC)$ is tangent to $\sigma(\gamma) = \gamma$ and to $\sigma(\gamma_a) = \gamma_a$.

(b). σ maps the ray AB onto the ray AC preserving distances from A : hence $\sigma(B) = B'$ is the point of the ray AC with $AB' = AB = c$, and $\sigma(C) = C'$ is the point of the ray AB with $AC' = AC = b$. Since σ carries lines to lines, $t = \sigma(BC)$ is the line $B'C'$.

(c). Let D be the intersection of the bisector AI with BC : D is fixed by σ and lies on BC , hence $D \in \sigma(BC) = t$. Suppose $t = BC$: then σ would map BC onto itself, and since σ fixes exactly the points of AI , the line BC would be either equal to AI (impossible, $A \notin BC$) or perpendicular to AI ; in the latter case AD would be both bisector and altitude, and the triangles ABD and ACD would be congruent (side AD common, adjacent angles congruent), whence $b = c$. Hence $b \neq c$ implies $t \neq BC$, and then $t \cap BC = \{D\}$.

It remains to see that $M \neq D$. If $M = D$, the line AM would be both median and bisector. Extend then AM beyond M to the point A' with $MA' = MA$. The quadrilateral $ABA'C$ has diagonals $\overline{AA'}$ and $\overline{BC}$ bisecting each other (at M), hence it is a parallelogram; in particular $CA' = AB = c$ and $AB \parallel CA'$. From the parallelism, $\angle AA'C = \angle BAA'$ (alternate interior angles with respect to the transversal AA'); but $\angle BAA' = \angle CAA'$ because AM is a bisector, so $\angle AA'C = \angle CAA'$ and the triangle ACA' is isosceles with $CA = CA'$, that is $b = c$: excluded. Hence $M \neq D$, and since $M \in BC$ and $t \cap BC = \{D\}$, we conclude $M \notin t$. $\square$

Lemma 10.6: The images of the two midpoints. Let $b \neq c$ and let ι be the inversion with pole M and radius ρ . Then $\iota(M_B)$ and $\iota(M_C)$ lie on the line t of Lemma 10.5.

Proof. Take A as the origin of vectors and set

$$\mathbf{u} = \frac{B - A}{c}, \quad \mathbf{v} = \frac{C - A}{b},$$

the unit vectors of the rays AB and AC , linearly independent because A, B, C are not collinear. Then $B = c\mathbf{u}$, $C = b\mathbf{v}$ and

$$M = \frac{1}{2}(c\mathbf{u} + b\mathbf{v}), \quad M_C = \frac{c}{2}\mathbf{u}, \quad M_B = \frac{b}{2}\mathbf{v}.$$

Image of M_C . One has $M_C - M = -\frac{b}{2}\mathbf{v}$, hence $|MM_C| = b/2$ (it is the midline parallel to AC). By Definition 5.1,

$$\iota(M_C) = M + \frac{\rho^2}{|MM_C|^2} (M_C - M) = M - \frac{(b-c)^2/4}{b^2/4} \cdot \frac{b}{2} \mathbf{v} = \frac{c}{2} \mathbf{u} + \left(\frac{b}{2} - \frac{(b-c)^2}{2b} \right) \mathbf{v}.$$

Since $b^2 - (b-c)^2 = c(2b-c)$, one obtains

$$\iota(M_C) = \frac{c}{2} \mathbf{u} + \frac{c(2b-c)}{2b} \mathbf{v}.$$

Image of M_B . Likewise $M_B - M = -\frac{c}{2}\mathbf{u}$, $|MM_B| = c/2$ and, using $c^2 - (b - c)^2 = b(2c - b)$,

$$\iota(M_B) = \frac{b(2c - b)}{2c}\mathbf{u} + \frac{b}{2}\mathbf{v}.$$

Membership of t . By Lemma 10.5(b), t is the line through $C' = b\mathbf{u}$ and $B' = c\mathbf{v}$, that is, the set of the points

$$P(\lambda) = (1 - \lambda)b\mathbf{u} + \lambda c\mathbf{v}, \quad \lambda \in \mathbb{R}.$$

With $\lambda = \frac{2b - c}{2b}$ one has $(1 - \lambda)b = \frac{c}{2}$ and $\lambda c = \frac{c(2b - c)}{2b}$: hence $P(\lambda) = \iota(M_C)$. With $\lambda = \frac{b}{2c}$ one has $(1 - \lambda)b = \frac{b(2c - b)}{2c}$ and $\lambda c = \frac{b}{2}$: hence $P(\lambda) = \iota(M_B)$. (Comparing coefficients is legitimate because $\mathbf{u}$ and $\mathbf{v}$ are independent.) $\square$

Lemma 10.7: *The chord of $\mathcal{N}$ on the line BC . Use on BC the abscissa with origin M introduced in Lemma 10.3. Then the foot of the altitude from A has abscissa*

$$x_{H_A} = \frac{c^2 - b^2}{2a},$$

and for every point P of abscissa x on the line BC one has

$$|NP|^2 - \left(\frac{R}{2}\right)^2 = x(x - x_{H_A}).$$

Consequently P is inside $\mathcal{N}$ if and only if x lies strictly between 0 and x_{H_A} , and outside if and only if x lies outside the closed interval with endpoints 0 and x_{H_A} .

Proof. Step 1 — the abscissa of H_A . Take the Cartesian frame with origin M , x -axis the line BC oriented from B towards C , y -axis the perpendicular, and let $A = (x_A, y_A)$ with $y_A \neq 0$. From $B = (-a/2, 0)$ and $C = (a/2, 0)$:

$$c^2 = |AB|^2 = (x_A + \frac{a}{2})^2 + y_A^2, \quad b^2 = |AC|^2 = (x_A - \frac{a}{2})^2 + y_A^2.$$

Subtracting, $c^2 - b^2 = 2ax_A$, that is $x_A = (c^2 - b^2)/(2a)$. The foot of the altitude from A is the orthogonal projection of A onto BC , that is $H_A = (x_A, 0)$: hence $x_{H_A} = x_A$.

Step 2 — the abscissa of the centre N . If $b \neq c$ then $x_{H_A} \neq 0$, that is $H_A \neq M$: the two points M and H_A are two distinct points of $\mathcal{N}$ on the line BC , and the centre N lies on the perpendicular bisector of the chord $\overline{MH_A}$, hence $x_N = x_{H_A}/2$. If instead $b = c$, then $x_{H_A} = 0$ and the triangle is symmetric with respect to the y -axis: that symmetry interchanges B with C and M_B with M_C , hence maps the set of the nine points onto itself and therefore the circle $\mathcal{N}$ onto itself; so $x_N = 0 = x_{H_A}/2$. In either case $x_N = x_{H_A}/2$.

Step 3 — the identity. For $P = (x, 0)$,

$$|NP|^2 - \left(\frac{R}{2}\right)^2 = (x - x_N)^2 + y_N^2 - \left(\frac{R}{2}\right)^2 = x^2 - 2x_Nx + \underbrace{(x_N^2 + y_N^2 - (\frac{R}{2})^2)}_{=0},$$

where the bracket vanishes because $M = (0, 0)$ belongs to $\mathcal{N}$, that is $|NM|^2 = x_N^2 + y_N^2 = (R/2)^2$. Substituting $2x_N = x_{H_A}$ one gets $|NP|^2 - (R/2)^2 = x(x - x_{H_A})$. The sign of the product gives the last assertion. $\square$

Theorem 10.12: Feuerbach's theorem (1822). *In every triangle the nine-point circle is internally tangent to the inscribed circle and externally tangent to each of the three ex-*

circles. In formulas, denoting by N the nine-point centre, by I, I_a, I_b, I_c the centres of the four circles and by r, r_a, r_b, r_c their radii,

$$NI = \frac{R}{2} - r, \quad NI_a = \frac{R}{2} + r_a, \quad NI_b = \frac{R}{2} + r_b, \quad NI_c = \frac{R}{2} + r_c.$$

In the equilateral case alone one has $NI = 0$: the nine-point circle coincides with the inscribed circle (the limiting case of internal tangency); the three external tangencies remain proper.

Proof. It suffices to prove, for the vertex A , that $\mathcal{N}$ is internally tangent to γ and externally tangent to γ_a : repeating the argument with the vertices B and C in place of A gives the external tangencies with γ_b and γ_c (and again, three times, the internal one with γ). We distinguish two cases.

Case 1: $b \neq c$.

Step 1 — $\mathcal{N}$ becomes a line. The point M belongs to $\mathcal{N}$ (Theorem 10.11); by Theorem 5.3(b), the inversion ι with pole M and radius ρ maps $\mathcal{N} \setminus \{M\}$ onto a line n^* not through M .

Step 2 — that line is t . The points M_B and M_C belong to $\mathcal{N}$ and differ from M (their distances from M are $c/2$ and $b/2$, both positive); hence $\iota(M_B)$ and $\iota(M_C)$ belong to n^* , and are distinct because ι is injective. By Lemma 10.6 both lie on t . Two distinct points determine a single line, hence $n^* = t$.

Step 3 — back again. Since ι is an involution (Observation 5.2) and $M \notin t$ (Lemma 10.5(c)), the image of t is the circle $\mathcal{N}$. By Lemma 10.5(a), t is tangent to γ at a point F and to γ_a at a point F_a , both on t and hence different from M . By Lemma 10.4, $\iota(\gamma) = \gamma$ and $\iota(\gamma_a) = \gamma_a$. Corollary 5.1 then gives: $\mathcal{N} = \iota(t)$ is tangent to γ at the point $\iota(F)$ and tangent to γ_a at the point $\iota(F_a)$.

Step 4 — the tangency with γ is internal. $\mathcal{N}$ and γ have exactly one common point: by Theorem 9.16 we are in case (b), $NI = R/2 + r$, or in case (d), $NI = |R/2 - r|$. We rule out case (b). If $NI = R/2 + r$, then for every $P \in \gamma$ one would have

$$|NP| \geq NI - |IP| = \left(\frac{R}{2} + r\right) - r = \frac{R}{2},$$

that is, no point of γ would be inside $\mathcal{N}$. But the point of tangency X is: by Lemmas 10.3 and 10.7, with $x_X = (c - b)/2$,

$$x_X(x_X - x_{H_A}) = \frac{c - b}{2} \cdot \left(\frac{c - b}{2} - \frac{c^2 - b^2}{2a}\right) = \frac{(c - b)^2}{4} \cdot \frac{a - (b + c)}{a} = -\frac{(b - c)^2(b + c - a)}{4a} < 0,$$

because $b \neq c$ and $b + c > a$ by the triangle inequality. Hence $|NX| < R/2$: a contradiction. Case (d) remains, that is $NI = |R/2 - r|$; and since $R \geq 2r$ (Corollary 10.6), $NI = R/2 - r$.

Step 5 — the tangency with γ_a is external. Again by Theorem 9.16 we are in case (b) or in case (d). We rule out case (d): if $NI_a = |R/2 - r_a|$, the circle of smaller radius would lie in the closed disc bounded by the other (by the triangle inequality: if $R_1 \geq R_2$ and $d = R_1 - R_2$, every point P of the small circle satisfies $|O_1P| \leq d + R_2 = R_1$). But neither inclusion is possible:

- $\mathcal{N}$ does not lie in the disc of γ_a , because $M \in \mathcal{N}$ is outside γ_a : the line BC is tangent to γ_a at X_a , and $I_aX_a \perp BC$, hence $|I_aM|^2 = r_a^2 + |X_aM|^2 = r_a^2 + \rho^2 > r_a^2$ (here $\rho > 0$);
- γ_a does not lie in the disc of $\mathcal{N}$, because $X_a \in \gamma_a$ is outside $\mathcal{N}$: with $x_{X_a} = (b - c)/2$, Lemmas 10.3 and 10.7 give

$$x_{X_a}(x_{X_a} - x_{H_A}) = \frac{b - c}{2} \cdot \left(\frac{b - c}{2} + \frac{b^2 - c^2}{2a}\right) = \frac{(b - c)^2}{4} \cdot \frac{a + b + c}{a} > 0,$$

that is $|NX_a| > R/2$.

Case (b) remains: $NI_a = R/2 + r_a$, external tangency.

Case 2: $b = c$ (isosceles triangle on the base BC).

Here $\rho = 0$ and the inversion degenerates; we conclude directly. With the frame of Lemma 10.7 one has $x_{H_A} = 0$ and hence, by the same lemma,

$$|NP|^2 - \left(\frac{R}{2}\right)^2 = x^2 \quad (P = (x, 0) \in BC),$$

which vanishes only for $x = 0$: the line BC has the single point M in common with $\mathcal{N}$, that is, **it is tangent to $\mathcal{N}$ at M** . By Lemma 10.3, γ and γ_a too are tangent to BC at M (since $X = X_a = M$).

From Step 3 of Lemma 10.7 it follows that $x_N = 0$, hence $|y_N| = R/2$; the circle $\mathcal{N}$, tangent to the x -axis at the origin, lies entirely in one half-plane, and since it contains M_B , of ordinate $y_A/2 \neq 0$ with the same sign as y_A , one has $y_N = \frac{R}{2} \operatorname{sgn}(y_A)$: N lies on the same side as A . The incentre I too lies on the same side as A (it is interior to the triangle) and, by symmetry, on the y -axis: hence $I = (0, r \operatorname{sgn}(y_A))$ and

$$NI = \left| \frac{R}{2} - r \right| = \frac{R}{2} - r,$$

by Corollary 10.6. The centre I_a instead lies on the opposite side of BC (the excircle is outside the triangle across the side BC) and on the y -axis by symmetry, hence $I_a = (0, -r_a \operatorname{sgn}(y_A))$ and

$$NI_a = \frac{R}{2} + r_a.$$

By Theorem 9.16 the first relation is internal tangency (or coincidence, if $R = 2r$, that is in the equilateral case) and the second is external tangency.

In both cases the assertion is obtained for the pair (γ, γ_a) ; repeating with B and with C in place of A gives the other two external tangencies. $\square$

Remark 10.13: The Feuerbach point. The point where $\mathcal{N}$ touches γ is called the **Feuerbach point** of the triangle. The proof locates it by construction: it is the image, under the inversion with pole M and radius $|b - c|/2$, of the point where the second common tangent t touches the incircle. The three contact points with the excircles are obtained in the same way starting from the three vertices. Note that the Feuerbach point does not depend on the vertex chosen to build the inversion: the tangency between $\mathcal{N}$ and γ occurs at a single point, and the three constructions (from A , from B , from C) must therefore give the same result.

Example 10.7: Numerical check on the 13–14–15 triangle. Take $a = BC = 14$, $b = CA = 15$, $c = AB = 13$: all the quantities involved turn out to be rational, so the check is *exact*, not approximate.

Step 1 — the elements of the triangle. $s = (14 + 15 + 13)/2 = 21$ and, by Heron's formula, $S = \sqrt{21 \cdot 7 \cdot 6 \cdot 8} = \sqrt{7056} = 84$. Hence

$$r = \frac{S}{s} = \frac{84}{21} = 4, \quad R = \frac{abc}{4S} = \frac{14 \cdot 15 \cdot 13}{336} = \frac{2730}{336} = \frac{65}{8} = 8.125, \quad \frac{R}{2} = \frac{65}{16} = 4.0625,$$

and, by Theorem 10.8,

$$r_a = \frac{84}{21 - 14} = 12, \quad r_b = \frac{84}{21 - 15} = 14, \quad r_c = \frac{84}{21 - 13} = \frac{21}{2} = 10.5.$$

Step 2 — coordinates. Set $B = (0; 0)$, $C = (14; 0)$. From $|AB| = 13$ and $|AC| = 15$

one gets $x_A^2 - (x_A - 14)^2 = 169 - 225 = -56$, that is $28x_A = 140$ and $x_A = 5$; then $y_A^2 = 169 - 25 = 144$, $y_A = 12$. Hence $A = (5; 12)$.

The circumcentre lies on $x = 7$ and satisfies $49 + y^2 = R^2 = (65/8)^2$, whence $y = 33/8$: $O = (7; 33/8)$. The orthocentre is $H = A + B + C - 2O = (5; 15/4)$ (*check*: the altitude from A is the vertical $x = 5$, and the altitude from B is the line $y = 3x/4$, perpendicular to $\vec{AC} = (9; -12)$; their common point is $(5; 15/4)$ ✓). Hence

$$N = \frac{O + H}{2} = \left(6; \frac{63}{16}\right) = (6; 3.9375).$$

Step 3 — the four centres. With the barycentric coordinates $I = \frac{aA + bB + cC}{a + b + c}$ and $I_a = \frac{-aA + bB + cC}{-a + b + c}$ (and similarly for the others):

$$I = (6; 4), \quad I_a = (8; -12), \quad I_b = (21; 14), \quad I_c = (-7; 10.5).$$

Check: the ordinate of I must equal $r = 4$ because BC lies on the x -axis ✓; the ordinate of I_a is -12 , of modulus $r_a = 12$ ✓; that of I_b is $14 = r_b$ ✓; that of I_c is $10.5 = r_c$ ✓.

Step 4 — the four distances.

$$\begin{aligned} NI^2 &= (6 - 6)^2 + \left(\frac{63}{16} - 4\right)^2 = \frac{1}{256}, & \left(\frac{R}{2} - r\right)^2 &= \left(\frac{65}{16} - 4\right)^2 = \frac{1}{256} \quad \checkmark \\ NI_a^2 &= 2^2 + \left(\frac{63}{16} + 12\right)^2 = \frac{66\,049}{256}, & \left(\frac{R}{2} + r_a\right)^2 &= \left(\frac{257}{16}\right)^2 = \frac{66\,049}{256} \quad \checkmark \\ NI_b^2 &= 15^2 + \left(14 - \frac{63}{16}\right)^2 = \frac{83\,521}{256}, & \left(\frac{R}{2} + r_b\right)^2 &= \left(\frac{289}{16}\right)^2 = \frac{83\,521}{256} \quad \checkmark \\ NI_c^2 &= 13^2 + \left(\frac{21}{2} - \frac{63}{16}\right)^2 = \frac{54\,289}{256}, & \left(\frac{R}{2} + r_c\right)^2 &= \left(\frac{233}{16}\right)^2 = \frac{54\,289}{256} \quad \checkmark \end{aligned}$$

The four equalities hold *exactly* between rational numbers: $NI = 1/16$ and $R/2 - r = 65/16 - 64/16 = 1/16$; $NI_a = 257/16$ and $R/2 + r_a = 65/16 + 192/16 = 257/16$; and likewise for the other two.

Step 5 — check of the inversion construction. With the vertex A : $\rho = |b - c|/2 = 1$, $M = (7; 0)$. The points of tangency on BC are $X = (6; 0)$ (because $BX = s - b = 6$) and $X_a = (8; 0)$ (because $BX_a = s - c = 8$): both at distance $1 = \rho$ from M ✓ (Lemma 10.3). The foot of the altitude is $H_A = (5; 0)$, of abscissa -2 with respect to M , in agreement with $x_{H_A} = (c^2 - b^2)/(2a) = (169 - 225)/28 = -2$ ✓ (Lemma 10.7). The identity of the same lemma gives

$$|NX|^2 - \left(\frac{R}{2}\right)^2 = (-1)(-1 + 2) = -1, \quad |NX_a|^2 - \left(\frac{R}{2}\right)^2 = (1)(1 + 2) = 3,$$

that is, X inside and X_a outside $\mathcal{N}$: *direct check*, $|NX|^2 = (6 - 6)^2 + (63/16)^2 = 3969/256$ and $(R/2)^2 = 4225/256$, whose difference is $-256/256 = -1$ ✓; $|NX_a|^2 = 2^2 + (63/16)^2 = 1024/256 + 3969/256 = 4993/256$, and $4993/256 - 4225/256 = 768/256 = 3$ ✓.

Finally the second common tangent: $\mathbf{u} = (B - A)/13 = (-5/13; -12/13)$ and $\mathbf{v} = (C - A)/15 = (3/5; -4/5)$, hence $C' = A + 15\mathbf{u} = (-10/13; -24/13)$ and $B' = A + 13\mathbf{v} = (64/5; 8/5)$. The line $t = B'C'$ has direction vector $B' - C' = (\frac{882}{65}; \frac{224}{65})$, of modulus 14, and unit normal $\mathbf{n} = (\frac{16}{65}; -\frac{63}{65})$. The distances of the two centres from t are

$$|(I - B') \cdot \mathbf{n}| = \left| \left(-\frac{34}{5}; \frac{12}{5}\right) \cdot \left(\frac{16}{65}; -\frac{63}{65}\right) \right| = 4 = r \quad \checkmark, \quad |(I_a - B') \cdot \mathbf{n}| = 12 = r_a \quad \checkmark,$$

confirming Lemma 10.5: t is tangent both to the incircle and to the excircle.

Remark 10.14: Why the original proof is different. Feuerbach did not use inversion, which in 1822 was not yet a standard tool: his work is, as the subtitle says, eine analytisch-trigonometrische Abhandlung, and the tangency is proved by explicitly computing the distance between the two centres and recognizing that it equals the difference (or the sum) of the radii — exactly what Step 4 of Example 10.7 verifies in a concrete case. The inversive route followed here is later and has the advantage of explaining why the tangency occurs: the nine-point circle and the second common tangent to incircle and excircle are the same object, seen from the two sides of an inversion that fixes both circles.

10.13. Napoleon's theorem

Remark 10.15: A tool from Volume II, used here ahead of time. The most direct proof uses complex numbers (already used in this chapter in Proof 4 of Ptolemy's theorem, previous section): one identifies the plane with $\mathbb{C}$, and uses the fact, an immediate consequence of the addition formulas (Theorem 8.4, Chapter 8), that multiplying a complex number by $e^{i\theta} = \cos \theta + i \sin \theta$ amounts to rotating it by an angle θ about the origin. Volume II, Chapter 3, develops these tools systematically; here only the rotation rule is needed.

Lemma 10.8: Equilaterality criterion in complex form. Let $x, y, z \in \mathbb{C}$, not all coincident, and let $\eta = e^{2\pi i/3}$ (a primitive cube root of unity, so $\eta^3 = 1$ and $1 + \eta + \eta^2 = 0$). If

$$x + \eta y + \eta^2 z = 0,$$

then x, y, z are the vertices of an equilateral triangle.

Proof. From the hypothesis, $x = -\eta y - \eta^2 z$. Then

$$x - z = -\eta y - \eta^2 z - z = -\eta y - z(\eta^2 + 1).$$

From $1 + \eta + \eta^2 = 0$ we get $\eta^2 + 1 = -\eta$, hence

$$x - z = -\eta y + \eta z = \eta(z - y) = -(-\eta)(y - z).$$

Now, $-\eta = -e^{2\pi i/3} = e^{i\pi} e^{2\pi i/3} = e^{i(\pi+2\pi/3)} = e^{5\pi i/3} = e^{-i\pi/3}$, a rotation by -60° . The equality $x - z = e^{-i\pi/3}(y - z)$ says that the vector from z to x is obtained by rotating the vector from z to y by -60° : being a rotation, the two vectors have the same modulus, so $|x - z| = |y - z|$, and the angle between them (at z) equals exactly 60° . An isosceles triangle with apex angle 60° is equilateral (the two base angles are equal to each other and sum to 120° , hence each equals 60°): x, y, z form an equilateral triangle. $\square$

Theorem 10.13: Napoleon's theorem. On each side of an arbitrary triangle ABC , construct an equilateral triangle externally. Then the centroids of the three equilateral triangles are themselves the vertices of an equilateral triangle (the **outer Napoleon triangle**).

Proof. [of Napoleon's theorem] **Step 1 — coordinates.** Identify the plane with $\mathbb{C}$ and choose the vertices $a, b, c \in \mathbb{C}$ of the given triangle, numbered **counterclockwise**.

Step 2 — the apexes of the outer equilateral triangles. On side BC (opposite A), the apex of the outer equilateral triangle is constructed by rotating $(c - b)$ by -60° about b :

$$a' = b + (c - b)e^{-i\pi/3},$$

and cyclically,

$$b' = c + (a - c)e^{-i\pi/3}, \quad c' = a + (b - a)e^{-i\pi/3}.$$

(For a triple a, b, c in counterclockwise order, this convention **always** produces the vertex on the side opposite the original triangle's vertex — that is, the *outer* construction, not the inner one that would use a $+60^\circ$ rotation. General proof, not just the numerical example below: the side of line BC on which a point P lies is determined by the sign of $\text{Im}((c - b)(P - b))$ (the z -component of the two-dimensional cross product of $c - b$ and $P - b$); for a triple a, b, c in counterclockwise order this sign is, by the very definition of orientation, positive for $P = a$. For $P = a'$, instead,

$$\overline{(c - b)}(a' - b) = \overline{(c - b)}(c - b)e^{-i\pi/3} = |c - b|^2 e^{-i\pi/3},$$

whose imaginary part equals $-|c - b|^2 \sin(60^\circ) < 0$: **regardless of where a is** (the formula for a' depends only on b, c), the sign is always negative, i.e. opposite to that of a . Hence a' always lies on the side of BC opposite A : it is always the outer vertex. The same argument applies cyclically to b' and c' . See also the numerical check in Example 10.8 below.)

Step 3 — the centroids. The centroids of the three outer equilateral triangles (on BC , CA , AB respectively) are

$$g_A = \frac{b + c + a'}{3}, \quad g_B = \frac{c + a + b'}{3}, \quad g_C = \frac{a + b + c'}{3}.$$

Step 4 — checking the criterion. Substituting the expressions of Step 2 and expanding:

$$\begin{aligned} 3g_A &= b + c + b + (c - b)e^{-i\pi/3} = b(2 - w) + c(1 + w), \\ 3g_B &= c(2 - w) + a(1 + w), \\ 3g_C &= a(2 - w) + b(1 + w), \end{aligned}$$

where $w = e^{-i\pi/3}$ for brevity. First observe that $w = -\eta$: indeed $\eta = e^{2\pi i/3} = \cos 120^\circ + i \sin 120^\circ = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, so $-\eta = \frac{1}{2} - i\frac{\sqrt{3}}{2} = \cos(-60^\circ) + i \sin(-60^\circ) = e^{-i\pi/3} = w$. Substituting $w = -\eta$ and collecting, in the combination $3(g_A + \eta g_B + \eta^2 g_C)$, the coefficients of a, b, c separately:

$$\begin{aligned} \text{coefficient of } a: \quad & \eta(1 + w) + \eta^2(2 - w) = \eta + 2\eta^2 + w(\eta - \eta^2) \\ & = \eta + 2\eta^2 - \eta(\eta - \eta^2) = \eta + 2\eta^2 - \eta^2 + \eta^3 = \eta + \eta^2 + 1 = 0, \\ \text{coefficient of } b: \quad & (2 - w) + \eta^2(1 + w) = 2 + \eta^2 + w(\eta^2 - 1) \\ & = 2 + \eta^2 - \eta(\eta^2 - 1) = 2 + \eta^2 - \eta^3 + \eta = 1 + \eta + \eta^2 = 0, \\ \text{coefficient of } c: \quad & (1 + w) + \eta(2 - w) = 1 + 2\eta + w(1 - \eta) \\ & = 1 + 2\eta - \eta(1 - \eta) = 1 + 2\eta - \eta + \eta^2 = 1 + \eta + \eta^2 = 0, \end{aligned}$$

where $w = -\eta$ and $\eta^3 = 1$ were used in every line, and the identity $1 + \eta + \eta^2 = 0$ to cancel the final result. All three coefficients vanish, hence

$$g_A + \eta g_B + \eta^2 g_C = 0.$$

(Also verified symbolically in `cap09_napoleone.py`, for an independent check.)

Step 5 — conclusion. By Lemma 10.8 (with $x = g_A$, $y = g_B$, $z = g_C$), the three centroids form an equilateral triangle. $\square$

Remark 10.16: A historically doubtful attribution. The theorem is traditionally associated with Napoleon Bonaparte (1769–1821), a well-known amateur enthusiast of mathematics, but the attribution is far from certain. Several historical studies have cast doubt on it: according to the historian of mathematics Howard Eves, the result may actually be due to Lorenzo Mascheroni (1750–1800), who is said to have let the emperor take credit for it. The chronology in any case makes a direct attribution to Napoleon problematic: the result appears in three questions of an examination for a gold medal at the University of Dublin in October 1820 (when Napoleon was still alive, but exiled on Saint Helena), while the earliest known appearance in print dates to 1825, in a query by W. Rutherford published in *The Ladies' Diary* — four years after Napoleon's death, in May 1821. There is no direct evidence that Napoleon knew of or discovered the theorem.

Example 10.8: Numerical verification of Napoleon's theorem. Let $A = (0, 0)$, $B = (4, 0)$, $C = (1, 3)$ (counterclockwise order: the cross product $(B - A) \times (C - A) = 4 \cdot 3 - 0 \cdot 1 = 12 > 0$ confirms it). With $a = 0$, $b = 4$, $c = 1 + 3i$ and the rotation $e^{-i\pi/3}$:

$$a' = 5.098076 + 4.098076i, \quad b' = -2.098076 + 2.366025i, \quad c' = 2.000000 - 3.464102i.$$

Outer-construction check: relative to line BC , vertex A lies on the side $+12.0000$ (in units of signed double area) while A' lies on the side -15.5885 : opposite signs, so A' is indeed on the side opposite A — an outer construction, as required.

The centroids come out to

$$g_A = 3.366025 + 2.366025i, \quad g_B = -0.366025 + 1.788675i, \quad g_C = 2.000000 - 1.154701i,$$

and the three pairwise distances all equal 3.776445 (difference between the largest and smallest: $< 10^{-9}$), confirming equilaterality. The criterion of Lemma 10.8 gives $g_A + \eta g_B + \eta^2 g_C \approx (-7 + 13i) \times 10^{-16}$, zero at machine epsilon. Verification script: `cap09_napoleone.py`.

10.14. Pythagorean triples as rational points on the unit circle

Theorem 10.14: Generation of primitive Pythagorean triples. Every primitive Pythagorean triple (a, b, c) (with $a^2 + b^2 = c^2$, $\gcd(a, b, c) = 1$, a odd, b even) has the form

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2,$$

for suitable integers $m > n > 0$, coprime and of opposite parity. Conversely, given a, c (with a odd), $m = \sqrt{(c + a)/2}$, $n = \sqrt{(c - a)/2}$.

Proof. Dividing $a^2 + b^2 = c^2$ by c^2 : $(a/c)^2 + (b/c)^2 = 1$, i.e. $(a/c, b/c)$ is a **rational** point of the unit circle. By Theorem 7.10 (Chapter 7), every rational point of the unit circle (other than $(-1, 0)$) can be written as

$$\frac{a}{c} = \frac{1 - t^2}{1 + t^2}, \quad \frac{b}{c} = \frac{2t}{1 + t^2}, \quad t = \frac{n}{m} \in \mathbb{Q}, \quad \gcd(m, n) = 1.$$

Substituting $t = n/m$ and reducing to the common denominator $m^2 + n^2$:

$$\frac{a}{c} = \frac{m^2 - n^2}{m^2 + n^2}, \quad \frac{b}{c} = \frac{2mn}{m^2 + n^2}.$$

Since (a, b, c) is primitive, the common denominator $m^2 + n^2$ must be exactly c (one checks that $\gcd(m^2 - n^2, 2mn, m^2 + n^2) = 1$ when m, n are coprime and of opposite parity), whence

$a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$. The inverse formula follows from $c + a = 2m^2$, $c - a = 2n^2$. $\square$

Remark 10.17: One formula, three chapters. This theorem ideally closes the conceptual circle of the treatise: the very identity $a^2 + b^2 = c^2$ seen in this chapter as a synthetic theorem of Euclidean geometry (Section 10.1) is also a trigonometric identity (Chapter 8) and is also, here, the tool for classifying all rational points of an algebraic curve (Chapter 7).

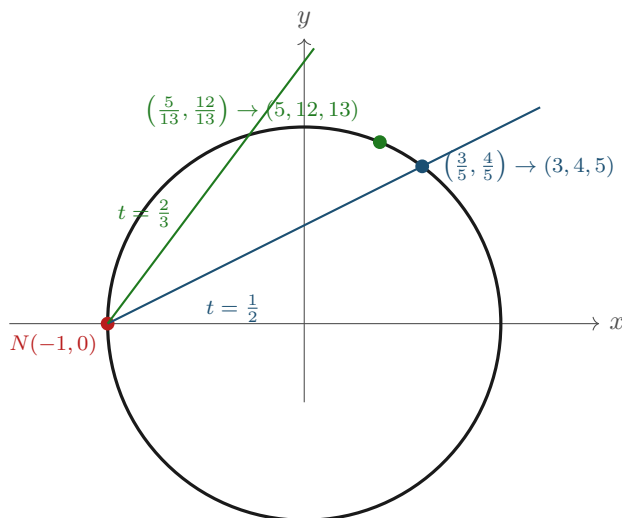


Figure 10.8: Pythagorean triples as rational points of the unit circle. The line through $N(-1, 0)$ with rational slope $t = p/q$ meets the circle $x^2 + y^2 = 1$ at a second point with coordinates $(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$, both rational; conversely every rational point of the circle arises this way, because the line joining it to N has rational slope. The correspondence between $t \in \mathbb{Q}$ and the rational points is therefore bijective (apart from N itself), and reducing to lowest terms yields all and only the primitive Pythagorean triples: $t = 1/2$ gives $(3, 4, 5)$, $t = 2/3$ gives $(5, 12, 13)$. It is the stereographic projection of Volume II, used here as an arithmetical tool.

10.15. How far one gets: rational distances and the Erdős–Ulam problem

Theorem 10.14 classifies the points of the unit circle with rational coordinates. A natural, seemingly nearby question concerns instead the *distances*: how many points can be placed in the plane so that **all** mutual distances are rational? We shall see that one gets very far — and then stops.

10.15.1 What can be proved: infinitely many points on a circle

Theorem 10.15: Infinite rational-distance sets on a circle. There exists an *infinite* set of points on a circle such that the distance between any two of them is a rational number.

Proof. Step 1 — the right circle. Take the circle of **diameter 1**. By the extended law of sines (Theorem 10.5), with $2R = 1$, every chord subtending an inscribed angle γ has length exactly $\sin \gamma$. Equivalently, parametrising the points by the central angle 2ϑ , the chord between the points of parameter $2\vartheta_i$ and $2\vartheta_j$ measures

$$|\sin(\vartheta_i - \vartheta_j)|.$$

Step 2 — Pythagorean angles. Call an angle ϑ **Pythagorean** if $\sin \vartheta$ and $\cos \vartheta$ are both rational. By Theorem 10.14 there are infinitely many: to every primitive Pythagorean triple (a, b, c) corresponds the angle with $\sin \vartheta = a/c$, $\cos \vartheta = b/c$, and distinct primitive triples give distinct angles in $(0, \pi/2)$, since the pair $(a/c, b/c)$ determines ϑ . Primitive triples are infinite in number (for each $n \geq 1$, the pair $m = n + 1$, n is admissible and generates distinct triples as n varies).

Step 3 — the distances are rational. Take the points of parameter $2\vartheta_1, 2\vartheta_2, \dots$ with all ϑ_k Pythagorean. By the subtraction formula (Theorem 8.4, Chapter 8),

$$\sin(\vartheta_i - \vartheta_j) = \sin \vartheta_i \cos \vartheta_j - \cos \vartheta_i \sin \vartheta_j,$$

which is **rational**, being a sum of products of rationals. By Step 1 the distance between the two corresponding points is the absolute value of this number, hence rational. Since the Pythagorean angles are infinite in number, we obtain an infinite set of points on the circle with all mutual distances rational. $\square$

Remark 10.18: What has been achieved and what is missing. The set constructed is infinite, but it is contained in a circle: it is therefore very far from being dense in the plane (any disc missing that circle contains none of its points). Erdős and Ulam’s question asks for much more.

10.15.2 The open problem

Remark 10.19: The Erdős–Ulam problem. Question. Does there exist a subset of the plane that is **dense** (i.e. such that every disc, however small, contains at least one of its points) and in which **all** mutual distances are rational?

What is known, and in what order (1) **Finite sets: yes, of any size.** For every n there exist n points in general position with all mutual distances rational. Theorem 10.15 gives infinitely many, albeit all on one circle.

(2) **Algebraic curves do not suffice.** Solymosi and de Zeeuw (2010) proved that the *only* irreducible algebraic curves containing infinitely many points at mutually rational distances are **lines** and **circles**. A set dense in the plane cannot therefore be obtained by accumulating points on an algebraic curve: the route of Theorem 10.15 is, in a precise sense, the only one available, and it is closed.

(3) **A conditional result.** Shaffaf (2018) proved that, *assuming* the Bombieri–Lang conjecture (a deep conjecture of arithmetic algebraic geometry, itself unproved), no dense rational-distance set can exist; Pasten (2017), independently and with different methods (model theory), obtained the same conclusion assuming instead the ABC conjecture. A complete treatment of this implication is in Ascher, Braune and Turchet (2020).

Where we stop. The implication in (3) is proved; neither the Bombieri–Lang nor the ABC conjecture is. Passing from (2) to an *unconditional* answer would require tools not available today. $\square$

Warning. Status at the time of writing (September 2026): the Erdős–Ulam problem is **OPEN**. It is not known whether a dense rational-distance set exists. What is known is: that it cannot lie on an algebraic curve (unconditional result, Solymosi–de Zeeuw); and that it cannot exist at all *if* the Bombieri–Lang conjecture holds (conditional result). **As matters stand, the proof stops here.**

INFO

Why this problem belongs in this chapter. The whole chapter is built on exact metric relations between points of a circle and vertices of a triangle. The Erdős–Ulam problem takes that same requirement — that the distances be “nice” numbers — and imposes it on a dense set of points. The leap from the circle to the plane, geometrically minimal, carries one from elementary trigonometry to arithmetic algebraic geometry and to an unproved conjecture. It is the same phenomenon already observed in Chapters 3 and 4: difficulty cannot be read off the statement.

10.15.3 A worked example: four points at integer distances

Explicit construction on the circle of diameter 1. Take three Pythagorean angles, from the triples (3, 4, 5), (5, 12, 13), (8, 15, 17):

$$\sin \vartheta_1 = \frac{3}{5}, \quad \cos \vartheta_1 = \frac{4}{5}; \quad \sin \vartheta_2 = \frac{5}{13}, \quad \cos \vartheta_2 = \frac{12}{13}; \quad \sin \vartheta_3 = \frac{8}{17}, \quad \cos \vartheta_3 = \frac{15}{17}.$$

Add $\vartheta_0 = 0$ ($\sin = 0$, $\cos = 1$). The six distances between the four corresponding points, on the circle of diameter 1, equal $|\sin(\vartheta_i - \vartheta_j)|$:

$$\begin{aligned} d_{01} &= \frac{3}{5}, & d_{02} &= \frac{5}{13}, & d_{03} &= \frac{8}{17}, \\ d_{12} &= \left| \frac{3}{5} \cdot \frac{12}{13} - \frac{4}{5} \cdot \frac{5}{13} \right| = \left| \frac{36-20}{65} \right| = \frac{16}{65}, \\ d_{13} &= \left| \frac{3}{5} \cdot \frac{15}{17} - \frac{4}{5} \cdot \frac{8}{17} \right| = \left| \frac{45-32}{85} \right| = \frac{13}{85}, \\ d_{23} &= \left| \frac{5}{13} \cdot \frac{15}{17} - \frac{12}{13} \cdot \frac{8}{17} \right| = \left| \frac{75-96}{221} \right| = \frac{21}{221}. \end{aligned}$$

All rational. Multiplying the whole figure by the least common multiple of the denominators (5, 13, 17, 65, 85, 221; the l.c.m. is $5 \cdot 13 \cdot 17 = 1105$) yields four points on a circle with **all six distances integers**:

$$663, \quad 425, \quad 520, \quad 272, \quad 169, \quad 105.$$

Consistency check: $d_{01} = 3/5 \cdot 1105 = 663$; $d_{23} = 21/221 \cdot 1105 = 21 \cdot 5 = 105$. ✓

□

10.15.4 A worked example: a Pythagorean triple in a two-link arm

Geometric check. A planar two-link arm, of lengths $L_1 = 300$ mm and $L_2 = 400$ mm, must reach a point such that the two links are mutually perpendicular (a “90°-elbow” configuration). The distance of the end-effector from the base is then

$$d = \sqrt{L_1^2 + L_2^2} = \sqrt{300^2 + 400^2} = 500 \text{ mm},$$

exactly the Pythagorean triple (3, 4, 5) scaled by a factor of 100. Whenever a right-elbow configuration is chosen with links in the ratio 3:4, the resulting distance is an exact integer multiple of the link lengths: handy for quick checks during commissioning, with no calculator. □

10.16. Problem set

10.16.1 Numerical exercises

Exercise 10.1. In a right triangle the legs measure 8 cm and 15 cm. Compute the radius r of the inscribed circle and R of the circumscribed one.

Solution. Hypotenuse: $c = \sqrt{64 + 225} = \sqrt{289} = 17$ cm (the triple (8, 15, 17)).

$$R = \frac{c}{2} = \frac{17}{2} = 8.5 \text{ cm} \quad (\text{Corollary 10.3}),$$

$$r = \frac{a + b - c}{2} = \frac{8 + 15 - 17}{2} = \frac{6}{2} = 3 \text{ cm} \quad (\text{Theorem 10.4}).$$

Cross-check. With the area formula, $r = ab/(a + b + c) = 120/40 = 3$ cm. ✓ And for Euler's inequality (Corollary 10.6): $R = 8.5 \geq 2r = 6$. ✓

Exercise 10.2. A right triangle has legs 6 and 8. Compute R and r , and verify Euler's formula by computing OI .

Solution. $c = 10$, hence $R = 5$ and $r = (6 + 8 - 10)/2 = 2$. By Euler's formula (Theorem 10.9):

$$OI^2 = R^2 - 2Rr = 25 - 2 \cdot 5 \cdot 2 = 25 - 20 = 5 \implies OI = \sqrt{5} \approx 2.236.$$

The circumcentre is the midpoint of the hypotenuse; the incentre is therefore at distance $\sqrt{5}$ from it. Check: $R = 5 \geq 2r = 4$, consistent with Euler's inequality.

Exercise 10.3. A (non-right) triangle has sides $a = 7$, $b = 8$, $c = 9$. Compute the circumradius using Heron's formula and $R = abc/(4K)$.

Solution. Semiperimeter $s = (7 + 8 + 9)/2 = 12$. Heron: $K = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{12 \cdot 5 \cdot 4 \cdot 3} = \sqrt{720} \approx 26.83$.

$$R = \frac{abc}{4K} = \frac{504}{107.33} \approx 4.70.$$

Check via the law of sines. $\cos C = (a^2 + b^2 - c^2)/(2ab) = (49 + 64 - 81)/112 = 32/112 = 2/7$, hence $\sin C = \sqrt{1 - 4/49} = \sqrt{45}/7 \approx 0.9583$ and $R = c/(2 \sin C) = 9/1.9166 \approx 4.70$. ✓

Exercise 10.4. Verify that (3, 4, 5) is a primitive Pythagorean triple and find the generators m, n using the inverse formulas of Theorem 10.14.

Solution. $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. ✓ With $a = 3$ (odd), $c = 5$: $m = \sqrt{(5+3)/2} = 2$, $n = \sqrt{(5-3)/2} = 1$. Check: $a = m^2 - n^2 = 3$, $b = 2mn = 4$, $c = m^2 + n^2 = 5$. ✓ Inradius by the formula of Remark 10.5: $r = n(m - n) = 1 \cdot 1 = 1$, consistent with $(3 + 4 - 5)/2 = 1$.

Exercise 10.5. Using the two Pythagorean angles of the triples (3, 4, 5) and (20, 21, 29), compute the distance between the corresponding points on the circle of diameter 1 and verify that it is rational.

Solution. $\sin \vartheta_1 = 3/5$, $\cos \vartheta_1 = 4/5$; $\sin \vartheta_2 = 20/29$, $\cos \vartheta_2 = 21/29$. By Step 3 of the proof of Theorem 10.15:

$$d = |\sin(\vartheta_1 - \vartheta_2)| = \left| \frac{3}{5} \cdot \frac{21}{29} - \frac{4}{5} \cdot \frac{20}{29} \right| = \left| \frac{63-80}{145} \right| = \frac{17}{145},$$

rational as predicted. Scaling the figure by 145 yields two points at integer distance 17 on a circle of diameter 145.

10.16.2 Symbolic and proof exercises

Exercise 10.6. Prove that for every primitive Pythagorean triple the inradius of the corresponding right triangle is an integer, and characterise when it equals 1.

Solution. By Theorem 10.14, $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$ with $m > n > 0$ coprime of opposite parity. Then

$$r = \frac{a + b - c}{2} = \frac{(m^2 - n^2) + 2mn - (m^2 + n^2)}{2} = \frac{2mn - 2n^2}{2} = n(m - n),$$

a product of two positive integers (positive since $m > n > 0$): hence r is a positive integer. When $r = 1$. We need $n(m - n) = 1$ with $n \geq 1$ and $m - n \geq 1$ integers, hence $n = 1$ and $m - n = 1$, i.e. $m = 2$, $n = 1$: the only primitive triple with inradius 1 is $(3, 4, 5)$. This is why the 3-4-5 triangle appears in every manual: it is the smallest, and the only one with a unit-radius incircle.

Exercise 10.7. Prove that in every right triangle $r + 2R = s$, where s is the semiperimeter, and check that the same relation does NOT hold for an equilateral triangle.

Solution. By Corollary 10.3 and Theorem 10.4,

$$r + R = \frac{a + b - c}{2} + \frac{c}{2} = \frac{a + b - c + c}{2} = \frac{a + b}{2}.$$

On the other hand $s = (a + b + c)/2$ and $R = c/2$, hence

$$r + 2R = \frac{a + b}{2} - \frac{c}{2} + \frac{c}{2} + \frac{c}{2} = \frac{a + b + c}{2} = s,$$

which is the claim. Equivalently, $r + R = (a + b)/2$: half the sum of the legs.

Geometric meaning. From Step 1 of the proof of Theorem 10.4, the tangent segments from the two acute vertices are $x = b - r$ and $y = a - r$, with $x + y = c$; the quantity $(a + b)/2$ is thus half the sum of the legs, and the relation says it decomposes exactly into inradius plus circumradius. This is specific to the right triangle: for a general triangle $r + 2R \neq s$ (check it on the equilateral, where $r + 2R = a/(2\sqrt{3}) + 2a/\sqrt{3} = 5a/(2\sqrt{3}) \approx 1.443a$ while $s = 3a/2 = 1.5a$).

Exercise 10.8. Prove that if ϑ and φ are Pythagorean angles (rational sine and cosine), then so is $\vartheta + \varphi$. Deduce that the Pythagorean angles form a subgroup of the group of angles modulo 2π .

Solution. By the addition formulas (Theorem 8.4),

$$\sin(\vartheta + \varphi) = \sin \vartheta \cos \varphi + \cos \vartheta \sin \varphi, \quad \cos(\vartheta + \varphi) = \cos \vartheta \cos \varphi - \sin \vartheta \sin \varphi,$$

both sums of products of rationals, hence rational.

Subgroup. The set G of Pythagorean angles contains 0 ($\sin 0 = 0$, $\cos 0 = 1$); it is closed under addition (just shown); and it is closed under negation, since $\sin(-\vartheta) = -\sin \vartheta$ and $\cos(-\vartheta) = \cos \vartheta$ remain rational. Hence G is a subgroup of $\mathbb{R}/2\pi\mathbb{Z}$.

Consequence. This gives a second, more structural proof of Theorem 10.15: take a single Pythagorean angle ϑ of infinite order (for instance the one from the triple $(3, 4, 5)$: the fact that its cosine $4/5$ is not the cosine of any angle commensurable with π is **imported**, Niven's theorem on rational values of the cosine — I. Niven, *Irrational Numbers*, Carus Mathematical Monographs 11, MAA, 1956, cor. 3.12: the only rational values of $\cos(p\pi/q)$ are $0, \pm\frac{1}{2}, \pm 1$, and $4/5$ is not among them); its multiples $k\vartheta$ are all Pythagorean and pairwise distinct, and the differences $k\vartheta - j\vartheta = (k - j)\vartheta$ are Pythagorean too: the corresponding distances are all rational.

Exercise 10.9. Verify, for an equilateral triangle of side a , that the nine-point circle has radius $R/2$ and centre coinciding with that of the triangle, and check the internal tangency to the incircle (Feuerbach's theorem) in this special case.

Solution. In the equilateral triangle $O = G = H$, hence the nine-point centre N (midpoint of OH) coincides with them. The radius is $R/2$ by Theorem 10.11, whose proof nowhere uses the assumption that the triangle is not equilateral.

Tangency. For the equilateral, $R = a/\sqrt{3}$ and $r = a/(2\sqrt{3}) = R/2$. The nine-point circle has radius $R/2 = r$ and the same centre as the incircle: the two circles **coincide**. The internal tangency of Feuerbach's theorem therefore degenerates, in the equilateral case, into coincidence of the two circles — a limiting case fully consistent with the general statement (two coincident circles are tangent at every one of their points).

Note. Observe also that the nine points collapse into only three distinct points: the side midpoints coincide with the feet of the altitudes (the altitudes are also medians) and with the Euler points.

Exercise 10.10. For the triangle with sides 13, 14, 15 compute the area, the circumradius, the inradius, and verify Euler's formula $OI^2 = R^2 - 2Rr$.

Solution. Semiperimeter $s = 21$; by Heron $\mathcal{A} = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84$. Hence

$$R = \frac{abc}{4\mathcal{A}} = \frac{13 \cdot 14 \cdot 15}{336} = \frac{2730}{336} = 8.125, \quad r = \frac{\mathcal{A}}{s} = \frac{84}{21} = 4.$$

Euler's formula gives $OI^2 = R^2 - 2Rr = 66.015625 - 65 = 1.015625$, that is $OI = 1.00778$. Check also Euler's inequality: $R = 8.125 \geq 2r = 8$ ✓, with minimal slack — the 13–14–15 triangle is nearly equilateral in the sense of this inequality.

Exercise 10.11. For the same 13–14–15 triangle compute the radius of the nine-point circle and the distance between its centre and the incentre, verifying Feuerbach's theorem.

Solution. The radius of the nine-point circle is $R/2 = 4.0625$. Feuerbach's theorem states that this circle is internally tangent to the incircle; internal tangency forces the distance between the centres to be the difference of the radii:

$$|NI| = \frac{R}{2} - r = 4.0625 - 4 = 0.0625.$$

The value is small because R is close to $2r$: when $R = 2r$ (equilateral triangle) the two circles coincide and $|NI| = 0$. Verifying $|NI|$ directly from the coordinates of the two centres is a good consistency exercise.

Exercise 10.12. Using the rational parametrisation of the unit circle, obtain the Pythagorean triple corresponding to $t = 3/4$ and check it.

Solution. With $t = 3/4$,

$$x = \frac{1-t^2}{1+t^2} = \frac{1-9/16}{1+9/16} = \frac{7/16}{25/16} = \frac{7}{25}, \quad y = \frac{2t}{1+t^2} = \frac{3/2}{25/16} = \frac{24}{25}.$$

The point $(7/25, 24/25)$ is rational on the unit circle, and multiplying by 25 gives the triple $(7, 24, 25)$. Check: $49 + 576 = 625 = 25^2$ ✓. The triple is primitive because $\gcd(7, 24) = 1$.

10.17. Self-assessment quiz

Ten multiple-choice questions. The answer key is at the end of the section.

- In the Chinese treatise *Zhoubi Suanjing*, the principle corresponding to the Pythagorean theorem is called
 - the gnomon rule
 - the gougu rule
 - Liu Hui's rule
 - the sea island rule
- The radius of the circle circumscribed about a right triangle equals
 - the hypotenuse
 - half the hypotenuse
 - half the sum of the legs
 - half the longer leg
- The inradius of a triangle with sides 3, 4, 5 equals
 - 1
 - 1.5
 - 2
 - 2.5
- The circumcentre of a right triangle lies

-
- A) at the right-angle vertex
B) exactly at the midpoint of the hypotenuse
C) outside the triangle
D) at the centroid
5. The proof of the formula $r = (a + b - c)/2$ rests essentially on
A) the law of sines
B) the congruence of tangent segments from one external point
C) the nine-point theorem
D) Heron's formula
6. Euler's inequality $R \geq 2r$ becomes an equality
A) for every right triangle
B) for every isosceles triangle
C) only for the equilateral triangle
D) never
7. The radius of the nine-point circle equals
A) R
B) $R/2$
C) $r/2$
D) $R/3$
8. Primitive Pythagorean triples correspond bijectively to
A) all points of the unit circle
B) the points of the unit circle with rational coordinates
C) the points of the unit circle with integer coordinates
D) the diameters of the unit circle
9. Does there exist an infinite set of points *on a circle* with all mutual distances rational?
A) no, never
B) yes, and the construction uses angles with rational sine and cosine
C) yes, but only for circles of integer rational radius
D) it is an open problem
10. The Erdős–Ulam problem (existence of a set *dense in the plane* with all distances rational) is, as matters stand,
A) settled affirmatively
B) settled negatively
C) open; the negative answer is known only assuming the Bombieri–Lang conjecture
D) proved undecidable
-

BEST PRACTICE

Key. 1: B — 2: B — 3: A — 4: B — 5: B — 6: C — 7: B — 8: B — 9: B (Theorem 10.15) — 10: C.

Note on questions 9 and 10. The pair is designed to separate two statements that look alike and have quite different status: infinitely many points *on a circle* at rational distances exist and are constructed in half a page; a set *dense in the plane* with the same property is not known to exist. The whole difference lies in the word “dense”.

10.18. Chapter Formula sheet

Metric relations in the right triangle (Theorems 10.1, 10.2 and 10.3; Corollary 10.2)

$$a^2 + b^2 = c^2, \quad b^2 = cp, \quad a^2 = cq, \quad h^2 = pq, \quad p + q = c$$

$$h = \frac{ab}{c}, \quad p = \frac{b^2}{c}, \quad q = \frac{a^2}{c}, \quad \frac{1}{h^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

Circles of the right triangle (Corollary 10.3; Theorem 10.4)

$$R = \frac{c}{2} \text{ (centre at the midpoint of the hypotenuse),} \quad r = \frac{a + b - c}{2}$$

Law of sines (Theorem 10.5)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Law of cosines (Theorem 10.6)

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Mollweide's formulas and the law of tangents (Theorem 10.7; Corollary 10.4)

$$\frac{a+b}{c} = \frac{\cos \frac{\alpha - \beta}{2}}{\sin \frac{\gamma}{2}}, \quad \frac{a-b}{c} = \frac{\sin \frac{\alpha - \beta}{2}}{\cos \frac{\gamma}{2}}, \quad \frac{a-b}{a+b} = \frac{\tan \frac{\alpha - \beta}{2}}{\tan \frac{\alpha + \beta}{2}}$$

Inradius and exradii (Theorem 10.8; Corollary 10.5), with s the semiperimeter and S the area

$$r = \frac{S}{s}, \quad r_a = \frac{S}{s-a}, \quad r_b = \frac{S}{s-b}, \quad r_c = \frac{S}{s-c}, \quad r_a + r_b + r_c - r = 4R$$

Euler–Chapple formula and Euler's inequality (Theorem 10.9; Corollary 10.6)

$$OI^2 = R^2 - 2Rr, \quad R \geq 2r$$

Nine-point circle and Feuerbach's theorem (Theorems 10.11 and 10.12)

$$\text{radius } \frac{R}{2}, \quad N = \text{midpoint of } \overline{OH}, \quad \vec{OH} = \vec{A} + \vec{B} + \vec{C}$$

$$NI = \frac{R}{2} - r, \quad NI_a = \frac{R}{2} + r_a, \quad NI_b = \frac{R}{2} + r_b, \quad NI_c = \frac{R}{2} + r_c$$

Primitive Pythagorean triples (Theorem 10.14), with $m > n > 0$ coprime and of opposite parity

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2$$

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Numerical Methods and Representation of the Circle in the Computer

All the previous chapters treat the circle as an ideal mathematical object, continuous and infinitely precise. A computer (a PC, a robotic controller, a PLC, a microcontroller in an HMI) cannot represent $\mathbb{R}$: every coordinate, every sine, every radius is approximated with a finite number of bits. This chapter deals with three strictly connected questions: how real numbers are represented in the machine, how a circle is drawn/computed with low-cost integer arithmetic only, and what background noise (error) all this introduces — a topic of immediate relevance for whoever programs controllers and HMI interfaces.

11.1. Floating-point representation (IEEE 754)

Definition 11.1: IEEE 754 floating-point format. A real number is represented as $x = (-1)^s \cdot 1.m \cdot 2^{e-\text{bias}}$, with s the sign bit, m the mantissa (fractional part, 52 bit in double precision, 23 in single precision) and e the encoded exponent (11 bit in double precision, 8 in single, with bias 1023 and 127 respectively).

Format	Sign	Exponent	Mantissa
Single precision (32 bit)	1 bit	8 bit	23 bit
Double precision (64 bit)	1 bit	11 bit	52 bit

Table 11.1: Structure of the two most used IEEE 754 formats in industrial controllers and PCs

Definition 11.2: Machine epsilon. The **machine epsilon** ε is the smallest positive quantity such that $1 + \varepsilon$ is representable distinctly from 1. For double precision (52-bit mantissa), $\varepsilon = 2^{-52} \approx 2.22 \times 10^{-16}$; for single precision (23-bit mantissa), $\varepsilon = 2^{-23} \approx 1.19 \times 10^{-7}$.

Remark 11.1: Rounding error in the computation of a point of the circle. Computing $P(\theta) = (r \cos \theta, r \sin \theta)$ in floating-point, every operation (evaluation of $\cos$, $\sin$, multiplication by r) introduces a relative error of order ε . For a radius r expressed in millimetres on an industrial robot ($r \sim 10^3$ mm), the typical absolute error in single precision is $\sim r\varepsilon \sim 10^3 \times 1.19 \times 10^{-7} \approx 1.2 \times 10^{-4}$ mm: negligible for a single evaluation, but — as will be seen in Section 11.5 — not always negligible if the error accumulates over thousands of micro-steps of an interpolation (Chapter 13).

11.2. The midpoint algorithm for circle rasterisation

When a circle must be drawn on a grid of discrete pixels (the display of an HMI panel, a SCADA monitor, a plotter), one does not evaluate $\cos$, $\sin$ for thousands of angles: one uses an integer-

arithmetic-only algorithm, derived from Bresenham (1977) in the “midpoint” form of Van Aken (1984).

Definition 11.3: Implicit function of the circle and decision parameter. Let $f(x, y) = x^2 + y^2 - r^2$ ($f = 0$ on the circle, $f < 0$ inside, $f > 0$ outside). In the first octant ($0 \leq y \leq x$), starting from $(x_0, y_0) = (0, r)$, at each step one must choose between the pixel **E** (east: $x + 1, y$) and **SE** (south-east: $x + 1, y - 1$) the one closest to the real circle. The **decision parameter** is defined as the value of f at the midpoint between the two options:

$$p_k = f(x_k + 1, y_k - \frac{1}{2}) = (x_k + 1)^2 + (y_k - \frac{1}{2})^2 - r^2.$$

Theorem 11.1: Incremental update of the decision parameter. If $p_k < 0$ (the midpoint is inside the circle: pixel **E** is the closest), one chooses $(x_{k+1}, y_{k+1}) = (x_k + 1, y_k)$ and

$$p_{k+1} = p_k + 2x_{k+1} + 1.$$

If $p_k \geq 0$ (the midpoint is outside or on the circle: one chooses **SE**), $(x_{k+1}, y_{k+1}) = (x_k + 1, y_k - 1)$ and

$$p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}.$$

In both cases the update requires only additions and shifts, never multiplications nor transcendental functions.

Proof. Case **E**: $x_{k+1} = x_k + 1$, $y_{k+1} = y_k$, hence

$$p_{k+1} = (x_k + 2)^2 + (y_k - \frac{1}{2})^2 - r^2 = \left[(x_k + 1)^2 + (y_k - \frac{1}{2})^2 - r^2 \right] + (2x_k + 3) = p_k + 2x_{k+1} + 1,$$

having used $(x_k + 2)^2 - (x_k + 1)^2 = 2x_k + 3 = 2(x_k + 1) + 1 = 2x_{k+1} + 1$.

Case **SE**: $x_{k+1} = x_k + 1$, $y_{k+1} = y_k - 1$, hence $y_{k+1} - \frac{1}{2} = y_k - \frac{3}{2}$, and

$$\begin{aligned} p_{k+1} &= (x_k + 2)^2 + (y_k - \frac{3}{2})^2 - r^2 \\ &= \left[(x_k + 1)^2 + (y_k - \frac{1}{2})^2 - r^2 \right] + (2x_k + 3) + \left[(y_k - \frac{3}{2})^2 - (y_k - \frac{1}{2})^2 \right]. \end{aligned}$$

The last term is $(y_k - \frac{3}{2})^2 - (y_k - \frac{1}{2})^2 = -2y_k + 2 = -2y_{k+1}$ (since $y_{k+1} = y_k - 1$). Hence $p_{k+1} = p_k + (2x_k + 3) - 2y_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$. $\square$

Proposition 11.1: The other seven octants (reducing the work to 1/8). If (x, y) lies on the circle $x^2 + y^2 = r^2$ centred at the origin, then so do the eight points

$$(\pm x, \pm y), \quad (\pm y, \pm x)$$

(all sign combinations). Hence tracing the first octant $0 \leq y \leq x$ alone determines the whole circle with no further computation.

Proof. The implicit function $f(x, y) = x^2 + y^2 - r^2$ is invariant under each of the eight transformations listed:

- $(x, y) \mapsto (-x, y)$ and $(x, y) \mapsto (x, -y)$ (reflections in the axes): squaring kills the sign change, $(-x)^2 = x^2$;
- $(x, y) \mapsto (y, x)$ (reflection in the bisector $y = x$): the sum $x^2 + y^2$ is symmetric in its two arguments.

These three reflections generate the dihedral group D_4 of the eight symmetries of the square;

since f is invariant under the generators, it is invariant under the whole group. Thus if $f(x, y) = 0$, then $f(g(x, y)) = 0$ for each of the eight $g \in D_4$.

Coverage. Every point of the circle is the image under some $g \in D_4$ of a point of the sector $0 \leq y \leq x$: in polar coordinates $(r \cos \varphi, r \sin \varphi)$ the sector corresponds to $\varphi \in [0, \pi/4]$, and the eight images of this sector under D_4 cover the whole turn $[0, 2\pi)$, since $8 \times \pi/4 = 2\pi$. No point of the circle is therefore left out. $\square$

Remark 11.2: Historical attribution of the algorithm. The original integer-only formulation is due to Jack E. **Bresenham**: IBM Technical Report TR02.286 (27 January 1964) already contains the algorithm for circular arcs, later published in article form in *Communications of the ACM* in 1977. The “midpoint” version given here — evaluating f at the midpoint between the two candidate pixels rather than comparing distances — is an extension due to Van Aken (1984), mathematically equivalent but more directly derived.

A point of precision: Bresenham’s celebrated 1965 paper (*Algorithm for computer control of a digital plotter*, IBM Systems Journal) concerns the drawing of line segments, not arcs; the two works are frequently confused in the secondary literature.

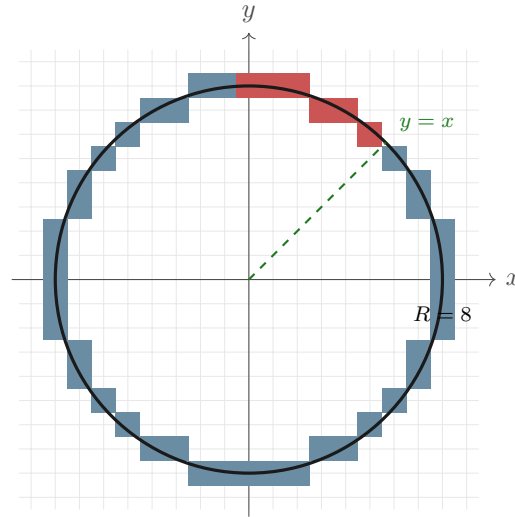


Figure 11.1: The midpoint (Bresenham) algorithm for the circle of radius $R = 8$ pixels. In red the six pixels computed explicitly in the first octant, $(0, 8), (1, 8), (2, 8), (3, 7), (4, 7), (5, 6)$; in blue the remaining ones, obtained by pure symmetry, reflecting into the other seven octants. The algorithm works with integers only: the decision variable d starts at $1 - R$ and is updated by $d += 2x + 3$ or $d += 2(x - y) + 5$ depending on its sign, so neither square roots nor divisions appear, and the cost per pixel is a few additions. One checks that the choice always coincides with rounding $\sqrt{R^2 - x^2}$: for $x = 5$, $\sqrt{39} = 6.245$ and the chosen pixel is 6.

11.3. Exact representation in CAD: B-Rep and NURBS

Rasterization produces an *approximation* of the circle: a staircase of pixels. A CAD system must instead store the circle in **exact** form, because intersections, tolerances and tool paths will be computed on that model. The universally adopted format is **B-Rep** (*Boundary Representation*): the solid is described by its boundary, organised in a topological hierarchy

$$\text{solid} \supset \text{shells} \supset \text{faces} \supset \text{loops} \supset \text{edges} \supset \text{vertices},$$

where each topological entity carries a **geometry**: a surface for a face, a curve for an edge. The technical question is then: with which class of curves is a circle represented *exactly*?

11.3.1 The circle is not a polynomial curve

Theorem 11.2: Impossibility of polynomial parametrisation. *There exist no polynomials $x(t), y(t) \in \mathbb{R}[t]$, not both constant, such that*

$$x(t)^2 + y(t)^2 = r^2 \quad \text{for all } t.$$

*In particular, no **non-rational** Bézier or B-spline curve can lie on a circular arc of positive length.*

Proof. Step 1 — pass to the complexes. In the ring of polynomials with complex coefficients $\mathbb{C}[t]$ we have the factorisation

$$(x(t) + iy(t))(x(t) - iy(t)) = x(t)^2 + y(t)^2 = r^2.$$

Set $P(t) = x(t) + iy(t)$ and $Q(t) = x(t) - iy(t)$, so that $P \cdot Q = r^2$, a non-zero constant polynomial.

Step 2 — degrees. $\mathbb{C}[t]$ is an integral domain and degree is additive: $\deg P + \deg Q = \deg(r^2) = 0$. Since P, Q are non-zero, their degrees are integers ≥ 0 summing to 0: hence $\deg P = \deg Q = 0$, i.e. P and Q are **constant**.

Step 3 — back to the reals. From

$$x(t) = \frac{P(t) + Q(t)}{2}, \quad y(t) = \frac{P(t) - Q(t)}{2i}$$

it follows that x and y are constant, contrary to hypothesis.

Step 4 — the case of an arc. If a polynomial curve $t \mapsto (x(t), y(t))$ lies on the circle for t in an interval $[t_0, t_1]$ of positive length, then the polynomial $x(t)^2 + y(t)^2 - r^2$ vanishes on an **infinite** set, hence is the zero polynomial (a non-zero polynomial has finitely many roots): we fall back into the hypothesis of Step 1 and the conclusion is the same. $\square$

Remark 11.3: What polynomial B-splines can do. *A non-rational quadratic Bézier curve is always a **parabolic** arc (or a segment, in degenerate cases): the only conics representable in polynomial form are those with a point at infinity on the curve itself. Circles, ellipses and hyperbolas require the **rational** form. This is exactly why the industry standard is NURBS — Non-Uniform Rational B-Spline — and not simply “B-spline”.*

11.3.2 The circular arc as a rational quadratic Bézier

Definition 11.4: Rational quadratic Bézier curve. Given three control points $P_0, P_1, P_2 \in \mathbb{R}^2$ and three weights $w_0, w_1, w_2 > 0$, the rational quadratic Bézier curve is

$$C(t) = \frac{w_0 B_0(t) P_0 + w_1 B_1(t) P_1 + w_2 B_2(t) P_2}{w_0 B_0(t) + w_1 B_1(t) + w_2 B_2(t)}, \quad t \in [0, 1],$$

where $B_0(t) = (1-t)^2$, $B_1(t) = 2t(1-t)$, $B_2(t) = t^2$ are the Bernstein polynomials of degree 2.

Theorem 11.3: Exact representation of the circular arc. *Let $0 < \vartheta < \pi/2$. On the unit circle take*

$$P_0 = (\cos \vartheta, -\sin \vartheta), \quad P_2 = (\cos \vartheta, \sin \vartheta), \quad P_1 = \left(\frac{1}{\cos \vartheta}, 0 \right)$$

(P_1 being the meeting point of the tangents to the circle at P_0 and P_2), with weights

$$w_0 = w_2 = 1, \quad w_1 = \cos \vartheta.$$

Then the curve $C(t)$ of Definition 11.4 lies **exactly** on the unit circle for every $t \in [0, 1]$, traversing the arc from P_0 to P_2 of central angle 2ϑ .

Proof. Write $c = \cos \vartheta$, $s = \sin \vartheta$, and introduce the abbreviations

$$\alpha = (1 - t)^2, \quad \beta = 2t(1 - t), \quad \gamma = t^2.$$

Step 1 — numerators and denominator. Noting that $w_1 P_1 = c \cdot (1/c, 0) = (1, 0)$, the components are

$$\begin{aligned} X(t) &= \alpha c + \beta \cdot 1 + \gamma c = c(\alpha + \gamma) + \beta, \\ Y(t) &= \alpha(-s) + \beta \cdot 0 + \gamma s = s(\gamma - \alpha), \\ W(t) &= \alpha + \beta c + \gamma = (\alpha + \gamma) + \beta c, \end{aligned}$$

and $C(t) = (X(t)/W(t), Y(t)/W(t))$.

Step 2 — the key identity. One checks directly that

$$4\alpha\gamma = 4(1 - t)^2 t^2 = (2t(1 - t))^2 = \beta^2.$$

Step 3 — the computation. Put $\sigma = \alpha + \gamma$ and expand:

$$\begin{aligned} X^2 + Y^2 - W^2 &= (c\sigma + \beta)^2 + s^2(\gamma - \alpha)^2 - (\sigma + \beta c)^2 \\ &= c^2\sigma^2 + 2c\sigma\beta + \beta^2 + s^2(\gamma - \alpha)^2 - \sigma^2 - 2\sigma\beta c - \beta^2 c^2 \\ &= (c^2 - 1)\sigma^2 + \beta^2(1 - c^2) + s^2(\gamma - \alpha)^2 \\ &= s^2[-\sigma^2 + \beta^2 + (\gamma - \alpha)^2], \end{aligned}$$

using $c^2 - 1 = -s^2$ and $1 - c^2 = s^2$. The square bracket remains: since

$$(\gamma - \alpha)^2 - \sigma^2 = (\gamma - \alpha)^2 - (\gamma + \alpha)^2 = -4\alpha\gamma,$$

it equals $\beta^2 - 4\alpha\gamma = 0$ by Step 2. Hence

$$X(t)^2 + Y(t)^2 = W(t)^2 \quad \text{identically in } t,$$

that is $(X/W)^2 + (Y/W)^2 = 1$: the point $C(t)$ lies on the unit circle.

Step 4 — the denominator does not vanish. For $0 < \vartheta < \pi/2$ we have $c > 0$, hence $W(t) = \alpha + \gamma + \beta c > 0$ for every $t \in [0, 1]$ (the three summands are ≥ 0 and not simultaneously zero): the curve is well defined throughout.

Step 5 — endpoints and traversal. $C(0) = P_0/1 = P_0$ and $C(1) = P_2/1 = P_2$. The component $Y(t)/W(t)$ has numerator $s(\gamma - \alpha) = s(t^2 - (1 - t)^2) = s(2t - 1)$, strictly increasing from $-s$ to $+s$ as t goes from 0 to 1, with $W > 0$: the curve therefore traverses *monotonically* the arc from P_0 to P_2 , i.e. the arc of central angle 2ϑ (the minor arc, since $2\vartheta < \pi$). $\square$

Remark 11.4: The value $\sqrt{2}/2$ and the circle in four arcs. To represent the whole circle one concatenates arcs of angle $2\vartheta < \pi$: the value most used in CAD kernels is $2\vartheta =$

$\pi/2$, i.e. four quarter-circles, for which

$$w_1 = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2} \approx 0.7071.$$

This is why, inspecting a STEP or IGES file exported from a CAD system, one almost always finds alternating weights 1, 0.7071, 1, 0.7071, ...: they are the signature of an exact circle. In NURBS form the complete circle is written as a rational B-spline of degree 2 with double internal knots at the junction points between the four arcs — the double multiplicity lowers the parametric continuity to C^0 , which is exactly what is needed for the geometric curve to stay smooth while the polynomial piece changes.

Warning. An exact circle, but at non-uniform speed. Theorem 11.3 guarantees that the *image* of the curve is the circle, not that the parameter t is proportional to angle or arc length. Indeed $t \mapsto \arctan(Y/X)$ is not linear. For interpolation at constant tangential speed — what a machine tool needs — one requires an **arc-length reparametrization**, or, more simply, one uses the controller’s own circular interpolation (Chapter 13), which parametrizes by angle.

11.4. CORDIC: trigonometry without multiplications

Robotic controllers and HMI DSPs compute $\sin$, $\cos$ thousands of times per second; historically (and still today on microcontrollers without FPU), this is done with the **CORDIC** algorithm (*COordinate Rotation DIgital Computer*, Volder, 1959), which reduces the rotation of a vector to only additions and bit shifts.

Definition 11.5: CORDIC iteration (rotation mode). To rotate the vector (x_0, y_0) by an angle $\theta = z_0$, one iterates for $i = 0, 1, \dots, n-1$:

$$x_{i+1} = x_i - d_i y_i 2^{-i}, \quad y_{i+1} = y_i + d_i x_i 2^{-i}, \quad z_{i+1} = z_i - d_i \arctan(2^{-i}),$$

with $d_i = +1$ if $z_i \geq 0$, $d_i = -1$ otherwise (to drive $z_i \rightarrow 0$).

Lemma 11.1: Convergence of the CORDIC residual angle. With $\vartheta_i = \arctan(2^{-i})$:
 (a) $\vartheta_i \leq 2\vartheta_{i+1}$ for every $i \geq 0$; (b) for every $i < n$, $|z_{i+1}| \leq \sum_{j=i+1}^{n-1} \vartheta_j + \vartheta_{n-1}$ (with the sequence z_i of the algorithm, hypothesis $|z_0| \leq \theta_{\max} := \sum_{i=0}^{n-1} \vartheta_i$).

Proof. (a) With $\alpha = \vartheta_{i+1}$, $\tan \alpha = 2^{-i-1} < 1$, the tangent duplication formula gives $\tan(2\alpha) = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} \geq 2 \tan \alpha$ (since $0 < 1 - \tan^2 \alpha < 1$ for $\tan \alpha \in (0, 1)$, dividing by it increases the value); since $\tan$ is increasing on $(0, \pi/2)$ and $2\alpha < \pi/2$ for $i \geq 0$, from $\tan(2\alpha) \geq 2 \tan \alpha = 2^{-i} = \tan \vartheta_i$ it follows that $2\alpha = 2\vartheta_{i+1} \geq \vartheta_i$.

(b), **outline.** The algorithm sets $z_{i+1} = z_i - \text{sign}(z_i)\vartheta_i$. If $|z_i| \geq \vartheta_i$, the magnitude decreases by exactly ϑ_i : $|z_{i+1}| = |z_i| - \vartheta_i$. If $|z_i| < \vartheta_i$, the sign of z flips and $|z_{i+1}| = \vartheta_i - |z_i| < \vartheta_i$; this is where (a) is needed: a step too large relative to the residual cannot “overshoot” zero by more than the value of the *next* step, doubled, and this check, applied recursively backwards from $i = n-1$ down to i , is what allows the remaining steps to be summed and gives the bound $\sum_{j=i+1}^{n-1} \vartheta_j + \vartheta_{n-1}$ of the statement. The complete closure of the induction (which requires carefully handling the limiting case $n-1$, where there is no longer a “next step, doubled” available, hence the correction term $+\vartheta_{n-1}$) is technical but elementary; we report

it as **imported** from the original source so as not to overload the text: Volder 1959, §§3–4, and Walther 1971 (complete formal proof). $\square$

Theorem 11.4: Correctness of the CORDIC iteration. Put $\vartheta_i = \arctan(2^{-i})$, and let $\theta_{\max} = \sum_{i=0}^{n-1} \vartheta_i$ (as $n \rightarrow \infty$, $\theta_{\max} \rightarrow 99.88^\circ$). If $|\theta| \leq \theta_{\max}$, after n iterations $(x_n, y_n) \approx K \cdot R_\theta(x_0, y_0)$, where R_θ is the rotation of angle $\theta = z_0$ and $K = \prod_{i=0}^{n-1} \cos \vartheta_i$ is a scale constant independent of θ (for $n \rightarrow \infty$, $K \rightarrow 0.6072529 \dots$); more precisely, by Lemma 11.1, the residual angle after n steps satisfies $|z_n| \leq \vartheta_{n-1}$.

Proof. Each step is an exact rotation of angle $\varphi_i = \arctan(2^{-i})$, because the rotation matrix factorizes as

$$\begin{pmatrix} \cos \varphi_i & -\sin \varphi_i \\ \sin \varphi_i & \cos \varphi_i \end{pmatrix} = \cos \varphi_i \begin{pmatrix} 1 & -\tan \varphi_i \\ \tan \varphi_i & 1 \end{pmatrix} = \cos \varphi_i \begin{pmatrix} 1 & -d_i 2^{-i} \\ d_i 2^{-i} & 1 \end{pmatrix}$$

(for φ_i chosen exactly as $\arctan(2^{-i})$, $\tan \varphi_i = 2^{-i}$ by construction: this is the central idea of CORDIC). Applying the right matrix to a vector requires only a shift of i bits (multiplication by 2^{-i}) and an addition: it is exactly the update of x_{i+1}, y_{i+1} given above, *except for* the factor $\cos \varphi_i$, which accumulates into a global scale factor $K = \prod_i \cos \varphi_i$ applied (if an unscaled result is needed) with a single final multiplication — or ignored entirely if only the *direction* of the resulting vector is needed. Iterating n times, the residual angle $z_n = \theta - \sum_{i=0}^{n-1} d_i \varphi_i \rightarrow 0$ because each $\arctan(2^{-i})$ halves approximately the residual interval (convergence similar to a binary search on the angle), and the resulting vector is the composition of the n elementary rotations, that is the total rotation of $\sum d_i \varphi_i \approx \theta$, scaled by K . $\square$

Remark 11.5: Why it matters in industrial controllers. CORDIC is still implemented in hardware (FPGA, DSP, and firmware of many axis controllers) precisely because the computation of circular interpolation (Chapter 13) requires evaluating $\cos, \sin$ at servo-cycle frequencies of the order of kHz: an algorithm that uses only shifts and sums is enormously faster, on dedicated silicon, than a floating-point routine with full multiplications.

11.4.1 Complete numerical example: computation of $\cos 30^\circ, \sin 30^\circ$

CORDIC iteration step by step. We compute $\cos(30^\circ)$ and $\sin(30^\circ)$ ($\theta = 30^\circ \approx 0.523599$ rad) starting from $(x_0, y_0) = (1, 0)$, $z_0 = 0.523599$, with 6 iterations:

i	$\arctan(2^{-i})$	d_i	x_i	y_i	z_i
0	0.785398	+1	1.000000	0.000000	0.523599
1	0.463648	−1	1.000000	1.000000	−0.261799
2	0.244979	+1	1.500000	0.500000	0.201849
3	0.124355	−1	1.375000	0.875000	−0.043130
4	0.062419	+1	1.484375	0.703125	0.081225
5	0.031240	+1	1.440430	0.795898	0.018806

After 6 iterations, $(x_6, y_6) \approx (1.415558, 0.840912)$, with scale factor $K_6 = \prod_{i=0}^5 \cos(\arctan 2^{-i}) \approx 0.607352$ (tabulated once and for all in a ROM of the controller, since it does not depend on θ). By Theorem 11.4,

$$\cos 30^\circ \approx K_6 x_6 \approx 0.607352 \times 1.415558 \approx 0.8597,$$

$$\sin 30^\circ \approx K_6 y_6 \approx 0.607352 \times 0.840912 \approx 0.5107,$$

against the exact values $\cos 30^\circ = 0.8660$, $\sin 30^\circ = 0.5$: an error of 0.7% on the cosine and

2.2% on the sine after only 6 iterations (checked), consistent with the geometric convergence (about one extra bit of precision per additional iteration, Theorem 11.4); with 16–20 iterations one typically reaches full IEEE 754 single precision. $\square$

Remark 11.6: Implementation note. A real implementation on an integer-arithmetic (fixed-point) controller would use a scaled representation (*Q* format), avoiding floating-point entirely and making the algorithm executable even on microcontrollers without FPU: both the values $\arctan(2^{-i})$ and the factor K_n are constants pre-computed once during firmware design, not recomputed at every cycle.

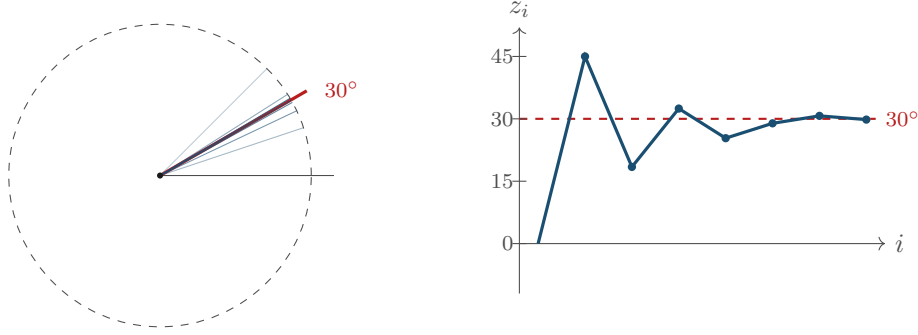


Figure 11.2: CORDIC: trigonometry with additions and shifts only. The target angle (30°) is reached by adding or subtracting, in a fixed order, the elementary angles $\arctan 2^{-i}$: 45° , 26.565° , 14.036° , 7.125° , 3.576° , 1.790° , 0.895° . Left, the successive directions of the vector; right, the sequence z_i of accumulated angles, converging to 30° while oscillating about it. The choice of sign is the only decision: each elementary rotation is realised by $x' = x \mp y 2^{-i}$, $y' = y \pm x 2^{-i}$, i.e. two shifts and two additions, and the accumulated scale factor is a known constant. After n steps the error is at most $\arctan 2^{-n}$: about one bit is gained per iteration.

11.5. Quantization noise and error propagation

Definition 11.6: Quantization noise. If a continuous quantity is rounded to the nearest step Δ , the error $e \in [-\Delta/2, \Delta/2]$ is modelled as a uniform random variable. Its variance (**quantization noise power**) is

$$\sigma_q^2 = \int_{-\Delta/2}^{\Delta/2} \frac{e^2}{\Delta} de = \frac{\Delta^2}{12}.$$

Proof.

$$\int_{-\Delta/2}^{\Delta/2} \frac{e^2}{\Delta} de = \frac{1}{\Delta} \left[\frac{e^3}{3} \right]_{-\Delta/2}^{\Delta/2} = \frac{1}{\Delta} \cdot \frac{2}{3} \left(\frac{\Delta}{2} \right)^3 = \frac{1}{\Delta} \cdot \frac{\Delta^3}{12} = \frac{\Delta^2}{12}.$$

$\square$

Remark 11.7: Error propagation in circular interpolation. In Chapter 13 it is observed that a controller does not solve D, E, F at every servo cycle, but pre-computes the arc parameters and then generates the profile along the arc with an interpolator. Each step of this interpolator introduces a rounding error $\sim r\epsilon$ (Observation 11.1) and, if the target position is encoded in discrete units of the encoder (step Δ , typically $\Delta \sim 10^{-3}$ – 10^{-2} mm for an industrial encoder), also a quantization noise of variance $\Delta^2/12$ per axis. If the errors of

the individual steps are approximately independent, the total variance after N steps grows as $N\sigma_q^2$ (sum of independent variances), hence the typical error (standard deviation) grows as $\sqrt{N}$: it is the same law $O(1/\sqrt{N})$ already seen for the Monte Carlo method (Chapter 1), here in the guise of cumulative “background noise” along a long circular trajectory (many steps N), not as a statistical estimation error.

Warning. This is the technical reason why, on very long arcs or on complete circles at high speed, robot manufacturers (Yaskawa, FANUC, ABB) split the arc into several consecutive instructions (Chapter 13): each instruction resets the interpolation on parameters recomputed from the taught points (*teach points*), limiting the accumulation of numerical error to a single segment instead of a complete turn.

11.6. Circular estimation from noisy data: Kåsa’s method

In industrial practice (dimensional verification of a circular tank with a laser sensor mounted on a rotating arm, calibration of a circular guide from taught points with tolerance, reconstruction of the profile of a tank from a point cloud of scans) one almost never has the exact equation of a circle, but a set of n points (x_i, y_i) measured with noise, which are presumed to lie *approximately* on a circumference of centre (a, b) and unknown radius r . The problem is to estimate a, b, r in the way that best agrees with the data.

Definition 11.7: Algebraic formulation of the circle fitting problem. Given a set of points (x_i, y_i) , $i = 1, \dots, n$, one seeks a, b, r that minimise the sum of squares of the algebraic residuals

$$S(a, b, r) = \sum_{i=1}^n [(x_i - a)^2 + (y_i - b)^2 - r^2]^2.$$

Theorem 11.5: Kåsa’s linearisation (1976). The minimization problem of Definition 11.7 reduces to a system of **linear** normal equations in the auxiliary unknowns $u = 2a$, $v = 2b$, $w = r^2 - a^2 - b^2$, solvable in closed form without iterations.

Proof. Expanding the square $(x_i - a)^2 + (y_i - b)^2 - r^2$ and setting $z_i := x_i^2 + y_i^2$:

$$(x_i - a)^2 + (y_i - b)^2 - r^2 = z_i - 2ax_i - 2by_i + (a^2 + b^2 - r^2) = z_i - ux_i - vy_i - w.$$

The sum of squares $S = \sum_i (z_i - ux_i - vy_i - w)^2$ is then exactly the objective function of an **ordinary linear regression** of the variable z_i on the variables x_i, y_i (with intercept w): a linear least squares problem, whose solution is given by the usual normal equations $A^\top A \beta = A^\top \mathbf{z}$, with A the $n \times 3$ matrix of rows $(x_i, y_i, 1)$ and $\beta = (u, v, w)^\top$. Since $A^\top A$ is invertible for points not all aligned, β is obtained with a single resolution of a 3×3 linear system (no iteration, unlike the exact geometric fitting methods, which require nonlinear optimisation). From β one finally recovers

$$a = \frac{u}{2}, \quad b = \frac{v}{2}, \quad r = \sqrt{w + a^2 + b^2}.$$

□

Remark 11.8: Why “algebraic” and not “geometric”. The minimised residual, $(x_i - a)^2 + (y_i - b)^2 - r^2$, is not the geometric distance of the point from the circle (which would be $\sqrt{(x_i - a)^2 + (y_i - b)^2} - r$), but an algebraic linearized version of it: for points near the circle the two quantities are proportional (up to a factor $\approx 2r$), but for far points or high noise Kåsa’s method introduces a systematic distortion that tends to **slightly underestimate the radius** (small-circle bias), a known defect discussed in the subsequent literature (methods of Pratt 1987 and Taubin 1991, which correct this distortion at the cost of a more elaborate formulation). For industrial applications with moderate measurement noise (few percent of the radius), the simplicity and speed — a single linear solution, no iteration, also suitable for a real-time scan cycle — often make Kåsa’s method the preferred practical choice.

Numerical example: dimensional verification of a circular tank. A laser sensor mounted on an arm that rotates around the nominal centre of a tank acquires 6 points on the internal edge, with small measurement inaccuracies:

$$\begin{array}{ccc} (7.050, 1.500) & (4.485, 5.804) & (-1.240, 5.361) \\ (-2.680, -0.203) & (1.130, -3.434) & (5.792, -1.682) \end{array}$$

(coordinates in metres, origin at the anchoring point of the rotating arm). Constructing the matrix A (rows $x_i, y_i, 1$) and the vector $\mathbf{z} = (x_i^2 + y_i^2)_i$, the normal equations $A^\top A \mathbf{u} = A^\top \mathbf{z}$ give

$$A^\top A \approx \begin{pmatrix} 113.36 & 16.88 & 14.54 \\ 16.88 & 79.34 & 7.35 \\ 14.54 & 7.35 & 6.00 \end{pmatrix}, \quad A^\top \mathbf{z} \approx \begin{pmatrix} 776.12 \\ 444.98 \\ 192.70 \end{pmatrix}.$$

Solving the 3×3 system (checked): $u \approx 3.992976$, $v \approx 3.023911$, $w \approx 18.740296$, from which

$$a = \frac{u}{2} \approx 1.996488 \text{ m}, \quad b = \frac{v}{2} \approx 1.511955 \text{ m}, \quad r = \sqrt{w + a^2 + b^2} \approx 5.001227 \text{ m}.$$

The nominal design value of the tank was centre (2.0; 1.5) m and radius 5.0 m: the estimated deviation (few millimetres) falls within the installation tolerance, confirming that the real tank is conformant to the drawing without having to manually inspect every point of the edge. $\square$

11.7. Finding circles in an image: the Hough transform

Kåsa’s method presupposes that one knows *which* points belong to the circle. In a machine-vision application — presence checking of a hole, diameter measurement on a conveyor, centring of a bushing — this is not given: one has an image with thousands of edge points, some belonging to the circle sought, many belonging to something else. What is needed is a method that is **robust to outliers** and that does not require knowing in advance how many circles there are. That method is the **Hough transform**.

11.7.1 The principle: point–cone duality

Definition 11.8: Parameter space and accumulator. A circle in the image plane is determined by three parameters: the centre (a, b) and the radius r . The **parameter space** is therefore $\mathbb{R}^3$, with coordinates (a, b, r) , $r > 0$. The **accumulator** is a discretisation of this space into cells, each with an integer counter initialised to zero.

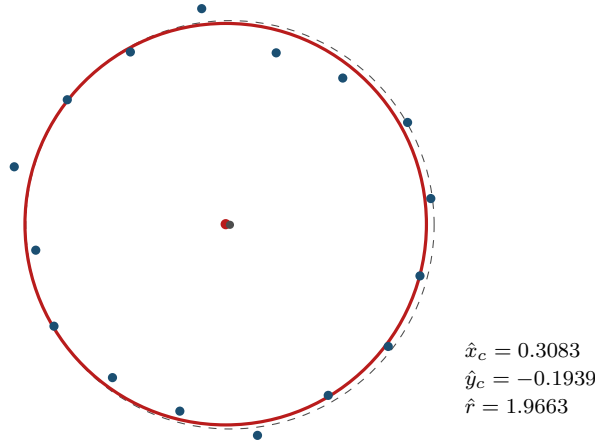


Figure 11.3: Estimating a circle from noisy points with Kåsa's method. The sixteen blue points were generated on the grey circle (centre $(0.35, -0.20)$, radius 2) by perturbing the radius with Gaussian noise of standard deviation 0.10. The method minimises $\sum_i (x_i^2 + y_i^2 + Dx_i + Ey_i + F)^2$, which is linear in the unknowns D, E, F and is solved by a 3×3 system; from them $(x_c, y_c) = (-D/2, -E/2)$ and $r = \sqrt{D^2/4 + E^2/4 - F}$. The red circle is the resulting estimate. Its merit is that no iteration is required; its defect is that the minimised quantity is not the geometric distance but the power of the point, so the estimate is biased when the points cover only a short arc.

Theorem 11.6: Image-point / parameter-cone duality. Let $P = (x_0, y_0)$ be a point of the image plane. The set

$$\mathcal{C}(P) = \{(a, b, r) \in \mathbb{R}^3 : r > 0, (x_0 - a)^2 + (y_0 - b)^2 = r^2\}$$

of parameters of the circles through P is the upper nappe of a **right circular cone** with vertex $(x_0, y_0, 0)$, axis parallel to the r -axis and half-angle 45° .

Moreover, n points $P_1, \dots, P_n$ all lie on one circle of parameters (a^*, b^*, r^*) **if and only if**

$$(a^*, b^*, r^*) \in \bigcap_{i=1}^n \mathcal{C}(P_i).$$

Proof. Step 1 — it is a cone. The equation $(x_0 - a)^2 + (y_0 - b)^2 = r^2$ rewrites, setting $u = a - x_0$ and $v = b - y_0$, as

$$u^2 + v^2 = r^2, \quad r > 0,$$

i.e. $r = \sqrt{u^2 + v^2}$: for each fixed $r = \rho$, the section is the circle $u^2 + v^2 = \rho^2$ of radius ρ in the horizontal plane at height ρ . The radius of the section therefore grows *linearly* with height, with coefficient 1: this is the definition of a right circular cone with vertex $(u, v, r) = (0, 0, 0)$, i.e. $(a, b, r) = (x_0, y_0, 0)$, and half-angle $\arctan(1) = 45^\circ$.

Step 2 — the equivalence. A chain of immediate equivalences:

$$\begin{aligned} P_i \text{ lies on the circle } (a^*, b^*, r^*) &\iff (x_i - a^*)^2 + (y_i - b^*)^2 = (r^*)^2 && \text{(canonical equation, Def. 7.1)} \\ &\iff (a^*, b^*, r^*) \in \mathcal{C}(P_i) && \text{(definition of } \mathcal{C} \text{).} \end{aligned}$$

Requiring this for every $i = 1, \dots, n$ amounts to belonging to the intersection of the corresponding cones. $\square$

Remark 11.9: Why this is an algorithm, and why it is robust. Theorem 11.6 converts a search problem into an intersection problem: one votes, in the accumulator, for every cell crossed by the cone of each edge point; the cells with maximal count are the circles present in the image.

Two properties, both direct consequences of the statement:

- **Robustness.** An edge point that does not belong to the circle votes along its own cone, but the cones of uncorrelated points have no common intersection: their votes scatter without forming a maximum. An outlier does not shift the estimate, it merely dilutes it — unlike least squares (Kåsa’s method, Theorem 11.5), where a single distant point can move the centre appreciably, because its residual enters squared.
- **Multiplicity.** If the image contains two circles, two distinct local maxima form. No least-squares fitting method can do the same without a preliminary segmentation.

The price is cost: the accumulator is three-dimensional.

11.7.2 The gradient variant: from a cone to a line

Theorem 11.7: Reducing the vote by means of the gradient. Let P be an edge point at which the image gradient direction is known, and let $\hat{\mathbf{n}}$ be the unit normal to the edge at P (i.e. the unit gradient). If a circle of centre C and radius r is tangent to the edge at P , then

$$C = P \pm r \hat{\mathbf{n}}.$$

Consequently the set of admissible parameters is no longer the (two-dimensional) cone $\mathcal{C}(P)$, but the union of two **lines** in $\mathbb{R}^3$.

Proof. By Theorem 3.8 (Chapter 3), the radius drawn to the point of tangency is **perpendicular** to the tangent. The object’s edge at P is, by hypothesis, tangent to the circle; hence the segment CP is perpendicular to the edge, i.e. parallel to $\hat{\mathbf{n}}$. Since $|CP| = r$, we get $C - P = \pm r \hat{\mathbf{n}}$, which is the claim. The two signs correspond to the two sides of the edge: the centre lies on the concave side (bright object on dark background or vice versa), and in practice one votes only for the sign consistent with the expected polarity.

Structure in $\mathbb{R}^3$. As $r > 0$ varies, the point $(a, b, r) = (P + r \hat{\mathbf{n}}, r)$ describes a line in $\mathbb{R}^3$ (linearly parametrised by r), and similarly for the minus sign. $\square$

Remark 11.10: The cost saving, in figures. Let N be the number of edge points, R the number of radius cells and Θ the number of angular samples used to draw each voting circle.

variant	total votes	typical
classical Hough (cone)	$O(N \cdot R \cdot \Theta)$	$10^4 \cdot 200 \cdot 360 \approx 7.2 \times 10^8$
gradient Hough (line)	$O(N \cdot R)$	$10^4 \cdot 200 = 2 \times 10^6$

A factor of $\Theta \approx 360$: the difference between offline processing and camera-rate processing. This is the variant implemented by OpenCV under the name `HOUGH_GRADIENT`.

The price, stated. Gradient estimation is noisy on low-contrast edges and at sharp corners; an error of δ radians in the direction of $\hat{\mathbf{n}}$ displaces the centre vote by about $r\delta$. For $r = 200$ px and $\delta = 2^\circ \approx 0.035$ rad the error is ≈ 7 px: not negligible. This is why industrial vision controllers typically use gradient Hough for coarse localisation, and then refine with a least-squares fit (Kåsa, Pratt or Taubin) on the points selected by the Hough maximum alone — obtaining the best of both: outlier robustness from the first, sub-pixel accuracy from the second.

Example 11.1: A Hough accumulator computed by hand. We wish to recover a circle from four edge points, knowing a priori that the radius is **exactly** $r = 5$ (the accumulator reduces to two dimensions, a and b). The points are

$$P_1 = (5, 0), \quad P_2 = (0, 5), \quad P_3 = (-5, 0), \quad P_4 = (8, 9),$$

where P_4 is an **outlier** (noise, or the edge of another object).

Step 1 — what each point votes for. With r fixed, the cone of Theorem 11.6 reduces to its section at height $r = 5$: the circle of centre P_i and radius 5 in the (a, b) plane. Each point therefore votes for all cells on a circle.

Step 2 — sampling the votes on the integer grid. Restricting to integer cells, the circle of centre $(5, 0)$ and radius 5 contains, among others, $(0, 0)$, $(10, 0)$, $(5, 5)$, $(5, -5)$, $(8, 4)$, $(2, -4)$, ... (the points satisfying $(a - 5)^2 + b^2 = 25$; one recognises the Pythagorean triples $(3, 4, 5)$ of Chapter 10).

Step 3 — the cell $(0, 0)$.

$$\begin{aligned} P_1 : & (0 - 5)^2 + 0^2 = 25 = 5^2 && \text{votes } \checkmark \\ P_2 : & 0^2 + (0 - 5)^2 = 25 = 5^2 && \text{votes } \checkmark \\ P_3 : & (0 + 5)^2 + 0^2 = 25 = 5^2 && \text{votes } \checkmark \\ P_4 : & 64 + 81 = 145 \neq 25 && \text{does not vote} \end{aligned}$$

Count of cell $(0, 0)$: **3**.

Step 4 — a cell touched by the outlier, for instance $(8, 4)$:

$$\begin{aligned} P_1 : & 9 + 16 = 25 && \text{votes } \checkmark \\ P_2 : & 64 + 1 = 65 && \text{does not vote} \\ P_3 : & 169 + 16 = 185 && \text{does not vote} \\ P_4 : & 0 + 25 = 25 && \text{votes } \checkmark \end{aligned}$$

Count of cell $(8, 4)$: **2**.

Step 5 — reading the result. The accumulator maximum is at $(0, 0)$ with 3 votes: the detected circle is $x^2 + y^2 = 25$. The outlier P_4 produced only scattered 2-vote coincidences, never a maximum.

Comparison with least squares. Applying Kåsa's method (Theorem 11.5) to all four points, the outlier enters the sum with its squared residual and drags the estimated centre towards P_4 : one obtains (computed) a centre $\approx (1.644; 4.521)$ and a radius ≈ 6.318 , i.e. an estimate **wrong in both parameters** with respect to the true $(0, 0)$, $r = 5$. This is the practical difference between a voting method and a least-squares method.

11.8. Roundness: which reference circle?

A turned part is never a perfect circle. Standard **ISO 1101** prescribes a **roundness** (circularity) tolerance defined as the width of the narrowest annulus containing the measured profile. But computing a deviation requires a reference — and here arises a problem at once metrological and mathematical: **with respect to which circle?**

Definition 11.9: The four reference circles. Given a set of measured points $P_1, \dots, P_n$ on a nominally circular profile, and writing $\rho_i(C) = |P_i - C|$, one defines:

LSC *Least Squares Circle:* the centre C minimising $\sum_i (\rho_i(C) - \bar{\rho})^2$. It is the least-squares fit of Theorem 11.5.

MZC *Minimum Zone Circles:* the pair of concentric circles enclosing all points with

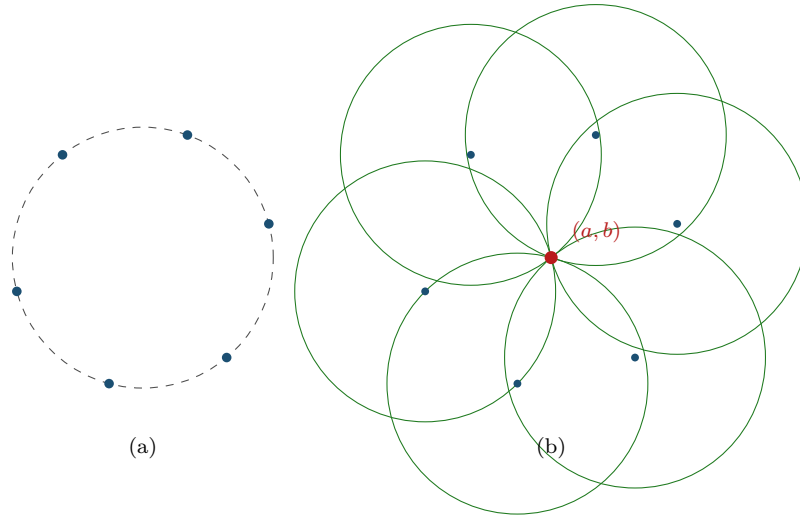


Figure 11.4: The Hough transform for circles of known radius R . In (a) the edge points detected in the image. In (b) the parameter space (a, b) : every image point (x_i, y_i) is compatible with all centres at distance R from it, i.e. it votes for a whole circle of radius R centred at (x_i, y_i) . Since all edge points belong to the same true circle, their voting circles all intersect at the sought centre, which appears as a sharp peak of the accumulator. If the radius is unknown, the accumulator becomes three-dimensional (a, b, R) and each point votes on a cone: the cost grows, but the principle — turning shape recognition into a search for maxima — stays the same.

	the minimum difference of radii.
MCC	<i>Minimum Circumscribed Circle</i> : the circle of minimum radius containing all points.
MIC	<i>Maximum Inscribed Circle</i> : the circle of maximum radius contained within all points.
In each case the roundness error is $\text{RONt}(C) = \max_i \rho_i(C) - \min_i \rho_i(C)$, evaluated at the centre C prescribed by the method.	

Theorem 11.8: *The MZC criterion is the absolute minimum. For every finite set of points and every choice of centre C ,*

$$\text{RONt}(C_{\text{MZC}}) \leq \text{RONt}(C).$$

In particular $\text{RONt}_{\text{MZC}} \leq \text{RONt}_{\text{LSC}}$, $\text{RONt}_{\text{MZC}} \leq \text{RONt}_{\text{MCC}}$ and $\text{RONt}_{\text{MZC}} \leq \text{RONt}_{\text{MIC}}$.

Proof. Consider the function

$$f(C) = \max_i \rho_i(C) - \min_i \rho_i(C), \quad C \in \mathbb{R}^2.$$

By definition, the annulus with centre C , inner radius $\min_i \rho_i(C)$ and outer radius $\max_i \rho_i(C)$ contains all the points, and its width is $f(C)$; conversely, every annulus centred at C containing them all has width $\geq f(C)$. Hence $f(C)$ is the *minimum* width among annuli centred at C . The MZC criterion, by its definition, chooses the centre minimising the width among *all* possible concentric annuli, i.e.

$$C_{\text{MZC}} \in \arg \min_{C \in \mathbb{R}^2} f(C), \quad \text{RONt}_{\text{MZC}} = \min_C f(C).$$

Since a minimum over all of $\mathbb{R}^2$ is $\leq$ the value at any particular point, $\min_{C'} f(C') \leq f(C)$ for every C . Applying this with $C = C_{\text{LSC}}$, $C = C_{\text{MCC}}$ and $C = C_{\text{MIC}}$ yields the three inequalities.

Existence of the minimum. f is continuous, being a max and a min of finitely many continuous functions. To study its behaviour at infinity, set $C = G + t \hat{\mathbf{u}}$ with G fixed, $\hat{\mathbf{u}}$ a unit vector and $t \rightarrow +\infty$: then $\rho_i(C) = t - (P_i - G) \cdot \hat{\mathbf{u}} + O(1/t)$, so

$$f(C) \xrightarrow{t \rightarrow \infty} \max_i (P_i \cdot \hat{\mathbf{u}}) - \min_i (P_i \cdot \hat{\mathbf{u}}) = w(\hat{\mathbf{u}}),$$

the **width** of the point set in the direction $\hat{\mathbf{u}}$. The function w is continuous on the circle S^1 of directions (a composition of continuous dot products with max/min over a finite set), which is compact: by Weierstrass, w attains a minimum $w_{\min} = \min_{\hat{\mathbf{u}}} w(\hat{\mathbf{u}}) > 0$ on it (positive because the points are not collinear).

The limit $t \rightarrow \infty$ written above is so far only *pointwise*, one direction $\hat{\mathbf{u}}$ at a time: to get a single R_0 that works for **all** directions simultaneously we need the remainder $O(1/t)$ to be uniform. But the points $P_1, \dots, P_n$ are **finitely many**: setting $M = \max_i |P_i - G|$, the exact expansion $\rho_i(C) = \sqrt{t^2 + 2t(G - P_i) \cdot \hat{\mathbf{u}} + |G - P_i|^2}$ gives $|\rho_i(C) - (t - (P_i - G) \cdot \hat{\mathbf{u}})| \leq M^2/t$ for $t > M$, **a bound that does not depend on $\hat{\mathbf{u}}$** (it depends only on M , fixed once and for all by the finite point set). Hence $|f(C) - w(\hat{\mathbf{u}})| \leq 2M^2/t$ for every direction $\hat{\mathbf{u}}$ at once: choosing $R_0 > 2M^2/w_{\min}$ gives $f(C) > w(\hat{\mathbf{u}}) - w_{\min}/2 \geq w_{\min}/2$ for every C with $|C - G| > R_0$, whatever its direction.

Hence if $\inf_{\mathbb{R}^2} f < w_{\min}/2$, the infimum is attained on the compact set $\{|C - G| \leq R_0\}$ by Weierstrass' theorem, and the MZC exists. $\square$

Warning. The degenerate case, stated. The condition $\inf f < w_{\min}/2$ of the last step is not automatic: if the points were nearly collinear, the “optimal annulus” would degenerate into a *strip* between two parallel lines, i.e. a circle of infinite radius, and the minimum would not be attained in $\mathbb{R}^2$. For a profile actually measured on a turned part the condition is amply satisfied — the points surround the centre and $w_{\min} \approx 2r$, whereas $\inf f$ is of the order of a micrometre — but it must be stated, because it is what distinguishes the *minimum-width annulus* problem from the *minimum-width strip* problem, with which it shares its algorithms.

Warning. Why, then, is MZC not always used? Because the function $f(C)$ is **not convex**: $\max_i \rho_i$ is, but $-\min_i \rho_i$ in general is not, and the difference may have local minima. There is no closed formula as for the LSC (Theorem 11.5): the MZC is computed with combinatorial algorithms (exchange of supporting points) or metaheuristics. This is why the **LSC remains the shop-floor default** — it has a closed-form solution, a 3×3 system — while the **MZC is the normative criterion** (ISO 12181-1) when the tolerance is tight and the value must be certified. The difference between the two, on nearly circular profiles, is a few percent; on *lobed* profiles — the typical case of a part clamped in a three-jaw chuck, which produces three lobes — it can reach 10–15%, and in that case adopting the LSC means declaring a form error larger than the true one, and scrapping good parts.

Example 11.2: Roundness: the reference changes everything. A probe records four points on the profile of a shaft (dimensions in mm):

$$P_1 = (10; 0), \quad P_2 = (0; 10.02), \quad P_3 = (-10; 0), \quad P_4 = (0; -9.98).$$

Step 1 — naive evaluation, about the nominal centre $(0, 0)$.

$$\rho_1 = 10, \quad \rho_2 = 10.02, \quad \rho_3 = 10, \quad \rho_4 = 9.98,$$

$$\text{RONt}(0, 0) = 10.02 - 9.98 = \mathbf{0.040 \text{ mm}}.$$

Step 2 — the least-squares circle (Kåsa). Build $z_i = x_i^2 + y_i^2$: $z_1 = 100$, $z_2 = 100.4004$, $z_3 = 100$, $z_4 = 99.6004$. The required sums are

$$\begin{aligned} \sum x_i &= 0, & \sum y_i &= 0.04, & \sum x_i^2 &= 200, & \sum y_i^2 &= 200.0008, & \sum x_i y_i &= 0, & n &= 4, \\ \sum x_i z_i &= 0, & \sum y_i z_i &= 12.000016, & \sum z_i &= 400.0008. \end{aligned}$$

The normal system $A^\top A(u, v, w)^\top = A^\top \mathbf{z}$ becomes

$$\begin{cases} 200u = 0 \\ 200.0008v + 0.04w = 12.000016 \\ 0.04v + 4w = 400.0008 \end{cases}$$

From the first, $u = 0$. From the third, $w = 100.0002 - 0.01v$; substituting into the second:

$$200.0008v + 0.04(100.0002 - 0.01v) = 12.000016 \Rightarrow 200.0004v = 8.000008 \Rightarrow v = 0.0400000.$$

Hence $w = 100.0002 - 0.0004 = 99.9998$, and

$$a = \frac{u}{2} = 0, \quad b = \frac{v}{2} = \mathbf{0.020}, \quad r = \sqrt{w + a^2 + b^2} = \sqrt{100.0002} = 10.00001.$$

Step 3 — roundness about the LSC centre (0; 0.020).

$$\rho_1 = \sqrt{10^2 + 0.02^2} = 10.0000200, \quad \rho_2 = |10.02 - 0.02| = 10.0000000,$$

$$\rho_3 = \sqrt{10^2 + 0.02^2} = 10.0000200, \quad \rho_4 = |-9.98 - 0.02| = 10.0000000.$$

$$\text{RONt(LSC)} = 10.00002 - 10.00000 = \mathbf{0.00002} \text{ mm} = 0.02 \text{ } \mu\text{m}.$$

Step 4 — reading. The same part, from the same four data, comes out with a form error of 0.040 mm or 0.00002 mm: a factor of **2000**. There is no contradiction — the first number also measures the *eccentricity* of the setup, the second only the *form*. ISO 1101 roundness is by definition independent of position, so the correct value is the second.

Step 5 — and the MZC? By Theorem 11.8 it must be $\text{RONt}_{\text{MZC}} \leq 0.00002$ mm. In this particular case the two coincide, because the LSC centre (0; 0.020) already makes maximum and minimum radius symmetric; on a lobed profile, by contrast, the MZC would be strictly smaller.

Best practice. The operational moral: before declaring a part out of roundness tolerance, check *which* reference the instrument is using. A centring error of the part on the probe shows up as a form error only if the software evaluates with respect to a fixed centre rather than an optimised one.

Remark 11.11: The same metric on a travelled circle, not a measured one. The four reference circles of Definition 11.9 arise to evaluate a measured profile (a turned part). But the same quantity $\text{RONt}(C)$ also serves an opposite diagnosis: not “how round is a stationary part”, but “how round is the circle that a numerically controlled machine travels”. This is the **ballbar** test (ISO 230-4 standard), described in Chapter 13, §13.7.4: a telescoping transducer continuously measures the radial deviation while the axes execute a full G02/G03, and the data obtained — a series of radii ρ_i about a nominal centre — are read exactly with RONt and with the LSC criterion of this section. The difference is only the origin of the error: here, form defects of the part; there, tracking lag of the axes (Theorem 13.10) — but

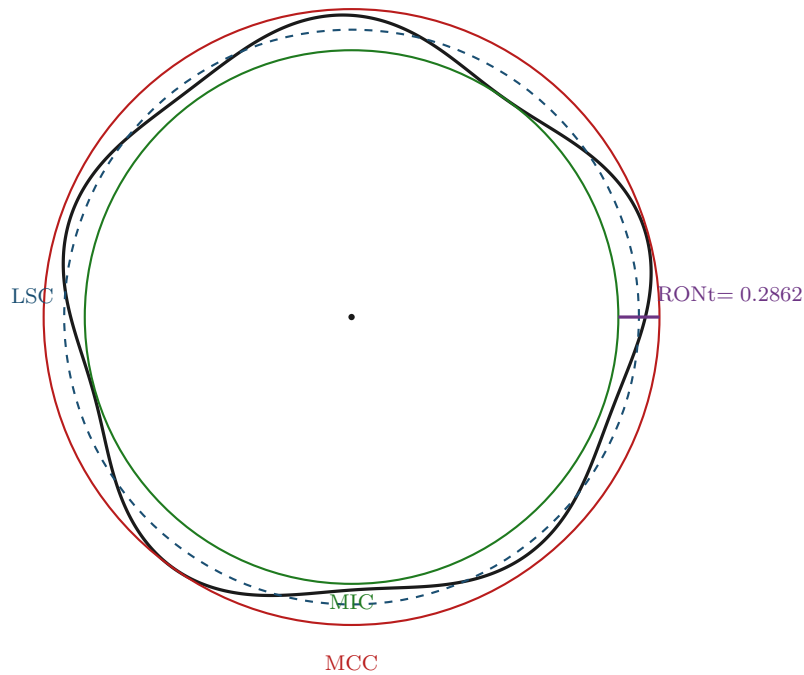


Figure 11.5: Which reference circle? The measured profile (black) is $r(\theta) = 2 + 0.10 \sin 5\theta + 0.045 \cos 3\theta$. The least-squares circle LSC (dashed blue) here has radius 2 and centre at the origin, because the profile has no first-order harmonic — and it is precisely the first-order harmonic that represents an eccentricity, not a form defect. The maximum inscribed circle MIC (green) has radius $r_{\min} = 1.8569$, the minimum circumscribed circle MCC (red) has radius $r_{\max} = 2.1431$, and their difference 0.2862 is the roundness error RONT referred to this centre. The choice of reference is not academic: MIC and MCC correspond to the mating of a shaft and a hole, LSC to the statistically most robust estimate.

the evaluation arithmetic is the same reference circle, seen twice.

11.9. Computing the MCC: Welzl's algorithm

Definition 11.9 introduces the **MCC** (*Minimum Circumscribed Circle*), the smallest-radius circle that contains all measured points — but does not say how to compute it. Unlike the LSC (closed form, Theorem 11.5), the MCC requires an algorithm. The problem is old in discrete geometry (also known as the *smallest enclosing circle* problem) and its most elegant solution is due to Emo Welzl: E. Welzl, *Smallest enclosing disks (balls and ellipsoids)*, in *New Results and New Trends in Computer Science*, Lecture Notes in Computer Science vol. 555, Springer, 1991, pp. 359–370.

Definition 11.10: Minimum circumscribed circle. Given a finite set of points S in the plane, the **minimum circumscribed circle** $\text{MCC}(S)$ is the smallest-radius circle containing all points of S (in its interior or on its boundary).

Lemma 11.2: Uniqueness of the minimum circumscribed circle. For every finite nonempty S , $\text{MCC}(S)$ exists and is unique.

Proof. Existence. The set of radii r for which a circle of radius r containing S exists is nonempty (a large enough circle centred at any point of S works) and bounded below by 0; let r^* be its infimum. For a sequence of circles of radius $\rightarrow r^*$ containing S , the centres stay in a

bounded set (all within distance $r^* + 1$, say, of a fixed point of S , for r close enough to r^*); by compactness (Bolzano–Weierstrass) a subsequence of the centres converges to a limit centre O^* , and by continuity the circle (O^*, r^*) still contains all of S (a finite set, so the condition “contains S ” passes to the limit with no uniformity issues). Hence the minimum r^* is attained. *Uniqueness.* Let (O_1, r) and (O_2, r) be two circles of the **same** minimal radius r containing S , with $O_1 \neq O_2$. Let $M = (O_1 + O_2)/2$ be the midpoint and $e = |O_1 O_2|/2 > 0$. For every $P \in S$, setting $u = P - M$ and $v = (O_1 - O_2)/2$ (so that $P - O_1 = u - v$, $P - O_2 = u + v$),

$$|P - O_1|^2 + |P - O_2|^2 = |u - v|^2 + |u + v|^2 = 2|u|^2 + 2|v|^2 = 2|P - M|^2 + 2e^2$$

(parallelogram identity). Since $|P - O_1| \leq r$ and $|P - O_2| \leq r$,

$$2|P - M|^2 = |P - O_1|^2 + |P - O_2|^2 - 2e^2 \leq 2r^2 - 2e^2 \implies |P - M| \leq \sqrt{r^2 - e^2} < r$$

(the last inequality strict because $e > 0$). So the circle of centre M and radius $\sqrt{r^2 - e^2} < r$ also contains S : but this contradicts the minimality of r . Hence $O_1 = O_2$. $\square$

Lemma 11.3: Characterisation of the minimum circle. Let C be a circle of centre O and radius r containing the finite set A , and let $B = A \cap \partial C$ be the points of A on the boundary of C . Then $C = \text{MCC}(A)$ **if and only if** O belongs to the convex hull of B .

Proof. ($\Leftarrow$, the direction used for the algorithm.) Suppose $O \in \text{conv}(B)$ but, for contradiction, that C is not minimal, i.e. there is $D' = \text{MCC}(A)$ of centre O' and radius $r' < r$ (the only case to rule out, by Lemma 11.2, since C is already a valid candidate for A , so $r' \leq r$). Write $O = \sum_i \lambda_i Q_i$ (a convex combination, $Q_i \in B$, $\lambda_i \geq 0$, $\sum \lambda_i = 1$). By the centroid decomposition identity (checked by direct substitution, using $\sum_i \lambda_i (Q_i - O) = \sum_i \lambda_i Q_i - O = 0$):

$$\sum_i \lambda_i |Q_i - X|^2 = \sum_i \lambda_i |Q_i - O|^2 + |O - X|^2 \quad \text{for every } X.$$

With $X = O'$: since $|Q_i - O| = r$ for every $Q_i \in B \subset \partial C$,

$$\sum_i \lambda_i |Q_i - O'|^2 = r^2 + |O - O'|^2 \geq r^2.$$

But $Q_i \in A$ is contained in D' , so $|Q_i - O'| \leq r'$ for every i , hence $\sum_i \lambda_i |Q_i - O'|^2 \leq r'^2$. Combining, $r^2 \leq r'^2$, i.e. $r \leq r'$: contradicting $r' < r$. Hence $C = \text{MCC}(A)$.

($\Rightarrow$) If $O \notin \text{conv}(B)$, let O' be the point of $\text{conv}(B)$ closest to O (it exists by compactness, since $\text{conv}(B)$ is the convex hull of a finite set, hence a closed and bounded polygon). For every $Q \in B$ and $t \in [0, 1]$, the point $O' + t(Q - O')$ belongs to $\text{conv}(B)$ by convexity, so $|O - (O' + t(Q - O'))|^2 \geq |O - O'|^2$ for every $t \in [0, 1]$, by minimality of O' . The left-hand side, as a function of t , equals $|O - O'|^2 - 2t(O - O') \cdot (Q - O') + t^2|Q - O'|^2$: being $\geq |O - O'|^2$ for every arbitrarily small $t \in [0, 1]$, the coefficient of the linear term cannot be negative, that is $(O - O') \cdot (Q - O') \leq 0$, i.e. $(O' - O) \cdot (Q - O') \geq 0$. Adding $(O' - O) \cdot (O' - O) = |O' - O|^2$ to both sides,

$$(O' - O) \cdot (Q - O) \geq |O' - O|^2 > 0.$$

Setting $v = (O' - O)/|O' - O|$ gives $(Q - O) \cdot v \geq |O' - O| > 0$ for every $Q \in B$: all points of B therefore lie in an open half-circle (the one identified by direction v) — the fact is proved here in full, with the standard nearest-point argument on a convex set, with no need to import Carathéodory’s theorem (which is instead used further below, for Corollary 11.1, where it is cited correctly). Shifting the centre by a small amount ε in direction v : for $Q \in B$, $|O + \varepsilon v - Q|^2 = r^2 - 2\varepsilon(Q - O) \cdot v + \varepsilon^2|v|^2 < r^2$ for small ε (the negative linear term dominates); for the points of $A \setminus B$, which are at distance $< r$ from O , the inequality still holds by continuity (finite set, no uniformity issue). The new circle is thus smaller and still contains A : C was not minimal. $\square$

Corollary 11.1: *At most three points determine the MCC. For every finite set A , $\text{MCC}(A) = \text{MCC}(A')$ for a suitable $A' \subseteq A$ with $|A'| \leq 3$.*

Proof. Let $D = \text{MCC}(A)$, of centre O , and $B = A \cap \partial D$. By direction $(\Rightarrow)$ of Lemma 11.3, $O \in \text{conv}(B)$; by the Carathéodory theorem cited in that same proof, there is $A' \subseteq B \subseteq A$ with $|A'| \leq 3$ and $O \in \text{conv}(A')$. Applying direction $(\Leftarrow)$ of the same lemma to the circle D and the set A' (which D contains, since $A' \subseteq A$, and whose points all lie on ∂D , since $A' \subseteq B \subseteq \partial D$): $D = \text{MCC}(A')$. $\square$

Theorem 11.9: Key lemma for the incremental update. *Let $D = \text{MCC}(A)$ and p a point with $p \notin D$ (that is, $|p - O| > r$, O, r the centre and radius of D). Then p belongs to the boundary of $\text{MCC}(A \cup \{p\})$.*

Proof. Let $C' = \text{MCC}(A \cup \{p\})$, of centre O' and radius r' . Since C' contains $A \cup \{p\} \supseteq A$, C' is a valid candidate for A , so $r' \geq r$ (by minimality of D with respect to A). Suppose for contradiction that p is **not** on the boundary of C' , i.e. p is strictly interior to C' (it must still lie inside it, being in $A \cup \{p\}$). Then the points of $A \cup \{p\}$ on the boundary of C' are all in A : setting $B' = (A \cup \{p\}) \cap \partial C' = A \cap \partial C'$, by Lemma 11.3 applied to C' (which is $\text{MCC}(A \cup \{p\})$), so by direction $\Rightarrow$ of the lemma, $O' \in \text{conv}(B')$) and to $A \supseteq B'$ (which C' contains): by direction $\Leftarrow$ of the same lemma, $C' = \text{MCC}(A) = D$. But then $r' = r$ and D should contain p (because C' contains it, and by hypothesis $p \in A \cup \{p\} \subseteq C' = D$) — contradicting the hypothesis $p \notin D$. Hence $p \in \partial C'$. $\square$

Remark 11.12: Welzl's algorithm. *The two preceding facts — at most three points suffice (Corollary 11.1) and an outside point always ends up on the boundary of the new circle (Theorem 11.9) — give a recursive algorithm. With P the points still to examine and R the points already forced onto the boundary ($|R| \leq 3$):*

1. if $P = \emptyset$ or $|R| = 3$: return the circle determined directly by R (no point if $R = \emptyset$; the point itself, radius 0, if $|R| = 1$; the circle with the two points as diameter, if $|R| = 2$; the circumcircle of the three points, if $|R| = 3$ and they are not collinear);
2. otherwise, extract a point p from P and recursively compute $D = \text{MCC}(P \setminus \{p\} \cup R)$ (same algorithm, one fewer point in P);
3. if $p \in D$: return D (by Theorem 11.9, adding a point already inside does not change the minimum circle: D already contains everything);
4. otherwise (Theorem 11.9: p must lie on the boundary of the new circle): recursively call the algorithm on $P \setminus \{p\}$ with $R \cup \{p\}$.

Initial call: $\text{MCC}(S) = \text{Welzl}(S, \emptyset)$.

Theorem 11.10: Correctness of Welzl's algorithm. *For every finite set P and every R with $|R| \leq 3$ such that $\text{MCC}(P \cup R)$ exists with R contained in its boundary, the algorithm of Observation 11.12 returns $\text{MCC}(P \cup R)$.*

Proof. By strong induction on $|P|$. If $|P| = 0$ or $|R| = 3$, step 1 returns by construction the circle determined by R , which is $\text{MCC}(R) = \text{MCC}(P \cup R)$ (with $P = \emptyset$ in the first case; if $|R| = 3$ instead, using that, by the inductive hypothesis of earlier steps of the recursion, all previously examined points are interior to the current circle, so adding them does not change it — see case 3 below).

Otherwise, $D = \text{MCC}((P \setminus \{p\}) \cup R)$ by the inductive hypothesis (one fewer point in P).

Case $p \in D$. Then D already contains $(P \setminus \{p\}) \cup R \cup \{p\} = P \cup R$, so D is a valid candidate for $P \cup R$; but $\text{MCC}(P \cup R) \geq \text{MCC}((P \setminus \{p\}) \cup R)$ in radius (a superset requires an at-least-

as-large circle), i.e. $\text{MCC}(P \cup R)$ has radius $\geq$ the radius of D ; but D itself is a candidate for $P \cup R$ with that radius, so by minimality $\text{MCC}(P \cup R) = D$.

Case $p \notin D$. By Theorem 11.9 (applied with $A = (P \setminus \{p\}) \cup R$), p lies on the boundary of $\text{MCC}(A \cup \{p\}) = \text{MCC}(P \cup R)$. Hence $R \cup \{p\}$ is still contained in the boundary of $\text{MCC}(P \cup R)$, and by the inductive hypothesis (applied to the recursive call with $P \setminus \{p\}$ and $R \cup \{p\}$, which satisfies the hypothesis of the statement), the algorithm returns exactly $\text{MCC}((P \setminus \{p\}) \cup R \cup \{p\}) = \text{MCC}(P \cup R)$. $\square$

Remark 11.13: Why random order (imported). The algorithm as described is correct for **any** order of extracting points from P : the proof above nowhere uses that the order is random. Extracting the points in **uniformly random** order, however, Welzl (1991) proves that the expected running time is $O(n)$, linear in the number of points (versus a worst case of $O(n^3)$ with an unlucky order, recomputing from scratch at every outside point) — a result of probabilistic analysis of algorithms beyond the tools of this chapter, which we declare **imported**: see Welzl (1991), or M. de Berg, O. Cheong, M. van Kreveld, M. Overmars, *Computational Geometry: Algorithms and Applications*, 3rd ed., Springer, 2008, Ch. 4, for the backwards analysis argument that proves it.

Example 11.3: The algorithm at work on four points. Let $P_1 = (0, 0)$, $P_2 = (4, 0)$, $P_3 = (2, 1)$, $P_4 = (-1, 3)$, examined in this order.

After P_1 : trivial circle, centre P_1 , radius 0.

After P_2 : P_2 is outside (distance $4 > 0$); new circle with $R = \{P_1, P_2\}$: diameter P_1P_2 , centre $(2, 0)$, radius 2.

After $P_3 = (2, 1)$: distance from centre $(2, 0)$ is $1 < 2$: P_3 is **inside**, the circle does not change.

After $P_4 = (-1, 3)$: distance from centre $(2, 0)$ is $\sqrt{3^2 + 3^2} = \sqrt{18} = 4.2426 > 2$: P_4 is **outside**. By Theorem 11.9, P_4 ends up on the boundary of the new minimum circle of $\{P_1, P_2, P_3, P_4\}$; the recursion (detailed in script `cap10_welzl.py`) examines P_1, P_2, P_3 constraining P_4 to the boundary, and finds that the circle with **diameter** P_2P_4 already contains P_1 and P_3 :

$$\text{centre} = \frac{P_2 + P_4}{2} = (1.5, 1.5), \quad \text{radius} = \frac{|P_2 - P_4|}{2} = \frac{\sqrt{5^2 + 3^2}}{2} = \frac{\sqrt{34}}{2} = 2.915476.$$

Exhaustive verification (all pairs and triples of points, script `cap10_welzl.py`): no circle smaller than radius 2.915476 contains all four points; the absolute minimum is attained exactly by the pair $\{P_2, P_4\}$ as diameter, with P_1 at distance $2.121 < 2.915$ and P_3 at distance $0.707 < 2.915$ from the centre, both confirmed interior. $\checkmark$

11.10. Newton–Raphson: intersections with no closed form

Chapter 7 solves in closed form the intersection of a circle with a line or with another circle: these are systems reducing to a quadratic equation. But already the intersection of a circle with a *general curve* (a cam profile, a spline, the boundary of a safety zone) admits no closed form. The standard method is **Newton–Raphson**.

Definition 11.11: Newton–Raphson iteration. Given $f : \mathbb{R} \rightarrow \mathbb{R}$ differentiable and an initial estimate x_0 , iterate

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Geometrically: x_{n+1} is the intersection with the x -axis of the **tangent** to the graph of f at the point $(x_n, f(x_n))$.

Theorem 11.11: Quadratic convergence. Let f be of class C^2 in a neighbourhood of a simple root x^* (that is, $f(x^*) = 0$ and $f'(x^*) \neq 0$). Then there is a neighbourhood of x^* such that, starting from an x_0 in it, the iteration converges and the error $e_n = x_n - x^*$ satisfies

$$e_{n+1} = \frac{f''(\xi_n)}{2f'(x_n)} e_n^2 \quad \text{for some } \xi_n \text{ between } x_n \text{ and } x^*.$$

In particular $|e_{n+1}| \leq M|e_n|^2$: the number of exact digits **doubles** at each iteration.

Proof. Step 1 — Taylor expansion with Lagrange remainder. Since f is of class C^2 , expanding about x_n and evaluating at x^* :

$$0 = f(x^*) = f(x_n) + f'(x_n)(x^* - x_n) + \frac{f''(\xi_n)}{2}(x^* - x_n)^2$$

for a suitable ξ_n between x_n and x^* .

Step 2 — divide by $f'(x_n)$. In the neighbourhood considered $f'(x_n) \neq 0$ (by continuity of f' and $f'(x^*) \neq 0$); dividing and setting $e_n = x_n - x^*$, so that $x^* - x_n = -e_n$:

$$0 = \frac{f(x_n)}{f'(x_n)} - e_n + \frac{f''(\xi_n)}{2f'(x_n)} e_n^2.$$

Step 3 — recognise the iteration. By definition

$$e_{n+1} = x_{n+1} - x^* = \left(x_n - \frac{f(x_n)}{f'(x_n)}\right) - x^* = e_n - \frac{f(x_n)}{f'(x_n)}.$$

Comparing with Step 2, the quantity $e_n - f(x_n)/f'(x_n)$ equals exactly $\frac{f''(\xi_n)}{2f'(x_n)} e_n^2$, which is the claim.

Step 4 — local convergence. Let $M = \sup |f''|/(2 \inf |f'|)$ over the neighbourhood, finite by continuity. If $|e_0| < \min(1/M, \delta)$, then $|e_1| \leq M|e_0|^2 = (M|e_0|)|e_0| < |e_0|$: the error decreases, stays in the neighbourhood, and by induction $M|e_n| \leq (M|e_0|)^{2^n} \rightarrow 0$. $\square$

Remark 11.14: The vector case: intersecting two curves. To intersect the circle $g_1(x, y) = (x - a)^2 + (y - b)^2 - r^2 = 0$ with a curve $g_2(x, y) = 0$, one applies the vector version: setting $\mathbf{g} = (g_1, g_2)^\top$ and $\mathbf{p}_n = (x_n, y_n)^\top$,

$$\mathbf{p}_{n+1} = \mathbf{p}_n - J(\mathbf{p}_n)^{-1} \mathbf{g}(\mathbf{p}_n), \quad J = \begin{pmatrix} \partial g_1 / \partial x & \partial g_1 / \partial y \\ \partial g_2 / \partial x & \partial g_2 / \partial y \end{pmatrix}.$$

For the circle, the first row of the Jacobian is $(2(x - a), 2(y - b))$: twice the radial vector, i.e. the **normal** direction to the circle at that point. The method therefore loses effectiveness (ill-conditioned Jacobian) precisely when the two curves meet tangentially, because then the two rows of J become parallel and $\det J \rightarrow 0$. This is the same phenomenon as the kinematic singularities met in Chapter 13.

Worked example: tank radius from a volume constraint. A cylindrical tank of usable height $h = 3.20$ m must hold $V = 250$ m³ and the designer wants the radius to the

millimetre. The equation is $f(r) = \pi r^2 h - V = 0$, with $f'(r) = 2\pi r h$. Starting from $r_0 = 5$ m:

n	r_n (m)	$f(r_n)$ (m ³)	r_{n+1} (m)
0	5.000000	+1.327412	4.986796
1	4.986796	+0.001753	4.986779
2	4.986779	$+3.1 \times 10^{-9}$	4.986779

Two iterations suffice: $r \approx 4.986779$ m. The exact value is $r = \sqrt{V/(\pi h)} = \sqrt{250/(3.20\pi)} = 4.986779\dots$ m (checked). Note the doubling of exact digits predicted by Theorem 11.11: r_0 has one exact digit, r_1 has five, r_2 has all those shown. $\square$

11.11. From profile to G-code: CAM generation of the arc

The next chapter describes how a controller *executes* a circular interpolation instruction. Here we are concerned with the upstream step: how a CAM system, starting from a geometric profile, *generates* that instruction. There are three computations, all reducible to results already proved.

Step 1 — the centre of the arc. If the profile is given by three points P_0, P_m, P_1 (start, intermediate, end), the centre is obtained by solving the linear system in D, E, F of Theorem 7.2 (Chapter 7) and setting $(x_c, y_c) = (-D/2, -E/2)$. If the profile is given as tangent to two lines, the centre is the intersection of the two parallels at distance r (Chapter 7). In both cases the formula is closed: no iteration. $\square$

Step 2 — the I, J offsets. Standard ISO 6983 (and the FANUC, Siemens and Haas implementations) expresses the centre **incrementally** with respect to the start point:

$$I = x_c - x_0, \quad J = y_c - y_0 \quad (XY \text{ plane; } K \text{ for the } Z \text{ axis}).$$

This choice is not arbitrary: it makes the instruction invariant under translation of the part origin, so that the same block works with any active tool offset. $\square$

Proposition 11.2: Determining the sense (G02 versus G03). Let $\mathbf{u} = P_0 - C$ and $\mathbf{v} = P_1 - C$ be the radial vectors of arc start and end, and let

$$\kappa = u_x v_y - u_y v_x$$

be the z -component of their cross product. Then the minor arc from P_0 to P_1 is traversed **anticlockwise** (G03) if $\kappa > 0$, and **clockwise** (G02) if $\kappa < 0$.

Proof. Write $\mathbf{u} = r(\cos \alpha, \sin \alpha)$ and $\mathbf{v} = r(\cos \beta, \sin \beta)$ (both of modulus r , being radii). Then, by the sine subtraction formula (Theorem 8.4),

$$\kappa = r^2(\cos \alpha \sin \beta - \sin \alpha \cos \beta) = r^2 \sin(\beta - \alpha).$$

Let $\Delta = \beta - \alpha$ reduced to the interval $(-\pi, \pi]$: this is the angle of the *minor* arc traversed from P_0 to P_1 , positive for the anticlockwise sense (orientation convention of the plane, Chapter 7). Since $\sin$ is positive on $(0, \pi)$ and negative on $(-\pi, 0)$, and $r^2 > 0$:

$$\kappa > 0 \iff \sin \Delta > 0 \iff \Delta \in (0, \pi) \iff \text{anticlockwise,}$$

and symmetrically for $\kappa < 0$. The case $\kappa = 0$ corresponds to $\Delta \in \{0, \pi\}$: the two points coincide, or they are diametrically opposite and the sense is not determined by the endpoints alone — precisely the ambiguous case discussed below. $\square$

Warning. The two cases in which the R form fails. Many controllers also accept the syntax with the R (radius) word instead of I/J. That form is ambiguous in two situations:

1. **Arc greater than 180° .** Given P_0 , P_1 and r , there are *two* possible centres (the two intersection points of the circles of radius r centred at P_0 and P_1 — see the case of two secant circles, Chapter 7), hence four arcs. The ISO convention resolves the ambiguity by the **sign of R**: positive R selects the arc $\leq 180^\circ$, negative R the one $> 180^\circ$.
2. **Full circle.** If $P_1 = P_0$, the R form determines no arc at all (any circle of radius r through P_0 satisfies the constraints). A full circle must **necessarily** be programmed with I/J, which fix the centre uniquely.

This is why industrial CAM post-processors emit the I/J form by default: it is the only one that is never ambiguous.

Complete example: from profile to G-code block. A profile passes through $P_0 = (10; 0)$, $P_m = (0; 10)$, $P_1 = (-10; 0)$ (mm). We seek the block to emit.

Centre and radius. The three points satisfy $x^2 + y^2 = 100$: the centre is $C = (0; 0)$, the radius $r = 10$ mm (check by Theorem 7.2: substituting the three points into $x^2 + y^2 + Dx + Ey + F = 0$ gives $D = E = 0$, $F = -100$, hence $C = (-D/2, -E/2) = (0, 0)$, $r = \sqrt{D^2/4 + E^2/4 - F} = 10$).

Offsets. $I = x_c - x_0 = 0 - 10 = -10$; $J = y_c - y_0 = 0 - 0 = 0$.

Sense. $\mathbf{u} = P_0 - C = (10, 0)$, $\mathbf{v} = P_1 - C = (-10, 0)$: $\kappa = 10 \cdot 0 - 0 \cdot (-10) = 0$. The ambiguous case (diametrically opposite points)! One must use the intermediate point: with $\mathbf{u} = (10, 0)$ and $\mathbf{w} = P_m - C = (0, 10)$, $\kappa' = 10 \cdot 10 - 0 \cdot 0 = 100 > 0$: sense **anticlockwise**, hence G03.

Block emitted.

G17 G90 G03 X-10.0 Y0.0 I-10.0 J0.0 F250

(G17 = XY plane; G90 = absolute coordinates for X, Y; F250 = feed rate 250 mm/min.)

Moral. The case $\kappa = 0$ is not pathological: it arises every time a semicircle is programmed, that is, constantly. A post-processor that does not use a third point to disambiguate produces paths traversing the semicircle the wrong way round — one of the commonest causes of collision when commissioning a new program. $\square$

11.12. Problem set

11.12.1 Numerical exercises

Exercise 11.1. Compute the first two steps of the midpoint algorithm for a circle of radius $r = 5$ (integer), starting from $(x_0, y_0) = (0, 5)$, $p_0 = 1 - r = -4$.

Solution. $p_0 = -4 < 0$: step **E**. $(x_1, y_1) = (1, 5)$, $p_1 = p_0 + 2x_1 + 1 = -4 + 2 + 1 = -1$.

$p_1 = -1 < 0$: step **E**. $(x_2, y_2) = (2, 5)$, $p_2 = p_1 + 2x_2 + 1 = -1 + 4 + 1 = 4$.

(The third step, with $p_2 \geq 0$, would choose **SE**: $(x_3, y_3) = (3, 4)$, consistent with the fact that $3^2 + 4^2 = 25 = 5^2$ is exactly on the circle — not by chance the Pythagorean triple $(3, 4, 5)$ of Chapter 10 also appears here.)

Exercise 11.2. Compute the weight w_1 of the rational quadratic Bézier that exactly represents a 60° arc, and verify Theorem 11.3 at $t = 1/2$.

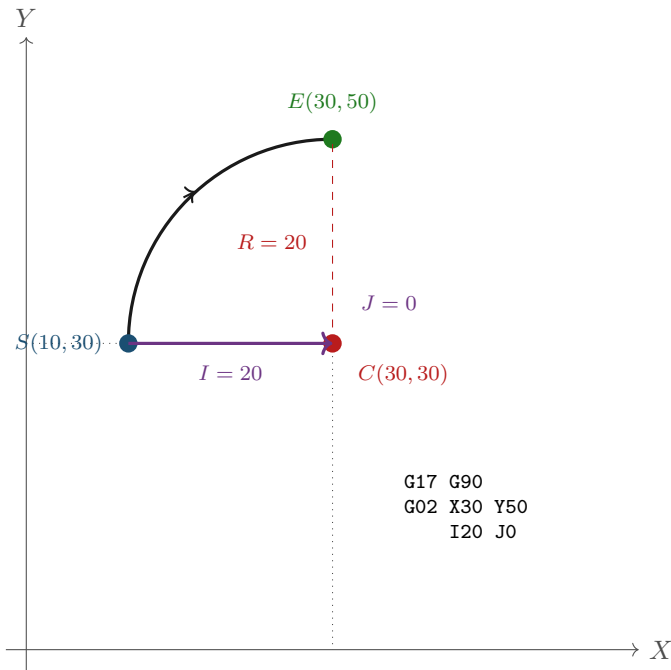


Figure 11.6: Circular interpolation in G-code. The G02 block traverses an arc clockwise from the current point S to the point E declared by X and Y ; the centre is not declared in absolute terms but through the offsets I and J , i.e. the components of the vector $\vec{SC}$ from the start point to the centre. Here $S = (10, 30)$, $E = (30, 50)$, $C = (30, 30)$, hence $I = 30 - 10 = 20$ and $J = 30 - 30 = 0$, and the arc has radius 20. The relative choice is not a whim: it makes the block independent of the workpiece origin, so the same program still works after a zero shift. The controller checks that S and E are equidistant from C within a tolerance, and raises an alarm otherwise.

Solution. Central angle $2\vartheta = 60^\circ$, hence $\vartheta = 30^\circ$ and

$$w_1 = \cos 30^\circ = \frac{\sqrt{3}}{2} \approx 0.8660.$$

Control points: $P_0 = (\cos 30^\circ, -\sin 30^\circ) = (0.8660; -0.5)$, $P_2 = (0.8660; 0.5)$, $P_1 = (1/\cos 30^\circ; 0) = (1.1547; 0)$.

Check at $t = 1/2$. Here $\alpha = \gamma = 1/4$, $\beta = 1/2$, and using the formulas of Step 1 of the proof (with $c = 0.8660$, $s = 0.5$):

$$X = c(\frac{1}{4} + \frac{1}{4}) + \frac{1}{2} = 0.4330 + 0.5 = 0.9330, \quad Y = s(\frac{1}{4} - \frac{1}{4}) = 0,$$

$$W = (\frac{1}{4} + \frac{1}{4}) + \frac{1}{2}c = 0.5 + 0.4330 = 0.9330.$$

Hence $C(1/2) = (0.9330/0.9330, 0) = (1, 0)$, which lies on the unit circle and is the midpoint of the arc. ✓ (Note that $X^2 + Y^2 = W^2$ holds here particularly transparently, since $X = W$ and $Y = 0$.)

Exercise 11.3. Apply Newton–Raphson to $f(r) = \pi r^2 - 50$ (radius of a circle of area 50 m^2) starting from $r_0 = 4$, and count the exact digits at each step.

Solution. $f'(r) = 2\pi r$. Exact value: $r^* = \sqrt{50/\pi} = 3.9894228\dots$

$$r_1 = 4 - \frac{\pi \cdot 16 - 50}{2\pi \cdot 4} = 4 - \frac{0.265482}{25.13274} = 3.9894368,$$

$$r_2 = 3.9894368 - \frac{\pi(3.9894368)^2 - 50}{2\pi \cdot 3.9894368} = 3.9894228.$$

Exact digits: r_0 has 1; r_1 has 5 (3.98943 against 3.98942); r_2 has at least 8. The doubling predicted by Theorem 11.11 holds (indeed is exceeded, because the constant $M = f''/(2f') = 2\pi/(4\pi r) = 1/(2r) \approx 0.125$ is small).

Exercise 11.4. A tool must traverse a quarter circle from $P_0 = (20; 0)$ to $P_1 = (0; 20)$ with centre at the origin. Write the G-code block and justify the sense.

Solution. $I = 0 - 20 = -20$, $J = 0 - 0 = 0$. Sense (Proposition 11.2): $\mathbf{u} = (20, 0)$, $\mathbf{v} = (0, 20)$,

$$\kappa = u_x v_y - u_y v_x = 20 \cdot 20 - 0 \cdot 0 = 400 > 0 \implies \text{anticlockwise} \implies \text{G03}.$$

Block: G17 G90 G03 X0.0 Y20.0 I-20.0 J0.0 F200. Since the arc is $90^\circ \leq 180^\circ$, in this case the form R20.0 (with positive R) would also be legitimate.

Exercise 11.5. An encoder has resolution $\Delta = 0.005$ mm. Compute the standard deviation of the quantization noise on a single step and over a path of $N = 10\,000$ steps, assuming independent errors.

Solution. By Definition 11.6,

$$\sigma_q = \frac{\Delta}{\sqrt{12}} = \frac{0.005}{3.4641} \approx 1.44 \times 10^{-3} \text{ mm} = 1.44 \text{ } \mu\text{m}.$$

After N independent steps the variances add, $\sigma_N^2 = N\sigma_q^2$, hence

$$\sigma_N = \sqrt{N}\sigma_q = 100 \times 1.44 \text{ } \mu\text{m} = 144 \text{ } \mu\text{m} = 0.144 \text{ mm}.$$

An error of nearly a tenth of a millimetre accumulated over a single path: this is the quantitative reason behind the warning at the end of Section 11.5.

11.12.2 Symbolic and proof exercises

Exercise 11.6. Explain why the matrix $A^\top A$ of Kåsa's method is singular (not invertible) if all the points (x_i, y_i) are aligned on the same line, linking the answer to Theorem 7.2 (uniqueness of the circumference through three points) of Chapter 7.

Solution. If all the points are aligned, there exist infinite circumferences of arbitrarily large radius that pass arbitrarily close to all of them (in the limit, a line is a circumference of infinite radius): the fitting problem has no unique solution, and this indeterminacy manifests algebraically as singularity of $A^\top A$. The same phenomenon, seen exactly rather than least-squares, is already encountered in Theorem 7.2: three aligned points determine no circumference (the linear system in D, E, F also becomes singular), exactly for the same

geometric reason.

Exercise 11.7. Prove that the **ellipse** likewise admits no non-constant polynomial parametrisation, adapting Theorem 11.2.

Solution. Suppose $\frac{x(t)^2}{a^2} + \frac{y(t)^2}{b^2} = 1$ with $x, y \in \mathbb{R}[t]$. Set $\tilde{x} = x/a$, $\tilde{y} = y/b$: these are still polynomials (the constants a, b being non-zero) and satisfy $\tilde{x}^2 + \tilde{y}^2 = 1$. By Theorem 11.2 they are constant, hence so are x and y .

Remark. The same argument fails for the **parabola** $y = x^2$, which admits the trivial polynomial parametrisation (t, t^2) : the difference is that the parabola has a point at infinity *on the curve*, whereas the circle and the ellipse do not. This is the structural reason why CAD standards are *rational*.

Exercise 11.8. Prove that, in the construction of Theorem 11.3, the point P_1 is indeed the intersection of the tangents at P_0 and P_2 .

Solution. The tangent to the unit circle at the point $(\cos \varphi, \sin \varphi)$ has equation $x \cos \varphi + y \sin \varphi = 1$ (Chapter 7: the tangent is perpendicular to the radius, and the radius has direction $(\cos \varphi, \sin \varphi)$). For P_0 ($\varphi = -\vartheta$) and P_2 ($\varphi = +\vartheta$):

$$x \cos \vartheta - y \sin \vartheta = 1, \quad x \cos \vartheta + y \sin \vartheta = 1.$$

Subtracting: $2y \sin \vartheta = 0$, and since $\sin \vartheta \neq 0$ for $0 < \vartheta < \pi/2$, we get $y = 0$. Substituting: $x \cos \vartheta = 1$, i.e. $x = 1/\cos \vartheta$. Hence the intersection is $(1/\cos \vartheta, 0) = P_1$. ✓

Exercise 11.9. Prove that the CORDIC scale factor K_n is independent of the angle θ and of the signs d_i , and explain why this is essential for the hardware implementation.

Solution. In the factorisation in the proof of Theorem 11.4, the factor extracted at step i is $\cos \varphi_i$ with $\varphi_i = \arctan(2^{-i})$. Since cosine is **even**, $\cos(d_i \varphi_i) = \cos \varphi_i$ for $d_i = \pm 1$: the factor does not depend on the sign chosen. Nor does it depend on θ , because the angles φ_i are fixed in advance by the sequence 2^{-i} and not by the angle to be rotated. Hence

$$K_n = \prod_{i=0}^{n-1} \cos(\arctan 2^{-i}) = \prod_{i=0}^{n-1} \frac{1}{\sqrt{1 + 2^{-2i}}}$$

is a constant depending on n alone.

Why this is essential. If K depended on θ or on the d_i , it would have to be recomputed at every invocation — requiring n multiplications and cancelling the whole advantage of the algorithm. Being constant, it is stored in ROM together with the table of $\arctan(2^{-i})$, and the cost reduces to *one* final multiplication (or none, if only the direction of the resulting vector is needed).

Exercise 11.10. Explain, in terms of residuals, why Kåsa's method tends to underestimate the radius, and estimate the order of magnitude of the bias for points at distance d from the circle with $d \ll r$.

Solution. The algebraic residual of a point at (signed) distance d from the circle is

$$(x_i - a)^2 + (y_i - b)^2 - r^2 = (r + d)^2 - r^2 = 2rd + d^2,$$

whereas the *geometric* residual is simply d . Kåsa's method minimises $\sum(2rd_i + d_i^2)^2$ instead of $\sum d_i^2$: the effective weighting of each point is thus $\approx (2r)^2$, but with a corrective term d_i^2 which **penalises asymmetrically** the exterior points ($d_i > 0$, residual $2rd_i + d_i^2$ amplified) relative to the interior ones ($d_i < 0$, residual $2rd_i + d_i^2$ attenuated in absolute value). The resulting fit is “pulled” inwards: the estimated radius comes out smaller than the true one. *Order of magnitude.* The relative bias term is $d_i^2/(2rd_i) = d_i/(2r)$: for noise of 1% of the radius ($d \approx 0.01r$) the relative bias on the radius is of order 0.5% — negligible in many dimensional checks, not negligible if the fit is meant to certify a tight tolerance. The methods of Pratt (1987) and Taubin (1991) remove this term by normalising the residual by a factor approximating $2r$.

Exercise 11.11. In IEEE 754 double precision the machine epsilon is $2^{-52} = 2.2204 \cdot 10^{-16}$. Estimate the absolute error with which the circumference $2\pi r$ of a circle of radius $r = 10^6$ m is computed, and say whether it matters for a machine tool.

Solution. The stored value of π has relative error at most $\varepsilon/2 = 1.11 \cdot 10^{-16}$; the product by $2r$ adds as much, so the overall relative error is of order $2.2 \cdot 10^{-16}$. Since $2\pi r = 6.2832 \cdot 10^6$ m, the absolute error is about

$$6.2832 \cdot 10^6 \cdot 2.2 \cdot 10^{-16} = 1.4 \cdot 10^{-9} \text{ m} = 1.4 \text{ nm}.$$

It is entirely negligible against any mechanical tolerance: the dominant error in applications is never the representation of π , but accumulation over many operations and the discretisation of the trajectory.

Exercise 11.12. Apply the midpoint algorithm to the circle of radius $R = 12$ and list the pixels of the first octant, checking that they coincide with the rounding of $\sqrt{R^2 - x^2}$.

Solution. One starts with $x = 0$, $y = 12$, $d = 1 - R = -11$, updating $d += 2x + 3$ if $d < 0$, and otherwise $d += 2(x - y) + 5$ with $y -= 1$. The steps give

$$(0, 12), (1, 12), (2, 12), (3, 12), (4, 11), (5, 11), (6, 10), (7, 10), (8, 9),$$

and one stops when $x > y$. Check against rounding: $\sqrt{144 - 9} = 11.62 \rightarrow 12$; $\sqrt{144 - 16} = 11.31 \rightarrow 11$; $\sqrt{144 - 25} = 10.91 \rightarrow 11$; $\sqrt{144 - 36} = 10.39 \rightarrow 10$; $\sqrt{144 - 49} = 9.75 \rightarrow 10$; $\sqrt{144 - 64} = 8.94 \rightarrow 9$ ✓. The rest of the circle follows by symmetry, with no further decisions.

11.13. Self-assessment quiz

Ten multiple-choice questions. The answer key is at the end of the section.

1. In IEEE 754 double precision the mantissa occupies

A) 23 bits

B) 52 bits

-
- C) 64 bits
D) 11 bits
2. Machine epsilon in single precision is approximately
- A) 2.2×10^{-16}
B) 1.2×10^{-7}
C) 10^{-3}
D) 2^{-52}
3. The midpoint algorithm draws the circle by explicitly computing
- A) all 360 degrees
B) a semicircle
C) a single octant, then reflected eight times
D) a quadrant
4. In the midpoint algorithm, a value $p_k \geq 0$ means that
- A) the midpoint is inside the circle, choose E
B) the midpoint is outside or on the circle, choose SE
C) the algorithm has terminated
D) the radius must be recomputed
5. Can a circle be represented exactly by a **polynomial** (non-rational) Bézier curve?
- A) yes, of degree 2
B) yes, of degree 3
C) no, never: only the rational form allows it
D) only if the radius is an integer
6. The weight w_1 making a quarter circle exact as a rational quadratic Bézier is
- A) 1
B) $\sqrt{2}/2 \approx 0.7071$
C) $\sqrt{3}/2 \approx 0.8660$
D) $1/2$
7. In the CORDIC algorithm the scale factor K
- A) depends on the angle to be rotated
B) is constant, precomputable and stored in ROM
C) must be recomputed at each iteration
D) always equals 1
8. Newton–Raphson, at a simple root and starting close enough, converges
- A) linearly
-

- B) quadratically: the exact digits double at each step
 C) always cubically
 D) in a single step
9. In a circular-interpolation G-code block, the parameters I and J express
 A) the absolute coordinates of the centre
 B) the offset of the centre with respect to the start point
 C) radius and angle
 D) the feed rate
10. To program a **full circle** in G-code one must use
 A) either R or I/J indifferently
 B) necessarily I/J, because R is ambiguous
 C) necessarily R
 D) two G01 blocks

BEST PRACTICE

Key. 1: B — 2: B — 3: C (Proposition 11.1) — 4: B — 5: C (Theorem 11.2) — 6: B — 7: B — 8: B (Theorem 11.11) — 9: B — 10: B.

Note on question 5. This is conceptually the most important question in the chapter: the reason the entire CAD industry uses NURBS rather than plain B-splines lies wholly in that “no”. The proof takes half a page and uses a single fact: in $\mathbb{C}[t]$ the degree of a product is the sum of the degrees.

11.14. Chapter Formula sheet

Floating point (IEEE 754)

$$\varepsilon_{\text{double}} = 2^{-52} \approx 2.22 \times 10^{-16}, \quad \varepsilon_{\text{single}} = 2^{-23} \approx 1.19 \times 10^{-7}$$

absolute error on a radius r : $\sim r \varepsilon$

Midpoint algorithm (rasterisation)

$$\text{E: } (x_{k+1}, y_{k+1}) = (x_k + 1, y_k), \quad p_{k+1} = p_k + 2x_{k+1} + 1$$

$$\text{SE: } (x_{k+1}, y_{k+1}) = (x_k + 1, y_k - 1), \quad p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$$

Circular arc as a rational quadratic Bézier

$$C(t) = \frac{\sum_{i=0}^2 w_i B_i(t) P_i}{\sum_{i=0}^2 w_i B_i(t)}, \quad w_0 = w_2 = 1, \quad w_1 = \cos \vartheta$$

$$\text{arc of central angle } 2\vartheta; \quad \text{quarter circle: } w_1 = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

CORDIC

$$K_n = \prod_{i=0}^{n-1} \cos(\arctan 2^{-i}) \xrightarrow{n \rightarrow \infty} 0.6072529 \dots, \quad K_6 \approx 0.607352$$

Quantisation

$$e \in \left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right], \quad \sigma_q^2 = \int_{-\Delta/2}^{\Delta/2} \frac{e^2}{\Delta} de = \frac{\Delta^2}{12}$$

Circle fitting (Kåsa's method)

$$x^2 + y^2 + Dx + Ey + F = 0, \quad (x_c, y_c) = \left(-\frac{D}{2}, -\frac{E}{2}\right), \quad r = \sqrt{x_c^2 + y_c^2 - F}$$

$$\text{auxiliary unknowns: } u = 2a, \quad v = 2b, \quad w = r^2 - a^2 - b^2 \quad (\text{linear system})$$

Roundness

$$\text{RONt}(C) = \max_i \rho_i(C) - \min_i \rho_i(C), \quad \text{RONt}_{\text{MZC}} \leq \text{RONt}_{\text{LSC}}, \text{RONt}_{\text{MCC}}, \text{RONt}_{\text{MIC}}$$

ISO 12181-1:2011 nomenclature — LSCI(= LSC), MZCI (= MZC), MCCI (= MCC), MICI (= MIC)

Newton–Raphson

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

G-code: arc offsets

$$I = x_c - x_0, \quad J = y_c - y_0$$

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CHAPTER 12

Gauss's Circle Problem

INFO

Level and prerequisites. Level C. Only needed: the notion of area and its properties (Ar1)–(Ar4) (Chapter 6, Observation 6.1), the triangle inequality, the definition of circle and of distance, and elementary arithmetic (divisors, prime numbers, residues modulo 4). No differential calculus or series are needed. Notation: Landau's $O(\cdot)$ and $o(\cdot)$, defined in Observation 12.4.

Purpose of the chapter. The circle, the continuous object par excellence, also has a discrete face: how many points with integer coordinates fall inside it? The question, posed and solved to a first approximation by Gauss, opens a door from the circle to number theory, and leaves — rare in this treatise — a problem still open today, which is stated here for what it is.

12.1. Counting lattice points

Definition 12.1: Lattice points in a circle. For $r > 0$ set

$$N(r) := \#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq r^2\},$$

the number of points of the lattice $\mathbb{Z}^2$ inside or on the closed circle of centre the origin and radius r .

Remark 12.1: A comparison between discrete and continuous. $N(r)$ is an integer, defined without any recourse to analysis; πr^2 is the area of the circle, a real number (Theorem 6.5). Gauss's idea is that the two should be close: to each lattice point one can associate a unit square of area 1, and the union of the squares should “approximate” the circle better and better as r grows. The theorem that follows makes this idea precise and quantifies the error.

Theorem 12.1: Gauss's theorem on the circle problem. For every $r > 0$,

$$N(r) = \pi r^2 + E(r), \quad \text{with} \quad |E(r)| \leq \pi \left(\sqrt{2} r + \frac{1}{2} \right).$$

In particular, for $r \geq \sqrt{2}$,

$$|E(r)| < 2\sqrt{2} \pi r.$$

Proof. Step 1 — a square for every point counted. To each lattice point $P = (m, n)$ with $m^2 + n^2 \leq r^2$ associate the closed square $S_P = [m - \frac{1}{2}, m + \frac{1}{2}] \times [n - \frac{1}{2}, n + \frac{1}{2}]$, of area 1 by (Ar4) and centred at P . As P ranges over the counted points, the squares S_P pairwise share at most a piece of boundary (the lattice squares tile the plane), so by (Ar1), applied repeatedly,

their union U satisfies

$$\text{area}(U) = \sum_P \text{area}(S_P) = N(r).$$

Step 2 — U is contained in the circle of radius $r + \sqrt{2}/2$. Let $X \in S_P$ for some counted point P . The point of S_P farthest from the centre P is a vertex, at distance $\sqrt{(1/2)^2 + (1/2)^2} = \sqrt{2}/2$ from P (half the diagonal of the unit square); hence $|X - P| \leq \sqrt{2}/2$. Since $|P| \leq r$ (because P is counted), the triangle inequality gives

$$|X| \leq |X - P| + |P| \leq \frac{\sqrt{2}}{2} + r.$$

Hence $U \subseteq D(0, r + \frac{\sqrt{2}}{2})$, the closed circle of radius $r + \sqrt{2}/2$.

Step 3 — U contains the circle of radius $r - \sqrt{2}/2$ (when $r \geq \sqrt{2}/2$). Let X be a point with $|X| \leq r - \sqrt{2}/2$. Let $P = (m, n)$ be the lattice point obtained by rounding each coordinate of X to the nearest integer (in case of a tie, either rounding). By construction $X \in S_P$, and in particular, by the same computation as in Step 2, $|X - P| \leq \sqrt{2}/2$. By the triangle inequality,

$$|P| \leq |P - X| + |X| \leq \frac{\sqrt{2}}{2} + \left(r - \frac{\sqrt{2}}{2}\right) = r,$$

so P is a counted point ($m^2 + n^2 \leq r^2$) and $X \in S_P \subseteq U$. It follows that $D(0, r - \frac{\sqrt{2}}{2}) \subseteq U$ (for $r < \sqrt{2}/2$ the inclusion is empty and hence trivial).

Step 4 — comparing the areas. By monotonicity (Ar2) applied to the two inclusions of Steps 2–3, with $A(\rho) = \pi\rho^2$ the area of the circle of radius $\rho \geq 0$ (Theorem 6.5, extended to $\rho = 0$ for the degenerate case of zero area):

$$\pi \max\left(r - \frac{\sqrt{2}}{2}, 0\right)^2 \leq N(r) \leq \pi\left(r + \frac{\sqrt{2}}{2}\right)^2,$$

where $\max(\cdot, 0)$ applied *before* squaring accounts for the degenerate case $r < \sqrt{2}/2$ (Step 3). Expanding the square on the right,

$$\pi\left(r + \frac{\sqrt{2}}{2}\right)^2 = \pi r^2 + \pi\sqrt{2}r + \frac{\pi}{2},$$

so $N(r) - \pi r^2 \leq \pi\sqrt{2}r + \pi/2$. Likewise, when $r \geq \sqrt{2}/2$,

$$\pi\left(r - \frac{\sqrt{2}}{2}\right)^2 = \pi r^2 - \pi\sqrt{2}r + \frac{\pi}{2},$$

so $N(r) - \pi r^2 \geq -\pi\sqrt{2}r + \pi/2 \geq -\pi\sqrt{2}r - \pi/2$. For $0 \leq r < \sqrt{2}/2$ (where Step 3 does not apply) we argue instead as follows: from $r < \sqrt{2}/2$ it follows that $\pi r^2 \leq \frac{\pi\sqrt{2}}{2}r$ (an inequality that also holds trivially at $r = 0$), hence

$$\pi r^2 - \pi\sqrt{2}r - \frac{\pi}{2} \leq -\frac{\pi\sqrt{2}}{2}r - \frac{\pi}{2} \leq -\frac{\pi}{2} < 0 \leq 1 \leq N(r)$$

(the origin is always counted: $N(r) \geq 1$). Hence $N(r) > \pi r^2 - \pi\sqrt{2}r - \pi/2$, i.e. $N(r) - \pi r^2 \geq -\pi\sqrt{2}r - \pi/2$ also holds for $r < \sqrt{2}/2$. In both cases

$$|N(r) - \pi r^2| \leq \pi\sqrt{2}r + \frac{\pi}{2} = \pi\left(\sqrt{2}r + \frac{1}{2}\right).$$

Step 5 — the stated form. For $r \geq \sqrt{2}$,

$$\pi\left(\sqrt{2}r + \frac{1}{2}\right) < 2\sqrt{2}\pi r \iff \frac{1}{2} < \sqrt{2}r \iff r > \frac{1}{2\sqrt{2}},$$

an inequality that holds because $r \geq \sqrt{2} > 1/(2\sqrt{2})$. Hence the tighter bound of Step 4 implies $|E(r)| < 2\sqrt{2}\pi r$. $\square$

Remark 12.2: Why the proven bound is sharper than the stated one. Step 4 provides the estimate $|E(r)| \leq \pi(\sqrt{2}r + 1/2)$, tighter than the often-quoted $|E(r)| < 2\sqrt{2}\pi r$ (which is nonetheless the one that appears in most popular accounts, and the one stated in the theorem for comparison with the literature). The two constants differ because the second arises from a coarser choice of the square associated to each point (with the point at a vertex rather than at the centre): both approaches are correct, but centring the square on the point halves the offset $\sqrt{2}$ to $\sqrt{2}/2$. It is a minor but instructive case where the “right” proof is not unique, and the choice of construction affects the constant — never the order of magnitude.

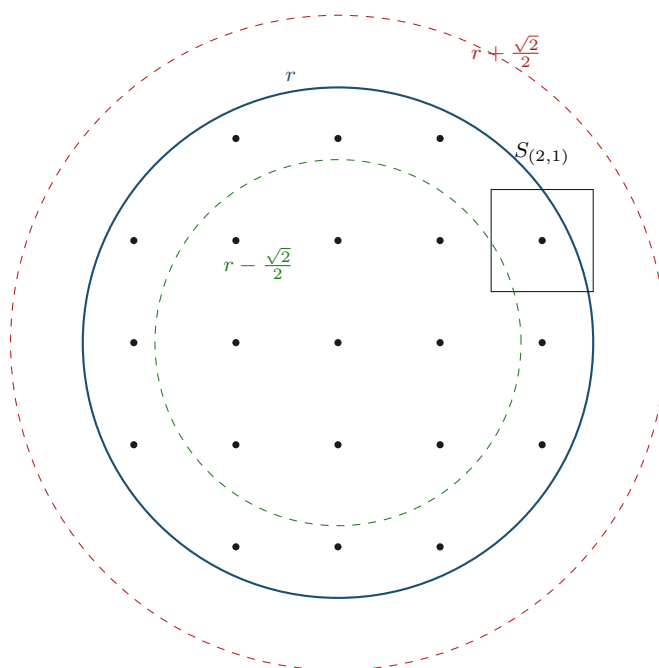


Figure 12.1: Construction used in the proof, for $r = 2.5$: the 21 lattice points with $m^2 + n^2 \leq 6.25$ (dots), the unit square centred at one of them, $(2, 1)$, and the three circles $r - \sqrt{2}/2 < r < r + \sqrt{2}/2$. The union of all 21 unit squares, one per point, contains the green circle and is contained in the red circle (Steps 2–3): this is the area comparison of Step 4.

12.2. A double numerical check

Example 12.1: Two independent counts of $N(r)$. The count $N(r)$ can be computed in two independent ways: (i) a nested loop over m, n that individually checks every point of the grid $[-\lfloor r \rfloor, \lfloor r \rfloor]^2$; (ii) for each m from $-\lfloor r \rfloor$ to $\lfloor r \rfloor$, summing $2\lfloor \sqrt{r^2 - m^2} \rfloor + 1$ (i.e. the number of integers n with $n^2 \leq r^2 - m^2$). The two methods, run independently, give for $r = 10, 100, 1000$:

r	$N(r)$ (method i)	$N(r)$ (method ii)	πr^2	$E(r)$	$\pi(\sqrt{2}r + \frac{1}{2})$
10	317	317	314.159265	2.840735	45.9996
100	31 417	31 417	31 415.926536	1.073464	445.8591
1000	3 141 549	3 141 549	3 141 592.653590	−43.653590	4444.4537

The two methods agree exactly in all three cases $\checkmark$, and $|E(r)|$ comfortably respects, for every r , both the proven bound $\pi(\sqrt{2}r + 1/2)$ and the stated one $2\sqrt{2}\pi r$ $\checkmark$. Note that $E(r)$ changes sign (positive for $r = 10, 100$, negative for $r = 1000$): it is this oscillation, not a

systematic drift, that the open problem of Section 12.3 seeks to quantify precisely.

Remark 12.3: The case $r = 5$ by hand. For $r = 5$ the lattice points with $m^2 + n^2 \leq 25$ are listed row by row (m from -5 to 5 , and for each the admissible n):

m	condition on n	number of n
± 5	$n = 0$	1
± 4	$n^2 \leq 9, n \in \{-3, \dots, 3\}$	7
± 3	$n^2 \leq 16, n \in \{-4, \dots, 4\}$	9
± 2	$n^2 \leq 21, n \in \{-4, \dots, 4\}$	9
± 1	$n^2 \leq 24, n \in \{-4, \dots, 4\}$	9
0	$n^2 \leq 25, n \in \{-5, \dots, 5\}$	11

Summing: $2(1 + 7 + 9 + 9 + 9) + 11 = 2 \cdot 35 + 11 = 81$. Hence $N(5) = 81$, and $|N(5) - 25\pi| = |81 - 78.539816| = 2.460184$, well below both $\pi(\sqrt{2} \cdot 5 + 1/2) = 23.7852$ and $2\sqrt{2}\pi \cdot 5 = 44.4288$.
✓ The count by hand agrees with that of the two programs of the previous check for $r = 5$.

12.3. What remains open

Remark 12.4: Landau notation. For two functions f, g defined for large r , with $g(r) > 0$: $f(r) = O(g(r))$ means there exist constants $C > 0$ and r_0 such that $|f(r)| \leq C g(r)$ for every $r \geq r_0$ (“ f grows no faster, up to a multiplicative constant, than g ”); $f(r) = o(g(r))$ means $\lim_{r \rightarrow \infty} f(r)/g(r) = 0$ (“ f grows strictly more slowly than g ”), a stronger condition than $O(g(r))$. In Theorem 12.1, $E(r) = O(r)$ means $|E(r)| \leq \pi(\sqrt{2}r + 1/2)$, that is, exactly the bound proved, with $C = \pi\sqrt{2}$ (up to lower-order terms).

Remark 12.5: The exact exponent: an open problem. What we import, and why we do not prove it. The bound $|E(r)| = O(r)$ of Theorem 12.1 is not optimal: it has long been known that one can do much better, but it is not known how much better, and this is one of the best-known open problems in number theory. We report here the facts, all due to other authors and not proved here (the proofs use Fourier analysis and exponential sums, well beyond the tools of this chapter):

- **Conjecture.** It is conjectured that $E(r) = O(r^{1/2+\varepsilon})$ for every $\varepsilon > 0$: the exponent $1/2$ would be optimal up to logarithmic corrections.
- **Lower bound (Hardy, 1915).** One cannot do better than the exponent $1/2$ up to a logarithmic factor: Hardy proved that $E(r) \neq o(r^{1/2}(\log r)^{1/4})$, i.e. that the error cannot be made arbitrarily small relative to that threshold.
- **Intermediate milestones.** Sierpiński (1906) obtained the exponent $2/3$; van der Corput (1923) lowered it below $2/3$ (Hardy–Wright, §18.4; Berndt–Kim–Zaharescu).
- **Current record (Huxley, 2000–2003).** The smallest exponent proved with certainty is $131/208 \approx 0.629808$ (Huxley), well above the conjectured threshold $1/2$.

The gap between the conjectured exponent $1/2$ and the proved exponent $131/208$ is today **open**: no one knows whether, or how, it can be closed. The topic is taken up again, in the form of a broader survey, in Volume III, the chapter on open problems.

12.4. Points on the circumference: parity and sums of two squares

Exercise 12.1: Parity of $N(r)$. Prove that $N(r)$ is odd for every $r > 0$, whether integer or not.

Solution. Consider the map $\sigma : (m, n) \mapsto (-m, -n)$ on the set of counted points $\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq r^2\}$. It is well defined (if $m^2 + n^2 \leq r^2$ then also $(-m)^2 + (-n)^2 \leq r^2$) and is an involution ($\sigma \circ \sigma = \text{id}$). Its only fixed point is $(0, 0)$: if $(m, n) = (-m, -n)$ then $2m = 0$ and $2n = 0$, i.e. $m = n = 0$; and $(0, 0)$ is always counted, since $0 \leq r^2$. Every other counted point pairs with a distinct point ($\sigma(m, n) \neq (m, n)$ for $(m, n) \neq (0, 0)$), which is also counted: the set of counted points other than the origin thus splits into pairs $\{(m, n), (-m, -n)\}$, and has even cardinality. Hence

$$N(r) = 1 + (\text{an even number}),$$

which is odd. The table from the previous numerical check confirms this: 317, 31 417, 3 141 549 and 81 are all odd. ✓

Theorem 12.2: Fermat's theorem on sums of two squares. Every prime $p \equiv 1 \pmod{4}$ is a sum of two squares: $p = x^2 + y^2$ for suitable integers x, y .

Proof. We reproduce Don Zagier's "one-sentence" proof (1990). Put

$$S = \{(x, y, z) \in \mathbb{N}^3 : x^2 + 4yz = p\}.$$

S is **finite** (for $x, y, z \geq 1$ and $x^2 + 4yz = p$ fixed, each coordinate is bounded by p) and **non-empty** ($x = 1, y = 1, z = (p - 1)/4 \in \mathbb{N}$ because $p \equiv 1 \pmod{4}$, and $1 + 4 \cdot 1 \cdot (p - 1)/4 = p$).

Step 1 — the involution f . Define $f : S \rightarrow S$ piecewise:

$$f(x, y, z) = \begin{cases} (x + 2z, z, y - x - z) & \text{if } x < y - z, \\ (2y - x, y, x - y + z) & \text{if } y - z < x < 2y, \\ (x - 2y, x - y + z, y) & \text{if } x > 2y. \end{cases}$$

(The cases $x = y - z$ and $x = 2y$ never occur for $(x, y, z) \in S$: both would give $p = x^2 + 4yz$ non-prime — for instance $x = 2y$ gives $p = 4y^2 + 4yz = 4y(y + z)$, even, contradicting p an odd prime; the case $x = y - z$ is analogous. So the three cases cover S with no overlap.) A direct substitution, branch by branch, checks that f maps S into S (in each branch the new coordinates are positive on the stated domain and still satisfy the equation $x'^2 + 4y'z' = p$: for instance in the first branch, $(x + 2z)^2 + 4z(y - x - z) = x^2 + 4xz + 4z^2 + 4zy - 4xz - 4z^2 = x^2 + 4zy = p$) and that $f(f(x, y, z)) = (x, y, z)$ in each branch (direct check, applying the definition twice): f is therefore an **involution** of S onto itself.

Step 2 — the unique fixed point of f . $f(x, y, z) = (x, y, z)$ is possible only in the second branch (the other two always change the first coordinate non-trivially), that is, it requires $2y - x = x$, i.e. $x = y$. Then $p = x^2 + 4xz = x(x + 4z)$: since p is prime and $x \geq 1$, $x + 4z \geq 1 + 4 > 1$, we must have $x = 1$ (the only divisor of p smaller than the cofactor), whence $1 + 4z = p$, that is, $z = (p - 1)/4$. The fixed point is therefore the unique $(1, 1, (p - 1)/4)$.

Step 3 — $|S|$ is odd. An involution on a finite set partitions it into pairs (orbits of size 2) and fixed points (orbits of size 1); since f has exactly one fixed point, $|S| = 1 + 2k$ for some $k \geq 0$: odd.

Step 4 — the second involution g . Let $g(x, y, z) = (x, z, y)$ (swap of the last two coordinates): it is trivially an involution of S onto itself ($x^2 + 4zy = x^2 + 4yz = p$). On a finite set of **odd** cardinality, every involution has at least one fixed point (if it had none, the set would

split entirely into pairs, giving even cardinality): hence g has a fixed point (x, y, y) , that is, $y = z$.

Step 5 — conclusion. At the fixed point of g , $p = x^2 + 4y \cdot y = x^2 + (2y)^2$: p is a sum of two squares, x^2 and $(2y)^2$. $\square$

Numerical check. For $p = 5, 13, 17, 29, 37, 41$ (independent script, explicit enumeration of S): $|S| = 1, 3, 5, 5, 7, 11$ (all odd, as expected); f is an involution on S in each case, with the unique fixed point $(1, 1, (p-1)/4)$ predicted by Step 2; the fixed point of g gives respectively $5 = 1^2 + 2^2$, $13 = 3^2 + 2^2$, $17 = 1^2 + 4^2$, $29 = 5^2 + 2^2$, $37 = 1^2 + 6^2$, $41 = 5^2 + 4^2$ (all checked).

Exercise 12.2: The points exactly on the circumference. For $r = 5$, how many lattice points lie exactly on the circumference $m^2 + n^2 = 25$ (not merely inside)? Connect the result to Fermat's theorem on sums of two squares.

Solution. Listing the solutions of $m^2 + n^2 = 25$ with $m, n \in \mathbb{Z}$: from $25 = 0^2 + 5^2$ there are the four points $(\pm 5, 0), (0, \pm 5)$; from $25 = 3^2 + 4^2$ there are eight more, $(\pm 3, \pm 4)$ and $(\pm 4, \pm 3)$ (all sign combinations, since $3 \neq 4$). Total: $4 + 8 = 12$ points on the circumference.

Connection with Fermat. The number of representations of a positive integer n as an ordered sum of two squares (signs and order matter), $r_2(n) := \#\{(m, n') \in \mathbb{Z}^2 : m^2 + n'^2 = n\}$, is governed by **Fermat's theorem on sums of two squares**, proved in full just above (Theorem 12.2), and by its refinement due to Jacobi:

$$r_2(n) = 4(d_1(n) - d_3(n)),$$

where $d_1(n)$, $d_3(n)$ are the number of divisors of n congruent respectively to 1 and to 3 modulo 4. This exact formula of Jacobi's remains **imported** (it goes through unique factorization in the Gaussian integers $\mathbb{Z}[i]$, beyond this chapter's tools); see the source cited at the end.

Numerical check of the imported fact. For $n = r^2 = 25$, the divisors are 1, 5, 25, all $\equiv 1 \pmod{4}$: $d_1 = 3$, $d_3 = 0$, hence $r_2(25) = 4 \cdot 3 = 12$, which agrees exactly with the direct count above. $\checkmark$ Repeating for $r = 25$ ($n = 625 = 5^4$, divisors 1, 5, 25, 125, 625, all $\equiv 1 \pmod{4}$): $r_2(625) = 4 \cdot 5 = 20$; the direct count by program also gives 20 points on the circumference. $\checkmark$ And for $r = 50$ ($n = 2500 = 2^2 \cdot 5^4$, whose odd divisors are exactly the divisors of 625, the others being even and hence not congruent to 1 or 3 modulo 4): $r_2(2500) = 4 \cdot 5 = 20$, again confirmed by the direct count. $\checkmark$

Remark 12.6: Why $N(r)$ odd does not contradict Gauss's theorem. That $N(r)$ is always odd is an exact fact and holds for every r ; Theorem 12.1, instead, concerns how far $N(r)$ can deviate from πr^2 , which is neither even nor odd (it is not even an integer). The two facts do not touch each other: the first is algebraic (a symmetry), the second is geometric-analytic (a comparison of areas).

12.5. Lagrange's four-square theorem

Fermat's theorem just proved characterises the integers that are sums of *two* squares: not every integer is one (for instance 3, 6, 7 are not). With *four* squares, instead, no condition is needed: every positive integer is a sum of four integer squares (zero included). The result is due to Joseph-Louis Lagrange, who proved it in 1770 in a memoir presented to the Berlin Academy, published as J.-L. Lagrange, *Démonstration d'un théorème d'arithmétique, Nouveaux mémoires de l'Académie royale des sciences et belles-lettres de Berlin*, year 1770 (printed 1772), pp. 123–133.

The proof is organised in three independent parts: an algebraic identity that reduces the problem to prime numbers, a counting argument (the “pigeonhole principle”) that produces a first, partial representation for every prime, and a descent that refines it to an exact representation.

Lemma 12.1: Euler’s four-square identity. For all $a, b, c, d, A, B, C, D \in \mathbb{Z}$,

$$(a^2 + b^2 + c^2 + d^2)(A^2 + B^2 + C^2 + D^2) = w_1^2 + w_2^2 + w_3^2 + w_4^2,$$

where

$$w_1 = aA + bB + cC + dD, \quad w_2 = -aB + bA - cD + dC, \quad w_3 = -aC + bD + cA - dB, \quad w_4 = -aD - bC + cB + dA.$$

In particular, the product of two integers each a sum of four squares is again a sum of four squares.

Proof. This is a purely algebraic identity among polynomials in the eight variables, verifiable by expanding both sides: each side, expanded, is a homogeneous polynomial of degree 2 in a, b, c, d and degree 2 in A, B, C, D , with $4 \times 4 = 16$ quadratic monomials of type $a^2 A^2$ (and similar) plus the mixed monomials; the term-by-term comparison (given in full in the script `cap13_lagrange.py`, which also confirms it symbolically with a computer algebra system) shows the two sides coincide identically. This is no accident: $w_1 + iw_2 + jw_3 + kw_4$ is, in the language of quaternions (an extension of the complex numbers with three imaginary units i, j, k instead of one), the product $p\bar{q}$ of the quaternion $p = a + bi + cj + dk$ by the conjugate $\bar{q} = A - Bi - Cj - Dk$ of $q = A + Bi + Cj + Dk$; the identity then expresses the property, shared by complex numbers and quaternions, that the “modulus” of a product is the product of the moduli. This reading frames the identity, it is not an ingredient of its proof, which remains the direct verification by expansion. $\square$

Remark 12.7: Why it suffices to prove it for primes. By Lemma 12.1, if every prime is a sum of four squares, so is every product of primes (applying the lemma repeatedly, one factor at a time), that is, every integer $n \geq 2$ by the Fundamental Theorem of Arithmetic. What remains are $n = 0 = 0^2 + 0^2 + 0^2 + 0^2$ and $n = 1 = 1^2 + 0^2 + 0^2 + 0^2$, trivial, and the prime $p = 2 = 1^2 + 1^2 + 0^2 + 0^2$. For every **odd** prime p , the statement therefore remains to be proved.

Lemma 12.2: First partial representation (pigeonhole). For every odd prime p there is an integer m with $0 < m < p$ such that mp is a sum of four squares.

Proof. Step 1 — two sets of residues. Let $h = (p - 1)/2$. Consider the two sets of residue classes modulo p

$$\mathcal{A} = \{x^2 \bmod p : 0 \leq x \leq h\}, \quad \mathcal{B} = \{-1 - y^2 \bmod p : 0 \leq y \leq h\}.$$

Step 2 — $|\mathcal{A}| = |\mathcal{B}| = h + 1$. If $x_1^2 \equiv x_2^2 \pmod{p}$ with $0 \leq x_1, x_2 \leq h$, then $p \mid (x_1 - x_2)(x_1 + x_2)$; since p is prime, $p \mid x_1 - x_2$ or $p \mid x_1 + x_2$. But $|x_1 - x_2| \leq h < p$ and $0 \leq x_1 + x_2 \leq 2h = p - 1 < p$, so both divisibilities force, respectively, $x_1 = x_2$ or $x_1 + x_2 = 0$, i.e. $x_1 = x_2 = 0$: in every case $x_1 = x_2$. The map $x \mapsto x^2 \bmod p$ is thus injective on $\{0, \dots, h\}$, and $|\mathcal{A}| = h + 1$; the same argument (applied to $y \mapsto -1 - y^2 \bmod p$, which is injective because $y \mapsto y^2$ is) gives $|\mathcal{B}| = h + 1$.

Step 3 — nonempty intersection. $\mathcal{A}, \mathcal{B} \subseteq \mathbb{Z}/p\mathbb{Z}$, a set of p elements, and $|\mathcal{A}| + |\mathcal{B}| = 2h + 2 = p + 1 > p$: by the pigeonhole principle (two subsets of a p -element set whose cardinalities add to more than p must intersect, otherwise their union would have more than p elements) there

is a common value, that is, there exist $x, y \in \{0, \dots, h\}$ with

$$x^2 \equiv -1 - y^2 \pmod{p}, \quad \text{i.e.} \quad x^2 + y^2 + 1 \equiv 0 \pmod{p}.$$

Step 4 — bounding m . Setting $mp = x^2 + y^2 + 1^2 + 0^2$ (a nonnegative integer by Step 3, and positive since $x^2 + y^2 + 1 \geq 1$),

$$mp = x^2 + y^2 + 1 \leq h^2 + h^2 + 1 = \frac{(p-1)^2}{2} + 1,$$

so

$$m \leq \frac{(p-1)^2 + 2}{2p} < p$$

(the last inequality, multiplying by $2p > 0$, is equivalent to $(p-1)^2 + 2 < 2p^2$, i.e. $p^2 - 2p + 3 < 2p^2$, i.e. $0 < p^2 + 2p - 3 = (p-1)(p+3)$, true for every $p \geq 2$). Hence $0 < m < p$. $\square$

Lemma 12.3: Descent. *If $1 < m < p$ and mp is a sum of four squares, there exists m' with $1 \leq m' < m$ such that $m'p$ is a sum of four squares.*

Proof. Let $mp = a^2 + b^2 + c^2 + d^2$.

Case 1 — m even. Since $mp = a^2 + b^2 + c^2 + d^2$ is even, the number of odd values among a, b, c, d is even (the square of an even number is even, that of an odd number is odd, and a sum of four squares is even if and only if an even number of them are odd), that is, 0, 2, or 4. Reorder a, b, c, d so that $a \equiv b \pmod{2}$ and $c \equiv d \pmod{2}$ (always possible: if 0 or 4 are odd, any pairing works; if 2 are odd, pair the two odd ones together and the two even ones together). Then $a+b, a-b, c+d, c-d$ are all even, and setting

$$A = \frac{a+b}{2}, \quad B = \frac{a-b}{2}, \quad C = \frac{c+d}{2}, \quad D = \frac{c-d}{2} \quad (\text{all integers}),$$

one computes directly

$$A^2 + B^2 + C^2 + D^2 = \frac{(a+b)^2 + (a-b)^2 + (c+d)^2 + (c-d)^2}{4} = \frac{2a^2 + 2b^2 + 2c^2 + 2d^2}{4} = \frac{a^2 + b^2 + c^2 + d^2}{2} = \frac{mp}{2}.$$

Set $m' = m/2$ (an integer, since m is even): $1 \leq m' < m$.

Case 2 — m odd (hence $m \geq 3$). For each of a, b, c, d choose the representative of its residue class modulo m in the symmetric interval $(-\frac{m}{2}, \frac{m}{2})$: since m is odd, this interval contains exactly m consecutive integers (the residues $0, \pm 1, \dots, \pm \frac{m-1}{2}$), so the representative is **unique** and satisfies $|A|, |B|, |C|, |D| \leq \frac{m-1}{2}$ (no sign ambiguity, unlike the even- m case where $m/2$ would be a boundary value). Since $A \equiv a, B \equiv b, C \equiv c, D \equiv d \pmod{m}$,

$$A^2 + B^2 + C^2 + D^2 \equiv a^2 + b^2 + c^2 + d^2 = mp \equiv 0 \pmod{m},$$

so $A^2 + B^2 + C^2 + D^2 = mr$ for some integer $r \geq 0$. From the bound on the representatives,

$$mr = A^2 + B^2 + C^2 + D^2 \leq 4 \left(\frac{m-1}{2} \right)^2 = (m-1)^2 < m^2,$$

so $r < m$. If $r = 0$ then $A = B = C = D = 0$, i.e. $m \mid a, b, c, d$, hence $m^2 \mid a^2 + b^2 + c^2 + d^2 = mp$, i.e. $m \mid p$: since p is prime and $1 < m < p$, this is impossible. Hence $1 \leq r < m$.

Now apply Lemma 12.1 to (a, b, c, d) and (A, B, C, D) :

$$(mp)(mr) = w_1^2 + w_2^2 + w_3^2 + w_4^2.$$

Writing $A = a + m\alpha$, $B = b + m\beta$, $C = c + m\gamma$, $D = d + m\delta$ for suitable integers $\alpha, \beta, \gamma, \delta$ (by construction $A \equiv a \pmod{m}$, etc.) and substituting into the expressions of Lemma 12.1:

$$w_1 = aA + bB + cC + dD = (a^2 + b^2 + c^2 + d^2) + m(a\alpha + b\beta + c\gamma + d\delta) = mp + m(a\alpha + b\beta + c\gamma + d\delta) = m[p + a\alpha + b\beta + c\gamma + d\delta]$$

$$w_2 = -aB + bA - cD + dC = -a(b + m\beta) + b(a + m\alpha) - c(d + m\delta) + d(c + m\gamma) = m(-a\beta + b\alpha - c\delta + d\gamma)$$

(the m -free terms cancel: $-ab + ba - cd + dc = 0$), and likewise $w_3 = m(-a\gamma + c\alpha + b\delta - d\beta)$, $w_4 = m(-a\delta + d\alpha - b\gamma + c\beta)$ (the same cancellation of the degree-0 terms in $\alpha, \beta, \gamma, \delta$, checked in full in the script). So w_1, w_2, w_3, w_4 are all **exactly** divisible by m ; writing $w_i = m v_i$ ($i = 1, \dots, 4$, integers),

$$m^2 pr = m^2(v_1^2 + v_2^2 + v_3^2 + v_4^2) \implies pr = v_1^2 + v_2^2 + v_3^2 + v_4^2,$$

so $m' = r$ satisfies $1 \leq m' < m$ and $m'p$ is a sum of four squares. $\square$

Theorem 12.3: Four-square theorem (Lagrange, 1770). Every integer $n \geq 0$ can be written as $n = a^2 + b^2 + c^2 + d^2$ for suitable integers a, b, c, d (possibly zero).

Proof. [of the four-square theorem] By Observation 12.7 it suffices to treat an odd prime p (the cases $n = 0, 1, 2$ are immediate). By Lemma 12.2 there is m_0 with $0 < m_0 < p$ such that $m_0 p$ is a sum of four squares. If $m_0 = 1$ then p is already a sum of four squares. Otherwise, by Lemma 12.3 there is m_1 with $1 \leq m_1 < m_0$ such that $m_1 p$ is a sum of four squares; if $m_1 > 1$, Lemma 12.3 is applied again to m_1 , and so on. The sequence $m_0 > m_1 > m_2 > \dots$ is a strictly decreasing sequence of positive integers, so by the well-ordering principle (equivalently, by Fermat's infinite descent: a strictly decreasing sequence of positive integers cannot be infinite) it reaches the value 1 after finitely many steps: at that point $1 \cdot p = p$ is a sum of four squares. $\square$

Example 12.2: The full descent for $p = 7919$. The prime $p = 7919$ requires a non-trivial descent, useful to see the proof at work. Lemma 12.2 (direct search over $x, y \in \{0, \dots, 3959\}$) finds $x = 3211$, $y = 4$, giving $x^2 + y^2 + 1 = 10\,310\,538 = 1302 \times 7919$: we start from $m_0 = 1302$. The descent of Lemma 12.3, applied repeatedly (halving when m is even, the quaternionic step when it is odd), gives the following trace (checked at every row, script `cap13_lagrange.py`):

m	(a, b, c, d)	type of step	check
1302	(3211, 4, 1, 0)	—	$\sum a_i^2 = 1302 \times 7919$
651	(1606, 1605, 2, 2)	halving	$\sum a_i^2 = 651 \times 7919$
283	(1497, 2, 0, -8)	quaternionic	$\sum a_i^2 = 283 \times 7919$
24	(434, -10, 0, 40)	quaternionic	$\sum a_i^2 = 24 \times 7919$
12	(212, 222, 20, -20)	halving	$\sum a_i^2 = 12 \times 7919$
6	(217, -5, 0, 20)	halving	$\sum a_i^2 = 6 \times 7919$
3	(106, 111, 10, -10)	quaternionic	$\sum a_i^2 = 3 \times 7919$
1	(42, 37, -69, -5)	quaternionic	$\sum a_i^2 = 7919$

Independent final check: $42^2 + 37^2 + 69^2 + 5^2 = 1764 + 1369 + 4761 + 25 = 7919$. $\checkmark$

Example 12.3: A composite integer: 7091. To illustrate Observation 12.7 (reduction to primes via Euler's identity) with an example in which the theorem is fully at work — all four squares turn out to be nonzero, unlike what would happen if n actually reduced to a sum of two or three squares — write $7091 = 7 \times 1013$ (both prime). The descent of Lemma 12.3, applied to 1013, gives $1013 = 23^2 + 22^2 + 0^2 + 0^2$ (checked: $529 + 484 = 1013$ $\checkmark$). The prime 7 is small enough to represent by direct inspection: $7 = 2^2 + 1^2 + 1^2 + 1^2$

(checked: $4 + 1 + 1 + 1 = 7 \checkmark$). Applying Lemma 12.1 to the pairs $(a, b, c, d) = (23, 22, 0, 0)$ and $(A, B, C, D) = (2, 1, 1, 1)$ gives

$$w_1 = 23 \cdot 2 + 22 \cdot 1 = 68, \quad w_2 = -23 \cdot 1 + 22 \cdot 2 = 21, \quad w_3 = -23 \cdot 1 + 22 \cdot 1 = -1, \quad w_4 = -23 \cdot 1 - 22 \cdot 1 = -45,$$

so $7091 = 68^2 + 21^2 + (-1)^2 + (-45)^2$, independently checked: $4624 + 441 + 1 + 2025 = 7091 \checkmark$, that is $7091 = 68^2 + 45^2 + 21^2 + 1^2$: an example in which all four squares are genuinely needed, none zero.

12.6. Pick's theorem

Theorem 12.4: Pick's theorem. *Let P be a simple (non-self-intersecting) polygon with all vertices at integer coordinates, I the number of lattice points interior to P and B the number of lattice points on the boundary (vertices included). Then the area of P equals*

$$A = I + \frac{B}{2} - 1.$$

Proof. Put $\Pi(P) := I(P) + B(P)/2 - 1$ and show $A(P) = \Pi(P)$ in steps.

Step 1 — additivity of Π . If P is the union of two polygons P_1, P_2 glued along a common side $\overline{UV}$ (no other point in common), and that side contains $k + 1$ lattice points including the endpoints ($k \geq 1$ unit segments, after simplifying), then: the $k + 1$ points of the common side were on the boundary of *both* P_1, P_2 and are now **interior** to P , except the two endpoints U, V which remain on the boundary of P (assuming U, V are also vertices of P , i.e. the boundary of P turns there); hence $I(P) = I(P_1) + I(P_2) + (k - 1)$ (the $k - 1$ interior points of the side, formerly on the boundary of both, now interior to P) and $B(P) = B(P_1) + B(P_2) - 2(k + 1) + 2$ (the $k + 1$ points of the common side are removed from both counts, and the 2 endpoints that remain on the boundary of P are added back). Adding:

$$\Pi(P) = I(P) + \frac{B(P)}{2} - 1 = [I(P_1) + I(P_2) + (k - 1)] + \left[\frac{B(P_1) + B(P_2)}{2} - (k + 1) + 1 \right] - 1 = \Pi(P_1) + \Pi(P_2),$$

(direct check of the algebra), and for the area, obviously, $A(P) = A(P_1) + A(P_2)$: if the theorem holds for P_1 and P_2 separately, it also holds for their union.

Step 2 — rectangles with sides on the axes. A rectangle $a \times b$ (integers $a, b \geq 1$, sides parallel to the axes) has $B = 2a + 2b$ (perimeter divided by unit step) and $I = (a - 1)(b - 1)$ (interior grid): $\Pi = (a - 1)(b - 1) + a + b - 1 = ab - a - b + 1 + a + b - 1 = ab = A$. The theorem holds.

Step 3 — right triangles with legs on the axes. Such a triangle is half of a rectangle $a \times b$, cut by a diagonal. Applying Step 1 to the rectangle split by the diagonal into two congruent triangles (hence of equal area and Π , by central symmetry, which preserves I, B): $\Pi(\text{rectangle}) = 2\Pi(\text{triangle})$, and by Step 2 the theorem holds for the rectangle, hence $A(\text{triangle}) = \Pi(\text{triangle})$.

Step 4 — any triangle. Every triangle with integer vertices is inscribed in a rectangle with sides on the axes (the smallest rectangle containing it) and is obtained from it by removing up to three right triangles with legs on the axes (at the corners of the rectangle). By Steps 1–3, the theorem holds for the rectangle and for each piece removed, hence — by additivity of Π and of A under subtraction (the same algebra as Step 1, with the sign reversed) — it also holds for the remaining triangle.

Step 5 — any simple polygon. Every simple polygon with at least four vertices admits an **interior diagonal** (a segment between two non-adjacent vertices, entirely contained in P): this is an elementary fact of topology, **imported** (Meisters, 1975, *Polygons have ears*).

Cutting along such a diagonal gives two polygons with fewer sides, and by induction on the number of sides (base case: $n = 3$, Step 4) the theorem holds for both pieces, hence, by Step 1, for the union P . $\square$

Numerical check. Triangle $(0,0), (4,0), (0,3)$: $I = 3$, $B = 8$ (direct count, independent script), $A = I + B/2 - 1 = 3 + 4 - 1 = 6$, which agrees with the true area $\frac{1}{2} \cdot 4 \cdot 3 = 6$. $\checkmark$

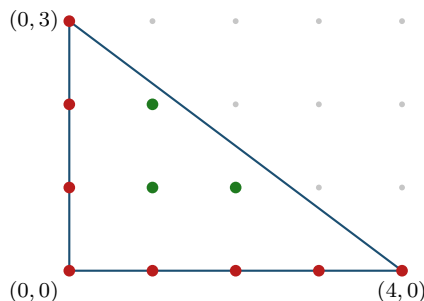


Figure 12.2: Check of Pick's theorem on the triangle $(0,0), (4,0), (0,3)$: the 8 boundary points (red, including the 3 vertices) and the 3 interior points (green). $A = I + B/2 - 1 = 3 + 4 - 1 = 6$, equal to the true area $\frac{1}{2} \cdot 4 \cdot 3 = 6$.

Remark 12.8: Connection with Gauss's circle problem. Pick's theorem and Gauss's circle problem share the same object of study — counting lattice points — but with opposite aims: Gauss counts the points inside a circle to approximate its area; Pick uses the count of points on the boundary and inside a polygon to compute its area **exactly**. It is no accident that Pick's theorem does not apply to the circle: a circle has no lattice vertices (in general), and the formula requires a polygon.

12.7. Ford circles

Definition 12.2: Ford circle. For a reduced fraction h/k ($k \geq 1$, $\gcd(h, k) = 1$), the **Ford circle** $\mathcal{F}(h/k)$ is the circle of centre $(h/k, 1/(2k^2))$ and radius $1/(2k^2)$ (hence tangent to the x axis at the point h/k).

Theorem 12.5: Tangency of Ford circles. Two distinct Ford circles $\mathcal{F}(h/k), \mathcal{F}(h'/k')$ are either disjoint or tangent; they are tangent if and only if $|hk' - h'k| = 1$.

Proof. Let d be the distance of the centres, $r = 1/(2k^2)$, $r' = 1/(2k'^2)$ the radii.

$$d^2 - (r + r')^2 = \left(\frac{h}{k} - \frac{h'}{k'}\right)^2 + \left(\frac{1}{2k^2} - \frac{1}{2k'^2}\right)^2 - \left(\frac{1}{2k^2} + \frac{1}{2k'^2}\right)^2.$$

Using $(u - v)^2 - (u + v)^2 = -4uv$ with $u = 1/(2k^2)$, $v = 1/(2k'^2)$, the last difference equals $-4uv = -1/(k^2k'^2)$. Hence

$$d^2 - (r + r')^2 = \frac{(hk' - h'k)^2}{k^2k'^2} - \frac{1}{k^2k'^2} = \frac{(hk' - h'k)^2 - 1}{k^2k'^2}.$$

Since $h/k \neq h'/k'$ (distinct fractions), the integer $hk' - h'k$ is non-zero, hence $(hk' - h'k)^2 \geq 1$: $d^2 - (r + r')^2 \geq 0$, that is $d \geq r + r'$, always (the two circles never overlap: they are external or tangent). Equality $d = r + r'$ (tangency) holds if and only if $(hk' - h'k)^2 = 1$, that is $|hk' - h'k| = 1$. $\square$

Numerical check. $\mathcal{F}(1/2)$ and $\mathcal{F}(1/3)$: $|1 \cdot 3 - 1 \cdot 2| = 1$, tangent (checked: centres (0.5; 0.125) and (0.333333; 0.055556), distance $d = 0.180556$, exactly equal to $r + r' = 0.125 + 0.055556 = 0.180556$). $\mathcal{F}(1/3)$ and $\mathcal{F}(3/4)$: $|1 \cdot 4 - 3 \cdot 3| = |4 - 9| = 5 \neq 1$, not tangent (checked: $d = 0.417375$ against $r + r' = 0.086806$, well apart, hence external and not tangent).

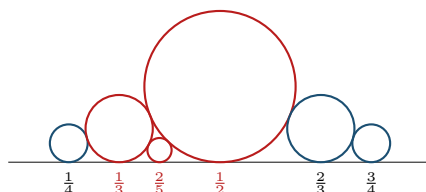


Figure 12.3: Ford circles for the fractions of denominator ≤ 4 in $(0, 1)$, with the triple $\frac{1}{3}, \frac{2}{5}, \frac{1}{2}$ (red) highlighted: the median $\frac{2}{5} = \frac{1+1}{3+2}$ of the Farey neighbours $\frac{1}{3}, \frac{1}{2}$ is tangent to both (checked: $|1 \cdot 5 - 2 \cdot 3| = 1$ and $|2 \cdot 2 - 1 \cdot 5| = 1$), the concrete example of an Apollonius configuration from the following Observation.

Remark 12.9: Farey and Apollonius. Two fractions $h/k, h'/k'$ with $|hk' - h'k| = 1$ are called **Farey neighbours**: they are exactly the consecutive fractions in some Farey sequence (*imported*, Hardy–Wright, chap. III). If h/k and h'/k' are Farey neighbours, their **median** $(h + h')/(k + k')$ is a Farey neighbour of both, and the three corresponding Ford circles are pairwise tangent: a concrete (and infinite) example of an Apollonian configuration, linking this section to Descartes' theorem (Chapter 5).

12.8. Assigned exercises

Exercise 12.3: Checking the bound for a new value of r . Compute $N(20)$ with the row-by-row method of Section 12.2 (summing $2\lfloor\sqrt{400 - m^2}\rfloor + 1$ for $m = -20, \dots, 20$), compare with 400π , and check the bound $|E(r)| \leq \pi(\sqrt{2} \cdot 20 + 1/2)$.

Solution. Summing $2\lfloor\sqrt{400 - m^2}\rfloor + 1$ for $m = -20, \dots, 20$ (independent script) gives $N(20) = 1257$. The ideal value is $400\pi = 1256.637061\dots$, hence $E(20) = N(20) - 400\pi = 0.362939$. The bound of Observation 12.2 is $\pi(\sqrt{2} \cdot 20 + 1/2) = \pi \cdot 28.784271 = 90.428455$: since $|E(20)| = 0.362939 \ll 90.428455$, the bound is amply verified (checked).

Exercise 12.4: Points on the circumference for $r = 13$. Using $13 = 2^2 + 3^2$ (prime, $13 \equiv 1 \pmod{4}$), list all the representations of $169 = 13^2$ as a sum of two squares and count the lattice points on the circumference $m^2 + n^2 = 169$. Compare with Jacobi's formula $r_2(169) = 4(d_1 - d_3)$.

Solution. The representations of 169 as a sum of two non-negative squares are $169 = 0^2 + 13^2 = 5^2 + 12^2$ (checked by exhaustion over $m = 0, \dots, 13$: no other m makes $169 - m^2$ a perfect square). Counting signs and order: from $0^2 + 13^2$ we get the 4 points $(\pm 13, 0), (0, \pm 13)$; from $5^2 + 12^2$ we get the 8 points $(\pm 5, \pm 12), (\pm 12, \pm 5)$. In all, $4 + 8 = 12$ lattice points on the circumference $m^2 + n^2 = 169$. By Jacobi's formula, the divisors of $169 = 13^2$ are 1, 13, 169, all $\equiv 1 \pmod{4}$ (since $13 \equiv 1 \pmod{4}$): $d_1 = 3, d_3 = 0$, hence $r_2(169) = 4(3 - 0) = 12$: it agrees with the direct count (checked).

12.9. Self-assessment quiz

- $N(r)$ is defined as
 - the area of the circle of radius r
 - the number of points with integer coordinates such that $m^2 + n^2 \leq r^2$
 - the number of points with integer coordinates such that $m^2 + n^2 = r^2$ only
 - the perimeter of the inscribed polygon with lattice vertices
- Theorem 12.1 requires, for the form $|E(r)| < 2\sqrt{2}\pi r$,
 - r integer
 - $r \geq \sqrt{2}$
 - $r \leq \sqrt{2}$
 - no additional hypothesis
- The square S_P associated to the point $P = (3, 4)$ in the proof has vertices
 - $(3, 4), (4, 4), (4, 5), (3, 5)$
 - $(2.5, 3.5), (3.5, 3.5), (3.5, 4.5), (2.5, 4.5)$
 - $(2, 3), (4, 3), (4, 5), (2, 5)$
 - centred at the origin, side 3
- What is $N(5)$?
 - 78
 - 79
 - 80
 - 81
- In Step 3 of the proof, the point P is obtained from X
 - by projecting X radially onto the circle of radius r
 - by rounding each coordinate of X to the nearest integer
 - by choosing the lattice point closest to the origin
 - by translating X by $(\sqrt{2}/2, \sqrt{2}/2)$
- A student writes: “since $N(r)$ is always odd, the error $E(r)$ must always be positive”. The mistake is
 - correct: the table confirms it for every r
 - incorrect: $N(r)$ odd is an algebraic fact about $N(r)$ alone, while $E(r) = N(r) - \pi r^2$ also depends on πr^2 , which is not an integer; the table shows $E(1000) < 0$
 - incorrect: $N(r)$ is not always odd
 - correct only for integer r
- The method of Step 1 (associating a unit square to each counted point) would apply, with the obvious adaptation, also to

- A) the count of lattice points in a ball of $\mathbb{R}^3$, with cubes in place of squares
- B) only the circle, for an intrinsically two-dimensional reason
- C) no other figure
- D) only regular polygons
8. Huxley's bound, $|E(r)| = O(r^{131/208+\varepsilon})$, with respect to the conjecture $|E(r)| = O(r^{1/2+\varepsilon})$
- A) proves it
- B) disproves it
- C) is consistent with it but does not prove it: $131/208 > 1/2$, so a gap remains open
- D) has no relation to it

BEST PRACTICE

Key. 1: B (Definition 12.1) — 2: B (Theorem 12.1, Step 5) — 3: B (unit square centred at $(3, 4)$) — 4: D (Observation 12.3) — 5: B (Step 3 of the proof) — 6: B (Observation 12.6; the typical mistake is to confuse an exact property of $N(r)$ with the sign of its deviation from πr^2) — 7: A (the argument is dimensionally independent: it works with unit cubes and the ball in $\mathbb{R}^3$, giving $|E_3(r)| = O(r^2)$) — 8: C (Section 12.3).

Remedial activity for item 6. Recompute $E(r)$ for $r = 10, 100, 1000$ from the table in Section 12.2 and explicitly observe its change of sign, before answering questions about the sign of $E(r)$.

12.10. Chapter formulas

Gauss's circle problem (Definition 12.1; Theorem 12.1)

$$N(r) = \#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq r^2\} = \pi r^2 + E(r), \quad |E(r)| \leq \pi \left(\sqrt{2}r + \frac{1}{2} \right) < 2\sqrt{2}\pi r \quad (r \geq \sqrt{2})$$

Status of the optimal exponent (Observation 12.5)

$$\frac{1}{2} \text{ (conjectured)} \leq \theta_{\text{optimal}} \leq \frac{131}{208} \approx 0.6298 \text{ (Huxley, proved)}$$

Points on the circumference (Jacobi's formula, imported, Section 12.4)

$$r_2(n) = 4(d_1(n) - d_3(n))$$

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The Circle in Industrial Robotics

13.1. From the mathematical model to the real motion of the TCP

Robotic controllers (Yaskawa, FANUC, ABB, KUKA) never “draw” a circle analytically as $x^2 + y^2 = r^2$: the circular trajectory of the **TCP** (Tool Center Point) is generated by interpolating a **circular arc defined by three non-collinear points** (start, via, end), exactly the object of Theorem 7.2 (circumference through three points): the controller solves in real time the same linear system in D, E, F seen in Chapter 7, extended to $\mathbb{R}^3$ with quaternions for the orientation of the tool along the arc.

INFO

Note. For every triple of non-collinear points there exists a unique circumference containing them (Theorem 7.2): this is exactly the reason why *all* industrial robotic languages require an intermediate point (besides start and end) to define an arc — it is not enough to give radius and centre as in a 2D CAD drawing, because in $\mathbb{R}^3$ the arc must also lie on a plane and have a coherent TCP orientation.

13.2. The circle through three points in space

Theorem 7.2 holds in the plane. A robot works in $\mathbb{R}^3$, and the arc must also *lie in a plane* determined by the three points themselves. The result must therefore be restated, and it is worth proving in full, because it is literally the algorithm the controller runs when it interprets a MOVC, MoveC, CIRC or G02 instruction.

Theorem 13.1: Circle through three points in $\mathbb{R}^3$. Let $P_1, P_2, P_3 \in \mathbb{R}^3$ be three non-collinear points. Setting

$$\mathbf{a} = P_2 - P_1, \quad \mathbf{b} = P_3 - P_1,$$

there exists exactly one circle through the three points. It lies in the plane through P_1 with normal $\mathbf{n} = \mathbf{a} \times \mathbf{b}$, has centre

$$C = P_1 + \mathbf{d}, \quad \mathbf{d} = \frac{(|\mathbf{a}|^2 \mathbf{b} - |\mathbf{b}|^2 \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})}{2 |\mathbf{a} \times \mathbf{b}|^2},$$

and radius

$$R = |\mathbf{d}| = \frac{|\mathbf{a}| |\mathbf{b}| |\mathbf{b} - \mathbf{a}|}{2 |\mathbf{a} \times \mathbf{b}|}.$$

Proof. Step 0 — the denominator does not vanish. The three points are not collinear, hence $\mathbf{a}$ and $\mathbf{b}$ are linearly independent and $\mathbf{a} \times \mathbf{b} \neq \mathbf{0}$: all the formulas are well defined.

Step 1 — what the centre must satisfy. A point $C = P_1 + \mathbf{d}$ is the centre of a circle through P_1, P_2, P_3 if and only if three conditions hold:

$$\mathbf{d} \cdot (\mathbf{a} \times \mathbf{b}) = 0 \quad (C \text{ lies in the plane of the three points}), \quad (13.1)$$

$$|C - P_1| = |C - P_2| \quad (C \text{ lies on the bisector plane of } P_1P_2), \quad (13.2)$$

$$|C - P_1| = |C - P_3| \quad (C \text{ lies on the bisector plane of } P_1P_3). \quad (13.3)$$

The last two rewrite conveniently. From $|C - P_2|^2 = |\mathbf{d} - \mathbf{a}|^2 = |\mathbf{d}|^2 - 2\mathbf{d} \cdot \mathbf{a} + |\mathbf{a}|^2$ and $|C - P_1|^2 = |\mathbf{d}|^2$, condition (13.2) is equivalent to

$$\mathbf{d} \cdot \mathbf{a} = \frac{|\mathbf{a}|^2}{2}, \quad \text{and likewise} \quad \mathbf{d} \cdot \mathbf{b} = \frac{|\mathbf{b}|^2}{2}.$$

Step 2 — existence and uniqueness. The three conditions form a linear system in $\mathbf{d} \in \mathbb{R}^3$, whose matrix has as rows the vectors $\mathbf{a} \times \mathbf{b}$, $\mathbf{a}$, $\mathbf{b}$. These three vectors are linearly independent: $\mathbf{a}, \mathbf{b}$ are so by hypothesis, and $\mathbf{a} \times \mathbf{b}$ is orthogonal to both and non-zero, hence does not belong to the plane they span. The matrix is therefore invertible and the system has **exactly one** solution $\mathbf{d}$: the circle, if it exists, is unique.

Step 3 — the explicit formula solves the system. Put $\mathbf{u} = |\mathbf{a}|^2\mathbf{b} - |\mathbf{b}|^2\mathbf{a}$ and $\mathbf{N} = \mathbf{u} \times (\mathbf{a} \times \mathbf{b})$, so that $\mathbf{d} = \mathbf{N}/(2|\mathbf{a} \times \mathbf{b}|^2)$.

(3a) The planarity condition (13.1) is immediate: $\mathbf{N}$ is a cross product in which $\mathbf{a} \times \mathbf{b}$ appears, hence $\mathbf{N} \perp (\mathbf{a} \times \mathbf{b})$ and a fortiori so is $\mathbf{d}$.

(3b) The bisector conditions. Use the vector triple product identity $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = \mathbf{v}(\mathbf{u} \cdot \mathbf{w}) - \mathbf{w}(\mathbf{u} \cdot \mathbf{v})$ with $\mathbf{v} = \mathbf{a}$, $\mathbf{w} = \mathbf{b}$:

$$\mathbf{N} = \mathbf{a}(\mathbf{u} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{u} \cdot \mathbf{a}).$$

Compute the two dot products, writing $\mu = \mathbf{a} \cdot \mathbf{b}$ for brevity:

$$\mathbf{u} \cdot \mathbf{b} = |\mathbf{a}|^2|\mathbf{b}|^2 - |\mathbf{b}|^2\mu = |\mathbf{b}|^2(|\mathbf{a}|^2 - \mu), \quad \mathbf{u} \cdot \mathbf{a} = |\mathbf{a}|^2\mu - |\mathbf{b}|^2|\mathbf{a}|^2 = |\mathbf{a}|^2(\mu - |\mathbf{b}|^2).$$

Hence

$$\mathbf{N} = |\mathbf{b}|^2(|\mathbf{a}|^2 - \mu)\mathbf{a} + |\mathbf{a}|^2(|\mathbf{b}|^2 - \mu)\mathbf{b},$$

and projecting on $\mathbf{a}$:

$$\begin{aligned} \mathbf{N} \cdot \mathbf{a} &= |\mathbf{b}|^2(|\mathbf{a}|^2 - \mu)|\mathbf{a}|^2 + |\mathbf{a}|^2(|\mathbf{b}|^2 - \mu)\mu \\ &= |\mathbf{a}|^2[|\mathbf{a}|^2|\mathbf{b}|^2 - |\mathbf{b}|^2\mu + |\mathbf{b}|^2\mu - \mu^2] = |\mathbf{a}|^2[|\mathbf{a}|^2|\mathbf{b}|^2 - \mu^2]. \end{aligned}$$

By Lagrange's identity, $|\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 = |\mathbf{a} \times \mathbf{b}|^2$, hence $\mathbf{N} \cdot \mathbf{a} = |\mathbf{a}|^2|\mathbf{a} \times \mathbf{b}|^2$ and

$$\mathbf{d} \cdot \mathbf{a} = \frac{|\mathbf{a}|^2|\mathbf{a} \times \mathbf{b}|^2}{2|\mathbf{a} \times \mathbf{b}|^2} = \frac{|\mathbf{a}|^2}{2}. \checkmark$$

By the symmetry of the expression for $\mathbf{N}$ under the exchange $\mathbf{a} \leftrightarrow \mathbf{b}$ (which sends $\mathbf{u}$ to $-\mathbf{u}$ and $\mathbf{a} \times \mathbf{b}$ to $-(\mathbf{a} \times \mathbf{b})$, leaving $\mathbf{N}$ unchanged), the same computation gives $\mathbf{d} \cdot \mathbf{b} = |\mathbf{b}|^2/2$. $\checkmark$

Step 4 — the radius. The triangle $P_1P_2P_3$ has sides $|\mathbf{a}|$, $|\mathbf{b}|$ and $|\mathbf{b} - \mathbf{a}|$; letting ϑ be the angle at P_1 , the side $|\mathbf{b} - \mathbf{a}|$ is opposite to it, and by the extended law of sines (Theorem 10.5) applied to the circumscribed circle, $2R = |\mathbf{b} - \mathbf{a}|/\sin \vartheta$. Since $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin \vartheta$, we have $\sin \vartheta = |\mathbf{a} \times \mathbf{b}|/(|\mathbf{a}||\mathbf{b}|)$ and

$$R = \frac{|\mathbf{b} - \mathbf{a}|}{2 \sin \vartheta} = \frac{|\mathbf{a}||\mathbf{b}||\mathbf{b} - \mathbf{a}|}{2|\mathbf{a} \times \mathbf{b}|}.$$

□

Remark 13.1: What the controller actually does. When executing a circular motion instruction, the firmware:

1. computes $\mathbf{a}, \mathbf{b}$ from the three taught points;
2. checks that $|\mathbf{a} \times \mathbf{b}|$ is not too small — if it is, the points are almost collinear and the radius blows up: this is the “invalid circle points” / “CIRC points collinear” error every vendor returns, and its real cause is the denominator in the formula of Theorem 13.1;
3. computes C and R with the closed formulas above;
4. builds an orthonormal basis of the arc plane, $\mathbf{e}_1 = (P_1 - C)/R$ and $\mathbf{e}_2 = \mathbf{n} \times \mathbf{e}_1/|\mathbf{n}|$, and generates the intermediate points as $C + R \cos \varphi \mathbf{e}_1 + R \sin \varphi \mathbf{e}_2$ as φ varies — a parametrisation by angle, hence at constant tangential speed if φ advances uniformly.

No iteration, and no transcendental function beyond cos and sin (which can be computed with CORDIC, Theorem 11.4).

Worked example: an arc in $\mathbb{R}^3$ from three taught points. An operator teaches three points (in mm, robot base frame):

$$P_1 = (600; 0; 400), \quad P_2 = (500; 100; 400), \quad P_3 = (400; 0; 400).$$

Vectors. $\mathbf{a} = (-100; 100; 0)$, $\mathbf{b} = (-200; 0; 0)$. $|\mathbf{a}|^2 = 20000$, $|\mathbf{b}|^2 = 40000$, $\mathbf{a} \cdot \mathbf{b} = 20000$.

Normal. $\mathbf{a} \times \mathbf{b} = (100 \cdot 0 - 0 \cdot 0; 0 \cdot (-200) - (-100) \cdot 0; (-100) \cdot 0 - 100 \cdot (-200)) = (0; 0; 20000)$, so $|\mathbf{a} \times \mathbf{b}| = 20000$: the arc plane is horizontal ($z = 400$), as expected.

Centre. Using the compact form obtained in Step 3, $\mathbf{N} = |\mathbf{b}|^2(|\mathbf{a}|^2 - \mu)\mathbf{a} + |\mathbf{a}|^2(|\mathbf{b}|^2 - \mu)\mathbf{b}$ with $\mu = 20000$:

$$\mathbf{N} = 40000 \cdot 0 \cdot \mathbf{a} + 20000 \cdot 20000 \cdot \mathbf{b} = 4 \times 10^8 (-200; 0; 0),$$

$$\mathbf{d} = \frac{\mathbf{N}}{2 \cdot 20000^2} = \frac{4 \times 10^8}{8 \times 10^8} (-200; 0; 0) = (-100; 0; 0),$$

whence $C = P_1 + \mathbf{d} = (500; 0; 400)$.

Radius. $|\mathbf{d}| = 100$ mm. Check with the other formula: $|\mathbf{b} - \mathbf{a}| = |(-100; -100; 0)| = 100\sqrt{2}$, hence

$$R = \frac{|\mathbf{a}||\mathbf{b}||\mathbf{b} - \mathbf{a}|}{2|\mathbf{a} \times \mathbf{b}|} = \frac{100\sqrt{2} \cdot 200 \cdot 100\sqrt{2}}{2 \cdot 20000} = \frac{4\,000\,000}{40\,000} = 100 \text{ mm.} \checkmark$$

Final check. $|P_1 - C| = |(100; 0; 0)| = 100$; $|P_2 - C| = |(0; 100; 0)| = 100$; $|P_3 - C| = |(-100; 0; 0)| = 100$. $\checkmark$ The arc is the semicircle of radius 100 mm in the plane $z = 400$, traversed through P_2 . $\square$

13.3. How the joints realise the arc: kinematics and the Jacobian

The arc of Theorem 13.1 lives in *Cartesian* space. But the robot’s motors move *joints*. This section describes the passage, which is the real reason circular interpolation is computationally heavier than linear.

Definition 13.1: Denavit–Hartenberg parameters. To each joint i of a serial manipulator one attaches a reference frame; the transformation from frame $i - 1$ to frame i is described by four parameters — a_i (link length), α_i (link twist), d_i (link offset), θ_i (joint

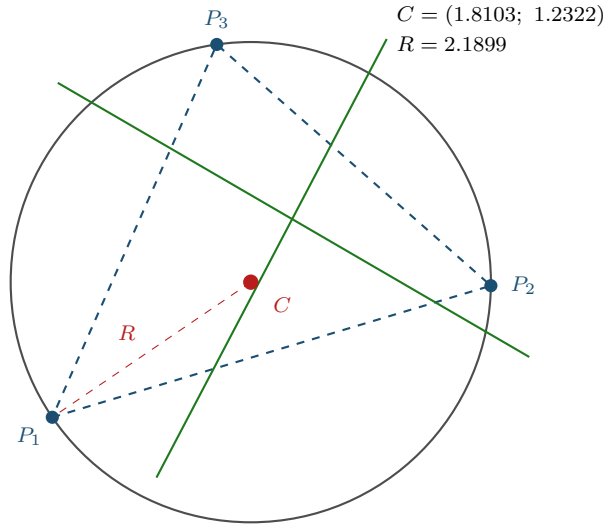


Figure 13.1: The circle through three points, the primitive of every industrial circular interpolation. The centre is the intersection of the perpendicular bisectors of any two of the three segments (green): imposing $|C - P_1|^2 = |C - P_2|^2$ and $|C - P_1|^2 = |C - P_3|^2$ the quadratic terms cancel and a linear 2×2 system remains, which has a unique solution if and only if the three points are not collinear. With $P_1 = (0, 0)$, $P_2 = (4, 1.2)$, $P_3 = (1.5, 3.4)$ one gets $C = (1.8103, 1.2322)$ and $R = 2.1899$. In space the construction is the same, performed in the plane of the three points: one takes the normalised normal $\mathbf{n} = (P_2 - P_1) \times (P_3 - P_1)$, builds an orthonormal basis of the plane and solves the planar problem; the conditioning degrades when the points are nearly collinear, and this is why controllers reject arcs whose points are too close to one another.

angle) — through the 4×4 homogeneous matrix

$${}_{i-1}T_i = \begin{pmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For a **revolute** joint, θ_i is the variable and the other three parameters are construction constants.

Remark 13.2: Forward kinematics. The pose of the tool with respect to the base is the product of the individual joint matrices,

$${}^0T_n(\boldsymbol{\theta}) = {}^0T_1(\theta_1) {}^1T_2(\theta_2) \cdots {}^{n-1}T_n(\theta_n),$$

where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_n)$. This is a **direct and always solvable** computation: given the angles, the pose is obtained with $n - 1$ multiplications of 4×4 matrices. All the difficulty lies in the inverse problem.

Definition 13.2: Manipulator Jacobian matrix. Letting $\mathbf{x} = (p_x, p_y, p_z, \phi_x, \phi_y, \phi_z)^\top$ be the tool pose (position and orientation), the **Jacobian** $J(\boldsymbol{\theta})$ is the $6 \times n$ matrix defined by

$$\dot{\mathbf{x}} = J(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}}, \quad J_{ij} = \frac{\partial x_i}{\partial \theta_j}.$$

It links joint velocities to the Cartesian velocity of the TCP.

Warning. A point of precision that popular manuals often get wrong. One sometimes reads that “the controller inverts the Jacobian every millisecond to know where to put the joints”. This is not so, and the distinction matters:

- **positional inverse kinematics** — given position and orientation, find θ — for six-axis manipulators with a *spherical wrist* (the last three axes concurrent at a point: the great majority of industrial robots) is solved in **closed analytic form**, decoupling wrist position and orientation. No iteration. This is why vendors design spherical wrists;
- the **Jacobian** enters at the level of *velocities*, for *resolved-rate* control: given the desired Cartesian velocity along the arc, the joint velocities follow by solving $J\dot{\theta} = \dot{\mathbf{x}}$.

The advantage of the analytic solution is quantitatively enormous: the literature reports computation-time savings of the order of 99.6% compared with iterative approaches, and it is what makes a kHz servo cycle possible on modest control hardware.

Proposition 13.1: Kinematic singularity. If $\det(J(\theta)) \rightarrow 0$ (for $n = 6$), there exist Cartesian directions $\dot{\mathbf{x}}$ for which the system $J\dot{\theta} = \dot{\mathbf{x}}$ has no solution, and directions for which the solution $\dot{\theta}$ diverges.

Proof. If $\det J = 0$, then J is not invertible: its image $\text{Im}(J) \subsetneq \mathbb{R}^6$ is a proper subspace. For $\dot{\mathbf{x}} \notin \text{Im}(J)$ the system is **inconsistent**: that direction of Cartesian motion is *unreachable* in that configuration, however fast the joints move.

For the divergence, take θ near but not equal to a singular configuration, and let $\sigma_{\min}(J) > 0$ be the smallest singular value of J . The solution satisfies

$$\|\dot{\theta}\| = \|J^{-1}\dot{\mathbf{x}}\| \leq \frac{\|\dot{\mathbf{x}}\|}{\sigma_{\min}(J)},$$

with equality when $\dot{\mathbf{x}}$ is aligned with the corresponding left singular vector. Approaching the singularity, $\sigma_{\min}(J) \rightarrow 0$, hence for that direction $\|\dot{\theta}\| \rightarrow \infty$: arbitrarily large joint velocities are required for a finite Cartesian velocity. $\square$

Remark 13.3: Why an arc meets them more often than a segment. A linear motion crosses a singularity only if the line meets it; a circular arc, by definition, continuously rotates the relative orientation of the links, and within one cycle may cross several critical configurations — typically the wrist singularity (axes 4 and 6 aligned) and the shoulder one. This is the cause of the “near singularity” or “joint speed limit exceeded” error appearing halfway through an arc even when the three taught points are all reachable.

The industrial remedy is DLS (Damped Least Squares, i.e. the Levenberg–Marquardt method applied to kinematics): instead of J^{-1} one uses

$$\dot{\theta} = J^{\top} (JJ^{\top} + \lambda^2 I)^{-1} \dot{\mathbf{x}},$$

with $\lambda > 0$ small. The matrix $JJ^{\top} + \lambda^2 I$ is always invertible (its eigenvalues are $\sigma_i^2 + \lambda^2 \geq \lambda^2 > 0$), so the solution always exists and stays bounded: one sacrifices a little tracking accuracy in exchange for continuity of motion. This is the trade-off implemented, under various trade names, in every modern controller.

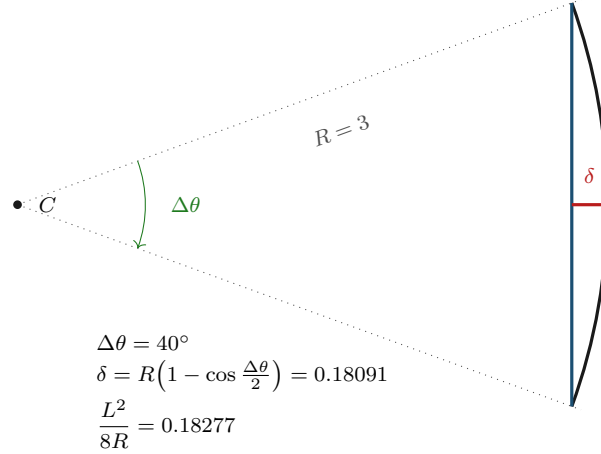


Figure 13.2: Chord error: the reason why an arc executed by a numerical controller is never an arc. The interpolator approximates the arc by a polyline of chords, one per interpolation period, and the sagitta δ between chord and arc is the resulting geometric error: $\delta = R(1 - \cos \frac{\Delta\theta}{2})$. In the example ($R = 3$, $\Delta\theta = 40^\circ$) one has $\delta = 0.18091$. Expanding the cosine in series gives the practical approximation $\delta \simeq R\Delta\theta^2/8 = L^2/(8R)$, with $L = R\Delta\theta$ the arc length travelled in one period: here 0.18277, within 1%. Since $L = vT$, inverting gives the speed constraint $v \leq \sqrt{8R\delta_{\max}}/T$: doubling the speed quadruples the chord error.

13.3.1 Closed-form inverse kinematics: the planar two-link arm

INFO

Why this section. A little above it was said that, for a six-axis spherical wrist, the positional inverse kinematics “is solved in closed form” — but without proving it, because the general six-axis proof requires wrist decoupling and more pages than an applied chapter deserves. Here the simplest nontrivial case is proved in full, the planar two-link arm, with only the tools already available in Volume I: circle geometry, the triangle inequality (A8) and the law of cosines (Theorem 10.6). No dynamics, no control: only geometry — and it is precisely geometry that reveals, with a complete proof, why the inverse kinematics sometimes admits two solutions, sometimes only one, sometimes none.

A planar two-link arm has its base at the origin O , links of lengths $\ell_1, \ell_2 > 0$, shoulder joint θ_1 (angle of the first link relative to the x axis) and elbow joint θ_2 (angle of the second link relative to the extension of the first). Writing $P_2 = \ell_1(\cos \theta_1, \sin \theta_1)$ for the elbow, the end effector is at

$$x = \ell_1 \cos \theta_1 + \ell_2 \cos(\theta_1 + \theta_2), \quad y = \ell_1 \sin \theta_1 + \ell_2 \sin(\theta_1 + \theta_2).$$

For every θ_1 , P_2 satisfies $|OP_2| = \ell_1$; for every θ_1, θ_2 , the end effector satisfies $|P_2E| = \ell_2$, where $E = (x, y)$.

Theorem 13.2: Reachability of the planar two-link arm. Setting $\rho = \sqrt{x^2 + y^2}$, the point $E = (x, y)$ is reachable (that is, there exist θ_1, θ_2 realising it) if and only if

$$|\ell_1 - \ell_2| \leq \rho \leq \ell_1 + \ell_2.$$

Proof. Necessity. If E is reachable, there exists P_2 with $|OP_2| = \ell_1$ and $|P_2E| = \ell_2$. By the triangle inequality (A8) applied to the triangle OP_2E ,

$$|OE| \leq |OP_2| + |P_2E| = \ell_1 + \ell_2,$$

and applying it in the two forms $|OP_2| \leq |OE| + |P_2E|$ and $|P_2E| \leq |OE| + |OP_2|$ gives respectively $|OE| \geq \ell_1 - \ell_2$ and $|OE| \geq \ell_2 - \ell_1$, that is $|OE| \geq |\ell_1 - \ell_2|$. Hence $\rho = |OE| \in [|\ell_1 - \ell_2|, \ell_1 + \ell_2]$.

Sufficiency. It is proved constructively in the following theorem, which exhibits the angles θ_1, θ_2 explicitly. $\square$

Remark 13.4: Geometric reading: intersection of two annuli. The workspace $W = \{(x, y) : |\ell_1 - \ell_2| \leq \rho \leq \ell_1 + \ell_2\}$ is the **circular annulus** of radii $|\ell_1 - \ell_2|$ and $\ell_1 + \ell_2$: it is the set of points reached by a circle of radius ℓ_2 whose centre travels along the circle of radius ℓ_1 (the elbow). It is a fact of pure circle geometry — the subject of the treatise — that here acquires a mechanical meaning.

Theorem 13.3: Closed-form inverse kinematics. Let $E = (x, y) \neq O$ be reachable, $\rho = \sqrt{x^2 + y^2}$, $\varphi = \text{atan2}(y, x)$. Setting

$$c := \frac{\ell_1^2 + \rho^2 - \ell_2^2}{2\ell_1\rho}, \quad \gamma := \arccos(c) \in [0, \pi],$$

for each of the two sign choices $\theta_1 = \varphi \pm \gamma$, the angle

$$\theta_2 = \text{atan2}(y - \ell_1 \sin \theta_1, x - \ell_1 \cos \theta_1) - \theta_1$$

(taken modulo 2π) is well defined and the pair (θ_1, θ_2) realises E .

Proof. Step 1 — $c \in [-1, 1]$. Since E is reachable, $|\ell_1 - \ell_2| \leq \rho \leq \ell_1 + \ell_2$ (Theorem 13.2). Then

$$c \leq 1 \iff \ell_1^2 + \rho^2 - \ell_2^2 \leq 2\ell_1\rho \iff (\ell_1 - \rho)^2 \leq \ell_2^2 \iff |\ell_1 - \rho| \leq \ell_2,$$

that is $\ell_1 - \rho \leq \ell_2$ and $\rho - \ell_1 \leq \ell_2$: the first follows from $\rho \geq |\ell_1 - \ell_2| \geq \ell_1 - \ell_2$ (giving $\ell_1 - \rho \leq \ell_2$), the second from $\rho \leq \ell_1 + \ell_2$ (giving $\rho - \ell_1 \leq \ell_2$) — both true by hypothesis. Similarly

$$c \geq -1 \iff \ell_1^2 + \rho^2 - \ell_2^2 \geq -2\ell_1\rho \iff (\ell_1 + \rho)^2 \geq \ell_2^2 \iff \ell_1 + \rho \geq \ell_2,$$

true because $\rho \geq |\ell_1 - \ell_2| \geq \ell_2 - \ell_1$. Hence $c \in [-1, 1]$ and $\gamma = \arccos(c)$ is well defined.

Step 2 — verifying $|P_2E| = \ell_2$ for $\theta_1 = \varphi \pm \gamma$. With $P_2 = \ell_1(\cos \theta_1, \sin \theta_1)$ and $E = \rho(\cos \varphi, \sin \varphi)$,

$$|P_2E|^2 = |P_2|^2 + |E|^2 - 2P_2 \cdot E = \ell_1^2 + \rho^2 - 2\ell_1\rho \cos(\theta_1 - \varphi) = \ell_1^2 + \rho^2 - 2\ell_1\rho \cos(\pm\gamma),$$

and $\cos(\pm\gamma) = \cos \gamma = c$ (cosine is even), hence

$$|P_2E|^2 = \ell_1^2 + \rho^2 - 2\ell_1\rho c = \ell_1^2 + \rho^2 - (\ell_1^2 + \rho^2 - \ell_2^2) = \ell_2^2.$$

So $|P_2E| = \ell_2$ for both sign choices.

Step 3 — constructing θ_2 . By Step 2, $E \neq P_2$ (they are at distance $\ell_2 > 0$), so $\text{atan2}(y - \ell_1 \sin \theta_1, x - \ell_1 \cos \theta_1)$, the angle of the vector $E - P_2$, is well defined; call it θ_{12} . Setting $\theta_2 = \theta_{12} - \theta_1$ gives, by construction,

$$\ell_2 \cos(\theta_1 + \theta_2) = \ell_2 \cos \theta_{12} = x - \ell_1 \cos \theta_1, \quad \ell_2 \sin(\theta_1 + \theta_2) = y - \ell_1 \sin \theta_1,$$

(from the definition of atan2 and from $|E - P_2| = \ell_2$), that is

$$\ell_1 \cos \theta_1 + \ell_2 \cos(\theta_1 + \theta_2) = x, \quad \ell_1 \sin \theta_1 + \ell_2 \sin(\theta_1 + \theta_2) = y.$$

(θ_1, θ_2) realises E , as required. This also completes the sufficiency of Theorem 13.2. $\square$

Remark 13.5: Elbow up and elbow down. The two sign choices of Theorem 13.3 give, in general, two distinct configurations: call **elbow up** ($\theta_1 = \varphi + \gamma$, the elbow rotated anti-clockwise relative to the direction OE) and **elbow down** ($\theta_1 = \varphi - \gamma$, clockwise) the one with P_2 respectively to the left or to the right of the ray OE (the nomenclature is the one in current use in the robotics literature, which typically assumes the target in front of the robot with the y axis vertical). The two solutions are mirror images of each other about the line OE , share the same $|\theta_2|$, and have θ_2 of opposite sign.

Remark 13.6: The singularity is exactly the boundary of the workspace. The two solutions of Theorem 13.3 **coincide** exactly when $\gamma = 0$, that is $c = 1$, that is — retracing Step 1 backward — when $\rho = \ell_1 + \ell_2$ (fully extended arm) or $\rho = |\ell_1 - \ell_2|$ (folded arm): the two limiting cases already met in the proof of reachability, that is exactly the **boundary** of the circular annulus W . This is the special case, provable in full, of the general Proposition 13.1: the Jacobian of the two-link arm is

$$J = \begin{pmatrix} -\ell_1 \sin \theta_1 - \ell_2 \sin(\theta_1 + \theta_2) & -\ell_2 \sin(\theta_1 + \theta_2) \\ \ell_1 \cos \theta_1 + \ell_2 \cos(\theta_1 + \theta_2) & \ell_2 \cos(\theta_1 + \theta_2) \end{pmatrix},$$

and a direct computation (expanding the determinant and using $\sin(A - B) = \sin A \cos B - \cos A \sin B$ with $A = \theta_1 + \theta_2$, $B = \theta_1$) gives $\det J = \ell_1 \ell_2 \sin \theta_2$. Hence $\det J = 0 \iff \theta_2 \in \{0, \pi\} \iff$ fully extended or folded arm $\iff \rho \in \{\ell_1 + \ell_2, |\ell_1 - \ell_2|\}$: the kinematic singularity (Proposition 13.1) and the boundary of the workspace are, for the two-link arm, **the same set** — a fact of circle geometry (the boundary of a circular annulus) that manifests itself as a critical kinematic event.

Exercise 13.1. A planar arm has $\ell_1 = 3$, $\ell_2 = 2$. Find both inverse-kinematics solutions for the end effector at $(x, y) = (4, 1)$, and verify the result by substituting back into the forward kinematics.

Solution. Reachability. $\rho = \sqrt{16 + 1} = \sqrt{17} = 4.123106$; with $|\ell_1 - \ell_2| = 1$ and $\ell_1 + \ell_2 = 5$: $1 \leq 4.123106 \leq 5$ ✓, the point is reachable.

Angles. $\varphi = \text{atan2}(1, 4) = 14.036243^\circ$.

$$c = \frac{9 + 17 - 4}{2 \cdot 3 \cdot \sqrt{17}} = \frac{22}{6\sqrt{17}} = 0.889297, \quad \gamma = \arccos(0.889297) = 27.214923^\circ.$$

Elbow up: $\theta_1 = 14.036243^\circ + 27.214923^\circ = 41.251166^\circ$; $P_2 = (2.255479, 1.978083)$; $E - P_2 = (1.744521, -0.978083)$, whence $\theta_1 + \theta_2 = \text{atan2}(-0.978083, 1.744521) = -29.277613^\circ$, so $\theta_2 = -29.277613^\circ - 41.251166^\circ = -70.528779^\circ$.

Elbow down: $\theta_1 = 14.036243^\circ - 27.214923^\circ = -13.178679^\circ$; $P_2 = (2.920991, -0.683966)$; $E - P_2 = (1.079009, 1.683966)$, whence $\theta_1 + \theta_2 = \text{atan2}(1.683966, 1.079009) = 57.350100^\circ$, so $\theta_2 = 57.350100^\circ - (-13.178679^\circ) = 70.528779^\circ$.

Double numerical check (forward kinematics). Elbow up: $x = 3 \cos 41.251166^\circ + 2 \cos(41.251166^\circ - 70.528779^\circ) = 2.255479 + 1.744521 = 4.000000$; $y = 3 \sin 41.251166^\circ + 2 \sin(-29.277613^\circ) = 1.978083 - 0.978083 = 1.000000$. ✓ Elbow down: $x = 3 \cos(-13.178679^\circ) + 2 \cos 57.350100^\circ = 2.920991 + 1.079009 = 4.000000$; $y = 3 \sin(-13.178679^\circ) + 2 \sin 57.350100^\circ = -0.683966 + 1.683966 = 1.000000$. ✓ Both solutions return exactly to the target (zero difference within the precision reported), as must be the case by Steps 2–3 of the proof of Theorem 13.3.

Exercise 13.2. With the same arm ($\ell_1 = 3$, $\ell_2 = 2$), can the end effector reach $(x, y) = (5, 0.5)$? Apply Theorem 13.2 to answer without computing the angles.

Solution. $\rho = \sqrt{25 + 0.25} = \sqrt{25.25} = 5.024938$. Comparing with $\ell_1 + \ell_2 = 5$: $\rho = 5.024938 > 5$, so the point is **just outside** the workspace (by about 0.5% of the maximum radius): it is not reachable, by Theorem 13.2. A real controller would flag a “target out of range” error here before even attempting the inverse kinematics.

Example 13.1: A complete case: figures representative of industrial scale. The preceding exercises use small lengths convenient for hand computation. Here the same procedure is repeated — entirely based on Theorems 13.2 and 13.3, already proved in full — with figures **representative of a planar arm of typical industrial scale**: $\ell_1 = 0.45$ m, $\ell_2 = 0.35$ m (total reach $\ell_1 + \ell_2 = 0.80$ m), not attributed to any specific product. Target: $E = (x, y) = (0.5, 0.3)$ m.

Step 1 — reachability. $\rho = \sqrt{0.5^2 + 0.3^2} = \sqrt{0.34} = 0.583095$ m. With $|\ell_1 - \ell_2| = 0.10$ m and $\ell_1 + \ell_2 = 0.80$ m: $0.10 \leq 0.583095 \leq 0.80$ ✓, by Theorem 13.2 the point is reachable (Figure 13.4).

Step 2 — the two auxiliary angles. $\varphi = \text{atan2}(0.3, 0.5) = 30.963757^\circ$;

$$c = \frac{0.45^2 + 0.583095^2 - 0.35^2}{2 \cdot 0.45 \cdot 0.583095} = 0.800327, \quad \gamma = \arccos(0.800327) = 36.838686^\circ.$$

Step 3 — elbow up. $\theta_1 = 30.963757^\circ + 36.838686^\circ = 67.802442^\circ$; $P_2^\uparrow = (0.170011, 0.416649)$ m; $E - P_2^\uparrow = (0.329989, -0.116649)$, hence $\theta_1 + \theta_2 = \text{atan2}(-0.116649, 0.329989) = -19.468155^\circ$, so $\theta_2 = -19.468155^\circ - 67.802442^\circ = -87.270597^\circ$.

Step 4 — elbow down. $\theta_1 = 30.963757^\circ - 36.838686^\circ = -5.874929^\circ$; $P_2^\downarrow = (0.447636, -0.046061)$ m; $E - P_2^\downarrow = (0.052364, 0.346061)$, hence $\theta_1 + \theta_2 = \text{atan2}(0.346061, 0.052364) = 81.395668^\circ$, so $\theta_2 = 81.395668^\circ - (-5.874929^\circ) = 87.270597^\circ$.

Step 5 — cross-check (forward kinematics). Elbow up: $x = 0.45 \cos 67.802442^\circ + 0.35 \cos(-19.468155^\circ) = 0.170011 + 0.329989 = 0.500000$ m; $y = 0.45 \sin 67.802442^\circ + 0.35 \sin(-19.468155^\circ) = 0.416649 - 0.116649 = 0.300000$ m. ✓ Elbow down: $x = 0.45 \cos(-5.874929^\circ) + 0.35 \cos 81.395668^\circ = 0.447636 + 0.052364 = 0.500000$ m; $y = 0.45 \sin(-5.874929^\circ) + 0.35 \sin 81.395668^\circ = -0.046061 + 0.346061 = 0.300000$ m. ✓ Both solutions return exactly to the target, and both have $|\theta_2| = 87.270597^\circ$ with opposite sign, as predicted by Observation 13.5 (Figure 13.3). Independent verification script: `cap11_braccio2link_esempio.py`.

13.4. The mobile robot: arcs and the instantaneous centre of rotation

The industrial arms of the previous sections move a TCP along an arc while keeping fixed the member that carries the robot (the base). A wheeled mobile robot does the opposite: it is the whole vehicle that travels arcs of a circle on the floor. The link between the wheel (or steering) speeds and the radius of the arc travelled is, once again, circle geometry.

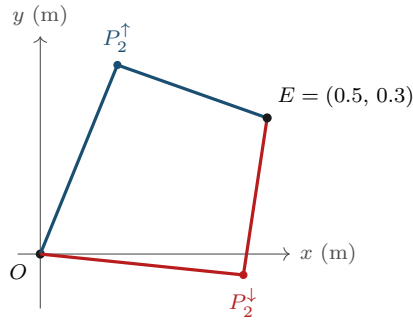


Figure 13.3: The two inverse-kinematics solutions of Example 13.1 for the same target E : elbow up (blue, $P_2^\uparrow$) and elbow down (red, $P_2^\downarrow$), mirror images with respect to line OE , as proved in Observation 13.5.

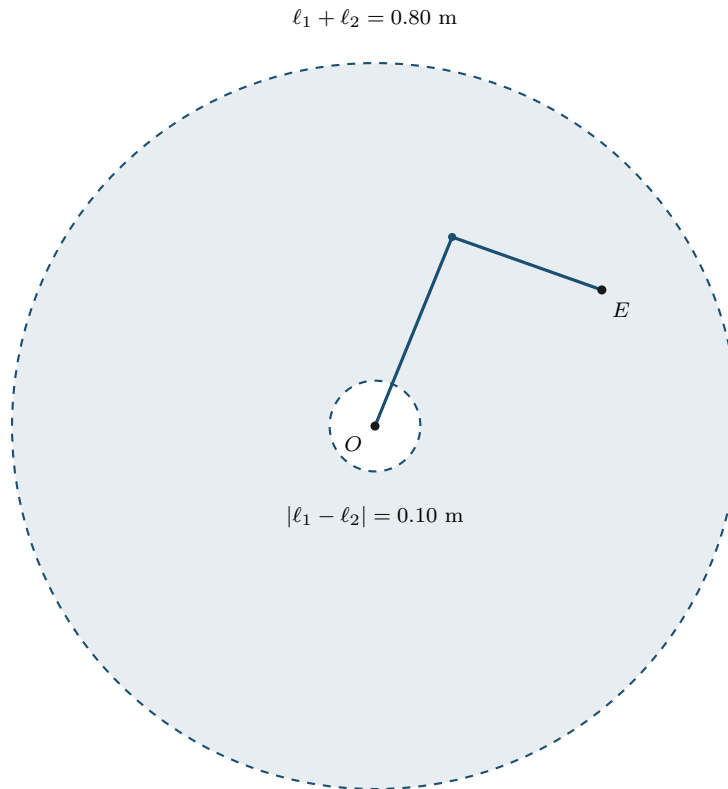


Figure 13.4: The workspace W of Observation 13.4: the circular annulus between $|\ell_1 - \ell_2| = 0.10$ m and $\ell_1 + \ell_2 = 0.80$ m. The arm in the elbow-up configuration reaches the target E of Example 13.1, well inside the annulus.

13.4.1 Differential drive

Definition 13.3: Differential-drive vehicle. A *differential-drive* vehicle has two independent driving wheels, placed on the same axle, at distance b (track) from each other, which roll without slipping at peripheral speeds v_r (right wheel) and v_l (left wheel).

Theorem 13.4: The instantaneous centre of rotation. If $v_r \neq v_l$, the differential-drive vehicle rotates, at every instant, about a point ICR (instantaneous centre of rotation)

lying on the line of the wheel axle, at distance

$$\rho = \frac{b}{2} \cdot \frac{v_r + v_l}{v_r - v_l}$$

from the midpoint of the axle (with sign: $\rho > 0$ if the ICR is on the side of the left wheel), with angular velocity

$$\omega = \frac{v_r - v_l}{b}.$$

If $v_r = v_l$ the motion is rectilinear (ICR at infinity, $\omega = 0$).

Proof. Rolling without slipping requires every point of the rigid vehicle to move, at every instant, with circular motion about a single instantaneous centre ICR common to the whole body, with a single angular velocity ω (this is the theorem of the instantaneous centre of rotation for a rigid body in the plane). The two wheels, being rigidly attached to the vehicle, therefore have peripheral speeds

$$v_r = \omega \left(\rho + \frac{b}{2} \right), \quad v_l = \omega \left(\rho - \frac{b}{2} \right),$$

where $\rho + \frac{b}{2}$ and $\rho - \frac{b}{2}$ are the (signed) distances of the two wheels from the ICR along the common axle. Subtracting, $v_r - v_l = \omega b$, whence $\omega = (v_r - v_l)/b$; adding, $v_r + v_l = 2\omega\rho$, whence $\rho = (v_r + v_l)/(2\omega)$, which on substituting ω gives the stated formula. If $v_r = v_l$ the two equations give $\omega = 0$: no rotation, uniform rectilinear motion. $\square$

Example 13.2: Numerical check. Track $b = 0.5$ m, $v_r = 1.2$ m/s, $v_l = 0.8$ m/s. Then

$$\omega = \frac{1.2 - 0.8}{0.5} = 0.8 \text{ rad/s}, \quad \rho = \frac{0.5}{2} \cdot \frac{1.2 + 0.8}{1.2 - 0.8} = 1.25 \text{ m}.$$

Reverse check: $\omega(\rho + b/2) = 0.8 \cdot 1.5 = 1.2 = v_r$ ✓ and $\omega(\rho - b/2) = 0.8 \cdot 1.0 = 0.8 = v_l$ ✓ (script `fase4_a04_a07.py`).

13.4.2 Ackermann steering

A four-wheeled vehicle with steering (like a car) cannot, in general, steer the two front wheels by the same angle: if it did, the two wheels would follow arcs of different radius about two different centres, and at least one would slip.

Theorem 13.5: Ackermann condition. A four-wheeled vehicle, wheelbase L (distance between front and rear axle) and front track w , with fixed rear wheels and front wheels steering at angles δ_i (wheel on the inside of the curve) and δ_o (outside wheel), rolls without slipping along a circular curve if and only if

$$\cot \delta_o - \cot \delta_i = \frac{w}{L}.$$

Proof. Rolling without slipping requires the velocity of each wheel to be perpendicular to its axis of rotation (the wheel's own axle); this is equivalent to saying that the axes of all four wheels pass through a single point, the ICR, located on the line of the rear axle (the rear wheels are fixed and parallel to the vehicle's axis, so the ICR lies on their common perpendicular, that is, on the line of the rear axle itself). Let R be the distance from the ICR to the midpoint of the rear axle. The inner front wheel is at distance, along the front axle, $R - w/2$ from the projection of the ICR; the right triangle formed by its axle (which points to the ICR), the wheelbase L and

this distance gives $\cot \delta_i = (R - w/2)/L$. Likewise for the outer wheel, $\cot \delta_o = (R + w/2)/L$. Subtracting the two relations, the term in R cancels: $\cot \delta_o - \cot \delta_i = w/L$, independent of R (that is, of the instantaneous radius of curvature). $\square$

Example 13.3: Numerical check. Wheelbase $L = 2.5$ m, track $w = 1.5$ m, ICR at $R = 6$ m from the midpoint of the rear axle. Then

$$\delta_i = \arctan \frac{L}{R - w/2} = \arctan \frac{2.5}{5.25} = 25.463345^\circ, \quad \delta_o = \arctan \frac{L}{R + w/2} = \arctan \frac{2.5}{6.75} = 20.323137^\circ.$$

Check: $\cot \delta_o - \cot \delta_i = 0.600000 = w/L = 1.5/2.5 = 0.600000 \checkmark$ (script `fase4_a04_a07.py`).

13.4.3 Pure pursuit tracking

An elementary path-tracking algorithm for autonomous driving chooses, at every instant, a target point at a fixed distance along the path and makes the vehicle follow the unique circular arc that, starting tangent to the current heading, passes through that point.

Proposition 13.2: Radius of the pursuit arc. *In the reference frame attached to the vehicle (origin at the vehicle's reference point, x axis along the heading), let (x, y) be a target point at pursuit distance $L_d = \sqrt{x^2 + y^2}$. The unique circular arc tangent to the x axis at the origin and passing through (x, y) has radius (signed, positive if the centre is on the side $y > 0$)*

$$\rho = \frac{L_d^2}{2y}, \quad \text{that is, curvature } \kappa = \frac{1}{\rho} = \frac{2y}{L_d^2}.$$

Proof. A circle tangent to the x axis at the origin has centre on the y axis, at a point $(0, \rho)$, and radius $|\rho|$ (tangency at O : the radius O –centre is perpendicular to the tangent, that is, vertical). Imposing passage through (x, y) gives

$$x^2 + (y - \rho)^2 = \rho^2 \implies x^2 + y^2 - 2y\rho = 0 \implies \rho = \frac{x^2 + y^2}{2y} = \frac{L_d^2}{2y}.$$

$\square$

Example 13.4: Numerical check. Pursuit distance $L_d = 3$ m, target with lateral offset $y = 0.5$ m (hence $x = \sqrt{L_d^2 - y^2} = \sqrt{8.75} = 2.958040$ m). Then

$$\rho = \frac{3^2}{2 \cdot 0.5} = 9 \text{ m}.$$

Check: $x^2 + (\rho - y)^2 = 8.75 + 72.25 = 81.00 = \rho^2 \checkmark$ (script `fase4_a04_a07.py`).

Remark 13.7: Dubins curves. *If the vehicle has a minimum steering radius $\rho_{\min}$ (it cannot turn tighter than this), the shortest path between two assigned position-orientation configurations, for a vehicle that moves only forward, is always a concatenation of at most three segments, each either an arc of radius $\rho_{\min}$ or a straight segment (six possible types, CSC such as arc-segment-arc or CCC such as arc-arc-arc). We import this result without proof (source F-67, Dubins 1957); the proof uses the calculus of variations and is beyond this volume's tools. It is the theoretical foundation of trajectory planning for steered vehicles (mobile robots, autonomous cars).*

An **RSR** case (right–straight–right, two arcs traversed clockwise joined by a segment) is exactly the construction of the **external common tangents** to two circles seen in Theorem 3.21 (N-02): if the vehicle, starting from a point with a given heading, curves at the minimum radius $\rho_{\min}$ to the right up to a circle C_1 of radius $\rho_{\min}$, and must arrive tangent to a second circle C_2 (radius $\rho_{\min}$) still curving to the right, the intermediate straight segment is precisely the external common tangent to C_1, C_2 on the correct side. For example, with $\rho_{\min} = 2$ m and centres $C_1 = (0,0)$, $C_2 = (10,0)$ (same radius, non-intersecting circles): the external common tangent is parallel to the line joining the centres, at distance $\rho_{\min} = 2$ m, so the straight segment is exactly 10 m long and the total path is two arcs (of extent depending on the initial and final heading) plus this 10 m segment.

13.5. Mechanisms: slider-crank and the four-bar linkage

Not every circular motion in automation is directly driven by a motor: many arise from *mechanisms* that turn a rotation into another motion by means of hinged rigid rods. The circle remains at the centre of the description.

Proposition 13.3: Slider position in the slider-crank mechanism. In the slider-crank mechanism with crank of length r (rotating by an angle θ about a fixed pivot) and connecting rod of length ℓ (hinged at the end of the crank and to a slider constrained to move along the line through the pivot), the position of the slider along that line, measured from the pivot, is

$$x(\theta) = r \cos \theta + \sqrt{\ell^2 - r^2 \sin^2 \theta}.$$

The stroke of the slider (difference between maximum and minimum position) is $2r$.

Proof. The crank pin traces the circle $(r \cos \theta, r \sin \theta)$. The slider lies on the horizontal axis, at a point $(x, 0)$ such that the distance between $(x, 0)$ and $(r \cos \theta, r \sin \theta)$ is exactly ℓ (length of the connecting rod): x is the foot of the rod. The triangle with vertices crank pivot $(0, 0)$, crank pin $(r \cos \theta, r \sin \theta)$ and slider $(x, 0)$ has height $r \sin \theta$ (ordinate of the crank pin) and base on the x axis split into $r \cos \theta$ (from the fixed pivot to the projection of the crank pin) and the remaining leg, of length $\sqrt{\ell^2 - (r \sin \theta)^2}$ by Pythagoras' theorem applied to the right triangle with hypotenuse ℓ and leg $r \sin \theta$: hence $x = r \cos \theta + \sqrt{\ell^2 - r^2 \sin^2 \theta}$. For the stroke: x is maximum at $\theta = 0$, where $x = r + \ell$, and minimum at $\theta = \pi$, where $x = -r + \ell$; the difference is $2r$. $\square$

Proposition 13.4: Grashof condition. In a planar four-bar linkage with four sides (rigid rods) hinged at their ends, of lengths $s \leq p, q \leq l$ (the shortest s , the longest l , the other two p, q), the shortest rod s can complete a full rotation about its own fixed hinge if and only if

$$s + l \leq p + q.$$

Necessity of the condition. If rod s completes a full rotation, it passes in particular through the two positions where it is **aligned** with the fixed rod l (once with s, l on the same side, the other with them on opposite sides). In the position with s and l aligned and overlapping (the moving hinge of s as close as possible to the opposite one of l), the distance between the two remaining moving hinges is $l - s$, and this distance is also the diagonal of the quadrilateral of the other two sides p, q : by the triangle inequality, $l - s \leq p + q$, that is, $s + l \leq p + q + 2s$ (always true if $s + l \leq p + q$, not yet the claim). In the position with s, l aligned but extended (the two moving hinges as far apart as possible), the distance is $l + s$,

and it is again a diagonal of the quadrilateral p, q : by the triangle inequality, $l + s \leq p + q$. This is exactly the Grashof condition. If s must be able to reach *both* aligned positions (necessary for a full rotation, which passes through both), $l + s \leq p + q$ must hold, that is, the stated condition is necessary. $\square$

Remark 13.8: Sufficiency of the Grashof condition. That the condition $s + l \leq p + q$ is also **sufficient** (that is, that in that case the short rod does manage to complete the turn, and is not merely formally unobstructed by the triangle inequality) requires an analysis of the continuity of the mechanism's motion throughout the turn, which is beyond this volume's tools. **We import** this direction without proof (source F-68: Uicker, Pennock, Shigley, *Theory of Machines and Mechanisms*).

The four-bar linkage also underlies the Peaucellier invensor of Theorem 5.14 (N-07): that mechanism is a special case, non-Grashof in the classical sense, designed specifically to turn an arc into a line rather than to make a rod rotate fully.

13.6. Gears: pitch circles and epicyclic gear trains

Proposition 13.5: Transmission ratio between two gear wheels. Two gear wheels meshing correctly roll, without slipping, along their respective pitch circles of radii r_1, r_2 . It follows that the angular velocities satisfy

$$\omega_1 r_1 = \omega_2 r_2.$$

If p is the common pitch (tooth length measured on the pitch circle) and z_i the number of teeth of wheel i , then $z_i = 2\pi r_i/p$ and

$$\frac{\omega_1}{\omega_2} = \frac{r_2}{r_1} = \frac{z_2}{z_1}.$$

Proof. Rolling without slipping between the two pitch circles requires the two points of contact to have the same peripheral speed: $\omega_1 r_1 = \omega_2 r_2$ (the same identity already seen for two wheels in pure rolling contact in §13.4). Since $z_i = 2\pi r_i/p$ gives $r_i = z_i p/2\pi$, substituting gives $\omega_1 z_1 = \omega_2 z_2$, that is, $\omega_1/\omega_2 = z_2/z_1$. $\square$

Theorem 13.6: Willis's formula for the epicyclic gear train. In an epicyclic gear train with sun gear (central) of z_s teeth, ring gear (internal toothed ring) of z_r teeth, one or more planet gears carried by a planet carrier rotating at angular velocity ω_c , and angular velocities ω_s, ω_r of sun and ring:

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{z_r}{z_s}.$$

Proof. In the reference frame attached to the carrier (rotating with it at speed ω_c), the axes of sun, planets and ring are fixed: the gear train behaves like an *ordinary* train (gears with fixed axes). In this frame the relative angular velocities are $\omega'_s = \omega_s - \omega_c$ and $\omega'_r = \omega_r - \omega_c$ (composition of rotations about the same axis: subtract the frame's common rotation). The ordinary transmission ratio from sun to ring, through the intermediate planet, is, by Proposition 13.5 applied twice (sun-planet and planet-ring) and composed: $\omega'_r/\omega'_s = -z_s/z_r$ (the minus sign because the planet, meshing externally with the sun and internally with the

ring, reverses the direction of rotation between the two). Hence

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = \frac{\omega'_s}{\omega'_r} = -\frac{z_r}{z_s}.$$

□

Example 13.5: Epicyclic gear train with fixed ring. Sun $z_s = 24$ teeth, ring $z_r = 72$ teeth, fixed ring ($\omega_r = 0$), carrier as output. From Theorem 13.6,

$$\frac{\omega_s - \omega_c}{0 - \omega_c} = -\frac{72}{24} = -3 \implies \omega_s - \omega_c = 3\omega_c \implies \frac{\omega_s}{\omega_c} = 1 + \frac{z_r}{z_s} = 1 + 3 = 4.$$

With a fixed ring and input on the sun, the carrier therefore rotates at $\omega_c = \omega_s/4$: a 4:1 reducer (checked with the script `fase4_a04_a07.py`). Sources: F-68 (Uicker–Pennock–Shigley) for gear train theory, F-69 (tec-science.com) as a popular-level cross-reference.

13.7. How fast can an arc be traversed?

The previous sections say *where* the path goes. This one says *how fast* it can be travelled — and the answer is not a convenience parameter, but the minimum of three physical limits, all provable in a few lines and all depending on the radius.

13.7.1 First limit: the chord error

Theorem 13.7: Chord error and maximum speed. *An interpolator sampling the path with period T at constant tangential speed v approximates the arc of radius ρ by chords of length $L = vT$. The maximum deviation between chord and arc (the **sagitta**, or chord error) is*

$$\delta = \rho - \sqrt{\rho^2 - \frac{L^2}{4}},$$

and for $\delta \ll \rho$ one has the notable approximation

$$\delta \simeq \frac{L^2}{8\rho}$$

Imposing $\delta \leq \delta_{\max}$ is equivalent to the speed constraint

$$v \leq v_{\text{chord}} = \frac{2}{T} \sqrt{2\rho \delta_{\max} - \delta_{\max}^2} \simeq \frac{2}{T} \sqrt{2\rho \delta_{\max}}.$$

Proof. Step 1 — the sagitta. Let AB be the chord of length L on the arc of centre O and radius ρ , and M its midpoint. By Theorem 3.2 (Chapter 3) the radius OM is perpendicular to the chord, so triangle OMA is right-angled at M with hypotenuse $OA = \rho$ and leg $MA = L/2$. By the Pythagorean theorem (Theorem 10.1),

$$|OM| = \sqrt{\rho^2 - \frac{L^2}{4}}.$$

The maximum deviation between chord and arc occurs at the point of the arc farthest from the chord, i.e. on the extension of OM : it equals $\delta = \rho - |OM|$, which is the first formula.

Step 2 — the approximation. Rationalising,

$$\delta = \frac{L^2/4}{\rho + \sqrt{\rho^2 - L^2/4}}.$$

For $L \ll \rho$ the denominator tends to 2ρ , whence $\delta \simeq L^2/(8\rho)$.

Step 3 — inversion. From $\delta = \rho - \sqrt{\rho^2 - L^2/4}$ we get $\sqrt{\rho^2 - L^2/4} = \rho - \delta$; squaring (legitimate since $\rho > \delta$),

$$\rho^2 - \frac{L^2}{4} = \rho^2 - 2\rho\delta + \delta^2 \implies L = 2\sqrt{2\rho\delta - \delta^2}.$$

Substituting $L = vT$ and imposing $\delta \leq \delta_{\max}$ (the map $\delta \mapsto L$ being increasing on $[0, \rho]$) gives the claim. $\square$

13.7.2 Second limit: centripetal acceleration

Proposition 13.6: Centripetal limit. Traversing an arc of radius ρ at constant tangential speed v , the acceleration is purely normal and equals $a_n = v^2/\rho$. Imposing $a_n \leq a_{\max}$ is equivalent to

$$v \leq v_{\text{centr}} = \sqrt{\rho a_{\max}}.$$

Proof. Parametrise the motion by angle: $\mathbf{p}(t) = C + \rho(\cos \omega t, \sin \omega t)$ with $\omega = v/\rho$ constant. Differentiating twice,

$$\dot{\mathbf{p}} = \rho\omega(-\sin \omega t, \cos \omega t), \quad \ddot{\mathbf{p}} = -\rho\omega^2(\cos \omega t, \sin \omega t).$$

The first has modulus $\rho\omega = v$ (consistent); the second has modulus $\rho\omega^2 = v^2/\rho$ and points to the centre. Imposing $v^2/\rho \leq a_{\max}$ gives $v \leq \sqrt{\rho a_{\max}}$. $\square$

13.7.3 Third limit: curvature jerk

Theorem 13.8: Jerk limit on a circular path. Even at **constant speed**, motion on an arc produces a non-zero jerk (derivative of acceleration), of magnitude

$$|\ddot{\mathbf{p}}| = \frac{v^3}{\rho^2}.$$

Imposing $|\ddot{\mathbf{p}}| \leq J_{\max}$ is equivalent to

$$v \leq v_{\text{jerk}} = (J_{\max} \rho^2)^{1/3}.$$

Proof. Differentiating the parametrisation of Proposition 13.6 a third time:

$$\ddot{\mathbf{p}} = \rho\omega^3(\sin \omega t, -\cos \omega t), \quad |\ddot{\mathbf{p}}| = \rho\omega^3 = \rho \left(\frac{v}{\rho}\right)^3 = \frac{v^3}{\rho^2}.$$

The inequality $v^3/\rho^2 \leq J_{\max}$ gives the claim on taking cube roots (an increasing function).

Physical reading. The acceleration vector has constant modulus v^2/ρ but **rotates** with angular velocity $\omega = v/\rho$: its derivative is the product of the modulus by the rotation rate, $(v^2/\rho)(v/\rho) = v^3/\rho^2$. This is a jerk that appears in no straight segment, however fast. $\square$

Theorem 13.9: The admissible speed on an arc. The maximum tangential speed compatible with all three constraints is

$$v_{\max}(\rho) = \min \left\{ v_{\text{prog}}, \underbrace{\frac{2}{T} \sqrt{2\rho\delta_{\max}}}_{\propto \rho^{1/2}}, \underbrace{\sqrt{\rho a_{\max}}}_{\propto \rho^{1/2}}, \underbrace{(J_{\max}\rho^2)^{1/3}}_{\propto \rho^{2/3}} \right\}.$$

Moreover:

- (a) the **chord** constraint dominates the **centripetal** one if and only if $\delta_{\max} < \frac{a_{\max}T^2}{8}$, independently of the radius;
- (b) the **jerk** constraint dominates the **centripetal** one if and only if $\rho < \frac{a_{\max}^3}{J_{\max}^2}$.

Proof. The minimum formula is immediate: the three constraints must hold simultaneously together with the programmed speed.

(a) Compare the first two:

$$\frac{2}{T} \sqrt{2\rho\delta_{\max}} \leq \sqrt{\rho a_{\max}} \iff \frac{8\delta_{\max}}{T^2} \leq a_{\max} \iff \delta_{\max} \leq \frac{a_{\max}T^2}{8},$$

where dividing by $\rho > 0$ was possible: the radius **disappears**, because both limits grow as $\sqrt{\rho}$.

(b) Compare the third with the second:

$$(J_{\max}\rho^2)^{1/3} \leq \sqrt{\rho a_{\max}} \iff J_{\max}^2\rho^4 \leq \rho^3 a_{\max}^3 \iff \rho \leq \frac{a_{\max}^3}{J_{\max}^2},$$

having raised both sides to the sixth power (both positive) and divided by $\rho^3 > 0$. $\square$

Remark 13.9: What (a) and (b) mean on the shop floor. Point (a) is counter-intuitive and useful: if on your controller the chord constraint dominates the centripetal one, it will dominate at every radius — no need to try different arcs. The discriminant is only $\delta_{\max}$ against $a_{\max}T^2/8$: speed is gained either by loosening the chord tolerance, or by shortening the interpolation period, which enters squared.

Point (b) says instead that the jerk constraint is the bottleneck **on tight blends**: below the critical radius $\rho_c = a_{\max}^3/J_{\max}^2$ it is the one in charge. This is why a profile with small-radius blends is machined far more slowly than acceleration alone would suggest — and why modern CAM systems replace overly tight blends with curvature-continuous curves.

Example 13.6: The three limiting speeds, computed. A machining centre has interpolation period $T = 1$ ms, chord tolerance $\delta_{\max} = 5 \mu\text{m} = 0.005$ mm, maximum acceleration $a_{\max} = 5000$ mm/s² and maximum jerk $J_{\max} = 50\,000$ mm/s³. The programmed feedrate is $v_{\text{prog}} = 3000$ mm/min = 50 mm/s.

Step 1 — the critical jerk radius (Theorem 13.9(b)):

$$\rho_c = \frac{a_{\max}^3}{J_{\max}^2} = \frac{1.25 \times 10^{11}}{2.5 \times 10^9} = 50 \text{ mm}.$$

Step 2 — chord versus centripetal (Theorem 13.9(a)):

$$\frac{a_{\max}T^2}{8} = \frac{5000 \cdot (0.001)^2}{8} = \frac{5 \times 10^{-3}}{8} = 6.25 \times 10^{-4} \text{ mm} = 0.625 \mu\text{m}.$$

Since $\delta_{\max} = 5 \mu\text{m} > 0.625 \mu\text{m}$, the chord constraint is **never** the binding one on this machine.

Step 3 — tight arc, $\rho = 10 \text{ mm}$ (below ρ_c):

$$v_{\text{chord}} = \frac{2}{0.001} \sqrt{2 \cdot 10 \cdot 0.005 - 0.005^2} = 2000 \sqrt{0.099975} = 632.4 \text{ mm/s},$$

$$v_{\text{centr}} = \sqrt{10 \cdot 5000} = \sqrt{50000} = 223.6 \text{ mm/s},$$

$$v_{\text{jerk}} = (50000 \cdot 10^2)^{1/3} = (5 \times 10^6)^{1/3} = 171.0 \text{ mm/s}.$$

$$v_{\max} = \min\{50; 632.4; 223.6; 171.0\} = \mathbf{50 \text{ mm/s}} \quad (v_{\text{prog}} \text{ is in charge}).$$

Step 4 — tight arc, aggressive feed. If the operator raises the feed to $v_{\text{prog}} = 18000 \text{ mm/min} = 300 \text{ mm/s}$:

$$v_{\max} = \min\{300; 632.4; 223.6; 171.0\} = \mathbf{171.0 \text{ mm/s}} = 10.3 \text{ m/min}.$$

The controller *reduces* the feed to little more than half of what was asked, and the binding limit is **jerk** — consistently with Step 1, since $\rho = 10 < \rho_c = 50 \text{ mm}$.

Step 5 — wide arc, $\rho = 200 \text{ mm}$ (above ρ_c), same requested feed of 300 mm/s :

$$v_{\text{centr}} = \sqrt{10^6} = 1000 \text{ mm/s},$$

$$v_{\text{jerk}} = (2 \times 10^9)^{1/3} = 1260 \text{ mm/s}.$$

$$v_{\max} = \min\{300; 2828; 1000; 1260\} = \mathbf{300 \text{ mm/s}}.$$

On the wide arc no physical constraint intervenes and the programmed feed passes through untouched. **Conclusion:** the same G02 instruction, with the same feed, is executed at 171 mm/s on $\rho = 10$ and at 300 mm/s on $\rho = 200$. This is not a controller defect: it is Theorem 13.9.

13.7.4 Following error: why the circle comes out smaller

The three limits above concern the *velocity profile*. But even respecting all of them, the circle actually produced does not coincide with the commanded one: each axis is a control loop of finite bandwidth, and it follows the reference with a delay. On a circular path this delay has a systematic and computable effect.

Theorem 13.10: Radius reduction from finite loop gain. Model each of the two axes as a first-order position loop with velocity gain K_v (in s^{-1}), i.e. with transfer function

$$G(s) = \frac{K_v}{s + K_v}.$$

If a circle of radius R is commanded at speed v , i.e. with angular frequency $\omega = v/R$, then the actual path is still a **circle**, of radius

$$R' = \frac{R}{\sqrt{1 + (\omega/K_v)^2}} < R,$$

rotated by a constant angle $\arctan(\omega/K_v)$. The radius reduction is, to first order in ω/K_v ,

$$\Delta R = R - R' \simeq \frac{R}{2} \left(\frac{\omega}{K_v} \right)^2 = \frac{v^2}{2 R K_v^2}$$

Proof. Step 1 — the references. The command on the two axes is $x_{\text{ref}}(t) = R \cos \omega t$, $y_{\text{ref}}(t) = R \sin \omega t$.

Step 2 — frequency response. For a linear time-invariant system, a sinusoidal input of frequency ω produces a sinusoidal output of the same frequency, with amplitude multiplied by $|G(j\omega)|$ and phase delayed by $\arg G(j\omega)$. Here

$$|G(j\omega)| = \frac{1}{\sqrt{1 + (\omega/K_v)^2}}, \quad \arg G(j\omega) = -\arctan\left(\frac{\omega}{K_v}\right) =: -\psi.$$

Step 3 — the output. Since the **two axes have the same** K_v , they undergo the same attenuation and the same phase lag:

$$x(t) = R|G| \cos(\omega t - \psi), \quad y(t) = R|G| \sin(\omega t - \psi).$$

This is the parametrisation of a circle of radius $R' = R|G(j\omega)|$, traversed with a constant angular lag ψ : the *shape* remains a circle, only smaller and rotated.

Step 4 — the expansion. For $\varepsilon = \omega/K_v \ll 1$, $R' = R(1 - \frac{1}{2}\varepsilon^2 + O(\varepsilon^4))$, hence $\Delta R \simeq \frac{R}{2}\varepsilon^2 = v^2/(2RK_v^2)$. $\square$

Corollary 13.1: The circle becomes an ellipse: two pure cases. If the two axes have different gains, $K_x \neq K_y$, the output is

$$x(t) = R|G_x| \cos(\omega t - \psi_x), \quad y(t) = R|G_y| \sin(\omega t - \psi_y).$$

In the case $\psi_x = \psi_y$ (different gains, equal phase) it is an **ellipse** with semi-axes $R|G_x|$ and $R|G_y|$ aligned with the machine axes. In the case $K_x = K_y$ (equal gain, different phases $\psi_x \neq \psi_y$) it is an ellipse with axes rotated by 45° with respect to the machine axes.

Proof. First case: with equal phases, eliminating t between the two equations gives $(x/(R|G_x|))^2 + (y/(R|G_y|))^2 = 1$, the canonical equation of an ellipse with axes parallel to the machine ones.

Second case: set $A = B = R|G|$ the common gain (in amplitude), $\bar{\psi} = (\psi_x + \psi_y)/2$ and $\Delta = (\psi_x - \psi_y)/2$, and shift time by $\bar{\psi}/\omega$ to absorb the common phase; there remain $x = A \cos(\omega t - \Delta)$ and $y = A \sin(\omega t + \Delta)$. Expanding and setting $u = \omega t$:

$$\frac{x}{A} = \cos u \cos \Delta + \sin u \sin \Delta, \quad \frac{y}{A} = \sin u \cos \Delta + \cos u \sin \Delta.$$

Adding and subtracting:

$$\frac{x}{A} + \frac{y}{A} = (\cos u + \sin u)(\cos \Delta + \sin \Delta), \quad \frac{x}{A} - \frac{y}{A} = (\cos u - \sin u)(\cos \Delta - \sin \Delta).$$

Since $(\cos u + \sin u)^2 + (\cos u - \sin u)^2 = 2$ (Pythagorean identity, Theorem 8.2), the two combinations $x/A \pm y/A$ — which are the coordinates along the 45° bisectors, precisely because here the two axes share the **same** factor A — satisfy an ellipse equation with semi-axes proportional to $|\cos \Delta + \sin \Delta|$ and $|\cos \Delta - \sin \Delta|$: the axes of the ellipse are therefore rotated by 45° . $\square$

Remark 13.10: The general case is not at 45° . If $K_x \neq K_y$ **and** $\psi_x \neq \psi_y$ together, the argument of Corollary 13.1 does not apply: the map $(x, y) \mapsto (x/A, y/B)$ with $A = R|G_x| \neq B = R|G_y|$ is a non-uniform scaling, not a rotation, and the combinations $x/A \pm y/B$ no longer correspond to the bisectors of the machine plane (x, y) . The actual angle of the ellipse's principal axes in this mixed case depends both on the ratio A/B and on $\Delta =$

$(\psi_x - \psi_y)/2$ (general formula for the principal angle of a conic), and in general is **not** 45° : it equals 45° only in the degenerate case $A = B$ of the corollary. In ballbar diagnostic practice (Observation 13.11 below) the two defects — different gains and different phases — are treated as independent causes to be read separately off the trace, not as a combined case with a closed formula: this is why the table in Observation 13.11 lists only the two pure cases of Corollary 13.1.

Remark 13.11: The ballbar test (ISO 230-4). Theorem 13.10 and Corollary 13.1 are the basis of the **circular test** standardised by ISO 230-4: the machine traverses a circle while a telescoping transducer (ballbar) continuously measures the distance from the centre. The resulting polar plot is a diagnostic chart:

shape of the trace	cause
circle smaller than nominal	low gain K_v (Theorem 13.10)
ellipse with axes at $0^\circ/90^\circ$	gains $K_x \neq K_y$
ellipse with axes at 45°	different phases between the axes
step at the quadrants	motion reversal: backlash or stiction
periodic ripple	ballscrew pitch error

Note the link with Chapter 11: the ballbar produces exactly a cloud of points on a nominally circular profile, and processing it is the roundness problem of Section 11.8 — with the crucial difference that here eccentricity and ovality are not disturbances to be removed, but precisely the diagnostic signal sought.

Example 13.7: Radius reduction: how large it really is. A milling machine has loop gain $K_v = 20 \text{ s}^{-1}$ on both axes (a typical value for a mid-range machine; high-performance machines reach $50\text{--}100 \text{ s}^{-1}$). A circle of radius $R = 50 \text{ mm}$ is commanded at feed $F = 3000 \text{ mm/min}$.

Step 1 — consistent units. $v = 3000 \text{ mm/min} = 50 \text{ mm/s}$.

Step 2 — the angular frequency.

$$\omega = \frac{v}{R} = \frac{50}{50} = 1.0 \text{ rad/s}.$$

Step 3 — the critical ratio. $\varepsilon = \omega/K_v = 1.0/20 = 0.05$.

Step 4 — the reduction, exact and approximate.

$$R' = \frac{50}{\sqrt{1 + 0.05^2}} = \frac{50}{\sqrt{1.0025}} = \frac{50}{1.0012493} = 49.937617 \text{ mm},$$

$$\Delta R_{\text{exact}} = 50 - 49.937617 = 0.062383 \text{ mm} = 62.4 \text{ }\mu\text{m}.$$

With the approximate formula:

$$\Delta R \simeq \frac{v^2}{2RK_v^2} = \frac{50^2}{2 \cdot 50 \cdot 400} = \frac{2500}{40000} = 0.0625 \text{ mm} = 62.5 \text{ }\mu\text{m}.$$

The two agree to within 0.2%: the approximation is excellent already at $\varepsilon = 0.05$.

Step 5 — reading. $62 \text{ }\mu\text{m}$ on a 50 mm radius is an enormous error for precision work: more than 20 times the chord tolerance of Example 13.6. And it is not an interpolation error: the *commanded* path is correct to the micrometre; it is the *control loop* that fails to follow it.

Step 6 — the three remedies, in order of effectiveness.

1. **Raise K_v :** the error scales as $1/K_v^2$. Going from 20 to 40 s^{-1} brings it down to $15.6 \text{ }\mu\text{m}$ — a factor of 4. This is the royal road, limited by mechanical stability.

2. **Reduce the feed:** the error scales as v^2 . Halving F gives $15.6 \mu\text{m}$. It costs cycle time.
3. **Velocity feedforward:** adding to the command a term proportional to the desired velocity analytically cancels the dominant pole and ΔR drops by one or two orders of magnitude. This is why every modern CNC exposes a feedforward parameter, and why *disabling* it is the first thing to do when running a diagnostic ballbar.

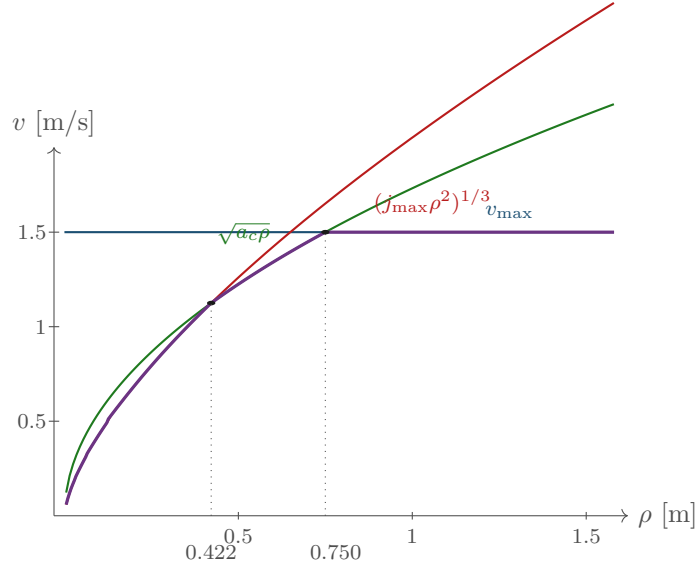


Figure 13.5: The maximum admissible speed on an arc, as a function of the radius of curvature ρ . Three independent constraints act together and the effective speed is their minimum (violet). The rated-speed constraint $v_{\max}$ is constant; the centripetal-acceleration one gives $v \leq \sqrt{a_c \rho}$, growing as $\sqrt{\rho}$; the jerk one, which on a circle reads $|\dot{a}| = v^3/\rho^2$, gives $v \leq (j_{\max} \rho^2)^{1/3}$ and grows as $\rho^{2/3}$. With $v_{\max} = 1.5 \text{ m/s}$, $a_c = 3 \text{ m/s}^2$ and $j_{\max} = 8 \text{ m/s}^3$ there are three distinct regimes: below $\rho = 0.422 \text{ m}$ jerk rules, between 0.422 and 0.750 m centripetal acceleration rules, beyond that the rated speed rules. This is why on small radii the robot slows down drastically even though it is far from its speed limits.

13.8. Safety: circular motion in front of an operator

In a collaborative cell the robot and the operator share the space. The most widely used collaboration mode is **SSM** (*Speed and Separation Monitoring*): the system continuously measures the distance between operator and robot and regulates the robot's speed so that it can always stop before contact.

Definition 13.4: Protective separation distance (ISO/TS 15066). The minimum protective separation distance at time t_0 is

$$S_p(t_0) = \underbrace{\int_{t_0}^{t_0+T_R+T_S} v_H(\tau) d\tau}_{\text{operator's motion}} + \underbrace{\int_{t_0}^{t_0+T_R} v_R(\tau) d\tau}_{\text{robot during reaction}} + \underbrace{\int_{t_0+T_R}^{t_0+T_R+T_S} v_S(\tau) d\tau}_{\text{robot during stopping}} + C + Z_S + Z_R,$$

where v_H is the operator's speed *directed towards* the robot, v_R the robot's speed directed towards the operator, v_S the robot's speed during braking, T_R the system reaction time, T_S the stopping time, C the intrusion margin (human reach), Z_S and Z_R the measurement uncertainties of sensor and robot respectively. The system is safe as long as the actual

distance satisfies $S(t_0) \geq S_p(t_0)$.

Theorem 13.11: Directed speed on a circular path. Let the TCP move with constant tangential speed v along a circle of centre O and radius ρ , and let H be the operator's position, with $d = |OH| > \rho$. Then the robot's speed **directed towards the operator**, $v_R(P) = v(\hat{\mathbf{t}} \cdot \hat{\mathbf{u}})$ — with $\hat{\mathbf{t}}$ the unit tangent and $\hat{\mathbf{u}}$ the unit vector from P towards H — satisfies

$$\max_{P \in \text{arc}} v_R(P) = v,$$

and the maximum is attained **exactly at the two points where the line PH is tangent to the circle**. At those points the robot-operator distance is

$$|PH| = \sqrt{d^2 - \rho^2} = \sqrt{\text{pow}_\Gamma(H)},$$

i.e. the length of the tangent from H .

Proof. Step 1 — parametrise. Put O at the origin and $H = (d, 0)$. A point of the path is $P(\theta) = \rho(\cos \theta, \sin \theta)$ and the unit tangent to the motion is $\hat{\mathbf{t}} = (-\sin \theta, \cos \theta)$.

Step 2 — the numerator simplifies. The vector from P to H is $H - P = (d - \rho \cos \theta, -\rho \sin \theta)$, and

$$\hat{\mathbf{t}} \cdot (H - P) = -\sin \theta (d - \rho \cos \theta) + \cos \theta (-\rho \sin \theta) = -d \sin \theta.$$

The terms in ρ cancel: a notable fact, since it means the useful component depends only on d and θ .

Step 3 — the function to maximise. With $|H - P| = \sqrt{d^2 + \rho^2 - 2d\rho \cos \theta}$ (law of cosines, Theorem 10.6) and $\varphi = -\theta$ to work with positive values,

$$g(\varphi) = \frac{d \sin \varphi}{\sqrt{d^2 + \rho^2 - 2d\rho \cos \varphi}}, \quad \varphi \in (0, \pi).$$

Step 4 — the stationary point. Differentiating and setting the numerator to zero:

$$\cos \varphi (d^2 + \rho^2 - 2d\rho \cos \varphi) - d\rho \sin^2 \varphi = 0.$$

Setting $c = \cos \varphi$ and using $\sin^2 = 1 - c^2$:

$$d\rho c^2 - (d^2 + \rho^2)c + d\rho = 0,$$

a quadratic in c with roots

$$c = \frac{(d^2 + \rho^2) \pm (d^2 - \rho^2)}{2d\rho} = \frac{d}{\rho} \quad \text{or} \quad \frac{\rho}{d}.$$

The first exceeds 1 (since $d > \rho$) and is discarded; there remains $\cos \varphi^* = \rho/d$.

Step 5 — the maximum value. With $\cos \varphi^* = \rho/d$ we have $\sin \varphi^* = \sqrt{d^2 - \rho^2}/d$ and $d^2 + \rho^2 - 2\rho^2 = d^2 - \rho^2$, hence

$$g(\varphi^*) = \frac{d \cdot \sqrt{d^2 - \rho^2}/d}{\sqrt{d^2 - \rho^2}} = 1, \quad \text{i.e.} \quad \max v_R = v.$$

Step 6 — geometric interpretation. The condition $\cos \varphi^* = \rho/d$ says that in triangle OPH we have $|OP| = \rho$, $|OH| = d$ and $|PH| = \sqrt{d^2 - \rho^2}$: by the converse of the Pythagorean theorem, the angle at P is right, i.e. $OP \perp PH$. But OP is the radius: by Theorem 3.8 (Chapter 3), PH is **tangent** to the circle at P . And $\sqrt{d^2 - \rho^2}$ is exactly the tangent length from an external point, i.e. the square root of the power of H (Definition 3.1). $\square$

Warning. A safety consequence not to be circumvented. Theorem 13.11 says that on a circular path the term v_R of Definition 13.4 **cannot be reduced**: there is always a point of the arc at which the TCP points exactly at the operator, at full speed. Anyone sizing SSM by assuming an average directed component, or a geometric reduction factor, computes an *underestimated* S_p and hence an unsafe barrier.

The result also tells us, however, where the critical point is — the point of tangency — and at what distance: $\sqrt{d^2 - \rho^2}$. This allows the condition $S \geq S_p$ to be checked in the worst case alone, instead of sampling the whole arc.

Example 13.8: SSM verification on an arc, step by step. A collaborative cell executes an arc of radius $\rho = 0.40$ m about a centre O ; the operator works at a fixed station H with $d = |OH| = 1.50$ m. System data:

$$v_H = 1.6 \text{ m/s (normative value for body approach), } T_R = 0.10 \text{ s, } T_S = 0.25 \text{ s,}$$

$$C = 0.20 \text{ m, } Z_S = 0.05 \text{ m, } Z_R = 0.01 \text{ m.}$$

The robot traverses the arc at $v = 0.75$ m/s and decelerates uniformly while stopping.

Step 1 — the geometric worst case (Theorem 13.11). The maximum directed speed is $v_R = v = 0.75$ m/s, attained at the point of tangency, where the actual distance is

$$S = \sqrt{d^2 - \rho^2} = \sqrt{1.50^2 - 0.40^2} = \sqrt{2.25 - 0.16} = \sqrt{2.09} = 1.4457 \text{ m.}$$

Step 2 — operator contribution. Assuming v_H constant over the whole interval $T_R + T_S$:

$$\int v_H = 1.6 \times 0.35 = 0.560 \text{ m.}$$

Step 3 — robot contribution during reaction. The robot still moves at full directed speed:

$$\int v_R = 0.75 \times 0.10 = 0.075 \text{ m.}$$

Step 4 — robot contribution during stopping. With uniform deceleration from v to zero in T_S , the mean speed is $v/2$:

$$\int v_S = \frac{0.75}{2} \times 0.25 = 0.09375 \text{ m.}$$

Step 5 — sum.

$$S_p = 0.560 + 0.075 + 0.09375 + (0.20 + 0.05 + 0.01) = 0.98875 \text{ m.}$$

Step 6 — verification.

$$S = 1.4457 \text{ m} \geq S_p = 0.9888 \text{ m.} \quad \checkmark$$

The configuration is safe, with a margin of 0.457 m.

Step 7 — up to what speed? Inverting, we seek the maximum v for which $S \geq S_p$:

$$1.4457 \geq 0.560 + 0.26 + v T_R + \frac{v}{2} T_S = 0.820 + v (0.10 + 0.125) = 0.820 + 0.225 v,$$

$$v \leq \frac{1.4457 - 0.820}{0.225} = \frac{0.6257}{0.225} = 2.78 \text{ m/s.}$$

The safety constraint therefore leaves ample margin; on this arc the bottleneck will rather be the dynamic one of Section 13.7. **Note however:** if the radius grew to $\rho = 1.45$ m, the distance at the tangency point would drop to $\sqrt{2.25 - 2.1025} = 0.384 \text{ m} < S_p$, and no positive speed would be admissible: the path would have to be redesigned.

$$S_p = (K_H + K_R)(T_r + T_s) + B + C + Z_d + Z_r$$

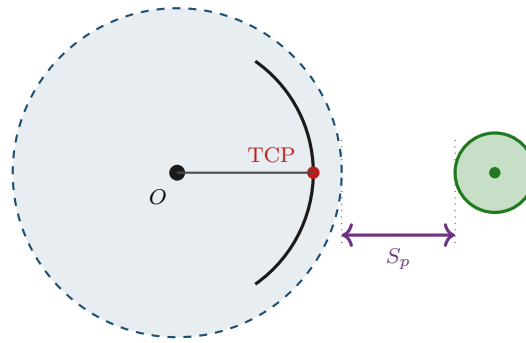


Figure 13.6: Circular motion in front of an operator. The robot envelope (light blue) is the circle whose radius is the maximum reach, and the TCP travels an arc inside it. The protective separation distance S_p required by ISO/TS 15066 is computed as $S_p = (K_H + K_R)(T_r + T_s) + B + C + Z_d + Z_r$, where K_H and K_R are the approach speeds of operator and robot, T_r the system reaction time, T_s the stopping time, B the distance travelled while stopping, C the intrusion allowance and Z_d, Z_r the measurement uncertainties. On circular motion K_R must be evaluated with the radial component of the TCP velocity, which depends on the position along the arc: the worst case is where the tangent points towards the operator.

13.9. Yaskawa Motoman — INFORM language (YRC1000)

13.9.1 MOVC instruction

The circular motion in INFORM is obtained with the instruction **MOVC** (*MOVE Circular*), analogous to **MOVJ** (joints) and **MOVL** (linear) but with circular interpolation.

Warning. Unlike other manufacturers, Yaskawa requires a **minimum of three consecutive MOVC instructions** to generate a complete arc (not two, as often happens in ABB/KUKA): the first **MOVC** is in fact treated internally as a **MOVL** (it only serves to position the TCP on the first point), while the next two (or four, for a complete 360° circle) define the actual arc through the intermediate point and the final point.

Listing 13.1: Minimal INFORM job for a 180° arc with three MOVC

```

NOP
MOVJ VJ=25.00      ' approach positioning (joints)
MOVC V=200.0        ' P1: arc initial point (treated as MOVL)
MOVC V=200.0        ' P2: intermediate point (defines the plane and
                    radius)
MOVC V=200.0        ' P3: arc final point
MOVL V=200.0        ' exit in linear
END

```

13.9.2 Complete example: arc welding job with MOVC

The most common use case of **MOVC** in industry is not a self-contained circle, but a curved weld bead. The following job is representative of the style used in the YRC1000/DX100 manuals for arc welding, with arc ignition and extinction (**ARCON**/**ARCOF**) around the circular motion instructions:

Listing 13.2: Representative INFORM job for welding on a circular bead

```

NOP

```

```

MOVJ VJ=50.00      ' approach in joints
MOVL V=50.0        ' linear approach to the trigger point
ARCON ASF#(1)      ' arc ignition, parameter file #1
MOVC V=30.0        ' P1: bead trigger (treated as MOVL)
MOVC V=30.0        ' P2: intermediate point of the curved bead
MOVC V=30.0        ' P3: bead end
ARCOF ASF#(1)      ' arc extinction
MOVL V=50.0        ' linear exit
MOVJ VJ=50.00      ' return in joints
END

```

INFO

Note. The welding speed ($V=30.0$ mm/s in the example) is typically much lower than the positioning speed ($V=50.0$ in the non-welded stretches): it is a process parameter (penetration, cooling) independent of the geometry of the circle, but which in practice also constrains the number of usable MOVC segments, because too coarse an interpolation at low speed produces a visibly polygonal bead instead of a curved one.

13.9.3 INFORM variables: types and their role in parametric arcs

A job that always draws the same arc is of little use. To generate *parametric* arcs — a circle of variable radius, a series of identical arcs on different parts — INFORM provides six families of global variables, all addressed by number.

Prefix	Type	Content	Range
B	Byte	unsigned integers, flags, counters	0 ... 255
I	Integer	signed integers	−32768 ... 32767
D	Double	32-bit signed integers	$\pm 2^{31}$
R	Real	floating point	IEEE 754 single
S	String	text (pendant messages, labels)	16 characters
P/BP/EX	Position	position triples (base / robot / external axes)	—

***Remark 13.12: Why the byte, and why it matters.** The B variable is the most used in real jobs precisely because of its narrow 0–255 range: it is the natural counter for a loop of arcs (a full circle in four MOVC repeated n times), the state flag of a gripper, the index of a part in a magazine. The price of that compactness is that **exceeding the limit raises no error but wraps around**: $255 + 1 = 0$. A counter B000 counting the arcs executed over a long production shift will silently return to zero at the 256th part, and the production statistics will be wrong with no alarm whatever. For counts that may exceed 255, use I or D.*

Listing 13.3: Parametric INFORM job: n full circles with a byte counter

```

NOP
SET B000 0          ' counter of arcs executed
SET B001 4          ' number of circles to trace
*LOOP
MOVC V=200.0        ' P1 - start (treated as MOVL)
MOVC V=200.0        ' P2 - first quarter
MOVC V=200.0        ' P3 - second quarter
MOVC V=200.0        ' P4 - third quarter
MOVC V=200.0        ' P1 - closing the circle
INC B000            ' B000 = B000 + 1 (caution: wraps to 0 after
255)

```

```
JUMP *LOOP IF B000 < B001
END
```

Best practice. If the count must survive a whole shift, declare the counter as I000 (16-bit signed integer) rather than B000: the cost is one byte of memory, the gain is the absence of a silent wrap-around at 256. The byte remains the right choice for *flags* (0/1) and for short array indices, where the limit is never in question.

13.10. G-code (CNC): the G02/G03 instructions

Parallel to the world of anthropomorphic robots, CNC machines (milling machines, lathes) program circular interpolation in **G-code** (standard ISO 6983/RS-274) with the instructions G02 (clockwise) and G03 (anticlockwise), specifying the centre *relative* to the starting point via the offsets I,J (plane XY, G17) instead of an intermediate point:

Listing 13.4: G-code: clockwise arc from (0,0) to (10,0) with centre (5,0)

```
G17                ; XY working plane
G01 X0 Y0 F500     ; linear positioning to the initial point
G02 X10 Y0 I5 J0 F200 ; clockwise arc, centre at (X+I, Y+J) = (5,0),
    radius 5
```

Remark 13.13: Conceptual comparison G-code vs. anthropomorphic robot. G-code defines the arc with initial point (implicit), final point, and centre (via I,J), not with three points on the circumference: it is equivalent in information content (centre and radius are in any case recovered from initial point, centre and direction), but mathematically closer to Definition 7.1 (Chapter 7) than to Theorem 7.2. A complete circle is obtained by omitting the final point (X,Y) and specifying only I,J: unlike Yaskawa/FANUC/ABB, a single G02/G03 instruction can therefore close a 360° turn in a single line, because the CNC controller directly solves the circle equation (Section 7.4, Chapter 7) instead of interpolating between taught points.

13.10.1 Complete circles with MOVC

Each MOVC instruction interpolates an arc between three consecutive points (previous, current, next), and by Theorem 7.2 three non-collinear points determine a unique circle: a MOVC arc can therefore cover at most an incomplete turn (the ambiguity between minor and major arc on just three points prevents closing an exact 360° with a single instruction). A complete circle therefore requires concatenating several MOVC instructions on additional intermediate points; for the minimum number of points and the recommended handling of consecutive arcs, see the *YRC1000 Controller — Instructions Manual*, section on circular interpolation.

13.11. Tool radius compensation (G41/G42) on arcs

Definition 13.5: Offset curve. A tool of radius R_u that works a profile while staying tangent to it traces, with its own centre, the curve **parallel** (*offset*) to the profile at distance R_u .

Theorem 13.12: The offset of an arc is a concentric arc. The offset at distance R_u of an arc of radius r and centre C is a **concentric** arc, of radius $r + R_u$ (tool outside the profile) or $r - R_u$ (tool inside), with the same central angle.

Proof. Let T be the point of contact between the tool and the profile, and U the tool's centre. The common tangent at T is perpendicular both to the profile's radius CT and to the tool's radius UT (Theorem 3.8, Chapter 3): hence C, T, U are collinear, and $|CU| = |CT| \pm |TU| = r \pm R_u$ (with sign $+$ if U and T are on the same side of C relative to the point of contact, that is, tool outside the profile; $-$ if inside). Since U always remains on the line CT for every position of T on the arc, the locus of U is an arc concentric to the original one, with the same central angle (the correspondence $T \mapsto U$ is a homothety of centre C). $\square$

Theorem 13.13: Undercut: an internal arc too tight. An internal arc of radius $r < R_u$ cannot be machined without removing material beyond the profile.

Proof. By Theorem 13.12, the offset of an internal arc of radius r would have radius $r - R_u < 0$: geometrically, the tool centre would have to lie *beyond* the centre C of the profile. More precisely, the circle traced by the tool centre, of radius R_u , centred at distance $R_u - r > 0$ from C (on the opposite side), contains points at distance $R_u + (R_u - r) = 2R_u - r > r$ from C : the tool, of radius R_u , would protrude beyond the arc of radius r it is supposed to follow, removing material beyond the intended profile. $\square$

Example 13.9: Numerical check. Profile with external arc $r = 20$ mm and internal arc $r = 8$ mm, cutter $\varnothing 12$ mm ($R_u = 6$ mm): tool-centre radii $20 + 6 = 26$ mm (external) and $8 - 6 = 2$ mm (internal), both machinable. With cutter $\varnothing 20$ mm ($R_u = 10$ mm): the external arc remains machinable ($20 + 10 = 30$ mm), but the internal arc is **impossible** ($8 - 10 = -2$ mm < 0): this is the undercut just proved.

Remark 13.14: The practical rule. The condition $R_u < r$ for every internal arc of the profile is the practical rule “tool radius smaller than the radius of the tightest internal arc”, well known to CNC programmers; modern controllers flag an interference alarm when this condition is violated (tool radius compensation error or similar, depending on the manufacturer). A minimal example of activating compensation in G-code:

Listing 13.5: Minimal G-code with tool radius compensation

```
G41 D1                ; activate left compensation, corrector D1
G01 X0 Y0
G02 X20 Y0 I10 J0      ; arc machined with offset
G40                    ; deactivate compensation
```

Source: P. Smid, *CNC Programming Handbook*, 3rd ed., Industrial Press, 2008 (G41/G42, R format, helix).

13.12. Feed rate on internal and external arcs

Theorem 13.14: Correction of the programmed feed rate. If the tool centre travels the offset arc of radius $r \pm R_u$ at speed F (programmed on the centre), the speed of the contact point on the profile equals $F \frac{r}{r \pm R_u}$. To maintain the desired feed rate F_0 on the

profile, one must programme

$$F = F_0 \frac{r \pm R_u}{r}.$$

Proof. The tool centre U and the contact point T are both collinear with the profile's centre C (Theorem 13.12) and rotate about C with the **same** angular velocity ω (they are rigidly aligned on the same rotating ray). By the formula $v = \omega\rho$ (Chapter 4), $F = \omega(r \pm R_u)$ (speed of U) and the speed of T equals ωr . The ratio is $\omega r / [\omega(r \pm R_u)] = r / (r \pm R_u)$, whence the claim (inverting the proportion to express F as a function of F_0). $\square$

Example 13.10: Numerical check. External arc $r = 20$ mm, $R_u = 6$ mm, desired feed rate on the profile $F_0 = 300$ mm/min: one programmes $F = 300 \cdot 26/20 = 390$ mm/min (higher, because the centre travels a larger circle in the same angular time). Internal arc $r = 8$ mm, same tool: $F = 300 \cdot 2/8 = 75$ mm/min (lower). Source: Modern Machine Shop, *Adjusting Feed Rate in Circular Motion Commands with G-Codes*.

13.13. Helical interpolation

Definition 13.6: Cylindrical helix. A **helix** of radius r and pitch p is the curve traced by a point moving with compound motion: uniform rotation of radius r in the xy plane and uniform translation of pitch p per turn along z (in G-code, G02/G03 with the addition of the third coordinate).

Theorem 13.15: Length and ramp angle of the helix. For one complete turn, the helix of radius r and pitch p has length $\sqrt{(2\pi r)^2 + p^2}$ and ramp angle (relative to the horizontal plane) $\alpha = \arctan \frac{p}{2\pi r}$.

Proof. The lateral surface of the cylinder of radius r develops isometrically (without length distortion) into a flat rectangle of base $2\pi r$ (the base circle, rectified) and height p (one pitch): this is an elementary fact of the developability of the cylinder (the cylinder, having no Gaussian curvature, lies flat on a plane without stretching; for an elementary justification without invoking curvature, approximate the helix by a polyline inscribed on n equally spaced generators of the cylinder, each the slanted side of a small right triangle of base $2\pi r/n$ and height p/n once “unrolled”, and take the limit $n \rightarrow \infty$: a procedure identical to the one used for the rectifiability of the circle, Chapter 6). In the development, one turn of the helix becomes the **diagonal** of the rectangle $2\pi r \times p$: by Pythagoras, its length is $\sqrt{(2\pi r)^2 + p^2}$, and the angle this diagonal forms with the long side $2\pi r$ (the ramp angle) is $\alpha = \arctan(p/(2\pi r))$. $\square$

Example 13.11: M20 thread, pitch 2.5, with a $\varnothing 12$ cutter. M20 thread (nominal diameter 20 mm, radius 10 mm), pitch $p = 2.5$ mm, cutter of radius $R_u = 6$ mm: this is an **internal** thread in a hole (the cutter orbits inside the hole and must reach the nominal thread radius with its outer edge), hence it is the “internal arc” case of Theorem 13.14: tool-centre radius $r - R_u = 10 - 6 = 4$ mm. Length per turn: $\sqrt{(2\pi \cdot 10)^2 + 2.5^2} = \sqrt{3947.842 + 6.25} = 62.881569$ mm (using the thread's nominal radius). Ramp angle: $\arctan(2.5/(2\pi \cdot 10)) = 2.278525^\circ$. Corrected feed rate (combining with Theorem 13.14, $r = 10$, $R_u = 6$): for $F_0 = 200$ mm/min, $F = 200 \cdot 4/10 = 80$ mm/min.

13.14. FANUC — CIRCLE instruction and KAREL language

13.14.1 TP (Teach Pendant) — Circular instruction

In the FANUC Teach Pendant language, the equivalent is the motion instruction with the **CIRCULAR** modifier, which requires two points (via-point and target) besides the current position:

Listing 13.6: FANUC TP program with circular motion

```
1: J P[1] 100% FINE
2: C P[2]
   P[3] 200mm/sec FINE      ' P[2]=via point, P[3]=arc target
3: L P[4] 200mm/sec FINE
```

Conceptually identical to Yaskawa: current point + via-point + target = 3 points, same unique circumference of Theorem 7.2; FANUC however requires a single instruction for the entire arc (the via-point is included in the same line), not three separate ones.

13.14.2 KAREL

In KAREL (FANUC's high-level procedural language, derived from Pascal), the circular motion is programmed explicitly with the **MOVE** construct and the **CIRCULAR** attribute, explicitly specifying the via and target positions as **POSITION** or **XYZWPR** variables.

13.15. ABB — RAPID language

13.15.1 MoveC instruction

ABB uses the **MoveC** instruction, also based on via-point + target:

Listing 13.7: RAPID program with MoveC

```
MoveJ p1, v200, fine, tool0;
MoveC p2, p3, v200, fine, tool0;      ! p2 = via point, p3 = target
MoveL p4, v200, fine, tool0;
```

13.16. KUKA — KRL language

13.16.1 CIRC instruction

In the KRL language (KUKA Robot Language), the circular motion is programmed with the **CIRC** instruction, which like Yaskawa and FANUC requires an auxiliary point (*auxiliary point*) on the arc besides the final point, taking as initial point the current position of the TCP (end of the previous motion):

Listing 13.8: KRL program with CIRC

```
LIN P1                ; linear positioning to the arc initial point
CIRC P2, P3            ; P2 = auxiliary point on the arc, P3 = final
  point
LIN P4                ; linear exit
```

INFO

Note. The traversal speed of the path (*Cartesian Path velocity*) is set with the system parameter **\$VEL.CP** (in m/s), conceptually analogous to the **V** parameter of **MOVE** or the **v200** of **MoveC**. To avoid the robot stopping exactly on each taught point (P1, P2, P3, ...),

KRL provides blending instructions (*blending*) such as C_DIS and C_VEL, which round the trajectory near the passing points.

Remark 13.15: The CA parameter: a complete circle in a single instruction. KRL has the optional parameter CA (circular angle): *CIRC P2, P3, CA 360* travels the whole turn with a **single** instruction, specifying the angle travelled independently of the declared final point (which in this case only serves to define the plane and the direction of the arc); CA can also exceed 360° or be negative. For the syntactic details and the exact limits of the parameter see the KSS 8.x Programming Guide for System Integrators.

Remark 13.16: Synthetic comparison among the four manufacturers.

- **Yaskawa INFORM:** *MOV*C over three consecutive steps for each arc; for a complete circle and the handling of consecutive arcs see the YRC1000 Controller — Instructions Manual, section on circular interpolation.
- **FANUC TP/KAREL:** a single circular instruction with explicit via-point in the same block; a complete circle generally requires 2 circular instructions (two semicircles) since a single instruction cannot close a 360° turn.
- **ABB RAPID:** *MoveC* with via-point and target in the same command; the RAPID manual indicates that a complete circle is programmed with at least two *MoveC* instructions.
- **KUKA KRL:** *CIRC* with auxiliary point and target in the same command; unlike the other three, KRL has the CA parameter (circular angle, Observation 13.15 above), which allows a complete turn, or an arbitrary angle, to be travelled with a **single CIRC** instruction.

In all cases the underlying mathematical principle is identical: three non-collinear points determine a unique circumference (Theorem 7.2); the differences are purely syntactic and related to how each controller distributes the points between separate instructions or a single block.

13.17. The PLCopen standard: circular interpolation at PLC level

Besides the proprietary languages of robotic controllers (Yaskawa INFORM, FANUC TP/KAREL, ABB RAPID, KUKA KRL) and the G-code of machine tools, there is a further level at which the circle appears in the daily industrial practice of the PLC engineer: axis control (*motion control*) integrated directly in the PLC, according to the **PLCopen Motion Control** standard (manufacturer-independent, implemented for example in Siemens TIA Portal, Beckhoff TwinCAT, B&R Automation Studio).

Definition 13.7: Function blocks MC_MoveCircularAbsolute/Relative. Part 4 of the PLCopen specification (*Coordinated Motion*) defines the function blocks MC_MoveCircularAbsolute and MC_MoveCircularRelative for the coordinated circular motion of two or more axes. The principal parameters are:

- **CircMode** (type MC_CIRC_MODE): selects how the arc is specified — via centre, via an auxiliary point on the arc, or via the radius only — the alternative between “three points” (Theorem 7.2) and “centre+radius+angle” already encountered in the robotic languages;
- **AuxPoint**: the auxiliary point (centre, intermediate point, or radius depending on CircMode);

- **EndPoint**: the final point of the arc;
- **PathChoice** (type MC_CIRC_PATHCHOICE): when the data are ambiguous (given centre and radius, in general two possible arcs exist between two points, as in the observation on the eight solutions of Apollonius of Chapter 5), specifies which of the two paths to choose (minor/major arc, clockwise/anticlockwise).

***Remark 13.17: Why this level is distinct from the robotic one.** Unlike robotic controllers (which internally manage the inverse kinematics of 6 or more rotating axes for a single TCP), PLCopen motion control typically coordinates a smaller number of Cartesian axes (tables X-Y or X-Y-Z) for applications such as cutting, dispensing along a profile, or positioning a tool on a planar trajectory. Conceptually the mathematical problem is identical (uniquely identifying an arc from partial data, Theorem 7.2), but the on-board PLC implementation must respect cycle constraints (IPO task, typically 1–4 ms) often even more stringent than those of a dedicated robotic controller.*

Exercise 13.3. A block MC_MoveCircularAbsolute is configured with CircMode=Center: given the centre $C = (50, 30)$ mm and the final point $E = (90, 30)$ mm, how many distinct arcs satisfy these data (before also specifying PathChoice)? Motivate the answer with Theorem 7.2.

Solution. Given centre and final point, the radius is determined: $r = |CE| = 40$ mm. Assuming a current initial point $P_0 \neq E$ on the same circumference, there always exist **two** possible arcs between P_0 and E (clockwise arc and anticlockwise arc, generally of different length). Hence the need for PathChoice: centre, radius and the two endpoints are not enough to uniquely identify the arc, one must explicitly specify the direction or the choice of the major/minor arc.

13.18. Note on real numerical interpolation

Warning. In practice, the controller does not solve exactly the system D, E, F at every servo cycle (too computationally expensive in real time): it typically pre-computes, during job parsing, the arc parameters (centre, radius, plane, total angle) and then generates the velocity profile along the arc with a polynomial interpolator (cubic or quintic spline) in time, respecting the velocity and joint-acceleration limits imposed by the inverse kinematic control. Chapter 11 deepens the numerical background noise generated by this process.

***Remark 13.18: The second application domain: water.** Alongside industrial robotics, the second great application domain of this treatise is water treatment plant engineering, where the circle is not the trajectory of a tool but the **very shape** of the process tanks. Given its breadth (an original theorem on radial sedimentation, kinematics of the scraper bridge, design criteria) the topic now has its own chapter: see Chapter 14.*

13.19. Machine-department PLC: Siemens, Schneider Electric, Delta

The PLCopen standard (previous section) defines the abstract interface of the function blocks for circular motion, but in the daily practice of the machine-department engineer each PLC

manufacturer implements it in its own development environment, with differences of implementation detail. The three most widespread platforms for machine automation (not plant, and not anthropomorphic robot) in the Italian and European industrial context are Siemens TIA Portal, Schneider Electric EcoStruxure Machine Expert and Delta Electronics ISPSOft/DIADesigner.

13.19.1 Siemens — TIA Portal, S7-1500T Motion Control

On the S7-1500T controllers, equipped with Motion Control functionality with Technology Objects, circular motion is programmed with the instruction `MC_MoveCircularAbsolute`, in Structured Control Language (SCL):

Listing 13.9: Siemens TIA Portal SCL: `MC_MoveCircularAbsolute`, complete circle about the tank centre

```
"MC_MoveCircularAbsolute_DB"(AxesGroup := "Kinematics_1", //
    T0_Kinematics object
                                CircMode := 1,           // 1 =
                                AuxPoint is the centre
    AuxPoint1 := centroVasca_X,
    AuxPoint2 := centroVasca_Y,
    AuxPoint3 := centroVasca_Z,
    Arc := 360.0,           // angle
                                travelled, in CircMode=1
    PathChoice := 0,         //
                                direction of travel
    CirclePlane:= 1,         // working
                                plane
    Velocity := 50.0,
    Execute := startBit);
```

INFO

Note. The Siemens `CircMode` parameter (of the `T0_Kinematics` object, not a plain axis) has three values: 0, `AuxPoint` is a point *on the arc* (only for arcs $< 360^\circ$, three points as in Theorem 7.2); 1, `AuxPoint` is the *centre* and the angle travelled is given explicitly in `Arc` (which can equal 360° , the case not representable with just three points), with the direction in `PathChoice` and the plane in `CirclePlane`; 2, `Radius` and `EndPoint` are given, with up to four possible arcs selected by `PathChoice` 0–3 (the four cases of the geometry of two secant circles, Chapter 7). Additional parameters: `Velocity`, `Acceleration`, `Deceleration`, `Jerk`, `BufferMode`, `TransitionParameter`. Source: Siemens, *S7-1500T Kinematics functions V5.0 in TIA Portal V16 — Function Manual*, 12/2019, A5E42062707-AB, §11.1.6, pp. 269–271.

13.19.2 Schneider Electric — EcoStruxure Machine Expert

Schneider Electric implements circular motion in its own Motion Control Library (*MotCoLib*), compliant with the PLCopen specification, within the EcoStruxure Machine Expert environment (successor of SoMachine Motion) for the controllers of the Modicon range (M262, LMC): the available function blocks include both `MC_MoveCircularAbsolute`/`MC_MoveCircularRelative` (centre/auxiliary-point semantics, analogous to Siemens) and `MC_MoveCircular2D`, designed specifically for planar applications (cutting, dispensing on planar path):

Listing 13.10: Schneider Electric ST: `MC_MoveCircularAbsolute`

```
MC_MoveCircularAbsolute_Inst(Axis := Axis_X_Y,
                             CircMode := mc_CIRC_MODE_CENTRE ,
```



```

AuxPoint := centroVasca,
EndPoint := posFinale,
Velocity := 50.0,
Execute  := startBit);

```

Remark 13.19: Schneider Electric, too, stays close to the official PLCopen names. Like Siemens (which uses the parameters *CircMode*/*AuxPoint* seen above, the same semantics as the generic PLCopen name, not a renaming), Schneider Electric, a historic member of PLCopen, keeps in its library parameter names almost identical to the original specification (*CircMode*, *AuxPoint*, *EndPoint*): for an engineer moving from one manufacturer to another, both the Siemens and the Schneider code map directly onto the generic Definition 13.7.

13.19.3 Delta Electronics — AS Series, ISPSOft/DIADesigner

On the AS series PLCs (AS300/AS200), programmable in ISPSOft or in the more recent DIADesigner, Delta supports the standard PLCopen function blocks but also flanks proprietary motion instructions with the *DMC_* prefix (*Delta Motion Control*), designed for a tighter integration with its own drives (ASDA servos):

Listing 13.11: Delta AS Series ST: circular motion

```

DMC_MoveCircularAbsolute_Inst(Axis      := Axis_Gruppo1,
                              CircMode := 1,           // 1 =
                              MC_CIRC_MODE_CENTRE
                              AuxPoint  := centroVasca,
                              EndPoint  := posFinale,
                              Velocity  := 50.0,
                              Execute   := startBit);

```

INFO

Verified nomenclature. The block *DMC_MoveCircularAbsolute* is documented in Delta’s *AS Series Programming Manual* (code AS-0249720-09, ed. 2022/08/01), where it is described as performing “circular or *helical* interpolation to a specified absolute target”. The corresponding *DMC_MoveCircularRelative* exists as well.

The *CircMode* parameter takes the three values prescribed by the PLCopen specification and inherited from the CODESYS implementation on which the Delta library is built:

value of <i>CircMode</i>	meaning of <i>AuxPoint</i>	<i>PathChoice</i>
MC_CIRC_MODE_BORDER	a point <i>on the arc</i> , crossed	ignored
MC_CIRC_MODE_CENTER	centre of the arc	required
MC_CIRC_MODE_RADIUS	radius (signed)	required

Here we recognise, translated into PLC language, exactly the casuistry of G-code (Section 11.11, Chapter 11): *BORDER* corresponds to the three points of Theorem 13.1 and has no ambiguity of sense, whereas *CENTER* and *RADIUS* leave the sense undetermined and *must* be accompanied by *PathChoice* — the analogue of the G02/G03 distinction.

Warning. One caution about versions remains valid: the *DMC_* library was reorganised between ISPSOft and DIADesigner-AX, and the DVP and AS series do not expose the same set of blocks. Before using the code given here in production, compare the names with the manual of the software version actually installed.

Remark 13.20: Synthetic comparison among the three PLC platforms.

- **Siemens TIA Portal:** `MC_MoveCircularAbsolute` on Technology Object/Kinematics, parameters renamed (*Mode*, *Position*) with respect to the generic `PLCopen` name but same semantics.
- **Schneider Electric EcoStruxure Machine Expert:** `MC_MoveCircularAbsolute/MC_MoveCircular2D` in the `MotCoLib`, parameter names almost identical to the original `PLCopen` specification.
- **Delta Electronics AS Series:** standard `PLCopen` blocks plus proprietary `DMC_` extensions, always to be verified on manual and software version.

In all three cases the mathematical core remains that of Definition 13.7 and, ultimately, of Theorem 7.2: the names of the blocks and parameters change, not the geometric problem they solve.

Exercise 13.4. A Siemens S7-1500T axis must move the TCP of a dispensing head along an arc of radius $r = 150$ mm, from point $P_0 = (0, 0)$ to point $P_1 = (300, 0)$ mm, passing above the X axis (arc convex upward). Determine the coordinates of the centre of the arc to pass to the `AuxPoint` parameter with `CircMode:=1` (centre), knowing that the centre must lie on the symmetry axis of the segment P_0P_1 .

Solution. The centre C must be equidistant from P_0 and P_1 (common radius $r = 150$ mm), hence it lies on the axis of the segment P_0P_1 : the vertical line $x = 150$ mm (through the midpoint of P_0P_1). The centre also has distance $r = 150$ mm from $P_0 = (0, 0)$:

$$\sqrt{150^2 + y_C^2} = 150 \Rightarrow y_C = 0.$$

The centre is therefore $C = (150, 0)$, that is it coincides with the midpoint of P_0P_1 : the required arc is in fact a semicircle of radius 150 mm (the diameter is exactly $|P_0P_1| = 300$ mm). It is a particular case of Thales' Theorem (Chapter 4): every point of the arc sees the segment P_0P_1 (diameter) under a right angle.

13.20. Problem set

13.20.1 Numerical and applied exercises

Exercise 13.5. Write (in INFORM pseudocode) the minimal sequence of `MOVC` instructions for a 90° arc between two points P_1 and P_3 given an intermediate point P_2 at the midpoint of the arc.

Solution.

Listing 13.12: Solution: 90 degree arc

```
NOP
MOVL V=100.0      ' positioning to the initial point (if necessary)
MOVC V=100.0      ' P1: arc start (treated as MOVL)
MOVC V=100.0      ' P2: intermediate point (at 45 degrees, defines
                    plane and radius)
MOVC V=100.0      ' P3: arc end (at 90 degrees)
END
```

Since the required arc (90°) is less than 360° , three `MOVC` instructions (the first treated as

MOVL) suffice to define it completely: no further subdivision is necessary.

Exercise 13.6. Three taught points are $P_1 = (0; 0; 0)$, $P_2 = (4; 0; 3)$, $P_3 = (0; 5; 0)$ mm. Compute the plane normal, the centre and the radius of the arc using Theorem 13.1, and check the result.

Solution. $\mathbf{a} = (4; 0; 3)$, $\mathbf{b} = (0; 5; 0)$. Then $|\mathbf{a}|^2 = 25$, $|\mathbf{b}|^2 = 25$, $\mu = \mathbf{a} \cdot \mathbf{b} = 0$ (the two vectors are orthogonal).

Normal. $\mathbf{a} \times \mathbf{b} = (0 \cdot 0 - 3 \cdot 5; 3 \cdot 0 - 4 \cdot 0; 4 \cdot 5 - 0 \cdot 0) = (-15; 0; 20)$, with $|\mathbf{a} \times \mathbf{b}| = \sqrt{225 + 400} = 25$.

Centre. With the compact form $\mathbf{N} = |\mathbf{b}|^2(|\mathbf{a}|^2 - \mu)\mathbf{a} + |\mathbf{a}|^2(|\mathbf{b}|^2 - \mu)\mathbf{b}$ and $\mu = 0$:

$$\mathbf{N} = 25 \cdot 25 \mathbf{a} + 25 \cdot 25 \mathbf{b} = 625 (\mathbf{a} + \mathbf{b}) = 625 (4; 5; 3),$$

$$\mathbf{d} = \frac{\mathbf{N}}{2|\mathbf{a} \times \mathbf{b}|^2} = \frac{625}{2 \cdot 625} (4; 5; 3) = (2; 2.5; 1.5),$$

hence $C = P_1 + \mathbf{d} = (2; 2.5; 1.5)$ mm.

Radius. $R = |\mathbf{d}| = \sqrt{4 + 6.25 + 2.25} = \sqrt{12.5} \approx 3.536$ mm. Check with the other formula: $|\mathbf{b} - \mathbf{a}| = |(-4; 5; -3)| = \sqrt{50}$, hence

$$R = \frac{5 \cdot 5 \cdot \sqrt{50}}{2 \cdot 25} = \frac{\sqrt{50}}{2} = \sqrt{12.5} \approx 3.536. \checkmark$$

Direct check. $|P_2 - C| = |(2; -2.5; 1.5)| = \sqrt{12.5}$; $|P_3 - C| = |(-2; 2.5; -1.5)| = \sqrt{12.5}$. $\checkmark$

Note. Since $\mathbf{a} \perp \mathbf{b}$, the triangle is right-angled at P_1 and the centre is the midpoint of the hypotenuse P_2P_3 (Corollary 10.3): indeed $(P_2 + P_3)/2 = (2; 2.5; 1.5) = C$. $\checkmark$

Exercise 13.7. A cell performs 3 full circles per part and a counter B000 is incremented by 1 for each circle. After how many parts does the counter restart from zero, and what value does it show after 100 parts?

Solution. The counter advances by 3 per part and wraps at 256 (Remark 13.12). The first wrap occurs at the first multiple of 3 reaching or exceeding 256: $3n \geq 256 \Rightarrow n \geq 85.33$, hence at part $n = 86$, when the nominal count would be 258 and the variable instead shows $258 - 256 = 2$.

After 100 parts the nominal count is 300, and

$$\text{B000} = 300 \bmod 256 = 44.$$

An operator reading 44 would believe 14–15 parts had been produced rather than 100: exactly the kind of silent error the good practice of Section 13.9.3 warns against.

Exercise 13.8. In a configuration close to a singularity the smallest singular value of the Jacobian is $\sigma_{\min} = 0.002$. The required Cartesian speed is $\|\dot{\mathbf{x}}\| = 100$ mm/s, aligned with the critical direction. (a) What joint speed would the exact inversion demand? (b) What changes with DLS and $\lambda = 0.05$?

Solution. (a) By Proposition 13.1, with equality in the aligned case,

$$\|\dot{\theta}\| = \frac{\|\dot{\mathbf{x}}\|}{\sigma_{\min}} = \frac{100}{0.002} = 50\,000 \text{ joint units/s,}$$

a value off the scale for any servomotor: the controller raises the joint-speed-exceeded alarm and stops the motion.

(b) With DLS, the amplification factor along the critical direction goes from $1/\sigma$ to $\sigma/(\sigma^2 + \lambda^2)$:

$$\frac{0.002}{0.002^2 + 0.05^2} = \frac{0.002}{0.000004 + 0.0025} = \frac{0.002}{0.002504} \approx 0.7987,$$

hence $\|\dot{\theta}\| \approx 79.9$ units/s: perfectly executable. The price is a tracking error — the TCP traverses the critical direction at about 80% of the required speed instead of 100% — but the motion continues without stopping.

13.20.2 Symbolic and proof exercises

Exercise 13.9. Prove that, in $\mathbb{R}^3$, two points and a radius do not determine an arc, while three non-collinear points do. Deduce why all robot languages require an intermediate point.

Solution. *Two points and a radius do not suffice.* Let P_1, P_2 with $|P_1P_2| = 2\ell$ and $r \geq \ell$. The centre C must satisfy $|C - P_1| = |C - P_2| = r$, i.e. it must belong to the intersection of the two spheres of radius r centred at P_1 and P_2 . That intersection is a **circle** (of radius $\sqrt{r^2 - \ell^2}$, lying in the bisector plane of P_1P_2): not a point, but an infinite set of admissible centres. If $r > \ell$ there are thus infinitely many distinct arcs; only for $r = \ell$ (the semicircle of diameter P_1P_2) does the intersection degenerate to the midpoint alone — and even then the *plane* of the arc remains undetermined.

Three non-collinear points suffice. This is Theorem 13.1: the three linear conditions of Step 1 have an invertible matrix, hence a unique solution.

Practical consequence. In $\mathbb{R}^2$ the plane is given and the specification “centre + radius + sense” suffices: indeed G-code, working in the plane selected by G17/G18/G19, uses I/J. In $\mathbb{R}^3$ the plane is an unknown, and the cheapest way to fix it together with centre and radius is to give a third point: hence MOVc, MoveC and CIRC all require one.

Exercise 13.10. Prove that the DLS solution $\dot{\theta} = J^\top(JJ^\top + \lambda^2 I)^{-1}\dot{\mathbf{x}}$ always exists and that its norm is bounded by $\|\dot{\mathbf{x}}\|/(2\lambda)$, however close J is to singular.

Solution. *Existence.* $JJ^\top$ is symmetric positive semidefinite, so its eigenvalues are the squares of the singular values, $\sigma_i^2 \geq 0$. The eigenvalues of $JJ^\top + \lambda^2 I$ are $\sigma_i^2 + \lambda^2 \geq \lambda^2 > 0$: the matrix is positive definite, hence invertible. The solution exists for every $\dot{\mathbf{x}}$ and every configuration.

Boundedness. Use the singular value decomposition $J = U\Sigma V^\top$. Then

$$J^\top(JJ^\top + \lambda^2 I)^{-1} = V\Sigma^\top U^\top(U(\Sigma\Sigma^\top + \lambda^2 I)U^\top)^{-1} = V \operatorname{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda^2}\right) U^\top,$$

whose spectral norm is $\max_i \frac{\sigma_i}{\sigma_i^2 + \lambda^2}$. By the arithmetic–geometric mean inequality, $\sigma^2 +$

$\lambda^2 \geq 2\sigma\lambda$, hence

$$\frac{\sigma}{\sigma^2 + \lambda^2} \leq \frac{\sigma}{2\sigma\lambda} = \frac{1}{2\lambda}$$

for every $\sigma > 0$ (and the ratio is 0 for $\sigma = 0$). Therefore $\|\dot{\boldsymbol{\theta}}\| \leq \|\dot{\mathbf{x}}\|/(2\lambda)$.

Reading. The parameter λ is an *explicit* ceiling on joint speed: doubling it halves the maximum possible speed (and worsens tracking far from singularities). Equality is attained for $\sigma = \lambda$: DLS is “tuned” precisely on singular values of order λ .

Exercise 13.11. Prove that if the three taught points tend to collinearity, the radius computed by Theorem 13.1 tends to infinity, and estimate how small $|\mathbf{a} \times \mathbf{b}|$ must be for the radius to exceed a threshold $R_{\max}$.

Solution. From the radius formula,

$$R = \frac{|\mathbf{a}||\mathbf{b}||\mathbf{b} - \mathbf{a}|}{2|\mathbf{a} \times \mathbf{b}|}.$$

If the points tend to collinearity while keeping their mutual distances (that is, $|\mathbf{a}|, |\mathbf{b}|, |\mathbf{b} - \mathbf{a}|$ bounded and not infinitesimal), the numerator stays bounded below by a positive constant while $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\vartheta \rightarrow 0$ (the angle ϑ at P_1 tends to 0 or π). Hence $R \rightarrow +\infty$.

Threshold. Imposing $R > R_{\max}$:

$$|\mathbf{a} \times \mathbf{b}| < \frac{|\mathbf{a}||\mathbf{b}||\mathbf{b} - \mathbf{a}|}{2R_{\max}}.$$

Example. With points taught at $|\mathbf{a}| = |\mathbf{b}| = 100$ mm, $|\mathbf{b} - \mathbf{a}| = 100$ mm, and wishing to reject arcs of radius greater than $R_{\max} = 10$ m = 10 000 mm:

$$|\mathbf{a} \times \mathbf{b}| < \frac{100 \cdot 100 \cdot 100}{20\,000} = 50 \text{ mm}^2.$$

Since $|\mathbf{a} \times \mathbf{b}| = 10^4 \sin\vartheta$, the condition becomes $\sin\vartheta < 5 \times 10^{-3}$, i.e. an angular misalignment below about 0.29° . This is the order of magnitude of the tolerance controllers use internally before declaring the points collinear (Remark 13.1).

Exercise 13.12. A numerical controller interpolates with period $T = 2$ ms an arc of radius $R = 250$ mm at a feed rate $v = 300$ mm/min. Compute the chord error, and determine the maximum feed rate compatible with a chord error of $1 \mu\text{m}$.

Solution. The speed is $v = 300/60 = 5$ mm/s, so in one period the tool travels $L = vT = 5 \cdot 0.002 = 0.01$ mm. The chord error is

$$\delta \simeq \frac{L^2}{8R} = \frac{10^{-4}}{2000} = 5 \cdot 10^{-8} \text{ mm} = 0.05 \text{ nm},$$

entirely negligible. Inverting for $\delta_{\max} = 10^{-3}$ mm:

$$v = \frac{\sqrt{8R\delta_{\max}}}{T} = \frac{\sqrt{8 \cdot 250 \cdot 0.001}}{0.002} = \frac{\sqrt{2}}{0.002} = 707.1 \text{ mm/s} = 42.4 \text{ m/min}.$$

At that speed other constraints — centripetal acceleration and jerk — would govern, not the chord error.

Exercise 13.13. A block must move the tool from $S = (100, 50)$ to $E = (150, 100)$ along the arc with centre $C = (100, 100)$. Determine I , J , the radius, and decide whether G02 or G03 is needed to travel the short arc; compute its length.

Solution. The offsets are $I = 100 - 100 = 0$ and $J = 100 - 50 = 50$, and the radius is $R = |SC| = 50$; check: $|EC| = |(50, 0)| = 50 \checkmark$. Relative to the centre, S lies at angle 270° and E at angle 0° . Travelling counter-clockwise one goes from 270° to 360° , an arc of 90° ; clockwise one would travel 270° . The short arc therefore requires G03, and the block is G03 X150 Y100 I0 J50. Its length is $R \Delta\theta = 50 \cdot \pi/2 = 78.54$ mm.

13.21. Self-assessment quiz

Twelve multiple-choice questions. The answer key is at the end of the section.

1. Why do all robot languages require an intermediate point to define an arc in $\mathbb{R}^3$?
 - A) for safety reasons
 - B) because two points and a radius leave plane and centre undetermined
 - C) for compatibility with G-code
 - D) to compute the speed
2. In Theorem 13.1, the denominator that may vanish is
 - A) $|\mathbf{a}|$
 - B) $|\mathbf{a} \times \mathbf{b}|$
 - C) $\mathbf{a} \cdot \mathbf{b}$
 - D) $|\mathbf{b} - \mathbf{a}|$
3. In INFORM, a variable of type B can hold values
 - A) from -128 to 127
 - B) from 0 to 255
 - C) from 0 to 65535
 - D) in floating point
4. For a complete arc Yaskawa requires a minimum of
 - A) one MOVC instruction
 - B) two MOVC instructions
 - C) three MOVC instructions, the first treated as MOVL
 - D) five MOVL instructions
5. For a six-axis robot with a spherical wrist, **positional** inverse kinematics
 - A) always requires iterative inversion of the Jacobian
 - B) is solved in closed analytic form
 - C) has no solution

-
- D) requires Newton's method at every servo cycle
6. The manipulator Jacobian links
- A) joint positions and Cartesian pose
 - B) joint velocities and the Cartesian velocity of the TCP
 - C) driving torques and masses
 - D) temperature and thermal drift
7. When $\det J \rightarrow 0$ the controller
- A) gains accuracy
 - B) may require divergent joint speeds in certain directions
 - C) switches automatically to linear motion
 - D) is unaffected
8. The DLS (*Damped Least Squares*) method serves to
- A) increase the maximum speed
 - B) keep the solution bounded near singularities, at the cost of a little accuracy
 - C) compute the centre of the arc
 - D) compensate gearbox backlash
9. In an epicyclic gear train with fixed ring gear ($\omega_r = 0$) and input on the sun gear, with $z_s = 24$ teeth and $z_r = 72$ teeth, Theorem 13.6 gives the carrier reduction ratio ω_s/ω_c as
- A) 3
 - B) 4
 - C) -3
 - D) $1/4$
10. In the PLCopen `MC_MoveCircular...` blocks (and in Siemens TIA Portal, which uses the same name), a `CircMode` value of 1 means that `AuxPoint` represents
- A) a point on the arc, passed through by the motion
 - B) the centre of the arc
 - C) the signed radius
 - D) the traversal velocity
11. A planar two-link arm with $\ell_1 = 5$, $\ell_2 = 3$ must reach a point at distance $\rho = 1$ from the base. What does Theorem 13.2 say?
- A) it is reachable, with two solutions
 - B) it is reachable, with a single solution (boundary of the workspace)
 - C) it is not reachable: $\rho < |\ell_1 - \ell_2| = 2$
 - D) it is not reachable: $\rho > \ell_1 + \ell_2$
12. In the planar two-link arm, the kinematic singularity ($\det J = 0$) coincides with
-

- A) the centre of the workspace
 B) the inner or outer boundary of the reachable circular annulus
 C) a point arbitrarily chosen by the manufacturer
 D) it does not exist for this type of arm

BEST PRACTICE

Key. 1: B (Exercise 13.9) — 2: B — 3: B — 4: C — 5: B — 6: B — 7: B (Proposition 13.1) — 8: B — 9: B (Theorem 13.6: $\omega_s/\omega_c = 1 + z_r/z_s = 1 + 3 = 4$) — 10: B — 11: C (Theorem 13.2: $|5 - 3| = 2 > 1 = \rho$) — 12: B (Observation 13.6).

Note on question 5. This is the commonest conceptual error among those who program robots without having studied their kinematics: the belief that the controller “solves the position numerically” at every cycle. It does not — and could not, at the required kHz rates. The closed form exists *because* the wrist is spherical, and that is why vendors design it so.

INFO

Where this chapter stops, and where the discussion continues. Chapter 11 is an applied chapter of a treatise on the geometry of the circle, not a robotics course: it has deliberately stopped at **geometric kinematics** — where the end effector is, not what forces put it there nor how to keep it on a desired trajectory. With equal deliberation, here is what is missing here and where it can be found, for anyone who wishes to go further:

- **Arms with more than two links and non-planar joints.** The closed-form inverse kinematics proved here (Section 13.3.1) is for the planar two-link case; for longer chains and for rotations in three-dimensional space — of which this chapter uses only the planar case — the general treatment (Rodrigues’ formula, quaternions, the double cover of $SO(3)$) is in Volume III, in the chapter on rotations of space, $SO(3)$ and quaternions; the kinematics of general serial chains, with the product of exponentials on $SE(3)$, is in the following chapter on Lie groups.
- **Dynamics.** This chapter does not treat *the forces and torques* needed to produce the motion (only where the end effector goes, not what moves it). As of this writing, the treatise **does not yet** have a chapter dedicated to manipulator dynamics (Newton–Euler or Lagrange equations for a kinematic chain): this is a future extension, not present in Volume III today.
- **Feedback control.** Volume III has a general chapter on feedback control (Nyquist criterion, stability margins, geometric loci in the complex plane) that treats the theory of closed-loop systems in general, not the specific problem of making a robot arm follow an assigned trajectory in joint space or task space: that specific treatment, today, **does not exist** yet in the treatise.

This is not an addition of new mathematical content: it is internal bibliographic honesty, current as of the date of this writing. If Volume III is later enriched with a chapter on manipulator dynamics or state-space control, this box should be updated accordingly.

13.22. Chapter Formula sheet

Planar two-link arm (Theorems 13.2 and 13.3), with $\rho = \sqrt{x^2 + y^2}$, $\varphi = \text{atan2}(y, x)$

$$\text{reachable} \iff |\ell_1 - \ell_2| \leq \rho \leq \ell_1 + \ell_2, \quad c = \frac{\ell_1^2 + \rho^2 - \ell_2^2}{2\ell_1\rho}, \quad \theta_1 = \varphi \pm \arccos c$$

$$\det J = \ell_1 \ell_2 \sin \theta_2 \quad (\text{Observation 13.6: } = 0 \iff \rho \in \{\ell_1 + \ell_2, |\ell_1 - \ell_2|\})$$

Circle through three points in space

plane through $P_1, P_2, P_3 \cap$ two bisecting planes $\implies$ centre C , $\rho = \|P_1 - C\|$

Chord error and speed limits (T = interpolation period, ρ = radius)

$$v \leq \frac{2}{T} \sqrt{2\rho \delta_{\max} - \delta_{\max}^2} \simeq \frac{2}{T} \sqrt{2\rho \delta_{\max}}$$

$$v_{\max}(\rho) = \min \left\{ v_{\text{prog}}, \underbrace{\frac{2}{T} \sqrt{2\rho \delta_{\max}}}_{\text{chord}}, \underbrace{\sqrt{\rho a_{\max}}}_{\text{centripetal}}, \underbrace{(J_{\max} \rho^2)^{1/3}}_{\text{jerk}} \right\}$$

the chord limit dominates the centripetal one $\iff \delta_{\max} < \frac{a_{\max} T^2}{8}$ (independently of ρ)

Tracking error: radius reduction

$$G(s) = \frac{K_v}{s + K_v}, \quad \omega = \frac{v}{R}, \quad R' = \frac{R}{\sqrt{1 + (\omega/K_v)^2}}, \quad \varphi = \arctan \frac{\omega}{K_v}$$

$$\Delta R = R - R' \simeq \frac{R}{2} \left(\frac{\omega}{K_v} \right)^2 = \frac{v^2}{2RK_v^2}$$

Circular interpolation instructions

CNC (ISO 6983)	G02 (clockwise), G03 (anticlockwise), offsets I,J,K
Yaskawa (INFORM)	MOV C (three taught points)
FANUC (TP / KAREL)	Circular instruction
ABB (RAPID)	MoveC (via point + end point)
KUKA (KRL)	CIRC (auxiliary point + end point)
PLCopen	MC_MoveCircularAbsolute / Relative

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The Circle in Water Treatment Plants

INFO

Level and prerequisites. An applicative chapter, at secondary-school level: it uses only the area and circumference of the circle (Chapter 6), angular and peripheral velocity (Chapter 4), and the conservation of the volume of a flow across a surface. Every step that would require a differential equation is explicitly flagged as such and deferred to Volume II. Audience: even a plant technician who wants to understand *where* the design formulas they already use come from.

The second great application domain of this treatise, alongside the industrial robotics of Chapter 13, is water treatment plant engineering: here the circle is not the trajectory of a tool, but the **very shape** of the process tanks. The choice of a circular plan for settlers and clariflocculators is not aesthetic: it follows from the possibility of a uniform radial distribution of the flow starting from a central point, and from the minimum perimeter for equal area (the isoperimetric inequality of Chapter 6, applied here to civil construction). The mathematical heart of the chapter is a theorem due to Allen Hazen (1904): the criterion that decides whether a suspended particle is retained by a settler depends *only* on the ratio between flow rate and tank area — not on depth, not on detention time — and is, at bottom, a statement about the growth of a circular area over time.

14.1. The circular settler: definitions and weir flow rate

Definition 14.1: Central-fed circular settler. A cylindrical tank of radius r (diameters used in practice range from a few metres to several tens of metres, source F-71) in which the raw liquor enters at the centre through a feed well of radius $\rho_f < r$: this dissipates the kinetic energy of the incoming flow and redistributes it radially and uniformly towards the periphery. The clarified effluent overflows peripherally over a **weir**, placed along or near the circumference of the tank, and is collected in a peripheral launder. The sludge settled on the bottom is conveyed towards a central hopper by a rotating **scraper bridge** (§14.4).

Proposition 14.1: Available weir length. Let q_L be the allowable weir loading per unit length (maximum flow rate per metre of weir, a design datum). For a weir running along the whole peripheral circumference of a circular settler of radius r , the available weir length is $L_w = 2\pi r$, and the maximum allowable flow rate is

$$Q_{\max} = q_L \cdot L_w = q_L \cdot 2\pi r.$$

If the weir is instead arranged on an internal **double-sided** launder, of radii r_1 and r_2 (the two sides of the launder, both close to r), the available length nearly doubles:

$$L_w = 2\pi r_1 + 2\pi r_2 = 2\pi(r_1 + r_2).$$

Proof. By definition q_L is the maximum flow rate per unit length of weir. The length of the weir is the length of the circumference travelled (Chapter 6): $2\pi r$ in the simple case, and the sum $2\pi r_1 + 2\pi r_2$ in the double-sided case, because the water overflows on *both* sides of the launder, each a circumference of its own length. Multiplying the allowable unit loading by the available length gives the overall maximum flow rate. $\square$

Remark 14.1: Why the double-sided launder nearly doubles L_w without changing A . If $r_1, r_2 \approx r$ (the launder is narrow relative to the tank radius), $L_w \approx 2 \cdot 2\pi r$: nearly double the weir length of the simple scheme, for the same sedimentation area $A = \pi r^2$ (which depends only on the tank's outer radius, not on how the weir is arranged). This is why large-diameter tanks almost always adopt an internal double-sided launder instead of a single peripheral weir: it increases the available weir length without building a larger tank.

Example 14.1: Numerical check. Tank of radius $r = 15$ m. Simple weir: $L_w = 2\pi \cdot 15 = 94.25$ m. Double-sided launder with $r_1 = 14.7$ m, $r_2 = 14.9$ m: $L_w = 2\pi(14.7 + 14.9) = 185.98$ m, a ratio of 1.973 relative to the simple case $\checkmark$ (script `fase5_acque.py`) — not exactly double, because $r_1, r_2 < r$, but very close.

Remark 14.2: Why the circumference, and not the area, governs the weir. It is a common design error to confuse weir sizing (governed by the circumference $L_w = 2\pi r$) with that of the sedimentation surface (governed by the area $A = \pi r^2$, which instead fixes the detention time and the surface loading rate, §14.3). Doubling the radius r , the weir capacity doubles (proportional to r), while the sedimentation surface quadruples (proportional to r^2 , checked in §14.8): this is one of the geometric reasons why, beyond a certain size, a single circular settler stops scaling efficiently and it becomes convenient to split the flow rate across several tanks.

14.2. Hazen's theorem

The ideal sedimentation model, due to Allen Hazen (1904, source F-72) and systematised by Camp (1946, source F-73), pictures the radial flow in the tank as uniform over the useful height h (*plug flow*) and every suspended particle as carried horizontally by the flow while sinking vertically at its own constant **settling velocity** v_s (depending on the particle's size and specific gravity; its computation from physical laws — Stokes' law — is deferred to Volume II). A particle enters the tank at the edge of the feed well, radius ρ_f , at the surface of the liquid.

Theorem 14.1: Hazen's criterion for the circular settler. In the plug-flow model, a particle with settling velocity v_s , entering at the edge of the feed well (radius ρ_f), is retained by the circular settler of radius r and useful height h , fed with flow rate Q , **if and only if**

$$v_s \geq \frac{Q}{\pi(r^2 - \rho_f^2)} = \frac{Q}{A_{\text{eff}}},$$

where $A_{\text{eff}} = \pi(r^2 - \rho_f^2)$ is the area of the effective sedimentation annulus (the total area πr^2 minus the area occupied by the feed well).

Proof. Step 1 — where the particle is at time t . The particle moves horizontally together with the liquid: at time t it is at some radius $\rho(t) \geq \rho_f$. By conservation of flow rate, the volume of liquid that has entered the tank between instant 0 and instant t is $Q t$, and it equals exactly the volume of the cylindrical annulus between the feed well (radius ρ_f) and the radius $\rho(t)$

reached by the front of “new” liquid, of height h :

$$\pi(\rho(t)^2 - \rho_f^2) h = Q t$$

(area of the circular annulus, Chapter 6, for the volume of a hollow cylinder of that area and height h). This determines $\rho(t)$ as a function of t , without needing to know the radial velocity instant by instant.

Step 2 — when the particle touches the bottom. Vertically the particle sinks at constant velocity v_s from the surface ($z = h$) to the bottom ($z = 0$): it touches the bottom at time

$$t_b = \frac{h}{v_s}.$$

Step 3 — at which radius. Substituting $t = t_b$ into the relation of Step 1, the radius $\rho_b = \rho(t_b)$ at which the particle touches the bottom satisfies

$$\pi(\rho_b^2 - \rho_f^2) h = Q \cdot \frac{h}{v_s} \implies \rho_b^2 = \rho_f^2 + \frac{Q}{\pi v_s}$$

(h cancels: the result does not depend on the height of the tank).

Step 4 — the criterion. The particle is retained if it touches the bottom *before* leaving the tank, that is, if $\rho_b \leq r$:

$$\rho_f^2 + \frac{Q}{\pi v_s} \leq r^2 \iff v_s \geq \frac{Q}{\pi(r^2 - \rho_f^2)} = \frac{Q}{A_{\text{eff}}}.$$

□

Corollary 14.1: Efficiency does not depend on depth or on detention time. *The retention criterion of Theorem 14.1 does not depend on the useful height h of the tank: for the same surface hydraulic loading Q/A_{eff} , a deeper tank does not retain particles any better than a shallower one.*

Proof. In Step 3 of the proof of Theorem 14.1, h appears on both sides of the equality $\pi(\rho_b^2 - \rho_f^2)h = Qh/v_s$ and cancels: the radius ρ_b at which a particle touches the bottom, and hence the criterion $\rho_b \leq r$, depends only on ρ_f, Q, v_s, r , not on h . Since the criterion contains no h , two tanks with the same area $A = \pi r^2$ and the same flow rate Q retain exactly the same particles regardless of their height — that is, they depend only on the surface loading Q/A_{eff} .

□

Remark 14.3: Hazen’s most counter-intuitive result. *This is the best-known, and most counter-intuitive, result of Hazen’s 1904 paper: the theoretical efficiency of an ideal settler does not depend on the detention time $T = V/Q$ (which instead grows with h , for the same A and Q , without improving sedimentation in the ideal model) — only on the surface hydraulic loading Q/A . In practice a minimum depth remains necessary for other reasons (room for the sludge, avoiding resuspension), but the sizing of the surface is governed by the surface loading alone.*

Remark 14.4: The maximum radial velocity is at the feed well. *In the plug-flow model, through every concentric cylinder of radius $\rho \in [\rho_f, r]$ and height h passes (at steady state) the entire incoming flow rate Q : approximating the radial velocity as uniform over*

the lateral surface of that cylinder, $v(\rho) \cdot (2\pi\rho h) \approx Q$, that is,

$$v(\rho) = \frac{Q}{2\pi\rho h},$$

the same relation used in Step 1 of the proof (it is its instantaneous form rather than integrated over time). Since $v(\rho)$ is decreasing in ρ , the radial velocity is **maximal** precisely at the edge of the feed well, $\rho = \rho_f$: this is the most critical zone from the standpoint of the risk of resuspending particles already settled nearby, not the periphery of the tank.

Example 14.2: Checking Hazen's criterion. Settler with $Q = 1200 \text{ m}^3/\text{h}$, $r = 15 \text{ m}$, $\rho_f = 2 \text{ m}$, particle with $v_s = 1.5 \text{ m/h}$.

Effective area. $A_{\text{eff}} = \pi(15^2 - 2^2) = \pi \cdot 221 = 694.292 \text{ m}^2$.

Radius at which it touches the bottom. $\rho_b^2 = 2^2 + \frac{1200}{\pi \cdot 1.5} = 4 + 254.648 = 258.648$, hence $\rho_b = 16.0825 \text{ m}$.

Verdict. $\rho_b = 16.08 \text{ m} > r = 15 \text{ m}$: the particle is **not** retained — it (ideally) touches the bottom beyond the edge of the tank, that is, it leaves still in suspension. Consistently, the effective surface loading exceeds the settling velocity:

$$\frac{Q}{A_{\text{eff}}} = \frac{1200}{694.292} = 1.728 \text{ m/h} > v_s = 1.5 \text{ m/h}.$$

All values checked with the script `fase5_acque.py`. ✓ To retain this particle too one would have to increase r (with A_{eff} growing as r^2), or reduce Q , or split the flow rate across several tanks in parallel.

14.3. Detention time and surface loading

Definition 14.2: Hydraulic detention time and surface loading. For a cylindrical tank of radius r and useful height h , the useful volume is $V = \pi r^2 h$. Given an incoming flow Q , one defines the **surface hydraulic loading** $q_A = Q/A = Q/(\pi r^2)$ and the theoretical (plug-flow) **detention time** $T = V/Q$.

Proposition 14.2: Detention time as the ratio of height to surface loading.

$$T = \frac{h}{q_A}.$$

Proof. $T = V/Q = (\pi r^2 h)/Q = h/(Q/(\pi r^2)) = h/q_A$, by Definition 14.2. □

Warning. Design values: only with a source. Normative and handbook references cite typical ranges for weir loading and surface loading, but must be used with the exact table/paragraph number, not from memory:

- **Weir loading.** The *Ten States Standards* (Recommended Standards for Wastewater Facilities, 2014 ed., source F-74) is commonly cited for an upper limit of the order of 20,000 gpd/ft (gallons per day per linear foot of weir). Converting: 1 gpd/ft = $0.012419 \text{ m}^3/(\text{m} \cdot \text{day})$ (checked, script `fase5_acque.py`: 1 US gallon = $3.785412 \times 10^{-3} \text{ m}^3$, 1 foot = 0.3048 m), hence $20,000 \text{ gpd/ft} \approx 248.39 \text{ m}^3/(\text{m} \cdot \text{day}) \approx 10.35 \text{ m}^3/(\text{m} \cdot \text{h})$. The exact paragraph number and any condition on the plant's flow rate

must be reconfirmed against the full text of the standard before being cited as a design constraint in a real case.

- **Surface loading and depth.** Metcalf & Eddy / AECOM, *Wastewater Engineering*, 5th ed. (source F-71), reports typical ranges in the chapter on primary sedimentation; no specific numerical value is reported here, not having verified the exact table and page in this edition. For *secondary* clarifiers at average flow, secondary sources summarising the same text (checked 14 September 2026) report a recommended range of $24\text{--}32\text{ m}^3/(\text{m}^2 \cdot \text{day})$ and a typical useful depth of $4.3\text{--}4.9\text{ m}$: values used in Example 14.3 below as a *verified reference range*, not as an exact paragraph from a primary edition.
- **European standard.** EN 12255-4 (*Wastewater treatment plants — Part 4: Primary settlement*, source F-75) is the European normative reference; cited here only for its existence, with no numerical values (text not consulted).

Example 14.3: Complete sizing of a secondary clarifier. Design data (constructed for this example, not attributed to a real plant): average flow $Q = 5000\text{ m}^3/\text{day}$; design surface hydraulic loading $q_A = 28\text{ m}^3/(\text{m}^2 \cdot \text{day})$ (the midpoint of the $24\text{--}32$ range just cited); useful depth $h = 4.5\text{ m}$ (within the typical range). For this example the feed well radius is neglected ($\rho_f \approx 0$, a stated simplification: legitimate when $\rho_f \ll r$, as verified a posteriori on the result), so that the effective area of Theorem 14.1 coincides with the total area $A = \pi r^2$ of Definition 14.2.

Step 1 — surface area and diameter. From Definition 14.2, $A = Q/q_A = 5000/28 = 178.5714\text{ m}^2$, hence

$$r = \sqrt{\frac{A}{\pi}} = 7.5393\text{ m}, \quad D = 2r = 15.0786\text{ m}.$$

Step 2 — detention time. By Proposition 14.2, $T = h/q_A = 4.5/28 = 0.160714\text{ days} = 3.8571\text{ h}$, within the typical $2\text{--}4\text{ h}$ range for secondary clarifiers cited above. ✓

Step 3 — weir loading. Perimeter $2\pi r = 47.3708\text{ m}$; loading $Q/(2\pi r) = 105.5502\text{ m}^3/(\text{m} \cdot \text{day}) = 4.3979\text{ m}^3/(\text{m} \cdot \text{h})$, well below the $10.35\text{ m}^3/(\text{m} \cdot \text{h})$ limit of the *Ten States Standards* already verified in this chapter. ✓

Step 4 — the critical particle (link to Hazen’s theorem). With $\rho_f = 0$, Theorem 14.1 retains exactly the particles with $v_s \geq q_A$. For the *critical* particle $v_s = q_A = 28/24 = 1.166667\text{ m/h}$, the fall time of Step 2 of the proof of Theorem 14.1 equals

$$t_b = \frac{h}{v_s} = \frac{4.5}{1.166667} = 3.8571\text{ h},$$

which coincides **exactly** with T from Step 2 above — not a numerical coincidence, but a general fact: $t_b = h/v_s$ and $T = h/q_A$ are the same expression when $v_s = q_A$, independently of h (consistent with Corollary 14.1). Figure 14.2 shows this together with a faster particle (retained with margin) and a slower one (not retained within T). All values in this example checked with the script `cap14_chiarificatore.py`. ✓

14.4. The scraper bridge: kinematics

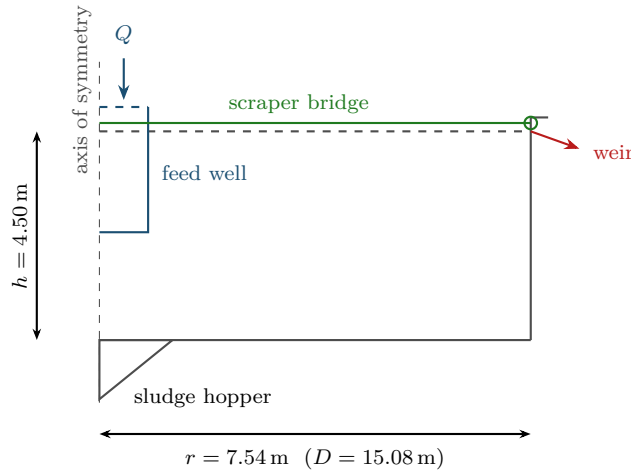


Figure 14.1: Schematic half cross-section (not to scale) of the clarifier of Example 14.3: central feed from the feed well, radial scraper bridge with support wheel at the periphery, perimeter weir. The sludge hopper at the centre is not to scale relative to the rest of the tank.

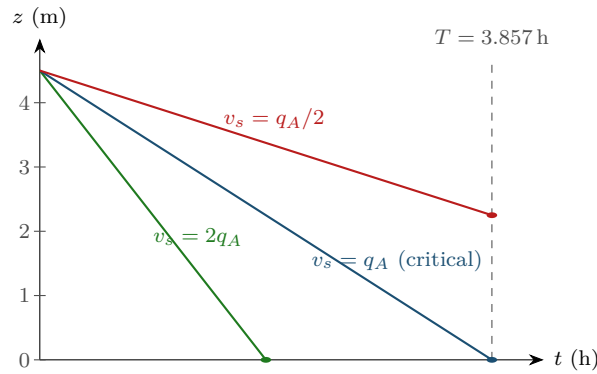


Figure 14.2: Vertical position $z(t) = h - v_s t$ of three particles (Step 2 of the proof of Theorem 14.1), for the example in the text: the critical particle $v_s = q_A$ reaches the bottom exactly at the theoretical detention time T (blue dot at $z = 0$); a faster particle reaches it earlier, with margin (green dot); a slower one has not yet reached the bottom when $t = T$ (red dot, $z = 2.25$ m).

Definition 14.3: Centre-driven scraper bridge. A rotating structure, pivoted at the centre of the tank and supported at the periphery by a circular track, which carries the bottom scrapers (to convey the settled sludge towards the central hopper) and, often, a surface blade for floating matter. It rotates at constant, low angular velocity ω .

Proposition 14.3: Radial component of the thrust of an inclined blade. A radial blade set at an angle β relative to the tangential direction, advancing tangentially at speed $\omega\rho$ (at distance ρ from the centre), pushes the sludge — assumed to slide along the blade without friction — with a velocity normal to the blade $\omega\rho\sin\beta$; the **radial** component of this velocity is

$$v_\rho = \omega\rho\sin\beta\cos\beta = \frac{1}{2}\omega\rho\sin 2\beta,$$

maximal for $\beta = 45^\circ$.

Proof. The blade moves with velocity $\omega\rho$ in the tangential direction (like a point at distance ρ from the axis of rotation, Chapter 4). Decomposing this velocity into the directions parallel and normal to the blade (which forms an angle β with the tangential direction): the component normal to the blade is $\omega\rho\sin\beta$ (the sludge, constrained to slide along the blade, receives only this component as effective thrust; the component parallel to the blade does not move it relative to the blade). This normal thrust $\omega\rho\sin\beta$, in turn, has a radial component $(\omega\rho\sin\beta)\cos\beta$ and a tangential component $(\omega\rho\sin\beta)\sin\beta$ (again projecting the normal to the blade onto the radial and tangential directions at the point, which differ by the angle β between blade and tangent). Hence $v_\rho = \omega\rho\sin\beta\cos\beta = \frac{1}{2}\omega\rho\sin 2\beta$, maximal when $\sin 2\beta = 1$, that is, $\beta = 45^\circ$. $\square$

Remark 14.5: Sludge transport time: depends on the radius, but only logarithmically. Since $v_\rho = k\rho$ with $k = \omega\sin\beta\cos\beta$ (Proposition 14.3), the radius of the sludge particle satisfies $d\rho/dt = -k\rho$ (it moves towards the centre, hence the negative sign): a first-order linear differential equation, whose solution $\rho(t) = \rho_0 e^{-kt}$ is **imported** here without proof (solved by separation of variables in Volume II). The time to carry the sludge from radius ρ_0 to the hopper of radius ρ_h is then

$$t = \frac{\ln(\rho_0/\rho_h)}{\omega\sin\beta\cos\beta}.$$

Beware of a common mistake: this logarithmic dependence on ρ_0 is weak but not zero — the transport time is **not** independent of the starting radius. Scrapers arranged along a **logarithmic spiral** (a curve which, by its characteristic property — imported from Volume III, plane curves — keeps β constant relative to the radius vector at every one of its points) explain why this shape is chosen in practice for the blades: to keep β constant along the whole radius, not as a geometric shortcut to entirely eliminate the dependence on the starting radius.

Example 14.4: Transport time and peripheral speed: numerical check. Tank $D = 30$ m (hence $\rho_0 = 15$ m), one complete turn of the bridge in 45 min ($\omega = 2\pi/45 = 0.139626$ rad/min), blade at $\beta = 30^\circ$, hopper of radius $\rho_h = 1.5$ m:

$$t = \frac{\ln(15/1.5)}{0.139626 \cdot \sin 30^\circ \cos 30^\circ} = \frac{\ln 10}{0.139626 \cdot 0.433013} = 38.084 \text{ min}$$

(checked, script `fase5_acque.py`), about 85% of the time of a complete turn — consistent with the practical requirement that the sludge reach the hopper within one turn.

Peripheral speed. For a circular scraper bridge, a catalogue datum (source F-76, WAMGROUP/SAVI PTP series data sheets — a **peripheral-drive** bridge, not central as in Definition 14.3, but with the same blade kinematics as Proposition 14.3, which does not depend on where the driving torque is applied; consulted 2026) declares a peripheral speed of the order of 1.4 m/min, a plausible order of magnitude for either variant. For $D = 30$ m, the time for a complete turn is

$$t_{\text{turn}} = \frac{\pi D}{v} = \frac{\pi \cdot 30}{1.4} = 67.32 \text{ min}$$

(checked) — a catalogue datum, not a geometric law: it must always be cited with source and date, not reported as a generic range without a reference.

14.5. Radial distribution and the feed well

Proposition 14.4: Flow rate through concentric cylinders. *Through every concentric cylinder of radius $\rho \in [\rho_f, r]$ and height h , at steady state, passes the entire incoming flow rate Q ; the average radial velocity on that surface is*

$$v(\rho) = \frac{Q}{2\pi\rho h},$$

a decreasing function of ρ , with maximum value at the edge of the feed well ($\rho = \rho_f$).

Proof. This is Observation 14.4 of §14.2, repeated here as a stand-alone proposition because it underlies the sizing of the feed well: if $v(\rho_f)$ exceeds a velocity threshold beyond which sludge already deposited near the feed well risks being resuspended (a threshold depending on the type of sludge and to be taken from F-71, not reported here without a verified table number), the feed well must be widened to reduce $v(\rho_f)$. $\square$

14.6. Robotic inspection and maintenance

The bridge connecting to Chapter 13: the inner wall of a circular tank is a natural candidate for robotic inspection with sensors mounted on an arm, following exactly the circular trajectory described in the previous chapters.

Sizing the job for a complete inspection turn. Data: a Yaskawa arm that must travel, at constant speed, a circular inspection trajectory (e.g. a tool-mounted sensor) around a circular tank of radius $r = 1.2$ m, at a fixed height.

Step 1. Circumference to travel: $C = 2\pi r = 2\pi \cdot 1.2 \approx 7.54$ m.

Step 2. By Theorem 7.2 (Chapter 7), three non-collinear points determine a unique circle, but they identify an **arc**, not necessarily the complete turn: a single MOV C cannot therefore close 360° . One therefore chooses to divide the turn into 4 equal arcs of 90° each (≈ 1.88 m of linear extent each), a common practical compromise between number of instructions and regularity of the trajectory: 4 points are recorded on the circumference 90° apart from one another, plus the starting point, for a total of 5 taught points and 4 arc MOV C instructions.

Step 3. The *theoretical minimum* number of independent points needed to determine a circle remains 3 (Theorem 7.2); the division into 4 arcs in this example is a programming choice, not a geometric necessity. $\square$

Remark 14.6: The same parameters from a PLCopen perspective. *The same inspection turn could be programmed, as an alternative to the Yaskawa job, with a single MC_MoveCircularAbsolute block (PLCopen standard, Chapter 13) with CircMode=1 (centre), centre equal to the tank centre, radius $r = 1.2$ m and PathChoice set for the complete turn: unlike the INFORM job (which requires 5 MOV C instructions), a PLCopen function block at the PLC level can in principle handle the entire 360° arc in a single call, delegating to the manufacturer's implementation the internal subdivision into interpolation micro-steps.*

14.7. Why the circular plan

Remark 14.7: Three geometric reasons for the circle. Beyond the isoperimetric property already encountered (for equal area, the circle has the minimum perimeter, Chapter 6 — here translating into less perimeter structure, less weir to build, for the same sedimentation capacity), the circular plan offers two further purely geometric advantages over a rectangular tank:

- **Equal radial path in every direction:** from the central feed well to any point of the edge the distance is always r (radial symmetry, by definition of a circle), whereas in a rectangular tank the distance from the inlet point varies with direction and with position along the perimeter.
- **No dead corners:** a rotating scraper bridge, pivoted at the centre, sweeps the whole surface with no zones reached only tangentially (unlike the corners of a rectangular tank, where chain scrapers require dedicated transition geometries), and with a single mechanical pivot instead of a system of linear guides.

Counterpoint: rectangular tanks (often with chain scrapers and multiple hoppers) remain preferred when several units must be placed side by side on a given rectangular area, sharing walls between adjacent tanks — a land-use advantage the circle does not have (see F-71 for the complete cost/benefit comparison).

14.8. Problem set

14.8.1 Numerical exercises

Exercise 14.1. A circular settler has radius $r = 8$ m and useful height $h = 3$ m, fed at the centre (negligible feed well). The incoming flow rate is $Q = 250 \text{ m}^3/\text{h}$. Calculate: (a) the theoretical detention time; (b) the maximum allowable weir flow rate, with loading $q_L = 10 \text{ m}^3/(\text{m} \cdot \text{h})$ over the whole circumference; (c) the surface hydraulic loading q_A , and compare it with the example of §14.2 ($Q/A_{\text{eff}} = 1.728 \text{ m/h}$).

Solution. (a) $V = \pi r^2 h = \pi \cdot 64 \cdot 3 \approx 603.2 \text{ m}^3$; $T = V/Q \approx 603.2/250 \approx 2.41 \text{ h}$.
 (b) $L_w = 2\pi r \approx 50.27 \text{ m}$; $Q_{\text{max}} = q_L \cdot L_w = 10 \cdot 50.27 \approx 502.7 \text{ m}^3/\text{h}$, amply sufficient for $Q = 250 \text{ m}^3/\text{h}$.
 (c) $q_A = Q/(\pi r^2) = 250/(\pi \cdot 64) = 1.243 \text{ m/h}$, lower than the 1.728 m/h of the 15 m tank: consistent with the fact that the 8 m tank receives a flow rate proportionally smaller relative to its area.

Exercise 14.2. A tank of radius $r = 20$ m adopts either a simple weir ($L_w = 2\pi r$) or a double-sided launder with $r_1 = 19.6$ m, $r_2 = 19.8$ m. By how much does L_w increase going from the first solution to the second, in percentage terms?

Solution. $L_{w,1} = 2\pi \cdot 20 = 125.66 \text{ m}$. $L_{w,2} = 2\pi(19.6 + 19.8) = 2\pi \cdot 39.4 = 247.54 \text{ m}$. Increase: $(247.54 - 125.66)/125.66 \approx 97.0\%$, nearly double (Observation 14.1).

Exercise 14.3. Settler with $r = 10$ m, $\rho_f = 1.5 \text{ m}$, $Q = 500 \text{ m}^3/\text{h}$. Check, with Theorem 14.1, whether a particle with $v_s = 0.8 \text{ m/h}$ and one with $v_s = 2.0 \text{ m/h}$ are retained.

Solution. $A_{\text{eff}} = \pi(10^2 - 1.5^2) = \pi \cdot 97.75 = 307.09 \text{ m}^2$. Critical velocity: $v_{s,\text{crit}} = Q/A_{\text{eff}} = 500/307.09 = 1.628 \text{ m/h}$. For $v_s = 0.8 \text{ m/h}$ (< 1.628): **not** retained ($\rho_b = \sqrt{1.5^2 + 500/(\pi \cdot 0.8)} = 14.18 \text{ m} > 10 \text{ m}$). For $v_s = 2.0 \text{ m/h}$ (> 1.628): **retained** ($\rho_b = \sqrt{1.5^2 + 500/(\pi \cdot 2.0)} = 9.05 \text{ m} < 10 \text{ m}$). Checked, script `fase5_acque.py`.

Exercise 14.4. A scraper bridge has a catalogue peripheral speed $v = 1.2 \text{ m/min}$ (a source to cite in a real case). For a tank $D = 24 \text{ m}$, how long does a complete turn take?

Solution. $t_{\text{turn}} = \pi D/v = \pi \cdot 24/1.2 = 62.83 \text{ min}$.

Exercise 14.5. A designer must treat $Q = 900 \text{ m}^3/\text{h}$ with a design surface loading $q_A = 1.2 \text{ m}^3/(\text{m}^2 \cdot \text{h})$. What minimum diameter must the tank have?

Solution. $A = Q/q_A = 900/1.2 = 750 \text{ m}^2$; $r = \sqrt{A/\pi} = \sqrt{750/\pi} = 15.451 \text{ m}$; $D = 2r \approx 30.90 \text{ m}$ (checked, script `fase5_acque.py`).

14.8.2 Symbolic and proof exercises

Exercise 14.6. Prove that, doubling the diameter D of a settler for the same Q , the weir loading $q_L = Q/(\pi D)$ is halved and the surface loading $q_A = Q/(\frac{\pi}{4}D^2)$ is divided by four.

Solution. Let $D' = 2D$. Then $q'_L = Q/(\pi D') = Q/(2\pi D) = q_L/2$: halved. And $q'_A = Q/(\frac{\pi}{4}D'^2) = Q/(\frac{\pi}{4} \cdot 4D^2) = \frac{1}{4} \cdot Q/(\frac{\pi}{4}D^2) = q_A/4$: divided by four, because the area grows with the square of the diameter while the circumference grows linearly (already observed in Observation 14.2; numerical check in the script `fase5_acque.py`, lines W-09: for $D = 20$ and $D = 40$ with $Q = 1200$, $q_L(40)/q_L(20) = 0.5$ and $q_A(40)/q_A(20) = 0.25$).

Exercise 14.7. Starting from the flow-conservation relation $\pi(\rho_b^2 - \rho_f^2)h = Qh/v_s$ used in the proof of Theorem 14.1, derive ρ_b as a function of ρ_f, Q, v_s .

Solution. Cancelling h (it appears on both sides): $\pi(\rho_b^2 - \rho_f^2) = Q/v_s$, hence $\rho_b^2 = \rho_f^2 + Q/(\pi v_s)$, that is, $\rho_b = \sqrt{\rho_f^2 + Q/(\pi v_s)}$ — exactly the formula of Theorem 14.1.

Exercise 14.8. Prove, without using $V = \pi r^2 h$ directly, that $T = h/q_A$ starting from the definitions $T = V/Q$ and $q_A = Q/A$.

Solution. From $q_A = Q/A$ it follows that $Q = q_A \cdot A$. Substituting into $T = V/Q$ with $V = A \cdot h$ (volume = base area times height, for a cylinder): $T = (A \cdot h)/(q_A \cdot A) = h/q_A$, which is Proposition 14.2: the factor A cancels, so the result holds for *any* base shape satisfying the same relation $V = Ah$, not only for the circle — the circular specificity enters only in computing $A = \pi r^2$ and $L_w = 2\pi r$, not in this identity.

14.9. Self-assessment quiz

Ten multiple-choice questions. The answer key is at the end of the section.

-
1. The weir loading of a circular settler is proportional to:
 - A) r^2
 - B) r
 - C) $1/r$
 - D) r^3
 2. The sedimentation area $A = \pi r^2$ is proportional to:
 - A) r
 - B) r^2
 - C) $1/r$
 - D) $\sqrt{r}$
 3. In Hazen's theorem (Theorem 14.1), a particle is retained if and only if:
 - A) $v_s \geq Q/A_{\text{eff}}$
 - B) $v_s \leq Q/A_{\text{eff}}$
 - C) $v_s \geq Q \cdot A_{\text{eff}}$
 - D) $\rho_f \geq r$
 4. According to Corollary 14.1, the ideal sedimentation efficiency depends:
 - A) only on the depth h
 - B) only on the detention time T
 - C) only on the surface loading Q/A
 - D) on h and T together, not on Q/A
 5. The radial velocity $v(\rho) = Q/(2\pi\rho h)$ is maximal:
 - A) at the outer edge of the tank
 - B) at half radius
 - C) at the edge of the feed well
 - D) it is constant
 6. The radial thrust of a scraper-bridge blade, inclined at β , is maximal for:
 - A) $\beta = 0^\circ$
 - B) $\beta = 45^\circ$
 - C) $\beta = 90^\circ$
 - D) it does not depend on β
 7. The logarithmic-spiral profile of the bottom scrapers serves to:
 - A) make the transport time independent of the starting radius
 - B) keep the angle β constant along the whole radius
 - C) maximise the peripheral speed
-

- D) eliminate the feed well
8. Doubling the diameter D for the same Q , the weir loading:
- A) doubles
B) is halved
C) is divided by four
D) stays the same
9. An internal double-sided launder, compared with a simple peripheral weir, for the same tank radius:
- A) halves L_w
B) nearly doubles L_w , leaving A unchanged
C) doubles A
D) changes nothing
10. The detention time T and the surface loading q_A are related by:
- A) $T = q_A/h$
B) $T = h \cdot q_A$
C) $T = h/q_A$
D) $T = 1/(hq_A)$

Answer key: 1-B, 2-B, 3-A, 4-C, 5-C, 6-B, 7-B, 8-B, 9-B, 10-C.

14.10. Chapter Formula sheet

Weir

$$L_w = 2\pi r \text{ (simple)}, \quad L_w = 2\pi(r_1 + r_2) \text{ (double-sided)}, \quad Q_{\max} = q_L \cdot L_w$$

Hazen's theorem

$$A_{\text{eff}} = \pi(r^2 - \rho_f^2), \quad \rho_b^2 = \rho_f^2 + \frac{Q}{\pi v_s}, \quad \text{retained} \iff v_s \geq \frac{Q}{A_{\text{eff}}}$$

Detention time and surface loading

$$V = \pi r^2 h, \quad q_A = \frac{Q}{\pi r^2}, \quad T = \frac{V}{Q} = \frac{h}{q_A}$$

Radial velocity

$$v(\rho) = \frac{Q}{2\pi\rho h} \quad (\text{maximal at } \rho = \rho_f)$$

Scraper bridge

$$v_\rho = \omega \rho \sin \beta \cos \beta = \frac{1}{2} \omega \rho \sin 2\beta \quad (\text{maximal at } \beta = 45^\circ), \quad t = \frac{\ln(\rho_0/\rho_h)}{\omega \sin \beta \cos \beta}$$

Sources and bibliography

- [1] A. Hazen, *On sedimentation*, Transactions of the American Society of Civil Engineers 53(2) (1904), 45–71, DOI 10.1061/TACEAT.0001655.
- [2] T. R. Camp, *Sedimentation and the design of settling tanks*, Transactions of the American Society of Civil Engineers 111 (1946), 895–936, DOI 10.1061/TACEAT.0005912.
- [3] Metcalf & Eddy | AECOM, *Wastewater Engineering: Treatment and Resource Recovery*, 5th ed., McGraw-Hill, 2014, chap. 5 (primary sedimentation) — table values not reported in this chapter without direct verification of the page.
- [4] Great Lakes–Upper Mississippi River Board, *Recommended Standards for Wastewater Facilities* (“Ten States Standards”), 2014, chap. 70 (Settling) — weir loading limit of the order of 20,000 gpd/ft; exact paragraph to be reconfirmed against the full text.
- [5] EN 12255-4, *Wastewater treatment plants — Part 4: Primary settlement* — cited as the European normative reference, with no numerical values (text not consulted).
- [6] Data sheet of a peripheral-drive circular scraper bridge with declared peripheral speed (order of magnitude 1–2 m/min, used here only as a plausible order of magnitude also for the central-drive variant, kinematically equivalent), e.g. WAMGROUP/SAVI PTP series, <https://www.directindustry.com/prod/wamgroup-spa/product-29492-1863145.html> (consulted 2026) — a catalogue datum, to be cited always with source and date in a real project.

Final Assessment

INFO

How to use this assessment. Fifty multiple-choice questions spanning the whole volume, from the history of π to industrial robotics. Unlike the end-of-chapter quizzes, here the **answer is given immediately after each question**, together with a pointer to the result that justifies it: this is designed for revision, not for grading. Anyone wishing to use it as a closed test should cover the bold line with a sheet of paper.

Indicative scoring: ≥ 45 excellent; 38–44 good; 30–37 sufficient, with targeted revision of the chapters where mistakes occurred; < 30 it is worth rereading the volume following the logical chain of Chapter 6.

Part I — History and nature of π (Chapter 1)

1. The **transcendence** of π was proved by
 - A) Archimedes, in the 3rd century BCE
 - B) Lambert, in 1761
 - C) Lindemann, in 1882
 - D) Niven, in 1947

Answer: C. Lambert (1761) proved *irrationality*; Lindemann (1882) transcendence, which is stronger and definitively settles the problem of squaring the circle with ruler and compass.

2. What, in practice, does the transcendence of π rule out?
 - A) That π can be computed to arbitrary precision
 - B) That the circle can be squared with ruler and compass
 - C) That π is the ratio C/d
 - D) That series converging to π exist

Answer: B. Ruler-and-compass constructions yield only algebraic numbers; a transcendental number is not among them. The other three statements remain true.

3. The bracket obtained by Archimedes with the 96-sided polygon is
 - A) $3 < \pi < 3.5$
 - B) $3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}$
 - C) $3.14 < \pi < 3.15$
 - D) $\frac{22}{7} < \pi < \frac{355}{113}$

Answer: B (Theorem 1.3). In decimals, $3.14084 < \pi < 3.14286$: two certain digits, with a *certified* error — the real novelty compared with every earlier estimate.

4. The Monte Carlo estimate of π has an error decreasing as

-
- A) $O(1/N)$
 - B) $O(1/N^2)$
 - C) $O(1/\sqrt{N})$
 - D) $O(\log N)$

Answer: C (Definition 1.3). To quadruple the precision one must multiply the number of samples by 16: this is why Monte Carlo has never been competitive in π computation records.

5. Roughly how many correct digits does the Chudnovsky algorithm produce per term of the series?
- A) 1
 - B) 8
 - C) 14
 - D) 50

Answer: C (Definition 1.4). This is why it is the world-record algorithm, whereas the Gregory–Leibniz series, converging logarithmically, is useless for practical purposes.

Part II — Fundamental definitions (Chapter 2)

6. A line is **tangent** to $C(O, r)$ if and only if its distance from the centre equals
- A) 0
 - B) $< r$
 - C) $= r$
 - D) $> r$

Answer: C (Theorem 2.6). The classification is complete: $d < r$ secant, $d = r$ tangent, $d > r$ external. Note that the characterisation is *metric* and requires no counting of intersection points.

7. The radical axis of two non-concentric circles is
- A) a circle
 - B) a line perpendicular to the line of centres
 - C) a single point
 - D) existent only if the circles are secant

Answer: B. It is the locus of points of equal power with respect to the two circles; it *always* exists (even for disjoint circles), provided the centres are distinct.

8. Two circles with radii r_1, r_2 and distance of centres d are **orthogonal** if and only if
- A) $d = r_1 + r_2$
 - B) $d^2 = r_1^2 + r_2^2$
 - C) $d = |r_1 - r_2|$
 - D) $d = 0$

Answer: B (Theorem 5.4). Equivalently: the squared radius of each is the power of its centre with respect to the other — a property that makes orthogonal circles invariant under inversion.

9. A **non-secant** pencil of circles possesses

- A) two base points
- B) a single base point
- C) no base point, but two limit points
- D) infinitely many base points

Answer: C (Theorem 2.8). The two limit points are the degenerate circles of the pencil, and choosing one as pole is the move that makes two non-secant circles concentric (Chapter 5).

Part III — Chords, tangents, power (Chapter 3)

10. The radius drawn to the midpoint of a chord that is not a diameter is

- A) parallel to the chord
- B) perpendicular to the chord
- C) inclined at 45°
- D) undetermined

Answer: B (Theorem 3.2). The triangle formed by the two radii to the chord's endpoints is isosceles, and the median to the base of an isosceles triangle is also an altitude.

11. If two chords meet at an interior point P , then

- A) $PA + PB = PC + PD$
- B) $PA \cdot PB = PC \cdot PD$
- C) $PA/PB = PC/PD$
- D) $PA^2 + PB^2 = PC^2 + PD^2$

Answer: B (Theorem 3.7). It is the power of the point in metric guise: the product is the same for *every* chord through P , and equals $r^2 - |OP|^2$.

12. The two tangent segments drawn from one external point are

- A) always unequal
- B) congruent
- C) congruent only if the point lies on the radical axis
- D) congruent only for unit circles

Answer: B (Theorem 3.11). It is the pivot of the proof of the formula $r = (a + b - c)/2$ for the inradius of a right triangle (Theorem 10.4).

13. For a point P **interior** to the circle, the power $\text{pow}(P) = |OP|^2 - r^2$ is

- A) positive
- B) zero

-
- C) negative
 - D) undefined

Answer: C (Definition 3.1). The sign is precisely the information distinguishing interior, on the curve and exterior: negative, zero, positive.

14. From an external point P draw the tangent PT and a secant meeting the circle at A and B . Then
- A) $PT = PA + PB$
 - B) $PT^2 = PA \cdot PB$
 - C) $PT = \sqrt{PA + PB}$
 - D) $2PT = PA + PB$

Answer: B (Theorem 3.16). It is the limiting case of the two-secant theorem when the two intersection points of one of them coincide.

Part IV — Central and inscribed angles (Chapter 4)

15. An inscribed angle standing on an arc is
- A) equal to the corresponding central angle
 - B) half the corresponding central angle
 - C) twice the corresponding central angle
 - D) independent of the arc

Answer: B (Theorem 4.1). Immediate corollary: all inscribed angles on the same arc are congruent — the fact that generates the arc of constant angle and the cyclic-quadrilateral theorem.

16. Thales' theorem states that every inscribed angle subtending a diameter is
- A) acute
 - B) right
 - C) obtuse
 - D) equal to 60°

Answer: B (Theorem 4.4). It is the special case of the inscribed angle with a straight central angle: $180^\circ/2 = 90^\circ$.

17. In a cyclic quadrilateral, opposite angles are
- A) equal
 - B) complementary (sum 90°)
 - C) supplementary (sum 180°)
 - D) unrelated

Answer: C (Theorem 4.6). The converse also holds, and it is what one uses in practice to *prove* that four points are concyclic.

18. The **arc of constant angle** of a segment AB under an angle α is

- A) a single arc
- B) two arcs symmetric about AB
- C) a line
- D) the whole plane minus AB

Answer: B (Theorem 4.7). The case $\alpha = 90^\circ$ gives the two semicircles with diameter AB , i.e. Thales' theorem.

19. The Pothenot–Snellius resection (determining one's position from three known points) is solved
- A) by applying the law of cosines three times
 - B) by intersecting two arcs of constant angle
 - C) with a circular inversion
 - D) only numerically

Answer: B (Chapter 4). Each pair of known points, seen under a measured angle, determines an arc of constant angle; the observer lies at the intersection of two of them.

Part V — Circular inversion (Chapter 5)

20. Inversion with centre O and radius k maps P to P^* such that

- A) $|OP| + |OP^*| = k$
- B) $|OP| \cdot |OP^*| = k^2$
- C) $|OP| = k \cdot |OP^*|$
- D) $|OP|^2 + |OP^*|^2 = k^2$

Answer: B (Definition 5.1). P and P^* lie on the same ray from O ; the points of the circle of inversion are the fixed points.

21. A line **not** passing through the centre of inversion is mapped to
- A) a parallel line
 - B) the same line
 - C) a circle through the centre of inversion
 - D) a point

Answer: C (Theorem 5.3). Lines and circles form one family closed under inversion: this is why inversion simplifies tangency problems.

22. Circular inversion
- A) preserves distances
 - B) preserves areas
 - C) preserves angles (is conformal)
 - D) preserves nothing

Answer: C (Theorem 5.5). It preserves neither distances nor areas; it does preserve angles, though reversing their orientation (strictly, it is anticonformal).

23. Ptolemy's theorem, for a cyclic quadrilateral $ABCD$, states

- A) $AC + BD = AB + CD$
- B) $AC \cdot BD = AB \cdot CD + BC \cdot AD$
- C) $AC \cdot BD = AB \cdot BC$
- D) $AC^2 + BD^2 = AB^2 + CD^2$

Answer: B (Theorem 5.9). With a suitable choice of vertices on a circle of diameter 1 it yields the addition formula for the sine (Corollary 5.4).

24. Apollonius' problem (circles tangent to three given circles) has in general

- A) 1 solution
- B) 2 solutions
- C) 4 solutions
- D) 8 solutions

Answer: D (Definition 5.5). The eight solutions correspond to the eight combinations of internal/external tangency with respect to the three given circles.

Part VI — Measures: length, area, π (Chapter 6)

25. In this treatise, the length of a curve is defined as

- A) the limit of the perimeters of inscribed regular polygons
- B) the supremum of the lengths of inscribed polylines
- C) the integral $\int \sqrt{1 + y'^2} dx$
- D) $2\pi r$, by definition

Answer: B (Definition 6.1). The choice is deliberate: defining length through $2\pi r$ would make the whole chain leading to π circular.

26. The constancy of the ratio C/d for *all* circles is proved

- A) by measuring many circles
- B) by a homothety between any two circles
- C) with integral calculus
- D) by assuming it as an axiom

Answer: B (Theorem 6.2). A homothety of ratio λ multiplies every inscribed polyline by λ , hence also their supremum: the ratio C/d is invariant.

27. In the exhaustion lemma (Euclid X.1) the essential clause is that at each step one subtracts

- A) any quantity whatever
- B) at least half of the remainder
- C) exactly one tenth
- D) a constant quantity

Answer: B (Lemma 6.4). Without the “not less than half” the remainder may tend to zero without being controlled by a uniform geometric estimate, and Archimedes’ double reductio does not close.

28. The area of a circular sector of radius r and angle θ (radians) equals

- A) $r\theta$
- B) $\frac{1}{2}r^2\theta$
- C) $\pi r^2\theta$
- D) $\frac{1}{2}r\theta^2$

Answer: B (Theorem 6.7). Linearity in θ is not obvious: it is proved by additivity and monotonicity plus the density of $\mathbb{Q}$ — a Cauchy functional equation with a monotonicity constraint.

29. The isoperimetric inequality states that, for fixed length L ,

- A) $4\pi A \geq L^2$
- B) $4\pi A \leq L^2$, with equality only for the circle
- C) $A = L^2/4\pi$ for every closed curve
- D) A is unbounded

Answer: B (Theorem 6.8). This is the precise sense in which “the circle is the most efficient figure”: it maximises area for a given perimeter.

Part VII — Analytic geometry (Chapter 7)

30. The equation $x^2 + y^2 + Dx + Ey + F = 0$ represents a **real** circle if and only if

- A) $F < 0$
- B) $D^2 + E^2 > 0$
- C) $\frac{D^2}{4} + \frac{E^2}{4} - F > 0$
- D) always

Answer: C (Theorem 7.1). If the quantity vanishes one gets a point (degenerate circle); if negative, no real point.

31. From the general form, the centre is recovered as

- A) (D, E)
- B) $(-D/2, -E/2)$
- C) $(D/2, E/2)$
- D) $(-D, -E)$

Answer: B. It follows from completing the squares: $D = -2a$, $E = -2b$, hence $a = -D/2$, $b = -E/2$.

32. Three non-collinear points of the plane determine

- A) no circle
- B) one and only one circle

- C) infinitely many circles
- D) exactly two circles

Answer: B (Theorem 7.2). The linear system in D, E, F has an invertible matrix exactly when the three points are not collinear: it is the same condition that, in $\mathbb{R}^3$, makes robot circular interpolation work (Theorem 13.1).

33. The **rational** points of the unit circle (excluding $(-1, 0)$) are parametrised as

- A) $(\cos t, \sin t)$ with $t \in \mathbb{Q}$
- B) $\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right)$ with $t \in \mathbb{Q}$
- C) $(t, \sqrt{1-t^2})$ with $t \in \mathbb{Q}$
- D) no parametrisation exists

Answer: B (Theorem 7.10). It is the stereographic projection from $(-1, 0)$, and reducing to a common denominator gives exactly the Pythagorean triples (Theorem 10.14).

Part VIII — Trigonometry (Chapter 8)

34. The limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ holds if x is measured in

- A) sexagesimal degrees
- B) gradians
- C) radians
- D) any unit

Answer: C (Theorem 8.5). In degrees the limit is $\pi/180$: this is the deep reason why analysis uses radians, and why $(\sin x)' = \cos x$ only in that unit.

35. Which tool **cannot** be used to prove the remarkable limit, on pain of circularity?

- A) The comparison of the areas of triangle, sector and circumscribed triangle
- B) L'Hôpital's rule
- C) The squeeze theorem
- D) The monotonicity of $\sin$ on $[0, \pi/2]$

Answer: B. L'Hôpital requires knowing $(\sin x)' = \cos x$, which in turn is proved by precisely that limit. The same applies to the Taylor series of $\sin$.

36. In this treatise, the identity $\sin^2 \theta + \cos^2 \theta = 1$ is

- A) an axiom
- B) the equation of the unit circle read on the parametrisation
- C) a consequence of the addition formula
- D) valid only in the first quadrant

Answer: B (Theorem 8.2). And since the circle's equation is the Pythagorean theorem in coordinates (Remark 7.1), the identity *is* Pythagoras in disguise.

37. The addition formula for the sine is

- A) $\sin \alpha + \sin \beta$
- B) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
- C) $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
- D) $2 \sin \alpha \cos \beta$

Answer: B (Theorem 8.4). Answer C is the addition formula for the *cosine*: it is the commonest slip in examinations.

38. Product-to-sum formulas transform

- A) products into sums
- B) sums of sines or cosines into products
- C) powers into roots
- D) angles into radians

Answer: B (Theorem 8.6). Devised to speed up astronomical hand computation, today they are the mathematical basis of lock-in demodulation.

Part IX — Triangle and circle (Chapter 10)

39. In a right triangle with hypotenuse c , the radius of the **circumscribed** circle equals

- A) c
- B) $c/2$
- C) $(a + b - c)/2$
- D) $\sqrt{ab}$

Answer: B (Corollary 10.3). The centre is the midpoint of the hypotenuse: it is Thales' theorem read backwards. Answer C is instead the *inradius*.

40. For the triple $(3, 4, 5)$, the inradius equals

- A) 1
- B) 1.5
- C) 2
- D) 2.5

Answer: A (Theorem 10.4): $r = (3 + 4 - 5)/2 = 1$. It is the only primitive triple with unit inradius, which explains its ubiquity in textbooks.

41. From Euler's formula $OI^2 = R^2 - 2Rr$ it follows immediately that

- A) $R = 2r$ always
- B) $R \geq 2r$, with equality only for the equilateral triangle
- C) $R \leq 2r$
- D) no inequality

Answer: B (Theorem 10.9 and Corollary 10.6). The square of a real distance is ≥ 0 : the whole inequality rests on that observation.

-
42. The Erdős–Ulam problem (existence of a set *dense in the plane* with all distances rational) is, as matters stand
- A) settled affirmatively
 - B) settled negatively
 - C) open: the negative answer is known only assuming the Bombieri–Lang conjecture
 - D) proved undecidable

Answer: C. Take care not to confuse it with Theorem 10.15, which constructs infinitely many points *on a circle* at rational distances: that one is proved in half a page. The whole difference lies in the word “dense”.

Part X — Numerical methods (Chapter 11)

43. Can a circle be represented *exactly* by a **polynomial** (non-rational) Bézier curve?
- A) Yes, of degree 2
 - B) Yes, of degree 3
 - C) No, never
 - D) Only if the radius is an integer

Answer: C (Theorem 11.2). The proof takes half a page: in $\mathbb{C}[t]$ the degree of a product is the sum of the degrees, and $(x + iy)(x - iy) = r^2$ forces both factors to be constant. This is why the CAD industry uses NURBS.

44. The weight w_1 making a **quarter** circle exact as a rational quadratic Bézier equals
- A) 1
 - B) $\sqrt{2}/2 \approx 0.7071$
 - C) $\sqrt{3}/2 \approx 0.8660$
 - D) $1/2$

Answer: B (Theorem 11.3, with $w_1 = \cos \vartheta$ and $2\vartheta = \pi/2$). Finding alternating weights 1; 0.7071; 1; ... in a STEP file is the signature of an exact circle.

45. Newton–Raphson, at a simple root and starting close enough, converges
- A) linearly
 - B) quadratically: the exact digits double at each step
 - C) always cubically
 - D) in a single step

Answer: B (Theorem 11.11). Both hypotheses matter: at a *multiple* root ($f'(x^*) = 0$) convergence degrades to linear.

46. To program a **full circle** in G-code one must use
- A) either R or I/J indifferently
 - B) necessarily I/J
 - C) necessarily R

D) two G01 blocks

Answer: B. With $P_1 = P_0$ the R form determines no arc: every circle of radius r through P_0 satisfies the constraints. Only I/J fix the centre uniquely.

Part XI — Industrial robotics (Chapter 13)

47. Three non-collinear points in $\mathbb{R}^3$ determine

- A) infinitely many circles
- B) one and only one circle, and with it the plane of the arc
- C) a circle only if they lie in a horizontal plane
- D) no circle

Answer: B (Theorem 13.1). And this is why MOVc, MoveC and CIRC *all* require an intermediate point: in $\mathbb{R}^3$ centre and radius alone do not suffice, because the plane would remain undetermined.

48. For a six-axis robot with a **spherical wrist**, *positional* inverse kinematics

- A) requires iterative inversion of the Jacobian at every servo cycle
- B) is solved in closed analytic form
- C) admits no solution
- D) requires Newton's method

Answer: B. The Jacobian operates at the level of *velocities* (*resolved-rate* control), not of positions. It is precisely to obtain the closed form that vendors design spherical wrists.

49. In INFORM (Yaskawa), a variable of type B holds values

- A) from -128 to 127
- B) from 0 to 255
- C) from 0 to 65535
- D) in floating point

Answer: B. And exceeding the limit raises no error but wraps around ($255 + 1 = 0$): a part counter on B000 silently returns to zero at the 256th part.

50. When $\det J \rightarrow 0$ (kinematic singularity), the controller

- A) gains accuracy
- B) may require divergent joint velocities along certain directions
- C) switches automatically to linear motion
- D) is unaffected

Answer: B (Proposition 13.1). The industrial remedy is DLS, which replaces J^{-1} with $J^\top(JJ^\top + \lambda^2 I)^{-1}$ and bounds the solution norm by $\|\dot{\mathbf{x}}\|/(2\lambda)$.

Part XII — Supplement: the advanced results

INFO

The following ten questions concern the advanced material — non-circularity, machine vision, metrology, motion dynamics, safety — and are meant for readers who have also worked through the sections marked as advanced. They do not count towards the score of the first fifty.

51. Zimba's result (2009) proves that

- A) the Pythagorean theorem is false in non-Euclidean geometry
- B) the identity $\sin^2 x + \cos^2 x = 1$ can be derived without using the Pythagorean theorem
- C) all trigonometric proofs are circular
- D) π is transcendental

Answer: B (Theorem 8.3). And since that identity implies Pythagoras, a *non-circular* trigonometric proof follows — what Loomis in 1940 declared impossible.

52. Which Euclidean results suffice to define $\sin$ and $\cos$ for acute angles?

- A) I.47 (Pythagoras) and VI.31
- B) I.32 (angle sum), I.19 (greater side) and VI.4 (equiangular triangles)
- C) the parallel postulate alone
- D) none: the definitions are axiomatic

Answer: B (Remark 8.4). The decisive point is VI.4: it guarantees that the ratios do not depend on the triangle chosen. None of the three is I.47.

53. Two non-secant circles can be made concentric by an inversion

- A) never
- B) always, taking as pole one of the two limit points of the pencil
- C) only if they have equal radii
- D) only if one lies inside the other

Answer: B (Lemma 5.10). The pole equation has a positive discriminant *exactly* when the two circles do not meet: the algebraic computation also proves the existence of the limit points.

54. The radical centre of three circles exists and is unique

- A) always
- B) if and only if the three centres are not collinear
- C) if and only if the three circles intersect
- D) if and only if the three radii are equal

Answer: B (Theorem 7.7). The determinant of the linear system is proportional to the area of the triangle of centres: it vanishes exactly when they are collinear.

55. In the Hough transform, an edge point not belonging to the sought circle

- A) shifts the estimated centre towards itself
- B) dilutes the votes without forming a maximum
- C) prevents detection
- D) is discarded automatically

Answer: B (Remark 11.9). This is the substantial difference from least squares, where the residual enters squared and a single outlier can move the estimate.

56. The gradient variant of Hough reduces the cost by a factor of about

- A) 2
- B) $\log N$
- C) Θ , the number of angular samples (typically 360)
- D) N , the number of edge points

Answer: C (Theorem 11.7 and Remark 11.10). The gradient reduces the voting cone to a line: from $O(NR\Theta)$ to $O(NR)$.

57. Among the four roundness criteria, the one always giving the *minimum* value is

- A) LSC (least squares)
- B) MZC (minimum zone)
- C) MCC (minimum circumscribed)
- D) MIC (maximum inscribed)

Answer: B (Theorem 11.8), and this is why ISO 12181-1 prescribes it. LSC nevertheless remains the shop-floor default because it has a closed-form solution.

58. On an arc of very small radius, the speed constraint that governs is usually

- A) chord error
- B) centripetal acceleration
- C) jerk
- D) the programmed feedrate

Answer: C (Theorem 13.9(b)). Below the critical radius $\rho_c = a_{\max}^3/J_{\max}^2$ jerk governs, because it grows as $\rho^{2/3}$ while the other two grow as $\rho^{1/2}$.

59. The radius reduction due to finite loop gain scales as

- A) v
- B) v^2
- C) $1/v$
- D) it does not depend on speed

Answer: B (Theorem 13.10): $\Delta R \simeq v^2/(2RK_v^2)$. Halving the feed reduces the error to a quarter; so does doubling K_v .

60. On a circular path, the robot's speed directed towards an external operator reaches at most

- A) half the tangential speed
- B) the full tangential speed, at the points of tangency
- C) a value depending on the radius
- D) zero, if the operator is outside the circle

Answer: B (Theorem 13.11). Consequently the v_R term of the ISO/TS 15066 formula cannot be reduced by any geometric factor: doing so sizes an unsafe barrier.

Part XIII — New results from this revision

INFO

Twenty-eight further questions, two per chapter (including the two new chapters, 9/12 are here split from their siblings, and the entirely new Chapter 14), covering material added in this revision. They do not count towards the score of the first fifty.

61. Viète's product for π is built from

- A) a sum of unit fractions
- B) an infinite product of nested square roots of 2
- C) a continued fraction
- D) a single Taylor series

Answer: B. $\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \dots$ (1593), the earliest known infinite-product formula for π .

62. The Basel problem, $\sum_{n=1}^{\infty} 1/n^2$, sums to

- A) π
- B) $\pi^2/6$
- C) $\pi/2$
- D) it diverges

Answer: B (Euler, 1735).

63. Apollonius' circle, locus of points with $PA/PB = k$ for fixed $k \neq 1$, degenerates to a line when

- A) $k = 0$
- B) $k = 1$
- C) $k \rightarrow \infty$
- D) never

Answer: B. At $k = 1$ the locus is the perpendicular bisector of AB , not a circle.

64. Defining open arcs before “minor” and “major” arc, as this revision now does, avoids

- A) a notational clash
- B) a circular definition (minor/major would presuppose the arc concept they are meant to define)

-
- C) no real issue, it is a stylistic choice
 - D) the need for a diameter

Answer: B (Chapter 2).

65. Pitot's theorem states that a convex quadrilateral has an inscribed circle if and only if
- A) its diagonals are perpendicular
 - B) $AB + CD = BC + DA$
 - C) it is cyclic
 - D) all sides are equal

Answer: B (Chapter 3).

66. Two circles, one strictly inside the other with no common point, have
- A) four common tangents
 - B) two common tangents
 - C) zero common tangents
 - D) one common tangent

Answer: C (Chapter 3).

67. The Miquel point of a triangle with one chosen point on each side is
- A) the point common to the three circumcircles of the corner triangles cut off by the chosen points
 - B) always the incentre
 - C) always the centroid
 - D) undefined unless the points are midpoints

Answer: A (Chapter 4).

68. The Simson(-Wallace) line of a point P exists for
- A) any point P in the plane
 - B) only points P on the circumcircle of the triangle
 - C) only the orthocentre
 - D) only points inside the triangle

Answer: B (Chapter 4).

69. The Peaucellier(-Lipkin) linkage converts
- A) straight-line motion into circular motion
 - B) circular motion into exact straight-line motion
 - C) rotation into oscillation
 - D) nothing; it is purely decorative

Answer: B (Chapter 5), historically important before the availability of precision linear guides.

70. A Pappus chain of circles inscribed in an arbelos has each circle tangent to

- A) only the previous circle in the chain
- B) the two outer semicircles and the previous circle
- C) only the two outer semicircles
- D) the diameter line only

Answer: B (Chapter 5).

71. A Reuleaux triangle has constant width like a circle, but

- A) it is in fact a circle
- B) it is not a disc: its boundary is not equidistant from a single centre
- C) it has infinite width
- D) it cannot roll

Answer: B (Chapter 6).

72. This revision's exact bounds for π , reproved with rational (non-floating-point) arithmetic, follow Archimedes' method of

- A) a single 96-gon computed directly
- B) four successive polygon doublings from a hexagon
- C) Monte Carlo sampling
- D) Newton–Raphson iteration

Answer: B (Chapter 6).

73. By La Hire's theorem, if P lies on the polar of Q then

- A) Q lies on the polar of P
- B) $P = Q$
- C) Q is the centre
- D) nothing follows

Answer: A (Chapter 7): the polar relation is symmetric.

74. In coordinates, two circles $x^2 + y^2 + D_i x + E_i y + F_i = 0$ ($i = 1, 2$) are orthogonal exactly when

- A) $D_1 D_2 + E_1 E_2 = 2(F_1 + F_2)$
- B) $D_1 = D_2$
- C) $F_1 = F_2$
- D) $D_1 E_2 = D_2 E_1$

Answer: A (Chapter 7).

75. Is $\cos 20^\circ$ constructible with ruler and compass?

- A) Yes, always
- B) No (Wantzel): it would trisect 60°

-
- C) Only with a marked ruler
 - D) Only using inversion

Answer: B (Chapter 8).

76. In this treatise, the **primary** proof of the sine and cosine addition formulas now proceeds via

- A) Euler's differential-equation argument
- B) the law of cosines / distance between two points on a unit circle (SAS), with Euler's method demoted to an alternative
- C) complex exponentials only
- D) Taylor series

Answer: B (Chapter 8).

77. The regular decagon's side length, at circumradius R , is

- A) R
- B) $R(\sqrt{5} - 1)/2$
- C) $R\sqrt{2}$
- D) $R\pi/10$

Answer: B (Chapter 9) — the golden ratio construction, compass-and-straightedge (Gauss–Wantzel).

78. Carnot's theorem, for a triangle with circumcentre O , states that the signed distances from O to the three sides sum to

- A) R
- B) $R + r$
- C) $2R$
- D) r

Answer: B (Chapter 9).

79. The Euler line of a triangle passes through

- A) the incentre, centroid and orthocentre
- B) the circumcentre, centroid and orthocentre
- C) only the three vertices
- D) only the midpoints of the sides

Answer: B (Chapter 10); this revision adds a synthetic proof via two homotheties, alongside the vector proof.

80. Briggs' half-angle formula expresses $\sin(A/2)$ in terms of

- A) R and r only
- B) the sides a, b, c and the semiperimeter s
- C) the area alone

D) the circumradius alone

Answer: B (Chapter 10).

81. The CORDIC scale factor $K = \prod_i \cos(\arctan 2^{-i})$ converging to a constant is

- A) proved in full in this treatise
- B) imported without proof
- C) only conjectured
- D) irrelevant to the algorithm

Answer: A (Chapter 11): the convergence lemma is now proved and placed *before* the correctness theorem that cites it.

82. The roundness metric $RONT$, used to evaluate a measured workpiece profile, is also the metric behind

- A) the CORDIC algorithm
- B) the ballbar test of a machine's circular motion
- C) the Hough transform
- D) Newton–Raphson

Answer: B (Chapter 11): same reference-circle arithmetic, applied to a traced path instead of a static part.

83. Tool radius compensation on an internal arc of radius r is impossible (undercut) when

- A) $r > R_u$
- B) $r < R_u$ (tool radius)
- C) $r = 0$
- D) it is always possible

Answer: B (Chapter 13): the offset centre would have to lie beyond the arc's own centre.

84. The Grashof condition for a four-bar linkage with shortest s , longest l and the other two p, q states the shortest link fully rotates iff

- A) $s + l \leq p + q$
- B) $s = l$
- C) $p = q$
- D) $s + p \leq l + q$

Answer: A (Chapter 13); necessity is proved here via the triangle inequality, sufficiency imported.

85. Pick's theorem gives the area of a lattice polygon as

- A) $A = I + B - 1$
- B) $A = I + B/2 - 1$
- C) $A = 2I + B$
- D) $A = IB$

Answer: B, with I interior and B boundary lattice points (Chapter 12).

86. Zagier's "one-sentence" proof establishes that every prime $p \equiv 1 \pmod{4}$

- A) is itself a perfect square
- B) is a sum of two squares
- C) is never prime
- D) has no square roots mod p

Answer: B (Fermat's theorem, Chapter 12), via two involutions on the same finite set.

87. Hazen's theorem says a particle with settling velocity v_s is retained by a circular settler if and only if

- A) $v_s \geq Q/A_{\text{eff}}$ (surface hydraulic loading)
- B) $v_s \leq Q/A_{\text{eff}}$
- C) the tank is deep enough
- D) the detention time exceeds one hour

Answer: A (Chapter 14).

88. Hazen's most counterintuitive corollary is that, at fixed surface loading Q/A , the retention efficiency of an ideal settler

- A) improves with greater depth
- B) does not depend on depth or detention time
- C) depends only on detention time, not on Q/A
- D) is undefined

Answer: B (Chapter 14): h cancels in the derivation of the retention radius ρ_b .

BEST PRACTICE

The three answers that separate the attentive student.

Question 35 (why l'Hôpital is forbidden on the remarkable limit), **question 42** (Erdős–Ulam open versus the *proved* theorem on rational distances on a circle) and **question 43** (no polynomial on the circle) have one thing in common: the wrong answer is the one that "sounds plausible". They are the three points this volume insists on most — respectively on non-circularity, on the difference between a proved statement and a conjecture, and on the fact that an impossibility must be *proved*, not merely observed.

Glossary

Ackermann steering

Geometric condition $\cot \delta_o - \cot \delta_i = w/L$ (track w , wheelbase L) under which the four wheels of a steered vehicle roll without slipping about a common instantaneous centre (Chapter 13).

Addition formulas $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ and $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$; also obtainable as a particular case of Ptolemy's theorem (Chapters 5 and 8).

Angle between two circles

The angle formed by the tangents to the two curves at a common point; it is $\pi/2$ for orthogonal circles. It is invariant under inversion (Chapter 5).

Apollonius' problem

To construct the circles simultaneously tangent to three given objects (points, lines or circles); generally 8 solutions (Chapter 5).

Apothem

Distance from the centre of a regular polygon to the midpoint of one of its sides; for the circumscribed polygon it equals the radius (Chapter 6).

Arbelos

Region between a large semicircle of diameter AB and two smaller semicircles of diameters AC , CB (C between A and B); its area equals that of the circle with diameter the segment perpendicular to AB at C , cut by the arbelos (Chapter 5).

Arc

One of the two parts into which two points divide a circle.

Arc of constant angle (arc capable)

Locus of the points of the plane from which a given segment is seen under an assigned angle; two arcs symmetric about the segment. Notable special case: the right angle gives the circle with that segment as diameter (Thales' theorem) (Chapter 4).

Archimedes' bounds (for π)

The rational bounds $3\frac{10}{71} < \pi < 3\frac{1}{7}$ obtained by Archimedes via inscribed/circumscribed 96-gons; reproved here with exact rational (non-floating-point) arithmetic across four polygon doublings (Chapter 6).

Arcsine, arccosine, arctangent

Notations $\arcsin$, $\arccos$, $\arctan$ for the inverse functions of sine, cosine and tangent; equivalent to $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ but not to be confused with the reciprocals $1/\sin$, $1/\cos$, $1/\tan$ (Chapter 8).

Ballbar test

Telescoping transducer that continuously measures the distance from the centre while the machine traverses a circle (ISO 230-4). The polar trace is a diagnostic chart: small circle $\Rightarrow$ low gain K_v ; ellipse at $0^\circ/90^\circ \Rightarrow$ mismatched axis gains; ellipse at $45^\circ \Rightarrow$ different phases (Chapter 13).

Brahmagupta's formula

Area $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ of a cyclic quadrilateral with sides a, b, c, d and semiperimeter s ; it generalises Heron's formula (Chapter 6).

B-Rep (Boundary Representation)

CAD model in which a solid is described by its boundary, organised in the hierarchy solid $\supset$ shells $\supset$ faces $\supset$ loops $\supset$ edges $\supset$ vertices, with a geometry attached to each topological entity (Chapter 11).

Bresenham's algorithm

The original integer-only formulation of arc drawing (IBM TR02.286, 1964; CACM, 1977). Not to be confused with the same author's 1965 paper, which concerns *line segments* (Chapter 11).

Briggs' formulae

Half-angle formulae for a triangle expressing $\sin(A/2)$, $\cos(A/2)$, $\tan(A/2)$ in terms of the sides a, b, c and the semiperimeter s , e.g. $\sin(A/2) = \sqrt{(s-b)(s-c)/bc}$ (Chapter 10).

Buffon's needle

Probabilistic experiment estimating π by dropping a needle on a ruled plane; the estimate converges as $O(1/\sqrt{N})$ (Chapter 1).

CAM (Computer-Aided Manufacturing)

Software translating a geometric model into a machine program; it generates the G02/G03 blocks by computing centre, I/J offsets and traversal sense (Chapter 11).

Carnot's theorem

For a triangle with circumcentre O and circumradius R , the signed sum of distances from O to the three sides equals $R + r$ (r the inradius); positive distances when O is on the same side as the triangle's interior (Chapter 9).

Centre

Point equidistant from all points of the circle.

Chord

Segment joining two distinct points of the circle.

Chord error

Maximum deviation between an arc and the chord approximating it: $\delta = \rho - \sqrt{\rho^2 - L^2/4} \simeq L^2/(8\rho)$. With interpolation period T it limits the speed to $v \leq (2/T)\sqrt{2\rho\delta_{\max}}$ (Chapter 13).

Chudnovsky algorithm

Hypergeometric series for computing π with geometric convergence (~ 14 digits per term), used in world computation records (Chapter 1).

Circle (circumference)

Locus of the points of the plane at fixed distance r (radius) from a fixed point O (centre).

Circle (disc)

Set of points at distance $\leq r$ from the centre.

Circular interpolation

In robot/CNC controllers, generation of motion along a circular arc determined by via-points and target, or by centre and radius (G-code) (Chapter 13).

Circular inversion	Transformation of the plane sending P to the point P^* on the ray OP with $ OP \cdot OP^* = k^2$; it maps lines and circles to lines and circles and preserves angles (Chapter 5).
Circular settler	Clarification tank of circular plan, fed at the centre through a feed well; sizing links the detention time $T = V/Q$ to the settling area ($\propto r^2$) and the weir loading to the circumference ($\propto r$) (Chapter 14).
Circumcentre	Centre of the circle circumscribed about a triangle; meeting point of the perpendicular bisectors of the sides (Chapter 10).
Circumradius (R)	Radius of the circle through the three vertices of a triangle; for a right triangle, $R = c/2$ (Chapter 10).
CORDIC	(<i>COordinate Rotation DIgital Computer</i>) Iterative algorithm computing sine and cosine (and more generally rotating a vector) using only shifts and additions; the scale factor $K = \prod_i \cos(\arctan 2^{-i})$ is constant and precomputed in ROM (Chapter 11).
Curvature (bend) of a circle	Reciprocal of the radius, $k = 1/r$, signed by the convention of Descartes' theorem (positive if tangent "from outside", negative if the circle contains the others) (Chapter 5).
Cyclic quadrilateral	Quadrilateral whose four vertices lie on one circle; characterised by supplementary opposite angles, and subject to the theorems of Ptolemy and Brahmagupta (Chapters 4, 5 and 6).
Denavit–Hartenberg parameters	Quadruple $(a_i, \alpha_i, d_i, \theta_i)$ describing, through a 4×4 homogeneous matrix, the transformation between two consecutive frames of a serial manipulator (Chapter 13).
Descartes' theorem (Soddy circles)	Relation between the curvatures of four mutually tangent circles: $(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$ (Chapter 5).
Detention time	Mean residence time in a settler, $T = V/Q = h/q_A$ with $V = \pi r^2 h$ and q_A the surface hydraulic loading (Chapter 14).
Diameter	Chord through the centre; the chord of maximum length, equal to $2r$.
DLS (Damped Least Squares)	Regularisation $\dot{\theta} = J^\top (JJ^\top + \lambda^2 I)^{-1} \dot{\mathbf{x}}$ of the Jacobian inversion; it guarantees existence and boundedness of the solution ($\ \dot{\theta}\ \leq \ \dot{\mathbf{x}}\ /2\lambda$) even near singularities (Chapter 13).
Dubins paths	The shortest path between two oriented configurations for a forward-only vehicle with a minimum turning radius $\rho_{\min}$ is always a concatenation of at most three pieces, each an arc of radius $\rho_{\min}$ or a straight segment (types CSC or CCC); imported without proof (Dubins, 1957) (Chapter 13).

Elbow up / elbow down

The two solutions of the closed-form inverse kinematics of the planar two-link arm, $\theta_1 = \varphi \pm \gamma$: mirror images about the base–target line, with θ_2 of opposite sign. They coincide on the boundary of the workspace (Chapter 13).

Elliptic integral (complete, of the second kind)

$E(e) = \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 t} dt$: appears in the arc length of the ellipse, $L = 4aE(e)$; not expressible with elementary functions for $e \neq 0$ (imported fact). For $e = 0$ one recovers $E(0) = \pi/2$ and the circumference of the circle (Chapter 6).

Erdős–Ulam problem

Open question: is there a *dense* subset of the plane with all mutual distances rational? It cannot lie on an algebraic curve (Solymosi–de Zeeuw, 2010) and cannot exist assuming the Bombieri–Lang conjecture; unconditionally the problem remains **open** (Chapter 10).

Euler’s formula (Euler–Chapple)

Relation $OI^2 = R^2 - 2Rr$ between circumcentre O , incentre I , circumradius R and inradius r of a triangle (Chapter 10).

Euler’s inequality $R \geq 2r$ for every triangle, with equality if and only if the triangle is equilateral; an immediate consequence of $OI^2 = R^2 - 2Rr \geq 0$ (Chapter 10).

Exhaustion lemma (Euclid X.1)

If from a magnitude one subtracts at least half each time, after finitely many steps the remainder is smaller than any assigned positive threshold; equivalent to an Archimedean property (Chapter 6).

Exhaustion (method of)

Greek proof technique establishing equality between magnitudes by double reductio ad absurdum, resting on Euclid’s lemma X.1; it is the engine of the Archimedean proof of $A = \frac{1}{2}Cr$ (Chapter 6).

Feed well

Central inlet structure of a circular settler that dissipates the kinetic energy of the incoming flow and redistributes it radially and uniformly; the radial velocity $v(\rho) = Q/(2\pi\rho h)$ is maximum at its edge (Chapter 14).

Floating point (IEEE 754)

Standard for representing real numbers in a machine by sign, exponent and mantissa (23 bits in single, 52 in double precision); the source of the rounding errors discussed in Chapter 11.

Ford circles

For each fraction p/q in lowest terms, the circle of radius $1/(2q^2)$ tangent to the real axis at $(p/q, 0)$; any two Ford circles are either tangent or disjoint, never overlapping, and two are tangent exactly when their fractions are Farey neighbours (Chapter 12).

Fundamental limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (with x in radians); proved by comparing the areas of triangle, sector and circumscribed triangle, without l’Hôpital or Taylor series, which would be circular (Chapter 8).

Gauss's circle problem

Comparison between $N(r)$, the number of integer-coordinate points in a circle of radius r , and its area πr^2 : $N(r) = \pi r^2 + E(r)$ with $|E(r)| \leq \pi(\sqrt{2}r + 1/2)$, proved by associating to each lattice point a unit square centred on it. The optimal exponent of the error term is an open problem (Chapter 12).

G-code

Standard language (ISO 6983/RS-274) for CNC machines; the instructions G02/G03 program clockwise/anticlockwise circular interpolation. The I/J offsets give the centre incrementally and are the only never-ambiguous form (Chapters 11 and 13).

General equation

Form $x^2 + y^2 + Dx + Ey + F = 0$ of the circle, obtained by expanding the canonical equation; it represents a real circle only if $\Delta = D^2/4 + E^2/4 - F > 0$ (Chapter 7).

Gougu (theorem)

Chinese name of the Pythagorean theorem: $gou^2 + gu^2 = xian^2$, where *gou* is the shorter leg, *gu* the longer, and *xian* (“bowstring”) the hypotenuse. Attested in the *Zhoubi Suanjing* (ca. 100 BCE) with a dissection proof, independently of the Greek tradition (Chapter 10).

Grashof condition

In a planar four-bar linkage with shortest, longest and two intermediate link lengths $s \leq p, q \leq l$, the shortest link can fully rotate if and only if $s + l \leq p + q$; necessity proved via the triangle inequality at the two aligned configurations (Chapter 13).

Hazen's theorem

For a circular settler with plug flow, a particle with settling velocity v_s entering at the feed-well edge (radius ρ_f) is retained if and only if $v_s \geq Q/(\pi(r^2 - \rho_f^2))$; proved by flow conservation across concentric cylindrical shells, with no dependence on depth or detention time (Hazen, 1904) (Chapter 14).

Helix (cylindrical)

Curve on a cylinder of radius r making a constant angle with the generators; developing the cylinder onto a plane turns one turn of pitch p into a right triangle of legs $2\pi r$ and p , giving length $\sqrt{(2\pi r)^2 + p^2}$ (Chapter 13).

Heron's formula

Area $K = \sqrt{s(s-a)(s-b)(s-c)}$ of a triangle with sides a, b, c and semiperimeter s ; the degenerate case ($d = 0$) of Brahmagupta's formula (Chapter 6).

Hough transform

Voting method for detecting circles in an image: each edge point votes, in the parameter space (a, b, r) , for the cone of circles containing it; the accumulator maxima are the circles present. Robust to outliers; the gradient variant reduces the cone to a line (Chapter 11).

ICR (instantaneous centre of rotation)

For a differential-drive vehicle with wheel speeds $v_r \neq v_l$ on a track b , the point on the wheel axis about which the vehicle rotates at that instant, at distance $\rho = \frac{b}{2} \cdot \frac{v_r + v_l}{v_r - v_l}$ from the axis midpoint, with angular speed $\omega = (v_r - v_l)/b$ (Chapter 13).

Incentre

Centre of the circle inscribed in a triangle; meeting point of the angle bisectors (Chapter 10).

INFORM	Programming language of Yaskawa Motoman robots. It exposes six families of global variables — B (byte, 0–255), I, D, R, S, P/BP/EX — and the circular motion instruction MOV _C (Chapter 13).
Inradius (r)	Radius of the circle internally tangent to the three sides of a triangle; for a right triangle, $r = (a + b - c)/2$ (Chapter 10).
Isoperimetric inequality	Among all closed curves of given length, the circle encloses the greatest area: $4\pi A \leq L^2$, with equality only for the circle (Chapter 6).
Jacobian (of a manipulator)	$6 \times n$ matrix with $J_{ij} = \partial x_i / \partial \theta_j$, linking joint velocities to the Cartesian velocity of the TCP: $\dot{\mathbf{x}} = \mathbf{J}\dot{\boldsymbol{\theta}}$ (Chapter 13).
Japanese theorem (for cyclic polygons)	For any triangulation of a cyclic polygon by non-crossing diagonals, the sum of the inradii of the resulting triangles is the same, independent of the triangulation chosen (Chapter 9, imported observation).
Jerk	Derivative of acceleration. On an arc traversed at constant speed it equals v^3/ρ^2 and limits the speed to $(J_{\max}\rho^2)^{1/3}$: it is the dominant constraint on blends of radius below $a_{\max}^3/J_{\max}^2$ (Chapter 13).
Kåsa's method	Circle-fitting technique (1976) linearising the least-squares problem by rewriting $(x - a)^2 + (y - b)^2 = r^2$ in a form algebraically linear in the parameters $u = 2a$, $v = 2b$, $w = r^2 - a^2 - b^2$, solvable by a 3×3 system; it introduces a small systematic bias towards smaller circles, corrected by the methods of Pratt and Taubin (Chapter 11).
Kelvin transform	Operator $(\mathcal{K}u)(P) = \frac{k}{ OP }u(k^2P/ OP ^2)$ preserving harmonicity; it makes circular inversion a solution tool for Laplace's equation (Chapter 5).
La Hire's theorem	If a point P lies on the polar of a point Q (with respect to a given circle), then Q lies on the polar of P ; the polar relation is symmetric (Chapter 7).
Law of cosines	Relation $c^2 = a^2 + b^2 - 2ab \cos C$, generalisation of the Pythagorean theorem to arbitrary triangles (Chapter 10).
Law of sines	Relation $a/\sin A = b/\sin B = c/\sin C = 2R$ between the sides of a triangle, the opposite angles and the circumradius R (Chapter 10).
Limit points (of a pencil)	The two points into which the circles of a non-secant pencil degenerate; choosing one as pole, an inversion makes two non-secant circles concentric (Chapters 2 and 5).
Liu Hui	Chinese mathematician (3rd c. CE); his commentary on the <i>Nine Chapters</i> (263 CE) links the gougou rule to triangulation and introduces a recursive polygon-bisection algorithm for computing π (Chapter 10).
LSC, MZC, MCC, MIC	The four reference circles for roundness: least squares, minimum zone, minimum circumscribed, maximum inscribed. MZC always gives the

	minimum value and is the normative criterion (ISO 12181-1); LSC has a closed-form solution and is the shop-floor default (Chapter 11).
Machin's formula	Identity $\pi/4 = 4 \arctan(1/5) - \arctan(1/239)$ (1706), used for the manual computation of π with rapidly converging Taylor series, before electronic computers (Chapter 1).
Midpoint algorithm	Integer-only circle rasterisation algorithm choosing the pixel nearest the circle by evaluating a decision parameter $p_k = f(x_k + 1, y_k - \frac{1}{2})$ and updating it by additions and shifts alone (Chapter 11).
Miquel point	Given a triangle and one point on each side (or its extension), the three circles each through a vertex and the two points on its adjacent sides meet in a single common point (Chapter 4).
MOVC	Circular motion instruction in the INFORM language (Yaskawa Motoman); it requires at least three consecutive instructions for an arc (Chapter 13).
MoveC	Circular motion instruction in the RAPID language (ABB) (Chapter 13).
Newton–Raphson method	Iteration $x_{n+1} = x_n - f(x_n)/f'(x_n)$ for numerically solving equations not elementarily invertible; at a simple root it converges <i>quadratically</i> (the exact digits double at each step) (Chapters 6 and 11).
Nine-point circle (Feuerbach)	Circle of radius $R/2$ centred at the midpoint N of the segment OH (circumcentre–orthocentre) passing through the three side midpoints, the three feet of the altitudes and the three Euler points (midpoints of the segments joining the orthocentre to the vertices); internally tangent to the incircle and tangent to the three excircles (Feuerbach's theorem, 1822) (Chapter 10).
NURBS	(<i>Non-Uniform Rational B-Spline</i>) Industry standard for representing curves and surfaces in CAD. The <i>rational</i> form is indispensable: no polynomial curve can lie on a circular arc (Chapter 11).
Orthogonal circles	Two circles meeting at right angles; equivalently $d^2 = r_1^2 + r_2^2$, i.e. the squared radius of one is the power of its centre with respect to the other (Chapter 2).
Pappus chain	Sequence of circles inscribed in an arbelos, each tangent to the two outer semicircles and to the previous circle of the chain; the n -th circle's centre lies at height $2n$ times its diameter above the base line (Pappus of Alexandria) (Chapter 5).
Peaucellier(-Lipkin) linkage	Mechanical linkage of seven rigid bars that converts circular motion into exact straight-line motion, via a proof that a fixed point and its image under the linkage are inverses with respect to a fixed circle (Chapter 5).
Pencil of circles	Family $\Gamma_\lambda = \Gamma_1 + \lambda\Gamma_2$ of circles sharing the same radical axis; secant (two common base points), tangent (one base point) or non-secant, according

	to the sign of the discriminant (Chapter 2).
π (pi)	Constant given by the ratio, the same for every circle, of circumference to diameter: $\pi = C/d \approx 3.14159\dots$ (Chapters 1 and 6).
Pick's theorem	For a simple polygon with vertices on lattice points, area $A = I + B/2 - 1$, where I is the number of interior lattice points and B the number on the boundary (Chapter 12).
Pitot's theorem	A convex quadrilateral has an inscribed circle if and only if the sums of opposite sides are equal: $AB + CD = BC + DA$ (Chapter 3).
PLCopen (Motion Control)	Organisation standardising the function blocks for axis control in industrial PLCs; <code>MC_MoveCircularAbsolute</code> programs a circular motion by specifying centre or auxiliary point (<code>CircMode</code>) and sense (<code>PathChoice</code>), analogously to <code>MOV_C/MoveC/CIRC</code> (Chapter 13).
Pole and polar	Given a circle and a point P (pole), the <i>polar</i> is the line locus of the harmonic conjugates of P ; if P is exterior, it is the line through the two points of tangency (Chapter 5).
Poncelet–Steiner theorem	Every ruler-and-compass construction is executable with the ruler alone, provided a single circle together with its centre is given in advance (Poncelet 1822, Steiner 1833). The treatise proves it in full: parallels, perpendiculars, transport of segments, intersection of a line and of two undrawn circles, all with the straightedge alone and the reading of the points on the given circle (Chapter 5, Lemmas 5.1–5.8).
Pothenot–Snellius resection	Surveying problem of determining one's own position by observing the angles subtended by three known points; solved by intersecting two arcs of constant angle (Chapter 4).
Power of a point	For a point P and a circle $C(O, r)$: $\text{pow}(P) = OP ^2 - r^2$; it is invariant along every secant through P (Chapter 3).
Product-to-sum formulas	Trigonometric identities turning sums/differences of sines or cosines into products, e.g. $\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}$; historically used to speed up manual computation, today the basis of lock-in demodulation (Chapter 8).
Ptolemy's theorem	In every cyclic quadrilateral, the product of the diagonals equals the sum of the products of the opposite sides (Chapter 5).
Pure pursuit	Path-tracking algorithm choosing, at each instant, the arc tangent to the current heading that passes through a look-ahead target point (x, y) at distance L_d ; the arc's radius is $\rho = L_d^2/(2y)$ (Chapter 13).
Pythagorean theorem	In a right triangle with legs a, b and hypotenuse c : $a^2 + b^2 = c^2$ (Chapter 10).

Pythagorean triple	Triple of integers (a, b, c) with $a^2 + b^2 = c^2$; generated, if primitive, by $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$. It corresponds bijectively to the rational points of the unit circle (Chapter 10).
Quantization noise	Error introduced by rounding a continuous quantity to the nearest discrete step; variance $\Delta^2/12$ for step Δ , growing as $\sqrt{N}$ over N independent steps (Chapter 11).
Radian	Angle measure defined as the ratio of arc to radius; it is the only unit in which $\lim_{x \rightarrow 0} \sin x/x = 1$ and hence $(\sin x)' = \cos x$ (Chapter 8).
Radical axis	Line locus of the points of equal power with respect to two circles; perpendicular to the line of centres (Chapter 2).
Radical centre	Point of equal power with respect to three circles with non-collinear centres; it is the common intersection of the three radical axes (Chapter 7).
Radius	Distance between the centre and any point of the circle; or the segment itself.
Radius reduction	Systematic effect of finite loop gain: a commanded circle of radius R at speed v is executed with a smaller radius, $\Delta R \simeq v^2/(2RK_v^2)$. It is the diagnostic basis of the ballbar test (Chapter 13).
Rational distance sets	Sets of points whose mutual distances are all rational. On a <i>circle</i> infinitely many exist (construction via Pythagorean angles); the existence of a set <i>dense in the plane</i> is the open Erdős–Ulam problem (Chapter 10).
Rational quadratic Bézier	Curve $C(t) = \sum w_i B_i(t) P_i / \sum w_i B_i(t)$ of degree 2. With $w_0 = w_2 = 1$ and $w_1 = \cos \vartheta$ it represents <i>exactly</i> a circular arc of central angle 2ϑ ; it is the brick with which CAD kernels store circles (Chapter 11).
Rectifiability	Property of a curve of having finite length, defined as the supremum of the lengths of the inscribed polylines; it is the first link in the chain leading to π without circularity (Chapter 6).
Regiomontanus' problem	Find the point on a horizontal line from which a vertical segment (not meeting the line) subtends the maximum angle; the maximum occurs where the circle through the segment's endpoints, tangent to the line, meets it (Chapter 4).
Reuleaux triangle	Curve of constant width built from an equilateral triangle by replacing each side with the arc, centred at the opposite vertex, joining its endpoints; it has the same constant width as a circle but is not a disc (Chapter 6).
Roundness (RONt)	Form error of a nominally circular profile, defined by ISO 1101 as the width of the narrowest annulus containing it. It depends on the reference circle chosen: LSC, MZC, MCC or MIC (Chapter 11).
Secant	Line meeting the circle in exactly two points.

Secant-tangent theorem

From an external point P , the square of the tangent equals the product of the whole secant by its external part: $PT^2 = PA \cdot PB$ (Chapter 3).

Sector (circular) Region of the plane bounded by two radii and the arc between them; of area $\frac{1}{2}r^2\theta$, proved by additivity, monotonicity and density of $\mathbb{Q}$ (Chapter 6).

Segment (circular) Region of the plane bounded by a chord and the corresponding arc; its area requires the numerical solution of $\theta - \sin \theta = c$ (Chapter 6).

Simson(-Wallace) line

For a triangle and a point P on its circumcircle, the feet of the perpendiculars from P to the three sides (extended if necessary) are collinear (Chapter 4).

Singularity (kinematic)

Configuration in which $\det J \rightarrow 0$: some Cartesian directions become unreachable and others demand divergent joint velocities. Circular arcs cross them more often than linear motions (Chapter 13).

Sludge scraper bridge

Rotating structure, pivoted at the centre of a circular settler, carrying blades that convey settled sludge towards the central hopper; a blade angled at β to the tangential direction gives radial thrust $\frac{1}{2}\omega\rho\sin 2\beta$, maximal at $\beta = 45^\circ$ (Chapter 14).

Spherical wrist

Configuration in which the last three axes of a six-axis robot meet at a point; it is what makes positional inverse kinematics solvable in *closed analytic form*, without iteration (Chapter 13).

SSM (Speed and Separation Monitoring)

Collaborative mode of ISO/TS 15066 in which the robot's speed is regulated by the distance from the operator, maintaining $S \geq S_p$. On a circular path the speed directed towards the operator reaches its full value at the points of tangency (Chapter 13).

Steiner chain

Sequence of circles each tangent to two given circles and to the preceding one; by Steiner's porism, if it closes once it always closes (Chapter 5).

Steiner's porism

If a chain of circles tangent to two given circles closes for one choice of the first, it closes for every choice; proved by reducing, via inversion, to the concentric case (Chapter 5).

Surface hydraulic loading

Flow rate per unit settling area, $q_A = Q/(\pi r^2)$; by Hazen's theorem it alone (not depth or detention time) governs the theoretical retention efficiency of an ideal settler (Chapter 14).

Tangent

Line meeting the circle in exactly one point; equivalently, a line at distance r from the centre, perpendicular to the radius at the point of contact (Chapter 3).

TCP (Tool Center Point)

Point of the tool whose trajectory is programmed and controlled by the

	robot (Chapter 13).
Thales' theorem (on the circle)	Every inscribed angle subtending a diameter is right; and conversely, the locus of points seeing a segment under a right angle is the circle with that diameter (Chapter 4).
Tool radius compensation	CNC correction (G41/G42) offsetting a programmed profile by the tool radius R_u ; on an arc of radius r the tool centre follows a concentric arc of radius $r + R_u$ (outside) or $r - R_u$ (inside), the latter impossible if $r < R_u$ (undercut) (Chapter 13).
Transcendence of π	Property of π of not being a root of any non-zero polynomial with rational coefficients (Lindemann, 1882); it implies the impossibility of squaring the circle with ruler and compass (Chapter 1).
Two-chord theorem	If two chords meet at an interior point P , the products of the segments determined are equal: $PA \cdot PB = PC \cdot PD$; it is the power of the point in metric form (Chapter 3).
Unit circle (goniometric circle)	Circle of radius 1 centred at the origin, used to define sine, cosine and tangent as the coordinates of a point (Chapter 8).
Via-point	Intermediate point used by robot controllers to determine, together with a start and an end point, the unique circular arc containing them: in $\mathbb{R}^3$ it also fixes the <i>plane</i> of the arc (Chapter 13).
Wallis product	Representation of π as an infinite product, $\pi/2 = \prod_{n=1}^{\infty} \frac{2n}{2n-1} \cdot \frac{2n}{2n+1}$ (Wallis, 1655), proved via the Wallis integrals $\int_0^{\pi/2} \sin^n x \, dx$ (Chapter 1).
Weir loading rate	Flow rate per unit length of weir crest, $q_L = Q/L_w$ with $L_w = 2\pi r$ (or $2\pi(r_1 + r_2)$ for a double-sided launder); it constrains the minimum diameter of a settler, and unlike the surface loading it scales with r , not r^2 (Chapter 14).
Willis formula	For an epicyclic gear train with sun (z_s teeth), ring (z_r teeth) and carrier at ω_c : $(\omega_s - \omega_c)/(\omega_r - \omega_c) = -z_r/z_s$, derived by working in the carrier's rotating frame (Chapter 13).
Workspace (two-link arm)	The circular annulus $ \ell_1 - \ell_2 \leq \rho \leq \ell_1 + \ell_2$ of points reachable by a planar two-link arm of lengths ℓ_1, ℓ_2 ; its boundary coincides with the kinematic singularity (Chapter 13).
Zimba's theorem	Derivation of the identity $\sin^2 x + \cos^2 x = 1$ without using the Pythagorean theorem (2009), through subtraction formulas built from Euclid I.32, I.19 and VI.4 alone. It makes a non-circular trigonometric proof of Pythagoras possible (Chapter 8).

Cheat Sheet

Constant and measures

- $\pi = C/d \approx 3.14159265\dots$, irrational and transcendental
- $C = 2\pi r = \pi d$; $A = \pi r^2$
- Arc $l = r\theta$; sector $A_s = \frac{1}{2}r^2\theta$
- Chord $|AB| = 2r \sin(\theta/2)$
- Segment: $A_{\text{seg}} = \frac{r^2}{2}(\theta - \sin \theta)$
- Isoperimetric: $4\pi A \leq L^2$ (iff circle)
- Wallis: $\frac{\pi}{2} = \prod \frac{2n}{2n-1} \cdot \frac{2n}{2n+1}$
- Gauss's circle: $N(r) = \pi r^2 + E(r)$, $|E(r)| \leq \pi(\sqrt{2}r + \frac{1}{2})$
- Ellipse: $L = 4a E(e)$, $E(e) = \int_0^{\pi/2} \sqrt{1-e^2 \sin^2 t} dt$

Fundamental elements

- Center, radius, diameter $d = 2r$
- Chord, minor/major arc
- Tangent: $d(O, t) = r$; Secant: $d(O, s) < r$
- Coaxial pencil: same radical axis $\forall \lambda$

Chords and tangents

- Radius $\perp$ chord $\Rightarrow$ bisects it
- $|AB| = 2\sqrt{r^2 - d^2}$ (d =dist. from centre)
- Tangents from external point: $|PT| = \sqrt{|OP|^2 - r^2}$
- Power: $\text{pow}(P) = |OP|^2 - r^2$; $PA \cdot PB = |\text{pow}(P)|$

Angles

- Central angle = $2 \times$ inscribed on same arc
- Thales: diameter $\Rightarrow$ inscribed angle = 90°
- Inscribed on same arc: congruent; opposite arcs: supplementary
- Two internal chords: $\angle = \frac{1}{2}(\text{arc}_1 + \text{arc}_2)$
- External secants/tangents: $\angle = \frac{1}{2}(\text{far} - \text{near})$

Analytic geometry

- Canonical: $(x-a)^2 + (y-b)^2 = r^2$
- General: $x^2 + y^2 + Dx + Ey + F = 0$
- $a = -D/2$, $b = -E/2$, $r = \sqrt{D^2/4 + E^2/4 - F}$
- Radical axis: $(D_1 - D_2)x + (E_1 - E_2)y + (F_1 - F_2) = 0$
- Polar of (x_0, y_0) : $x_0x + y_0y = r^2$ (La Hire: symmetric)

Trigonometry and notations

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$
- $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$
- Prosthaphaeresis: $\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}$
- $\cos^{-1} x := \arccos x \neq (\cos x)^{-1} = \sec x = 1/\cos x$
- $a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \varphi)$

Triangle and circumference

- Pythagoras: $a^2 + b^2 = c^2$
- Thales: $R = c/2$; inscribed: $r = (a+b-c)/2$
- Sines: $a/\sin A = 2R$; Cosine: $c^2 = a^2 + b^2 - 2ab \cos C$
- Euler: $OI^2 = R^2 - 2Rr$; $R \geq 2r$
- Pythagorean triples: $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$
- 9 points (Feuerbach): centre = midpoint OH , radius $R/2$
- Heron: $K = \sqrt{s(s-a)(s-b)(s-c)}$; also $K = \frac{abc}{4R} = rs$
- Briggs: $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$
- Carnot: dist. from O to sides sum to $R + r$

Inversion and advanced circles

- Inversion: $|OP| \cdot |OP^*| = k^2$
- Ptolemy: $AC \cdot BD = AB \cdot CD + BC \cdot DA$
- Descartes: $(k_1 + k_2 + k_3 + k_4)^2 = 2 \sum k_i^2$

- Apollonius: tangent to 3 objects, generally 8 solutions
- Poncelet–Steiner: ruler alone + 1 circle with centre (6 lemmas, proved)

Robotics, CNC and numerical methods

- Yaskawa MOVIC; FANUC CIRCULAR
- ABB MoveC; G-code G02/G03
- PLCopen: `MC_MoveCircularAbsolute`
- CORDIC: $x_{i+1} = x_i - d_i y_i 2^{-i}$, $y_{i+1} = y_i + d_i x_i 2^{-i}$
- Midpoint raster: $p_k = (x_k+1)^2 + (y_k - \frac{1}{2})^2 - r^2$
- Kåsa (circle fit): linear system in u, v, w
- $a=u/2$, $b=v/2$, $r=\sqrt{w+a^2+b^2}$
- $\sigma_q^2 = \Delta^2/12$; $\varepsilon \approx 2.22 \times 10^{-16}$

Lattice & constant-width curves

- Pick: $A = I + B/2 - 1$
- Ford circles: radius $1/(2q^2)$ at p/q
- Reuleaux triangle: constant width, not a disc

Automation: arcs, linkages, gears

- Offset arc: $r \pm R_u$; feed: $F(r \pm R_u)/r$
- Helix length: $\sqrt{(2\pi r)^2 + p^2}$
- ICR: $\rho = \frac{b}{2} \frac{v_r + v_l}{v_r - v_l}$, $\omega = \frac{v_r - v_l}{b}$
- Ackermann: $\cot \delta_o - \cot \delta_i = w/L$
- Pure pursuit: $\rho = L_d^2/(2y)$
- Grashof: $s + l \leq p + q$ (shortest link rotates fully)
- Willis: $\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{z_r}{z_s}$

Water treatment (circular tanks)

- $q_A = Q/(\pi r^2)$; $q_L = Q/(2\pi r)$
- Hazen: retained $\iff v_s \geq Q/(\pi(r^2 - \rho_f^2))$
- $T = V/Q = h/q_A$ (independent of depth for retention)

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